FORM 7A

THE PATENTS ACT, 1970

(39 of 1970)

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THE PATENTS RULES, 2003

Representation for Opposition to Grant of Patent

[See Section 25(1) and rule 55]

We, **Pinakin Engineers Private Limited**, having our office at Plot No. 249, Shed No. A1/29, Phase 2, GIDC, Vapi, Gujarat – 396195, India,

do hereby give a representation by way of opposition to the grant of patent under application No. **202141018455** dated 21.04.2021, titled "*CLEAN ROOM GLASS LINED REACTOR*", published on 28/10/2022 in the name of Standard Glasslining Technology Private Limited, having its address at D1-12, Phase I, IDA, Jeedimetla, Hyderabad, Telangana- 500055, India.

on the grounds:-

1. **Section 25(1)(b)**: that the invention as far as claimed in any claim of the complete specification has been published before the priority date of the claim: -
    i. in any specification filed in pursuance of an application for a patent made in India on or after the 1st day of January, 1912; or
    ii. in India or elsewhere, in any other document: Provided that the ground specified in sub-clause (ii) shall not be available where such publication does not constitute an anticipation of the invention by virtue of sub- section (2) or sub-section (3) of section 29.

2. **Section 25(1)(e)**: that the invention so far as claimed in any claim of the complete specification is obvious and clearly does not involve any inventive step, having regard to the matter published as mentioned in clause (b) or having to regard to what was used in India before the priority date of the applicant's claim

3. **Section 25(1)(f):** that the subject of any claim of the complete specification is not an invention within the meaning of this Act or is not Patentable under this Act.

4. **Section 25(1)(g)**: that the complete specification does not sufficiently and clearly describe the invention or the method by which it is to be performed.

Our address for service for this opposition is as follows:

Hemant Thadhani (IN/PA 4057)
M/s. Crawford Bayley & Co.
State Bank of India Building,
N.G.N. Vaidya Marg, Fort, Mumbai – 400 023,
Maharashtra, India,
Email : hemant.thadhani@crawfordbayley.com

Dated this 14th day of November 2024

Thanking you.
Yours faithfully,

For Pinakin Engineers Private Limited
By its Agent

HEMANT THADHANI
Digitally signed by HEMANT THADHANI
Date: 2024.11.18 15:27:13 +05'30'

Hemant Thadhani
Patent Agent ((IN/PA-4057)
Crawford Bayley & Co.

To
Controller of Patents
The Patent Office, Mumbai

Date: 14 November 2024

ONLINE SUBMISSION

To,
The Controller of Patents,
The Patent Office,

Dear Sir,

Representation for an Opposition under Section 25(1) of the Patents Act, 1970 read with rule 55 of the Patent Rules, 2003 against Application No. 202141018455

Reference:
i. Application No.: **202141018455** filed on **21.04.2021**
ii. Title: **"CLEAN ROOM GLASS LINED REACTOR"**
iii. Applicant: **Standard Glasslining Technology Private Limited**

We, **Pinakin Engineers Private Limited**, submit herewith a representation for an under Section 25(1) of the Patents Act, 1970 read with rule 55 of the Patent Rules, 2003, on **Form 7A** and Exhibits A through G in respect of the captioned subject matter.

We enclose herewith a copy of Power of Attorney in our favour to act on behalf of Pinakin Engineers Private Limited, the Opponent herein.

We humbly request you to please take the representation on record and take necessary steps as stipulated under the Patents Act, 1970.

Yours sincerely,

FOR **Pinakin Engineers Pvt. Ltd.**

By its Agent

HEMANT THADHANI
Digitally signed by HEMANT THADHANI
Date: 2024.11.18 15:24:41 +05'30'

Hemant Thadhani
Patent Agent (IN/PA-4057)
M/s. Crawford Bayley & Co.
State Bank Buildings,
NGN Vaidya Marg
Fort, Mumbai-400037,
Maharashtra, India

ENCLOSED
1) Form 7A;
2) Written Statement a/w evidence; &
3) Power of Attorney

BEFORE THE CONTROLLER OF PATENTS

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Dated this 14th day of November 2024

For Pinakin Engineers Pvt. Ltd.

By its Agent

HEMANT THADHANI

Digitally signed by HEMANT THADHANI
Date: 2024.11.18 15:25:13 +05'30'

Hemant Thadhani

Patent Agent ((IN/PA-4057)

Crawford Bayley & Co.

FORM 7A

THE PATENTS ACT, 1970

(39 of 1970)

&

THE PATENTS RULES, 2003

Representation for Opposition to Grant of Patent

[See Section 25(1) and rule 55]

We, **Pinakin Engineers Private Limited**, having our office at Plot No. 249, Shed No. A1/29, Phase 2, GIDC, Vapi, Gujarat – 396195, India,

do hereby give a representation by way of opposition to the grant of patent under application No. **202141018455** dated 21.04.2021, titled "*CLEAN ROOM GLASS LINED REACTOR*", published on 28/10/2022 in the name of Standard Glasslining Technology Private Limited, having its address at D1-12, Phase I, IDA, Jeedimetla, Hyderabad, Telangana- 500055, India.

on the grounds:-

1. **Section 25(1)(b)**: that the invention as far as claimed in any claim of the complete specification has been published before the priority date of the claim: -
    i. in any specification filed in pursuance of an application for a patent made in India on or after the 1st day of January, 1912; or
    ii. in India or elsewhere, in any other document: Provided that the ground specified in sub-clause (ii) shall not be available where such publication does not constitute an anticipation of the invention by virtue of sub- section (2) or sub-section (3) of section 29.

2. **Section 25(1)(e)**: that the invention so far as claimed in any claim of the complete specification is obvious and clearly does not involve any inventive step, having regard to the matter published as mentioned in clause (b) or having to regard to what was used in India before the priority date of the applicant's claim

3. **Section 25(1)(f):** that the subject of any claim of the complete specification is not an invention within the meaning of this Act or is not Patentable under this Act.

4. **Section 25(1)(g)**: that the complete specification does not sufficiently and clearly describe the invention or the method by which it is to be performed.

Our address for service for this opposition is as follows:

Hemant Thadhani (IN/PA 4057)
M/s. Crawford Bayley & Co.
State Bank of India Building,
N.G.N. Vaidya Marg, Fort, Mumbai – 400 023,
Maharashtra, India,
Email : hemant.thadhani@crawfordbayley.com

Dated this 14th day of November 2024

Thanking you.
Yours faithfully,

For Pinakin Engineers Private Limited
By its Agent

HEMANT THADHANI
Digitally signed by HEMANT THADHANI
Date: 2024.11.18 15:25:38 +05'30'

Hemant Thadhani
Patent Agent ((IN/PA-4057)
Crawford Bayley & Co.

To
Controller of Patents
The Patent Office, Chennai

BEFORE THE CONTROLLER OF PATENTS,
THE PATENT OFFICE, CHENNAI

In the matter of Section 25(1) of The Patents Act, 1970, as amended by The Patent (Amendment) Act, 2005

-And-

In the matter of Patent Rules, 2003 as amended by The Patent (Amendment) Rules, 2021

-And-

In the matter of
Patent application No.
202141018455 dated 21.04.2021,
titled "*Clean room glass lined reactor*", in the name of Standard Glasslining Technology Private Limited, D1-12, Phase I, IDA, Jeedimetla, Hyderabad, Telangana- 500055, India,
……Applicant

-And-

In the matter of representation by way of pre-grant opposition by
Pinakin Engineers Private Limited,
having our office at Plot No. 249, Shed No. A1/29, Phase 2, GIDC, Vapi, Gujarat – 396195, India,
……Opponent

IN THE MATTER OF

THE PATENTS ACT, 1970

THE PATENTS RULES 2003

IN THE MATTER OF a pre-grant opposition under Section 25(1)

AND

IN THE MATTER OF Patent Application No. **202141018455** dated 21.04.2021

AND

IN THE MATTER OF:

Pinakin Engineers Private Limited … Opponent

Vs.

Standard Glasslining Technology Pvt. Ltd. …Applicant

WRITTEN STATEMENT OF OPPOSITION

1. We, **Pinakin Engineers Private Limited**, having our office at Plot No. 249, Shed No. A1/29, Phase 2, GIDC, Vapi, Gujarat – 396195, India, (hereinafter referred to as the "**Opponent**") hereby file a representation challenging the grant of Patent under application No. 202141018455 (hereinafter referred to as the "**impugned patent application**") dated 21.04.2021 in the name of Standard Glasslining Technology Pvt. Ltd, of D1-12, Phase I, IDA, Jeedimetla, Hyderabad, Telangana- 500055, India. The impugned application was published on **28/10/2022** and is not granted yet, accordingly, the present representation is filed prior to the grant under Section 25(1) of the Patents Act, 1970 (hereinafter to as "**the Act**").

ABOUT THE OPPONENT

2. The Opponent is a company incorporated in the year 2023 under the Companies Act, 2013. The Opponent operated as a partnership firm since 2006 prior to its incorporation in 2023. The Opponent is a leading manufacturer of a range of filtration and drying equipment, Glass Lined

Equipment, etc., with global presence. The Opponent is specialized in glass lined equipment, including glass lined reactors.

LOCUS STANDI

3. Under Section 25(1) of the Patents Act, 1970, any person is entitled to file a representation seeking rejection of a pending patent application. The Opponent, therefore, qualifies to be a person under the requirements of Section 2(1)(t) of the Patents Act, 1970 and challenges the application for grant of the impugned patent application.

The impugned patent application (202141018455)

4. For the purposes of this opposition, the Opponent humbly seeks to rely upon the copy of the complete specification of the impugned patent application available at and downloaded from the InPASS database made available by the Patent Office website.

JURISDICTION

5. The Application for the grant of the impugned patent application was filed at the Patent Office in Chennai. Therefore, the Chennai Patent Office has jurisdiction to take cognizance of the present pre-grant opposition and adjudicate the matter relying on the present representation on Form 7A of the Patent Rules, 2003 and the evidence submitted thereof.

6. **GROUNDS OF OPPOSITION - The Patent is opposed on the following grounds under Section 25(1):**

    i. **Section 25(1)(b)**: that the invention as far as claimed in the claim of the complete specification has been published before the priority date of the claim: -
        a. in any specification filed in pursuance of an application for a Patent made in India on or after the 1st day of January 1912; or
        b. in India or elsewhere in any other document.

ii. **Section 25(1)(e)**: that the invention so far as claimed in the claims of the complete specification is obvious and clearly does not involve any inventive step, having regard to matter published in clause 25(1)(b) or having regard to what is used in India before the priority date of the Applicant's claim.

iii. **Section 25(1)(f):** that the subject of any claim of the complete specification is not an invention within the meaning of this Act or is not Patentable under this Act.

iv. Section 25(1)(g): that the complete specification does not sufficiently and clearly describe the invention or the method by which it is to be performed

THE CLAIMS ON FILE OF IMPUGNED PATENT APPLICATION TITLED "CLEAN ROOM GLASS LINED REACTOR"

7. *The as filed claims of the impugned patent application filed at the date of application are reproduced below for reference and convenience:-*

1. An improved mild steel reactor comprising of:

i. an internally glass lined reactor sandwiched between a top flange and a bottom flange, wherein fluidic channels are machined on said glass lined reactor.

ii. a glass lined metal flat surface below the top flange that allows to observe flow regime in the glass lined reactor or mixer.

iii. a cavity in the bottom flange that facilitates flow of heat transfer fluid through an inlet and an outlet nozzles and flanges, wherein said cavity can be closed with a bottom plate having a metal surface, wherein the metal surface of said bottom plate can touch the heat transfer fluid;

iv. a connection plate connecting the inlet and the outlet to the bottom flange of the glass lined stainless steel reactor; and

v. multiple bolts to hold tightly the top flange and the bottom flange; wherein the external surface of the said reactor is covered and overlaid with stainless steel sheet by welding.

2. *The reactor as claimed in claim 1, wherein the outer wall of the reactor consists of a base body of high mechanical strength, namely steel or stainless steel, the base body having a reaction chamber of stainless steel with a layer thickness > 1 mm, preferably in the range from essentially 1 mm to 5 mm and most preferably 2mm.*

3. *The reactor as claimed in claim 1, wherein the said SS body is made by welding the entire surface with SS welding electrodes after removing proportionate metal from the external surface of the reactor.*

4. *The reactor as claimed in claim 1, wherein the said nozzles or flanges made on or in the reactor are internally glass lined.*

5. *The reactor as claimed in claim 1, wherein the glass lining is configured for a non-reactive cladding, pressure-welded coating.*

6. *The reactor as claimed in claim 1, wherein the base body is designed as a doublewalled vessel.*

7. *The glass lined Stainless steel-reactor as claimed in claim 1, wherein the surface of the glass lining / coating is smooth as whole.*

8. *The reactor as claimed in claim 1, wherein said reactor has capacity ranging from 1Litre to 12500litre for AE type reactor and 1000Litre to 32000Litre for CE type of reactor and heat transfer area ranging from 0.5 m2 TO 34.6 m2.*

9. *The reactor as claimed in claim 1, wherein said micro-reactor can withstand pressure up to 6kg/cm2 and temperature ranging from −60° to 343° C.*

.

10. *The reactor as claimed in claim 1, wherein thickness of said glass lining ranges from 1 mm to 2 mm."*

BIBLIOGRAPHIC DATA OF THE IMPUGNED PATENT APPLICATION

8. It is submitted that the impugned patent application relates to an invention titled *"CLEAN ROOM GLASS LINED REACTOR"*. The details of the impugned patent application are as follows:

Title	CLEAN ROOM GLASS LINED REACTOR
Applicant	Standard Glass Lining Technology Private Limited
Application No.	202141018455
Application filed on	21.04.2021
Request for Examination filed on	21.04.2021
Application published on	28.10.2022
First Examination Report (FER) issued on	10.11.2022
Current Status	Reply Filed. Application in amended examination

BACKGROUND OF THE CASE

9. The Patent application bearing no. **202141018455** (hereinafter referred to as the "impugned **Patent Application**") titled as "CLEAN ROOM GLASS LINED REACTOR" was filed by Standard Glass Lining Technology Private Limited, (hereinafter referred to as the "**Applicant**") on 21.04.2021. The Patent Application was filed on 21.04.2021 as a complete application, and the as-filed Complete Application, including the description, claims, abstract and drawings, is annexed herein as **Exhibit – A (colly)**.

10. The request for examination in the imagined Patent Application was filed on 21.04.2021 and a First Examination Report (FER) was issued on

10.11.2022 whereby the Patent Office raised several objections *inter alia* that

i. Claims 1-9 lack inventive step as being obvious to a person skilled in the art in view of the cited Document D1: CN1583239A 23/02/2005; D2: CN103691349A 02/04/2014; and D3: WO2021063359A1 08/04/2021;

i. The statement of claims should, therefore, be revised & all essential features of the invention should be brought in the principal claim 1 while subsidiary features of the invention may be claimed in dependent claims;

ii. Independent/ Principal Claims have to be duly characterized in order to delineate the inventive part from the known features;

iii. Claim 1 should be drafted in such a manner that it describes the all-main feature of invention and then dependent claims elaborate these features

iv. New subject matter added in dependent claims make them unclear and indefinite, so all the main features of the invention must be added in independent claim;

v. The relevant prior art documents known to the applicant should be identified in the description and the distinguishing features of the present application with reference to those prior art should be briefly discussed

11. In response, the Applicant filed a response dated 10.08.2023 to the FER and the Applicant:-

i. amended the original claims 1-8 and 12;

ii. deleted claims 11, and 13-25;

iii. characterized the claim 1;

iv. made all the dependent claims dependent to claim 1 in one way or the other;

As per the current status and documents on record, the impugned Patent application is pending in amended examination.

A copy of the FER dated 23.03.2023 and Response to the FER dated 23.09.2023 is annexed herein as **Exhibit B (colly)**. A copy of the amended complete specification including amended description, amended claims and amended drawings, is annexed herein as **Exhibit C (colly).**

12. It is submitted that the amended specification and claims submitted by the Applicant are violative of the Patents Act and as such, claim or describe matters that are in substance not disclosed in the original specification and/or claims. On this ground itself, the amendments are liable to be rejected.

13. Even otherwise and without prejudice to the above said, the as filed amended claims 1-8 of the impugned patent application lack novelty and inventive step and as such are liable to be rejected. The as filed amended claims are reproduced below for reference and convenience:-

"We Claim :

1. A Mild Steel (MS) Glass-Lined reactors with stainless steel overlays on nozzles comprising of::

i. an internally glass lined reactor sandwiched between a top flange and a bottom flange, wherein fluidic channels are machined on said glass lined reactor.

ii. a glass lined metal flat surface below the top flange that allows to observe flow regime in the glass lined reactor or mixer.

iii. a cavity in the bottom flange that facilitates flow of heat transfer fluid through an inlet and an outlet nozzles and flanges, wherein said cavity can be closed with a bottom plate having a metal surface, wherein the metal surface of said bottom plate can touch the heat transfer fluid;

iv. a connection plate connecting the inlet and the outlet to the bottom flange of the glass lined stainless steel reactor; and

v. multiple bolts to hold tightly the top flange and the bottom flange; wherein the external surface of the said reactor is covered and overlaid with stainless steel sheet by welding;

wherein coated nozzle is seamlessly welded to the top cover's cladding sheet, fully enclosing the reactor in stainless steel tackling rust challenges in critical zones resulting a resilient barrier that upholds reactor integrity, longevity, and product purity

and ***Characterized in that said reactor*** *offers excellent resistance to oxidation at the nozzle and in the event of an accidental spillage of corrosive liquid onto the nozzle surface, which might otherwise lead to oxidation on the metal surface, the glass lining which is coated inside would be damaged as the base metal is oxidated and would result in distortion and wherein the the reactor is fully covered with stainless steel, enhancing its resilience against external damage and thereby increasing its overall safety.*

2. *The reactor as claimed in claim 1, wherein the outer wall of the reactor consists of a base body of high mechanical strength, namely steel or stainless steel, the base body having a reaction chamber of stainless steel with a layer thickness > 1 mm, preferably in the range from essentially 1 mm to 5 mm and most preferably 2mm.*

3. *The reactor as claimed in claim 1, wherein the said SS body is made by welding the entire surface with SS welding electrodes after removing proportionate metal from the external surface of the reactor.*

4. *The reactor as claimed in claim 1, wherein the said nozzles or flanges made on or in the reactor are internally glass lined.*

5. *The reactor as claimed in claim 1, wherein the glass lining is configured for a non-reactive cladding, pressure-welded coating.*

6. *The reactor as claimed in claim 1, wherein the base body is designed as a doublewalled vessel.*

7. *The reactor as claimed in claim 1, wherein the surface of the glass lining / coating is smooth as whole.*

8. *The reactor as claimed in claim 1, wherein said reactor has capacity ranging from 1Litre to 12500litre for AE type reactor and 1000Litre to 32000Litre for CE type of reactor and heat transfer area ranging from 0.5 m2 TO 34.6 m2.*

ABOUT THE IMPUGNED PATENT APPLICATION

14. The impugned application primarily relates to a "*Mild Steel (MS) Glass-Lined reactors with stainless steel overlays on nozzles"*. It is pertinent to note that claim 1 has been characterized to pin-point the only alleged novel and inventive features of the invention i.e. "*Characterized in that said reactor offers excellent resistance to oxidation at the nozzle and in the event of an accidental spillage of corrosive liquid onto the nozzle surface, which might otherwise lead to oxidation on the metal surface, the glass lining which is coated inside would be damaged as the base metal is oxidated and would result in distortion and wherein the the reactor is fully covered with stainless steel, enhancing its resilience against external damage and thereby increasing its overall safety".*

15. In this connection, the Opponent states that it is a well-established law, that the pre-characterized portion is considered to be part of the prior art and it is only the post-characterized part which is considered to be the novel and inventive step. This has been clearly envisaged in the Guidelines for Search and Examination of Patent, 2014 issued by the *Office of Controller General of Patents, which reads as follows:-*

> *"e) If the invention is an improvement to an existing product,* ***the claims should set the boundary very clearly by characterizing the invention with respect to the prior art****. In those cases, the claim will have two parts separated by the word 'characterized by' or 'wherein'.* ***The part coming before 'characterized by' is the prior art while that comes after will be the features of the invention.*** *It is equally applicable in the case of a process which is modification of the existing process."*

A copy of Guideline issued by the Patent Office is annexed herein as **Exhibit D**. The Opponent craves leave to refer to and rely upon judgments / further guidelines issued by USPTO and EPO in this regard, if necessary, at a later stage.

16. In view of the above, the Opponent submits that the novelty and inventive step for the impugned patent application has to be seen from the part of claim appearing after the characterising portion in independent claim 1 and the features preceding the characterising portion of claim 1 are all part of prior art / common knowledge. A bare reading of the claim would show that the characterising portion only discloses that the reactor disclosed in the impugned application offers excellent resistance to oxidation. It is submitted that the characterising portion of the claim, thus, discloses no features which may qualify to be novel or inventive. On this ground itself, the present opposition should be allowed and the principal independent claim 1 is liable to be rejected. Further, claims 2 to 8 are dependent on claim 1 and do not have any independent existence.

17. Without prejudice to the above said, the so called novel and inventive feature appearing after the characterising portion in claim 1 is deemed anticipated in view of prior being D1, more particularly set-out hereinafter.

18. For the purpose of this opposition, the Opponent opposes the grant of amended claims 1 to 8, which were amended by the Applicant in response to the objections raised in the FER.

GROUNDS OF OPPOSITION

A. Subject matter of the Impugned patent application lacks novelty as the invention as far as claimed in the claim of the complete specification has been published before the priority date of the claims in a document – Lack of novelty and Anticipation under Section 25(1)(b)

19. It is submitted that Section 2(1)(j) of the Act defines an ***"invention"*** *as "a new product or process involving an inventive step and capable of industrial application"* (emphasis added). Section 25(1)(b)(ii) of the Act provides ground for opposing an application for patent if the alleged invention, as claimed in any claim of the complete specification has been published before the priority date of the claim in India or elsewhere, in any other document. A patent application shall be invalid for anticipation if a single prior art reference discloses each and every limitation of the claimed invention. Further, the prior art reference may anticipate without disclosing a feature of the claimed invention if that missing characteristic is necessarily present, or inherent, in the single anticipating prior art.

DOCUMENT D1

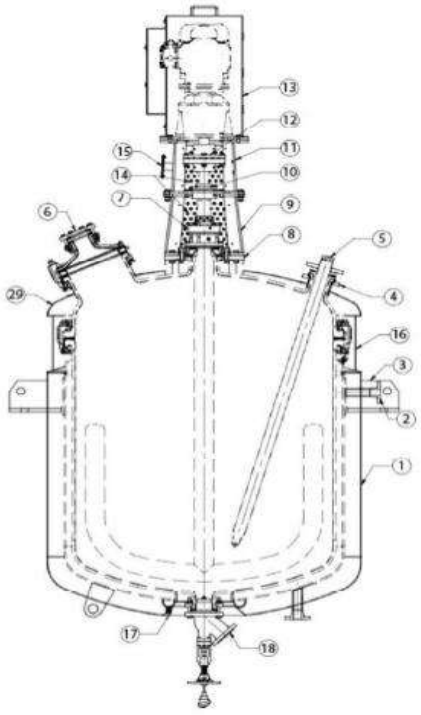
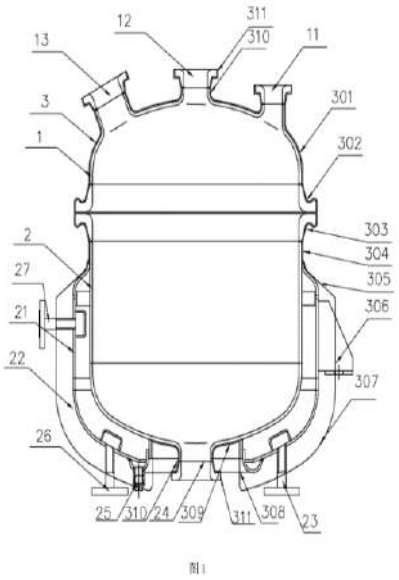
20. Without prejudice to other grounds raised herein, it is submitted that claims 1-8 of the Impugned patent application lack novelty in light of the prior publication discussed in detail below.

21. The Opponent refers to and relies upon the granted Chinese Patent **CN103820781A** having priority date 12.03.2014 titled ***"Glass lined equipment provided with outer profiling cladding and manufacturing method of glass lined equipment"*** available at https://worldwide.espacenet.com/patent/search?q=pn%3DCN103820781A

[hereinafter referred to as "**Document D1"**]. The same is annexed herein as **Exhibit 'E'**.

22. The Opponent submits that D1 in Figure 1 (see below) discloses or teaches a glass lined equipment provided with an outer profiling cladding and a manufacturing method of the glass lined equipment. The glass lined equipment mainly comprises a tank cover and a tank body, wherein an outer stainless steel cladding layer is arranged on the outer side of the tank cover, outer stainless steel cladding layers are arranged at the upper part of the outer side of the tank body and the outer side of a jacket, a gap between the jacket and the stainless steel outer cladding layers is filled with a thermal insulation material, and a thermal insulation layer is formed. GMP (good manufacturing practice) certification requirements that the production equipment cannot have any adverse effect on quality of medicine and the surface of production equipment directly contacted with medicine is required to be flat, smooth, easy to clean or disinfect and corrosion-resistant are met, and the production cost is lower than that of austenitic stainless steel glass lined equipment". In view of D1, the impugned Patent application that claimed a *"Mild Steel (MS) Glass-Lined reactors with stainless steel overlays on nozzles"* as claimed in claim 1 of the impugned Patent application is deemed anticipated. A comparison of the individual features of independent claims of the Impugned patent application and disclosure of D1 is given in Table below:

Claim	**Key Features of impugned Patent application**	**Relevant Text in Document D1 CN103820781A**
1	A Mild Steel (MS) Glass-Lined reactors with stainless steel overlays on nozzles	Glass lined equipment provided with outer profiling cladding and manufacturing method of glass lined equipment

	 Figure 1	
Indep. Claim 1	**Claim 1 claims "A mild steel Glass-Lined Reactors with stainless steel overlays on nozzles** comprising" i. an internally glass lined reactor sandwiched between a top flange and a bottom flange, wherein fluidic channels are machined on said glass lined reactor.	The description of D1 discloses / teaches that *"<u>The utility model relates to a profiled outer glass-lined device</u>, which mainly comprises a can lid and a can body, wherein the can lid is provided with a nozzle, a manhole and a sealing mouth; the outer part of the can body is covered with a jacket, the can body and the clip The sleeve is welded into a unitary structure with a cavity therebetween, a jacket flange is provided on one side of the tank body, and a discharge port, a condensate outlet, a steam inlet and*

		a steam outlet are arranged at a lower portion;" *"[The outer side of the outer side of the can body and the outer side of the jacket have a stainless steel outer layer, and the upper part of the stainless steel outer layer of the outer side of the can body is covered with a can body flange surface and a can body surface I stainless steel outer layer, and the can body and the jacket welding portion There is a canned transition surface I stainless steel outer layer, the outer side of the jacket is covered with a can body surface II stainless steel outer layer, and the lower part of the can body is covered with a stainless steel outer layer of the can body sealing surface, and the lower part of the can body and the jacket welding portion are covered The can body transition surface II stainless steel outer layer, the bottom of the can body is covered with a can body surface III stainless steel outer layer*" In view of the above, the features disclosed in claim 1 are anticipated.
Indep. Claim 1	ii. a glass lined metal flat surface below the top flange that allows to observe flow regime in the glass lined reactor or mixer.	The claimed feature is disclosed in D1 as a manhole, which provides access to the interior of the reactor.
Indep. Claim 1	iii. a cavity in the bottom flange that facilitates flow of heat transfer fluid through an inlet and an outlet nozzles and flanges, wherein said	The Description of D1 discloses that *"As shown in FIG. 1-2, a profiled outer glass-lined device mainly*

	cavity can be closed with a bottom plate having a metal surface, wherein the metal surface of said bottom plate can touch the heat transfer fluid;	*comprises a can lid 1 and a can body 2, and the can lid 1 is provided with a nozzle 11, a manhole 13 and a sealing port 12; 2 The outer lower part is covered with a jacket 21, and the can body 2 and the jacket 21 are welded into an integral structure with a cavity therebetween, and a side of the can body 2 is provided with a jacket flange 27 and a lower portion is disposed."* In view of the above, the features disclosed in claim 1 are anticipated.
Indep. Claim 1	iv. a connection plate connecting the inlet and the outlet to the bottom flange of the glass lined stainless steel reactor; and	There is no novelty or inventive step in providing a connection plate connecting the inlet and the outlet to the bottom flange of the glass lined stainless steel reactor and as such, the claimed connection plate lacks novelty or inventive step.
Indep. Claim 1	v. multiple bolts to hold tightly the top flange and the bottom flange; wherein the external surface of the said reactor is covered and overlaid with stainless steel sheet by welding.	Claim 1 discloses that *"The nozzle, the manhole, the sealing port and the discharge opening are covered with a stainless steel outer cladding composed of a flange high neck surface and a flange surface;* *The stainless steel outer cladding is a welded joint;"*

		In view of the above, the features disclosed in claim 1 are anticipated.
Indep. Claim 1	wherein coated nozzle is seamlessly welded to the top cover's cladding sheet, fully enclosing the reactor in stainless steel tackling rust challenges in critical zones resulting a resilient barrier that upholds reactor integrity, longevity, and product purity	Claim 1 discloses that *"The nozzle, the manhole, the sealing port and the discharge opening are covered with stainless steel outer cladding composed of a flange high neck surface and a flange surface;* *The stainless steel outer cladding is a welded joint;"* In view of the above, the features disclosed in claim 1 are anticipated.
Indep. Claim 1	and Characterized in that said reactor offers excellent resistance to oxidation at the nozzle and in the event of an accidental spillage of corrosive liquid onto the nozzle surface, which might otherwise lead to oxidation on the metal surface, the glass lining which is coated inside would be damaged as the base metal is oxidated and would result in distortion and wherein the the reactor is fully covered with stainless steel, enhancing its resilience against external damage and thereby increasing its overall safety.	Claim 1 of D1 discloses "The nozzle, the manhole, the sealing port and the discharge opening are covered with a stainless steel outer cladding com posed of a flange high neck surface and a flange surface;" Further, Description of D1 discloses that: *"The invention has the beneficial effects of providing a profiled outsourcing glass-lined device* ***with a stainless steel outer layer on the outside of the can lid and the can body****. …… The surface of the*

		production equipment should be smooth, smooth, easy to clean or disinfect, ***and corrosion resistant****, and the production cost is lower than that of austenitic stainless steel enamel equipment."* In view of the above, the features disclosed in claim 1 are anticipated.
Depen. Claim 2	2. The reactor as claimed in claim 1, wherein the outer wall of the reactor consists of a base body of high mechanical strength, namely steel or stainless steel, the base body having a reaction chamber of stainless steel with a layer thickness > 1 mm, preferably in the range from essentially 1 mm to 5 mm and most preferably 2mm.	Claim 2 of D1 discloses:- 2. A contoured outer glass-lined device according to claim 1, wherein <u>the outside of the can lid has a stainless steel outer layer having a thickness of 1.5 to 3 mm, and the outside of the can body has a thickness of stainless steel outer layer of 3 mm or more</u>. In view of the above, the features disclosed in claim 2 are anticipated.
Depen. Claim 3	3. The reactor as claimed in claim 1, wherein the said SS body is made by welding the entire surface with SS welding electrodes after removing proportionate metal from the external surface of the reactor	D1 discloses that *"The invention relates to a glass-lined device and a manufacturing method thereof, in particular to a glass-lined device which has a stainless steel profiled outsourcing and meets the requirements of pharmaceutical GMP cleanliness."*

		In view of the above, the features disclosed in claim 3 are anticipated.
Depen. Claim 4	4. The reactor as claimed in claim 1, wherein the said nozzles or flanges made on or in the reactor are internally glass lined.	D1 discloses that *"The utility model relates to a profiled outer glass-lined device, which mainly comprises a can lid and a can body, wherein the can lid is provided with a nozzle, a manhole and a sealing mouth; the outer part of the can body is covered with a jacket, the can body and the clip The sleeve is welded into a unitary structure with a cavity therebetween, a jacket flange is provided on one side of the tank body, and a discharge port, a condensate outlet, a steam inlet and a steam outlet are arranged at a lower portion;"* In view of the above, the features disclosed in claim 4 are anticipated
Depen. Claim 5	5. The reactor as claimed in claim 1, wherein the glass lining is configured for a non-reactive cladding, pressure-welded coating.	D1 discloses that *"The nozzle, the manhole, the sealing port and the discharge opening are covered with a stainless steel outer cladding comp osed of a flange high neck surface and a flange surface; The stainless steel outer cladding is a welded joint;".* Thus, claim 5 is neither novel nor inventive and liable to be rejected.

Depen. Claim 6	6. The reactor as claimed in claim 1, wherein the base body is designed as a double- walled vessel.	D1 also discloses a doubled walled vessel structure of the reactor in its Description. The relevant portion of the description is reproduced as follows:- *"The can lid surface has a stainless steel outer layer, and the stainless steel outer layer upper can lid head surface and the can lid flange surface together form a can lid stainless steel outer layer;* *The outer side of the outer side of the can body and the outer side of the jacket have a stainless steel outer layer, and the upper part of the stainless steel outer layer of the outer side of the can body is covered with a can body flange surface and a can body surface I stainless steel outer layer, and the can body and the jacket welding portion"* In view of the above, the features disclosed in claim 6 are anticipated
Depen. Claim 7	7. The reactor as claimed in claim 1, wherein the surface of the glass lining / coating is smooth as whole.	Claim 7 of D1 discloses:- *"7） The stainless steel weld is also polished to 320 mesh and then polished to make the surface finish of*

		the stainless steel plate meet the requirements;" Thus, claim 7 of the impugned application is anticipated by the disclosure contained in D1.
Depen. Claim 8	8. The reactor as claimed in claim 1, wherein said reactor has capacity ranging from 1Litre to 12500litre for AE type reactor and 1000Litre to 32000Litre for CE type of reactor and heat transfer area ranging from 0.5 m2 TO 34.6 m2.	The Claim 8 of the impugned patent application merely discloses the capacity of the reactor. It has no inventive features. In view of the above, the features disclosed in claim 8 are anticipated

23. Therefore, in view of the disclosure(s) in D1, the features recited in independent claims 1 of the impugned application are anticipated and therefore claim 1 ought to be rejected on this ground alone.

24. The Opponent submits that in view of D1 anticipating independent claim 1, none of the dependent claims stand on their own, which are anyway discloses in D1, and therefore ought to be summarily rejected. In the event the Applicant elects to amend the independent claim 1 by introducing any feature based on any of the dependent claims, or based on the specification, the Opponent requests the Learned Controller to allow the Opponent to make further necessary submissions to rebut the Applicant amendments.

25. The Opponent further submits that the subject matter of one or more of the dependent claims are also anticipated by prior arts cited in the present representation, which are elucidated in objection under Section 25(1)(e) and not reproduced herein for the sake of brevity and to avoid repetition. The same is incorporated herein by reference and the Opponent reserves its right to contest the novelty of the said dependent claims.

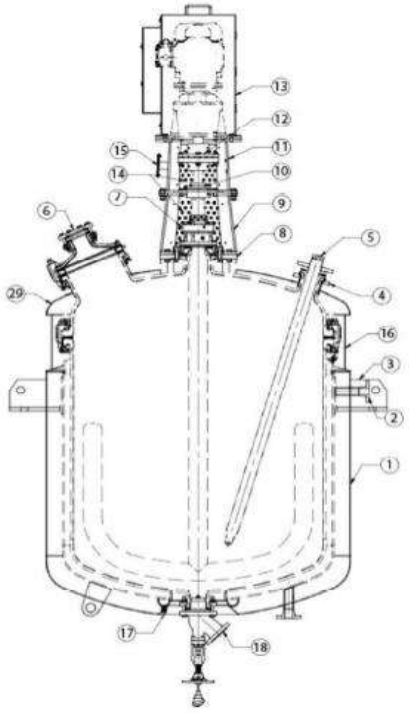
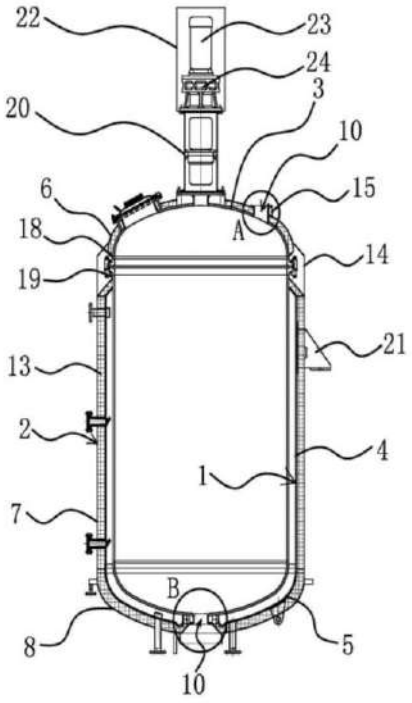
DOCUMENT D2

26. Without prejudice to other grounds raised herein, it is submitted that claims 1-8 of the Impugned patent application lack novelty in light of the prior publication discussed in detail below.

27. The Opponent refers to and relies upon the granted Chinese Patent **CN105148816A** having priority date 08.09.2015 titled ***"Manufacturing method of glass-lined reaction kettle"*** available at https://worldwide.espacenet.com/patent/search/family/054790040/publication/CN105148816A?q=CN105148816A [hereinafter referred to as "**Document D2"**]. The same is annexed herein as **Exhibit 'F'**.

28. The Opponent submits that D2 in Figure 1 (see below) discloses or teaches a manufacturing method of a glass-lined reaction kettle which solves the problems that the outer surface of the glass-lined reaction kettle is poor in corrosion resistance. The glass-lined reaction kettle comprises an upper end enclosure, a barrel and a lower end enclosure. The manufacturing method comprises seven steps of assembling, manufacturing a wrapper, perforating, fixing, packing, polishing and manufacturing and fixing a cover. The manufactured glass-lined reaction kettle has advantages of good corrosion resistance, high safety, attractive appearance and the like. In view of D2, the impugned Patent application that claimed a *"Mild Steel (MS) Glass-Lined reactors with stainless steel overlays on nozzles"* as claimed in claim 1 of the impugned Patent application is deemed anticipated.

29. A comparison of the individual features of independent claims of the Impugned patent application and disclosure of D1 is given in Table below:

Claim	Key Features of impugned Patent application	Relevant Text in Document D2 CN105148816A
1	A Mild Steel (MS) Glass-Lined reactors with stainless steel overlays on nozzles  Figure 1	Manufacturing method of glass-lined reaction kettle 
Indep. Claim 1	**Claim 1 claims "A mild steel Glass-Lined Reactors with stainless steel overlays on nozzles** comprising" i. an internally glass lined reactor sandwiched between a top flange and a bottom flange, wherein fluidic channels are machined on said glass lined reactor.	The description of D2 discloses / teaches that:- "d. Fixed: a stainless steel flange is placed on the nozzle flange of each process pipe, and the upper cover is covered from the top to the upper cover, so that the process pipe protrudes from the through hole of the upper foreskin, and then The upper foreskin is welded with the stainless steel flange, and the cylinder is wrapped with the middle foreskin, and

		the lower cover is covered from the bottom with the lower foreskin, so that the process tube protrudes from the through hole of the lower foreskin, and the lower foreskin is welded with the stainless steel flange. Finally, the lower foreskin and the middle foreskin are sealed and welded;" Further, claim 1 of D2 teaches *"1.A method for manufacturing a glass-lined reactor, the glass-lined (1) reactor comprises an upper head (3), a cylinder (4) and a lower head (5),……..."* In view of the above, the features disclosed in claim 1 are anticipated.
Indep. Claim 1	ii. a glass lined metal flat surface below the top flange that allows to observe flow regime in the glass lined reactor or mixer.	The claimed feature is disclosed in D2 as a manhole, which provides access to the interior of the reactor for observing the flow regime. In view of the above, the features disclosed in claim 1 are anticipated.
Indep. Claim 1	iii. a cavity in the bottom flange that facilitates flow of heat transfer fluid through an inlet and an outlet nozzles and flanges, wherein said cavity can be closed with a bottom plate having a metal surface,	The Description of D2 discloses that *"……...the outer surface of the kettle body is provided with an interlayer, the upper end of the kettle body is provided with a feeding port,*

	wherein the metal surface of said bottom plate can touch the heat transfer fluid;	*and the bottom of the kettle body is connected with a discharging port."* In view of the above, the features disclosed in claim 1 are anticipated.
Indep. Claim 1	iv. a connection plate connecting the inlet and the outlet to the bottom flange of the glass lined stainless steel reactor; and	There is no novelty or inventive step in providing a connection plate connecting the inlet and the outlet to the bottom flange of the glass lined stainless steel reactor and as such, the claimed connection plate lacks novelty or inventive step.
Indep. Claim 1	v. multiple bolts to hold tightly the top flange and the bottom flange; wherein the external surface of the said reactor is covered and overlaid with stainless steel sheet by welding.	The Description of D2 discloses that the reactor is covered and overlaid with stainless steel sheet by welding, which discloses :- "*Since the carbon steel of the high-efficiency glass-lined reactor is directly exposed to the outside, the outer surface of the reactor is easily corroded and not only affects the appearance of the reactor, but also the corroded rust stains may fall into the process of opening the manhole. In the glass-lined reactor, the reaction in the glass-lined reactor was affected.* *Therefore, it is necessary to coat a layer of stainless steel in carbon steel to isolate the carbon steel from the*

		outside, so that the carbon steel will not be corroded,……" In view of the above, the features disclosed in claim 1 are anticipated.
Indep. Claim 1	wherein coated nozzle is seamlessly welded to the top cover's cladding sheet, fully enclosing the reactor in stainless steel tackling rust challenges in critical zones resulting a resilient barrier that upholds reactor integrity, longevity, and product purity	*D2 discloses that :-* *"Corrosion resistance of the reactor; filling the gap between the stainless steel sheath and the bismuth glass with insulation material, and the operation of sealing welding can One step is to isolate the carbon steel on the glass from the outside, so that the corrosion resistance of the glass reactor is better; polishing the outer surface of the stainless steel sheath can make the glass reactor look more beautiful;"* In view of the above, the features disclosed in claim 1 are anticipated.
Indep. Claim 1	and Characterized in that said reactor offers excellent resistance to oxidation at the nozzle and in the event of an accidental spillage of corrosive liquid onto the nozzle surface, which might otherwise lead to oxidation on the metal surface, the glass lining which is coated inside would be damaged as the base metal is oxidated and would	D2 discloses that: *"The present invention provides a method for producing a glass-lined reactor according to the above problems in the prior art, and the technical problem to be solved by the present invention is how to produce a glass-lined reactor which is also resistant to corrosion on the outer*

	result in distortion and wherein the the reactor is fully covered with stainless steel, enhancing its resilience against external damage and thereby increasing its overall safety.	*surface."* In view of the above, the features disclosed in claim 1 are anticipated.

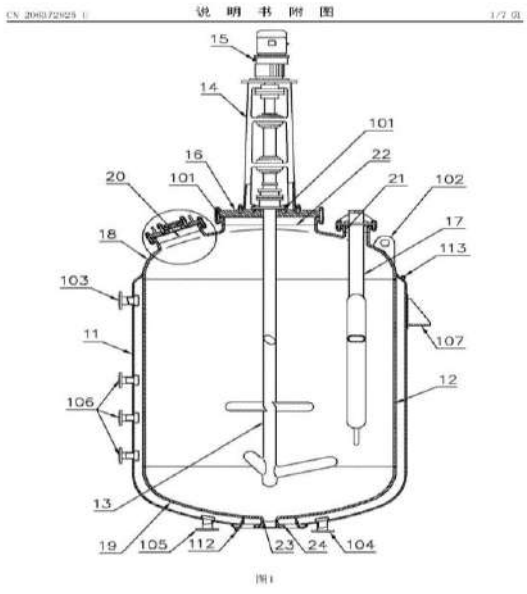
30. Therefore, in view of the disclosure(s) in D2, the features recited in independent claims 1 of the impugned application are anticipated and therefore claim 1 ought to be rejected on this ground alone.

31. The Opponent submits that in view of D2 anticipating independent claim 1, none of the dependent claims 2-8 stand on their own, which are anyway discloses in D2, and therefore ought to be summarily rejected. In the event the Applicant elects to amend the independent claim 1 by introducing any feature based on any of the dependent claims, or based on the specification, the Opponent requests the Learned Controller to allow the Opponent to make further necessary submissions to rebut the Applicant amendments.

32. The Opponent further submits that the subject matter of one or more of the dependent claims are also anticipated by prior arts cited in the present representation, which are elucidated in objection under Section 25(1)(e) and not reproduced herein for the sake of brevity and to avoid repetition. The same is incorporated herein by reference and the Opponent reserves its right to contest the novelty of the said dependent claims.

33. The Opponent further refers to and relies upon the granted Chinese Patent ***CN206372825U*** having priority date 12.01.2017 titled ***"Corrosion - resistant glass reactor that wards off of high strength"*** available at https://worldwide.espacenet.com/patent/search/family/059404215/publication/CN206372825U?q=CN206372825U [hereinafter referred to as "**Document D3"**]. The same is annexed herein as **Exhibit 'G'**.

34. The Opponent submits that D3 in Figure 1 (see below) discloses or teaches *"a corrosion -resistant glass reactor that wards off of high strength, including pressing from both sides cover, jar body, agitator, upper cover, low head, sight glass manhole, the thermometer mouth of pipe, feed inlet, drain hole, the external side of jar is equipped with presss from both sides the cover, it is equipped with a plurality of installation positioner to press from both sides angles such as overlapping the outside, pressing from both sides cover upper portion and being equipped with steam inlet, steam inlet one side is equipped with the relief port, presss from both sides the cover bottom and is equipped with water export with fixed attention, the welding has high -neck flange on the feed inlet, the welding has high -neck flange on the sight glass manhole, and the welding has high -neck flange on the thermometer mouth of pipe, and the welding has high -neck flange on the drain hole, be equipped with the ground coat on the internal lateral wall of jar, be equipped with cover glaze layer on the ground coat, be equipped with the ground coat on high -neck flange inside wall, terminal surface, the recess, be equipped with cover glaze layer on the ground coat. The utility model discloses trompil department welding has increased high -neck flange, has reduced the quantity at stamping forming trompil R position, has directly increased the effective thickness of this equipment, and compressive strength improves relatively, ensures the safe operation of equipment*.".



35. In view of D3, the impugned Patent application that claimed a *"Mild Steel (MS) Glass-Lined reactors with stainless steel overlays on nozzles"* as claimed in claim 1 of the impugned Patent application is deemed anticipated. Therefore, the features recited in independent claims 1 of the impugned application are anticipated and therefore claim 1 ought to be rejected on this ground alone.

36. The Opponent submits that in view of D3 anticipating independent claim 1, none of the dependent claims 2-8 stand on their own, which are anyway discloses in D3, and therefore ought to be summarily rejected. In the event the Applicant elects to amend the independent claim 1 by introducing any feature based on any of the dependent claims, or based on the specification, the Opponent requests the Learned Controller to allow the Opponent to make further necessary submissions to rebut the Applicant amendments.

37. The Opponent further submits that the subject matter of one or more of the dependent claims are also anticipated by prior arts cited in the present representation, which are elucidated in objection under Section 25(1)(e) and not reproduced herein for the sake of brevity and to avoid repetition. The same is incorporated herein by reference and the Opponent reserves its right to contest the novelty of the said dependent claims

B. Subject matter of the Impugned patent application is obvious in nature and does not involve any inventive step having regard to what was publicly known or used or published in India or elsewhere – Lack of inventive step and obviousness under Section 25(1)(e)

38. WITHOUT PREJUDICE to the fact that the Impugned patent application is not novel and was already known to a person skilled in the art in view of the prior art as discussed hereinabove, the invention as claimed in the Impugned patent application does not have any technical advancement over the prior art which was known at or before the date of filing of the

patent application and hence, the Impugned patent application would have been obvious to a person skilled in the art at the time of filing as set out herein under.

39. It is submitted that, the Division Bench of the Hon'ble Delhi High Court in the case of Roche v. Cipla, (2015) 225 DLT391 (DB) which judgment is a binding precedent in this proceeding, has interpreted the statutory provisions related to lack of inventive step and enunciated the following five-step test for assessment of inventive step:

    a) To identify the 'inventive concept' embodied and claimed in the Patent;
    b) To identify the 'person skilled in the art';
    c) To identify the relevant common general knowledge of such skilled person at the priority date;
    d) To identify the differences, if any, between the matter cited in the alleged invention as forming part of state of the art and the inventive concept;
    e) To decide whether those differences constituted steps that would have been obvious to a person skilled in the art.

40. Pertinently, the above test has also been provided in the "MANUAL OF PATENT OFFICE PRACTICE AND PROCEDURE" as the important points which need to be objectively judged to ascertain whether, looking at the invention as a whole, the invention does have inventive step.

41. The Opponent submits that in view of D1 anticipating the features of claim 1, the recited subject matter is consequently deemed obvious to a person skilled in the art and therefore lacks inventive step and ought to be rejected on this ground alone. The submissions made u/S 25(1)(b) are incorporated herein and not reproduced for the sake of brevity.

42. The Opponent has evidenced obviousness of claimed features of Impugned patent application compared to conjoint reading of prior art, i.e. document D1 to D3 as set-out hereinabove.

43. In this case, D1 to D3 disclose/teach the features known in the art. Considering the aspect that documents at Exhibits E to G disclose the key features of the impugned patent application, inventive step of the impugned patent application has to be assessed keeping in mind such disclosure. In the impugned patent application, the features described/taught/disclosed in the prior art are described as the feature of the invention in the description as also from Applicant's own stand these features are considered the differentiating aspect vis-à-vis the prior arts. Since the prior art discloses all the features of the impugned patent application, there is virtually no differentiating feature in the impugned patent application.

44. Therefore, in light of the submissions herein above and the documents produced, the Opponent submits that a person skilled in the art could and would readily arrive at the present invention, finding sufficient motivation from the cited prior arts to reliably arrive at the present invention without requiring any intellectual creativity. Thus, the claims of the impugned application are obvious and lack inventive step, in view of which, the claims ought to be rejected for failing to comply with the provisions of Section 2(1)(ja) of the Act.

C. SECTION 25(1)(f): Impugned patent application is not an invention or is not Patentable under this Act – Section 3(f)

Impugned Application is invalid in the light of Section 3(f) of the Patents Act

45. The Opponent submits that the features of the impugned patent application were previously known in the art and is a mere arrangement of the known devices/features and each functioning independently of one another in a

known way. Hence, it attracts the provisions of Section 3(f) of the Act. The section has been reproduced herein for ease of reference and reads as follows:

> *Section 3 – What are not inventions:-*
> *The following are not inventions within the meaning of this Act:-*
> *……*
> 3(d) "the mere use of a known process, machine or apparatus"
> *3(f) "the mere arrangement or re-arrangement or duplication of known devices each functioning independently of one another in a known way".*

Section 3(d)

46. The Opponent submits that in view of independent claim 1 being anticipated by D1 to D3 independently, the provision of Section 3(d) is necessarily attracted, under which, "mere use of known product" is not patentable and ought to be rejected on this ground alone.

Section 3(f)

47. The Opponent submits that in view of independent claims 1 and 8 being anticipated, the claimed product is a mere arrangement/re-arrangement/duplication of known components, functioning in a known way, which is not patentable and ought to be rejected on this ground alone.

48. The Opponent submits that in view of the independent claims of the impugned patent application being anticipated and therefore being non-patentable under Section 3 of the Act, the dependent alone do not stand separately. However, in the event the Applicant amends any of the independent claims to attempt to overcome Section 3 objection, the Opponent reserves its right and craves the leave of the Learned Controller to amend its objection raised under Section 25(1)(f).

D. <u>Section 25(1)(g): that the complete specification does not sufficiently and clearly describe the invention or the method by which it is to be performed</u>

49. Aspects/features of the claims have not been enabled clearly in the complete specification and the same in contravention of Section 10(4) of the Act. Relevant provisions of the Act has been reproduced herein below for the sake of convenience of the Ld. Controller:

Section 10(4): "Every complete specification shall-

i. **<u>fully and particularly describe</u>** the invention and **<u>its operation or use</u>** and **<u>the method by which it is to be performed</u>**;
ii. disclose the **<u>best method of performing the invention</u>** which is known to the applicant and for which he is entitled to claim protection; and ... "
iii. (c) end with a claim or claims defining the scope of the invention for which protection is claimed;
iv. (d) be accompanied by an abstract to provide technical information on the invention

Section 10(5):

The claim or claims of a complete specification shall relate to a single invention, or to a group of inventions linked so as to form a single inventive concept, shall be clear and succinct and shall be fairly based on the matter disclosed in the specification."

50. Opponent states that, claims of the Impugned patent application are not sufficiently and clearly described in the complete specification of the Impugned patent application. For this reason alone, the scope of the claims is indeterminate and the claims are not based on the disclosure made in the description and ought to be rejected. Therefore, a person skilled in art would not be able to reasonably ascertain the scope of the invention.

51. Further, with reference to the claim of 100% glass lined surface made by the Applicant in the description, the Applicant claims the followings:-

> *"Because of this invention the **heat exchanger has 100% glass lined surface** inside of it.*
>
> *As we said earlier **100% glass lined surface** will help customers to use HCL or any other core chemicals to use our heat exchanger's work up to 220 deg centigrade which is not possible in any shell and tube heat exchanger which uses PTFE or Hastelloy materials.*
>
> ***Standard glass has achieved 100% glass lining inside surface** of standard glass heat exchanger with unique design of shell and tube heat exchanger."*

It is submitted that there is no data or proof submitted by the Applicant that the heat exchanger disclosed in the present invention is 100% glass lined. In absence of any data or material to support such claim, the Applicant's claim is ambiguous, unsupported and ought to be rejected.

52. No "best-mode": The Opponent submits that upon a reading of the complete specification of the impugned patent application, it is apparent that the complete description is merely descriptive and generic, and that the Applicant has not provided a single identifiable embodiment, which provides an exemplary "heat exchanger", which would be representative of "best mode" of practicing the invention. A person skilled in the art would have to make an exceedingly large number of varying specifications to identify the invention as intended by the Applicant. The lack of (exemplary) working description of the claimed apparatus suggests that even the Applicant is not aware of the best mode and is not in possession of the invention. Therefore, the invention is in blatant violation of Section 10(4)(b) of the Act and ought to be rejected on this ground alone.

53. The Opponent submits that in view of any of the aforementioned objections, if the Applicant amends any of the pending claims in any manner whatsoever, the Opponent reserves its right and prays the Learned Controller to grant leave to the Opponent in the interest of natural justice to accordingly modify the present presentation, including introducing new prior arts and/or arguments.

54. In view of the above, the Opponent submits that the complete specification does not sufficiently and clearly describe the invention or the method by which it is to be performed and thus, the Impugned patent application is liable to be rejected on this ground alone.

Hearing requested

55. The Opponent hereby requests a hearing under Section 25(1) of the Patents Act and Rule 55 of the Patents Rules, before any further steps or decision are taken in this matter.

Reliefs sought

56. In light of the foregoing facts and submissions, the Opponent humbly prays that the Ld. Controller be pleased to:
    i. Take the present representation by way of opposition on record;
    ii. Issue notice to the Applicant;
    iii. Grant leave to file further evidence, if required;
    iv. Refuse the grant of impugned patent application no. **202141018455** dated 21.04.2021, titled "*Clean room glass lined reactor*";
    v. grant a hearing to the Opponent before passing any adverse order; and
    vi. grant such other relief or reliefs as the Ld. Controller may deem fit in the fact and circumstance of the present Opposition.

Dated this 14th day of November 2024

HEMANT THADHANI
Digitally signed by HEMANT THADHANI
Date: 2024.11.18 15:26:11 +05'30'

Hemant Thadhani (IN/PA-4057)
Crawford Bayley & Co.
Agent for the Opponent

FORM - 2

THE PATENTS ACT, 1970
(39 of 1970)
&
THE PATENT RULES, 2003

Complete specification
[See Section 10; Rule 13]

1. TITLE OF THE INVENTION

CLEAN ROOM GLASS LINED REACTOR

2.NAME OF APPLICANTS:

1) Standard Glass Lining Technology Private Limited a Indian company of D.12, PHASE I, IDA, JEEDIMETLA HYDERABAD Hyderabad Telangna 500055 INDIA

3. PREAMBLE TO THE DESCRIPTION:

The following specification describes the nature of the invention

FIELD OF INVENTION

The present invention relates to corrosion resistant, clean room glass lined overlaid stainless steel reactor. More particularly, the present invention discloses a reactor specifically designed for clean and polished working environment in food and medical industries, which don't require any paint coat the look clean as mirror finished. These reactors help to keep the production floor and its surroundings less dusty and prevent contamination of the product with enhanced heat transfer efficiency.

BACKGROUND ART

Conventional reactors are widely used by fine chemicals and pharmaceutical industry. Corrosion, cleaning and contamination are one of the key criteria that determine the efficiency of a process, which in turn depends on the type of equipment used. Stirred tank reactors (STRS) used in pharmaceutical and fine chemical industries are designed for providing and removing heat. This is done with coil or half coil with coolant passing through it, or using jacketed tanks.

Standard glass lining technology wants to make reactors for easy maintenance and suitable for clean environments. Standard glass has designed and modified its existing large variety of reactors to make them well suited for the clean room and exclusive process areas.

Glass Lined Reactors are generally designed as per ASME SECTION-8 DIVISION-I, Dimensions and capacities are as per DIN-28136 with Mild Steel Material with Glass Coating on the Inner side of the reactor. They can be of AE Type or CE Type and both types can be made as clean room reactors.

The Reactor's Mild Steel outer surface is also prone to corrosion by the spilled material during charging the reactor, Over a period of time there is a high probability that the corrosive particles contaminate the charge thus causing rejections of the material and also costly Regulatory Compliance Issues.

Chemical resistant paints are used to protect the reactor outer surface from exposing it to harsh chemicals and acidic substances. Special procedures and paint materials are used to coat the reactor parts. The paint will also protect the surface from getting rust so that, the rust will not convert to dust or debris around the reactor. Also, the paint itself may peel off after certain time in its lifetime which again form as dirt, paint is prone to damage for impacts cause by any tools or objects. So, taking care of the paint itself a big task and any damage to paint will create dirt and allows the material to expose and material to form rust over a period. It is also proven that Paint will attract dust. There are many possibilities for the traditional methods of protecting surfaces (special paint) may produce dust, dirt and rust. The lifetime of paint is also less.

SUMMARY OF INVENTION

The invention discloses mild steel Glass Lined Reactors overlaid with Stainless Steel material of approximately 2mm depth in some places and manufacturing directly with SS Parts in other places thereby giving the reactor Protection Corrosion, easy maintainability thereby extending the life of the Glass Lined reactor.

Therefore such as herein described there is provided an improved mild steel reactor comprising of an internally glass lined reactor sandwiched between a top flange and a bottom flange, wherein fluidic channels are machined on said glass lined reactor, a glass lined metal flat surface below the top flange that allows to observe flow regime in the glass lined reactor or mixer, a cavity in the bottom flange that facilitates flow of heat transfer fluid through an inlet and an outlet nozzles and flanges, wherein said cavity can be closed with a bottom plate having a metal surface, wherein the metal surface of said bottom plate can touch the heat transfer fluid, a connection plate connecting the inlet and the outlet to the bottom flange of the glass lined stainless steel reactor, and multiple bolts to hold tightly the top flange and the bottom flange wherein the external surface of the said reactor is covered and overlaid with stainless steel sheet by welding.

In an embodiment the outer wall of the reactor consists of a base body of high mechanical strength, namely steel or stainless steel, the base body having a reaction chamber of stainless steel with a layer thickness > 1 mm, preferably in the range from essentially 1 mm to 5 mm and most preferably 2mm.

In another embodiment the said SS body is made by welding the entire surface with SS welding electrodes after removing proportionate metal from the external surface of the reactor and the said nozzles or flanges are made on or in the reactor are internally glass lined.

10. The glass lined metal micro-reactor as claimed in claim 1, wherein thickness of said glass lining ranges from 1 mm to 2 mm.

[Fig.1] [Fig.2] represents the typical reactor in assembled state. Isometric view and basic front views.

BRIEF DESCRIPTION OF THE ACCOMPANYING DRAWINGS

[Fig.1] [Fig.2] illustrates a typical reactor in assembled state. Isometric view and basic front views.

[Fig.3] [Fig.4] [Fig.5] represents the detailed parts of the clean room reactors with differentiating line types for MS and SS. The component names are also described in the images.

DESCRIPTION OF INVENTION

Standard glass technical team has changed the material of construction from MS to SS in the places wherever the reactors outer side is exposed to ambient light and the result is it looks like it is made of SS completely from outside but, the reactor core is same and efficient that of MS reactor since MS reactors are better in process efficiency than SS reactors in places where heating is the primary process requirement. Entire reactor is completely covered with SS materials. Few places the components parts are made with SS, few components are completely wrapped with SS dome. The critical areas like nozzles and thermowells are overlaid with SS metal sheet by welding process.

In an exemplary embodiment, the said overlaying of SS is made by removing MS surface up to 2mm of depth and this 2mm is again covered with SS by welding the entire surface with SS welding electrodes. The thickness is proportional such that the total thickness is

maintained and is a constant. The removal of the top surface of the MS reactor can be mechanical, chemical and / or both. A person skilled in the art will be able to understand a plurality of method available for the removal of the tor surface. This welding surface then goes under grinding process to attain its original dimensions and original shape of the component.

MAIN COMPONENTS OF CLEAN ROOM REACTORS

[Fig.3] [Fig.4] [Fig.5] represents the detailed parts of the clean room reactors with differentiating line types for MS and SS. The component names are also described in the said images. The continues line represent the SS and dashed line represent the MS material. The internal surface is cladded with glass lining using spraying method.

Overlay also mentioned in the images wherever it is applicable. This over laying of SS will be subjected to change for each reactor model based on it number of nozzles and type of thermowell that is required for the model. The glass cladding is made in range of 1mm to 2mm for smooth operation and durability. The said glass lining is configured for a non-reactive cladding, pressure-welded coating and the surface of the glass lining / coating is smooth as whole. The base body is designed as a double-walled vessel.

The disclosed modified reactor has capacity ranging from 1Litre to 12500litre for AE type reactor and 1000Litre to 32000Litre for CE type of reactor and heat transfer area ranging from 0.5 m^2 to 34.6 m^2. The said micro-reactor is configured to withstand pressure up to 6kg/cm^2 and temperature ranging from −60° to 343° C.

ADVANTAGES OF INVENTION

The disclosed clean room reactors come with many tangible advantages for the process and for the maintenance. Stainless steel gives many advantages. These rectors are placed at extreme working conditions which are typically high in temperature, exposed to harsh chemicals. At these conditions these reactors need to be protected from the damaging working conditions and retain its strength. Stainless steel strength to weight ratio is higher than other materials it can withstand more heat and stress compared to MS.

Clean room reactor eliminates the requirement of the painting of outer surface. Painting will attract the dust and over a time period the paint itself peal and becomes dust., so regular maintenance and check-ups are cleaning ups are required when painting comes to picture but clean room reactor doesn't require to paint on its surface. They are hygienic dust repellent and very easy to clean.

Clean rom reactors will be given mirror like finish to its surface thus makes it easy to remove dust or any liquid from its surface. It is as easy as cleaning a mirror. This clean and mirror like finish surface makes them look premium. This aesthetic appearance is attractive and gives advantage to identify any dirt or any waste particles on the surface. Easy to inspect the surface in maintenance exercise.

Stainless steel is highly corrosive resistant, because these reactors are employed in pharma and chemical processing industries, they are subjected to rigorous environments so, it must withstand corrosion from any medium. This is also a primary reason for developing clean room reactors.

Clean room reactors are built with SS on outside to resist scaling, scaling is common phenomenon with the substance used in chemical and pharma industries. Any chemical residue on the reactor surface will not only affect the reactor surface but can also cause problems with the underlying glass thereby reducing the usable life of the reactor. The scaling may affect the environment or labs where these equipment's are kept in operation. Clean room reactor will not have this problem because of appropriate use of SS material around it. Contamination will be lot less in SS when compared to MS.

Standard glass technology has designed and offering these clean room reactors to its customers to add value to their process by means of reducing the maintenance cost, helps maintaining the labs or the production floor clean, give the premium look to the equipment, withstand tough working conditions and increasing the life of the reactor.

The subject matter of embodiments of the present invention is described here with specificity to meet statutory requirements, but this description is not necessarily

intended to limit the scope of the claims. The claimed subject matter may be embodied in other ways, may include different elements or steps, and may be used in conjunction with other existing or future technologies. This description should not be interpreted as implying any particular order or arrangement among or between various steps or elements except when the order of individual steps or arrangement of elements is explicitly described.

We Claim

1. An improved mild steel reactor comprising of:

i. an internally glass lined reactor sandwiched between a top flange and a bottom flange, wherein fluidic channels are machined on said glass lined reactor.

ii. a glass lined metal flat surface below the top flange that allows to observe flow regime in the glass lined reactor or mixer.

iii. a cavity in the bottom flange that facilitates flow of heat transfer fluid through an inlet and an outlet nozzles and flanges, wherein said cavity can be closed with a bottom plate having a metal surface, wherein the metal surface of said bottom plate can touch the heat transfer fluid;

iv. a connection plate connecting the inlet and the outlet to the bottom flange of the glass lined stainless steel reactor; and

v. multiple bolts to hold tightly the top flange and the bottom flange; wherein the external surface of the said reactor is covered and overlaid with stainless steel sheet by welding.

2. The reactor as claimed in claim 1, wherein the outer wall of the reactor consists of a base body of high mechanical strength, namely steel or stainless steel, the base body having a reaction chamber of stainless steel with a layer thickness > 1 mm, preferably in the range from essentially 1 mm to 5 mm and most preferably 2mm.

3. The reactor as claimed in claim 1, wherein the said SS body is made by welding the entire surface with SS welding electrodes after removing proportionate metal from the external surface of the reactor.

4. The reactor as claimed in claim 1, wherein the said nozzles or flanges made on or in the reactor are internally glass lined.

5. The reactor as claimed in claim 1, wherein the glass lining is configured for a non-reactive cladding, pressure-welded coating.

6. The reactor as claimed in claim 1, wherein the base body is designed as a double-walled vessel.

7. The glass lined Stainless steel-reactor as claimed in claim 1, wherein the surface of the glass lining / coating is smooth as whole.

8. The reactor as claimed in claim 1, wherein said reactor has capacity ranging from 1Litre to 12500litre for AE type reactor and 1000Litre to 32000Litre for CE type of reactor and heat transfer area ranging from 0.5 m2 TO 34.6 m2.

9. The reactor as claimed in claim 1, wherein said micro-reactor can withstand pressure up to 6kg/cm^2 and temperature ranging from −60° to 343° C.

10. The reactor as claimed in claim 1, wherein thickness of said glass lining ranges from 1 mm to 2 mm.

Dated this 21st day of April , 2021

BHARAT DASWANI

of DASWANI & DASWANI

APPLICANTS' AGENT

Application No.

No. of sheets: 03

Name : **Standard Glass Lining Technology Private Limited**

Sheet No. : 01

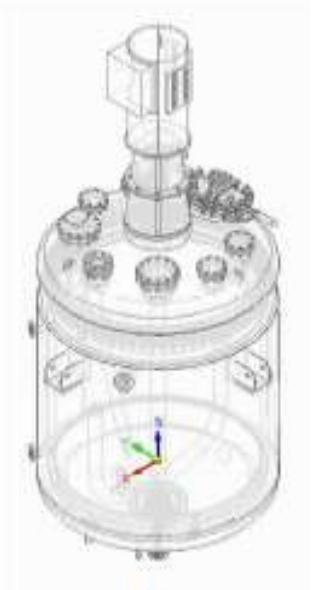


Fig.1

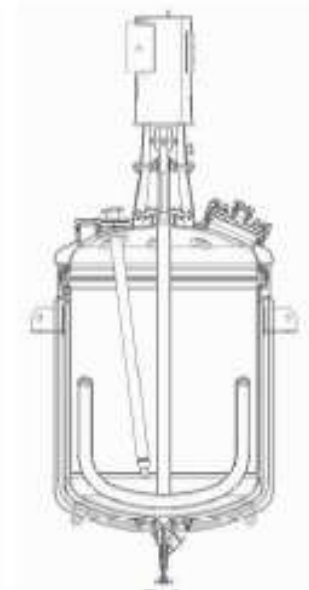


Fig.2

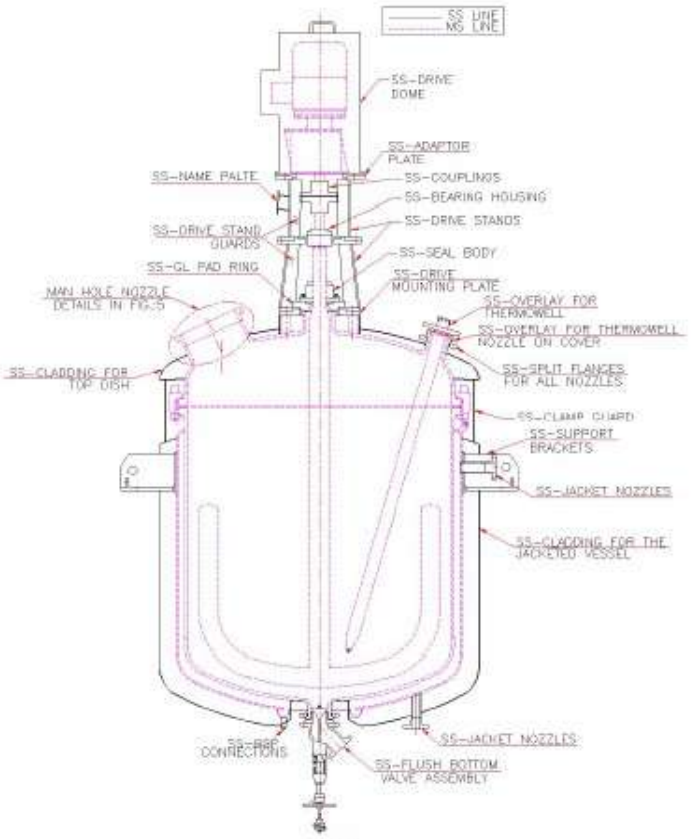
BHARAT DASWANI

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APPLICANTS' AGENT

Application No. No. of sheets: 03

Name : **Standard Glass Lining Technology Private Limited** **Sheet No. : 02**



[Fig.3]

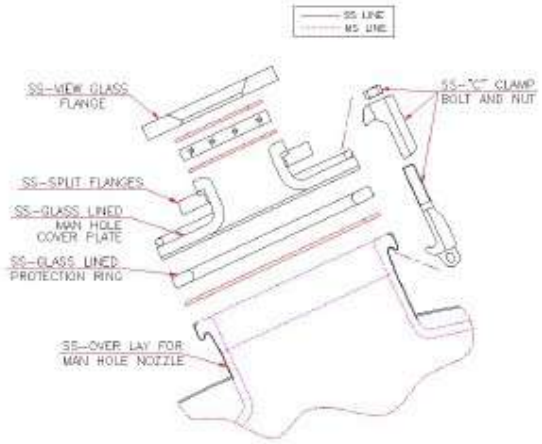
BHARAT DASWANI

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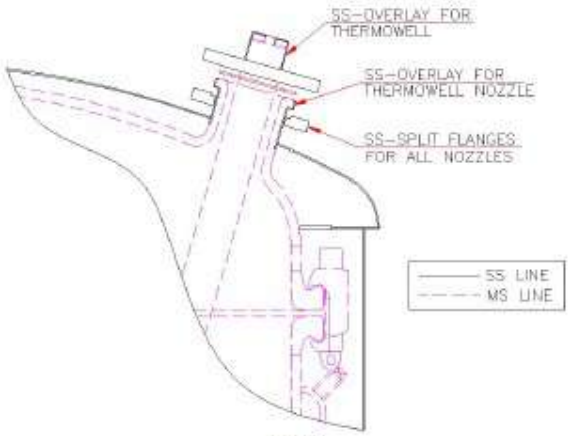
APPLICANTS' AGENT

Application No. No. of sheets: 03

Name : **Standard Glass Lining Technology Private Limited** **Sheet No. : 03**



[FIG. :4]



[FIG.: 5]

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APPLICANTS' AGENT

INTELLECTUAL PROPERTY INDIA
एकस्व/PATENTS|अभिकल्प/DESIGNS|
व्यापार चिह्न/TRADE MARKS|भौगोलिक
उपदर्शन/GEOGRAPHICAL INDICATIONS

सत्यमेव जयते
भारत सरकार
GOVERNMENT OF INDIA

एकस्व कार्यालय / THE PATENT OFFICE
बौद्धिक सम्पदा भवन / I.P.O. BUILDING
जी.एस.टी.रोड, गिंडी/ G.S.T.Road, Guindy,
चेन्नई/ Chennai- 600032
दूरभाष/ Tel. No. : (091)(044)22502081-84
फ़ैक्स/ Fax: 044 22502066

Email/ई मेल : Chennai-patent@nic.in
वेबसाइट /Website:http://ipindia.nic.in

सं.संख्या/Ref.No /आवेदन संख्या/Application No/ 202141018455 दिनांक/Date of Dispatch/Email: 10/11/2022

सेवा मे,/To
MR. BHARAT DASWANI,
DASWANI & DASWANI Patent & Trade Mark (Intellectual Property) Attorneys Daswani House, Green Acres 23B, Ahiripukur 1st Lane Kolkata 700 019
Email : kolkata@daswaniindia.net

विषय: एकस्व अधिनियम, 1970 की धारा 12 व 13 तथा एकस्व नियम, 2003 के अधीन परीक्षण रिपोर्ट
Subject: Examination report under sections 12 & 13 of the Patents Act, 1970 and the Patents Rules, 2003.

1. उपर्युक्त आवेदन के संदर्भ मे परीक्षण रिपोर्ट (अर्थात, एकस्व नियम, 2003 (यथा संशोधित) के नियम 24-ख(3) में विनिर्दिष्ट आपत्तियों का प्रथम कथन) इसके साथ संलग्न है। यह रिपोर्ट परीक्षण हेतु अनुरोध दिनांक 21/04/2021 के उत्तर मे जारी की गयी है। परीक्षण रिपोर्ट का उत्तर दाखिल करने की अंतिम तिथि (अर्थात, इस रिपोर्ट में लगाई गयी सभी आवश्यकताओं के अनुपालन की अवधि) आवेदक को आपत्तियों का प्रथम कथन जारी होने की तिथि से छः माह है।
   Please find enclosed herewith an Examination Report (i.e. a first statement of objections as specified in Rule 24-B(3) of The Patents Rules, 2003 (as amended)) in respect of above-mentioned application. This report is issued with reference to a request for examination dated 21/04/2021.The last date for filing a response to the Examination Report (i.e. a period to comply with all the requirements raised in this examination report) is six months from the date on which the first statement of objections is issued to the Applicant.

2. यदि रिपोर्ट के अंतर्गत लगाई गयी आवश्यकताओं का अनुपालन एकस्व नियम, 2003 (यथा संशोधित) के नियम 24 ख(5) में विनिर्दिष्ट अवधि के भीतर अंदर अनुपालन नहीं किया गया तो एकस्व अधिनियम 1970 की धारा 21(1) के अधीन वर्तमान आवेदन को परित्यक्त माना जाएगा।
   The instant application shall be deemed to have been abandoned under Section 21(1) of The Patents Act, 1970, unless all the requirements raised in this report are complied with in the period as specified in Rule 24-B (5) of The Patents Rules, 2003 (as amended).

3. आपका ध्यान एकस्व नियम, 2003 के नियम 24 ख(6) के प्रावधानों की ओर भी आमंत्रित किया जाता है।
   Your attention is also invited to the provisions of Rule 24-B (6) of the Patents Rules 2003.

4. आपको सलाह दी जाती है कि शीघ्र निपटान हेतु अपना उत्तर शीघ्र प्रस्तुत करें।
   You are advised to file the reply at the earliest for early disposal.

Santanu Dey
नियंत्रक पेटेंट/ Controller of Patents

संलग्न/Enclosed: अपरोक्त अनुसार/As above

टिप्पणी: यह इलेक्ट्रोनिक रूप से उत्पन्न रिपोर्ट है।
NOTE: This is an electronically generated report.

सभी पत्राचार नियंत्रक एकस्व को उपरोल्लिखित पते पर भेजा जाये।
All communications should be sent to the Controller of Patents at the above mentioned address.

परीक्षण रिपोर्ट /Examination Report

आवेदन संख्या /Application Number	202141018455
दाखिल करने की तिथि /Date of Filing	21/04/2021
पूर्विक्ता दिनांक /Date of Priority	--
पीसीटी अंतर्राष्ट्रीय आवेदन की संख्या व दिनांक / PCT International Application No. & Date	--
आवेदक /Applicant	Standard Glass Lining Technology Private Limited
परीक्षण हेतु अनुरोध की संख्या व दिनांक /Request for Examination No. & Date	R20214013940 21/04/2021
प्रकाशन की तिथि /Date of Publication	28/10/2022

इस परीक्षण रिपोर्ट के चार भाग हैं, अर्थात रिपोर्ट का सारांश, विस्तृत तकनीकी रिपोर्ट, औपचारिक आवश्यकताएँ तथा रिकॉर्ड मे दस्तावेज़ /
This examination report consists of four parts, namely summary of the report, detailed technical report, formal requirements and documents on record.

भाग -1: रिपोर्ट का सारांश
PART-I: SUMMARY OF THE REPORT

क्र. सं. /Sl. No.	अधिनियम के तहत आवश्यकताओं पर विस्तृत टिप्पणियां /Requirements under the Act		दावों की संख्या /Claim Numbers	टिप्पणी /Remarks
1.	धारा 2(1)(ज) के तहत आविष्कार /Invention u/s 2(1)(j)	नवीनता /Novelty	दावे /Claims: 1-9	हाँ /Yes
			दावे /Claims:	नहीं /No
		आविष्कारी कदम / Inventive step	दावे /Claims:	हाँ /Yes
			दावे /Claims: 1-9	नहीं /No
		औद्योगिक उपयोगिता /Industrial Applicability	दावे /Claims: 1-9	हाँ /Yes
			दावे /Claims:	नहीं /No
2.	[धारा 10(5) व 10(4) (ग)] के अधीन दावे /Claims [u/s 10(5) & 10(4) (c)]	स्पष्टता/ संक्षिप्तता /Clarity / Conciseness	दावे /Claims:	हाँ /Yes
			दावे /Claims: 1-9	नहीं /No
		विवरण द्वारा समर्थित /Supported by description	दावे /Claims:	हाँ /Yes
			दावे /Claims: 10	नहीं /No

भाग –II विस्तृत तकनीकी रिपोर्ट
PART-II: DETAILED TECHNICAL REPORT

क. उद्धरित दस्तावेजों की सूची /A.List of documents cited:

(क) पेटेंट साहित्य / (a). Patent Literature :

क्र. सं. / Sl.no	दस्तावेज़ों का विवरण /Details of documents	प्रकाशन तिथि(दिन/माह/वर्ष) / Publication date	उद्धरित दस्तावेज़ का प्रासंगिक विवरण (पृष्ठ व अनुच्छेद संख्या) / Relevant description (page and paragraph no.) of cited document	उद्धरित दस्तावेज़ के प्रासंगिक दावे / Relevant claims of cited document	अभिकथित आविष्कार के दावे /Claims of alleged invention
1	CN1583239A	23/02/2005	whole document		1-9

2	CN103691349A	02/04/2014	whole document		1-9
3	WO2021063359A1	08/04/2021	claim 9		1-9

(ख) गैर-पेटेंट साहित्य /(b).Non-patent literature

कोई दस्तावेज़ उद्धृत नहीं है /No Document Cited

ख. अधिनियम के तहत आवश्यकताओं पर विस्तृत टिप्पणियां /B. Detailed observations on the requirements under the Act:

(1).आविष्कारी कदम / INVENTIVE STEP:

(I) ऊपर उद्धरित दस्तावेज़(जों) के संदर्भ मे स्पष्ट अध्यापन(नों) को ध्यान मे रखते हुए, निम्नलिखित कारणों से दावा(वों) (1-9) मे आविष्कारी कदम की कमी है
Claim(s) (1-9) lack(s) inventive step, being obvious in view of teaching (s) of cited document(s) above under reference for the following reasons:

Features of claims are obvious over the documents as follows and lack in inventive step u/s 2(1)(ja) of The Patents Act 1970 as amended by the Patents (Amendment) Act 2005. (Reference numerals are same as mentioned in cited document)

D1: CN1583239A: (PBD: 23/02/2005; whole document)

D2: CN103691349A: (PBD: 02/04/2014; whole document)

D3: WO2021063359A1: (PBD: 08/04/2021; claim 9)

D1 discloses a glass enamelled reactor with high resistance to corrosion and high-temp oxidization has an internal cylinder made of steel plate and a glass enamel layer coated on its internal surface, and an external jacket made of stainless steel.

D2 discloses a stirring kettle with a polyester coating and belongs to the field of chemical equipment. An inner barrel body of the kettle is made of a steel plate, an internal wall is coated with the polyester coating, and an external jacket is made of stainless steel and is in an overall structure. Long-term use is effectively ensured, the thickness of the stainless steel external jacket of the kettle is not reduced for a long term after repair for many times, the external jacket can be repeatedly used, and the polyester coating can effectively prevent a heavy metal content in a product from being overproof at the same time. The kettle can be repaired for many times, so that the production cost is greatly lowered, many steel products are saved, and the kettle has very significant economic and social benefits, and has market competitiveness at home and abroad.

D3 discloses a method of lithium sulfate and sodium (potassium) carbonate directly producing lithium carbonate and reducing sulfate radical content. Starting from the step of thermal precipitation of a lithium sulfate purification liquid and a sodium (potassium) carbonate purification liquid to produce crude lithium carbonate until the production of a refined lithium carbonate wet product, most of the original methods of a manufacturer for removing silicon, aluminum, iron, magnesium, calcium, and heavy metals remain unchanged, and the detection method remains unchanged. The employment of "reverse feeding, non-circulating mother liquor", "pre-precipitation supplementary impurity removal" and "high-efficiency desorption" can reduce industrial grade lithium carbonate sulfate radicals to 0.03%, increase the main content to 2.5N, reduce battery grade sulfate radicals to 0.008%, and stably increase the main content to 3N, or even reach the limit of 3.5N-4N. "High-efficiency desorption" comprises "thermal precipitation with small temperature increases and thermal stirring washing", "medium-high temperature strong desorption" and "hydrocyclone separation", impurities such as sulfate radicals that are chemically adsorbed and deeply encapsulated in the peritectic core of lithium carbonate particles can be released into deionized water, which are then effectively carried away by a "hydrocyclone separation" liquid

phase.

In view of above documents, subject matter is obvious to the person skilled in art, therefore the claims do not constitute an invention under Section 2[1(ja)] of The Patents Act 1970 as amended by the Patents (Amendment) Act 2005.

(2).प्रकटन की दक्षता /SUFFICIENCY OF DISCLOSURE:

(I) दावा(वे) 10 विनिर्देश मे प्रकट विषय पर आधारित नहीं है अथवा निम्नलिखित कारणों से विनिर्देश मे प्रकटन द्वारा समर्थित नहीं है।
Claim(s)'10' are not fairly based on the matter disclosed in the specification or not supported by the disclosure in the specification for the following reasons:

1. Omnibus claim 10 does not define the scope of the invention for which protection is sought, hence it is not allowable u/s 10(4) (c) of the Patents Act 1970 (as amended).

(II) सार /Abstract:

Revised Abstract shall be provided along with concise summery of invention including technical problem & solution to problem [read rule 13(7) (a) with section 10(4) (d) of The Patents Act/ Rules (as Amended)]

(III) आविष्कार का शीर्षक /Title of Invention:

Title does not sufficiently indicate the subject. It should be suitably amended to sufficiently indicate the subject matter to which the invention relates, in accordance with the requirements of section 10(1) of The Patents Act 1970 (as amended).

(IV) पेटेंट नियम, 2003 के नियम 15 के प्रावधानों के अनुसार निम्नलिखित कारणों से रेखाचित्र तैयार नहीं किए गए हैं.
Drawings are not prepared in accordance with the provisions of Rule 15 of the Patents Rules, 2003 for the following reasons:

Reference numeral shall be marked in the drawings and reference of it shall be incorporated in the claims to enhance the intelligibility of Claims. No description matter is allowed on the drawings as per Rule 15(7) of The Patents Rule, 2003 (as amended).

(3).स्पष्टता एवं संक्षिप्तता /CLARITY AND CONCISENESS:

(I) दावा(वे) 1-9 के संबंध मे स्पष्ट रूप से परीभाषित नहीं हैं.
Claim(s) 1-9 are not clearly worded in respect of:

1. Claims do not sufficiently define the invention. The claims should be redrafted to make them sufficiently definitive and all constructional/Inventive features should be brought out clearly under characterized clause of claim 1 and clearly define the scope of the invention, in accordance with section 10(4) (c) and 10 (5) of The Patents Act 1970 (as amended).
2. Preamble of claims is broad in nature.
3. Claims are not clearly worded, expression "comprises/comprising" should be replaced with word "consisting of".
4. Dependency of dependent claim 9 as mentioned is unclear and vague in nature as “micro reactor” is not disclosed in independent claim 1
5. Claims are not clearly worded and indefinite in respect the words like “preferably, essentially, more preferably, entire, can, at least”. They should be reworded suitably.

6. The phrase “steel or stainless steel” in claim 2 do not have any limiting effect and not allowable u/s 10(4)(c) of The Patents Act 1970 (as amended).
7. The phrase "after removing proportionate metal from the external surface of the reactor" in the claim 3 is not clear and definitive, applicant is directed to file a proper technical clarification as on what criteria the amount of material to be removed depends.
8. Reference numerals should be supplemented in all claims to enhance the intelligibility of claims as prescribed in rule 13 (4) of Indian Patent Rules, 2016
9. Preamble of claims is inconsistent with the title of the alleged invention.

(4).अन्य आवश्यकताएँ /OTHERS REQUIREMENTS:

(I)

The phrase “10. The glass lined metal micro-reactor as claimed in claim 1, wherein thickness of said glass lining ranges from 1 mm to 2 mm.” in description is vague in nature and hence should be deleted.

भाग – III: औपचारिक आवश्यकताएँ /PART-III: FORMAL REQUIREMENTS

आपत्तियां /Objections	टिप्पणी /Remarks
Date and Signature of Applicant	All forms should be signed as per the second schedule of The Patents Act 1970 (amended 2005). (Stamped signature is not allowed)
Statement & Under Taking (Form 3 Details)	Details regarding application for Patents including prosecution details which may be filed outside India from time to time for the same or substantially the same invention if any should be furnished within Six months from the date of filing of the said application under Section 8 of The Patents Act.
Form 6	All forms should be filed in proper format as per the second schedule of The Patents Act 1970 (amended 2005). Form-1, 3, 5 and 18 has not been submitted in proper format as per the second schedule of The Patents Act 1970 (amended 2005). Further, applicant should tick/cross whichever is applicable /not applicable in declaration in para 12(iii) of form 1. Applicant/Agent of applicant shall not modify the forms as per their wish. Applicant is directed to resubmit the form as prescribed in the Act/Rule.
Power of Attorney (Whether GPA, SPA, Stamped, requisite fee etc.)	The Form-26(Authorization) filed is inadequately stamped, applicant failed to meet the requirements of rule 135(1) of The Patents Rule, 2003 (as amended).
Format of Specification (rule 13)	The statement in the description on page 5 last paragraph implies that the subject-matter for which protection is sought may be different to that defined by the claims thereby resulting in a lack of clarity of the claims when the description is used to interpret the claims. It is difficult to ascertain the subject matter for which protection is sought and hence do not fulfil the requirements of section 10(4)(c) of The Patents Act/ Rules (as Amended)]
Format of Drawings	The drawings prepared are not as per the format described in rule 15(6) of The Patents Rules, 2003. Drawings shall be sequentially or systematically numbered and shall bear the application number in the left hand top corner.

Other Deficiencies	1. Page numbers in the complete specification may be written at the bottom middle of each page. (Page 1 shall start from opening page of form-2) 2. Applicant during reply to FER/SER shall reply to each and every objection individually in detail, reply shall contain detailed submission for each cited document. 3. The applicant is requested to clearly identify/mark the amendments carried out (if any) in a separate copy (from the original filed set); irrespective of either they concern amendments by addition, correction, replacement or deletion in the specification. 4. Details should be furnished within Six months from the date of filing of the same or substantially same application in other Patent office's under section 8(1)(b)and rule 12(1) of Indian Patent Act. 5. Documents regarding the search and/or examination report as referred to in Rule 12(3) of the Patent Rule, 2003, in respect of same or substantially the same invention filed in all the other Patent offices along with appropriate English translation should be submitted within a prescribed time period as suggested in section 8(2) of the Indian Patents Act. 6. Whenever any agent is signing on any form, full name of agent with his/her registered agent number shall be provided. 7. Every fifth line of each page of the description and each page of claims should be serially numbered as per rule 9(1)(d) of Indian Patent Rules, 2003.

भाग-IV: रिकॉर्ड मे दस्तावेज़ /PART-IV: DOCUMENTS ON RECORD

निम्नलिखित दस्तावेज़ों के आधार पर यह परीक्षण रिपोर्ट तैयार की गयी है
The examination report has been prepared based on the following documents:

कार्यसूची तिथि / Docket Date	कार्यसूची संख्या /Docket Number	प्रविष्टि संख्या विवरण /Entry Number Description
21 Apr 2021	34742	1-New Application For Patent With Provisional /Complete Specification
21 Apr 2021	34742	28(i)-Request For Examination After 18 months Publication - Form 18
28 Apr 2021	36871	OTHERS(NON CASH)

नियंत्रक का नाम /Name of the Controller: Santanu Dey

नियंत्रक स्थान /Controller Location: Kolkata

टिप्पणी: परीक्षण रिपोर्ट का उत्तर दाखिल करने की अंतिम तिथि / Note: Last date for filing response to the Examination Report: 10/05/2023

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August 10, 2023 Rs. 4,000/-
Your ref : 202141018455 Kind Attention :
Our Ref : 202141018455 Dr. Santanu Dey
Controller of Patents

To
The Controller of Patents,
The Patent Office,
Chennai

Dear Sir,
Re. Indian Patent Application No. 202141018455 of 21/04/2021 in relation to "Glass Lined Reactor with Full Cladding and Nozzle Overlay" in the name of Standard Glass Lining Technology Private Limited

We refer to your letter dated 10.11.2022 regarding First Examination Report.

Regarding objection of INVENTIVE STEP under part II B(1) :

Claims have been amended as required.

D1 discloses a glass enamelled reactor with high resistance to corrosion and high-temp oxidization has an internal cylinder made of steel plate and a glass enamel layer coated on its internal surface, and an external jacket made of stainless steel.

In present invention, the claim pertains to the application of a stainless steel overlay onto the reactor's nozzle—an area where conventional stainless steel cladding is typically unfeasible. The outer jacket of the reactor is composed of mild steel, while a thin stainless steel sheet is used as cladding to prevent rust formation and ensure hygiene. However, due to constraints in cladding the nozzle using traditional methods, an alternative approach involving the deposition of

stainless steel material is employed. This deposited material is carefully leveled to achieve complete coverage, effectively accomplishing full stainless steel encapsulation of the reactor.

While this D1 is explaining about the reactor with stainless steel jacket which is not related to present case.

D2 discloses a stirring kettle with a polyester coating and belongs to the field of chemical equipment. An inner barrel body of the kettle is made of a steel plate, an internal wall is coated with the polyester coating, and an external jacket is made of stainless steel and is in an overall structure. Long-term use is effectively ensured, the thickness of the stainless steel external jacket of the kettle is not reduced for a long term after repair for many times, the external jacket can be repeatedly used, and the polyester coating can effectively prevent a heavy metal content in a product from being overproof at the same time. The kettle can be repaired for many times, so that the production cost is greatly lowered, many steel products are saved, and the kettle has very significant economic and social benefits, and has market competitiveness at home and abroad.

In present invention, the claim pertains to the application of a stainless steel overlay onto the reactor's nozzle—an area where conventional stainless steel cladding is typically unfeasible. The outer jacket of the reactor is composed of mild steel, while a thin stainless steel sheet is used as cladding to prevent rust formation and ensure hygiene. However, due to constraints in cladding the nozzle using traditional methods, an alternative approach involving the deposition of stainless steel material is employed. This deposited material is carefully leveled to achieve complete coverage, effectively accomplishing full stainless steel encapsulation of the reactor.

While this D2 is explaining about the polyster reactor with stainless steel jacket which is not related to present case.

D3 discloses a method of lithium sulfate and sodium (potassium) carbonate directly producing lithium carbonate and reducing sulfate radical content. Starting from the step of thermal precipitation of a lithium sulfate purification liquid and a sodium (potassium) carbonate purification liquid to produce crude lithium carbonate until theproduction of a refined lithium carbonate wet product, most of the original methods of a manufacturer for removing silicon, aluminum,

iron, magnesium, calcium, and heavy metals remain unchanged, and the detection method remains unchanged. The employment of "reverse feeding, non-circulating mother liquor", "preprecipitation supplementary impurity removal" and "high-efficiency desorption" can reduce industrial grade lithium carbonate sulfate radicals to 0.03%, increase the main content to 2.5N, reduce battery grade sulfate radicals to 0.008%, and stably increase the main content to 3N, or even reach the limit of 3.5N-4N. "High-efficiency desorption" comprises "thermal precipitation with small temperature increases and thermal stirring washing"

"medium-high temperature strong desorption" and "hydrocyclone separation", impurities such as sulfate radicals that are chemically adsorbed and deeply encapsulated in the peritectic core of lithium carbonate particles can be released into deionized water, which are then effectively carried away by a "hydrocyclone separation" liquid phase.

In present invention, the claim pertains to the application of a stainless steel overlay onto the reactor's nozzle—an area where conventional stainless steel cladding is typically unfeasible. The outer jacket of the reactor is composed of mild steel, while a thin stainless steel sheet is used as cladding to prevent rust formation and ensure hygiene. However, due to constraints in cladding the nozzle using traditional methods, an alternative approach involving the deposition of stainless steel material is employed. This deposited material is carefully leveled to achieve complete coverage, effectively accomplishing full stainless steel encapsulation of the reactor.

While this D3 is about a completely different application which is not related to claims of present invention.

It is seen that D1, D2 and D3 either singly or in relevant combination cannot give rise to present invention.

Hence, present invention is not only novel but also has inventive step.

Regarding objection of SUFFICIENCY OF DISCLOSURE under part II B(2) :

Being omnibus claim, claim 10 is deleted.

Fresh abstract is submitted as required.

The title of the invention has been amended to read as "**Glass Lined Reactor with Full Cladding and Nozzle Overlay"**

Objection may kindly be waived.

Regarding objection of CLARITY AND CONCISENESS under part II B(3) :

Please find enclosed amended claims along with marked-up copy thereof.

Regarding objection of OTHERS REQUIREMENTS under part II B(4) :

Being omnibus claim, claim 10 is deleted.

Regarding objection under PART-III: FORMAL REQUIREMENTS :

We enclose herewith :

1. Fresh Form 1, Form 3, Form 5 and Form 18.

2. Fresh power of attorney in our favour with adequate stamp duty.

3. Fresh complete specification.

4. Fresh claims along with marked-up copy thereof.

5. Fresh drawing and abstract.

6. Form 13 together with Govt. fee Rs. 4,000/- for changes in claims and complete description.

In view of above, objections may kindly be waived.

Acceptance is requested.

Yours Sincerely
Of DASWANI & DASWANI

Ranjan Paul

(RANJAN PAUL)
APPLICANT'S AGENT

Exhibit 'C'

FIELD OF INVENTION

The present invention relates to corrosion resistant, clean room glass lined overlaid stainless steel reactor. More particularly, the present invention discloses a reactor specifically designed for clean and polished working environment in food and medical industries, which don't require any paint coat the look clean as mirror finished. These reactors help to keep the production floor and its surroundings less dusty and prevent contamination of the product with enhanced heat transfer efficiency.

The present inventions also presents Mild Steel (MS) Glass-Lined reactors with stainless steel overlays on nozzles, designed for sectors like pharmaceuticals, food, chemicals, and agrochemicals. It aims to elevate glass-lined reactors for cleanroom use, ensuring high cleanliness standards, less maintenance, and rust prevention. By integrating stainless steel overlays onto reactor nozzles with SS cladding around the vessel, the solution covers 100% of the reactor, offering durability for industries requiring stringent hygiene. Stainless steel enhances corrosion resistance, which is vital in industries where rust could compromise product quality.

BACKGROUND ART

Conventional reactors are widely used by fine chemicals and pharmaceutical industry. Corrosion, cleaning and contamination are one of the key criteria that determine the efficiency of a process, which in turn depends on the type of equipment used. Stirred tank reactors (STRS) used in pharmaceutical and fine chemical industries are designed for providing and removing heat. This is done with coil or half coil with coolant passing through it, or using jacketed tanks.

Standard glass lining technology wants to make reactors for easy maintenance and suitable for clean environments. Standard glass has designed and modified its existing large variety of reactors to make them well suited for the clean room and exclusive process areas.

Glass Lined Reactors are generally designed as per ASME SECTION-8 DIVISION-I, Dimensions and capacities are as per DIN-28136 with Mild Steel Material with

Glass Coating on the Inner side of the reactor. They can be of AE Type or CE Type and both types can be made as clean room reactors.

The Reactor's Mild Steel outer surface is also prone to corrosion by the spilled material during charging the reactor, Over a period of time there is a high probability that the corrosive particles contaminate the charge thus causing rejections of the material and also costly Regulatory Compliance Issues.Chemical resistant paints are used to protect the reactor outer surface from exposing it to harsh chemicals and acidic substances. Special procedures and paint materials are used to coat the reactor parts. The paint will also protect the surface from getting rust so that, the rust will not convert to dust or debris around the reactor. Also, the paint itself may peel off after certain time in its lifetime which again form as dirt, paint is prone to damage for impacts cause by any tools or objects. So, taking care of the paint itself a big task and any damage to paint will create dirt and allows the material to expose and material to form rust over a period. It is also proven that Paint will attract dust. There are many possibilities for the traditional methods of protecting surfaces (special paint) may produce dust, dirt and rust. The lifetime of paint is also less.

Glass-lined reactors play a vital role across diverse industry sectors that grapple with corrosive applications. The ingenious use of glass lining as a protective barrier on mild steel plates within these reactors renders them impervious to the corrosive agents present in their operational environment. This unique feature not only enhances the longevity of the reactors but also ensures the integrity of the substances they contain.

In particularly stringent settings, such as clean rooms where stringent safety and hygiene protocols are paramount, glass-lined reactors find special application. These controlled environments demand absolute rust prevention due to its potential to compromise both the safety of the operation and the purity of the products being processed. To address this, these reactors are meticulously engineered with an inner layer of glass lining on mild steel. The outer layer, known as the jacket, serves as a conduit for heat transfer processes.

However, the challenge arises in safeguarding the outer surface of these reactors against rust formation. To counteract this, a sheet of stainless steel is welded as cladding over the jacket. This layer, in addition to insulating the reactor, acts as a barrier against rust development.

Yet, it's crucial to note that this cladding has limitations—it only extends coverage up to the uppermost part of the reactor and the vessel itself. Regrettably, it leaves vulnerable areas such as the nozzles that are essential for ingress and egress.

Consequently, the nozzles are typically protected with specialized industrial paints. Over time, however, these paints can peel off due to various factors, including temperature variations and mechanical stress. This leads to the exposure of the underlying mild steel surface, setting the stage for potential rust formation and degradation.

To counteract this, a routine maintenance strategy involving periodic inspection and repainting becomes necessary. While this approach ensures the structural integrity of the reactor and prevents rust-related issues, it introduces concerns in clean room environments. The possibility of paint particles contaminating the products being processed poses a significant risk, warranting stringent quality control measures and further amplifying maintenance challenges.

In essence, the usage of glass-lined reactors exemplifies a delicate balance between material science, engineering ingenuity, and meticulous maintenance practices. While these reactors provide indispensable utility in handling corrosive substances, their effective and sustainable operation demands a multifaceted approach that encompasses materials, design, and stringent maintenance protocols to ensure both durability and product purity.

SUMMARY OF INVENTION

The invention discloses mild steel Glass Lined Reactors overlaid with Stainless Steel material of approximately 2mm depth in some places and manufacturing directly with SS Parts in other places thereby giving the reactor Protection Corrosion, easy maintainability thereby extending the life of the Glass Lined reactor.

Therefore such as herein described there is provided an improved mild steel reactor comprising of an internally glass lined reactor sandwiched between a top flange and a bottom flange, wherein fluidic channels are machined on said glass lined reactor, a glass lined metal flat surface below the top flange that allows to observe

flow regime in the glass lined reactor or mixer, a cavity in the bottom flange that facilitates flow of heat transfer fluid through an inlet and an outlet nozzles and flanges, wherein said cavity can be closed with a bottom plate having a metal surface, wherein the metal surface of said bottom plate can touch the heat transfer fluid, a connection plate connecting the inlet and the outlet to the bottom flange of the glass lined stainless steel reactor, and multiple bolts to hold tightly the top flange and the bottom flange wherein the external surface of the said reactor is covered and overlaid with stainless steel sheet by welding.

In an embodiment the outer wall of the reactor consists of a base body of high mechanical strength, namely steel or stainless steel, the base body having a reaction chamber of stainless steel with a layer thickness > 1 mm, preferably in the range from essentially 1 mm to 5 mm and most preferably 2mm.

In another embodiment the said SS body is made by welding the entire surface with SS welding electrodes after removing proportionate metal from the external surface of the reactor and the said nozzles or flanges are made on or in the reactor are internally glass lined.

In the realm of clean room reactors, stainless-steel cladding prevents rust, except for vulnerable nozzles. To counter this, a solution involves depositing a layer of stainless steel onto the outer nozzle surface, followed by precise grinding for uniformity. This innovation forms a stainless-steel shield, permanently safeguarding against rust. This coated nozzle is then seamlessly welded to the top cover's cladding sheet, fully enclosing the reactor in stainless steel. This dynamic approach harmonizes material science and engineering, tackling rust challenges in critical zones. The result is a resilient barrier that upholds reactor integrity, longevity, and product purity. This cutting-edge technique epitomizes the fusion of advanced technology and craftsmanship, underscoring a commitment to stringent clean room standards.

A glass-lined reactor offers excellent resistance to oxidation at the nozzle. In the event of an accidental spillage of corrosive liquid onto the nozzle surface, which might otherwise lead to oxidation on the metal surface, the glass lining which is coated inside would be damaged as the base metal is oxidated and would result in distortion. The claim asserts that the reactor is fully covered with stainless steel, enhancing its resilience against external damage and thereby increasing its overall safety.

~~10. The glass lined metal micro reactor as claimed in claim 1, wherein thickness of said glass lining ranges from 1 mm to 2 mm.~~

~~[Fig.1] [Fig.2] represents the typical reactor in assembled state. Isometric view and basic front views.~~

BRIEF DESCRIPTION OF THE ACCOMPANYING DRAWINGS

[Fig.1] [Fig.2] illustrates a typical reactor in assembled state. Isometric view and basic front views.

[Fig.3] [Fig.4] [Fig.5] represents the detailed parts of the clean room reactors with differentiating line types for MS and SS. The component names are also described in the images.

Figure 1: Clean Room Reactor with SS cladding and nozzles covered with SS overlay.
Figure 2: Manhole cover with all SS components and SS overlay on the nozzles.
Figure 3: Nozzle with SS overlay and thermowell.

The invention will now be described, by way of examples only, with reference to the accompanying drawings.

Figure 1 depicts a Glass-lined reactor integrated with Stainless Steel (SS) cladding, showcasing pivotal components that contribute to its functionality and structural integrity. At the core of this assembly is the Bottom Vessel (1), which serves as the foundational chamber for chemical reactions. The Clamps (20) are strategically positioned to secure various components in place, ensuring stability and safety during operation. Positioned above the vessel is the Top Dish (16), which encapsulates the reactor's contents.

The reactor's drive mechanism is supported by an array of SS components, including the Pad Plate (8), Bottom Drive Stand (9), Bearing Housing (10), Top Drive Stand (11), Moto Flange (12), Motor Cover (13), and Mechanical Seals (14 and 7). The Name Plate (15) provides crucial identification and information about the reactor setup.

The reactor design also accommodates accessibility and monitoring. A Manhole Assembly (6) offers a portal for maintenance and inspections, while the Thermowell (5) and its associated Nozzle (4) facilitate temperature measurements and control. This is shown in **Figure 2.**

Figure 2 offers an intricate view of the SS cladding application, particularly on Nozzle (26). This SS cladding extends along the entire height of both the manhole and thermowell nozzles, ensuring corrosion resistance and structural integrity. Complementing this cladding are SS Clamps (27), the Manhole Cover (24), and the Sight Glass Flange (22), all meticulously constructed from stainless steel.

The overarching design of the reactor involves the strategic integration of glass-lined and stainless steel components, striking a balance between the specialized properties of each material. The SS cladding provides robust protection against corrosive substances and external elements, while the glass lining offers superior chemical resistance for the reactor's interior. This synthesis of materials and components underscores the reactor's efficiency, durability, and operational safety within various industrial applications.

DESCRIPTION OF INVENTION

Standard glass technical team has changed the material of construction from MS to SS in the places wherever the reactors outer side is exposed to ambient light and the result is it looks like it is made of SS completely from outside but, the reactor core is same and efficient that of MS reactor since MS reactors are better in process efficiency than SS reactors in places where heating is the primary process requirement. Entire reactor is completely covered with SS materials. Few places the

components parts are made with SS, few components are completely wrapped with SS dome. The critical areas like nozzles and thermowells are overlaid with SS metal sheet by welding process.

In an exemplary embodiment, the said overlaying of SS is made by removing MS surface up to 2mm of depth and this 2mm is again covered with SS by welding the entire surface with SS welding electrodes. The thickness is proportional such that the total thickness is maintained and is a constant. The removal of the top surface of the MS reactor can be mechanical, chemical and / or both. A person skilled in the art will be able to understand a plurality of method available for the removal of the tor surface. This welding surface then goes under grinding process to attain its original dimensions and original shape of the component.

MAIN COMPONENTS OF CLEAN ROOM REACTORS

[Fig.3] [Fig.4] [Fig.5] represents the detailed parts of the clean room reactors with differentiating line types for MS and SS. The component names are also described in the said images. The continues line represent the SS and dashed line represent the MS material. The internal surface is cladded with glass lining using spraying method.

Overlay also mentioned in the images wherever it is applicable. This over laying of SS will be subjected to change for each reactor model based on it number of nozzles and type of thermowell that is required for the model. The glass cladding is made in range of

1mm to 2mm for smooth operation and durability. The said glass lining is configured for a non-reactive cladding, pressure-welded coating and the surface of the glass lining / coating is smooth as whole. The base body is designed as a double-walled vessel.

The disclosed modified reactor has capacity ranging from 1Litre to 12500litre for AE type reactor and 1000Litre to 32000Litre for CE type of reactor and heat transfer area ranging from 0.5 m^2 to 34.6 m^2. The said micro-reactor is configured to withstand pressure up to 6kg/cm^2 and temperature ranging from −60° to 343° C.

ADVANTAGES OF INVENTION

The disclosed clean room reactors come with many tangible advantages for the process and for the maintenance. Stainless steel gives many advantages. These rectors are placed at extreme working conditions which are typically high in temperature, exposed to harsh chemicals. At these conditions these reactors need to be protected from the damaging working conditions and retain its strength. Stainless steel strength to weight ratio is higher than other materials it can withstand more heat and stress compared to MS.

Clean room reactor eliminates the requirement of the painting of outer surface. Painting will attract the dust and over a time period the paint itself peal and becomes dust., so regular maintenance and check-ups are cleaning ups are required when painting comes to picture but clean room reactor doesn't require to paint on its surface. They are hygienic dust repellent and very easy to clean.

Clean rom reactors will be given mirror like finish to its surface thus makes it easy to remove dust or any liquid from its surface. It is as easy as cleaning a mirror. This clean and mirror like finish surface makes them look premium. This aesthetic appearance is attractive and gives advantage to identify any dirt or any waste particles on the surface. Easy to inspect the surface in maintenance exercise.

Stainless steel is highly corrosive resistant, because these reactors are employed in pharma and chemical processing industries, they are subjected to rigorous environments so, it must withstand corrosion from any medium. This is also a primary reason for developing clean room reactors.

Clean room reactors are built with SS on outside to resist scaling, scaling is common phenomenon with the substance used in chemical and pharma industries. Any chemical residue on the reactor surface will not only affect the reactor surface but can also cause problems with the underlying glass thereby reducing the usable life of the reactor. The scaling may affect the environment or labs where these equipment's are kept in operation. Clean room reactor will not have this problem because of appropriate use of SS material around it. Contamination will be lot less in SS when compared to MS.

Standard glass technology has designed and offering these clean room reactors to its customers to add value to their process by means of reducing the maintenance cost, helps maintaining the labs or the production floor clean, give the premium look to the equipment, withstand tough working conditions and increasing the life of the reactor.

The subject matter of embodiments of the present invention is described here with specificity to meet statutory requirements, but this description is not necessarily intended to limit the scope of the claims. The claimed subject matter may be embodied in other ways, may include different elements or steps, and may be used in conjunction with other existing or future technologies. This description should not be interpreted as implying any particular order or arrangement among or between various steps or elements except when the order of individual steps or arrangement of elements is explicitly described.

We Claim :

1. A Mild Steel (MS) Glass-Lined reactors with stainless steel overlays on nozzles comprising of:

i. an internally glass lined reactor sandwiched between a top flange and a bottom flange, wherein fluidic channels are machined on said glass lined reactor.

ii. a glass lined metal flat surface below the top flange that allows to observe flow regime in the glass lined reactor or mixer.

iii. a cavity in the bottom flange that facilitates flow of heat transfer fluid through an inlet and an outlet nozzles and flanges, wherein said cavity can be closed with a bottom plate having a metal surface, wherein the metal surface of said bottom plate can touch the heat transfer fluid;

iv. a connection plate connecting the inlet and the outlet to the bottom flange of the glass lined stainless steel reactor; and

v. multiple bolts to hold tightly the top flange and the bottom flange; wherein the external surface of the said reactor is covered and overlaid with stainless steel sheet by welding.

wherein coated nozzle is seamlessly welded to the top cover's cladding sheet, fully enclosing the reactor in stainless steel tackling rust challenges in critical zones resulting a resilient barrier that upholds reactor integrity, longevity, and product purity

and Characterized in that

said reactor offers excellent resistance to oxidation at the nozzle and in the event of an accidental spillage of corrosive liquid onto the nozzle surface, which might otherwise lead to oxidation on the metal surface, the glass lining which is coated inside would be damaged as the base metal is oxidated and would result in distortion

and wherein the the reactor is fully covered with stainless steel, enhancing its resilience against external damage and thereby increasing its overall safety.

2. The reactor as claimed in claim 1, wherein the outer wall of the reactor consists of a base body of high mechanical strength, namely steel or stainless steel, the base body having a reaction chamber of stainless steel with a layer thickness > 1 mm, preferably in the range from essentially 1 mm to 5 mm and most preferably 2mm.

3. The reactor as claimed in claim 1, wherein the said SS body is made by welding the entire surface with SS welding electrodes after removing proportionate metal from the external surface of the reactor.

4. The reactor as claimed in claim 1, wherein the said nozzles or flanges made on or in the reactor are internally glass lined.

5. The reactor as claimed in claim 1, wherein the glass lining is configured for a non-reactive cladding, pressure-welded coating.

6. The reactor as claimed in claim 1, wherein the base body is designed as a double- walled vessel.

7. The reactor as claimed in claim 1, wherein the surface of the glass lining / coating is smooth as whole.

8. The reactor as claimed in claim 1, wherein said reactor has capacity ranging from 1Litre to 12500litre for AE type reactor and 1000Litre to 32000Litre for CE type of reactor and heat transfer area ranging from 0.5 m2 TO 34.6 m2.

Dated this 21st day of April 2021

Ranjan Paul

(Ranjan Paul)
Of Daswani & Daswani
Applicant's Agent

ABSTRACT

“Glass Lined Reactor with Full Cladding and Nozzle Overlay”

This invention introduces stainless steel overlays on glass-lined reactors, enhancing their functionality in cleanroom settings. By depositing the stainless steel layers onto vulnerable nozzle surfaces and seamlessly integrating them with the reactor's cladding, this innovation offers comprehensive rust prevention. This approach ensures reactor integrity, longevity, and product purity while meeting rigorous cleanroom standards. The combined use of glass-lined and stainless steel components provides a balanced solution for improved efficiency and durability across diverse industries.

Fig. 1

APPLICATION NO. 202141018455 NO. OF SHEETS : 03

NAME : Standard Glass Lining Technology Private Limited SHEET NO. : 01

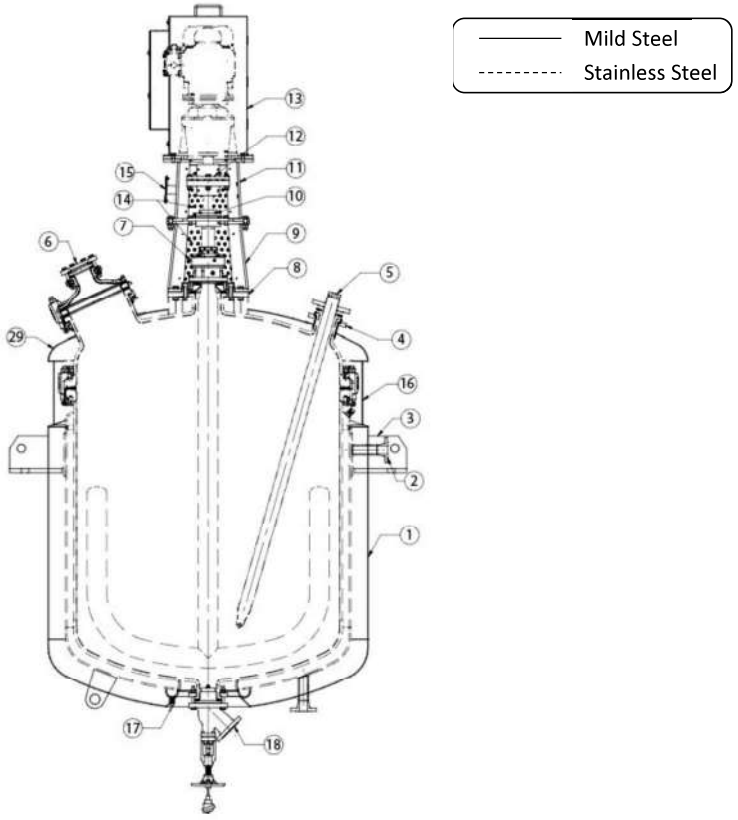


Figure 1

Ranjan Paul

(RANJAN PAUL)
Of DASWANI & DASWANI
APPLICANT'S AGENT

APPLICATION NO. 202141018455 NO. OF SHEETS : 03

NAME : Standard Glass Lining Technology Private Limited SHEET NO. : 02

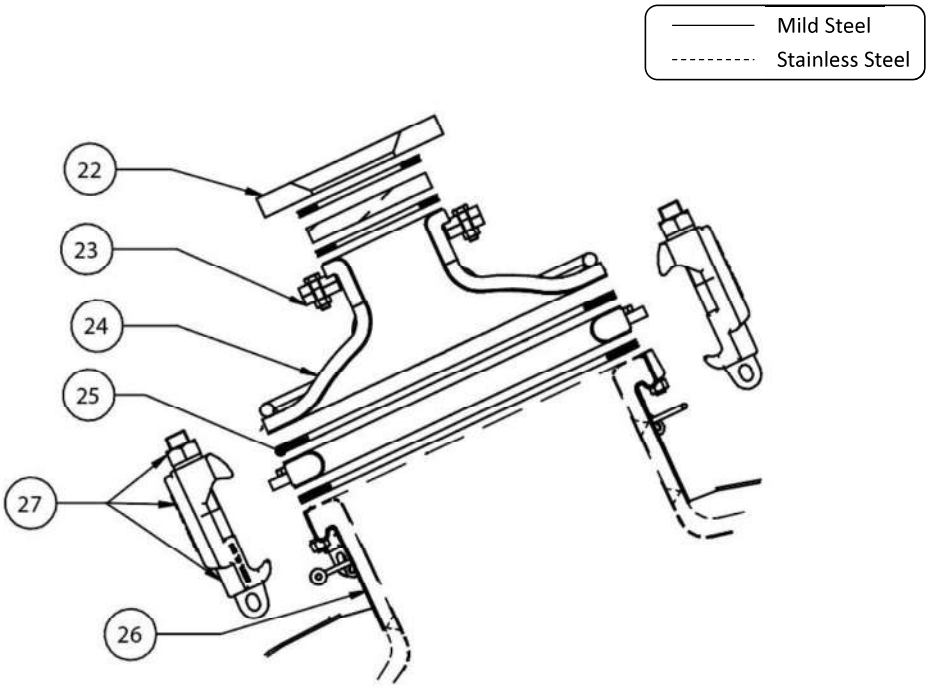


Figure 2

Ranjan Paul

(RANJAN PAUL)
Of DASWANI & DASWANI
APPLICANT'S AGENT

APPLICATION NO. 202141018455 NO. OF SHEETS : 03

NAME : Standard Glass Lining Technology Private Limited SHEET NO. : 03

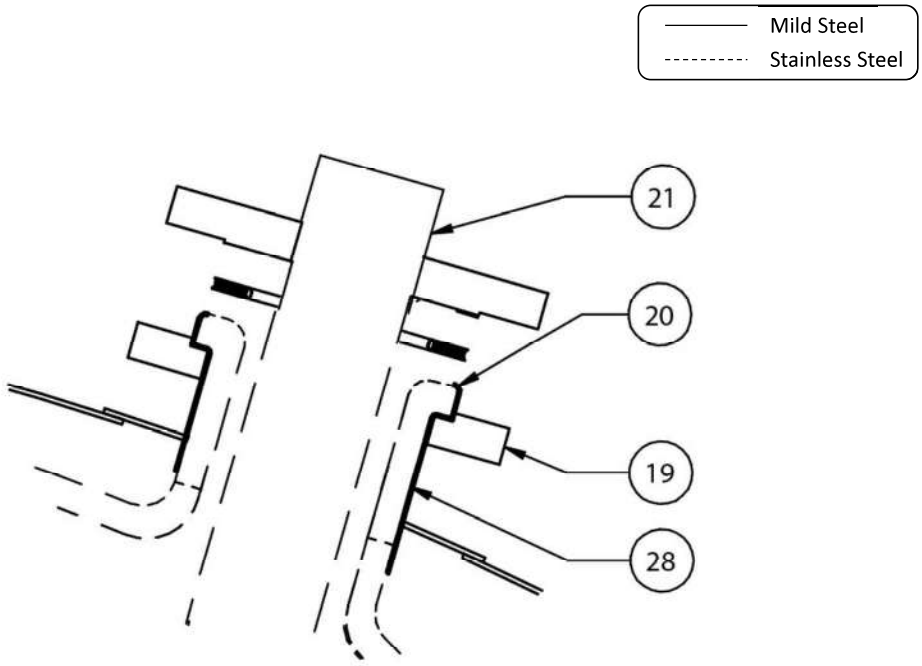


Figure 3

Ranjan Paul

(RANJAN PAUL)
Of DASWANI & DASWANI
APPLICANT'S AGENT

सत्यमेव जयते

GUIDELINES FOR SEARCH AND EXAMINATION OF PATENT APPLICATIONS

INDIAN PATENT OFFICE

INTELLECTUAL PROPERTY INDIA

OFFICE OF THE CONTROLLER GENERAL OF PATENTS, DESIGNS AND TRADEMARKS

2015

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"when required" or "any" are used, then it forces the reader to make a subjective judgment and not an objective observation, unless such expression follows any definite values.

f. A claim must be specific and not vague, ambiguous, speculative or hypothetical in nature. Each claim should be complete so that it covers the inventive feature and enough elements around it to put the invention in the proper context.

g. Trade marks / Trade names are not permitted in claims.

Structure of Claims

a) A claim usually consists of three parts:

- Introductory phrase,
- Body of the claim, and
- Link that joins the two segments.

b) The introductory phrase identifies the category of the invention and sometimes the purpose (For example, a machine for waxing paper, a composition for fertilizing soil).

c) The body of the claim is the specific legal description of the exact invention, which is sought to be protected.

d) The linking consists of words and phrases such as:

- Which comprises
- Including
-Consisting of
-Consisting essentially of

For Example: In the following example, "A data input device" is the introductory phrase, "comprising" is the linking word, and the rest of the claim is the body. "A data input device comprising; an input surface adapted to be locally exposed to a pressure or pressure force, a sensor means disposed below the input surface for detecting the position of the pressure or pressure force on the input surface and for outputting an output signal representing said position and; an evaluating means for evaluating the output signal of the sensor means."

e) If the invention is an improvement to an existing product, the claims should set the boundary very clearly by characterizing the invention with respect to the prior art. In those cases, the claim will have two parts separated by the word *'characterized by'* or *'wherein'*. The part coming before *'characterized by'* is the prior art while that comes

after will be the features of the invention. It is equally applicable in the case of a process which is modification of the existing process. f) Structure of Claims should be on the following lines:

i) Independent Claim: This is the first claim which is also called the 'Principal Claim' should clearly define the essential novel features of the most preferred embodiment of the process, apparatus, device or the product that constitutes the invention and should be properly characterized with respect to the 'prior art', defining all the technical features essential to the invention or inventive concept. This should include the core integers as well as sufficient details of interrelationship, operation or utility to establish that the invention achieves the intended objectives and

ii) Dependent Claim(s): Dependent claims should be clubbed with the independent claims (or within themselves) to include all the features of the independent claim with additional non-essential features and even the minute aspects and optional features.

iii) Further independent claims are only justified where the inventive concept covers more than one category, e.g. apparatus, process, product, complementary versions within one category constituting unity of invention, e.g. plug and socket, transmitter and receiver, which work only together. Therefore, wherever possible, claims should not contain:--

• Multiple unrelated inventions

• Dependent claims that are not fully limited by the terms of the preceding independent claim, e.g. dependent claims which omit or substitute a feature of an independent claim.

Certain statements are not to be regarded as claims :

i) The statements of the following form given are not to be regarded as claims, in as much as, they do not define the invention:-

a) I claim to be the inventor of this appliance,

b) I claim a patent and that no one else shall use my invention without leave.

c) I claim that the machine described above is quite new and has never been seen or used before.

d) I claim some reward.

ii) Also, the claims should not be made, as in the examples given below, for illustrating the efficiency or advantages of the invention:-

a) I claim that this device is better and cheaper and more effectual than anything known.

b) I claim that my process or machine will do such and such things.

c) I claim the following advantages.

d) I claim an improved sewing machine.

e) I claim a mechanism for converting heat into electrical energy without any loss of efficiency.

f) I claim a new method of making silk waterproof.

iii) Where products are claimed, the invention will not be properly defined if merely the properties of the products are referred to, as in the following example:-

"I claim a lubricating oil which is of specific gravity…. and boiling point."

iv) The claims, such as "I claim an improved sewing machine as described or as illustrated "or " I claim the invention described in the specification", which merely refer back to the description are not sufficiently definitive.

3.3.4 Single inventive concept

Section 10(5) mandates that the claim/ claims of the complete specification shall relate to a single invention, or to a group of inventions linked so as to form a single inventive concept. The MANUAL OF PATENT OFFICE PRACTICE AND PROCEDURE, at 05.03.16 allows that there may be more than one independent claim in a single application if the claims fall under a single inventive concept. In the Manual, it has been advised "While there is no restriction as to the number of claims, including independent claims, it is advisable to limit the number of claims, as well as the number of independent claims in a single application so that the claims are linked so as to form a single inventive concept. If claims relate to a plurality of distinct inventions, it may be objected on ground of lack of unity of invention".

In other words when there is a group of inventions in a specification they should be linked by a single concept which is inventive or there should be a technical relationship among the claimed inventions, which makes the inventive contribution over the prior art. To fulfil the requirement of unity of invention each claim of a complete specification should share a single common technical relationship which is inventive. The single common technical relationship which is inventive is called the "special technical feature". This determination should be done on the content of the claims supported by the description in the light of the prior art.

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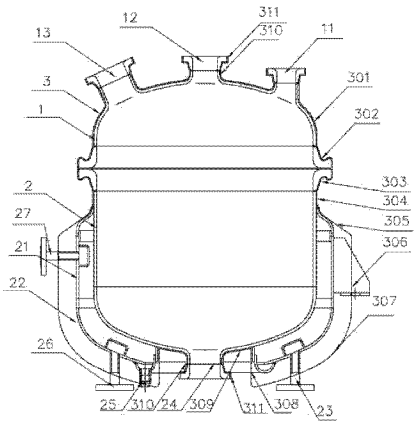
权利要求书2页 说明书3页 附图2页

(54)发明名称

一种具有仿形外包搪玻璃设备及其制造方法

(57)摘要

本发明提供的一种具有仿形外包搪玻璃设备及其制造方法，主要包括罐盖和罐身，罐盖外侧具有不锈钢外包层，罐身外侧上部和夹套外侧具有不锈钢外包层，夹套和不锈钢外包层之间填充有保温材料形成保温层，符合GMP认证中生产设备不得对药品质量产生任何不利影响，与药品直接接触的生产设备表面应当平整、光洁、易清洗或消毒、耐腐蚀要求，而且比奥氏体不锈钢搪玻璃设备生产成本较低。



CN 103820781 A

1. 一种具有仿形外包搪玻璃设备，主要包括罐盖和罐身，所述罐盖设有管口、人孔和密封口；所述罐身外侧下部包有夹套，所述罐身和夹套焊接成一体结构，两者之间为腔体，所述罐身一侧设有夹套接管法兰，下部分布设有放料口、冷凝水出口、蒸汽进口和蒸汽出口；其特征在于：

所述罐盖表面具有不锈钢外包层，所述不锈钢外包层上部罐盖封头面和罐盖法兰面共同构成罐盖不锈钢外包层；

所述罐身外侧上部和夹套外侧具有不锈钢外包层，所述罐身外侧不锈钢外包层上部包有罐身法兰面和罐身面Ⅰ不锈钢外包层，所述罐身和夹套焊接处包有罐身过渡面Ⅰ不锈钢外包层，所述夹套外侧包有罐身面Ⅱ不锈钢外包层，罐身下部包有罐身封头面不锈钢外包层，所述罐身下部和夹套焊接处包有罐身过渡面Ⅱ不锈钢外包层，所述罐身底部包有罐身面Ⅲ不锈钢外包层；

所述管口、人孔、密封口和放料口处包有法兰高颈面和法兰面组成的不锈钢外包层；

所述不锈钢外包层之间为焊接连接；

所述夹套和不锈钢外包层之间填充有保温材料形成保温层。

2. 如权利要求 1 所述的一种具有仿形外包搪玻璃设备，其特征在于所述罐盖外侧具有的不锈钢外包层厚度为 1.5-3mm，所述罐身外侧具有的不锈钢外包层厚度大于等于 3mm。

3. 如权利要求 1 所述的一种具有仿形外包搪玻璃设备，其特征在于所述保温层的保温材料为聚氨酯发泡材料或硅酸盐材料。

4. 如权利要求 1-3 任意所述的一种具有仿形外包搪玻璃设备的制造方法，具体包括如下步骤：

1)根据搪玻璃设备罐盖 1 和罐身 2 的实际尺寸对各个组成面在 304 不锈钢钢板上用划针画线，画线包括各个组成面具体尺寸、法兰和管口处开孔线，线画好后用钢剪刀按照线剪开；

2)把下完料的不锈钢钢板的罐盖封头面 301 和罐身封头面 307 用模具进行初步冲压成型，并在罐盖封头面 301 和罐身封头面 307 的法兰和管口处开孔；

3)把罐盖法兰面 301 放在搪玻璃罐盖 1 上边，从罐盖 1 上面向罐盖 1 周边展开，不锈钢钢板边铺展边整理，使外形和罐盖 1 外形一致，钢板遇到不平整或者凸鼓、起皱的地方，用橡胶锤敲打到和罐盖 1 贴合紧密，罐盖法兰面 302 通过折边初步成型后紧贴罐盖 1 下部的法兰，对接处用氩弧焊对焊，罐盖法兰面 302 与罐盖法兰面 301 用氩弧焊对焊，与罐盖 1 下部的法兰用氩弧焊均匀分布点焊；

4)罐身 2 外侧上部包有的罐身法兰面 303、罐身面Ⅰ 304 和罐身 2 和夹套 21 焊接处罐身过渡面Ⅰ 305 下好料后，边铺展边整理，对接处用氩弧焊对焊，使外形和罐盖 2 外形一致，钢板遇到不平整或者凸鼓、起皱的地方，用橡胶锤敲打到和罐盖 2 贴合紧密，夹套 21 外侧的罐身面Ⅱ 306、罐身 2 下部的罐身封头面 307、罐身 2 下部和夹套 21 焊接处包有罐身过渡面Ⅱ 308、罐身 2 底部包有罐身面Ⅲ 309 对接处用氩弧焊对焊，使外形和罐盖 2 外形一致；

5)法兰、管口处和罐盖 1 和罐身 2 周边贴合时，把法兰周边的不锈钢板用钢剪刀剪开钢板，成“V”型，尺寸大约为 3mm×（10-20）mm，不锈钢板翻上法兰面和罐盖 1 和罐身 2 周边后，用氩弧焊把“V”型缺口焊接起来，所用不锈钢焊丝直径 Φ2.0，焊接电流 70-80A，焊接电压 7-9V；

6）“V”型缺口用氩弧焊对焊，与罐盖 1 和罐身 2 法兰用氩弧焊均匀分布点焊，用橡胶锤等敲打使其和罐盖 1 和罐身 2 紧密贴合；

7）不锈钢焊缝还要进行打磨到 320 目后抛光处理，使不锈钢板的表面光洁度达到要求；

8）用钢剪刀修理钢板和罐盖 1 和罐身 2 紧密边不整齐处，使之外形一致。

一种具有仿形外包搪玻璃设备及其制造方法

技术领域

[0001] 本发明涉及一种搪玻璃设备及其制造方法，尤其涉及一种具有不锈钢仿形外包、符合制药 GMP 洁净要求的搪玻璃设备。

背景技术

[0002] 搪玻璃设备由含硅量高的玻璃质釉喷涂于金属胎表面， 经 920-960℃多次高温搪烧，使玻璃质釉密着于金属胎表面制成， 因此它具有类似玻璃的化学稳定性和金属的良好机械性能双重优点，在许多方面替代了不锈钢、有色金属、塑料等设备。

[0003] 随着生物、医药、食品等行业的迅速发展，不仅要求反应釜等设备内部具有抗腐蚀性而且要求外表洁净卫生，不锈钢材料使用寿命长、外形美观，但由于不锈钢对含氯离子或非氧化性酸的介质不耐腐蚀，还可能导致药品等特殊物料的重金属离子含量超标，不符合 GMP 认证中不得与药品发生化学反应、吸附药品或向药品中释放物质的要求；而且目前我国生产的搪玻璃反应釜的基体均为 Q235 系列或 20R 的碳钢或低合金钢，如果选用用奥氏体不锈钢作为搪玻璃设备的基体和设计的专用底釉，经特定的搪烧工艺方法制成在一定的工况下能安全运行的不锈钢搪玻璃设备，但是使用不锈钢作为搪玻璃设备的基体和设计的专用底釉成本较大，而且搪烧工艺比较复杂。

[0004] 现有技术搪玻璃设备无法满足洁净车间用设备的卫生要求，特别是无法满足制药企业实施《药品生产质量管理规范》(GMP) 的洁净要求。

发明内容

[0005] 为了解决现有技术中存在的难题，提供一种具有仿形外包搪玻璃设备解决搪玻璃设备不符合 GMP 洁净要求、生产成本较高、搪烧工艺复杂等问题。

[0006] 本发明解决其技术问题所采用的技术方案是：

一种具有仿形外包搪玻璃设备，主要包括罐盖和罐身，所述罐盖设有管口、人孔和密封口；所述罐身外侧下部包有夹套，所述罐身和夹套焊接成一体结构，两者之间为腔体，所述罐身一侧设有夹套接管法兰，下部分布设有放料口、冷凝水出口、蒸汽进口和蒸汽出口；

所述罐盖表面具有不锈钢外包层，所述不锈钢外包层上部罐盖封头面和罐盖法兰面共同构成罐盖不锈钢外包层；

所述罐身外侧上部和夹套外侧具有不锈钢外包层，所述罐身外侧不锈钢外包层上部包有罐身法兰面和罐身面 I 不锈钢外包层，所述罐身和夹套焊接处包有罐身过渡面 I 不锈钢外包层，所述夹套外侧包有罐身面 II 不锈钢外包层，罐身下部包有罐身封头面不锈钢外包层，所述罐身下部和夹套焊接处包有罐身过渡面 II 不锈钢外包层，所述罐身底部包有罐身面III不锈钢外包层；

所述管口、人孔、密封口和放料口处包有法兰高颈面和法兰面组成的不锈钢外包层；

所述不锈钢外包层之间为焊接连接；

所述夹套和不锈钢外包层之间填充有保温材料形成保温层。

[0007] 优选的，所述罐盖外侧具有的不锈钢外包层厚度为 1.5-3mm，所述罐身外侧具有的不锈钢外包层厚度大于等于 3mm。

[0008] 优选的，所述保温层的保温材料为聚氨酯发泡材料或硅酸盐材料。

[0009] 本仿形外包新搪玻璃设备的不锈钢外包层制造方法，具体包括如下步骤：

1)根据搪玻璃设备罐盖 1 和罐身 2 的实际尺寸对各个组成面在 304 不锈钢钢板上用划针画线，画线包括各个组成面具体尺寸、法兰和管口处开孔线，线画好后用钢剪刀按照线剪开；

2)把下完料的不锈钢钢板的罐盖封头面 301 和罐身封头面 307 用模具进行初步冲压成型，并在罐盖封头面 301 和罐身封头面 307 的法兰和管口处开孔；

3)把罐盖法兰面 301 放在搪玻璃罐盖 1 上边，从罐盖 1 上面向罐盖 1 周边展开，不锈钢钢板边铺展边整理，使外形和罐盖 1 外形一致，钢板遇到不平整或者凸鼓、起皱的地方，用橡胶锤敲打到和罐盖 1 贴合紧密，罐盖法兰面 302 通过折边初步成型后紧贴罐盖 1 下部的法兰，对接处用氩弧焊对焊，罐盖法兰面 302 与罐盖法兰面 301 用氩弧焊对焊，与罐盖 1 下部的法兰用氩弧焊均匀分布点焊；

4)罐身 2 外侧上部包有的罐身法兰面 303、罐身面Ⅰ 304 和罐身 2 和夹套 21 焊接处罐身过渡面Ⅰ 305 下好料后，边铺展边整理，对接处用氩弧焊对焊，使外形和罐盖 2 外形一致，钢板遇到不平整或者凸鼓、起皱的地方，用橡胶锤敲打到和罐盖 2 贴合紧密，夹套 21 外侧的罐身面Ⅱ 306、罐身 2 下部的罐身封头面 307、罐身 2 下部和夹套 21 焊接处包有罐身过渡面Ⅱ 308、罐身 2 底部包有罐身面Ⅲ 309 对接处用氩弧焊对焊，使外形和罐盖 2 外形一致；

5)法兰、管口处和罐盖 1 和罐身 2 周边贴合时，把法兰周边的不锈钢板用钢剪刀剪开钢板，成“V”型，尺寸大约为 3mm×（10-20）mm，不锈钢板翻上法兰面和罐盖 1 和罐身 2 周边后，用氩弧焊把“V”型缺口焊接起来，所用不锈钢焊丝直径 Φ2.0，焊接电流 70-80A，焊接电压 7-9V；

6)“V”型缺口用氩弧焊对焊，与罐盖 1 和罐身 2 法兰用氩弧焊均匀分布点焊，用橡胶锤等敲打使其和罐盖 1 和罐身 2 紧密贴合；

7)不锈钢焊缝还要进行打磨到 320 目后抛光处理，使不锈钢板的表面光洁度达到要求；

8)用钢剪刀修理钢板和罐盖 1 和罐身 2 紧密边不整齐处，使之外形一致。

[0010] 本发明的有益效果在于，提供的一种具有仿形外包搪玻璃设备，罐盖和罐身外侧具有不锈钢外包层，符合 GMP 认证中生产设备不得对药品质量产生任何不利影响，与药品直接接触的生产设备表面应当平整、光洁、易清洗或消毒、耐腐蚀要求，而且比奥氏体不锈钢搪玻璃设备生产成本较低。

附图说明

[0011] 下面结合附图对本发明一种具有仿形外包搪玻璃设备作进一步说明：

图 1 是本发明仿形外包搪玻璃容器的结构示意图；

图 2 是本发明仿形外包搪玻璃容器的法兰处包边放大结构示意图；

图中：1- 罐盖、11- 管口、12- 密封口、13- 人孔、2- 罐身、21- 夹套、22- 保温层、23- 蒸汽进口、24- 放料口、25- 冷凝水出口、26- 蒸汽出口、27- 夹套接管法兰、3- 不锈钢外包层、

301- 罐盖封头面、302- 罐盖法兰面、303- 罐身法兰面、304- 罐身面Ⅰ、305- 罐身过渡面Ⅰ、306- 罐身面Ⅱ、307- 罐身封头面、308- 罐身过渡面Ⅱ、309- 罐身面Ⅲ、310- 法兰高颈面、311- 法兰面。

具体实施方式

[0012] 下面结合附图对本发明作更进一步的说明：

如图 1-2 所示，一种具有仿形外包搪玻璃设备主要包括罐盖 1 和罐身 2，所述罐盖 1 设有管口 11、人孔 13 和密封口 12；所述罐身 2 外侧下部包有夹套 21，所述罐身 2 和夹套 21 焊接成一体结构，两者之间为腔体，所述罐身 2 一侧设有夹套接管法兰 27，下部分布设有放料口 24、冷凝水出口 25、蒸汽进口 23 和蒸汽出口 26；所述罐盖 1 表面具有不锈钢外包层 3，所述不锈钢外包层 3 上部罐盖封头面 301 和罐盖法兰面 302 共同构成罐盖 1 不锈钢外包层 3；所述罐身 2 外侧上部和夹套 21 外侧具有不锈钢外包层 3，所述罐身 2 外侧不锈钢外包层 3 上部包有罐身法兰面 303 和罐身面Ⅰ 304 不锈钢外包层 3，所述罐身 2 和夹套 21 焊接处包有罐身过渡面Ⅰ 305 不锈钢外包层 3，所述夹套 21 外侧包有罐身面Ⅱ 306 不锈钢外包层 3，罐身 2 下部包有罐身封头面 307 不锈钢外包层 3，所述罐身 2 下部和夹套 21 焊接处包有罐身过渡面Ⅱ 308 不锈钢外包层 3，所述罐身 2 底部包有罐身面Ⅲ 309 不锈钢外包层 3；所述管口 11、人孔 13、密封口 12 和放料口 24 处包有法兰高颈面 310 和法兰面 311 组成的不锈钢外包层 3；所述不锈钢外包层之间为焊接连接。

[0013] 上述虽然结合附图对本发明的具体实施方式进行了描述，但并非对本发明保护范围的限制，所属领域技术人员应该明白，在本发明的技术方案的基础上，本领域技术人员不需要付出创造性劳动即可做出的各种修改或变形仍在本发明的保护范围以内。

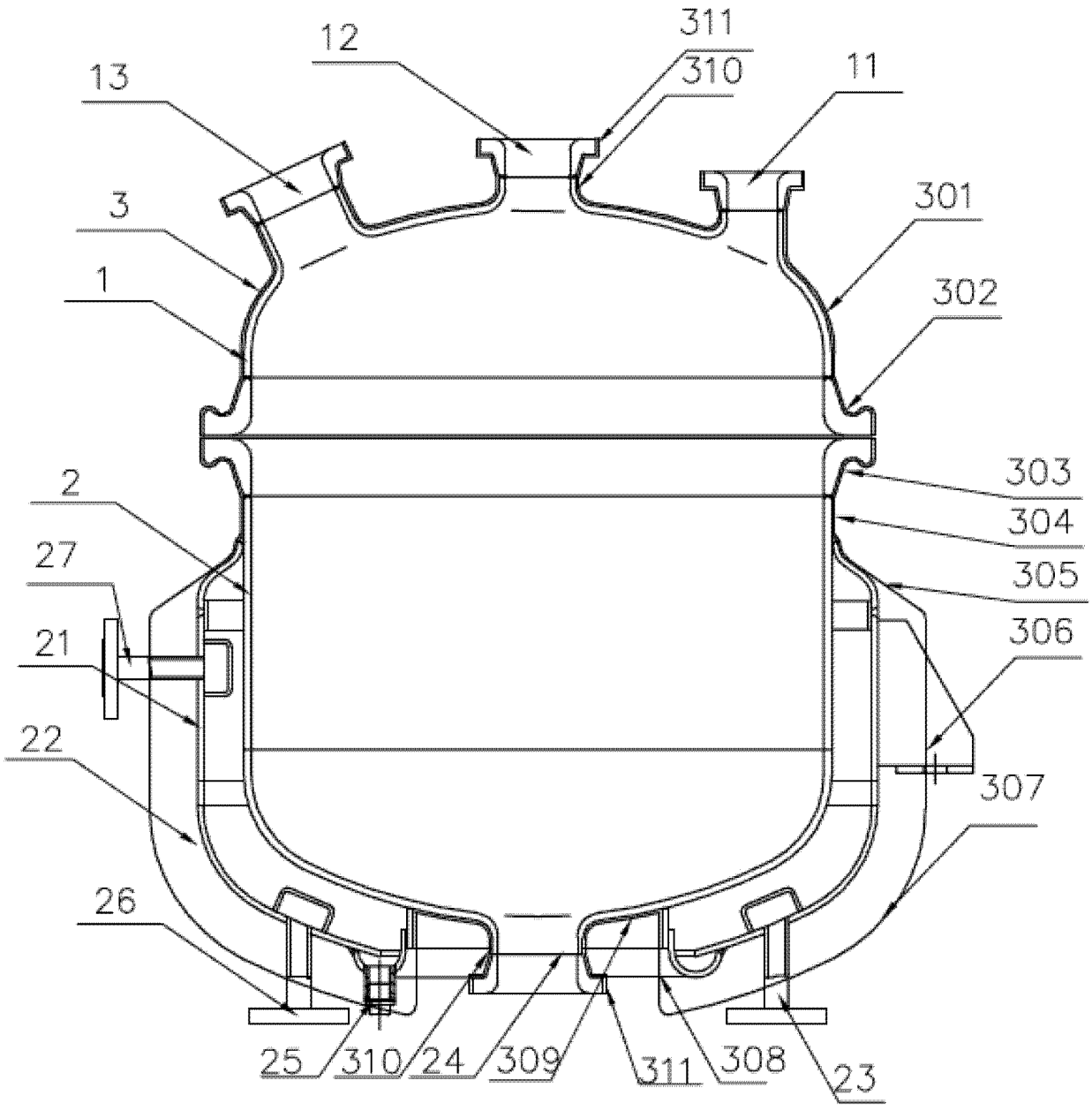

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图 1

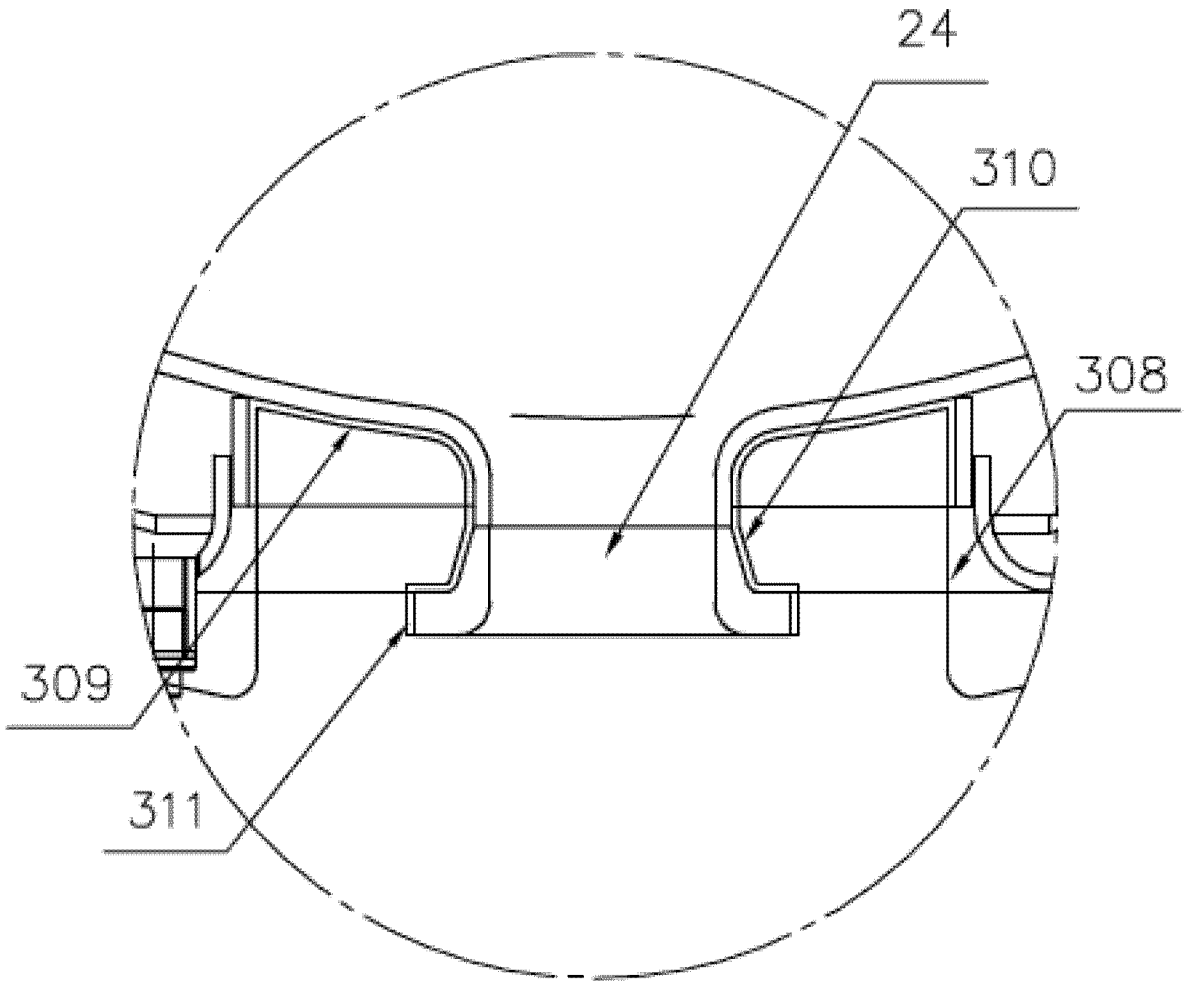


图 2

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CN103820781A Glass lined equipment provided with outer profiling cladding and manufacturing method of glass lined equipment

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Glass lined equipment provided with outer profiling cladding and manufacturing method of glass lined equipment

Abstract

The invention provides glass lined equipment provided with an outer profiling cladding and a manufacturing method of the glass lined equipment. The glass lined equipment mainly comprises a tank cover and a tank body, wherein an outer stainless steel cladding layer is arranged on the outer side of the tank cover, outer stainless steel cladding layers are arranged at the upper part of the outer side of the tank body and the outer side of a jacket, a gap between the jacket and the stainless steel outer cladding layers is filled with a thermal insulation material, and a thermal insulation layer is formed. GMP (good manufacturing practice) certification requirements that the production equipment cannot have any adverse effect on quality of medicine and the surface of production equipment directly contacted with medicine is required to be flat, smooth, easy to clean or disinfect and corrosion-resistant are met, and the production cost is lower than that of austenitic stainless steel glass lined equipment.

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Profiled outsourcing enamel glass device and manufacturing method thereof

Technical field

The invention relates to a glass-lined device and a manufacturing method thereof, in particular to a glass-lined device which has a stainless steel profiled outsourcing and meets the requirements of pharmaceutical GMP cleanliness.

Background technique

The glass-lined equipment is sprayed on the surface of the metal tire by a glassy glaze with high silicon content, and is fired several times at 920-960 °C to make the glass glaze adhere to the surface of the metal tire, so it has chemical stability similar to glass. It has the dual advantages of good mechanical properties and metal, and replaces stainless steel, non-ferrous metals, plastics and other equipment in many aspects.

With the rapid development of the bio-, pharmaceutical, food and other industries, not only the reactors and other equipment are required to have corrosion resistance and the appearance is clean and hygienic. The stainless steel material has a long service life and beautiful appearance, but the stainless steel is chlorine-containing or non-oxidizing. The acid medium is not resistant to corrosion, and may also cause the heavy metal ion content of special materials such as medicines to exceed the standard. It does not meet the requirements of GMP certification for chemical reaction with drugs, adsorption of drugs or release of substances into medicines; The base of the reactor is Q235 series or 20R carbon steel or low alloy steel. If austenitic stainless steel is used as the base of the glass-lined equipment and the special primer glaze is designed, it is made in a certain smoldering process. Stainless steel enamel equipment that can operate safely under working conditions, but the use of stainless steel as the base of the enamel glass equipment and the design of the special glaze is costly, and the simmering process is more complicated.

The prior art glass-lined equipment cannot meet the hygienic requirements of equipment used in clean workshops, and in particular, it cannot meet the cleanliness requirements of pharmaceutical companies for implementing the Good Manufacturing Practices (GMP).

Summary of the invention

In order to solve the problems existing in the prior art, the invention provides a method for copying the glass-lined equipment to solve the problem that the glass-lined equipment does not meet the GMP clean-up requirements, the production cost is high, and the simmering process is complicated.

The technical solution adopted by the present invention to solve the technical problem thereof is:

The utility model relates to a profiled outer glass-lined device, which mainly comprises a can lid and a can body, wherein the can lid is provided with a nozzle, a manhole and a sealing mouth; the outer part of the can body is covered with a jacket, the can body and the clip The sleeve is welded into a unitary structure with a cavity therebetween, a jacket flange is provided on one side of the tank body, and a discharge port, a condensate outlet, a steam inlet and a steam outlet are arranged at a lower portion;

The can lid surface has a stainless steel outer layer, and the stainless steel outer layer upper can lid head surface and the can lid flange surface together form a can lid stainless steel outer layer;

The outer side of the outer side of the can body and the outer side of the jacket have a stainless steel outer layer, and the upper part of the stainless steel outer layer of the outer side of the can body is covered with a can body

flange surface and a can body surface I stainless steel outer layer, and the can body and the jacket welding portion There is a canned transition surface I stainless steel outer layer, the outer side of the jacket is covered with a can body surface II stainless steel outer layer, and the lower part of the can body is covered with a stainless steel outer layer of the can body sealing surface, and the lower part of the can body and the jacket welding portion are covered The can body transition surface II stainless steel outer layer, the bottom of the can body is covered with a can body surface III stainless steel outer layer;

The nozzle, the manhole, the sealing port and the discharge opening are covered with a stainless steel outer cladding composed of a flange high neck surface and a flange surface;

The stainless steel outer cladding is a welded joint;

The insulating material is filled between the jacket and the stainless steel outer layer to form an insulating layer.

Preferably, the outer surface of the can lid has a stainless steel outer layer having a thickness of 1.5 to 3 mm, and the outside of the can body has a thickness of a stainless steel outer layer of 3 mm or more.

Preferably, the insulating material of the insulating layer is a polyurethane foaming material or a silicate material.

The method for manufacturing a stainless steel outer layer of the new glass-lined equipment of the contouring outsourcing comprises the following steps:

1）According to the actual size of the glass-lined equipment can lid 1 and the can body 2, the drawing lines are drawn on the 304 stainless steel plate by the respective constituent faces, and the drawn lines include the specific dimensions of the respective constituent faces, the flanges and the opening lines at the nozzles, and the line drawing. Afterwards, cut it with steel scissors according to the line;

2）The can lid head surface 301 and the can body head surface 307 of the finished stainless steel plate are preliminarily stamped with a mold, and holes are formed at the flange and the nozzle of the can lid head surface 301 and the can body head surface 307;

3）The can lid flange surface 301 is placed on the top of the glass can lid 1 , and the can lid 1 is unfolded toward the periphery of the can lid 1 , and the stainless steel plate is arranged side by side to make the shape conform to the shape of the can lid 1 , and the steel plate encounters unevenness or The drum and the wrinkle are hammered to the can lid 1 by a rubber hammer. The flange surface 302 of the can lid is initially formed by the hemming and then close to the flange of the lower part of the can lid 1. The butt joint is welded with argon arc welding. Welding, the can lid flange surface 302 and the can lid flange surface 301 are welded by argon arc welding, and the flange of the lower portion of the can lid 1 is uniformly spot welded by argon arc welding;

4）The can body flange surface 303, the can body surface I304 and the can body 2 and the jacket 21 of the can body 2 are welded under the transition surface I305 of the can body 2, and then the side is spread and finished, and the joint is argon-arc welded. Butt welding, the shape and the shape of the can lid 2 are the same, the steel plate meets the uneven or the convex drum, the wrinkle, the rubber hammer is used to hit the tightness with the can lid 2, the can body surface II306, the tank outside the jacket 21 The bottom of the body 2, the bottom of the can body 307, the lower part of the can body 2 and the jacket 21 are welded with a can body transition surface II308, and the bottom of the can body 2 is covered with a can body surface III309. The butt joint is welded by argon arc welding to make the shape and The can lid 2 has the same shape;

5）When the flange and the nozzle are attached to the periphery of the can lid 1 and the can body 2, the stainless steel plate around the flange is cut with a steel scissors to form a "V" shape, and the size is about 3 mm × (10-20) mm. After the stainless steel plate is turned over the flange surface and the can lid 1 and the periphery of the can body 2, the "V"-shaped notch is welded by argon arc welding. The diameter of the stainless steel wire used is Φ2.0, the welding current is 70-80A, and the welding voltage is 7- 9V;

6）The "V" type notch is welded by argon arc welding, and the can lid 1 and the can body 2 flange are evenly spot-welded by argon arc welding, and are knocked with a rubber hammer or the like to be closely adhered to the can lid 1 and the can body 2;

7）The stainless steel weld is also polished to 320 mesh and then polished to make the surface finish of the stainless steel plate meet the requirements;

8）Use steel scissors to repair the steel plate and the can lid 1 and the can body 2 are not tightly aligned, so that the shape is uniform.

The invention has the beneficial effects of providing a profiled outsourcing glass-lined device with a stainless steel outer layer on the outside of the can lid and the can body. The production equipment in accordance with the GMP certification must not have any adverse effect on the quality of the drug, and is in direct contact with the drug. The surface of the production equipment should be smooth, smooth, easy to clean or disinfect, and corrosion resistant, and the production cost is lower than that of austenitic stainless steel enamel equipment.

DRAWINGS

A further embodiment of the present invention with a profiled outsourcing glass-lined device will be further described below with reference to the accompanying drawings:

1 is a schematic structural view of a contoured outer glass container according to the present invention;

2 is a schematic view showing the enlarged structure of the flange at the flange of the profiled outer glass-lined container of the present invention;

In the picture: 1-can lid, 11-nozzle, 12-sealing port, 13-manhole, 2-can body, 21-jacket, 22-insulation layer, 23-steam inlet, 24-discharge port, 25 - Condensate outlet, 26-vapor outlet, 27-jacket adapter flange, 3-stainless steel outer cover, 301-can lid head face, 302-can lid flange face, 303-can body flange face, 304-can Body I, 305-can body transition surface I, 306-can body surface II, 307-can body head surface, 308-can body transition surface II, 309-can body surface III, 310-flange high neck surface, 311 - Flange face.

Detailed ways

The present invention will be further described below in conjunction with the accompanying drawings:

As shown in FIG. 1-2, a profiled outer glass-lined device mainly comprises a can lid 1 and a can body 2, and the can lid 1 is provided with a nozzle 11, a manhole 13 and a sealing port 12; 2 The outer lower part is covered with a jacket 21, and the can body 2 and the jacket 21 are welded into an integral structure with a cavity therebetween, and a side of the can body 2 is provided with a jacket flange 27 and a lower portion is disposed. There is a discharge port 24, a condensate outlet 25, a steam inlet 23 and a steam outlet 26; the surface of the can lid 1 has a stainless steel outer layer 3, the stainless steel outer layer 3 upper can lid head surface 301 and the can lid flange surface 302 Cooperating to form the stainless steel outer cover 3 of the can lid 1; the outer upper portion of the can body 2 and the outer side of the jacket 21 have a stainless steel outer layer 3, and the outer portion of the outer stainless steel outer layer 3 of the can body 2 is covered with a can body flange surface 303 and a can body I304 stainless steel outer cladding 3, the can body 2 and the jacket 21 welding portion is covered with a can body transition surface I305 stainless steel outer layer 3, the outer side of the jacket 21 is covered with a can body surface II306 stainless steel outer layer 3, the can body 2 The lower part is covered with a can body sealing surface 307 stainless steel outer layer 3, and the lower part of the can body 2 and the jacket 21 are welded with a can body transition surface II308 a stainless steel outer cladding 3, the bottom of the can body 2 is covered with a can body surface III309 stainless steel outer cladding layer 3; the nozzle opening 11, the manhole 13, the sealing opening 12 and the discharge opening 24 are covered with a flange high neck surface 310 A stainless steel outer cladding 3 composed of a flange surface 311; the stainless steel outer cladding is a welded joint.

The above description of the specific embodiments of the present invention has been described with reference to the accompanying drawings, but it is not intended to limit the scope of the present invention. Those skilled in the art should understand that the skilled in the art does not require the creative work on the basis of the technical solutions of the present invention. Various modifications or variations that can be made are still within the scope of the invention.

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1.The utility model relates to a profiled outer glass-lined device, which mainly comprises a can lid and a can body, wherein the can lid is provided with a nozzle, a manhole and a sealing mouth; the outer part of the can body is covered with a jacket, the can body and the clip The sleeve is welded into a unitary structure with a cavity therebetween, a jacket flange is provided on one side of the tank body, and a discharge port, a condensate outlet, a steam inlet and a steam outlet are arranged at a lower portion;

The can lid surface has a stainless steel outer layer, and the stainless steel outer layer upper can lid head surface and the can lid flange surface together form a can lid stainless steel outer layer;

The outer side of the outer side of the can body and the outer side of the jacket have a stainless steel outer layer, and the upper part of the stainless steel outer layer of the outer side of the can body is covered with a can body flange surface and a can body surface I stainless steel outer layer, and the can body and the jacket welding portion There is a canned transition surface I stainless steel outer layer, the outer side of the jacket is covered with a can body surface II stainless steel outer layer, and the lower part of the can body is covered with a stainless steel outer layer of the can body sealing surface, and the lower part of the can body and the jacket welding portion are covered The can body transition surface II stainless steel outer layer, the bottom of the can body is covered with a can body surface III stainless steel outer layer;

The nozzle, the manhole, the sealing port and the discharge opening are covered with a stainless steel outer cladding composed of a flange high neck surface and a flange surface;

The stainless steel outer cladding is a welded joint;

The insulating material is filled between the jacket and the stainless steel outer layer to form an insulating layer.

2.A contoured outer glass-lined device according to claim 1, wherein the outside of the can lid has a stainless steel outer layer having a thickness of 1.5 to 3 mm, and the outside of the can body has a thickness of stainless steel outer layer of 3 mm or more. .

3.A profiled outer glass-lined apparatus according to claim 1, wherein the heat insulating material of the heat insulating layer is a polyurethane foaming material or a silicate material.

4.A method for manufacturing a contoured outer glass-lined device according to any of claims 1-3, comprising the following steps:

1）According to the actual size of the glass-lined equipment can lid 1 and the can body 2, the drawing lines are drawn on the 304 stainless steel plate by the respective constituent faces, and the drawn lines include the specific dimensions of the respective constituent faces, the flanges and the opening lines at the nozzles, and the line drawing. Afterwards, cut it with steel scissors according to the line;

2）The can lid head surface 301 and the can body head surface 307 of the finished stainless steel plate are preliminarily stamped with a mold, and holes are formed at the flange and the nozzle of the can lid head surface 301 and the can body head surface 307;

3）The can lid flange surface 301 is placed on the top of the glass can lid 1 , and the can lid 1 is unfolded toward the periphery of the can lid 1 , and the stainless steel plate is arranged side by side to make the shape conform to the shape of the can lid 1 , and the steel plate encounters unevenness or The drum and the wrinkle are hammered to the can lid 1 by a rubber hammer. The flange surface 302 of the can lid is initially formed by the hemming and then close to the flange of the lower part of the can lid 1. The butt joint is welded with argon arc welding. Welding, the can lid flange surface 302 and the can lid flange surface 301 are welded by argon arc welding, and the flange of the lower portion of the can lid 1 is uniformly spot welded by argon arc welding;

4）The can body flange surface 303, the can body surface I304 and the can body 2 and the jacket 21 of the can body 2 are welded under the transition surface I305 of the can body 2, and then the side is spread and finished, and the joint is argon-arc welded. Butt welding, the shape and the shape of the can lid 2 are the same, the steel plate meets the uneven or the convex drum, the wrinkle, the rubber hammer is used to hit the tightness with the can lid 2, the can body surface II306, the tank outside the jacket 21 The bottom of the body 2, the bottom of the can body 307, the lower part of the can body 2 and the jacket 21 are welded with a can body transition surface II308, and the bottom of the can body 2 is covered with a can body surface III309. The butt joint is welded by argon arc welding to make the shape and The can lid 2 has the same shape;

5）When the flange and the nozzle are attached to the periphery of the can lid 1 and the can body 2, the stainless steel plate around the flange is cut with a steel scissors to form a "V" shape, and the size is about 3 mm × (10-20) mm. After the stainless steel plate is turned over the flange surface and the can lid 1 and the periphery of the can body 2, the "V"-shaped notch is welded by argon arc welding. The diameter of the stainless steel wire used is Φ2.0, the welding current is 70-80A, and the welding voltage is 7- 9V;

6）The "V" type notch is welded by argon arc welding, and the can lid 1 and the can body 2 flange are evenly spot-welded by argon arc welding, and are knocked with a rubber hammer or the like to be closely adhered to the can lid 1 and the can body 2;

7）The stainless steel weld is also polished to 320 mesh and then polished to make the surface finish of the stainless steel plate meet the requirements;

8）Use steel scissors to repair the steel plate and the can lid 1 and the can body 2 are not tightly aligned, so that the shape is uniform.

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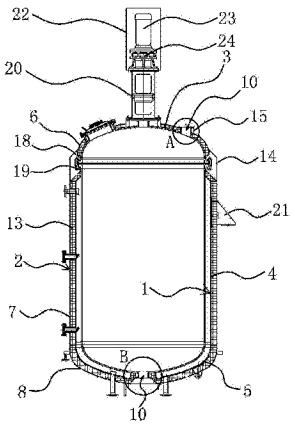
权利要求书2页 说明书5页 附图4页

(54)发明名称

一种搪玻璃反应釜的制作方法

(57)摘要

本发明提供了一种搪玻璃反应釜的制作方法，属于加工技术领域。它解决了现有一种搪玻璃反应釜的外表面耐腐蚀性较差等技术问题。本搪玻璃反应釜包括上封头、筒体和下封头，所述制作方法包括组装、制作包皮、开孔、固定、填料、抛光、罩子的制作与固定等七个步骤，本发明具有耐腐蚀性好、安全性高、外表美观等优点。



CN 105148816 A

1. 一种搪玻璃反应釜的制作方法，所述搪玻璃 (1) 反应釜包括上封头 (3)、筒体 (4) 和下封头 (5)，其特征在于，本制作方法包括如下步骤：

a、组装：将玻璃附着在碳钢上然后烧制成搪玻璃 (1)，所述上封头 (3)、筒体 (4) 和下封头 (5) 均由相应形状的搪玻璃 (1) 制成，将所述筒体 (4) 的下端与下封头 (5) 的上端焊接固定，将筒体 (4) 的上端与上封头 (3) 的下端可拆卸固定，将不锈钢材质的耳座 (21) 固定在筒体 (4) 的外侧，在上封头 (3) 的上端设置不锈钢材质的机架 (20)；

b、制作包皮：用胎具将不锈钢板压制成不锈钢包皮 (2)，所述不锈钢包皮 (2) 包括与上封头 (3) 对应的上包皮 (6)、与筒体 (4) 对应的中包皮 (7) 和与下封头 (5) 对应的下包皮 (8)；

c、开孔：在上包皮 (6) 上设有与设置在上封头 (3) 上的工艺管 (9) 对应的通孔 (10)，在下包皮 (8) 上设有与设置在下封头 (5) 上的工艺管 (9) 对应的通孔 (10)；

d、固定：在每个工艺管 (9) 的管口法兰 (11) 上均套设不锈钢法兰 (12)，把上包皮 (6) 从上往下罩住上封头 (3)，使工艺管 (9) 从上包皮 (6) 的通孔 (10) 中伸出，然后将上包皮 (6) 与不锈钢法兰 (12) 焊接，用中包皮 (7) 将筒体 (4) 包裹，用下包皮 (8) 从下往上罩住下封头 (5)，使工艺管 (9) 从下包皮 (8) 的通孔 (10) 中伸出，将下包皮 (8) 与不锈钢法兰 (12) 焊接，最后将下包皮 (8) 与中包皮 (7) 密封焊接；

e、填料：往不锈钢包皮 (2) 和搪玻璃 (1) 之间的间隙中填充保温材料 (13)，并密封焊接；

f、抛光：对不锈钢包皮 (2) 的外表面做抛光处理；

g、罩子的制作与固定：用不锈钢制作一环形的罩子 (14)，并将罩子 (14) 套设在上包皮 (6) 上，并使罩子 (14) 的上端与上包皮 (6) 密封焊接。

2. 根据权利要求 1 所述的一种搪玻璃反应釜的制作方法，其特征在于，所述步骤 d 的不锈钢法兰 (12) 外均套设有不锈钢筒体 (15)，所述不锈钢筒体 (15) 的上端与不锈钢法兰 (12) 密封焊接，所述不锈钢筒体 (15) 的下端位于通孔 (10) 内且与通孔 (10) 周边的不锈钢包皮 (2) 密封焊接。

3. 根据权利要求 1 所述的一种搪玻璃反应釜的制作方法，其特征在于，所述步骤 c 中，设置在上包皮 (6) 上并与上封头 (3) 的工艺管 (9) 对应的通孔 (10) 为第一通孔 (16)，设置在下包皮 (8) 上并与下封头 (5) 的工艺管 (9) 对应的通孔 (10) 为第二通孔 (17)，所述第一通孔 (16) 的孔径比套设在上封头 (3) 的工艺管 (9) 上的不锈钢法兰 (12) 的外径大 2mm-4mm，所述第二通孔 (17) 的孔径比套设在下封头 (5) 的工艺管 (9) 上的不锈钢法兰 (12) 的外径大 2mm-4mm。

4. 根据权利要求 1 所述的一种搪玻璃反应釜的制作方法，其特征在于，将所述步骤 e 中包有所述上包皮 (6) 的上封头 (3) 倒置，往上包皮 (6) 和上封头 (3) 之间的空隙中填充保温材料 (13)，使保温材料 (13) 将两者之间的间隙充满，然后将上包皮 (6) 的下端与设置在上封头 (3) 下端的第一不锈钢带环 (18) 密封焊接；往中包皮 (7) 和筒体 (4) 之间的空隙以及下包皮 (8) 和下封头 (5) 之间的空隙中填充保温材料 (13)，使保温材料 (13) 将上述的空隙充满，然后将中包皮 (7) 的上端与设置在筒体 (4) 上端的第二不锈钢带环 (19) 密封焊接，最后，将焊接好的上封头 (3) 与筒体 (4) 固连。

5. 根据权利要求 1 或 2 或 3 或 4 所述的一种搪玻璃反应釜的制作方法，其特征在于，所

述步骤 d 中，先将不锈钢法兰 (12) 对半切割，然后将切割后的不锈钢法兰 (12) 套在对应的工艺管 (9) 的管口法兰 (11) 上，再将对半切割的不锈钢法兰 (12) 焊接成整体。

6. 根据权利要求 1 或 2 或 3 或 4 所述的一种搪玻璃反应釜的制作方法，其特征在于，所述机架 (20) 上设有减速机 (24) 和电机 (23)，所述减速机 (24) 和电机 (23) 外均用不锈钢材质的外罩 (22) 罩住。

7. 根据权利要求 1 或 2 或 3 或 4 所述的一种搪玻璃反应釜的制作方法，其特征在于，所述步骤 g 中罩子 (14) 的上端内径小于上包皮 (6) 的最大直径，罩子 (14) 的下端内径大于上包皮 (6) 的最大直径且小于中包皮 (7) 的直径，罩子 (14) 的下端与中包皮 (7) 的外侧通过玻璃胶固定。

一种搪玻璃反应釜的制作方法

技术领域

[0001] 本发明属于加工技术领域，涉及一种搪玻璃反应釜的制作方法。

背景技术

[0002] 搪玻璃反应釜是将含高二氧化硅的玻璃，衬在钢制容器的内表面，经高温灼烧而牢固地密着于金属表面上成为复合材料制品，是一种优良的耐腐蚀设备。但由于煅烧后的玻璃无法附着在不锈钢上，因此，上述的钢制容器一般为碳钢，而裸露在外的碳钢又很容易被腐蚀。不仅影响反应釜的外观，被腐蚀后的锈渍还有可能在打开人孔的过程中飘进搪玻璃反应釜内，对搪玻璃反应釜内的反应造成影响。

[0003] 如申请号为“201520098060.1”的实用新型所公开的高效搪玻璃反应釜就是碳钢直接裸露在外部的，该高效搪玻璃反应釜包括釜体，其特征在于，所述的釜体上端连接有摆线减速机、电机，摆线减速机连接有搅拌轴，搅拌轴的中部连接有平直搅拌桨，搅拌轴的下端连接有叶轮搅拌桨，平直搅拌桨包括有内桨，内桨外部滑动连接有滑移桨，釜体的外表面设有夹层，釜体的上端设有进料口，釜体的底部连接有出料口。由于该高效搪玻璃反应釜的碳钢直接裸露在外部，因此反应釜的外表面很容易被腐蚀不仅影响反应釜的外观，被腐蚀后的锈渍还有可能在打开人孔的过程中飘进搪玻璃反应釜内，对搪玻璃反应釜内的反应造成影响。

[0004] 因此很有必要在碳钢外包上一层不锈钢，从而将碳钢与外部隔绝，使碳钢不会被腐蚀，但是由于碳钢与烧制后的玻璃是一体的，而不锈钢又无法与烧制后的玻璃直接焊接，因此在不锈钢和玻璃之间总会存在空隙，空气和水份会从该空隙中进入，碳钢依旧会被腐蚀，而如果用玻璃胶将不锈钢和玻璃之间的空隙堵住，由于玻璃胶的固定效果不是很好，仅通过玻璃胶固定，不锈钢和烧制后的玻璃之间很容易产生相对位移，导致空隙重新出现，影响搪玻璃反应釜的耐腐蚀效果。

发明内容

[0005] 本发明针对现有的技术存在的上述问题，提供一种搪玻璃反应釜的制作方法，本发明所要解决的技术问题是：如何制作外表面也能够耐腐蚀的搪玻璃反应釜。

[0006] 本发明的目的可通过下列技术方案来实现：

[0007] 一种搪玻璃反应釜的制作方法，所述搪玻璃反应釜包括上封头、筒体和下封头，其特征在于，本制作方法包括如下步骤：

[0008] a、组装：将玻璃附着在碳钢上然后烧制成搪玻璃，所述上封头、筒体和下封头均由相应形状的搪玻璃制成，将所述筒体的下端与下封头的上端焊接固定，将筒体的上端与上封头的下端可拆卸固定，将不锈钢材质的耳座固定在筒体的外侧，在上封头的上端设置不锈钢材质的机架；

[0009] b、制作包皮：用胎具将不锈钢板压制成不锈钢包皮，所述不锈钢包皮包括与上封头对应的上包皮、与筒体对应的中包皮和与下封头对应的下包皮；

[0010] c、开孔：在上包皮上设有与设置在上封头上的工艺管对应的通孔，在下包皮上设有与设置在下封头上的工艺管对应的通孔；

[0011] d、固定：在每个工艺管的管口法兰上均套设不锈钢法兰，把上包皮从上往下罩住上封头，使工艺管从上包皮的通孔中伸出，然后将上包皮与不锈钢法兰焊接，用中包皮将筒体包裹，用下包皮从下往上罩住下封头，使工艺管从下包皮的通孔中伸出，将下包皮与不锈钢法兰焊接，最后将下包皮与中包皮密封焊接；

[0012] e、填料：往不锈钢包皮和搪玻璃之间的间隙中填充保温材料，并密封焊接；

[0013] f、抛光：对不锈钢包皮的外表面做抛光处理；

[0014] g、罩子的制作与固定：用不锈钢制作一环形罩子，并将罩子套设在上包皮上，并使罩子的上端与上包皮密封焊接。

[0015] 其工作原理如下：本搪玻璃反应釜的制作方法中，由不锈钢制作的上包皮、中包皮和下包皮的直径要大于上封头、筒体和下封头的直径，该设计可以使不锈钢包皮能够更好将上封头、下封头和筒体包裹住，避免了因不锈钢包皮不能将上封头或下封头或筒体完全包裹而导致搪玻璃反应釜的耐腐蚀性降低等情况的出现；在每个工艺管的管口法兰上均套设不锈钢法兰，是因为不锈钢无法直接与玻璃焊接固定，而通过在管口法兰上套设不锈钢法兰，然后不锈钢包皮再与不锈钢法兰焊接固定的方式，实现了不锈钢法兰与工艺管的固定，增加了不锈钢包皮与搪玻璃连接的牢固度，并且不锈钢包皮还将搪玻璃上的碳钢部分完全包裹，提高了本搪玻璃反应釜的耐腐蚀性；而在不锈钢包皮和搪玻璃之间的间隙中填充保温材料，并密封焊接的操作能够进一步的将搪玻璃上的碳钢与外界进行隔绝，使本搪玻璃反应釜的耐腐蚀性更好；对不锈钢包皮的外表面做抛光处理，能够使本搪玻璃反应釜看起来更加美观；罩子的设计能够避免上封头与筒体分离时外界的异物进入筒体内，增加了本搪玻璃反应釜的安全性。

[0016] 在上述的一种搪玻璃反应釜的制作方法中，所述步骤 d 的不锈钢法兰外均套设有不锈钢筒体，所述不锈钢筒体的上端与不锈钢法兰密封焊接，所述不锈钢筒体的下端位于通孔内且与通孔周边的不锈钢包皮密封焊接。不锈钢筒体能够将不锈钢法兰与不锈钢包皮连接在一起，简化了不锈钢包皮的生产工艺，并使焊接后的结构强度更高，提高了不锈钢包皮与搪玻璃之间的密封性，有利于提高本搪玻璃反应釜的耐腐蚀性。

[0017] 在上述的一种搪玻璃反应釜的制作方法中，所述步骤 c 中，设置在上包皮上并与上封头的工艺管对应的通孔为第一通孔，设置在下包皮上并与下封头的工艺管对应的通孔为第二通孔，所述第一通孔的孔径比套设在上封头的工艺管上的不锈钢法兰的外径大 2mm-4mm，所述第二通孔的孔径比套设在下封头的工艺管上的不锈钢法兰的外径大 2mm-4mm。第一通孔的孔径比套设在上封头的工艺管上的不锈钢法兰的外径大 2mm-4mm，第二通孔的孔径比套设在下封头的工艺管上的不锈钢法兰的外径大 2mm-4mm，该设计为了方便在不锈钢法兰的外侧套设不锈钢筒体，使不锈钢包皮与搪玻璃之间的密封性得以提高，有利于提高本搪玻璃反应釜的耐腐蚀性。

[0018] 在上述的一种搪玻璃反应釜的制作方法中，将所述步骤 e 中包有所述上包皮的上封头倒置，往上包皮和上封头之间的空隙中填充保温材料，使保温材料将两者之间的间隙充满，然后将上包皮的下端与设置在上封头下端的第一不锈钢带环密封焊接；往中包皮和筒体之间的空隙以及下包皮和下封头之间的空隙中填充保温材料，使保温材料将上述的空

隙充满，然后将中包皮的上端与设置在筒体上端的第二不锈钢带环密封焊接，最后，将焊接好的上封头与筒体固连。保温材料的填充，并进行密封焊接，该设计能够进一步的将搪玻璃上的碳钢与外界进行隔绝，使本搪玻璃反应釜的耐腐蚀性更好。

[0019] 在上述的一种搪玻璃反应釜的制作方法中，所述步骤 d 中，先将不锈钢法兰对半切割，然后将切割后的不锈钢法兰套在对应的工艺管的管口法兰上，再将对半切割的不锈钢法兰焊接成整体。该设计能够在不损伤工艺管的情况下较好的将不锈钢法兰与工艺管进行固定，然后使不锈钢包皮与该不锈钢法兰固连，增加了不锈钢包皮与搪玻璃连接的牢固度，并且不锈钢包皮还将搪玻璃上的碳钢部分完全包裹，提高了本搪玻璃反应釜的耐腐蚀性。

[0020] 在上述的一种搪玻璃反应釜的制作方法中，所述机架上设有减速机和电机，所述减速机和电机外均用不锈钢材质的外罩罩住。该设计能够提高减速机和电机的耐腐蚀性，从而从局部上提高了搪玻璃反应釜的耐腐蚀性，并且使本搪玻璃反应釜的外观更加美观。

[0021] 在上述的一种搪玻璃反应釜的制作方法中，所述步骤 g 中罩子的上端内径小于上包皮的最大直径，罩子的下端内径大于上包皮的最大直径且小于中包皮的直径，罩子的下端与中包皮的外侧通过玻璃胶固定。罩子的上端内径小于上包皮的最大直径，能够使罩子较好的卡在上包皮上，并且与上包皮焊接固定，同时与中包皮用玻璃胶连接固定可以使罩子内部与外界隔绝，避免了异物进入筒体内，提高了本搪玻璃反应釜的安全性。

[0022] 与现有技术相比，本发明的优点如下：

[0023] 本搪玻璃反应釜的制作方法中在管口法兰上套设不锈钢法兰，然后不锈钢包皮再与不锈钢法兰焊接固定，实现了不锈钢法兰与工艺管的固定，增加了不锈钢包皮与搪玻璃连接的牢固度，并且不锈钢包皮还将搪玻璃上的碳钢部分完全包裹，提高了本搪玻璃反应釜的耐腐蚀性。

附图说明

[0024] 图 1 是本搪玻璃反应釜的整体结构示意图。

[0025] 图 2 是图 1 中 B 处的局部放大图。

[0026] 图 3 是图 1 中 A 处的局部放大图。

[0027] 图 4 是本搪玻璃反应釜的制作方法的流程图。

[0028] 图中，1 搪玻璃；2 不锈钢包皮；3 上封头；4 筒体；5 下封头；6 上包皮；7 中包皮；8 下包皮；9 工艺管；10 通孔；11 管口法兰；12 不锈钢法兰；13 保温材料；14 罩子；15 不锈钢筒体；16 第一通孔；17 第二通孔；18 第一不锈钢带环；19 第二不锈钢带环；20 机架；21 耳座；22 外罩；23 电机；24 减速机。

具体实施方式

[0029] 以下是本发明的具体实施例并结合附图，对本发明的技术方案作进一步的描述，但本发明并不限于这些实施例。

[0030] 如图 1-4 所示，本搪玻璃反应釜包括上封头 3、筒体 4 和下封头 5，本制作方法包括如下步骤：

[0031] a、组装：将玻璃附着在碳钢上然后烧制成搪玻璃 1，上封头 3、筒体 4 和下封头 5 均

由相应形状的搪玻璃 1 制成，将筒体 4 的下端与下封头 5 的上端焊接固定，将筒体 4 的上端与上封头 3 的下端可拆卸固定，将不锈钢材质的耳座 21 固定在筒体 4 的外侧，在上封头 3 的上端设置不锈钢材质的机架 20；机架 20 上设有减速机 24 和电机 23，减速机 24 和电机 23 外均用不锈钢材质的外罩 22 罩住。该设计能够提高减速机 24 和电机 23 的耐腐蚀性，从而从局部上提高了搪玻璃反应釜的耐腐蚀性，并且使本搪玻璃反应釜的外观更加美观；

[0032] b、制作包皮：用胎具将不锈钢板压制成不锈钢包皮 2，不锈钢包皮 2 包括与上封头 3 对应的上包皮 6、与筒体 4 对应的中包皮 7 和与下封头 5 对应的下包皮 8；本搪玻璃反应釜的制作方法中，由不锈钢制作的上包皮 6、中包皮 7 和下包皮 8 的直径要大于上封头 3、筒体 4 和下封头 5 的直径，该设计可以使不锈钢包皮 2 能够更好将上封头 3、下封头 5 和筒体 4 包裹住，避免了因不锈钢包皮 2 不能将上封头 3 或下封头 5 或筒体 4 完全包裹而导致搪玻璃反应釜的耐腐蚀性降低等情况的出现；

[0033] c、开孔：在上包皮 6 上设有与设置在上封头 3 上的工艺管 9 对应的通孔 10，在下包皮 8 上设有与设置在下封头 5 上的工艺管 9 对应的通孔 10；其中设置在上包皮 6 上并与上封头 3 的工艺管 9 对应的通孔 10 为第一通孔 16，设置在下包皮 8 上并与下封头 5 的工艺管 9 对应的通孔 10 为第二通孔 17，第一通孔 16 的孔径比套设在上封头 3 的工艺管 9 上的不锈钢法兰 12 的外径大 2mm-4mm，第二通孔 17 的孔径比套设在下封头 5 的工艺管 9 上的不锈钢法兰 12 的外径大 2mm-4mm，该设计为了方便在不锈钢法兰 12 的外侧套设不锈钢筒体 15，使不锈钢包皮 2 与搪玻璃 1 之间的密封性得以提高，有利于提高本搪玻璃反应釜的耐腐蚀性；

[0034] d、固定：在每个工艺管 9 的管口法兰 11 上均套设不锈钢法兰 12，在每个工艺管 9 的管口法兰 11 上均套设不锈钢法兰 12，是因为不锈钢无法直接与玻璃焊接固定，而通过在管口法兰 11 上套设不锈钢法兰 12，然后不锈钢包皮 2 再与不锈钢法兰 12 焊接固定的方式，实现了不锈钢法兰 12 与工艺管 9 的固定，增加了不锈钢包皮 2 与搪玻璃 1 连接的牢固度，并且不锈钢包皮 2 还将搪玻璃 1 上的碳钢部分完全包裹，提高了本搪玻璃反应釜的耐腐蚀性；在装设不锈钢法兰 12 时，先将不锈钢法兰 12 对半切割，然后将切割后的不锈钢法兰 12 套在对应的工艺管 9 的管口法兰 11 上，再将对半切割的不锈钢法兰 12 焊接成整体。该设计能够在不损伤工艺管 9 的情况下较好的将不锈钢法兰 12 与工艺管 9 进行固定，然后使不锈钢包皮 2 与该不锈钢法兰 12 固连，增加了不锈钢包皮 2 与搪玻璃 1 连接的牢固度，并且不锈钢包皮 2 还将搪玻璃 1 上的碳钢部分完全包裹，提高了本搪玻璃反应釜的耐腐蚀性。

[0035] 本搪玻璃反应釜的制作方法在固定上包皮 6 和上封头 3 时，把上包皮 6 从上往下罩住上封头 3，使工艺管 9 从上包皮 6 的通孔 10 中伸出，然后将上包皮 6 与不锈钢法兰 12 焊接，用中包皮 7 将筒体 4 包裹，用下包皮 8 从下往上罩住下封头 5，使工艺管 9 从下包皮 8 的通孔 10 中伸出，将下包皮 8 与不锈钢法兰 12 焊接，最后将下包皮 8 与中包皮 7 密封焊接；不锈钢法兰 12 外均套设有不锈钢筒体 15，不锈钢筒体 15 的上端与不锈钢法兰 12 密封焊接，不锈钢筒体 15 的下端位于通孔 10 内且与通孔 10 周边的不锈钢包皮 2 密封焊接，不锈钢筒体 15 能够将不锈钢法兰 12 与不锈钢包皮 2 连接在一起，简化了不锈钢包皮 2 的生产工艺，并使焊接后的结构强度更高，提高了不锈钢包皮 2 与搪玻璃 1 之间的密封性，有利于提高本搪玻璃反应釜的耐腐蚀性；

[0036] e、填料：往不锈钢包皮 2 和搪玻璃 1 之间的间隙中填充保温材料 13，并密封焊接；

将包有上包皮 6 的上封头 3 倒置，往上包皮 6 和上封头 3 之间的空隙中填充保温材料 13，使保温材料 13 将两者之间的间隙充满，然后将上包皮 6 的下端与设置在上封头 3 下端的第一不锈钢带环 18 密封焊接；往中包皮 7 和筒体 4 之间的空隙以及下包皮 8 和下封头 5 之间的空隙中填充保温材料 13，使保温材料 13 将上述的空隙充满，然后将中包皮 7 的上端与设置在筒体 4 上端的第二不锈钢带环 19 密封焊接，最后，将焊接好的上封头 3 与筒体 4 固连。保温材料 13 的填充，并进行密封焊接，该设计能够进一步的将搪玻璃 1 上的碳钢与外界进行隔绝，使本搪玻璃反应釜的耐腐蚀性更好；

[0037] f、抛光：对不锈钢包皮 2 的外表面做抛光处理；对不锈钢包皮 2 的外表面做抛光处理，能够使本搪玻璃反应釜看起来更加美观；

[0038] g、罩子 14 的制作与固定：用不锈钢制作一环形罩子 14，并将罩子 14 套设在上包皮 6 上，并使罩子 14 的上端与上包皮 6 密封焊接，罩子 14 的上端内径小于上包皮 6 的最大直径，罩子 14 的下端内径大于上包皮 6 的最大直径且小于中包皮 7 的直径，罩子 14 的下端与中包皮 7 的外侧通过玻璃胶固定。罩子 14 的上端内径小于上包皮 6 的最大直径，能够使罩子 14 较好的卡在上包皮 6 上，并且与上包皮 6 焊接固定，同时与中包皮 7 用玻璃胶连接固定可以使罩子 14 内部与外界隔绝，避免了异物进入筒体 4 内，提高了本搪玻璃反应釜的安全性。

[0039] 本文中所描述的具体实施例仅仅是对本发明精神作举例说明。本发明所属技术领域的技术人员可以对所描述的具体实施例做各种各样的修改或补充或采用类似的方式替代，但并不会偏离本发明的精神或者超越所附权利要求书所定义的范围。

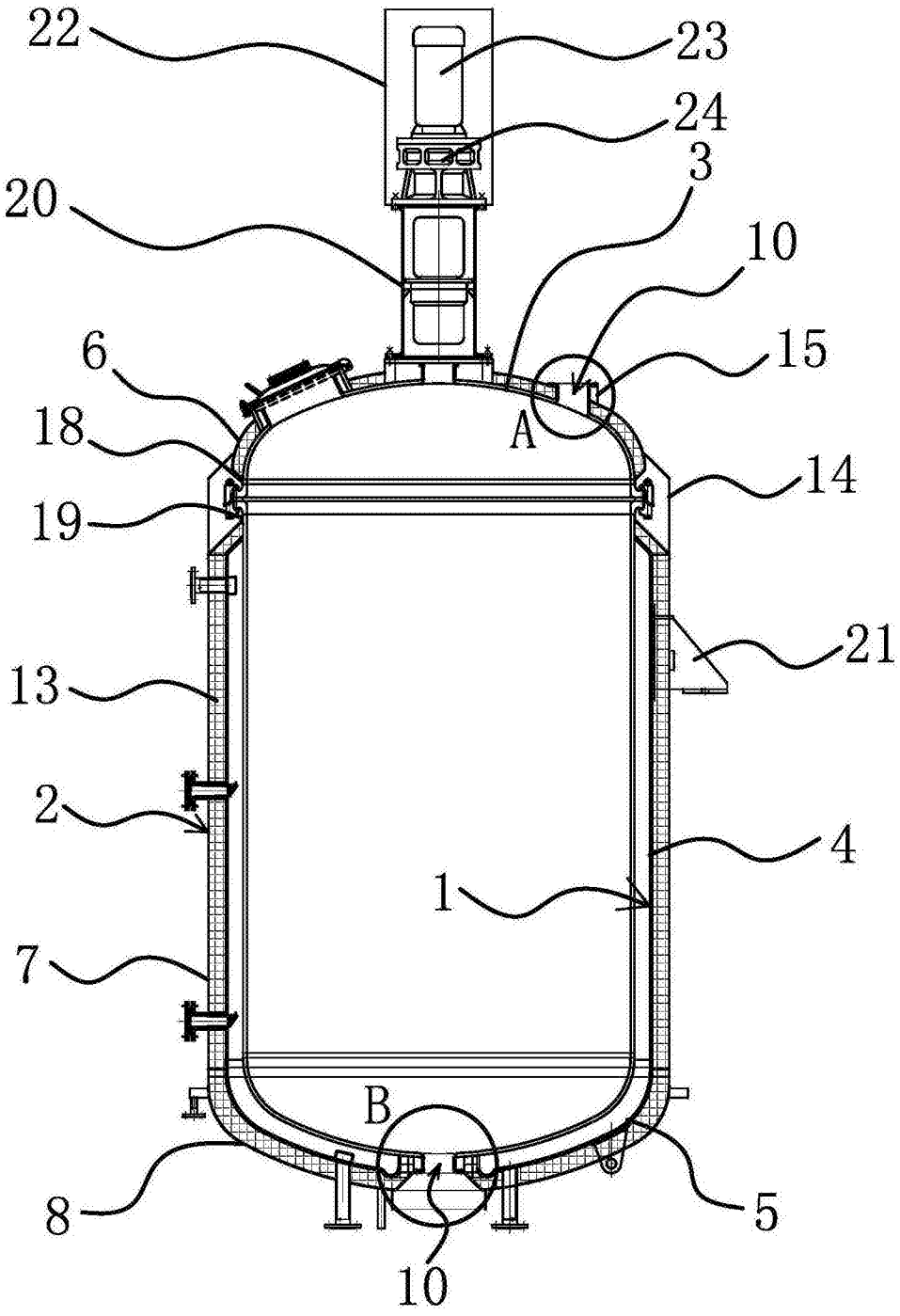


图 1

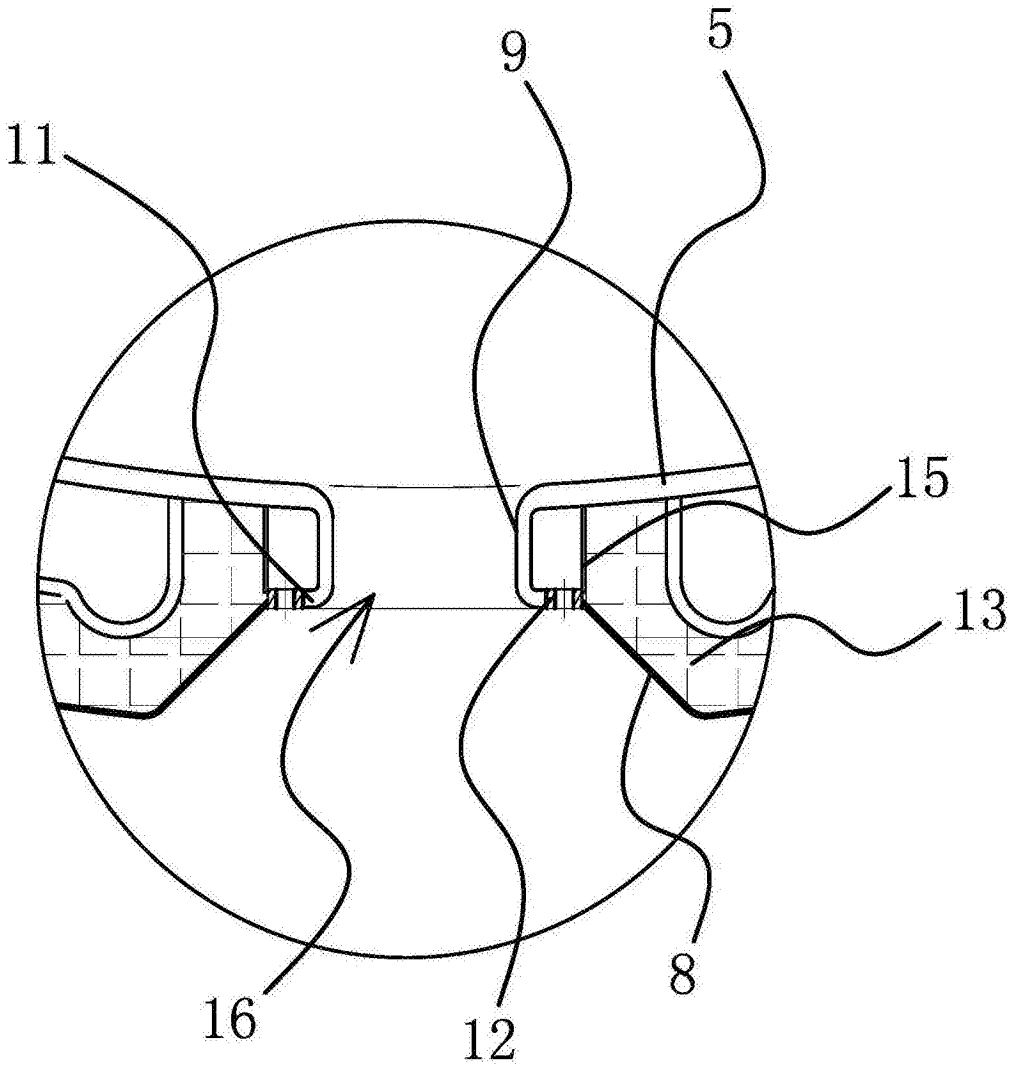


图 2

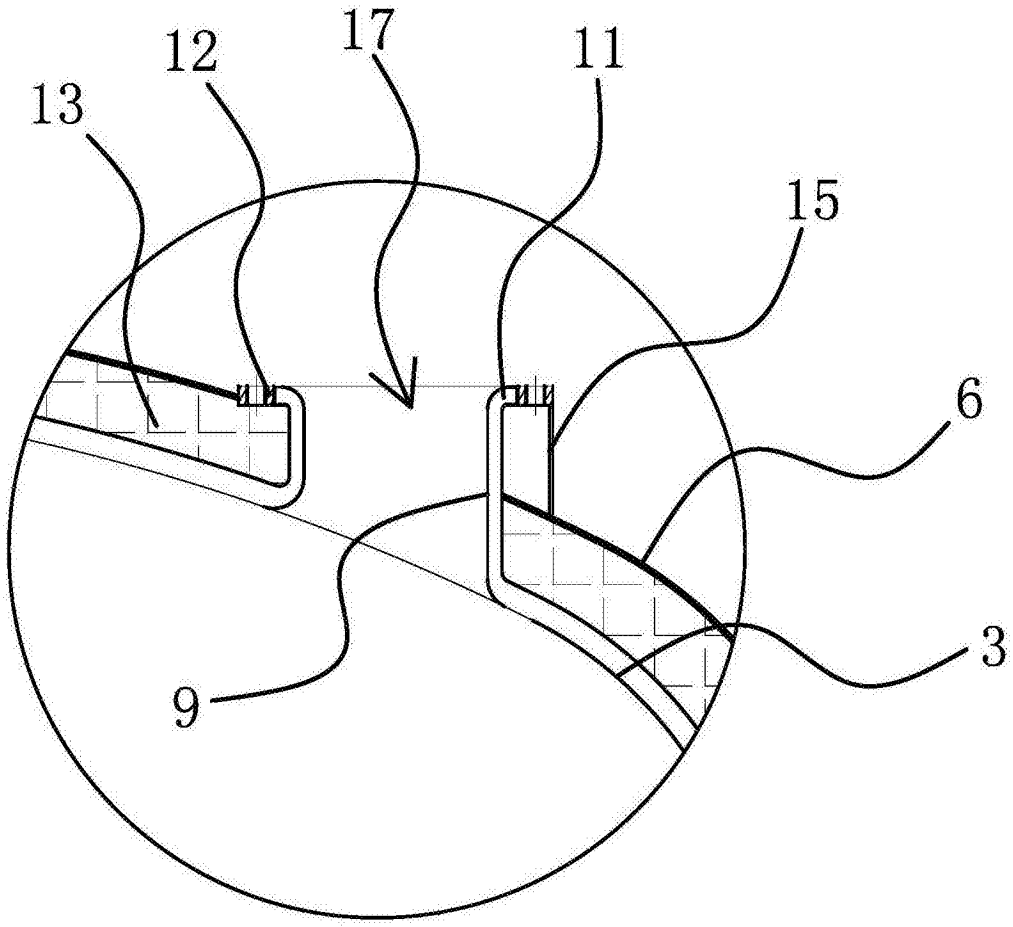


图 3

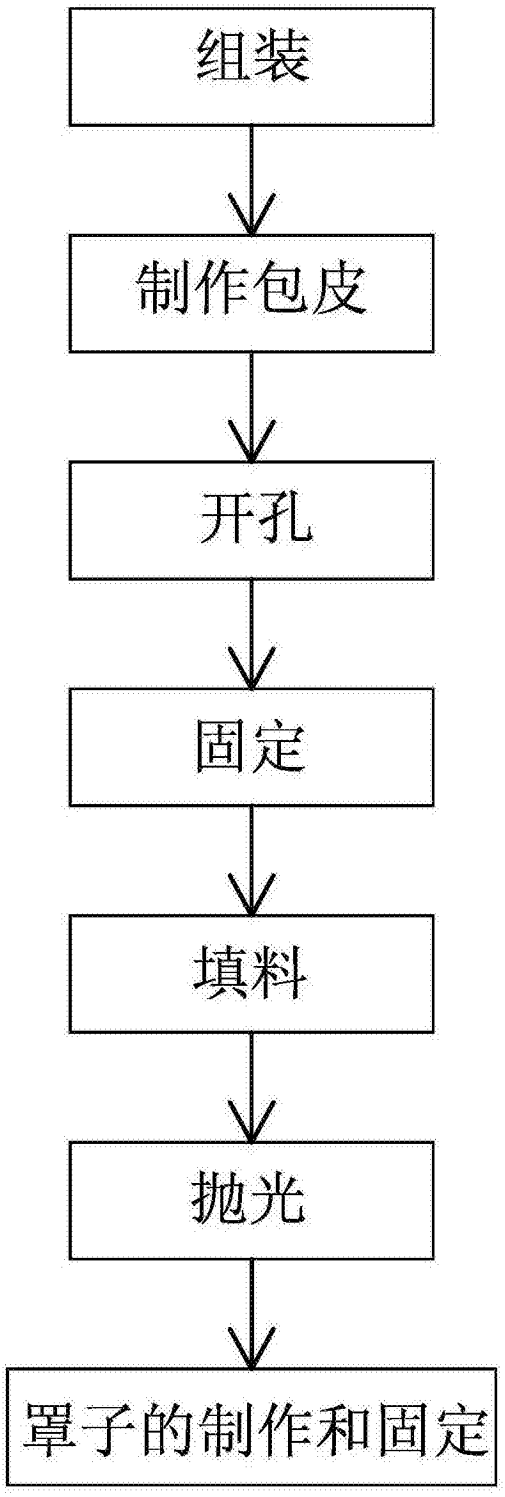


图 4

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CN105148816A Manufacturing method of glass-lined reaction kettle

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Manufacturing method of glass-lined reaction kettle

Abstract

The invention provides a manufacturing method of a glass-lined reaction kettle, and belongs to the technical field of processing, which solves the problems that the outer surface of the glass-lined reaction kettle is poor in corrosion resistance. The glass-lined reaction kettle comprises an upper end enclosure, a barrel and a lower end enclosure. The manufacturing method comprises seven steps of assembling, manufacturing a wrapper, perforating, fixing, packing, polishing and manufacturing and fixing a cover. The manufactured glass-lined reaction kettle has advantages of good corrosion resistance, high safety, attractive appearance and the like.

CN105148816A Manufacturing method of glass-lined reaction kettle

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Method for manufacturing bismuth glass reactor

Technical field

The invention belongs to the technical field of processing and relates to a method for preparing a glass-lined reactor.

Background technique

The glass-lined reactor is a high-silica-containing glass which is lining the inner surface of a steel container and is firmly adhered to the metal surface by high temperature burning to form a composite product. It is an excellent corrosion-resistant device. However, since the calcined glass cannot adhere to the stainless steel, the above-mentioned steel container is generally carbon steel, and the exposed carbon steel is easily corroded. Not only does it affect the appearance of the reactor, but the corroded rust stains may also drift into the glass-lined reactor during the opening of the manhole, which has an effect on the reaction in the glass-lined reactor.

The high-efficiency glass-lined reactor disclosed in the utility model as "201520098060.1" is a carbon steel directly exposed to the outside, and the high-efficiency glass-lined reactor comprises a kettle body, characterized in that the upper end of the kettle body is connected with a pendulum The line reducer, the motor, the cycloid reducer are connected with a stirring shaft, the middle of the stirring shaft is connected with a straight stirring paddle, the lower end of the stirring shaft is connected with an impeller stirring paddle, the straight stirring paddle includes an inner paddle, and the inner paddle is externally slidingly connected. There is a sliding paddle, and the outer surface of the kettle body is provided with an interlayer, the upper end of the kettle body is provided with a feeding port, and the bottom of the kettle body is connected with a discharging port. Since the carbon steel of the high-efficiency glass-lined reactor is directly exposed to the outside, the outer surface of the reactor is easily corroded and not only affects the appearance of the reactor, but also the corroded rust stains may fall into the process of opening the manhole. In the glass-lined reactor, the reaction in the glass-lined reactor was affected.

Therefore, it is necessary to coat a layer of stainless steel in carbon steel to isolate the carbon steel from the outside, so that the carbon steel will not be corroded, but because the carbon steel is integrated with the fired glass, the stainless steel cannot be fired. The rear glass is directly welded, so there will always be a gap between the stainless steel and the glass. Air and moisture will enter from the gap, and the carbon steel will still be corroded, and if the gap between the stainless steel and the glass is blocked with glass glue. Living, because the fixing effect of the glass glue is not very good, it is only fixed by the glass glue. The relative displacement between the stainless steel and the fired glass is easy to occur, resulting in the reappearance of the void and affecting the corrosion resistance of the glass-lined reactor.

Summary of the invention

The present invention provides a method for producing a glass-lined reactor according to the above problems in the prior art, and the technical problem to be solved by the present invention is how to produce a glass-lined reactor which is also resistant to corrosion on the outer surface.

The object of the invention can be achieved by the following technical solutions:

A method for fabricating a glass-lined reactor comprising an upper head, a barrel and a lower head, wherein the manufacturing method comprises the following steps:

a, assembly: the glass is attached to the carbon steel and then fired into a bismuth glass, the upper head, the cylinder and the lower head are made of the correspondingly shaped bismuth glass, the lower end and the lower

seal of the cylinder The upper end of the head is welded and fixed, and the upper end of the cylinder body and the lower end of the upper head are detachably fixed, the stainless steel ear seat is fixed on the outer side of the cylinder body, and the upper end of the upper head is provided with a stainless steel frame;

b. Making a foreskin: pressing a stainless steel plate into a stainless steel foreskin with a tire, the stainless steel sheath comprising an upper foreskin corresponding to the upper head, a middle fore corresponding to the barrel, and a lower fore covering corresponding to the lower head;

c. opening: a through hole corresponding to the process pipe disposed on the upper head is disposed on the upper cover, and a through hole corresponding to the process pipe disposed on the lower cover is disposed on the lower cover;

d. Fixed: a stainless steel flange is placed on the nozzle flange of each process pipe, and the upper cover is covered from the top to the upper cover, so that the process pipe protrudes from the through hole of the upper foreskin, and then The upper foreskin is welded with the stainless steel flange, and the cylinder is wrapped with the middle foreskin, and the lower cover is covered from the bottom with the lower foreskin, so that the process tube protrudes from the through hole of the lower foreskin, and the lower foreskin is welded with the stainless steel flange. Finally, the lower foreskin and the middle foreskin are sealed and welded;

e, filler: fill the gap between the stainless steel sheath and the enamel glass with insulation material, and seal welding;

f, polishing: polishing the outer surface of the stainless steel foreskin;

g. Making and fixing of the cover: an annular cover is made of stainless steel, and the cover is placed on the upper cover, and the upper end of the cover is sealed and welded to the upper cover.

The working principle is as follows: in the manufacturing method of the glass reactor, the diameters of the upper foreskin, the middle foreskin and the lower foreskin made of stainless steel are larger than the diameters of the upper head, the barrel body and the lower head, and the design can make the stainless steel foreskin It can better wrap the upper head, the lower head and the cylinder, avoiding the corrosion resistance of the glass-lined reactor caused by the stainless steel foreskin not being able to completely wrap the upper or lower head or the cylinder. Appears; stainless steel flanges are placed on the nozzle flange of each process tube because stainless steel cannot be directly fixed to the glass, but by setting a stainless steel flange on the nozzle flange, then stainless steel foreskin and stainless steel The method of flange welding fixes the fixing of the stainless steel flange and the process pipe, increases the firmness of the connection between the stainless steel sheath and the bismuth glass, and the stainless steel foreskin also completely wraps the carbon steel part of the enamel glass to improve the bismuth glass. Corrosion resistance of the reactor; filling the gap between the stainless steel sheath and the bismuth glass with insulation material, and the operation of sealing welding can One step is to isolate the carbon steel on the glass from the outside, so that the corrosion resistance of the glass reactor is better; polishing the outer surface of the stainless steel sheath can make the glass reactor look more beautiful; The design can avoid the foreign matter entering the cylinder when the upper head is separated from the cylinder, which increases the safety of the glass reactor.

In the above method for manufacturing a glass-lined reactor, the stainless steel flange of the step d is sleeved with a stainless steel cylinder, and the upper end of the stainless steel cylinder is sealed and welded with a stainless steel flange. The lower end is located in the through hole and is sealed and welded to the stainless steel sheath around the through hole. The stainless steel cylinder can connect the stainless steel flange with the stainless steel foreskin, which simplifies the production process of the stainless steel foreskin and makes the structural strength after welding higher, improving the sealing between the stainless steel sheath and the bismuth glass, which is beneficial to the improvement of the present. Corrosion resistance of glass-lined reactors.

In the above method for manufacturing a bismuth glass reactor, in the step c, the through hole disposed on the upper cover and corresponding to the process tube of the upper head is the first through hole, and is disposed on the lower foreskin and under The through hole corresponding to the process pipe of the head is a second through hole, and the diameter of the first through hole is larger than the outer diameter of the stainless steel flange sleeved on the process pipe of the upper head by 2 mm to 4 mm, the second The diameter of the through hole is 2 mm to 4 mm larger than the outer diameter of the stainless steel flange sleeved on the process pipe of the lower head. The diameter of the first through hole is larger than the outer diameter of the stainless steel flange sleeved on the process pipe of the upper head by 2 mm to 4 mm, and the diameter of the second through hole is

smaller than that of the stainless steel flange of the process pipe which is sleeved on the lower head The outer diameter is 2mm-4mm. This design is designed to facilitate the sealing of the stainless steel sheath and the enamel glass to facilitate the corrosion resistance of the glass reactor.

In the above method for manufacturing a glass-lined reactor, the upper head of the step e is covered with the upper cover, and the space between the upper cover and the upper cover is filled with a heat insulating material to make the heat preservation. The material fills the gap between the two, and then seals the lower end of the upper sheath with the first stainless steel belt loop disposed at the lower end of the upper head; the gap between the middle sheath and the cylinder and the lower and lower heads The gap is filled with the heat insulating material, so that the heat insulating material fills the above gap, and then the upper end of the middle sheath is sealed and welded with the second stainless steel belt ring disposed at the upper end of the cylinder. Finally, the welded upper head and the barrel are welded. Solid. The insulation material is filled and sealed and welded. This design can further insulate the carbon steel on the bismuth glass from the outside, so that the corrosion resistance of the bismuth glass reactor is better.

In the above method for manufacturing a bismuth glass reactor, in the step d, the stainless steel flange is first cut in half, and then the cut stainless steel flange is sleeved on the nozzle flange of the corresponding process tube. The semi-cut stainless steel flange is then welded into one piece. The design can better fix the stainless steel flange and the process tube without damaging the process tube, and then fix the stainless steel sheath to the stainless steel flange, thereby increasing the firmness of the stainless steel sheath and the bismuth glass connection, and the stainless steel The foreskin also completely encapsulates the carbon steel portion of the glass, which improves the corrosion resistance of the glass reactor.

In the above method for manufacturing a glass-lined reactor, the frame is provided with a reducer and a motor, and the reducer and the motor are covered with a stainless steel outer cover. This design improves the corrosion resistance of the reducer and the motor, thereby locally improving the corrosion resistance of the glass-lined reactor and making the appearance of the glass-lined reactor more aesthetically pleasing.

In the above method for manufacturing a bismuth glass reactor, the inner diameter of the upper end of the cover in the step g is smaller than the maximum diameter of the upper foreskin, and the inner diameter of the lower end of the cover is larger than the maximum diameter of the upper foreskin and smaller than the diameter of the middle foreskin, the lower end of the cover. The outer side of the middle foreskin is fixed by glass glue. The inner diameter of the upper end of the cover is smaller than the maximum diameter of the upper cover, so that the cover can be better clamped on the upper cover and fixed to the upper cover, and the middle cover can be sealed with the glass glue to isolate the inside of the cover from the outside, thereby avoiding foreign matter. Entering the cylinder improves the safety of the glass reactor.

The advantages of the present invention over the prior art are as follows:

In the manufacturing method of the glass reactor, a stainless steel flange is set on the nozzle flange, and then the stainless steel sheath is welded and fixed to the stainless steel flange to realize the fixing of the stainless steel flange and the process tube, and the stainless steel foreskin and the enamel glass are added. The tightness of the connection, and the stainless steel sheath also completely encapsulates the carbon steel portion of the glass, improving the corrosion resistance of the glass reactor.

DRAWINGS

Figure 1 is a schematic view showing the overall structure of the present glass reactor.

Figure 2 is a partial enlarged view of B in Figure 1.

Figure 3 is a partial enlarged view of a portion A in Figure 1.

4 is a flow chart of a method of fabricating the glass-lined reactor.

In the figure, 1 搪 glass; 2 stainless steel foreskin; 3 upper head; 4 cylinders; 5 lower heads; 6 upper foreskin; 7 medium foreskin; 8 lower foreskin; 9 process tubes; 10 through holes; 12 stainless steel flange; 13 insulation material; 14 cover; 15 stainless steel cylinder; 16 first through hole; 17 second through hole; 18 first stainless steel belt ring; 19 second stainless steel belt ring; 20 frame; Seat; 22 cover; 23 motor; 24 reducer.

Detailed ways

The technical examples of the present invention are further described below with reference to the specific embodiments of the present invention, but the present invention is not limited to the embodiments.

As shown in Figures 1-4, the present glass reactor comprises an upper head 3, a cylinder 4 and a lower head 5. The manufacturing method comprises the following steps:

a. Assembly: the glass is attached to the carbon steel and then fired into the bismuth glass 1. The upper head 3, the cylinder 4 and the lower head 5 are all made of the correspondingly shaped bismuth glass 1, and the lower end of the cylinder 4 is The upper end of the lower head 5 is welded and fixed, and the upper end of the cylinder 4 and the lower end of the upper head 3 are detachably fixed, and the stainless steel ear holder 21 is fixed to the outer side of the cylinder 4, and the upper end of the upper head 3 is provided with stainless steel. The frame 20 of the material; the frame 20 is provided with a reducer 24 and a motor 23, and the reducer 24 and the motor 23 are all covered with a stainless steel cover 22. The design can improve the corrosion resistance of the reducer 24 and the motor 23, thereby locally improving the corrosion resistance of the glass-lined reactor and making the appearance of the glass-lined reactor more beautiful;

b. Making a foreskin: The stainless steel sheet is pressed into a stainless steel sheath 2 by a tire, and the stainless steel sheath 2 includes an upper sheath 6 corresponding to the upper head 3, a middle sheath 7 corresponding to the cylinder 4, and a lower portion corresponding to the lower head 5. The foreskin 8; in the manufacturing method of the glass-lined reactor, the diameters of the upper fore skin 6, the middle fore skin 7 and the lower foreskin 8 made of stainless steel are larger than the diameters of the upper head 3, the barrel 4 and the lower head 5, the design The stainless steel sheath 2 can be better wrapped around the upper head 3, the lower head 5 and the cylinder 4, so that the stainless steel sheath 2 cannot completely wrap the upper head 3 or the lower head 5 or the cylinder 4 The occurrence of corrosion resistance of the glass-lined reactor is reduced;

c. Opening: a through hole 10 corresponding to the process pipe 9 provided on the upper head 3 is provided on the upper cover 6, and a corresponding passage corresponding to the process pipe 9 provided on the lower head 5 is provided on the lower cover 8. The hole 10; the through hole 10 disposed on the upper cover 6 and corresponding to the process tube 9 of the upper head 3 is the first through hole 16, and is disposed on the lower cover 8 and corresponding to the process tube 9 of the lower head 5. The hole 10 is a second through hole 17, and the diameter of the first through hole 16 is larger than the outer diameter of the stainless steel flange 12 sleeved on the process pipe 9 of the upper head 3 by 2 mm to 4 mm, and the aperture ratio of the second through hole 17 is The outer diameter of the stainless steel flange 12 sleeved on the process pipe 9 of the lower head 5 is 2 mm to 4 mm. This design is designed to facilitate the setting of the stainless steel cylinder 15 on the outer side of the stainless steel flange 12 so that the stainless steel sheath 2 and the bismuth glass 1 The sealing between the two is improved, which is beneficial to improving the corrosion resistance of the glass reactor;

d. Fixing: stainless steel flange 12 is sleeved on the nozzle flange 11 of each process tube 9, and stainless steel flange 12 is sleeved on the nozzle flange 11 of each process tube 9, because stainless steel cannot Directly fixed to the glass, and the stainless steel flange 12 and the process tube 9 are fixed by sheathing the stainless steel flange 12 on the nozzle flange 11 and then welding the stainless steel sheath 2 to the stainless steel flange 12. The firmness of the connection between the stainless steel sheath 2 and the bismuth glass 1 is increased, and the stainless steel sheath 2 completely wraps the carbon steel portion of the bismuth glass 1 to improve the corrosion resistance of the bismuth glass reactor; At the same time, the stainless steel flange 12 is first cut in half, and then the cut stainless steel flange 12 is placed on the nozzle flange 11 of the corresponding process pipe 9, and the semi-cut stainless steel flange 12 is welded integrally. The design can better fix the stainless steel flange 12 and the process tube 9 without damaging the process tube 9, and then fix the stainless steel sheath 2 to the stainless steel flange 12, adding the stainless steel sheath 2 and the bismuth glass 1 The firmness of the connection, and the stainless steel sheath 2 completely wraps the carbon steel portion of the bismuth glass 1 to improve the corrosion resistance of the bismuth glass reactor.

In the manufacturing method of the present glass reactor, when the upper cover 6 and the upper head 3 are fixed, the upper cover 6 is covered from the upper cover 3 to the upper cover 3, and the process tube 9 is extended from the through hole 10 of the upper cover 6. Then, the upper foreskin 6 is welded to the stainless steel flange 12, the cylinder 4 is wrapped with the middle sheath 7, and the lower head 5 is covered from the bottom with the lower foreskin 8 so that the process tube 9 is passed from the through hole 10 of the lower foreskin 8 In the middle extension, the lower foreskin 8 is welded with the stainless steel flange 12, and finally the lower foreskin 8 and the middle foreskin 7 are sealed and welded; the stainless steel flange 12 is sleeved with a stainless steel

cylinder 15, the upper end of the stainless steel cylinder 15 and the stainless steel method Lan 12 seal welding, the lower end of the stainless steel cylinder 15 is located in the through hole 10 and is sealed and welded with the stainless steel sheath 2 around the through hole 10. The stainless steel cylinder 15 can connect the stainless steel flange 12 and the stainless steel sheath 2, simplifying the stainless steel The production process of the foreskin 2, and the structural strength after welding is higher, the sealing property between the stainless steel sheath 2 and the bismuth glass 1 is improved, and the corrosion resistance of the bismuth glass reactor is improved;

e, filler: filling the gap between the stainless steel sheath 2 and the bismuth glass 1 with the heat insulating material 13 and sealing welding; the upper head 3 covered with the upper foreskin 6 is inverted, between the upper foreskin 6 and the upper head 3 The gap is filled with the heat insulating material 13, so that the heat insulating material 13 fills the gap between the two, and then the lower end of the upper cover 6 is sealed and welded with the first stainless steel belt ring 18 disposed at the lower end of the upper head 3; The gap between the cylinder 4 and the gap between the lower sheath 8 and the lower head 5 is filled with the heat insulating material 13 so that the heat insulating material 13 fills the above-mentioned gap, and then the upper end of the middle sheath 7 is placed on the cylinder 4 The second stainless steel belt ring 19 at the upper end is hermetically welded, and finally, the welded upper head 3 is fixed to the cylinder 4. The filling of the heat insulating material 13 and sealing welding, the design can further isolate the carbon steel on the bismuth glass 1 from the outside, so that the corrosion resistance of the bismuth glass reactor is better;

f. Polishing: Polishing the outer surface of the stainless steel sheath 2; polishing the outer surface of the stainless steel sheath 2 to make the glass reactor look more beautiful;

g. Making and fixing of the cover 14: an annular cover 14 is made of stainless steel, and the cover 14 is sleeved on the upper cover 6, and the upper end of the cover 14 is sealed and welded to the upper cover 6. The inner diameter of the upper end of the cover 14 is smaller than the upper cover. The maximum diameter of 6, the inner diameter of the lower end of the cover 14 is larger than the maximum diameter of the upper cover 6 and smaller than the diameter of the middle cover 7, and the lower end of the cover 14 and the outer side of the middle cover 7 are fixed by glass glue. The inner diameter of the upper end of the cover 14 is smaller than the maximum diameter of the upper cover 6, so that the cover 14 can be better caught on the upper cover 6, and can be welded and fixed to the upper cover 6, and the middle cover 7 can be fixed by glass glue to fix the inside of the cover 14. Isolated from the outside world, foreign matter is prevented from entering the cylinder 4, which improves the safety of the glass reactor.

The specific embodiments described herein are merely illustrative of the spirit of the invention. A person skilled in the art can make various modifications or additions to the specific embodiments described or in a similar manner, without departing from the spirit of the invention or as defined by the appended claims. The scope.

CN105148816A Manufacturing method of glass-lined reaction kettle

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The wording below is an initial machine translation of the original publication. To generate a version using the latest translation technology, go to the original language text and use Patent Translate.

1.A method for manufacturing a glass-lined reactor, the glass-lined (1) reactor comprises an upper head (3), a cylinder (4) and a lower head (5), characterized in that the manufacturing method comprises the following steps :

a. Assembly: The glass is attached to carbon steel and then fired into a bismuth glass (1), which is made up of correspondingly shaped bismuth glass (1) (the upper head (3), the cylinder (4) and the lower head (5) 1) made, the lower end of the cylinder (4) and the upper end of the lower head (5) are welded and fixed, and the upper end of the cylinder (4) and the lower end of the upper head (3) are detachably fixed, and the stainless steel is fixed. The material ear seat (21) is fixed on the outer side of the cylinder body (4), and a stainless steel frame (20) is arranged at the upper end of the upper head (3);

b. Making a foreskin: pressing a stainless steel plate into a stainless steel sheath (2) with a tire, the stainless steel sheath (2) including an upper sheath (6) corresponding to the upper head (3), corresponding to the cylinder (4) a middle foreskin (7) and a lower foreskin (8) corresponding to the lower head (5);

c. Opening: a through hole (10) corresponding to the process pipe (9) disposed on the upper head (3) is provided on the upper cover (6), and is disposed on the lower cover (8) and disposed on the lower cover a through hole (10) corresponding to the process tube (9) on the head (5);

d. Fixed: a stainless steel flange (12) is placed on the nozzle flange (11) of each process tube (9), and the upper cover (6) is covered from the top to the upper head (3). The process tube (9) is extended from the through hole (10) of the upper foreskin (6), and then the upper sheath (6) is welded to the stainless steel flange (12), and the cylinder (4) is used for the medium (4) Wrap, cover the lower head (5) from bottom to top with the lower foreskin (8), and extend the process tube (9) from the through hole (10) of the lower foreskin (8) to place the lower foreskin (8) with Stainless steel flange (12) is welded, and finally the lower foreskin (8) is sealed with the middle sheath (7);

e, filler: fill the gap between the stainless steel sheath (2) and the bismuth glass (1) with insulation material (13), and seal welding;

f, polishing: polishing the outer surface of the stainless steel sheath (2);

g. Manufacture and fixing of the cover: a ring-shaped cover (14) is made of stainless steel, and the cover (14) is sleeved on the upper cover (6), and the upper end of the cover (14) is sealed with the upper cover (6) welding.

2.The method for manufacturing a glass-lined reactor according to claim 1, wherein the stainless steel flange (12) of the step d is sleeved with a stainless steel cylinder (15), and the stainless steel cylinder (The upper end of 15) is hermetically welded to the stainless steel flange (12), and the lower end of the stainless steel cylinder (15) is located in the through hole (10) and is hermetically welded to the stainless steel sheath (2) around the through hole (10).

3.The method for fabricating a glass-lined reactor according to claim 1, wherein in the step c, the upper sheath (6) is disposed on the upper cover (3) and corresponds to the process tube (9) of the upper head (3). The through hole (10) is a first through hole (16), and the through hole (10) disposed on the lower cover (8) corresponding to the process tube (9) of the lower head (5) is a second through hole (17) The aperture of the first through hole (16) is larger than the outer diameter of the stainless steel flange (12) sleeved on the process tube (9) of the upper head (3) by 2 mm - 4 mm, the second pass The aperture (17) has an aperture that is 2 mm to 4 mm larger than the outer diameter of the stainless steel flange (12) that is placed over the process tube (9) of the lower head (5).

4.The method for fabricating a glass-lined reactor according to claim 1, wherein the upper head (3) of the step (e) covered in the step e is inverted, and the upper foreskin (6) Filling the gap between the upper head (3) with the insulating material (13), so that the insulating material (13) fills the gap between the two, and then placing

the lower end of the upper sheath (6) and the upper head (3) The first stainless steel belt loop (18) at the lower end is hermetically welded; the gap between the middle sheath (7) and the cylinder (4) and the gap between the lower sheath (8) and the lower head (5) Filling the heat insulating material (13), so that the heat insulating material (13) fills the above gap, and then sealing the upper end of the middle sheath (7) with the second stainless steel belt ring (19) disposed at the upper end of the cylinder (4), and finally , the welded upper head (3) is fixedly connected to the barrel (4).

5.The method for fabricating a glass-lined reactor according to claim 1 or 2 or 3 or 4, wherein in the step d, the stainless steel flange (12) is first cut in half, and then the cut The stainless steel flange (12) is placed over the nozzle flange (11) of the corresponding process tube (9), and the semi-cut stainless steel flange (12) is welded integrally.

6.The method for manufacturing a glass-lined reactor according to claim 1 or 2 or 3 or 4, wherein the frame (20) is provided with a reducer (24) and a motor (23), Both the reducer (24) and the motor (23) are covered with a stainless steel cover (22).

7.The method for fabricating a glass-lined reactor according to claim 1 or 2 or 3 or 4, wherein in the step g, the inner diameter of the upper end of the cover (14) is smaller than the maximum diameter of the upper foreskin (6), the cover The inner diameter of the lower end of (14) is larger than the maximum diameter of the upper sheath (6) and smaller than the diameter of the middle sheath (7), and the lower end of the cover (14) and the outer side of the middle cover (7) are fixed by glass glue.

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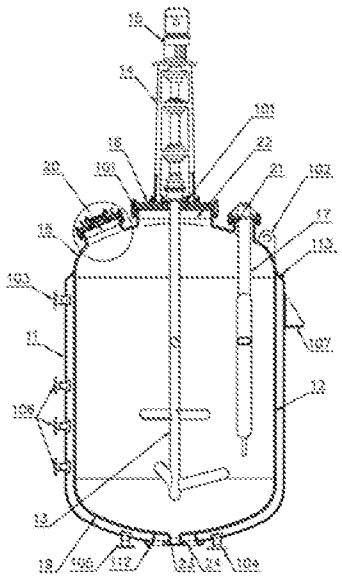
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(54)实用新型名称

一种高强度耐腐蚀搪玻璃反应釜

(57)摘要

本实用新型公开了一种高强度耐腐蚀搪玻璃反应釜，包括夹套、罐体、搅拌器、上封头、下封头、视镜人孔、温度计管口、进料口、放料口，所述罐体外侧设有夹套，所述夹套外部等角度设有若干安装定位装置，所述夹套上部设有蒸汽进口，蒸汽进口一侧设有放气口，夹套底部设有凝水出口；所述进料口上焊接有高颈法兰；所述视镜人孔上焊接有高颈法兰，温度计管口上焊接有高颈法兰，放料口上焊接有高颈法兰；所述罐体内侧壁上设有底釉层，底釉层上设有面釉层，高颈法兰内侧壁、端面、凹槽上设有底釉层，底釉层上设有面釉层。本实用新型开孔处焊接增加了高颈法兰，减少了冲压成型开孔R部位的数量，直接增加了本设备的有效厚度，受压强度相对提高，确保设备的安全运行。



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1.一种高强度耐腐蚀搪玻璃反应釜，包括夹套、罐体、搅拌器、上封头、下封头、视镜人孔、温度计管口、进料口、放料口，其特征在于所述罐体外侧设有夹套，所述夹套上部设有蒸汽进口，蒸汽进口一侧设有放气口，夹套底部设有凝水出口；

所述罐体上部设有进料口，所述进料口上焊接有高颈法兰，所述高颈法兰端面设有凹槽，凹槽上设有倒角，凹槽一侧设有螺栓过孔，高颈法兰凹槽处设有垫片，垫片上部设有顶盖，顶盖上设有螺栓过孔，顶盖与高颈法兰通过螺栓固定连接；顶盖上部设有固定架，固定架上设有电机，电机轴端设有搅拌器；

所述进料口一侧设有视镜人孔，所述视镜人孔上焊接有高颈法兰，所述高颈法兰端面设有凹槽，凹槽上设有倒角，凹槽一侧设有螺栓过孔，高颈法兰凹槽处设有垫片，垫片上部设有压板，压板与高颈法兰通过螺栓紧固连接，压板上设有透视孔，压板上表面设有垫片，垫片上设有视镜玻璃，视镜玻璃上设有垫片，垫片上部设有盖板，盖板通过螺钉固定连接在压板上；

所述进料口另一侧设有温度计管口，温度计管口上焊接有高颈法兰，所述高颈法兰端面设有凹槽，凹槽上设有倒角，凹槽一侧设有螺栓过孔，高颈法兰凹槽处设有垫片，垫片上部设有压板，压板与高颈法兰通过螺栓紧固连接，压板上设有温度计套，温度计套内设有温度计；

所述罐体底部设有放料口，放料口上焊接有高颈法兰，所述高颈法兰端面设有凹槽，凹槽上设有倒角，凹槽一侧设有螺栓过孔，高颈法兰凹槽处设有垫片，垫片上部设有底盖，底盖与高颈法兰通过螺栓紧固连接；

所述罐体内侧壁上设有底釉层，底釉层上设有面釉层，高颈法兰内侧壁、端面、凹槽上设有底釉层，底釉层上设有面釉层。

2.根据权利要求1所述的一种高强度耐腐蚀搪玻璃反应釜，其特征在于所述夹套底部设有液体进口，液体进口一侧设有放净口，所述夹套侧壁设有若干液体出口。

3.根据权利要求1所述的一种高强度耐腐蚀搪玻璃反应釜，其特征在于所述罐体上部设有若干备用口，备用口上上焊接有高颈法兰，所述高颈法兰端面设有凹槽，凹槽上设有倒角，凹槽一侧设有螺栓过孔，高颈法兰凹槽处设有垫片，垫片上部设有盖板，盖板与高颈法兰通过螺栓紧固连接。

4.根据权利要求3所述的一种高强度耐腐蚀搪玻璃反应釜，其特征在于所述视镜人孔、温度计管口、进料口、放料口、备用口采用无缝焊接高颈法兰，高颈法兰端面上等角度布置螺栓过孔，焊缝处设有底釉层，底釉层上设有面釉层。

一种高强度耐腐蚀搪玻璃反应釜

技术领域

[0001] 本实用新型属于搪玻璃化工设备技术领域，特别涉及一种高强度耐腐蚀搪玻璃反应釜。

背景技术

[0002] 对于搪玻璃反应釜来说，搪玻璃反应釜是将含高二氧化硅的玻璃，衬在钢制容器的内表面，经高温灼烧而牢固地密着于金属表面上成为复合材料制品，它具有玻璃的稳定性和金属强度的双重优点，是一种优良的耐腐蚀设备。

[0003] 现有的搪玻璃反应釜中上封头、下封头、视镜人孔、温度计管口、进料口、放料口等所有开孔部位处设有高颈法兰，获得高颈法兰的方式有以下两种：1)采用冲压方式得到高颈法兰口，冲压成型的开孔R部位，厚度由于延伸变薄，使得设备的受压力减弱；2)根据HG/T 2049-2012搪玻璃设备高颈法兰的标准，开孔部位焊接高颈法兰，但是HG/T 2049-2012公开的高颈法兰端口平面方向向外延伸构成外沿(说明书附图6-20所示)，通过卡箍(说明书附图7-301)固定连接，圆形的端口面通过若干卡箍固定连接，一般为目测等角度布置卡箍，这样的连接会使得垫片在各个点的受力不均匀，且釜体顶部的开孔部位的局部耐受压力大于釜体本身，使得设备整体在开孔部位的最高耐压值减小，使得釜体内部的耐压能力受到限制。

[0004] 搪玻璃设备鱼鳞爆瓷是搪玻璃行业的一个顽疾。鱼鳞爆瓷也叫氢爆，表现为瓷层以大小不等的鳞片状脱离的爆裂的现象。目前，本领域普遍认可的形成鱼鳞爆瓷的机理是：制造搪玻璃设备的钢板中有以各种形式存在的氢，在搪玻璃制品烧成过程中，氢在钢中达到了饱和，而当制品在空气中冷却到瓷釉凝固时，氢来不及逸出，以致在钢中达到了过饱和状态。过饱和氢产生一定的压力，不能及时在钢中逸出。当这个压力增大到会使瓷层脱落时，便形成了鱼鳞爆瓷。由于这种鱼鳞爆瓷具有一定的潜伏性，会在特定条件下出现，因而在搪玻璃设备制造完成后、安装中、使用时都会发生，严重困扰着搪玻璃设备在化学工业的应用。鱼鳞爆瓷严重影响着搪玻璃设备的使用的性能以及生产安全性。

[0005] 研究表明，搪玻璃釉的配方直接影响搪玻璃的性质。常规搪玻璃釉配方和熔制工艺与玻璃接近，搪玻璃釉有搪玻璃面釉和搪玻璃底釉。而底釉按密着剂种类可分为镍底釉、钴底釉、钴镍底釉、锑钼底釉和混合底釉等。搪玻璃底釉的功能：一是用作过渡层或者说密着层，使得面釉和钢连接起来牢固地粘在一起，二是起到储藏气体的作用，因为搪烧过程中会产生CO，CO_2，H_2等气体，这些气体有部分必须留在底釉的微气孔中才稳定，同时由于炉窑结构、产品结构、受热条件等因素，使得底釉要经受50-80℃的温差。一般要求约在840-860℃能玻璃化，到890-910℃还处于玻璃化而不过烧。本单位研究小组多年来致力于提高搪玻璃以及搪玻璃设备的性能，目前研发的产品已在抗鱼鳞爆瓷以及耐腐蚀性等方面有了较大的提高，例如CN103864303B、CN103863724B等实用新型提供的底釉，可以与胚体牢固结合，减少了鱼鳞爆瓷，表面美观洁净，具有较好的耐酸腐蚀性，但是耐碱腐蚀性能一般，限制了其应用范围。随着工业发展对于设备的要求越来越高，对于搪玻璃的性能要求也有了更高

的要求。

发明内容

[0006] 本实用新型的目的是克服现有技术中不足，提供一种高强度耐腐蚀搪玻璃反应釜，耐腐蚀，通过螺栓紧固连接，操作方便且密封性好，开孔处焊接增加了高颈法兰，减少了冲压成型开孔R部位的数量，直接增加了本设备有效厚度，受压强度相对提高，确保设备的安全运行。

[0007] 为了实现上述目的，本实用新型采用的技术方案是：

[0008] 一种高强度耐腐蚀搪玻璃反应釜，包括夹套、罐体、搅拌器、上封头、下封头、视镜人孔、温度计管口、进料口、放料口，所述罐体外侧设有夹套，所述夹套外部等角度设有若干安装定位装置，所述夹套上部设有蒸汽进口，蒸汽进口一侧设有放气口，夹套底部设有凝水出口，高温蒸汽通过蒸汽进口进入对夹层内进行换热，高温蒸汽变为低温蒸汽，低温蒸汽从放气口排出，蒸汽遇冷产生的冷凝水从凝水出口排出；

[0009] 所述夹套底部设有液体进口，液体进口一侧设有放净口，所述夹套侧壁设有若干液体出口，通过高温液体对罐体进行换热，高温液体通过液体进口进入夹层内进行换热，高温液体变为低温液体，低温液体通过液体出口排出，通过放净口对液体进行放空，避免液体残留在夹套内部；

[0010] 所述罐体上部设有进料口，所述进料口上焊接有高颈法兰，所述高颈法兰端面设有凹槽，凹槽上设有倒角，凹槽一侧设有螺栓过孔，高颈法兰凹槽处设有垫片，垫片上部设有顶盖，顶盖上设有螺栓过孔，顶盖与高颈法兰通过螺栓固定连接，顶盖通过垫片对进料口进行密封；顶盖上部设有固定架，固定架上设有电机，电机轴端设有搅拌器，搅拌器伸入罐体内，对罐体内物料或液体进行充分搅拌，混合均匀；

[0011] 所述进料口一侧设有视镜人孔，所述视镜人孔上焊接有高颈法兰，所述高颈法兰端面设有凹槽，凹槽上设有倒角，凹槽一侧设有螺栓过孔，高颈法兰凹槽处设有垫片，垫片上部设有压板，压板与高颈法兰通过螺栓紧固连接，压板上设有透视孔，压板上表面设有垫片，垫片上设有视镜玻璃，视镜玻璃上设有垫片，垫片上部设有盖板，盖板通过螺钉固定连接在压板上；

[0012] 所述进料口另一侧设有温度计管口，温度计管口上焊接有高颈法兰，所述高颈法兰端面设有凹槽，凹槽上设有倒角，凹槽一侧设有螺栓过孔，高颈法兰凹槽处设有垫片，垫片上部设有压板，压板与高颈法兰通过螺栓紧固连接，压板上设有温度计套，温度计套内设有温度计，温度计伸入罐体内进行温度检测；

[0013] 所述罐体底部设有放料口，放料口上焊接有高颈法兰，所述高颈法兰端面设有凹槽，凹槽上设有倒角，凹槽一侧设有螺栓过孔，高颈法兰凹槽处设有垫片，垫片上部设有底盖，底盖与高颈法兰通过螺栓紧固连接；

[0014] 所述罐体内侧壁上设有底釉层，底釉层上设有面釉层，高颈法兰内侧壁、端面、凹槽上设有底釉层，底釉层上设有面釉层，防止液体腐蚀罐体。

[0015] 优选的，所述罐体上部设有若干备用口，备用口上上焊接有高颈法兰，所述高颈法兰端面设有凹槽，凹槽上设有倒角，凹槽一侧设有螺栓过孔，高颈法兰凹槽处设有垫片，垫片上部设有盖板，盖板与高颈法兰通过螺栓紧固连接。

[0016] 优选的，所述视镜人孔、温度计管口、进料口、放料口、备用口采用无缝焊接高颈法兰，高颈法兰端面上等角度布置螺栓过孔，焊缝处设有底釉层，底釉层上设有面釉层，相比同等罐体壁厚的现有的搪玻璃压力容器中，减少了冲压成型开孔R部位的数量，通过等角度布置的螺栓固定连接，增大了开孔部位最高耐压值，直接增加了本设备有效厚度，受压强度相对提高，确保设备的安全运行。

[0017] 本实用新型还提供一种上述底釉层的新配方，该底釉层由底釉熔块引入混合密着剂研磨而成，按重量百分比计，底釉熔块的化学组成成分包括：二氧化硅40-45％，三氧化二铝3-7％，氧化钙5-10％，碳酸钾20-26％，氧化硼12-15％，氟硅酸钠3-15％，掺稀氧化铟2-4％；优选地，按重量百分比计，底釉熔块的化学组成成分包括：二氧化硅42-45％，三氧化二铝5-6％，氧化钙5-8％，碳酸钾25-26％，氧化硼13-15％，氟硅酸钠4-10％，掺稀氧化铟2-3％。目前，底釉熔块组成对于整个底釉的至关重要，现有技术中底釉、底釉熔块的组成五花八门，通过常规组合的底釉并不能满足更高要求的搪瓷产品，在本实用新型中，实用新型人对常规配方进行了优化，加入掺稀氧化铟后，意外发现，底釉更易熔，各种成分分散性和融合更好，该底釉烧成后，表面光亮、平整，形成的底釉无论抗鱼鳞爆瓷还是耐腐蚀性能都进一步的提高。

[0018] 金属和搪瓷是两种不同组成、结构各异的材料。它们的牢固结合是一个极其复杂的物理化学过程。因此混合密着剂关系着瓷釉和金属之间的扩散、融合等复杂过程，直接影响密着效果及容器的耐酸性、抗暴瓷效果。在本实用新型中，所述混合密着剂的组成成分为氟碳铈矿石粉45-55％，氧化锰10-15％，氧化铜2-6％，氧化钼25-35％，二氧化钛1-10％。进一步优选地，所述混合密着剂的组成成分为氟碳铈矿石粉45-48％，氧化锰10-14％，氧化铜4-8％，氧化钼28-30％，二氧化钛3-7％。进一步优选地，按重量百分比计，所述混合密着剂为底釉的1-2％，该范围下，密着效果最佳，减少用量则结合强度不够，容易出现爆瓷、开裂，而提高用量效果提升不明显并且提高成本。本实用新型的混合密着剂，在搪玻璃底釉的烧成过程中与底釉的其他组分发生复杂的物理化学反应，能够产生了更多的微气泡和化学键，这些微气泡在底釉层中形成了容纳和逸出钢板中饱和过饱和氢的巢穴和孔隙，降低了过饱和氢对瓷层的压力，本实用新型的混合密着剂的成分优化调整特别是二氧化钛的加入，无需特别高温即可使得底釉各成分以及釉与金属间形成强力密着，同时能够降低底釉的膨胀系数，减少釉在温变以及高压下开裂倾向。

[0019] 在本实用新型中，所述混合密着剂的制备方法可以根据本领域常规的方法进行，例如包括称料、粉料、混合、熔制、冷却、研磨、过筛过程，优选情况下，所述熔制过程包括：

[0020] 1)混合物料加热至130-160℃，搅拌，保温反应20min-40min；

[0021] 2)升温至500-600℃，搅拌保温30min-1h；

[0022] 3)升温至900-1000℃，搅拌、充入空气、烧结。

[0023] 在本实用新型中，所述的研磨可以通过常规的研磨机进行，例如采用球磨机进行。所述的过筛过程是指过80目的筛子。

[0024] 本领域的技术人员可以按照本领域常规密着剂的使用方法使用该混合密着剂，无需特殊操作步骤。同时，本实用新型所提供的底釉可以与市售的或已知配方的面釉一起使用。

[0025] 使用本实用新型的底釉，经过高温烧制，胚体上底釉全部熔融至胚胎凹凸不平的

表面上，搪瓷底釉与胚体的密着性好。在涂搪有底釉的胚体上涂搪上述面釉，烧制出的瓷层表面出现较强的光泽、平整，不会产生焦瓣、麻泡、爆瓷等缺陷。

[0026] 本实用新型与现有技术相比较有益效果表现在：

[0027] 1)罐体中进料口、视镜人孔、温度计管口、放料口分别焊接有高颈法兰，高颈法兰端面设有凹槽，高颈法兰凹槽处设有垫片，通过螺栓紧固连接，操作方便且密封性好，从而改变了原有设计的冲压形成的高颈法兰，增加了高颈法兰的壁厚，耐压强度相对增加；

[0028] 2)换热方式为两种方式，一是高温蒸汽通过蒸汽进口进入对夹层内进行换热，高温蒸汽变为低温蒸汽，二是通过高温液体对夹层内进行换热，高温液体变为低温液体；两种换热方式为可循环利用方式，降低生产成本；

[0029] 3)本设备所有开孔处焊接增加了高颈法兰，高颈法兰端面上等角度布置螺栓过孔，减少了冲压成型开孔R部位的数量，通过等角度布置的螺栓固定连接，并结合高颈法兰端面凹槽处的垫片，相比同等罐体壁厚的现有的搪玻璃压力容器，增大了开孔部位最高耐压值，直接增加了本设备的受压强度，确保受压安全，采用螺栓紧固连接，操作方便且密封性好；

[0030] 4)本实用新型提供的底釉，可以与胚体牢固结合，耐热冲击以及耐温差急变性能更好，减少了鱼鳞爆瓷；

[0031] 5)本实用新型提供的底釉，具有更优异的耐腐蚀性能，不但具有优异的耐酸腐蚀性而且具有优异的耐碱腐蚀性，扩展了搪玻璃设备的应用范围。

附图说明

[0032] 附图1是本实用新型一种高强度耐腐蚀搪玻璃反应釜剖切结构示意图；

[0033] 附图2是本实用新型一种高强度耐腐蚀搪玻璃反应釜俯视图；

[0034] 附图3是本实用新型一种高强度耐腐蚀搪玻璃反应釜中视镜人孔放大结构示意图；

[0035] 附图4是本实用新型一种高强度耐腐蚀搪玻璃反应釜中高颈法兰结构示意图；

[0036] 附图5是本实用新型一种高强度耐腐蚀搪玻璃反应釜中高颈法兰放大结构示意图；

[0037] 附图6是现有设计搪玻璃反应釜结构示意图；

[0038] 附图7是现有设计搪玻璃反应釜中高颈法兰放大结构示意图；

[0039] 图中：11-夹套，12-罐体，13-搅拌器，14-固定架，15-电机，16-顶盖，17-温度计，18-上封头，19-下封头，20-视镜人孔，21-温度计管口，22-进料口，23-放料口，24-高颈法兰，241-凹槽，242-螺栓过孔，243-倒角，25-备用口，101-垫片，102-吊耳，103-蒸汽进口，104-凝水出口，105-液体进口，106-液体出口，107-安装定位装置，108-螺栓，109-压板，110-视镜玻璃，111-盖板，112-放净口，113-放气口，301-卡箍。

具体实施方式

[0040] 为方便本技术领域人员的理解，下面结合附图1-7，对本实用新型的技术方案进一步具体说明。

[0041] 一种高强度耐腐蚀搪玻璃反应釜，包括夹套11、罐体12、搅拌器13、上封头18、下封

头19、视镜人孔20、温度计管口21、进料口22、放料口23，所述罐体12外侧设有夹套11，所述夹套11外部等角度设有若干安装定位装置107，所述夹套11上部设有蒸汽进口103，蒸汽进口103一侧设有放气口113，夹套11底部设有凝水出口104，高温蒸汽通过蒸汽进口103进入对夹层内进行换热，高温蒸汽变为低温蒸汽，低温蒸汽从放气口113排出，蒸汽遇冷产生的冷凝水从凝水出口104排出。

[0042] 所述夹套11底部设有液体进口105，液体进口105一侧设有放净口112，所述夹套11侧壁设有若干液体出口106，通过高温液体对罐体12进行换热，高温液体通过液体进口进入夹层内进行换热，高温液体变为低温液体，低温液体通过液体出口106排出，通过放净口112对液体进行放空，避免液体残留在夹套内部。

[0043] 所述罐体12上部设有进料口22，所述进料口22上焊接有高颈法兰24，所述高颈法兰24端面设有凹槽241，凹槽241上设有倒角243，凹槽241一侧设有螺栓过孔242，高颈法兰24凹槽241处设有垫片101，垫片101上部设有顶盖16，顶盖16上设有螺栓过孔242，顶盖16与高颈法兰24通过螺栓固定连接，顶盖16通过垫片101对进料口22进行密封；顶盖16上部设有固定架14，固定架14上设有电机15，电机15轴端设有搅拌器13，搅拌器13伸入罐体12内，对罐体12内物料或液体进行充分搅拌，混合均匀。

[0044] 所述进料口22一侧设有视镜人孔20，所述视镜人孔20上焊接有高颈法兰24，所述高颈法兰24端面设有凹槽241，凹槽241上设有倒角243，凹槽241一侧设有螺栓过孔242，高颈法兰24凹槽241处设有垫片101，垫片101上部设有压板109，压板109与高颈法兰24通过螺栓紧固连接，压板109上设有透视孔，压板109上表面设有垫片101，垫片101上设有视镜玻璃110，视镜玻璃110上设有垫片101，垫片101上部设有盖板111，盖板111通过螺钉固定连接在压板109上。

[0045] 所述进料口22另一侧设有温度计管口21，温度计管口21上焊接有高颈法兰24，所述高颈法兰24端面设有凹槽241，凹槽241上设有倒角243，凹槽241一侧设有螺栓过孔242，高颈法兰24凹槽241处设有垫片101，垫片101上部设有压板109，压板109与高颈法兰24通过螺栓紧固连接，压板109上设有温度计套，温度计套内设有温度计17，温度计17伸入罐体12内进行温度检测。

[0046] 所述罐体12底部设有放料口23，放料口23上焊接有高颈法兰24，所述高颈法兰24端面设有凹槽241，凹槽241上设有倒角243，凹槽241一侧设有螺栓过孔242，高颈法兰24凹槽241处设有垫片101，垫片101上部设有底盖，底盖与高颈法兰24通过螺栓紧固连接。

[0047] 所述罐体12内侧壁上设有底釉层，底釉层上设有面釉层，高颈法兰24内侧壁、端面、凹槽上设有底釉层，底釉层上设有面釉层，防止液体腐蚀罐体。

[0048] 所述罐体12上部设有若干备用口25，备用口25上上焊接有高颈法兰24，所述高颈法兰24端面设有凹槽241，凹槽241上设有倒角243，凹槽241一侧设有螺栓过孔242，高颈法兰24凹槽241处设有垫片101，垫片101上部设有盖板111，盖板111与高颈法兰24通过螺栓紧固连接。

[0049] 所述视镜人孔20、温度计管口21、进料口22、放料口23、备用口25采用无缝焊接高颈法兰24，焊缝处设有底釉层，底釉层上设有面釉层，相比同等罐体壁厚现有的搪玻璃压力容器中，减少了冲压成型开孔R部位的数量，直接增加了本设备的受压强度，确保受压安全。

[0050] 以下通过具体实施例和测试例对本实用新型的技术方案进一步具体说明。

[0051] 1、混合密着剂的制备

[0052] 制备例1

[0053] 1)称取氟碳铈矿石粉4.8Kg，氧化锰1.2Kg，氧化铜0.6Kg，氧化钼3Kg和二氧化钛0.4Kg并粉碎，混合物料加热至140℃，搅拌，保温反应40min；

[0054] 2)升温至550℃，搅拌保温1h；

[0055] 3)升温至900℃，搅拌、充入空气、烧结，冷却，球磨，过80目筛，得混合密着剂。

[0056] 制备例2

[0057] 1)称取氟碳铈矿石粉4.5Kg，氧化锰1.4Kg，氧化铜0.7Kg，氧化钼2.8Kg和二氧化钛0.6Kg并粉碎，混合物料加热至160℃，搅拌，保温反应40min；

[0058] 2)升温至600℃，搅拌保温1h；

[0059] 3)升温至950℃，搅拌、充入空气、烧结，冷却，球磨，过80目筛，得混合密着剂。

[0060] 制备例3

[0061] 1)称取氟碳铈矿石粉4.6Kg，氧化锰1.3Kg，氧化铜0.4Kg，氧化钼3Kg和二氧化钛0.7Kg并粉碎，混合物料加热至130℃，搅拌，保温反应30min；

[0062] 2)升温至500℃，搅拌保温1h；

[0063] 3)升温至1000℃，搅拌、充入空气、烧结，冷却，球磨，过80目筛，得混合密着剂。

[0064] 制备例4

[0065] 1)称取氟碳铈矿石粉5Kg，氧化锰1.5Kg，氧化铜0.2Kg，氧化钼3.2Kg和二氧化钛0.1Kg。并粉碎，混合物料加热至150℃，搅拌，保温反应40min；

[0066] 2)升温至600℃，搅拌保温1h；

[0067] 3)升温至900℃，搅拌、充入空气、烧结，冷却，球磨，过80目筛，得混合密着剂。

[0068] 制备例5

[0069] 1)称取氟碳铈矿石粉5.5Kg，氧化锰1.3Kg，氧化铜0.3Kg，氧化钼2.5Kg和二氧化钛0.3Kg并粉碎，混合物料加热至140℃，搅拌，保温反应30min；

[0070] 2)升温至500℃，搅拌保温1h；

[0071] 3)升温至1000℃，搅拌、充入空气、烧结，冷却，球磨，过80目筛，得混合密着剂。

[0072] 制备例6

[0073] 1)称取氟碳铈矿石粉5.5Kg，氧化锰1.3Kg，氧化铜0.3Kg和氧化钼2.5Kg并粉碎，混合物料加热至140℃，搅拌，保温反应30min；

[0074] 2)升温至500℃，搅拌保温1h；

[0075] 3)升温至1000℃，搅拌、充入空气、烧结，冷却，球磨，过80目筛，得混合密着剂。

[0076] 2、搪玻璃底釉得制备

[0077] 制备例7

[0078] 二氧化硅0.44Kg，三氧化二铝0.05Kg，氧化钙0.06Kg，碳酸钾0.25Kg，氧化硼0.13Kg，氟硅酸钠0.04Kg，掺稀氧化铟0.03Kg和制备例1中的混合密着剂0.01Kg，混合、研磨制成搪玻璃底釉。

[0079] 制备例8

[0080] 二氧化硅0.42Kg，三氧化二铝0.05Kg，氧化钙0.08Kg，碳酸钾0.25Kg，氧化硼0.13Kg，氟硅酸钠0.05Kg，掺稀氧化铟0.02Kg和制备例1中的混合密着剂0.01Kg，混合、研磨

制成搪玻璃底釉。

[0081] 制备例9

[0082] 二氧化硅0.45Kg，三氧化二铝0.05Kg，氧化钙0.05Kg，碳酸钾0.25Kg，氧化硼0.14Kg，氟硅酸钠0.04Kg和掺稀氧化铟0.02Kg和制备例1中的混合密着剂0.01Kg，混合、研磨制成搪玻璃底釉。

[0083] 制备例10

[0084] 二氧化硅0.44Kg，三氧化二铝0.05Kg，氧化钙0.06Kg，碳酸钾0.25Kg，氧化硼0.13Kg，氟硅酸钠0.04Kg，掺稀氧化铟0.03Kg和制备例1中的混合密着剂0.02Kg，混合、研磨制成搪玻璃底釉。

[0085] 制备例11

[0086] 二氧化硅0.44Kg，三氧化二铝0.05Kg，氧化钙0.06Kg，碳酸钾0.25Kg，氧化硼0.13Kg，氟硅酸钠0.04Kg，掺稀氧化铟0.03Kg和制备例2中的混合密着剂0.01Kg，混合、研磨制成搪玻璃底釉。

[0087] 制备例12

[0088] 二氧化硅0.44Kg，三氧化二铝0.05Kg，氧化钙0.06Kg，碳酸钾0.25Kg，氧化硼0.13Kg，氟硅酸钠0.04Kg，掺稀氧化铟0.03Kg和制备例3中的混合密着剂0.01Kg，混合、研磨制成搪玻璃底釉。

[0089] 制备例13

[0090] 二氧化硅0.44Kg，三氧化二铝0.05Kg，氧化钙0.06Kg，碳酸钾0.25Kg，氧化硼0.13Kg，氟硅酸钠0.04Kg，掺稀氧化铟0.03Kg和制备例4中的混合密着剂0.01Kg，混合、研磨制成搪玻璃底釉。

[0091] 制备例14

[0092] 二氧化硅0.44Kg，三氧化二铝0.05Kg，氧化钙0.06Kg，碳酸钾0.25Kg，氧化硼0.13Kg，氟硅酸钠0.04Kg，掺稀氧化铟0.03Kg和制备例5中的混合密着剂0.01Kg，混合、研磨制成搪玻璃底釉。

[0093] 制备例15

[0094] 二氧化硅0.44Kg，三氧化二铝0.05Kg，氧化钙0.06Kg，碳酸钾0.25Kg，氧化硼0.13Kg，氟硅酸钠0.04Kg，掺稀氧化铟0.03Kg和制备例6中的混合密着剂0.01Kg，混合、研磨制成搪玻璃底釉。

[0095] 制备例16

[0096] 二氧化硅0.44Kg，三氧化二铝0.05Kg，氧化钙0.06Kg，碳酸钾0.25Kg，氧化硼0.13Kg，氟硅酸钠0.04Kg和制备例1中的混合密着剂0.01Kg，混合、研磨制成搪玻璃底釉。

[0097] 制备例17

[0098] 二氧化硅0.44Kg，三氧化二铝0.05Kg，氧化钙0.06Kg，碳酸钾0.25Kg，氧化硼0.13Kg，氟硅酸钠0.04Kg和制备例6中的混合密着剂0.01Kg，混合、研磨制成搪玻璃底釉。

[0099] 制备例18

[0100] 二氧化硅0.44Kg，三氧化二铝0.05Kg，氧化钙0.06Kg，碳酸钾0.25Kg，氧化硼0.13Kg，氟硅酸钠0.04Kg，掺稀氧化铟0.03Kg和氧化镍密着剂0.01Kg，混合、研磨制成搪玻璃底釉。

[0101] 制备例19

[0102] 按照CN103863724B中实施例2-1-6制备搪玻璃底釉。

[0103] 测试例

[0104] 在测试例中使用的搪瓷面釉配方为:二氧化硅40份,三氧化二铝5.5份,氧化钠18份,氧化硼20.5份,氟化钙3份,氟硅酸钠7份,氧化钴0.01份,氧化锌1.09份,五氧化二磷4.9份。所用钢材试块为Q 235型,规格为80mm*80mm*5.5mm,其中C、Si、Mn、S、P的含量分别为0.16%、0.023%、0.41%、0.019%、0.017%。

[0105] 将上述制备例7-19中制得的搪玻璃底釉分别涂搪于上述钢材试块上,高温烧制。在涂搪有底釉的胚体上涂搪上述面釉,将已施釉且粉层干燥的胚胎置于高温环境下烧成,烧至瓷层全部熔融至表面出现较强的光泽,搪玻璃层厚度约为1.2mm。

[0106] 测试方法

[0107] 1、冲击试验:根据国家标准GB/T 7990-2013进行,以涂层被破坏所冲击的次数来评定涂层的密着性能,具体结果如表1所示。

[0108] 2、热震实验:根据国家标准GB/T 4547-2007进行,热疲劳试验的加热温度为400℃。耐温差急变性试验是将试块加热到200℃后放人室温下流动的水中急冷,若瓷层不开裂,递增20℃再加热、急冷,若不开裂继续递增温度加热、急冷,如此循环,直到开裂,具体结果如表1所示。

[0109] 3、耐腐蚀性能试验:根据国家标准GB/T7989-2013进行耐沸腾盐酸蒸气腐蚀性能的测定,连续测试168个小时,以质量损失率来评价耐酸腐蚀性,单位:g/(m^2.d);2)根据国家标准GB/T7988-2013进行耐碱腐蚀性能的测定,在0.1mol/LNaOH中80℃测试24个小时,以质量损失率来评价耐碱腐蚀性,单位:g/(m^2.d)。具体结果如表1所示。

[0110] 4、高压试验:根据国家标准GB/T 7991.6-2014进行,其中,试验电压为2.5kV,试验温度为20℃。结果显示,使用制备例7-15、制备例18和制备例19制备的搪玻璃层均没有观察到火花放电现象,搪玻璃层均没有缺陷或者薄弱点,而制备例16和制备例17各发现有一处火花放电。

[0111] 表1

[0112]

	冲击次数	热疲劳次数	耐急变温度差/℃	耐腐蚀性g/(m^2.d)	
				HCl (20%)	NaOH (0.1mol/L)
制备例7	59	22	480	0.23	2.21
制备例8	60	21	500	0.22	2.47
制备例9	59	24	460	0.21	2.36
制备例10	60	24	480	0.20	2.09
制备例11	56	22	440	0.33	2.46
制备例12	55	19	460	0.25	2.70
制备例13	47	16	380	0.27	3.15
制备例14	50	18	400	0.30	2.92

[0113]

制备例15	45	14	300	0.52	3.86
制备例16	36	12	280	1.15	4.71
制备例17	32	8	260	1.32	3.84
制备例18	25	11	360	1.07	3.42
制备例19	57	12	480	0.25	3.75

[0114] 通过表1可以看出，使用本实用新型提供的底釉能够与金属基体牢固结合，有效提高了试块的抗冲击能力和抗热震性能，实验过程未观察到鱼鳞爆瓷现象。并且，使用本实用新型的底釉结合常规面釉烧制后，不仅具有强的耐酸腐蚀性能同时也具有非常好的耐碱腐蚀性能。

[0115] 5、耐压力检测

[0116] 取制备例1-19中任意3种搪玻璃反应釜(命名为A1、A2、A3)与现有结构设计中同等罐体壁厚D1(25mm)、釜体公称直径DN(2200mm)、全容积V(10000L)、搪玻璃层厚度D2(2mm)的搪玻璃反应釜(命名为B1)进行单独对比耐压力测试(水压试验测试)，其中A1与B1采用的底釉、面釉一致，A2与B1采用的底釉、面釉一致，A3与B1采用的底釉、面釉一致，水压试验测试压力中罐体测试压力2.5Mpa，夹套测试压力0.8Mpa，记录实验数据如下；具体地：试验时应先缓慢升压至规定测试压力的10％，保压5min.并且对所有焊接接头和连接部位进行初次检查；确认无泄漏并检查无异常现象后，按测试压力的10％逐级升压，保压10min；保压足够时间进行检查，直至罐体或壳体首次出现渗漏、可见的变形和异常声响时的压力停止测试，并记录各测试压力。

[0117] 测试搪玻璃反应釜罐体与壳体的极限工作压力：表2

[0118]

名称	罐体极限工作压力(Mpa)	夹套极限工作压力(Mpa)
A1	2.1	0.5
A2	2.0	0.6
A3	2.2	0.6
B1	1.1	0.4

[0119] 注：上述水压试验根据GB/T 7994.1-2014进行，极限工作压力为罐体或壳体首次出现渗漏、可见的变形和异常声响时的压力。

[0120] 通过上述测试，不难看出，相比现有结构设计中同等罐体壁厚、釜体公称直径、全容积、搪玻璃层厚度现有的搪玻璃压力容器中，本实用新型通过等角度布置的螺栓固定连接，并结合高颈法兰端面凹槽处的垫片，增大了罐体的最高耐压值，直接增加了本设备的受压强度，确保受压安全。

[0121] 以上内容仅仅是对本实用新型的结构所作的举例和说明，所属本技术领域的技术人员对所描述的具体实施例做各种各样的修改或补充或采用类似的方式替代，只要不偏离实用新型的结构或者超越本权利要求书所定义的范围，均应属于本实用新型的保护范围。

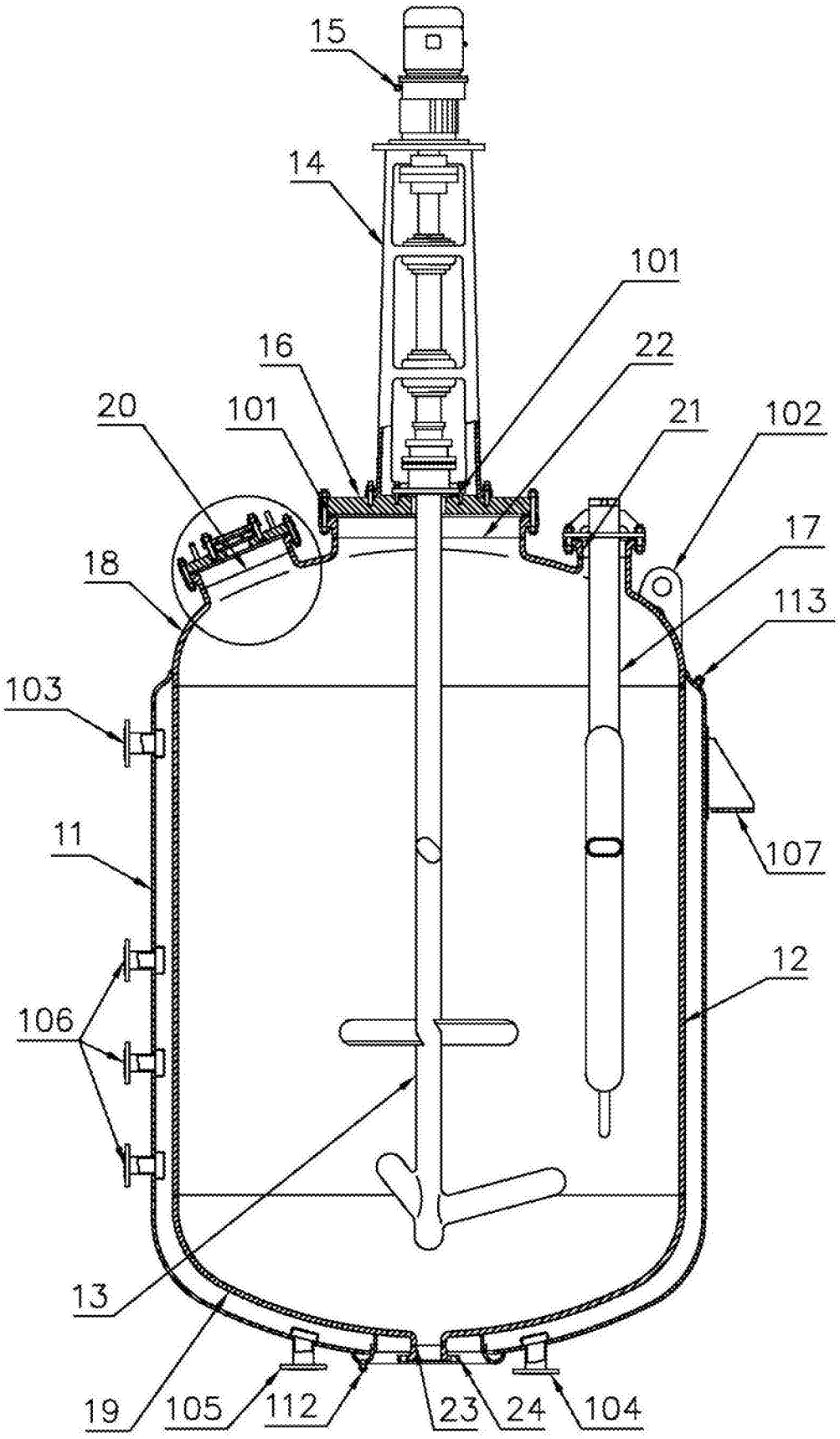

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图1

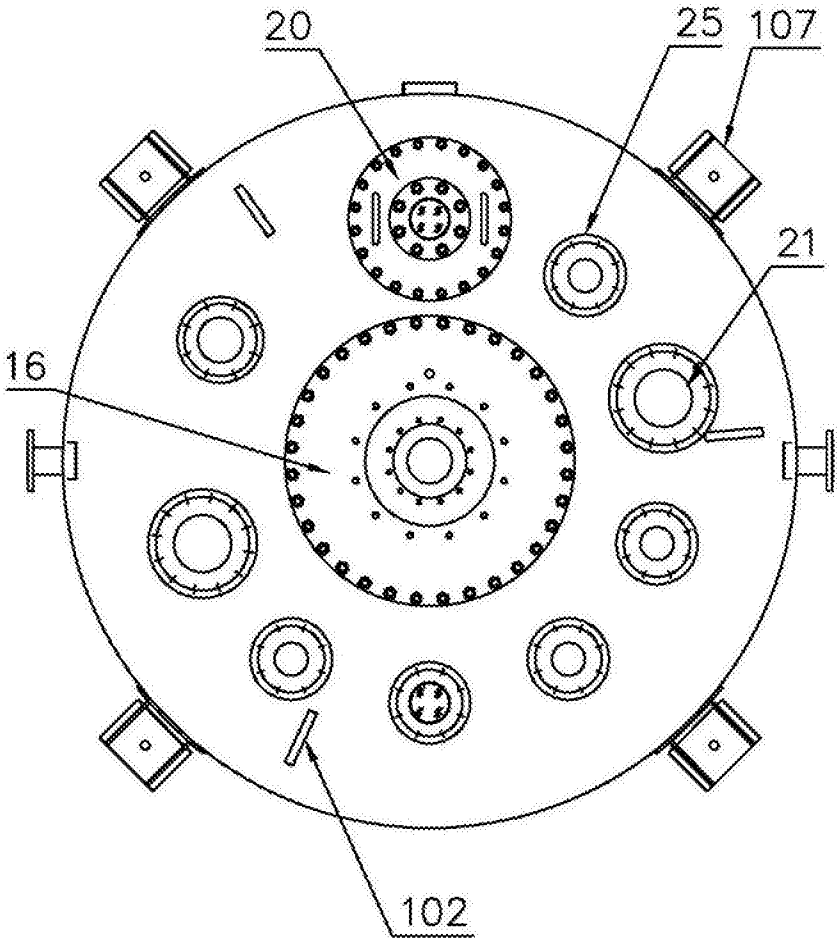


图2

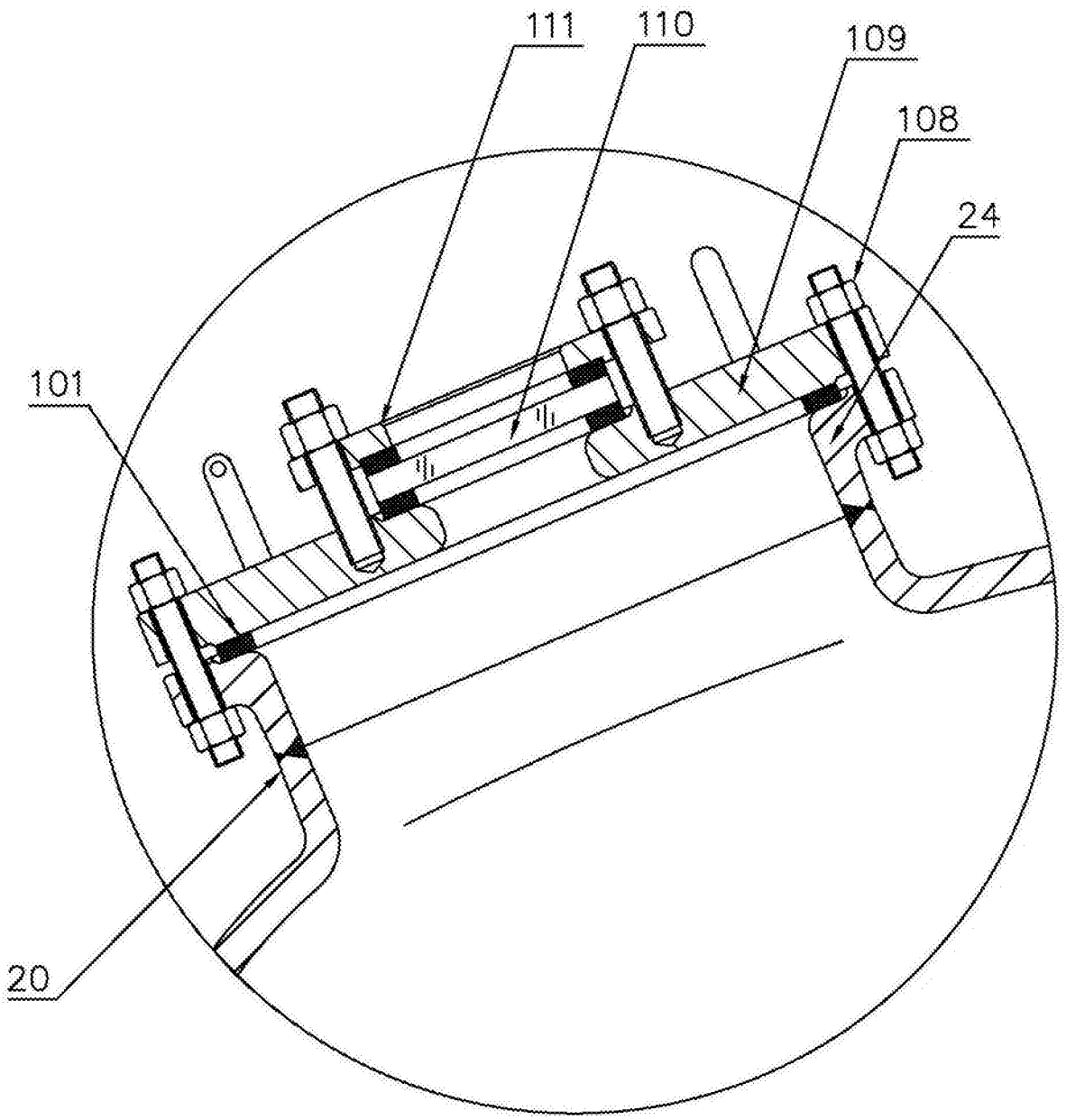


图3

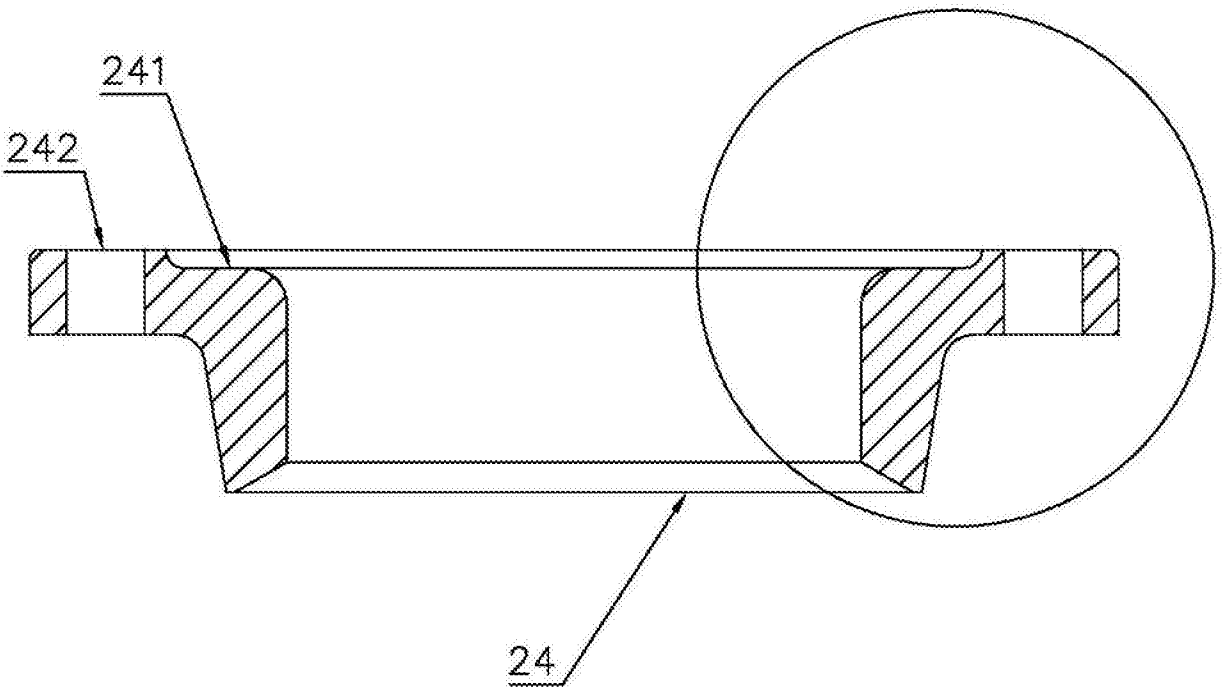


图4

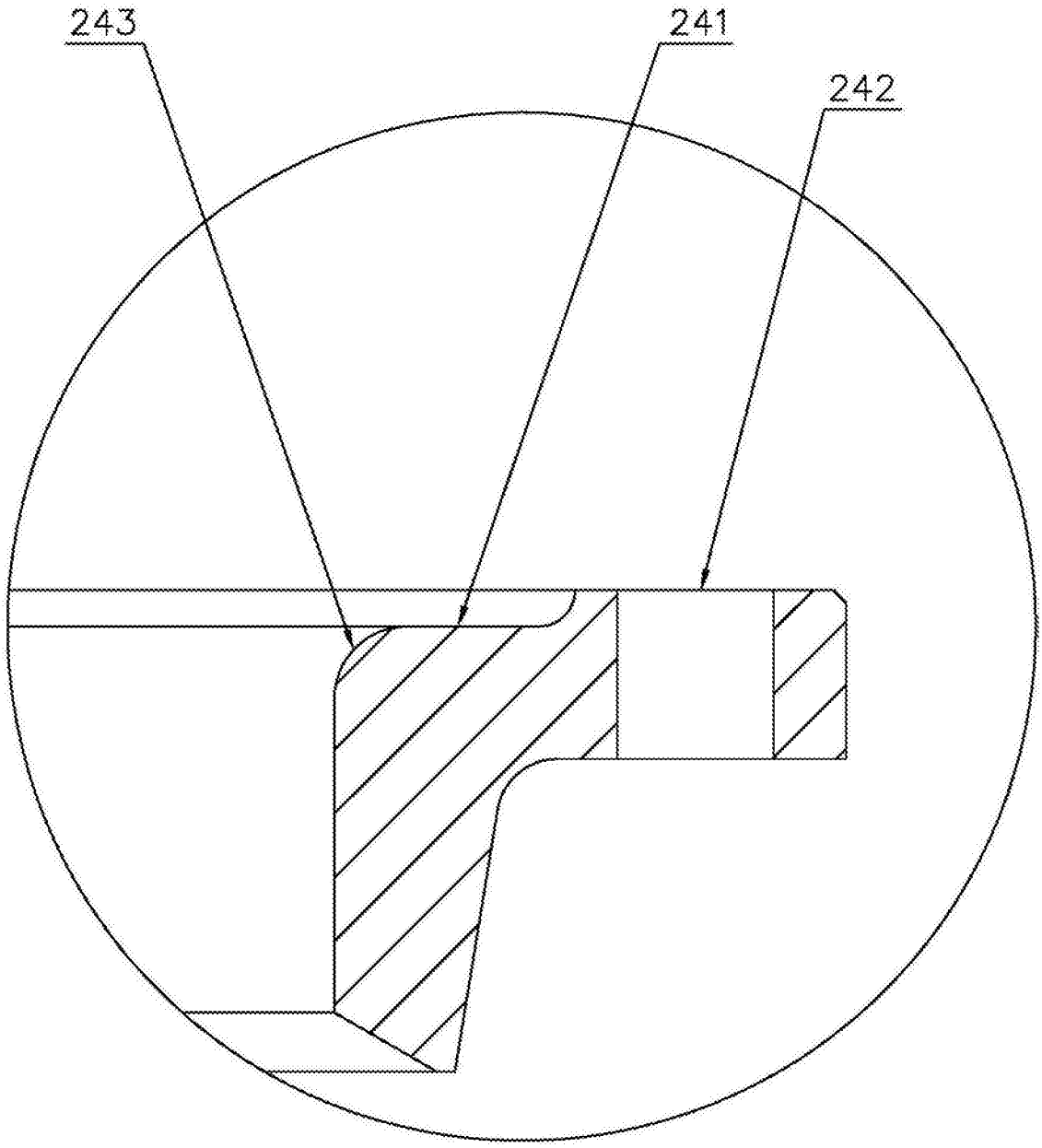


图5

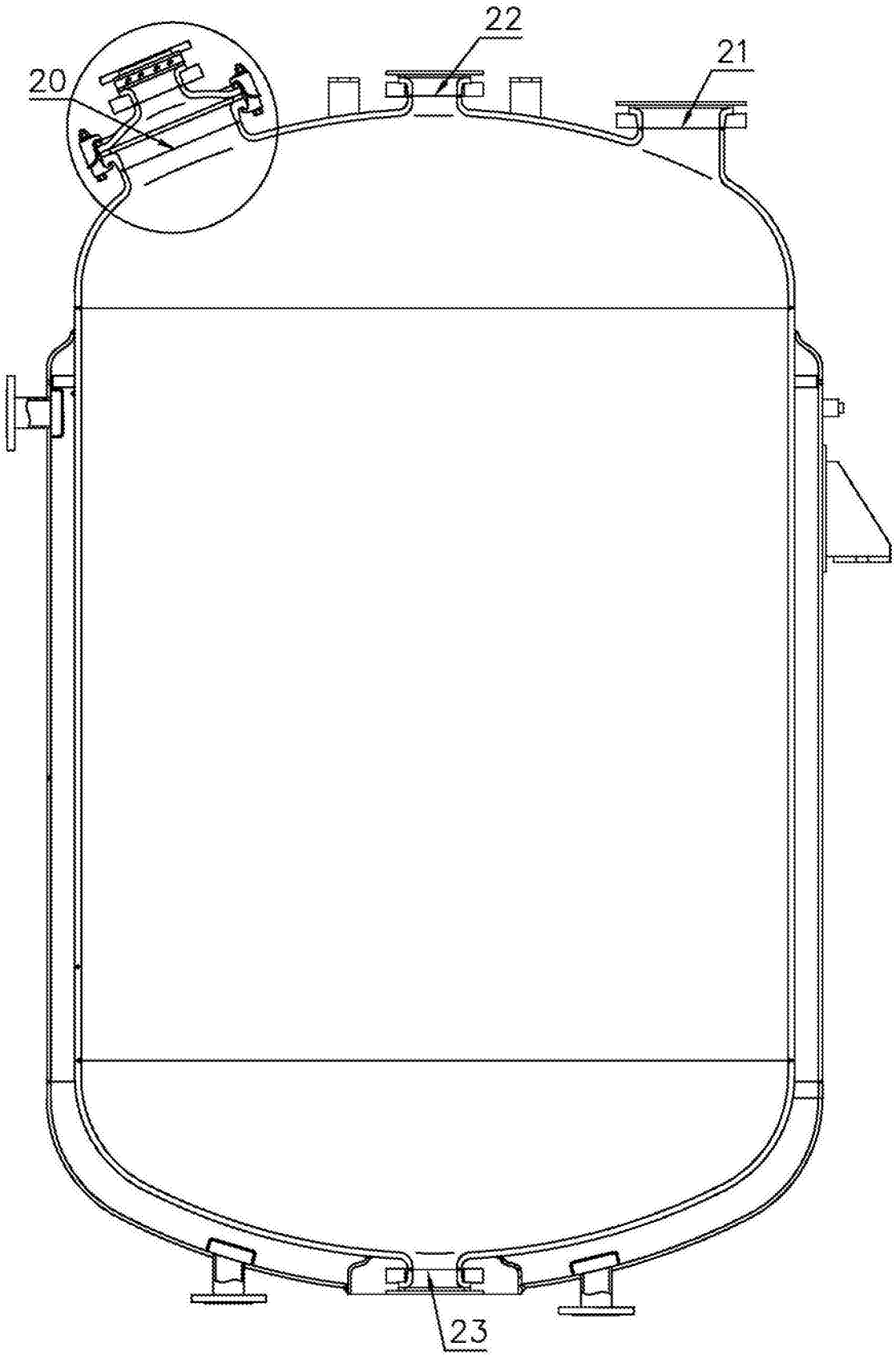


图6

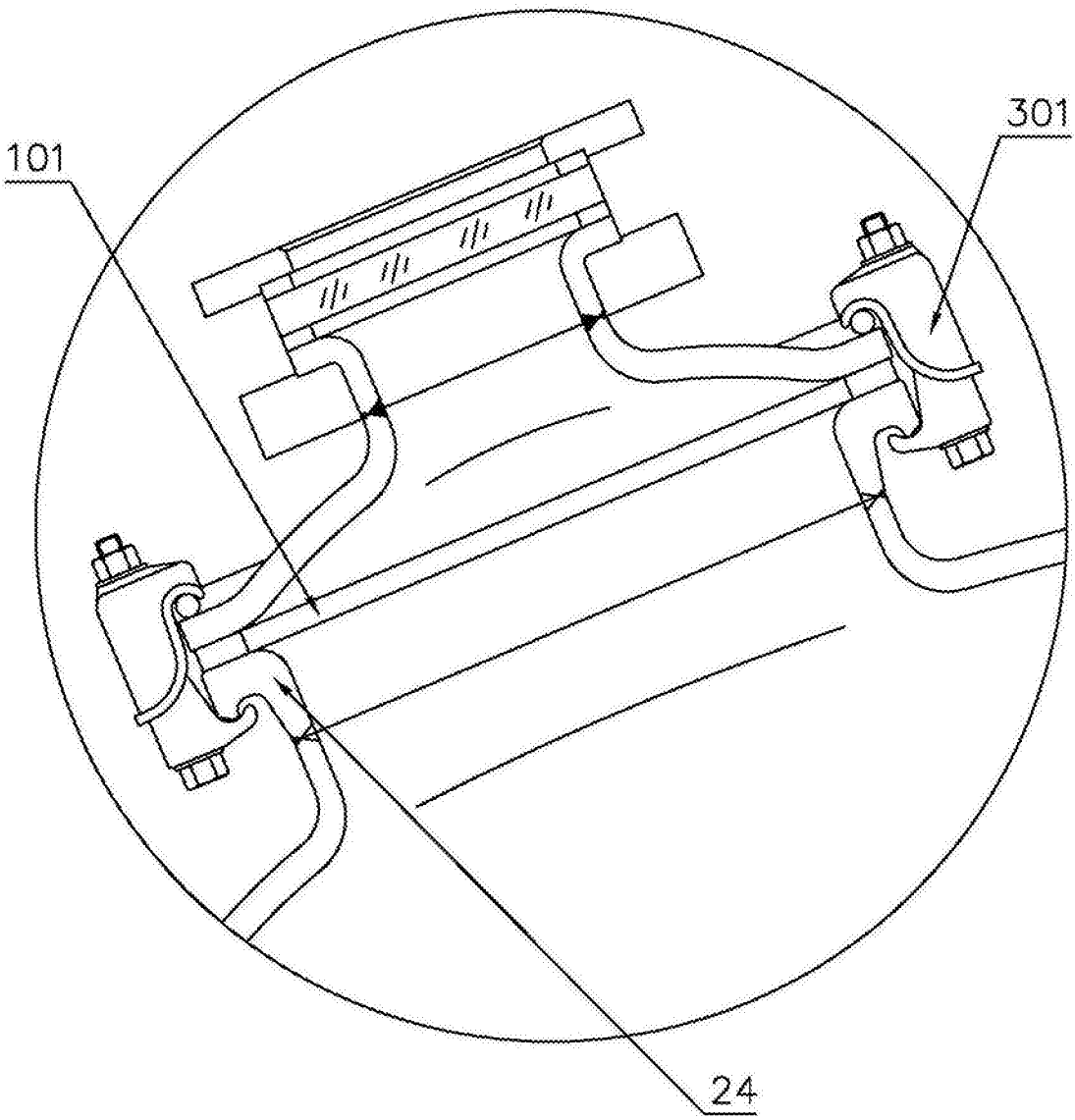


图7

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CN206372825U Corrosion -resistant glass reactor that wards off of high strength

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Corrosion -resistant glass reactor that wards off of high strength

Abstract

The utility model discloses a corrosion -resistant glass reactor that wards off of high strength, including pressing from both sides cover, jar body, agitator, upper cover, low head, sight glass manhole, the thermometer mouth of pipe, feed inlet, drain hole, the external side of jar is equipped with presss from both sides the cover, it is equipped with a plurality of installation positioner to press from both sides angles such as overlapping the outside, pressing from both sides cover upper portion and being equipped with steam inlet, steam inlet one side is equipped with the relief port, presss from both sides the cover bottom and is equipped with water export with fixed attention, the welding has high -neck flange on the feed inlet, the welding has high -neck flange on the sight glass manhole, and the welding has high -neck flange on the thermometer mouth of pipe, and the welding has high -neck flange on the drain hole, be equipped with the ground coat on the internal lateral wall of jar, be equipped with cover glaze layer on the ground coat, be equipped with the ground coat on high -neck flange inside wall, terminal surface, the recess, be equipped with cover glaze layer on the ground coat. The utility model discloses trompil department welding has increased high -neck flange, has reduced the quantity at stamping forming trompil R position, has directly increased the effective thickness of this equipment, and compressive strength improves relatively, ensures the safe operation of equipment.

CN206372825U Corrosion -resistant glass reactor that wards off of high strength

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A high - strength corrosion - resistant glass - lined reactor

Technical field

The utility model belongs to the technical field of glass-lined chemical equipment, and in particular relates to a high-strength corrosion-resistant glass-lined reactor.

Background technique

For the glass-lined reactor, the glass-lined reactor is a glass containing high silica, lining the inner surface of a steel container and being firmly adhered to the metal surface by high temperature burning to become a composite material. Stability and metal strength of the dual advantages, is an excellent corrosion-resistant equipment.

The existing glass-lined reactor in the upper head, under the head, mirror manhole, thermometer nozzle, feed port, discharge mouth and other openings at all parts of the high neck flange, access to high neck flange There are two ways: 1) the use of stamping way to get high-necked flange, stamping the opening of the R-site, the thickness of the thinning due to thinning, making the equipment by the pressure weakened; 2) according to HG / T2049-2012 glass The high-neck flange of the device is welded to the high-neck flange, but the HG / T 2049-2012 open high-neck flange port extends outwardly in the planar direction to form the outer edge (shown in Figures 6-20) Through the clamp (manual Figure 7-301) fixed connection, the circular port surface through a number of clamp fixed connection, usually for the visual angle of the arrangement of the clamp, such a connection will make the gasket at the various points of the uneven force , And the local resistance of the opening portion of the top of the kettle body is greater than that of the kettle body itself, so that the maximum pressure value of the entire apparatus at the opening portion is reduced, so that the pressure resistance inside the kettle body is limited.

Glass-making equipment is a glass of glass industry, a serious illness. Fish scale porcelain is also called hydrogen explosion, the performance of the porcelain layer to the size of the scales from the burst out of the phenomenon. At present, the mechanism for the formation of fish-scale porcelain is generally recognized in the art is that there are hydrogen in various forms of steel in the manufacture of glass-lined equipment, and hydrogen is saturated in the steel during the firing of the glass- When the product is cooled in the air to the enamel solidification, the hydrogen is too late to escape, so that the steel has reached a supersaturated state. Saturated hydrogen produces a certain pressure, can not be timely in the steel escape. When this pressure increases to make the porcelain layer off, they formed a fish scale porcelain. Because of this kind of fish scale porcelain has a certain latent, will appear under certain conditions, and thus in the glass-lined equipment manufacturing is completed, the installation, the use will occur, seriously troubled the glass-lined equipment in the chemical industry applications. Fish scales porcelain seriously affect the use of glass-lined equipment performance and production safety.

Studies have shown that the glass-lined glaze formula directly affects the properties of the glass-lined. Conventional glass-lined glaze formula and melting process and glass close to the glass-lined glaze with glass-lined glaze and glass-lined glaze. The bottom glaze by the type of adhesive can be divided into nickel glaze, cobalt bottom glaze, cobalt nickel glaze, antimony molybdenum glaze and mixed glaze and so on. Glass-lined glaze function: First, as a transition layer or close layer, making the glaze and steel connected together firmly stick together, the second is to play the role of storage gas, because the process of burning will produce CO, CO 2, H 2 and other gases, these gases must be part of the bottom glaze in the micro-pores to be stable, and because the furnace structure, product structure, heating conditions and other factors, making the bottom glaze to withstand the temperature difference of 50-80 ℃. General requirements of about 840-860 ℃ to vitrification, to 890-910 ℃ is still in the vitrification and not burn. Our research team has been working hard to improve the performance of glass-lined glass and glass-lined equipment for many years. Bottom glaze, can be firmly

combined with the embryo body, reducing the fish scales porcelain, the surface beautiful and clean, with good acid corrosion, but the corrosion resistance of alkali in general, limiting its scope of application. With the industrial development requirements for the equipment is getting higher and higher, for the performance requirements of glass-lined also have a higher demand.

The contents of the invention

The purpose of the utility model is to overcome the shortcomings of the prior art, to provide a high-strength corrosion-resistant glass-lined reactor, corrosion-resistant, fastened by bolts, easy to operate and good sealing, Blue, reducing the number of ram forming openings R, directly increase the effective thickness of the device, the relative increase in pressure to ensure the safe operation of the equipment.

In order to achieve the above object, the technical scheme adopted in the utility model is:

A high-strength corrosion-resistant glass-lined reactor comprising a jacket, a can body, a stirrer, an upper head, a lower head, a mirror manhole, a thermometer nozzle, an inlet, a discharge port, And the outer part of the jacket is provided with a plurality of mounting and positioning devices, the upper part of the jacket is provided with a steam inlet, the steam inlet side is provided with a vent port, the bottom of the jacket is provided with a condensate outlet, a high temperature steam Through the steam inlet into the sandwich for heat transfer, high temperature steam into low temperature steam, low temperature steam from the vent out, the steam generated by the condensate from the condensate condensate discharged from the condensate;

The bottom of the jacket is provided with a liquid inlet, the liquid inlet side is provided with a net opening, the jacket side wall is provided with a plurality of liquid outlets, the tank is heat-exchanged through a high-temperature liquid, and the high-temperature liquid enters the sandwich through the liquid inlet Heat exchange, high temperature liquid into a cryogenic liquid, low temperature liquid through the liquid outlet discharge, through the net to the liquid vent, to avoid the liquid residue in the jacket inside;

The upper part of the tank body is provided with a feed port, the inlet port is welded with a high neck flange, the high neck flange end face is provided with a groove on the groove, the groove is provided with a bolt The upper part of the gasket is provided with a cap, and the top cover is provided with a bolt through hole. The top cover and the high neck flange are fixed by bolts, The upper part of the top cover is provided with a fixed frame, the fixed frame is provided with a motor, the motor shaft end is provided with a stirrer, the stirrer is inserted into the tank body, the material or the liquid in the tank is fully stirred and mixed uniformly;

And the eyelet is provided with a high neck flange, the high neck flange end face is provided with a groove, the groove is provided with a chamfer, the groove One side is provided with a bolt through hole, the high neck flange groove is provided with a gasket, the upper part of the gasket is provided with a pressure plate, the pressing plate and the high neck flange are fastened by bolts, the pressure plate is provided with a viewing hole, There is a gasket on the glass, the glass of the sight glass is provided with a gasket, the upper part of the gasket is provided with a cover plate, the cover plate is fixedly connected with the pressure plate by screw;

And the other side of the feeding port is provided with a thermometer nozzle mouth, and the high neck flange is welded with a high neck flange on the thermometer nozzle, the groove is provided with a groove on the groove, The upper part of the gasket is provided with a pressure plate, the pressure plate and the high neck flange are fastened by bolts, the thermometer is set on the pressure plate, the thermometer is set with a thermometer, the thermometer and the thermometer Into the tank for temperature testing;

The bottom of the tank is provided with a discharge port on which the high neck flange is welded with a high neck flange, the groove end face is provided with a groove, the groove is provided with a chamfer, the groove side has a bolt hole , The high neck flange groove is provided with a gasket, the upper part of the gasket is provided with a bottom cover, the bottom cover and the high neck flange are fastened by bolts;

The bottom wall of the tank is provided with a bottom glaze layer, the glazed layer is provided with a glazed layer, an inner side wall of the high neck flange, an end face, a glazed layer on the groove, a glazed layer , To prevent liquid corrosion of the tank.

Preferably, the upper portion of the tank is provided with a plurality of spare ports on which the high neck flanges are welded, the groove having a chamfer on the groove, Bolt through hole, high neck flange groove with a gasket, the upper part of the gasket with cover, cover and high neck flange bolted fastening connection.

Preferably, the mirror manhole, the thermometer nozzle, the feed port, the discharge port, the spare port are welded with a high-necked flange, and the high-neck flange end face is arranged at an equal angle, There is a glazed layer, the glaze layer on the glaze layer, compared with the same tank wall thickness of the existing glass-lined pressure vessel, reducing the number of ram forming openings R parts, fixed by the angle of the bolt Connection, increasing the maximum pressure at the opening part of the value, a direct increase in the effective thickness of the device, the relative increase in compression strength to ensure the safe operation of equipment.

The utility model also provides a new formula of the bottom glaze layer, which is made of a bottom glaze frit by introducing a mixed agent into the following components. The chemical composition of the bottom glaze frit includes: silica 40 to 45%, 3-7% of aluminum oxide, 5-10% of calcium oxide, 20-26% of potassium carbonate, 12-15% of boron oxide, 3-15% of sodium fluorosilicate, 2- 4%; preferably, the chemical composition of the bottom glaze frit comprises 42-45% of silica, 5-6% of aluminum oxide, 5-8% of calcium oxide, 25-26% of potassium carbonate %, 13-15% of boron oxide, 4-10% of sodium fluorosilicate and 2-3% of indium oxide. At present, the bottom glaze frit composition is essential for the entire bottom glaze. In the prior art, the composition of the bottom glaze and bottom glaze fritters can not meet the higher requirements enamel products through the conventional combination glaze. In the new type, the utility model was optimized for the conventional formula. After adding the indium oxide, it was found that the bottom glaze was more fusible and the dispersibility and fusion of the various components were better. After the glaze was fired, the surface was bright, The formation of the glaze regardless of anti-fish scale porcelain or corrosion resistance are further improved.

Metal and enamel are two different compositions, different materials. Their solid combination is an extremely complex physical and chemical process. Therefore, the mixing agent is related to the process of diffusion and fusion between enamel and metal, which directly affects the adhesion effect and the acid resistance of the container and the effect of anti-porcelain. In the utility model, the composition of the mixed agent is 45-55% of fluorocarbonite ore powder, 10-15% of manganese oxide, 2-6% of copper oxide, 25-35% of molybdenum oxide, %. It is further preferred that the composition of the mixing agent is 45-48% of fluorocarbonite ore powder, 10-14% of manganese oxide, 4-8% of copper oxide, 28-30% of molybdenum oxide and 3-7% of titanium dioxide. It is further preferred that the mixing agent is in the range of 1 to 2% by weight of the bottom glaze, and the adhesion effect is optimum in the range, and the binding strength is insufficient, and the porcelain and cracking are liable to occur, The promotion is not obvious and the cost is increased. The mixing agent of the utility model has complicated physical and chemical reaction with the other components of the bottom glaze during the firing process of the glass-lined glaze, and can produce more micro-bubbles and chemical bonds, which are in the bottom glaze In the formation and retention of saturated steel in the saturation of the nest and pores, reducing the pressure of supersaturated hydrogen on the porcelain layer, the utility model of the mixing agent to optimize the composition of the adjustment, especially the addition of titanium dioxide, without special high temperature that Can make the bottom glaze components and glaze and the formation of strong adhesion between the metal, while reducing the expansion coefficient of the bottom glaze to reduce the glaze in the temperature and high pressure cracking tendency.

In the utility model, the preparation method of the mixing agent may be carried out according to a conventional method in the art, for example, including weighing, powder, mixing, melting, cooling, grinding and sieving process. Preferably, The melting process includes:

1)Mixed material heated to 130-160 ° C, stirring, incubation reaction 20min-40min;

2)Heating up to 500-600 ℃, stirring and holding 30min-1h;

3)Heating to 900-1000 ℃, stirring, filling the air, sintering.

In the utility model, the grinding may be carried out by a conventional grinding machine, for example, using a ball mill. The sieving process refers to a sieve of 80 mesh.

It will be apparent to those skilled in the art that the mixing agent may be used in accordance with the method of use of conventional binders in the art without the need for special procedures. At the same time, the glaze provided by the utility model can be used with commercially available or known recipe glazes.

The use of the utility model of the bottom glaze, after high temperature firing, embryo body on the bottom glaze all melted to the embryo uneven surface, enamel glaze and embryo body of the good adhesion. In the Tu Yan

with the bottom of the embryo body coated with the surface of the glaze, fired the surface of the porcelain layer appears strong luster, smooth, will not produce coke, soaked, porcelain and other defects.

Compared with the prior art, the utility model has the following advantages:

1)The mouth of the tank, the mirror manhole, the thermometer nozzle, the discharge port were welded with high neck flange, high neck flange with groove, high neck flange groove with gasket, through Bolt fastening connection, easy to operate and good sealing, which changed the original design of the formation of high-neck punching flange, increased the high-neck flange wall thickness, pressure strength is relatively increased;

2)Heat transfer method for the two ways, one high-temperature steam through the steam inlet into the sandwich for heat transfer, high temperature steam into low temperature steam, the second is through the high temperature liquid heat exchange within the sandwich, high temperature liquid into a low temperature liquid; Heat exchange method for the recyclable way to reduce production costs;

3)The welding of all the openings in the equipment increases the number of high-necked flanges and high-necked flanges on the end of the flange, reducing the number of R-shaped openings in the press-forming openings, fixed by bolts at equal angles, The existing glass-lined pressure vessel, which is thicker than that of the same tank, increases the maximum withstand voltage at the opening and increases the compressive strength Pressure safety, fastening with bolts, easy to operate and good sealing;

4)The utility model provides the bottom glaze, which can be firmly combined with the embryo body, the heat shock resistance and the temperature difference and the abrupt change performance, reduces the fish scale porcelain;

5)The glaze provided by the utility model has excellent corrosion resistance and not only has excellent acid corrosion resistance, but also has excellent alkali corrosion resistance and extends the application range of the glass-lined equipment.

Description of the drawings

BRIEF DESCRIPTION OF THE DRAWINGS Figure 1 is a schematic view of a cutting structure of a high strength corrosion resistant glass-lined reactor according to the utility model;

Figure 2 is a top view of a high strength corrosion resistant glass-lined reactor according to the utility model;

Figure 3 is a schematic view of a magnifying structure of a mirror in a high-strength corrosion-resistant glass-lined reactor according to the utility model;

Figure 4 is a schematic view of a high-neck flange structure of a high-strength corrosion-resistant glass-lined reactor according to the utility model;

Figure 5 is a schematic view of a high-neck flange enlargement structure of a high-strength corrosion-resistant glass-lined reactor according to the utility model;

Figure 6 is a schematic diagram of the existing design of glass-lined reactor structure;

Figure 7 is a schematic diagram of the enlarged structure of a high-neck flange in a conventional glass-lined reactor;

16-top cover, 17-thermometer, 18-top head, 19-under head, 20-motor, 16- 21- thermometer nozzle, 22 - feed port, 23 - discharge port, 24 - high neck flange, 241 - groove, 242 - bolt through hole, 243- chamfer, 25 - spare port 101-pad, 102-lug, 103- steam inlet, 104- condensate outlet, 105 - liquid inlet, 106 - liquid outlet, 107 - mounting fixture, 108 - bolt, 109 - platen, Glass, 111 - cover, 112 - open mouth, 113- vent, 301 - clamp.

detailed description

In order to facilitate the understanding of those skilled in the art, the technical solutions of the utility model will be described in further detail with reference to Figs. 1-7.

A high-strength corrosion-resistant glass-lined reactor includes a jacket 11, a can body 12, a stirrer 13, an upper header 18, a lower head 19, a mirror manhole 20, a thermometer nozzle 21, a feed port 22, A discharge port 23 is provided on the outside of the tank body 12. The jacket 11 is provided with a plurality of mounting and positioning devices 107 at an angle other than the outside of the jacket 11, and a steam inlet 103 and a steam inlet 103 There is provided a gas discharge port 113. At the bottom of the jacket 11 is provided with a condensate outlet 104 through which the high temperature steam enters the sandwich through the steam inlet 103, the high temperature steam becomes low temperature steam, the low temperature steam is discharged from the air discharge port 113, Of the condensed water is discharged from the condensate outlet 104.

The bottom of the jacket 11 is provided with a liquid inlet 105, and a liquid inlet 105 is provided with a plurality of liquid outlets 106, and the tank 11 is heat-exchanged by a high-temperature liquid. The liquid enters the interlayer through the liquid inlet for heat exchange, the high temperature liquid becomes a cryogenic liquid, the cryogenic liquid is discharged through the liquid outlet 106, and the liquid is vented by the discharge port 112 to prevent the liquid from remaining inside the jacket.

The upper part of the tank body 12 is provided with a feed port 22 on which a high neck flange 24 is welded, and the high end of the high neck flange 24 is provided with a groove 241 with chamfering 243 and a groove 241 are provided on one side of the groove 241. The upper surface of the gasket 101 is provided at the groove 241 of the high neck flange 24, and the top cover 16 is provided with a bolt through hole 242, The top cover 16 is fixed to the high-necked flange 24 by bolts, and the top cover 16 seals the feed port 22 through the washer 101. The upper portion of the top cover 16 is provided with a fixing frame 14 provided with a motor 15, The agitator 13 is provided at the shaft end, and the agitator 13 is inserted into the tank body 12, and the material or the liquid in the tank body 12 is sufficiently stirred and mixed uniformly.

The viewer manhole 20 is provided with a high-necked flange 24 on the side of the feed port 22, and the end face of the high-necked flange 24 is provided with a groove 241 and a groove 241 A chamfer 243 is provided on the side of the groove 241, and a spacer 101 is provided at the groove 241 of the high neck flange 24, and the upper surface of the gasket 101 is provided with a pressure plate 109, the pressure plate 109 and the high neck flange 24 The upper surface of the pressure plate 109 is provided with a gasket 101, which is provided with a glass mirror 110, which is provided with a spacer 101, an upper portion of the spacer 101 A cover plate 111 is provided, and the cover plate 111 is fixedly connected to the pressing plate 109 by screws.

The other side of the feed port 22 is provided with a thermometer nozzle 21 on which a high neck flange 24 is welded to the thermometer nozzle 21 and a recess 241 is provided on the end face of the high neck flange 24 and a recess 241 The horn 243 is provided with a bolt through hole 242 on the side of the groove 241. The gasket 101 is provided at the groove 241 of the high neck flange 24, and the upper surface of the gasket 101 is provided with a pressure plate 109, And a thermometer 17 is provided with a thermometer 17, and a thermometer 17 is inserted into the tank 12 for temperature detection.

The bottom of the can body 12 is provided with a discharge port 23 on which a high neck flange 24 is welded to the discharge port 23 and a groove 241 is provided on the end face of the high neck flange 24 and a chamfer 243 is provided on the groove 241, The groove 241 is provided with a bolt hole 242 and a high-neck flange 24 is provided with a gasket 101 at the groove 241. The upper part of the gasket 101 is provided with a bottom cover and the high-necked flange 24 is fastened by bolts.

The bottom wall of the can body 12 is provided with a bottom glaze layer, the glazed layer is provided with a glazed layer, an inner side wall of the high neck flange 24, an end face, a bottom glaze layer on the groove, Glaze layer, to prevent liquid corrosion tank.

The upper part of the tank body 12 is provided with a plurality of spare ports 25 on which the high neck flange 24 is welded and the back face 25 is provided with a groove 241, the groove 241 is provided with a chamfer 243, The groove 241 is provided with a bolt through hole 242, and the high-neck flange 24 is provided with a gasket 101 at the groove 241. The upper portion of the spacer 101 is provided with a cover plate 111, and the cover plate 111 and the high-necked flange 24 are fastened by bolts connection.

The goggle manhole 20, the thermometer nozzle 21, the feed port 22, the discharge port 23 and the spare port 25 are welded with a high-neck flange 24, and the bottom layer is provided with a glaze layer at the bottom There is a glazed layer, compared to the same tank wall thickness of the existing glass-lined pressure vessel, reducing the number of ram forming openings R, directly increase the pressure of the device to ensure compression and safety.

The technical solutions of the utility model will be described in further detail with reference to specific examples and test cases.

1、Preparation of mixed admixture

Preparation Example 1

1)Weigh the fluorocarbonite ore powder 4.8Kg, manganese oxide 1.2Kg, copper oxide 0.6Kg, molybdenum oxide 3Kg and titanium dioxide 0.4Kg and crushed, mixed material heated to 140 ℃, stirring, holding reaction 40min;

2)Heating up to 550 ℃, stirring and holding 1h;

3)Heating to 900 ° C, stirring, filling the air, sintering, cooling, ball milling, over 80 mesh sieve, get mixed with the agent.

Preparation 2

1)Weighing fluorocarbonite ore powder 4.5Kg, manganese oxide 1.4Kg, copper oxide 0.7Kg, molybdenum oxide 2.8Kg and titanium dioxide 0.6Kg and crushed, mixed material heated to 160 ℃, stirring, holding reaction 40min;

2)Heating up to 600 ℃, stirring and holding 1h;

3)Heating to 950 ° C, stirring, filling the air, sintering, cooling, ball milling, over 80 mesh sieve, get mixed with the agent.

Preparation Example 3

1)Weigh the fluorocarbonite ore powder 4.6Kg, manganese oxide 1.3Kg, copper oxide 0.4Kg, molybdenum oxide 3Kg and titanium dioxide 0.7Kg and crushed, mixed material heated to 130 ℃, stirring, holding reaction 30min;

2)Heating up to 500 ℃, stirring and holding 1h;

3)Heating up to 1000 ° C, stirring, filling the air, sintering, cooling, ball milling, over 80 mesh sieve, get mixed with the agent.

Preparation 4

1)Weigh the fluorocarbonite ore powder 5Kg, manganese oxide 1.5Kg, copper oxide 0.2Kg, molybdenum oxide 3.2Kg and titanium dioxide 0.1Kg. And crushed, mixed material heated to 150 ℃, stirring, incubation reaction 40min;

2)Heating up to 600 ℃, stirring and holding 1h;

3)Heating to 900 ° C, stirring, filling the air, sintering, cooling, ball milling, over 80 mesh sieve, get mixed with the agent.

Preparation Example 5

1)Weigh the fluorocarbonite ore powder 5.5Kg, manganese oxide 1.3Kg, copper oxide 0.3Kg, molybdenum oxide 2.5Kg and titanium dioxide 0.3Kg and crushed, mixed material heated to 140 ℃, stirring, holding reaction 30min;

2)Heating up to 500 ℃, stirring and holding 1h;

3)Heating up to 1000 ° C, stirring, filling the air, sintering, cooling, ball milling, over 80 mesh sieve, get mixed with the agent.

Preparation Example 6

1)Weigh the fluorocarbonite ore powder 5.5Kg, manganese oxide 1.3Kg, copper oxide 0.3Kg and molybdenum oxide 2.5Kg and crushed, the mixture heated to 140 ℃, stirring, holding reaction 30min;

2)Heating up to 500 ℃, stirring and holding 1h;

3)Heating up to 1000 ° C, stirring, filling the air, sintering, cooling, ball milling, over 80 mesh sieve, get mixed with the agent.

2、Glass-lined glaze was prepared

Preparation Example 7

0.44 Kg of silica, 0.05 kg of aluminum oxide, 0.06 Kg of calcium oxide, 0.25 Kg of potassium carbonate, 0.13 Kg of boron oxide, 0.04 Kg of sodium fluorosilicate, 0.03 kg of indium oxide doped with indium oxide, and a mixing agent 0.01Kg, mixed, ground made of glass-lined glaze.

Preparation Example 8

0.04 Kg of silica, 0.05 Kg of aluminum oxide, 0.08 Kg of calcium oxide, 0.25 Kg of potassium carbonate, 0.13 Kg of boron oxide, 0.05 kg of sodium fluorosilicate, 0.02 Kg of indium oxide doped with indium oxide, and mixed sealant in Preparation Example 1 0.01Kg, mixed, ground made of glass-lined glaze.

Preparation 9

0.45 Kg of silica, 0.05 kg of aluminum oxide, 0.05 Kg of calcium oxide, 0.25 Kg of potassium carbonate, 0.14 Kg of boron oxide, 0.04 Kg of sodium fluorosilicate and 0.02 Kg of indium oxide doped with indium oxide and the mixing agent 0.01Kg, mixed, ground made of glass-lined glaze.

Preparation Example 10

0.44 Kg of silica, 0.05 kg of aluminum oxide, 0.06 Kg of calcium oxide, 0.25 Kg of potassium carbonate, 0.13 Kg of boron oxide, 0.04 Kg of sodium fluorosilicate, 0.03 kg of indium oxide doped with indium oxide, and a mixing agent 0.02Kg, mixed, ground made of glass-lined glaze.

Preparation 11

0.44 Kg of silica, 0.05 kg of aluminum oxide, 0.06 Kg of calcium oxide, 0.25 Kg of potassium carbonate, 0.13 Kg of boron oxide, 0.04 Kg of sodium fluorosilicate, 0.03 kg of indium oxide doped with indium oxide, and mixed agent in Preparation 2 0.01Kg, mixed, ground made of glass-lined glaze.

Preparation Example 12

0.44 Kg of silica, 0.05 kg of aluminum oxide, 0.06 Kg of calcium oxide, 0.25 Kg of potassium carbonate, 0.13 Kg of boron oxide, 0.04 Kg of sodium fluorosilicate, 0.03 kg of indium oxide doped with indium oxide, and mixed sealant in Preparation 3 0.01Kg, mixed, ground made of glass-lined glaze.

Preparation Example 13

0.44 Kg of silica, 0.05 kg of aluminum oxide, 0.06 Kg of calcium oxide, 0.25 Kg of potassium carbonate, 0.13 Kg of boron oxide, 0.04 Kg of sodium fluorosilicate, 0.03 kg of indium oxide doped with indium oxide, and mixed sealant in Preparation 4 0.01Kg, mixed, ground made of glass-lined glaze.

Preparation 14

0.44 Kg of silica, 0.05 Kg of aluminum oxide, 0.06 Kg of calcium oxide, 0.25 Kg of potassium carbonate, 0.13 Kg of boron oxide, 0.04 Kg of sodium fluorosilicate, 0.03 kg of indium oxide doped with indium oxide, and mixed sealant in Preparation 5 0.01Kg, mixed, ground made of glass-lined glaze.

Preparation Example 15

0.44 Kg of silica, 0.05 kg of aluminum oxide, 0.06 Kg of calcium oxide, 0.25 Kg of potassium carbonate, 0.13 Kg of boron oxide, 0.04 Kg of sodium fluorosilicate, 0.03 kg of indium oxide doped with indium oxide, and mixed sealant in Preparation 6 0.01Kg, mixed, ground made of glass-lined glaze.

Preparation 16

0.44 Kg of silica, 0.05 kg of aluminum oxide, 0.06 Kg of calcium oxide, 0.25 Kg of potassium carbonate, 0.13 Kg of boron oxide, 0.04 Kg of sodium fluorosilicate and 0.01 Kg of the mixing agent in Preparation Example 1 were mixed and ground Into a glass-lined glaze.

Preparation 17

0.44 Kg of silica, 0.05 kg of aluminum oxide, 0.06 Kg of calcium oxide, 0.25 Kg of potassium carbonate, 0.13 Kg of boron oxide, 0.04 Kg of sodium fluorosilicate and 0.01 Kg of the mixed paste in Preparation 6 were mixed and ground Into a glass-lined glaze.

Preparation 18

0.44 Kg of silica, 0.05 kg of aluminum oxide, 0.06 Kg of calcium oxide, 0.25 Kg of calcium carbonate, 0.13 Kg of boron oxide, 0.04 Kg of sodium fluorosilicate, 0.03 kg of indium oxide oxide and 0.01 Kg of nickel oxide cement , Grinding made of glass-lined glaze.

Preparation 19

The glass-lined glaze was prepared according to Examples 2-1-6 in CN103863724B.

Test case

The porcelain glaze formulations used in the test examples were 40 parts of silica, 5.5 parts of aluminum oxide, 18 parts of sodium oxide, 20.5 parts of boron oxide, 3 parts of calcium fluoride, 7 parts of sodium fluorosilicate, Cobalt 0.01 part, 1.09 part of zinc oxide and 4.9 parts of phosphorus pentoxide. The contents of C, Si, Mn, S and P were 0.16%, 0.023%, 0.41%, 0.019% and 0.017%, respectively, and the specifications were 80mm * 80mm * 5.5mm.

The enamel glaze prepared in the above Preparation Examples 7 to 19 was coated on the above-mentioned steel test pieces and fired at a high temperature. In the coating with a glaze of the bottom of the embryo coated on the surface of the glaze, the glaze and the powder dry embryo placed in high temperature environment, firing to the porcelain layer all melted to the surface of a strong luster, Thickness of about 1.2mm.

Test Methods

1、Impact test: According to the national standard GB / T 7990-2013, the coating was damaged by the impact of the number of times to assess the coating of the adhesion performance, the specific results shown in Table 1.

2、Thermal shock experiment: According to the national standard GB / T 4547-2007, the thermal fatigue test of the heating temperature of 400 ℃. The temperature difference test is to test the block heated to 200 ℃ after the release of room temperature flow of water quenching, if the porcelain layer is not cracking, increasing by 20 ℃ and then heating, quenching, if not cracking continue to increase the temperature heating, quenching, so the cycle, Until the crack, the specific results shown in Table 1.

3、Corrosion resistance test: According to the national standard GB / T7989-2013 for boiling water vapor corrosion resistance of the determination of continuous testing 168 hours to the quality of the loss rate to assess the acid corrosion, unit: g / (m 2 .d); 2) Determination of alkali resistance according to the national standard GB / T7988-2013, test for 24 hours at 80 ° C in 0.1 mol / L NaOH, and to evaluate the corrosion resistance at the mass loss rate. Unit: g / (m 2. D). The specific results are shown in Table 1.

4、High pressure test: According to the national standard GB / T 7991.6-2014, where the test voltage is 2.5kV, the test temperature is 20 ℃. As a result, no spark discharge phenomenon was observed in the glass-lined layers prepared in Preparation Examples 7 to 15, Preparation Example 18 and Preparation Example 19, and the glass-lined layer had no defects or weaknesses, and both Preparation 16 and Preparation 17 were found There is a spark discharge.

Table 1

It can be seen from Table 1 that the bottom glaze provided by the utility model can be firmly bonded with the metal matrix, which effectively improves the impact resistance and thermal shock resistance of the test piece. No experimental phenomenon is observed in the experimental process. In addition, the use of the utility model of the glaze combined with conventional glaze fired, not only has a strong acid corrosion resistance but also has a very good alkali corrosion resistance.

5、Resistance to pressure detection

In addition, any three kinds of glass-lined reactor (named A1, A2, A3) are the same as those in the existing structural design, D1 (25mm), the nominal diameter DN (2200mm), the whole volume V (10000L), glass-lined layer thickness D2 (2mm) of the glass-lined reactor (named B1) for a separate comparison of pressure test (hydraulic test test), which A1 and B1 used in the bottom glaze, glaze consistent, A2 And B1 used in the glaze, glazed, A3 and B1 used in the bottom glaze, glazed consistent, the pressure test test pressure in the tank test pressure 2.5Mpa, jacket test pressure 0.8Mpa, record the experimental data as follows; : The test should be slowly boosted to the specified test pressure of 10%, holding pressure 5min. And all the welding joints and connection parts for the initial inspection; to confirm no leakage and check no abnormalities, according to the test pressure of 10% Pressurized, holding pressure 10min; pressure sufficient time to check until the tank or shell for the first time leakage, visible deformation and abnormal sound when the pressure to stop testing, and record the test pressure.

Test the ultimate working pressure of the tank and the shell of the glass-lined reactor: Table 2

Name Tank Extreme Operating Pressure (Mpa) Jacket Limit Operating Pressure (Mpa) A1 2.1 0.5 A2 2.0 0.6 A3 2.2 0.6 B1 1.1 0.4

Note: The above hydraulic pressure test according to GB / T 7994.1-2014, the ultimate working pressure for the tank or shell for the first time leakage, visible deformation and abnormal sound pressure.

Through the above test, it is not difficult to see that the utility model is arranged at equal angles in the existing glass-lined pressure vessel compared to the existing tank design in the same tank thickness, the nominal diameter of the kettle body, the whole volume, and the glass- Bolt fixed connection, and combined with high-neck flange groove at the end of the gasket, increasing the maximum pressure of the tank, directly increasing the pressure of the device to ensure compression and safety.

The foregoing is merely illustrative and illustrative of the structure of the utility model, and various modifications and additions of the specific embodiments described in the art to those skilled in the art may be substituted or modified in a similar manner without departing from the utility model Or beyond the scope of the claims, should fall within the scope of the present invention.

CN206372825U Corrosion -resistant glass reactor that wards off of high strength

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The wording below is an initial machine translation of the original publication. To generate a version using the latest translation technology, go to the original language text and use Patent Translate.

1.A high-strength corrosion-resistant glass-lined reactor, which comprises a jacket, a tank body, a stirrer, an upper head, a lower head, a mirror manhole, a thermometer nozzle, an inlet and a discharge port, The upper part of the tank is provided with a jacket, the upper part of the jacket is provided with a steam inlet, the steam inlet side is provided with a vent port, the bottom of the jacket is provided with a condensate outlet;

The upper part of the tank body is provided with a feed port, the inlet port is welded with a high neck flange, the high neck flange end face is provided with a groove on the groove, the groove is provided with a bolt The top cover is provided with a bolt through hole, the top cover and the high neck flange are fixed by bolts; the upper part of the top cover is provided with a fixed cover Frame, fixed frame with a motor, the motor shaft with a stirrer;

And the eyelet is provided with a high neck flange, the high neck flange end face is provided with a groove, the groove is provided with a chamfer, the groove One side is provided with a bolt through hole, the high neck flange groove is provided with a gasket, the upper part of the gasket is provided with a pressure plate, the pressing plate and the high neck flange are fastened by bolts, the pressure plate is provided with a viewing hole, There is a gasket on the glass, the glass of the sight glass is provided with a gasket, the upper part of the gasket is provided with a cover plate, the cover plate is fixedly connected with the pressure plate by screw;

And the other side of the feeding port is provided with a thermometer nozzle mouth, and the high neck flange is welded with a high neck flange on the thermometer nozzle, the groove is provided with a groove on the groove, The upper part of the gasket is provided with a pressure plate, the pressure plate and the high neck flange are fastened by bolts, the thermometer is set on the pressure plate, the thermometer is set with the thermometer;

The bottom of the tank is provided with a discharge port on which the high neck flange is welded with a high neck flange, the groove end face is provided with a groove, the groove is provided with a chamfer, the groove side has a bolt hole , The high neck flange groove is provided with a gasket, the upper part of the gasket is provided with a bottom cover, the bottom cover and the high neck flange are fastened by bolts;

The bottom wall of the tank is provided with a bottom glaze layer, the glazed layer is provided with a glazed layer, an inner side wall of the high neck flange, an end face, a glazed layer on the groove, a glazed layer The

2.A high strength corrosion resistant glass-lined reaction vessel according to claim 1, wherein the bottom of said jacket is provided with a liquid inlet, and the liquid inlet side is provided with a discharge opening, said jacket side wall being provided with a plurality Liquid outlet.

3.A high strength corrosion resistant glass-lined reactor according to claim 1, wherein the upper part of the tank is provided with a plurality of spare ports on which a high neck flange is welded and a high neck flange There is a groove, the groove is provided with a chamfer, the groove side is provided with a bolt through hole, the high neck flange groove is provided with a gasket, the upper part is provided with a cover plate, the cover plate and the high neck flange pass Bolt fastening connection.

4.A high strength corrosion resistant glass-lined reaction vessel according to claim 3, characterized in that said mirror manhole, the thermometer nozzle, the feed port, the discharge port and the spare port are welded by seamless welding of the high neck flange , High-neck flange on the end of the angle of the layout of the bolt through the hole, the weld at the end of the glaze layer, the bottom of the glaze layer with a glaze layer.

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PINAKIN ENGINEERS PRIVATE LIMITED PINAKIN ENGINEERS PRIVATE LIMITED PINAKIN ENGINEERS PRIVATE LIMITED PINAKIN ENGINEERS PRIVATE LIMITED

GENERAL POWER OF ATTORNEY

We,**M/S PINAKIN ENGINEERS PVT LTD**,having its address atPlot No 249, Shed No A1/29, Phase 2, GIDC, Vapi, Gujarat, India - 396195, India, do hereby authorize *Mr. Rajendra Ambalal Shah, Mr. Hemraj Chaturbhuj Asher, Mr. Darius Cawasji Shroff, Mr. Sanjay Khatau. Asher, Ms. Zarine MinocherTalaty, Mr. Marco Philippus Ardeshir Wadia, Mr. Saumil Shantaram Rege, Mr. Sanjay Ramakant Buch and Mr. Prashant Khatau Asher all of Mumbai Advocates and Solicitors, partners and Mr. Hemant Thadhani (Advocate&Patent Agent IN-PA/4053) of Crawford Bayley & Co.*having address at *State Bank Buildings, N.G. N. Vaidya Marg, Fort, Mumbai 400 023, Maharashtra, India,*to act as our agents/attorneysin connection with filing, prosecution or maintenance of our applications or registrations, and to represent us in all matters, including any and all proceedings, opposition, applications, petitions, rectifications, cancellations, revocations, notices, renewals, requisitions, communications, etc. of all our Trademarks under The Trade Marks Act, 1999, Copyrights under The Copyright Act 1957, Patents under The Patents Act, 1970 and, Designs under The Designs Act, 2000 and to authorize to sign and execute such deeds and documents onour behalf and also the appointment of substitute(s) for the above purpose as may be necessary andexpedient; and to act as our agent/attorney and to receive all notices, requisitions andcommunications until further notice.

We request that all notices, requisitions and communications relating to the above may be sent to such agent/attorney at the above address.

All communications may be sent to the following address in India:
Crawford Bayley & Co
Solicitors and Advocates
State Bank Buildings, N.G. N. Vaidya Marg, Fort, Mumbai 400 023, Maharashtra, India
Email: hemant.thadhani@crawfordbayley.com

Dated this 23 day of September, 2024.

To,
The Controller of Patents, Designs & Trade Marks
India

For PINAKIN ENGINEERS PRIVATE LIMITED

FOR M/S PINAKIN ENGINEERS PVT LTD

DIRECTOR

Mr.Hasmukh Patel
(Director)

INTELLECTUAL PROPERTY INDIA

GOVERNMENT OF INDIA
MINISTRY OF COMMERCE & INDUSTRY
PATENT OFFICE, CHENNAI

Intellectual Property Building, G.S.T. Road,
Guindy, Chennai-600032
E-mail: chennai-patent@nic.in; kanhaiya207.ipo@gov.in

- 4 JUN 2025

Application No. 202141018455 M 1021 **Dated:-04/06/2025**

To.
Mr. Bharat Daswani
DASWANI & DASWANI,
Patent & Trade Mark (Intellectual Property) Attorneys,
Daswani House, Green Acres
23B, Ahiripukur 1st Lane
Kolkata, 700 019, India,

Sub: - Pre-grant Representations under section 25(1) of the Patent Application No. 202141018455

This is to bring to your notice that a pre-grant representation under section 25(1) has been filed in this instant application. A copy of the said pr-grant representation filed by Pinakin Engineers Private Limited, having address at Plot No. 249, Shed No. A1/29, Phase 2, GIDC, Vapi, Gujarat – 396195, India, is available online on https://iprsearch.ipindia.gov.in/PublicSearch/ The said opposition is considered and the Office is of the opinion that a *prima facie* case is present under rule 55(3) (b) of the Patents Rules, 2003 (as Amended). (Report has been enclosed)

In view of the above if you wish, you may file your statement and evidence if any in support of your application within **two month** from the date of this letter, as per Rule 55(4) of The Patents Rules, 2003, failing which the matter shall be treated accordingly.

Kanhaiya 4/6/2025

O/C

(Kanhaiya Lal Balai)
Asst. Controller of Patents & Designs

Encl:
1. Prime facie report

Prima Facie Report

Subject: Pre-Grant Opposition to Patent Application No. 202141018455 by Pinakin Engineers Private Limited

Introduction

This report summarizes the pre-grant opposition filed in respect of application no. 202141018455 dated 21/04/2021 made by STANDARD GLASS LINING TECHNOLOGY PRIVATE LIMITED and published on 28/10/2022 on the following grounds:

a. Section 25(1)(b): That the invention so far as claimed in any claim of the complete specification has been published before the priority date of the claim; i) in any specification filed in pursuance of an application for a patent made in India on or after the 1st day of January, 1912, or ii) in India or elsewhere, in any other document: Provided that the ground specified in sub-clause (ii) shall not be available where such publication does not constitute an anticipation of the invention by virtue of sub-section (2) or sub-section (3) of section 29.

b. Section 25(1)(e): That the invention so far as claimed in any claim of the complete specification is obvious and clearly does not involve any inventive step, having regard to the matter published as mentioned in clause (b) or having regard to what was used in India before the priority date of the applicant's claim.

c. Section 25(1)(f): That the subject of any claim of the complete specification is not an invention within the meaning of this Act, or is not patentable under this Act.

d. Section 25(1)(g): That the complete specification does not sufficiently and clearly describe the invention or the method by which it is to be performed.

Ground A. Section 25(1)(b):

The subject matter of the application is not novel and has been published in several publications and documents covering the subject matter in question. The Indian Patent Act; vide Sec 25(1) when read with Section 13, provides that in case of the applications claiming subject matter already disclosed previously in another specification or other document, no patent can be granted for that application. The subject matter of the impugned claims 1-8 already stands disclosed in found prior art D1: CN103820781A (annexed as Exhibit E), D2: CN105148816A (annexed as Exhibit F) and D3: CN206372825U (annexed as Exhibit G). All the aspects of impugned claims 1-8 are disclosed in one single document hence the impugned claims 1-8 are not novel at all.

Conclusion

In view of the arguments of the opponent and disclosure of D1 (page no. 14-23 of Opposition filed on 18/11/2024), D2 (page no. 24-29 of Opposition filed on 18/11/2024) and D3 (page no. 29-31 of Opposition filed on 18/11/2024) prima facie it seems that the prior art D1, D2,and D3 is relevant to the present application with respect to novelty in view of the Section 25(1)(b) with the Section 2(1)(j) of the Patents Act.

Ground B. Section 25{1}{e):

The subject matter of impugned claims is obvious to a person skilled in the art, and therefore the application should not be granted. It is submitted that the invention as claimed is obvious and lack inventive step in view of Prior art

D1: CN103820781A (annexed as Exhibit E),

D2: CN105148816A (annexed as Exhibit F)

D3: CN206372825U (annexed as Exhibit G).

Conclusion:

In view of the submissions by the opponent (page no. 31-33 of Opposition filed on 18/11/2024), prime facie it seems that the Documents D1-D3 are relavant to the present application with respect to inventive step in view of section 25(1)(c) with the section 2(1)(ja) of the Patents Act.

Ground C. Section 25{1}{f):

The Opponent states that the impugned claims fall under the purview of the Sections 3(d) and 3(f) of Indian Patents Act, 1970.

Conclusion:

In view of the submissions by the opponent (page no. 33-34 of Opposition filed on 18/11/2024), prime facie it seems that the claims of present application are not an invention or not patentable in view of section 25(1)(f) with the section 3(d) and 3(f) of the Patents Act.

Ground D. Section 25(1)(g):

The opponents states that the complete specification does not sufficiently and clearly describe the invention or the method by which it is to be performed.

Conclusion:

In view of the submissions by the opponent (page no. 35-37 of Opposition filed on 18/11/2024), prime facie it seems that the subject matter of present application is not sufficiently and clearly describe the invention in view of section 25(1)(g) with the section 10(4) of the Patents Act.

Final Conclusion:

The opposition is based on substantial prior art and common industry practices that question the inventive step of the patent application. The evidence presented supports the claims of prior knowledge, and obviousness.

This prima facie report provides a detailed overview of the grounds of opposition and the supporting evidence, forming the basis for further examination and decision-making by the patent office.

The said opposition is considered and the Office is of the opinion that a *prima facie* case is present under rule 55(3) (b) of the Patents Rules, 2003 (as Amended).

Kanhaiya 4/6/2025

(Kanhaiya Lal Balai)
Asst. Controller of Patents & Designs

Daswani & Daswani
Patent & Trade Mark (Intellectual Property) Attorneys

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To

Date: 04th August, 2025

THE CONTROLLER OF PATENTS & DESIGNS
Bhoudhik Sampada Bhavan,
Guindy Chennai

Re: Opposition to grant of Patent on Patent Application No. 202141018455 dated 21ST April 2021 in the name of Standard Glasslining Technology Private Limited, under Section 25(1) by Pinakin Engineers Private Limited.

Dear Sir,

We submit herewith our Reply Statement with respect to the written arguments filed by the opponent in the pre-grant opposition under Section 25(1) of the Patents Act, 1970 filed against the aforementioned patent Application.

The Applicant requests the Learned Controller to grant leave to file further evidence. The Applicant respectfully requests for an opportunity to be heard in the matter.

The Learned Controller is respectfully requested to take the Reply Statement on record.

Yours Sincerely,
Of Daswani & Daswani

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(Bharat Daswani)
IP Attorney
Regn. No: IN/PA/289
Agent for the Applicant
Email: kolkata@daswaniindia.net

Enclosures:

1. Reply Statement

BEFORE THE CONTROLLER OF PATENTS,
PATENT OFFICE BRANCH, CHENNAI

IN THE MATTER OF The Patents Act, 1970 as amended by the Patents (Amendment) Act 2005

And

IN THE MATTER OF The Patent Rules 2003 as amended by the patents (Amendment) Rules, 2006 and in other subsequent amendments

And

IN THE MATTER OF Patents Application No. 202141018455 dated 21st April 2021 of Standard Glasslining Technology Private Limited, an Indian Company of D1-12, Phase I, IDA, Jeedimetla, Hyderabad, Telangana- 500055,India.
.....Applicant

And

IN THE MATTER OF representation by way of opposition under section 25(1) of the Patents Act, 1970 as amended by the Patents (Amendment) Act, 2005 for opposition to the grant of patent application No. 202141018455 dated 21ST April 2021 by Pinakin Engineers Private Limited, an Indian Company, of Plot No. 249, Shed No. A1/29, Phase 2, GIDC, Vapi, Gujarat – 396195, India,....
Opponents.

REPLY STATEMENT OF THE APPLICANT

This Counterstatement is filed by the Applicant, Standard Glass Lining Technology Private Limited, in response to the pre-grant opposition filed by Pinakin Engineers Private Limited (hereinafter “the Opponent”) under Section 25(1) of the Patents Act, 1970, read with Rule 55 of the Patent Rules, 2003, against Patent Application No. 202141018455 titled “CLEAN ROOM GLASS LINED REACTOR” (hereinafter “the Impugned Application / instant patent Application or patent Application”).

At the outset, the Applicant denies and disputes all the averments and allegations made by the Opponent in the Pre-Grant Opposition, received by the Applicant on 04 June 2025 (hereinafter referred to as the "present Opposition") u/s 25(1) of the Patents Act 1970 (hereinafter referred to as the "Act"). It is submitted that all such averments and allegations are false, incorrect and baseless and hence denied in-toto. It is submitted that nothing should be deemed to be admitted for the want of traverse unless expressly admitted herein below.

Before proceeding to deal with each and every allegation raised in the purported Opposition, the Applicant craves leave of Ld. Controller to place on record the following preliminary submissions/objections, which are being urged in addition, and without prejudice, to each other, and the contents of the present written/Reply Statement.

PRELIMINARY OBJECTIONS AND SUBMISSIONS :

The opponent presents himself as an individual “ANY PERSON” with interest in “CLEAN ROOM GLASS LINED REACTOR”. The provisions of any opposition proceedings under the section 25(1) of the Act are only applicable for a “person interested", which includes a person engaged in, or in promoting, research in

the same field as that to which the invention relates. The opponent has failed to provide any evidence in support of his claim for the interest in the present field. The opponent seems to be fraudster who is interested to delaying the proceedings of the Patent Office for an early grant of the present case and fulfilling his malafide intentions better known to him. The applicant(s) in the present case call a strict and valid proof from the opponent and come clean with his intentions before the Ld. Tribunal.

The Applicant herein provides an exhaustive para-wise rebuttal to every statement in the Opponent's Statement of Case, a detailed claim-by-claim comparison with cited prior art (D1: CN103820781A, D2: CN105148816A, D3: CN2063728SU), a comprehensive table of distinguishing features, and a robust challenge to the Opponent's locus standi, emphasizing their malicious and mala fide intent to delay the patent grant. The Applicant further highlights the Opponent's poor technical understanding and inability to appreciate the invention's intended selection / technical features / embodiments, supported by technical arguments, jurisprudence, and evidence of the invention's novelty, inventive step, and economic significance. The Impugned Application fully complies with the Patents Act, 1970, and is entitled to proceed to grant.

The Ld. Tribunal herein is made aware of the facts that the impugned patent application no. 202141018455, (hereinafter referred to as 'subject application') was filed by "Standard Glasslining Technology Private Limited, an Indian company, on April 21st, 2021. The subject application was published in the Patent Office Journal on Oct 28th 2022. The 'Request for Examination (RFE)' in respect of the patent application was filed on April 21st, 2021 and subsequently a First Examination Report was issued on Nov 10th 2022. A Response to the First Examination Report was filed on 10/08/2023.

The Opponent has filed the pre-grant opposition alleging that the invention as claimed in the Application No. 202141018455 is neither Novel nor involves

any Inventive step, is not patentable under the provisions of the Act and does not sufficiently and clearly describe the invention as required under Section 10(4) of the Act, The Opponent has merely made bald allegations and vague interpretations without any substantiation whatsoever. The Opponent has failed to show how said grounds of opposition are sustainable with respect to the invention as claimed in the instant patent Application.

The Hon‟ble Supreme Court in Anil Rishi v. Gurbaksh Singh, (2006) 5 SCC 558 has held that as per Sections 101 and 102 of the Evidence Act, 1872, the burden of proving a fact rests on the party who substantially asserts in the affirmative and not the party who denies it.

Further, it is submitted that the Hon'ble Bombay High Court in F & H v. Unichem, AIR 1969 Bom 225, while relying on the same principles of burden of proof in a patent case, has held that the onus in regard to all objections to validity lies on the person who has challenged the validity of the Patent.

Illustratively, reference may also be had to the decisions of Hon‟ble Intellectual Property Appellate Board (IPAB) in Fresenius Kabi Oncology Limited vs. Glaxo Group Limited & Ors, (Order No. 161 of 2013) and The Travancore Mats & Matting Co. v. The Controller of Patents and Ors, (ORA/44/2009/PT/CH, Order No. 47 of 2012). The Applicant reserves the right to further substantiate this submission in the course of the oral hearing, if required, through additional arguments and case-laws. It is submitted that the principle enunciated above applies to the present proceedings as well. The Opponent cannot fulfill its burden of proof by merely making bald allegations, but must discharge the burden by pleading specific facts and support the same with evidences. Therefore, in the absence of evidence having been filed along with the representation, the statements made therein cannot be relied upon, proving beyond any doubt that the Opponent failed to discharge its onus of proving the contents of its representation. Therefore, the pre-grant opposition ought to be dismissed in-limine.

The Applicant asserts that, as established supra, the burden of proving a fact lies on a party moving the court (or the adjudicating authority), and hence, in the present proceedings, firstly, the burden of proving the facts stated in the representation lies on the Opponent; secondly, the onus of restraining only to the specific provision / grounds of opposition under section 25(1) relied upon by the Opponent in the instant representation, on record also lies on the Opponent; thirdly, the Opponent has to prove the contents of each of the alleged prior-art documents relied upon by him in the representation; and most importantly, the Opponent has to prove that each of the alleged prior-art documents relied upon by him in the present proceeding was publically available before the earliest priority date of the instant application.

BURDEN OF PROVING THE CONTRARY LIES UNTO THE APPLICANT :

Particularly, in the present proceedings, the Opponent has conveniently failed to –

1. Put-forth/furnish the verified translated copies of alleged prior-art documents relied upon by him in the representation;
2. Prove the contents of each of the alleged prior-art documents relied upon by him in the instant representation and how the teachings are integrated thereupon;
3. Prove that each of the alleged prior-art documents relied upon by him in the present proceeding was publically available before the earliest priority date of the instant application; and
4. Prove that any or each of the alleged prior-art documents is/are material for assessment of patentability of claims of the instant application.

It is submitted that, unless the burden of proving the facts has been discharged by the Opponent, the burden of proving the contrary does not lie on the Applicant. It is pertinent to note that the Opponent, in the present proceedings, has not discharged its burden, and for this reason alone, the whole representation is a meaningless tautology and, as such, liable to be

dismissed *in-limine*.

BURDEN OF PROVING LACK OF NOVELTY NOT DISCHARGED BY THE OPPONENT:

It is most respectfully submitted that it is a well settled principle of Patent Law that in order to challenge lack of novelty of the claimed invention on the basis of prior arts, it is necessary to prove that – a document relied upon is a prior-art document i.e. the document, being relied upon, was, in-fact, available to the public, at least, as of the earliest priority date of the patent application; and the prior art document relied upon to challenge the validity of a claimed invention unambiguously discloses each of the claimed features of the invention in such a manner that enables a skilled person to practice the invention without bearing the burden of further experimentation.

Reliance is being placed on the Manual of Patent Office Practice and Procedure published by The Office of Controller General of Patents, Designs & Trademarks, 2019, at page no. 93, that prescribes –

Novelty

1) An invention is considered as new if it is not anticipated by prior publication in patent and non- patent literature, prior use or prior public knowledge. An invention is new (novel) if it has not been disclosed in the prior art, where the prior art means everything that has been published, presented or otherwise disclosed to the public before the date of filing of complete specification.

Justice Vimadalal (High Court of Bombay) in Farbewerke Hoechst and Ors. v. Unichem Laboratories and Ors. (AIR 1969 Bom 255) held that –

15. That brings me to the next ground of alleged invalidity of the plaintiffs' patent viz. want of novelty. The test of novelty as formulated by Halsbury, (3rd edn.) Vol. 29 p. 27 para 58) is in the following terms: "To anticipate a patent, a prior publication or activity must contain the whole of the invention impugned;

i.e., all the features by which the particular claim attacked is limited. In other words, the anticipation must be such as to describe, or be an infringement of the claim attacked."

Based on the aforesaid, it is crystal clear that the Opponent in the representation ought to evidentially show and prove that each of the features of claims of the present application is explicitly disclosed in the prior-art document and they are published, presented or otherwise disclosed to the public before the date of filing (priority date) of complete specification, which the Opponent has miserably failed to do.

It is pertinent to note, at this juncture, that the Opponent has miserably failed to prove that the document D1 (CN103820781A), which claims “"Glass lined equipment provided with outer profiling cladding and manufacturing method of glass lined equipment", and based on this reason alone, the purported representation by way of opposition ought to be dismissed at its threshold for the want of evidence. The applicant has merely compared the figures without the application of mind and highlighted the prior art therein and put forward general remarks in support. The Applicant reserves the right to further substantiate this submission in the course of the oral hearing, if required, through additional arguments and case-laws.

BURDEN OF PROVING LACK OF INVENTIVE STEP NOT DISCHARGED BY THE OPPONENT:

Qua lack of inventive step, a party is required to establish that the claimed invention was obvious to a person skilled in the relevant art prior to the date of invention in view of the teachings of prior art documents, which are clearly taught and enabled so that a person skilled in the art would have been able to independently arrive at the claimed invention without any undue research & experimentation on his/her part. It is submitted that - the Opponent has failed to provide valid arguments as disclosure thereof being of such nature to

render the invention as claimed in the instant application obvious to a person skilled in the art (either alone or in combination with other prior-art documents relied upon in the present purported representation), and based on this reason alone, the present representation by way of opposition ought to be dismissed at its threshold.

Aforesaid submissions proves beyond any iota of doubt that the Opponent miserably failed to discharge its burden of proving that the claimed invention lacks inventive step, and based on this reason alone, the present representation by way of opposition ought to be dismissed at its threshold. The Applicant reserves the right to further substantiate this submission in the course of the oral hearing, if required, through additional arguments and case-laws.

4. BURDEN OF PROVING INSUFFICIENCY OF DISCLOSURE NOT DISCHARGED BY THE OPPONENT:

With regards the ground of insufficiency of disclosure, it is settled position in law that the specification and claims are addressed to those with a high degree of knowledge of the field of science to which they relate (i.e. a person skilled in the art), and accordingly, for challenging the validity of the claimed invention based on the ground of insufficiency of disclosure, the Opponent ought to prove that the description of instant patent Application is not sufficient to enable a person skilled in the art to carry it into effect and/or the description is difficult to follow, vide evidence(s). Considering the evidence(s), the averments of the Opponent are nothing but unsubstantiated statements. It is noteworthy that the Opponent has not filed any evidence in the present opposition and accordingly, the averments of the Opponent ought to be dismissed *in-limine* for the want of evidence.

5. OPPONENT'S PLEA TO FILE FURTHER SUBMISSIONS / EVIDENCE IS CONTRARY TO THE PROVISIONS OF THE PATENTS ACT AND THE LAW OF

THE LAND

The Applicant strongly opposes the Opponent‟s plea to be permitted to file further submissions, as allowing the Opponent to file further evidences/submissions would only result in undue delay in grant of a legitimate patent on the instant patent application, immensely prejudicial to the Applicant. It is settled position in law that amendments of the pleadings are only to be allowed when sufficient cause is shown by the party proving that the documents/evidences could not be filed earlier by the party enumerating detailed reasoning therefor. It is also a settled position in law that amendment of the pleadings is not to be granted as a matter of right to the party, as otherwise the proceeding gets prolonged ad-infinitum at the behest of mala-fide intention of a party to the proceeding. The Applicant reserves the right to further substantiate the aforesaid basis the case laws during the hearing.

Without prejudice to the aforesaid, the Applicant further submits that since a pre-grant opposition is just an aid in examination of the application, the Controller's consideration of any further amendments can easily be taken care of by the Controller without filing of further frivolous submissions which will only serve to prolong the instant proceedings and defeat the purpose of the whole pre-grant opposition system.

Without prejudice to the above, the Applicant reserves its right to defend its application by filing any additional pleadings and/or evidence in future as it may consider necessary. The Applicant also reserves its rights to specifically respond/rebut to any argument/document as may be relied upon by the Opponent which already does not form part of the records of the Patent Office in the present matter.

Without prejudice to the generality of the foregoing, the Applicant seeks to give para-wise response to the written statement in the succeeding part of the

present reply statement. The Applicant has responded therein to the meritless and frivolous challenges raised by the Opponent under each ground. The correct teachings of the cited prior art documents would categorically show the frivolity of the present opposition as well as erroneous reading of the prior art documents as filed and put forth by the Opponent. The Opponent has committed an abuse of the process by raising erroneous and wrong averments and the opposition is liable to be dismissed on this ground alone.

PARAWISE REPLY TO THE OPPOSITION ON MERITS :

Locus Standi

The Opponent claims locus standi under Section 2(1)(t) as “any person.” The Applicant submits that the Opponent’s status as a competitor, coupled with its failure to specify a direct technical or commercial interest affected by the patent grant, undermines its standing. In *Ajay Industrial Corporation v. Controller of Patents* (1984 PTC 234), the Delhi High Court required opponents to demonstrate a tangible interest beyond rivalry. The Opponent’s vague “stakeholder” claim, combined with its mala fide intent, does not satisfy this requirement. The Applicant requests the Controller to dismiss the opposition on this ground alone.

INVENTION

The Impugned Application discloses a novel and inventive “Clean Room Glass Lined Reactor” designed for pharmaceutical and food industries, where stringent hygiene, corrosion resistance, and contamination prevention are critical. The reactor comprises a double-walled mild steel (MS) vessel with internal glass lining, fluidic channels for optimized heat transfer, a glass-lined flat surface for real-time flow observation, a cavity in the bottom flange for heat transfer fluid circulation, and a seamless stainless steel (SS) overlay on nozzles to prevent oxidation and ensure compliance with ISO 14644-1 clean room standards (Exhibit A, pages 40–45). The intended selection of the

seamless SS overlay on nozzles, combined with other structural features, addresses a long-standing technical challenge of rust formation in conventional reactors, ensuring product purity and operational reliability in clean room environments.

The technical advancements of the invention include:

- Targeted Corrosion Resistance: The seamless SS overlay on nozzles, applied via a proprietary welding process (Exhibit A, page 41), prevents oxidation in high-humidity clean room environments, unlike conventional MS nozzles that corrode and compromise product purity.

- Clean Room Hygiene: The seamless design eliminates crevices, reducing microbial growth risks and ensuring compliance with Good Manufacturing Practices (GMP) and regulatory standards (e.g., USFDA, EMA).

- Enhanced Heat Transfer: Fluidic channels and the bottom flange cavity improve heat transfer efficiency by 15–20%, reducing processing time and energy consumption by approximately 10 kWh per batch (Exhibit A, page 42).

- Operational Precision: The glass-lined flat surface enables real-time flow observation, reducing process errors by 10% and ensuring consistent product quality.

- Extended Durability: The double-walled MS vessel and SS overlay extend reactor lifespan by 20–30%, minimizing maintenance frequency (Exhibit A, page 44).

The economic significance of the invention is substantial:

- Cost Savings: The SS overlay reduces maintenance and replacement costs by INR 5–10 lakhs annually for a mid-sized facility, due to extended reactor lifespan and reduced downtime (Exhibit A, page 44).

- Increased Productivity: Enhanced heat transfer and operational precision increase throughput by 10–15%, equivalent to 50–100 additional batches annually for a typical pharmaceutical plant.

- Market Competitiveness: Compliance with stringent regulatory standards enables access to high-value markets, potentially increasing revenue by 20–25% for manufacturers.

- Sustainability Impact: Improved energy efficiency reduces carbon emissions by approximately 0.5 tons per reactor annually, aligning with global sustainability goals.

Grounds:
It is submitted that the contents of the said paragraph are fallacious and thus denied. It is most respectfully submitted that the grounds (i.e. 25(1)(b), 25(1)(e), 25(1)(g), 25(1)(d) and 25(1)(f) of the Patents Act, 1970) enlisted in the present paragraph do not have any valid foundation, and are frivolous in nature. It is submitted with respect to the above grounds of challenge that the opponent has made bald allegations without any basis and has thus, merely enumerated the grounds of opposition under section 25(1) of the Act. The Opponent has failed to substantiate any of the aforementioned grounds under the Act.

It is further submitted that each of the grounds enumerated in the instant paragraph have been specifically dealt with and rebutted herein-below by the Applicant, and reliance is being placed on the same.

Further, it is submitted that the Opponent has conceded that the only

grounds upon which the Opponent is placing reliance for the present opposition are the ones enlisted in the paragraph under reply, and hence, the Opponent is barred from relying on any other grounds during the course of present proceedings.

The Opponent's opposition is a deliberate attempt to delay the grant of the Applicant's legitimate patent, driven by malicious intent and commercial rivalry. As a competitor in the glass-lined equipment market, the Opponent seeks to obstruct the Applicant's innovation to maintain its market position, exhibiting unclean hands. In *Hindustan Lever Ltd. v. Godrej Soaps Ltd.* (1996 PTC 141), the Bombay High Court dismissed an opposition motivated by commercial rivalry, emphasizing that patent proceedings must not be misused to hinder innovation. Similarly, in *Snehalata C. Patel v. Controller of Patents* (2001 PTC 121), the Controller rejected an opposition lacking technical merit and driven by mala fide intent. The Opponent's failure to provide certified translations of prior art (D1–D3) and reliance on vague, unsubstantiated allegations further evidence its intent to delay proceedings, burdening the Applicant with unnecessary costs. The Applicant submits that the Opponent's conduct warrants dismissal of the opposition and an award of costs under Section 77(1)(e).

The Opponent's allegations demonstrate a fundamental misunderstanding of the invention's intended selection—the seamless SS overlay on nozzles integrated with specific structural features for clean room applications. The Opponent's reliance on irrelevant prior art (D1: insulation-focused, D2: manufacturing-focused, D3: flange-focused) and misinterpretation of the specification (e.g., "100% glass-lined surface") reflect a lack of appreciation for the invention's technical nuances. In *Farbwerke Hoechst v. Unichem Laboratories* (AIR 1969 Bom 255), the Bombay High Court held that opponents must demonstrate a clear understanding of the invention to challenge its patentability. The Opponent's failure to address the synergistic effect of the claimed features underscores its technical inadequacy, further

weakening its opposition.

Validity of Prior Art Citations

The Opponent relies on foreign prior art (D1, D2, D3) without certified English translations, violating Rule 55(2) of the Patent Rules, 2003. In *Novartis AG v. Union of India* (2013) 6 SCC 1, the Supreme Court emphasized the need for verified documentation in patent proceedings. Similarly, in *Pfizer Inc. v. Controller of Patents* (2008 PTC 65), untranslated prior art was disregarded. The Applicant objects to the admissibility of D1–D3 and requests their exclusion unless certified translations are provided. The Opponent's reliance on machine-translated excerpts is unreliable and indicative of its frivolous approach.

Legal Framework

The Applicant additionally relies on the following case laws and guidelines:

- Novelty: *Bishwanath Prasad Radhey Shyam v. Hindustan Metal Industries* (AIR 1982 SC 1444) requires a single prior art document to disclose all essential features for anticipation.
- Inventive Step: *F. Hoffmann-La Roche Ltd. v. Cipla Ltd.* (2015 SCC Online Del 13658) holds that non-obviousness requires creative insight beyond routine knowledge.
- Patentability: *Novartis AG* clarifies that Section 3(d) applies to new forms of known substances, not new products.
- Sufficiency: *Press Metal Corporation v. Noshir Sorabji* (1983 PTC 121) requires an enabling specification, which the Applicant provides.
- Amendments: *Aventis Pharma S.A. v. Lupin Ltd.* (2012 SCC Online Del 1133) permits amendments within the original disclosure.
- Combination Inventions: *Bajaj Auto Ltd. v. TVS Motor Company Ltd.* (2009) 41 PTC 489 (Mad) supports patentability of synergistic combinations.

- Hindsight: *Technograph v. Mills & Rockley* (1972 RPC 346) prohibits hindsight in obviousness assessments.
- Mala Fide Conduct: *Hindustan Lever Ltd.* and *Snehalata C. Patel* condemn oppositions driven by commercial rivalry.
- Guidelines: The *Guidelines for Examination of Patent Applications (2016)* mandate clear, supported claims, which the Impugned Application satisfies.

Section 25(1)(b): Anticipation i.e. lacking novelty:

The contents of the said paragraphs are denied for being baseless, frivolous and without any valid foundation of law. It is a settled position in law that in order to challenge the validity of a patent/ patent Application on the ground of lack of Novelty on the basis of prior arts/documents, it is necessary to prove that the cited documents are published before the priority date of the complete specification in India or elsewhere in the world or fallen in public domain before the priority date of the complete specification.

Reference is being made to; Section 25(1)(b) of the Patents Act –

"25. Opposition to the patent.—(1) Where an application for a patent has been published but a patent has not been granted, any person may, in writing, represent by way of opposition to the Controller against the grant of patent on the ground—

(b) that the invention so far as claimed in any claim of the complete specification has been published before the priority date of the claim— (i) in any specification filed in pursuance of an application for a patent made in India on or after the 1st day of January, 1912; or (ii) in India or elsewhere, in any other document:

Provided that the ground specified in sub-clause (ii) shall not be available where such publication does not constitute an anticipation of the invention by virtue of sub-section (2) or subsection (3) of section 29;

Attention is further being drawn towards the relevant excerpts from Manual of

Patent Office Practice and Procedure published by The Office of Controller General of Patents, Designs & Trademarks, 2019, at page no. 93, that prescribes –

Novelty

1) An invention is considered as new if it is not anticipated by prior publication in patent and non- patent literature, prior use or prior public knowledge.

An invention is new (novel) if it has not been disclosed in the prior art, where the prior art means everything that has been published, presented or otherwise disclosed to the public before the date of filing of complete specification.

Based on submission made supra, it is clearly evident that the necessary pre-requisite for a document to be considered as a prior-art is that it should be published, presented or otherwise disclosed to the public before the date of filing (priority date) of complete specification.

The Applicant submits that, in the instant opposition the Opponent relied upon following documents:

Reference no. in instant Opposition	Article/Publication/Document	Publication Year/Date
(D1)	CN103820781A dated 12/03/2014	Not provided

Reference no. in instant Opposition	Article/Publication/Document	Publication Year/Date
(D2)	CN105148816A dated 08/09/2015	Not provided

Reference no. in instant Opposition	Article/Publication/Document	Publication Year/Date

(D3)	CN206372825U dated 12/01/2017	Not provided

Based on submission made supra, it is clearly evident that the necessary pre-requisite for a document to be considered as a prior-art is that it should be published, presented or otherwise disclosed to the public before the date of filing (priority date) of complete specification.

The Applicant submit that –

The "MANUAL OF PATENT OFFICE PRACTICE AND PROCEDURE" at page 83 reads as "The prior art should disclose the invention either in explicit or implicit manner.

• The purported prior-art document D1, (page 93 of opponent document) in the instant opposition discloses utility model, which is a profiled outer glass-lined device, which mainly comprises a can lid and a can body, wherein the can lid is provided with a nozzle, a manhole and a sealing mouth; the outer part of the can body is covered with a jacket, the can body and the clip The sleeve is welded into a unitary structure with a cavity there between, a jacket flange is provided on one side of the tank body, and a discharge port, a condensate outlet, a steam inlet and a steam outlet are arranged at a lower portion; The can lid surface has a stainless steel outer layer, and the stainless steel outer layer upper can lid head surface and the can lid flange surface together form a can lid stainless steel outer layer; The outer side of the outer side of the can body and the outer side of the jacket have a stainless steel outer layer, and the upper part of the stainless steel outer layer of the outer side of the can body is covered with a can body flange surface and a can body surface I stainless steel outer layer, and the can body and the jacket welding portion There is a canned transition surface I stainless steel outer layer, the outer side of the jacket is covered with a can body surface II stainless steel outer layer, and the lower part of the can body is covered with a stainless steel outer layer of the can body sealing surface, and the lower part of the can body and the jacket welding portion are covered The can body transition surface II

stainless steel outer layer, the bottom of the can body is covered with a can body surface III stainless steel outer layer; The nozzle, the manhole, the sealing port and the discharge opening are covered with a stainless steel outer cladding composed of a flange high neck surface and a flange surface; The stainless steel outer cladding is a welded joint; The insulating material is filled between the jacket and the stainless steel outer layer to form an insulating layer. The document is a prior-art in the eyes of law, as it discuss merely SS cladding on tank body/cover for insulation, not nozzles or clean room features, and hence, for this reason alone, document cannot be taken on record.

The purported cited document "D1", which have been provided by the opponent, without verified translation, does not disclose glass lining inside nozzles or SS welding onto machined MS. The present invention addresses long-felt needs in the cleanroom manufacturing industry, especially pharmaceuticals and food processing sectors. The integration of machined fluidic channels, weld-bonded stainless-steel overlays, and a novel bottom cavity structure provides improved heat transfer, cleanability, and corrosion resistance is not addressed in the recital of D1.

The purported cited document "D2", which have been provided by the opponent, without verified translation, discloses manufacturing method of a glass-lined reaction kettle, and belongs to the technical field of processing, which solves the problems that the outer surface of the glass-lined reaction kettle is poor in corrosion resistance. The glass-lined reaction kettle comprises an upper end enclosure, a barrel and a lower end enclosure. The manufacturing method comprises seven steps of assembling, manufacturing a wrapper, perforating, fixing, packing, polishing and manufacturing and fixing a cover. The manufactured glass-lined reaction kettle has advantages of good corrosion resistance, high safety, attractive appearance and the like. The teaching of D2 is different from that of the impugned application. No method of manufacturing has been claimed in the impugned invention. D2 describes a

multi-step wrapping technique using foreskins; SS sheath is not welded after MS removal; insulation used between SS and core body; prone to gap formation and corrosion; lacks bottom heat plate design as well.

The purported cited document “D3”, which have been provided by the opponent, without verified translation, discloses a corrosion -resistant glass reactor that wards off of high strength, including pressing from both sides cover, jar body, agitator, upper cover, low head, sight glass manhole, the thermometer mouth of pipe, feed inlet, drain hole, the external side of jar is equipped with press from both sides the cover, it is equipped with a plurality of installation positioner to press from both sides angles such as overlapping the outside, pressing from both sides cover upper portion and being equipped with steam inlet, steam inlet one side is equipped with the relief port, press from both sides the cover bottom and is equipped with water export with fixed attention, the welding has high -neck flange on the feed inlet, the welding has high -neck flange on the sight glass manhole, and the welding has high -neck flange on the thermometer mouth of pipe, and the welding has high -neck flange on the drain hole, be equipped with the ground coat on the internal lateral wall of jar, be equipped with cover glaze layer on the ground coat, be equipped with the ground coat on high -neck flange inside wall, terminal surface, the recess, be equipped with cover glaze layer on the ground coat. The utility model discloses trompil department welding has increased high -neck flange, has reduced the quantity at stamping forming trompil R position, has directly increased the effective thickness of this equipment, and compressive strength improves relatively, ensures the safe operation of equipment. The two documents D2 and D3 and Impugned application possess entirely different teachings altogether. The document D3 only concerns corrosion-resistant reactors, but fails to teach any of the structural innovations in the claims; no mention of overlay methods or nozzle integration.

Based on detailed submissions supra, it is evident that the Opponent failed to raise a credible challenge as to lack of Novelty, at least, for the reason that no

evidence has been led thereby. The Applicant submits that, this conduct of the Opponent makes it crystal clear that either the Opponent lacks even rudimentary knowledge of the prevailing patent laws or is raising such baseless grounds fraudulently and with mala-fide intention solely to cause delay in grant of the instant patent Application for the reasons best known to him, and based on this reason alone the instant Opposition should be dismissed in- limine.

Without prejudice to the above, the Applicant reserves its right to defend its application by filing any additional pleadings and/or evidence in future as it may consider necessary. The Applicant also reserves its rights to specifically respond/rebut to any argument/document as may be relied upon by the Opponent which already does not form part of the records of the Patent Office in the present matter.

COMPARISON OF SUBJECT MATTER AND CLAIMS :

Feature	Description in Impugned Application	Prior Art (D1, D2, D3)	Technical Significance	Opponent's Misunderstanding
Seamless SS Overlay on Nozzles	Seamless stainless steel overlay on nozzles, applied via proprietary welding (2 mm thickness, Exhibit A, page 41), to prevent oxidation	D1: SS cladding on tank body/cover for insulation, not nozzles (page 15, Opposition). D2: No SS overlay (page 27, Opposition). D3: High-neck flanges with glaze, not SS (page 33, Opposition)	Prevents rust in high-humidity clean rooms, ensuring GMP compliance and product purity (reduces contamination risk by 99.9%)	Opponent misinterprets SS overlay as equivalent to D1's cladding, ignoring its clean room-specific application.
Fluidic Channels	Channels within	D1–D3: No disclosure of	Enhances heat transfer	Opponent fails to recognize

	double-walled vessel for optimized heat transfer (Exhibit A, page 42)	fluidic channels	efficiency by 15–20%, reducing processing time by 10–15 minutes per batch and energy costs by 10 kWh.	channels' role in clean room efficiency, focusing on irrelevant prior art.
Glass-Lined Flat Surface	Flat surface for real-time flow observation, glass-lined for corrosion resistance (Exhibit A, page 45).	D1–D3: No disclosure of glass-lined flat surface for flow observation	Reduces process errors by 10%, ensuring consistent quality in pharmaceutical production	Opponent overlooks the surface's monitoring function, misinterpreting it as a standard feature.
Cavity in Bottom Flange	Cavity for heat transfer fluid circulation, integrated with fluidic channels (Exhibit A, page 42).	D1–D3: No disclosure of bottom flange cavity.	Improves thermal efficiency by 15%, minimizing energy consumption in clean room operations.	Opponent ignores the cavity's role in heat transfer, citing unrelated flange designs in D3.
Double-Walled MS Vessel	Double-walled structure with internal glass lining for structural integrity and clean room compatibility (Exhibit A, page 40).	D1–D3: No specific disclosure of double-walled MS vessel for clean room use.	Extends reactor lifespan by 20–30%, reducing maintenance costs by INR 5–10 lakhs annually.	Opponent fails to appreciate the vessel's clean room-specific design, citing generic reactor structures.

The opponent, who allegedly calls himself an interested person and a person skilled in art, with mischievous and malafide intent for diverting and delaying the process of grant filed the impugned pre-grant opposition proceedings by relying on documents not valid in the present proceedings as mentioned above. While adjudicating novelty, it is also well known that prior art should

disclose the invention either in explicit or implicit manner. Mosaicing of prior art documents is not allowed in determination of novelty.

Section 25(1)(e): Lacking inventive step:
The contents of the said paragraphs are denied for being baseless, frivolous and without any valid foundation of law. It is a settled position in law that in order to challenge the validity of a patent/patent Application on the ground of lack of inventive step/obviousness, it is necessary to prove that the document(s) relied upon was/were published before the earliest priority date of the complete specification in India or elsewhere in the world or fallen in public domain.

Claim-by-Claim Comparison with Prior Art
Claim 1: A clean room glass lined reactor comprising a double-walled MS vessel with internal glass lining, fluidic channels, a glass-lined flat surface for flow observation, a cavity in the bottom flange for heat transfer fluid, and seamless SS overlay on nozzles.

D1: SS cladding for insulation, not nozzles or clean room features (page 15, Opposition). Lacks fluidic channels, glass-lined flat surface, or cavity.

D2: Manufacturing method, not structural features like SS overlay or fluidic channels (page 27, Opposition).

D3: High-neck flanges for strength, not SS overlays or clean room features (page 33, Opposition).

Claim 2: SS overlay thickness of 2 mm.
D1–D3: No disclosure of SS overlay thickness.

Claim 3: Seamless welding of SS overlay.
D1–D3: No disclosure of SS overlay welding.

Claims 4–8: Specify dimensions, materials, capacities. D1–D3 do not disclose these features, focusing on unrelated aspects.

The present invention addresses long-felt needs in the cleanroom manufacturing industry, especially pharmaceuticals and food processing sectors. The technical distinction includes integration of machined fluidic channels, weld-bonded stainless-steel overlays, and a novel bottom cavity structure provides improved heat transfer, cleanability, and corrosion resistance. Further, by retaining mild steel as the core material while externally presenting a stainless steel overlay, the reactor delivers cost efficiencies of MS with the regulatory compliance of SS illustrates the economic significance. Unlike reactors that use paint or glue-based sheaths, the invention eliminates rust, reduces maintenance costs, and avoids surface chipping or dust accumulation. These outcomes are neither disclosed nor obvious from any of D1, D2, or D3, even if certified translations were available. Therefore, none of the cited documents, singly or in combination, teach or suggest the reactor construction as disclosed and claimed. Further, their machine translations, even if accepted, would fail to anticipate or render the claimed features obvious.

Reference is being made to; Section 25(1)(b) and 25(1)(e) of the Patents Act –

"*25. Opposition to the patent.—(1) Where an application for a patent has been published but a patent has not been granted, any person may, in writing, represent by way of opposition to the Controller against the grant of patent on the ground—*
(b) that the invention so far as claimed in any claim of the complete specification has been published before the priority date of the claim— (i) in any specification filed in pursuance of an application for a patent made in India on or after the 1st day of January, 1912; or (ii) in India or elsewhere, in any other document:
Provided that the ground specified in sub-clause (ii) shall not be available where

such publication does not constitute an anticipation of the invention by virtue of sub-section (2) or subsection (3) of section 29;
(e) that the invention so far as claimed in any claim of the complete specification is obvious and clearly does not involve any inventive step, having regard to the matter published as mentioned in clause (b) or having regard to what was used in India before the priority date of the applicant's claim;
Based on submission made supra, it is clearly evident that the necessary pre-requisite for a document to be considered as a prior-art is that – the document relied upon to challenge validity of claim(s) must be published, presented or otherwise disclosed to the public before the earliest priority date of the complete specification.

The present invention pertains to a clean-room compatible, corrosion-resistant glass-lined reactor with improved exterior surface treatment. It focuses on engineering solutions that meet the stringent requirements of cleanliness, durability, and functionality necessary for pharmaceutical, chemical, and food-grade processing environments. The invention employs a novel approach to providing a mirror-polished stainless steel (SS) overlay on mild steel (MS) vessels by carefully removing the MS substrate before overlaying with SS weld, ensuring uniform thickness, corrosion resistance, and long-term stability.

In evaluating its inventiveness, the claimed features are analyzed in comparison to prior art documents D1 (CN103820781A), D2 (CN105148816A), and D3 (CN206372825U).

D1 discloses a jacketed glass-lined reactor wherein the outer surface is equipped with stainless steel profiling cladding fixed by welding or bolts. Its primary goal is to meet the appearance and surface finish requirements stipulated under GMP norms, while maintaining resistance to corrosion and enhancing insulation via internal filling. However, the nature of D1's construction is largely superficial—it provides a stainless steel cover or foreskin externally over an MS reactor body without altering the structure of

the base metal. This approach, while aesthetically beneficial, does not structurally integrate the SS with the MS base nor does it cater to maintaining mechanical uniformity or pressure resistance post-application.

In contrast, the present invention does not apply a cladding sheet externally. Instead, it uses a highly engineered method wherein a 2 mm thick portion of the mild steel is precisely removed from the outer surface and then replaced with a weld-deposited stainless steel layer. This ensures that the overall wall thickness remains consistent throughout the reactor body, preserving both mechanical strength and dimensional stability. Unlike D1, which is confined to providing cosmetic enhancement, the current invention enables better heat dissipation, non-contaminating surfaces for clean-room conditions, and easier maintenance due to the welded overlay's mechanical permanence. D1 does not discuss any such overlay method, nor does it teach or suggest the necessity of material removal before overlay, thereby lacking the technical sophistication of the present approach.

D2 focuses on the use of a sheath-type stainless steel foreskin structure covering the upper and lower heads and barrel of a reactor. This foreskin is fabricated from pre-formed stainless steel sheets and mounted externally over the carbon steel reactor shell. The assembly involves welding the cladding sheets together and leaving a hollow gap which is typically filled with insulation material. While this concept of D2 appears similar to D1 in objective, D2 further emphasizes that the foreskin acts as a detachable or externally welded unit rather than an integrally bonded layer with the MS substrate.

The present invention sharply deviates from this concept by providing a weld-bonded SS overlay that forms a metallurgical bond with the underlying MS body after a precise removal of the MS surface. There is neither hollow gap nor any insulation in between; instead, the SS overlay is directly responsible for forming the external boundary of the reactor. This contributes to superior heat

transfer, mechanical stability, and corrosion protection. Unlike D2, which does not offer insight into ensuring uniform wall thickness or weld metallurgical bonding, the present invention integrates the overlay into the structural strength of the reactor and even applies the same methodology to the reactor's nozzles, thermowells, and bottom dish—something not addressed in D2. Furthermore, D2 fails to address the use of the external surface for flow observation purposes using glass-lining, which is a central innovation of the present disclosure.

D3 relates to a high-strength corrosion-resistant glass-lined reactor, with particular emphasis on flange configurations and enamel glaze composition to improve corrosion resistance and pressure holding capacity. D3 discloses features such as welded high-neck flanges, gasket sealing, and improved bonding of the bottom glaze layer with the base metal to combat issues such as fish scale porcelain formation caused by hydrogen saturation. The technical narrative in D3 is inward-looking, focusing primarily on enhancements to the interior surface and structural integrity of flanges, thermometer ports, manholes, and feed openings.

The present invention diverges from this approach by addressing an entirely different problem space—that of the external cleanliness, uniformity, and functionality of the reactor's outer surface. While D3 does not concern itself with the visual finish, dust accumulation, or clean-room compatibility of the reactor's external body, the current invention does so by ensuring that the entire outer surface is a mirror-polished stainless steel layer integrally welded and finished to prevent microbial growth, dust retention, and corrosion. Moreover, D3 lacks any teaching or suggestion regarding the removal of MS followed by SS overlay for maintaining uniformity in wall thickness or providing visual observation capabilities via glass-lining. Hence, the two inventions operate on different planes—D3 on internal corrosion resistance via chemical composition, and the present invention on external engineering design with aesthetic and operational significance.

Even when considering D1, D2, and D3 in combination, there is no enabling disclosure or reasonable motivation that would lead a skilled artisan to the construction techniques and objectives of the present invention. While D1 and D2 superficially discuss stainless steel covers, they do not address nor suggest removing any MS substrate before overlay, nor do they explore the importance of maintaining structural uniformity during the overlay process. Furthermore, these references are silent about extending the overlay to nozzles and thermowells, which are often points of contamination and difficult cleaning in conventional designs.

Similarly, while D3 provides a comprehensive discussion on high-pressure resistance and corrosion protection via enamel and flange configuration, it is entirely silent on surface overlay methods or clean-room surface optimization. The teachings of D3 would not lead one to modify the outer surface in the manner disclosed by the present invention, nor would it provide motivation to do so for the purpose of achieving GMP/clean-room standards in combination with high corrosion resistance and durability.

Therefore, the technical combination of D1–D3 fails to render the present invention obvious. The specific fabrication steps (material removal followed by welding of 2mm SS overlay), the coverage of complex geometries such as nozzles and thermowells, and the focus on external finish quality and clean-room suitability present a synergistic advancement that is neither taught nor suggested in the prior art.

In conclusion, the present invention achieves a significant advancement over existing technologies by introducing a precisely engineered method of achieving a stainless steel surface over a mild steel reactor body through mechanical removal and weld overlay. This not only maintains structural integrity and dimensional uniformity but also enhances corrosion resistance and clean-room compatibility. None of the cited prior art references either in

isolation or in combination anticipates or motivates this integrated approach. Consequently, the present invention is both novel and non-obvious, meriting recognition under the relevant provisions of patentability. Thus, the arguments of the opponent challenging the inventive step of the claims of the instant patent application deserve to be dismissed at its threshold.

It is further submitted that the inventive step determination is to be done in accordance with the test laid down by the Hon‟ble Delhi High Court in Roche v. Cipla. The opponent has failed to show any pleadings based on the above principles to show lack of inventive step. Thus, on this ground alone, the challenge under Section 25(1)(e) on lack of inventive step ought to be dismissed in-limine.

Needless to mention, the opponent has failed to define a person skilled in the art for the purpose of judging the inventive step of the claimed invention. It is a settled law that to analyze the inventive step of any claim of a patent application, one has to first define a person skilled in the art, and based on this reason alone, the instant opposition relying on the ground of lack of inventive step must fail.

The Applicant reserves the right to further substantiate the above submissions in the course of the oral hearing, if required, through additional arguments, documents and case- laws.

Section 25(1)(f): Not an invention:

The contents of Paragraphs under reply pertain to the Opponent's submissions under Section 25(1)(f) of the Act and the same are denied as being incorrect, vague and unsubstantiated.

Section 3(f) of the Act:

The contents of Paragraphs under reply pertain to the Opponent‟s submissions under Section 25(1)(f) of the Act and the same are denied as being incorrect, vague and unsubstantiated. It is submitted that Section

3(f) of the Act are not attracted in this case and the paragraphs under reply are liable to be summarily rejected as the Opponent has failed to establish as to how the claimed invention does not constitute an invention under Section 3 of the Patents Act 1970.

Patentability under Section 3(f) of the Act:

The Opponent‟s contention in the paragraph under reply that the subject matter of claims of the instant patent Application is not patentable under Section 3(f) of the Act is vehemently denied being baseless and frivolous in nature.

Opponent has challenged the validity of the instant patent Application on the ground of 3(f) of the Patents Act, 1970 by breaking down the plurality of features from the specification and referred documents D1 to D3 alleging mere arrangement/re-arrangement/duplication of known components, functioning in a known way in the instant Opposition.

FEATURE / CLAIM	IMPUGNED APPLICATION	D1	D2	D3
1 (i) – Internally glass-lined reactor sandwiched between top & bottom flanges	p. 44, Claim 1 lines 1-6	Jackets & flanges described – but only for cladding (p. 91)	Upper/lower head and barrel of glass-lined reactor, detachable (p. 108)	Glass-lined reactor body with upper & lower heads (p. 126-128)
1 (i) – *Fluidic channels machined on glass-lined reactor*	p. 44, Claim 1 lines 6-8	**Not disclosed**	**Not disclosed**	**Not disclosed**
1 (ii) – Glass-lined flat surface below top flange for flow observation	p. 44, Claim 1 lines 8-10	Not mentioned	Not mentioned	Not mentioned
1 (iii) – Bottom flange cavity for heat-transfer fluid; bottom plate can touch fluid	p. 44, Claim 1 lines 10-15	Conventional jacket only; no direct-contact bottom plate (p. 91)	Interlayer with insulation; no direct contact (p. 110)	Steam/liquid jacket; no bottom cavity plate (p. 139-141)
1 (iv) – Connection plate joining inlet & outlet to bottom flange	p. 44, Claim 1 lines 15-17	Not disclosed	Not disclosed	Not disclosed
1 (v) – Multiple bolts clamping top & bottom flanges	p. 44, Claim 1 lines 17-19	Bolt joints shown for cladding but not for MS-SS structure (p. 92)	Bolted foreskin flanges (step d, p. 112)	Numerous high-neck bolted flanges (p. 131-133)
Claim 2 – MS base, SS reaction chamber thickness > 1 mm (pref. 2 mm)	p. 40, lines 24-33	SS outer layer 1.5-3 mm but on the outside, not the chamber (p. 92)	Stainless-steel *sheath*, thickness not fixed for reaction chamber (p. 108)	Inner glazing focus; no SS reaction chamber (p. 126)
Claim 3 – Entire external surface welded with SS electrodes after removing ~2 mm MS	p. 41, lines 1-10	SS plates added; MS not removed (p. 91)	Foreskin wrapped; no MS removal (p. 110)	No overlay; only enamel & flanges (p. 129-133)
Claim 4 – Nozzles/flanges internally glass-lined while SS-overlaid outside	p. 41-42, lines 11-22	Clad only around external nozzle region; no internal glass lining discussed (p. 91-92)	Stainless flange sleeves; gap issues (p. 110-111)	High-neck flanges enamelled; but no SS overlay (p. 130-13

As stated earlier above, D1 – D3 document is totally irrelevant to the object of the invention and the solution disclosed thereupon and therefore liable to be strike out from the reference or citations.

Therefore, it is submitted that the allegations made by the opponent with regard to the patentability of the claimed subject matter are not only bald but are without any application of mind and hence, the opposition is liable to be dismissed on the said ground alone.

Patentability under Section 3(d) of the Act:

The Opponent"s contention in the paragraph under reply that the subject matter of claims of the instant patent Application is not patentable under Section 3(d) of the Act is vehemently denied being baseless and frivolous in nature.

Opponent has challenged the validity of the instant patent Application on the ground of 3(d) of the Patents Act, 1970 by breaking down the plurality of features from the specification and referred documents D1 to D3 alleging mere use of known product in the instant Opposition.

Feature	**Present Invention**	**D1**	**D2**	**D3**
Construction Base	Mild Steel (MS) with glass lining and SS overlay	Carbon steel / MS with glass lining and external cladding	Carbon steel with detachable SS foreskin	Carbon steel / MS with focus on flange design and enamel coating
SS Application Type	SS welded overlay (after removing MS surface)	SS cladding, welded or bolted externally	SS foreskin – external sheath with insulation	No external SS overlay – only inner coating and high-neck flanges
MS Removal Prior to SS Application	Yes, approx. 2mm removed to ensure thickness uniformity	No	No	No
Glass Lining	Internal, 1–2 mm, sprayed	Yes	Yes	Yes

Overlay of SS on Nozzles and Thermowells	Yes, welded and formed structurally	No	No	No
Clean-Room Suitability / Mirror Polish	Yes, mirror finish; dust-repellent; visually clean	Hygiene discussed, no mirror finish	Attractive appearance mentioned, but not mirror-polished	No mention
Heat Transfer Mechanism (Bottom Plate & Flow)	Bottom cavity with direct heat transfer fluid contact	Jacketed, no flow cavity	Interlayer with insulation	Jacketed; flow-through steam and liquid heat exchange
Visual Flow Observation Feature	Yes, glass-lined metal flat plate under top flange	No	No	No
Weld Integrity for SS Bonding	Full structural weld; continuous overlay	Spot welding of cladding sheets	Welded foreskin only; no metallurgical bond to MS	Bolt-based sealing, no overlay welding
Ease of Maintenance	High – no paint; less corrosion; easy inspection and cleaning	Moderate – paint peeling possible	Moderate	Internal focus only
Integration with Pressure Vessel Norms (ASME, DIN)	Yes, retains mechanical strength while meeting visual/clean standards	Yes	Not specified	Yes
Intended Applications	Clean rooms, pharma, food, and chemical with high hygiene, corrosion resistance	Pharma, bio, chemical – basic hygiene requirement	Pharma – protective foreskin	Chemical – strength, enamel bonding focus

Therefore, it is submitted that the allegations made by the opponent with regard to the patentability of the claimed subject matter are not only bald but are without any application of mind and hence, the opposition is liable to be dismissed on the said ground alone.

It is submitted that the sole aim of the opponent in raising such frivolous grounds in the opposition is merely to cause delay in grant of a patent and to

cause immense prejudice to the Applicant to satisfy some ulterior motives best known to the Opponent. It is submitted that the opponent has filed a frivolous opposition and failed to elucidate how the ground for opposition as given under Section 25(1)(f) is applicable in the present case.

Thus, the present pre-grant opposition is liable to be dismissed with heavy exemplary costs.

The Applicant submits that the Applicant reserves the right to further substantiate the submissions with regards to ground of the opposition as given under Section 25(1)(f) in the course of the oral hearing, if required, through additional arguments, documents and case-laws.

Section 25(1)(g): Insufficiency & clarity:
At the outset, it is submitted that the contents of the paragraph under reply are denied for being baseless, frivolous and for want of evidence. Attention of the Ld. Controller is further being drawn towards the fact that most of documents cited in the instant Opposition in the ground under reply are published after the priority date of the complete specification and hence, none of these later published documents can be taken on record.

It is settled position in law, as enunciated by Justice Vimadalal in Farbewerke Hoechst AG and Ors. v. Unichem Laboratories and Ors. (AIR 1969 Bom 255), that –
(i) The specification and claims are addressed to those with a high degree of knowledge of the field of science to which they relate;
(ii) The description must be sufficient to enable those in the industry concerned to carry it into effect "without their making further inventions", and the description must not be difficult to follow;
(iii) It is not necessary to describe processes on the Claims to a specification when they are part of the common knowledge available to those skilled in the science who can, after reading them, refer to the technical literature on the

subject for the purpose of carrying them into effect; and

(iv) The party pleading insufficiency of disclosure ought to adduce the evidences showing that the description of invention as contained in a patent application is insufficient to enable those in the industry concerned to carry it into effect "without their making further inventions".

Refer to the operative part of the decision of Justice Vimadalal in the aforesaid case, as reproduced herein-below:

14. Dealing first with the ground of insufficiency of description it is stated in Halsbury, (3rd edn.) Vol. 29 p. 64 para 131 that the claim need only be as clear as the subject admits, and that a patentee need not so simplify his claim as to make it easy for infringers to evade it. It is further stated in that passage in Halsbury that the patentee's duty is not to prevent all possible argument as to whether there is or is not infringement in particular cases, but to enable the court to formulate the questions of fact to be answered. It is further stated in the same Volume of Halsbury (p. 66 para 138) that insufficiency of description has two branches, (1) the complete specification must describe "an embodiment" of the invention claimed in each of the claims and that the description must be sufficient to enable those in the industry concerned to carry it into effect "without their making further inventions"; and (2) that the description must be fair i.e. it must not be unnecessarily difficult to follow.

There is no evidence led by the defendants to show that the statement in the body of the specification that the synthesis of the desired sulphonylurea may be obtained by eliminating sulphur with a heavy metal oxide or a salt thereof in an aqueous or alcoholic solution would present any difficulty. The specification and claims are addressed to those with a high degree of knowledge of the field of science to which they relate, particularly when they relate to chemistry and allied subjects. It is not necessary to describe processes on the Claims to a specification when they are part of the common knowledge available to those skilled in the science who can, after reading them, refer to the technical literature on the subject for the purpose of carrying them into effect. "An

embodiment" of the invention is, therefore, in my opinion, sufficiently described in the plaintiffs patent and that description is not unnecessarily difficult to follow, it being sufficient to enable the invention to be carried into effect "without making further inventions".

Keeping in view, the settled position in patent law, it is patently clear that the averments of the Opponent in the paragraphs under reply are nothing but mere figments of the imagination, and has no substance, whatsoever.

Specifically, as can be seen from the decision of the High Court, the onus of adducing the evidence lies on the Opponent to prove that the specification of the instant patent Application cannot enable a person skilled in the pertinent art to practice the claimed invention, which the Opponent has miserably failed, so much so, the Opponent has chosen not to define the person skilled in the art, the pre-requisite to establish the ground of insufficiency, and based on this ground alone, the averments in the paragraphs under reply ought to be rejected at its threshold.

The complete specification of the present invention, titled "CLEAN ROOM GLASS LINED REACTOR", provides a comprehensive and enabling disclosure that satisfies the mandatory requirements of Section 10(4) of the Indian Patents Act, 1970. This section imposes statutory obligations on an applicant to disclose the invention in a manner sufficiently clear and complete for it to be performed by a person skilled in the art, and further mandates that the best mode known to the applicant of carrying out the invention must be included.

The specification begins with a detailed narrative of the background of the invention, which discusses the deficiencies in conventional glass-lined reactors, particularly with respect to clean-room compliance, corrosion-prone painted external surfaces, and the lack of integrated design features suitable for pharmaceutical and food-grade environments. It describes the inadequacy

of traditional reactors where the mild steel (MS) shell is either left painted or externally clad with stainless steel (SS) sheets. The resultant surface does not meet the hygiene or visual inspection standards required in high-grade processing facilities. This identification of the technical problem is expressed with clarity and precision and demonstrates the inventor understands of existing limitations in the art.

Following the background, the complete specification clearly elaborates on the core inventive concept. The invention pertains to a structurally integrated clean-room reactor that combines internal glass lining with a mirror-finished, corrosion-resistant stainless steel external surface. What distinguishes the invention from prior art is not merely the use of SS on MS, but the method by which the SS overlay is applied—namely, by first removing a uniform 2 mm thickness from the MS shell and then welding stainless steel electrodes to build up a structurally bonded SS layer. This method ensures that the overall thickness of the reactor vessel remains unaltered, thereby preserving mechanical integrity and pressure resistance, while simultaneously providing a clean, durable, and maintenance-friendly surface.

The description meticulously discloses the step-by-step process of manufacturing the reactor. It begins with the removal of MS material, either mechanically or chemically, and proceeds to the SS welding step, followed by grinding and polishing. The method is not described in abstract or conceptual terms but is technically supported with specific material thicknesses, preferred overlay configurations, and suitable welding techniques. Moreover, the specification goes on to describe additional embodiments where the SS overlay is extended to the nozzles and thermowells, which are known high-risk points for microbial accumulation and contamination in traditional vessels. Such extensions of the overlay are critically important for ensuring uniform cleanliness and durability across the entire reactor body, and their inclusion evidences a deep understanding of the operational realities in clean-room equipment.

In compliance with Section 10(4)(b), the best mode known to the applicant for carrying out the invention has also been disclosed. The specification does not merely describe the structural features of the reactor but explains the preferred ranges for the SS overlay (approximately 2 mm), the internal glass lining thickness (1 to 2 mm), and the recommended welding and finishing techniques to achieve a mirror polish suitable for clean-room applications. The bottom flange cavity design is detailed to facilitate efficient heat exchange, and the inclusion of a glass-lined observation plate beneath the top flange allows visual monitoring of fluid dynamics within the vessel—further showcasing the innovation's practical relevance and functional integration. None of these features are speculative or conceptual; instead, they are implementable with existing industrial techniques and known fabrication tools.

Additionally, the claims are supported by the description in both breadth and depth. Claim 1 outlines the fundamental structural configuration of the reactor, while dependent claims elaborate on specific technical enhancements such as the SS overlay thickness, method of overlay application, and nozzle treatment. These claims are fully enabled by the teaching in the description, and a person skilled in the art, such as a mechanical or process engineer working in the field of pressure vessels or reactor fabrication, would be able to perform the invention without undue experimentation. The manufacturing steps rely on standard operations such as machining, welding, and polishing—all of which are well within the capabilities of professionals in this domain. Furthermore, the figures provided in the specification complement the textual disclosure by offering visual guidance on the structural relationships between components, facilitating an even more effective understanding of the invention.

The Indian judiciary has repeatedly clarified the scope of enablement under Section 10(4). In Enercon India Ltd. v. Aloys Wobben and other similar cases, it has been held that a specification is enabling if it provides sufficient

guidance for a skilled person to perform the invention without exercising inventive ingenuity or engaging in extensive trial and error. The present specification comfortably satisfies this benchmark. It defines the invention with clarity, provides technical details on implementation, includes the best mode known to the applicant, and supports all claims with sufficient descriptive basis.

In conclusion, the complete specification of the present invention demonstrates full compliance with Section 10(4) of the Patents Act. It furnishes a clear, complete, and sufficiently detailed disclosure that allows a skilled person in the relevant field to make and use the invention without ambiguity or undue burden. The invention is not merely theoretical; it is workable, enabled, and industrially applicable, as evidenced by the completeness of the specification and its alignment with established fabrication techniques. The claims are properly supported and clearly define the scope of protection sought. Therefore, observation of the opponent is unfounded and merit withdrawal.

Accordingly, in view of submission supra, the averments of the Opponent, with regards the ground under reply, are liable to be dismissed for being false and completely baseless, and for the want of evidence. The Applicant reserves the right to further substantiate each of the individual issues raised in the paragraphs under reply, if required, with aid of further case laws and/or documents and/or evidence(s).

The Applicant reserves the right to further substantiate this submission in the course of the oral hearing, if required, through additional arguments, documents and case-laws.

In view of the detailed submissions made as part of the present Reply Statement, the Applicant urges the Ld. Controller to summarily reject the instant frivolous representation filed by the Opponent.

It is submitted that the opponent is not entitled to the reliefs prayed for in the

prayer clause of the opposition. The opponent cannot be permitted to file further evidence, as the same would be against the principles of natural justice. It is further submitted that the opponent has filed a frivolous opposition with the sole intent to harass the Applicant.

PRAYER

In view of the submissions made above, we respectfully pray that:

a) the instant opposition be dismissed in-limine;

b) the Applicant be allowed to file further documents as evidence, if necessary to support their averments;

c) the cost including the compensatory costs be awarded in the favor of the Applicant;

d) an opportunity of being heard be provided to the Applicant; and

e) such other relief(s) be granted as the Ld. Controller may deem appropriate in the interest of justice.

For and on behalf of Standard Glass Lining Technology Private Limited

By its Agent



[Bharat Daswani]
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Email: [kolkata@daswaniindia.net]
Date: August 04, 2025

FORM - 2

THE PATENTS ACT, 1970
(39 of 1970)
&
THE PATENT RULES, 2003

Complete specification
[See Section 10; Rule 13]

1. TITLE OF THE INVENTION

CLEAN ROOM GLASS LINED REACTOR

2.NAME OF APPLICANTS:

1) Standard Glass Lining Technology Private Limited a Indian company of D.12, PHASE I, IDA, JEEDIMETLA HYDERABAD Hyderabad Telangna 500055 INDIA

3. PREAMBLE TO THE DESCRIPTION:

The following specification particularly describes the nature of the invention and the manner in which it is to be performed :

FIELD OF INVENTION

The present invention relates to corrosion resistant, clean room glass lined overlaid stainless steel reactor. More particularly, the present invention discloses a reactor specifically designed for clean and polished working environment in food and medical industries, which don't require any paint coat the look clean as mirror finished. These reactors help to keep the production floor and its surroundings less dusty and prevent contamination of the product with enhanced heat transfer efficiency.

The present inventions also presents Mild Steel (MS) Glass-Lined reactors with stainless steel overlays on nozzles, designed for sectors like pharmaceuticals, food, chemicals, and agrochemicals. It aims to elevate glass-lined reactors for cleanroom use, ensuring high cleanliness standards, less maintenance, and rust prevention. By integrating stainless steel overlays onto reactor nozzles with SS cladding around the vessel, the solution covers 100% of the reactor, offering durability for industries requiring stringent hygiene. Stainless steel enhances corrosion resistance, which is vital in industries where rust could compromise product quality.

BACKGROUND ART

Conventional reactors are widely used by fine chemicals and pharmaceutical industry. Corrosion, cleaning and contamination are one of the key criteria that determine the efficiency of a process, which in turn depends on the type of equipment used. Stirred tank reactors (STRS) used in pharmaceutical and fine chemical industries are designed for providing and removing heat. This is done with coil or half coil with coolant passing through it, or using jacketed tanks.

Standard glass lining technology wants to make reactors for easy maintenance and suitable for clean environments. Standard glass has designed and modified its existing large variety of reactors to make them well suited for the clean room and exclusive process areas.

Glass Lined Reactors are generally designed as per ASME SECTION-8 DIVISION-I, Dimensions and capacities are as per DIN-28136 with Mild Steel Material with Glass Coating on the Inner side of the reactor. They can be of AE Type or CE Type and both types can be made as clean room reactors.

The Reactor's Mild Steel outer surface is also prone to corrosion by the spilled material during charging the reactor, Over a period of time there is a high probability that the corrosive particles contaminate the charge thus causing rejections of the material and also costly Regulatory Compliance Issues.Chemical resistant paints are used to protect the reactor outer surface from exposing it to harsh chemicals and acidic substances. Special procedures and paint materials are used to coat the reactor parts. The paint will also protect the surface from getting rust so that, the rust will not convert to dust or debris around the reactor. Also, the paint itself may peel off after certain time in its lifetime which again form as dirt, paint is prone to damage for impacts cause by any tools or objects. So, taking care of the paint itself a big task and any damage to paint will create dirt and allows the material to expose and material to form rust over a period. It is also proven that Paint will attract dust. There are many possibilities for the traditional methods of

protecting surfaces (special paint) may produce dust, dirt and rust. The lifetime of paint is also less.

Glass-lined reactors play a vital role across diverse industry sectors that grapple with corrosive applications. The ingenious use of glass lining as a protective barrier on mild steel plates within these reactors renders them impervious to the corrosive agents present in their operational environment. This unique feature not only enhances the longevity of the reactors but also ensures the integrity of the substances they contain.

In particularly stringent settings, such as clean rooms where stringent safety and hygiene protocols are paramount, glass-lined reactors find special application. These controlled environments demand absolute rust prevention due to its potential to compromise both the safety of the operation and the purity of the products being processed. To address this, these reactors are meticulously engineered with an inner layer of glass lining on mild steel. The outer layer, known as the jacket, serves as a conduit for heat transfer processes.

However, the challenge arises in safeguarding the outer surface of these reactors against rust formation. To counteract this, a sheet of stainless steel is welded as cladding over the jacket. This layer, in addition to insulating the reactor, acts as a barrier against rust development. Yet, it's crucial to note that this cladding has limitations—it only extends coverage up to the uppermost part of the reactor and the vessel itself. Regrettably, it leaves vulnerable areas such as the nozzles that are essential for ingress and egress.

Consequently, the nozzles are typically protected with specialized industrial paints. Over time, however, these paints can peel off due to various factors,

including temperature variations and mechanical stress. This leads to the exposure of the underlying mild steel surface, setting the stage for potential rust formation and degradation.

To counteract this, a routine maintenance strategy involving periodic inspection and repainting becomes necessary. While this approach ensures the structural integrity of the reactor and prevents rust-related issues, it introduces concerns in clean room environments. The possibility of paint particles contaminating the products being processed poses a significant risk, warranting stringent quality control measures and further amplifying maintenance challenges.
In essence, the usage of glass-lined reactors exemplifies a delicate balance between material science, engineering ingenuity, and meticulous maintenance practices. While these reactors provide indispensable utility in handling corrosive substances, their effective and sustainable operation demands a multifaceted approach that encompasses materials, design, and stringent maintenance protocols to ensure both durability and product purity.

SUMMARY OF INVENTION

The invention discloses mild steel Glass Lined Reactors overlaid with Stainless Steel material of approximately 2mm depth in some places and manufacturing directly with SS Parts in other places thereby giving the reactor Protection Corrosion, easy maintainability thereby extending the life of the Glass Lined reactor.

Therefore such as herein described there is provided an improved mild steel reactor comprising of an internally glass lined reactor sandwiched between a top flange and a bottom flange, wherein fluidic channels are

machined on said glass lined reactor, a glass lined metal flat surface below the top flange that allows to observe flow regime in the glass lined reactor or mixer, a cavity in the bottom flange that facilitates flow of heat transfer fluid through an inlet and an outlet nozzles and flanges, wherein said cavity can be closed with a bottom plate having a metal surface, wherein the metal surface of said bottom plate can touch the heat transfer fluid, a connection plate connecting the inlet and the outlet to the bottom flange of the glass lined stainless steel reactor, and multiple bolts to hold tightly the top flange and the bottom flange wherein the external surface of the said reactor is covered and overlaid with stainless steel sheet by welding.

In an embodiment the outer wall of the reactor consists of a base body of high mechanical strength, namely steel or stainless steel, the base body having a reaction chamber of stainless steel with a layer thickness > 1 mm, preferably in the range from essentially 1 mm to 5 mm and most preferably 2mm.

In another embodiment the said SS body is made by welding the entire surface with SS welding electrodes after removing proportionate metal from the external surface of the reactor and the said nozzles or flanges are made on or in the reactor are internally glass lined.

In the realm of clean room reactors, stainless-steel cladding prevents rust, except for vulnerable nozzles. To counter this, a solution involves depositing a layer of stainless steel onto the outer nozzle surface, followed by precise grinding for uniformity. This innovation forms a stainless-steel shield, permanently safeguarding against rust. This coated nozzle is then seamlessly welded to the top cover's cladding sheet, fully

enclosing the reactor in stainless steel. This dynamic approach harmonizes material science and engineering, tackling rust challenges in critical zones. The result is a resilient barrier that upholds reactor integrity, longevity, and product purity. This cutting-edge technique epitomizes the fusion of advanced technology and craftsmanship, underscoring a commitment to stringent clean room standards.

A glass-lined reactor offers excellent resistance to oxidation at the nozzle. In the event of an accidental spillage of corrosive liquid onto the nozzle surface, which might otherwise lead to oxidation on the metal surface, the glass lining which is coated inside would be damaged as the base metal is oxidated and would result in distortion. The claim asserts that the reactor is fully covered with stainless steel, enhancing its resilience against external damage and thereby increasing its overall safety.

BRIEF DESCRIPTION OF THE ACCOMPANYING DRAWINGS

[Fig.1] [Fig.2] illustrates a typical reactor in assembled state. Isometric view and basic front views.

[Fig.3] [Fig.4] [Fig.5] represents the detailed parts of the clean room reactors with differentiating line types for MS and SS. The component names are also described in the images.

Figure 1: Clean Room Reactor with SS cladding and nozzles covered with SS overlay.

Figure 2: Manhole cover with all SS components and SS overlay on the nozzles.

Figure 3: Nozzle with SS overlay and thermowell.

The invention will now be described, by way of examples only, with reference to the accompanying drawings.

Figure 1 depicts a Glass-lined reactor integrated with Stainless Steel (SS) cladding, showcasing pivotal components that contribute to its functionality and structural integrity. At the core of this assembly is the Bottom Vessel (1), which serves as the foundational chamber for chemical reactions. The Clamps (20) are strategically positioned to secure various components in place, ensuring stability and safety during operation. Positioned above the vessel is the Top Dish (16), which encapsulates the reactor's contents.

The reactor's drive mechanism is supported by an array of SS components, including the Pad Plate (8), Bottom Drive Stand (9), Bearing Housing (10), Top Drive Stand (11), Moto Flange (12), Motor Cover (13), and Mechanical Seals (14 and 7). The Name Plate (15) provides crucial identification and information about the reactor setup.

The reactor design also accommodates accessibility and monitoring. A Manhole Assembly (6) offers a portal for maintenance and inspections, while the Thermowell (5) and its associated Nozzle (4) facilitate temperature measurements and control. This is shown in **Figure 2.**

Figure 2 offers an intricate view of the SS cladding application, particularly on Nozzle (26). This SS cladding extends along the entire height of both the manhole and thermowell nozzles, ensuring corrosion resistance and structural integrity. Complementing this cladding are SS Clamps (27), the Manhole

Cover (24), and the Sight Glass Flange (22), all meticulously constructed from stainless steel.

The overarching design of the reactor involves the strategic integration of glass-lined and stainless steel components, striking a balance between the specialized properties of each material. The SS cladding provides robust protection against corrosive substances and external elements, while the glass lining offers superior chemical resistance for the reactor's interior. This synthesis of materials and components underscores the reactor's efficiency, durability, and operational safety within various industrial applications.

DESCRIPTION OF INVENTION

Standard glass technical team has changed the material of construction from MS to SS in the places wherever the reactors outer side is exposed to ambient light and the result is it looks like it is made of SS completely from outside but, the reactor core is same and efficient that of MS reactor since MS reactors are better in process efficiency than SS reactors in places where heating is the primary process requirement. Entire reactor is completely covered with SS materials. Few places the

components parts are made with SS, few components are completely wrapped with SS dome. The critical areas like nozzles and thermowells are overlaid with SS metal sheet by welding process.

In an exemplary embodiment, the said overlaying of SS is made by removing MS surface up to 2mm of depth and this 2mm is again covered with SS by welding the entire surface with SS welding electrodes. The

thickness is proportional such that the total thickness is maintained and is a constant. The removal of the top surface of the MS reactor can be mechanical, chemical and / or both. A person skilled in the art will be able to understand a plurality of method available for the removal of the tor surface. This welding surface then goes under grinding process to attain its original dimensions and original shape of the component.

MAIN COMPONENTS OF CLEAN ROOM REACTORS

[Fig.3] [Fig.4] [Fig.5] represents the detailed parts of the clean room reactors with differentiating line types for MS and SS. The component names are also described in the said images. The continues line represent the SS and dashed line represent the MS material. The internal surface is cladded with glass lining using spraying method.

Overlay also mentioned in the images wherever it is applicable. This over laying of SS will be subjected to change for each reactor model based on it number of nozzles and type of thermowell that is required for the model. The glass cladding is made in range of
1mm to 2mm for smooth operation and durability. The said glass lining is configured for a non-reactive cladding, pressure-welded coating and the surface of the glass lining / coating is smooth as whole. The base body is designed as a double-walled vessel.

The disclosed modified reactor has capacity ranging from 1Litre to 12500litre for AE type reactor and 1000Litre to 32000Litre for CE type

of reactor and heat transfer area ranging from 0.5 m^2 to 34.6 m^2. The said micro-reactor is configured to withstand pressure up to 6kg/cm^2 and temperature ranging from −60° to 343° C.

ADVANTAGES OF INVENTION

The disclosed clean room reactors come with many tangible advantages for the process and for the maintenance. Stainless steel gives many advantages. These rectors are placed at extreme working conditions which are typically high in temperature, exposed to harsh chemicals. At these conditions these reactors need to be protected from the damaging working conditions and retain its strength. Stainless steel strength to weight ratio is higher than other materials it can withstand more heat and stress compared to MS.

Clean room reactor eliminates the requirement of the painting of outer surface. Painting will attract the dust and over a time period the paint itself peal and becomes dust., so regular maintenance and check-ups are cleaning ups are required when painting comes to picture but clean room reactor doesn't require to paint on its surface. They are hygienic dust repellent and very easy to clean.

Clean rom reactors will be given mirror like finish to its surface thus makes it easy to remove dust or any liquid from its surface. It is as easy as cleaning a mirror. This clean and mirror like finish surface makes them look premium. This aesthetic appearance is attractive and gives advantage to identify any dirt or any waste particles on the surface. Easy to inspect the surface in maintenance exercise.

Stainless steel is highly corrosive resistant, because these reactors are employed in pharma and chemical processing industries, they are subjected to rigorous environments so, it must withstand corrosion from any medium. This is also a primary reason for developing clean room reactors.

Clean room reactors are built with SS on outside to resist scaling, scaling is common phenomenon with the substance used in chemical and pharma industries. Any chemical residue on the reactor surface will not only affect the reactor surface but can also cause problems with the underlying glass thereby reducing the usable life of the reactor. The scaling may affect the environment or labs where these equipment's are kept in operation. Clean room reactor will not have this problem because of appropriate use of SS material around it. Contamination will be lot less in SS when compared to MS.

Standard glass technology has designed and offering these clean room reactors to its customers to add value to their process by means of reducing the maintenance cost, helps maintaining the labs or the production floor clean, give the premium look to the equipment, withstand tough working conditions and increasing the life of the reactor.

The subject matter of embodiments of the present invention is described here with specificity to meet statutory requirements, but this description is not necessarily intended to limit the scope of the claims. The claimed subject matter may be embodied in other ways, may

include different elements or steps, and may be used in conjunction with other existing or future technologies. This description should not be interpreted as implying any particular order or arrangement among or between various steps or elements except when the order of individual steps or arrangement of elements is explicitly described.

We Claim :

1. A Mild Steel (MS) Glass-Lined reactors with stainless steel overlays on nozzles comprising of:

i. an internally glass lined reactor sandwiched between a top flange and a bottom flange, wherein fluidic channels are machined on said glass lined reactor.

ii. a glass lined metal flat surface below the top flange that allows to observe flow regime in the glass lined reactor or mixer.

iii. a cavity in the bottom flange that facilitates flow of heat transfer fluid through an inlet and an outlet nozzles and flanges, wherein said cavity can be closed with a bottom plate having a metal surface, wherein the metal surface of said bottom plate can touch the heat transfer fluid;

iv. a connection plate connecting the inlet and the outlet to the bottom flange of the glass lined stainless steel reactor; and

v. multiple bolts to hold tightly the top flange and the bottom flange; wherein the external surface of the said reactor is covered and overlaid with stainless steel sheet by welding.

wherein coated nozzle is seamlessly welded to the top cover's cladding sheet, fully enclosing the reactor in stainless steel tackling rust challenges in critical zones resulting a resilient barrier that upholds reactor integrity, longevity, and product purity

and Characterized in that

said reactor offers excellent resistance to oxidation at the nozzle and in the event of an accidental spillage of corrosive liquid onto the nozzle surface, which might otherwise lead to oxidation on the metal surface, the glass lining which is coated inside would be damaged as the base metal is oxidated and would result in distortion

and wherein the the reactor is fully covered with stainless steel, enhancing its resilience against external damage and thereby increasing its overall safety.

2. The reactor as claimed in claim 1, wherein the outer wall of the reactor consists of a base body of high mechanical strength, namely steel or stainless steel, the base body having a reaction chamber of stainless steel with a layer thickness > 1 mm, preferably in the range from essentially 1 mm to 5 mm and most preferably 2mm.

3. The reactor as claimed in claim 1, wherein the said SS body is made by welding the entire surface with SS welding electrodes after removing proportionate metal from the external surface of the reactor.

4. The reactor as claimed in claim 1, wherein the said nozzles or flanges made on or in the reactor are internally glass lined.

5. The reactor as claimed in claim 1, wherein the glass lining is configured for a non-reactive cladding, pressure-welded coating.

6. The reactor as claimed in claim 1, wherein the base body is designed as a double- walled vessel.

7. The reactor as claimed in claim 1, wherein the surface of the glass lining / coating is smooth as whole.

8. The reactor as claimed in claim 1, wherein said reactor has capacity ranging from 1Litre to 12500litre for AE type reactor and 1000Litre to 32000Litre for CE type of reactor and heat transfer area ranging from 0.5 m2 TO 34.6 m2.

Dated this 21st day of April 2021

Ranjan Paul

(Ranjan Paul)
Of Daswani & Daswani
Applicant's Agent

ABSTRACT

“Glass Lined Reactor with Full Cladding and Nozzle Overlay”

This invention introduces stainless steel overlays on glass-lined reactors, enhancing their functionality in cleanroom settings. By depositing the stainless steel layers onto vulnerable nozzle surfaces and seamlessly integrating them with the reactor's cladding, this innovation offers comprehensive rust prevention. This approach ensures reactor integrity, longevity, and product purity while meeting rigorous cleanroom standards. The combined use of glass-lined and stainless steel components provides a balanced solution for improved efficiency and durability across diverse industries.

Fig. 1

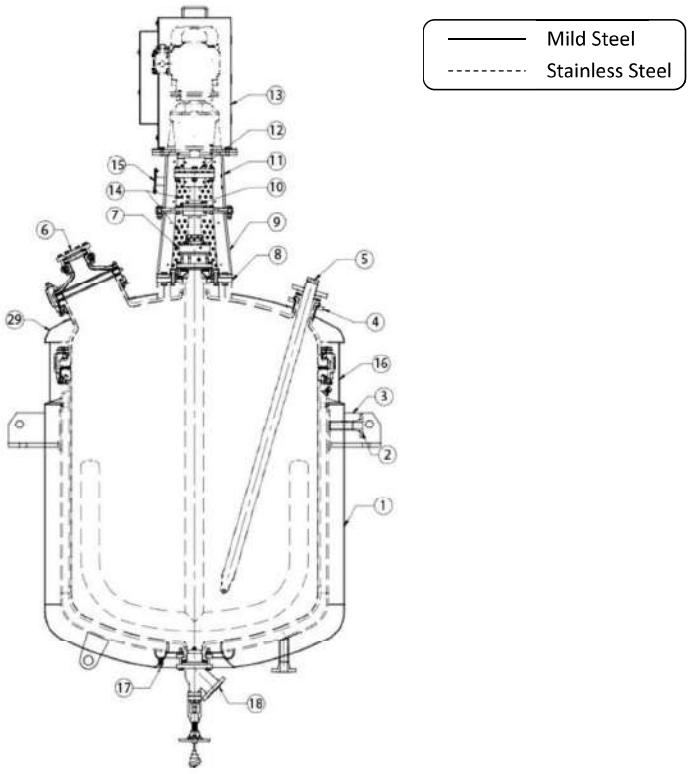


Figure 1

Ranjan Paul

(RANJAN PAUL)
Of DASWANI & DASWANI
APPLICANT'S AGENT

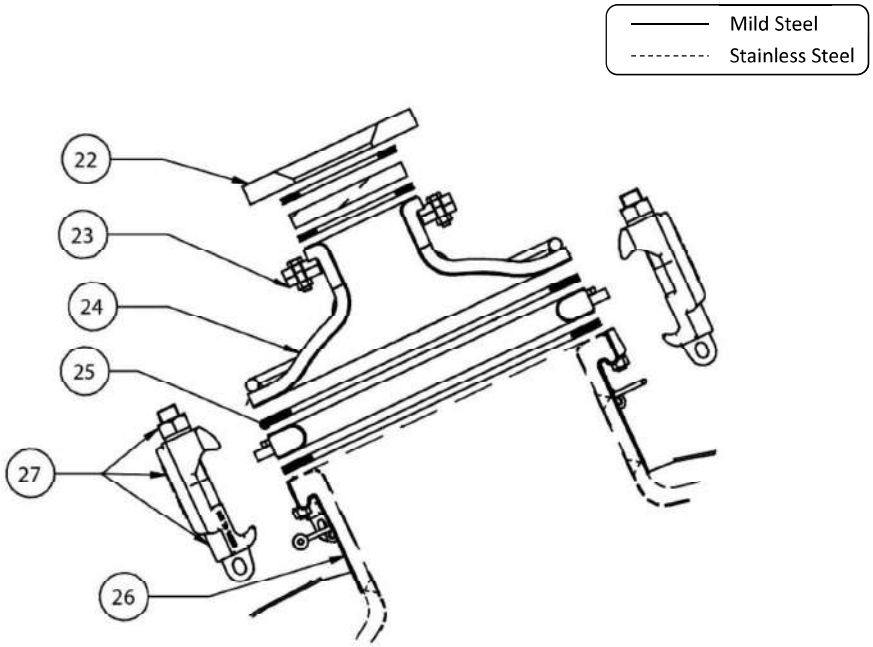


Figure 2

Ranjan Paul

(RANJAN PAUL)
Of DASWANI & DASWANI
APPLICANT'S AGENT

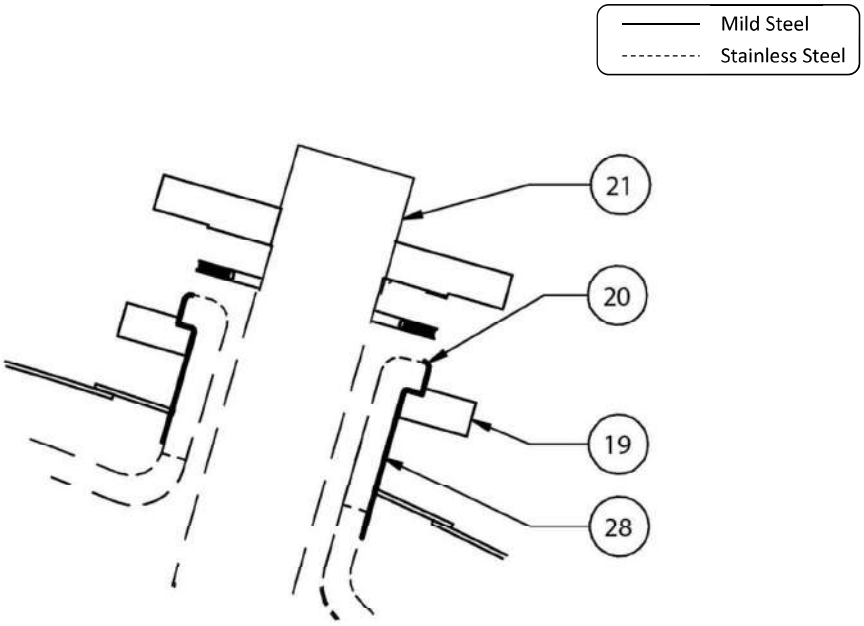


Figure 3

Ranjan Paul

(RANJAN PAUL)
Of DASWANI & DASWANI
APPLICANT'S AGENT

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सत्यमेव जयते
GOVERNMENT OF INDIA

INTELLECTUAL PROPERTY INDIA
PATENTS | DESIGNS | TRADE MARKS GEOGRAPHICAL INDICATIONS

संदर्भ सं. / Ref. No:
POC/Application No /202141018455

प्रेषण दिनांक / Date of Dispatch:
13-10-2025

सेवा मे, / To

आवेदक /Applicant:
Standard Glass Lining Technology Private Limited

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विषय: आवेदन संख्या 202141018455 के संदर्भ मे अनुदान-पूर्व सुनवाई नोटिस
Sub: Pre-Grant Hearing Notice in Reference of Application No. 202141018455

सुनवाई स्थल / Hearing Location: Through Video Conferencing
सुनवाई दिनांक व समय / Hearing Date & Time: 14/11/2025 / 11:30 HRS(IST) for (6h)
नियंत्रक ईमेल /Controller's EmailId: kanhaiya207.ipo@gov.in

आवेदन संख्या 202141018455 के अनुदान-पूर्व विरोध व उसके उत्तर मे दाखिल आवेदन के संदर्भ मे, इस मामले मे 14/11/2025 को 11:30 HRS(IST) for (6h) बजे विडियो कॉन्फ्रेंसिंग सुनवाई तय की गयी है। अतः, आपको उपरोक्त दिनांक व समय पर नियंत्रक के समक्ष सुनवाई हेतु उपस्थित होना है। आपको यह भी सलाह दी जाती है कि आप निर्धारित तिथि पर सुनवाई के लिए उपस्थित होने से एक सप्ताह पहले नियंत्रक को सूचित करें।

With reference to Pre-Grant Opposition filed in Application No 202141018455 and reply of same filed in the application, a hearing is fixed in the matter through Video Conferencing on14/11/2025 at 11:30 HRS(IST) for (6h). You are therefore, required to appear before the Controller for the hearing on said date and time. You are also advised to intimate the controller one week before for appearing in hearing on scheduled date.

इस आवेदन को पेटेंट अनुदान हेतु क्रम मे लाने की अंतिम तिथि से पूर्व / अंतिम तिथि के उपरांत, निम्नलिखित आपत्तियां अभी भी शेष हैं।
The following objection(s) are still outstanding before / after the expiry of last date for putting this application in order for grant of patent.

Kanhaiya Lal Balai
Assistant Controller of Patents & Designs

*दिनांक/समय, स्थल, स्थिति व सुनवाई के बारे में अन्य विवरण के लिए: कृपया निम्नलिखित यूआरएल देखें
https://iprsearch.ipindia.gov.in/DynamicUtility/CauseListForHearingCase/Index
Please refer to the following URL for: Date/Time, Venue, Status and other details about the Hearing
https://iprsearch.ipindia.gov.in/DynamicUtility/CauseListForHearingCase/Index
टिप्पणी:- उक्त सुनवाई में भाग लेने के लिए वीडियो कॉन्फ्रेंसिंग लिंक के संबंध में अलग से ईमेल भेजी जाएगी।
Note:- Separate email will be sent regarding the Video Conferencing Link to attend the said hearing.

* Hearing Objections are attached.

Objections

Clarity and Conciseness

1. The following objection raised in FER has still not been met.

   Reference numeral shall be marked in the drawings and reference of it shall be incorporated in the claims to enhance the intelligibility of Claims. No description matter is allowed on the drawings as per Rule 15(7) of The Patents Rule, 2003 (as amended).

2. 1. The reference numerals of the drawings are not incorporated in the claims; the applicant is required to amend the claims to include appropriate reference numerals corresponding to the features shown in the drawings to ensure clarity and consistency.

   2. Abbreviated forms of words have been used in the claims and/or description. The applicant is required to replace all such abbreviations with their complete forms to ensure clarity and proper understanding of the invention, as abbreviations may lead to ambiguity or misinterpretation of the technical features described.

Formal Requirement(s)

1. 1. The General Power of Attorney (GPA) filed by the opposition is found to be defective. It is observed that the Patent Agent number (IN/PA No.) mentioned in the GPA is incorrect and does not correspond to the registered number of the patent agent Mr. Hemant Thadhani, who is stated to be authorized to act on behalf of the opposition. Accordingly, the opposition is directed to submit a corrected and duly executed General Power of Attorney, wherein the correct Patent Agent number (IN/PA No.) of Mr. Hemant Thadhani is mentioned. The corrected GPA shall be properly executed, dated, and, if applicable, notarized or legalized in accordance with the prescribed procedure.

   2. It is observed that the applicant has submitted amendments to the claims in the present application by filing Form 13. As per Section 57 of the Patents Act, 1970, read with Rule 81(1) of the Patents Rules, 2003, any amendment to a patent application, complete specification, or any related document must be made by filing Form 13, accompanied by a clear statement showing the basis and support for the proposed amendment in the originally filed specification.In this regard, the applicant is directed to provide adequate justification and support for each amendment made, with specific references to the description, drawings, or original claims. Until such compliance is made, the submitted Form 13 and the amendments to the claims cannot be taken on record. The applicant is advised to ensure that the basis for each amendment is clearly documented to maintain conformity with the requirements of the Patents Act and Rules.

   3. It is observed that the applicant has filed Form 13 for amendment of the claims in the present application. As per Section 57(2) of the Patents Act, 1970, every application for leave to amend an application for a patent, a complete specification, or any related document must provide full particulars of the reasons for which the application is made. In the present submission, the reasons or justification for the amendments have not been clearly furnished. The applicant is therefore directed to submit a revised Form 13 clearly stating the detailed reasons and justification for making each amendment. Until compliance with the above requirements is made, the submitted Form 13 and the proposed amendments cannot be taken on record.

Invention u/s 2(1)(ja)

1. The applicant's reply to the objection raised in the First Examination Report (FER) under Section 2(1)(ja) of the Patents Act has been duly considered.

   Upon further examination, it has been observed that the following prior art documents disclose the features of the alleged invention:

   D4: CN103820781A (Publication Date: 25/08/2014)

   D5: CN105148816A (Publication Date: 16/12/2015)

   D6: CN206372825U (Publication Date: 04/08/2017)

   Doucmnet D4 discloses a profiled outer glass-lined device, which mainly comprises a can lid and a can body, wherein the can lid is provided with a nozzle, a manhole and a sealing mouth; the outer part of the can body is covered with a jacket, the can body and the clip The sleeve is welded into a unitary structure with a cavity therebetween, a jacket flange is provided on one side of the tank body, and a discharge port, a condensate outlet, a steam inlet and a steam outlet are arranged at a lower portion.

   The document further discloses the outer side of the outer side of the can body and the outer side of the jacket have a

stainless steel outer layer. The upper part of the stainless steel outer layer of the outer side of the can body is covered with a can body flange surface and a can body surface I stainless steel outer layer. The can body and the jacket welding portion There is a canned transition surface I stainless steel outer layer, the outer side of the jacket is covered with a can body surface II stainless steel outer layer, and the lower part of the can body is covered with a stainless steel outer layer of the can body sealing surface, and the lower part of the can body and the jacket welding portion are covered The can body transition surface II stainless steel outer layer, the bottom of the can body is covered with a can body surface III stainless steel outer layer.

D5 discloses the manufacturing method of a glass-lined reaction kettle and belongs to the technical field of processing, which solves the problem that the outer surface of the glass-lined reaction kettle is poor in corrosion resistance. A stainless steel flange is placed on the nozzle flange of each process pipe, and the upper cover is covered from the top to the upper cover, so that the process pipe protrudes from the through hole of the upper foreskin, and then The upper foreskin is welded with the stainless steel flange, and the cylinder is wrapped with the middle foreskin, and the lower cover is covered from the bottom with the lower foreskin, so that the process tube protrudes from the through hole of the lower foreskin, and the lower foreskin is welded with the stainless steel flange. Finally, the lower foreskin and the middle foreskin are sealed and welded.

D6 discloses a corrosion -resistant glass reactor that wards off of high strength, including pressing from both sides cover, jar body, agitator, upper cover, low head, sight glass manhole, the thermometer mouth of pipe, feed inlet, drain hole, the external side of jar is equipped with presss from both sides the cover, it is equipped with a plurality of installation positioner to press from both sides angles such as overlapping the outside, pressing from both sides cover upper portion and being equipped with steam inlet, steam inlet one side is equipped with the relief port, presss from both sides the cover bottom and is equipped with water export with fixed attention, the welding has high -neck flange on the feed inlet, the welding has high -neck flange on the sight glass manhole, and the welding has high -neck flange on the thermometer mouth of pipe, and the welding has high -neck flange on the drain hole, be equipped with the ground coat on the internal lateral wall of jar, be equipped with cover glaze layer on the ground coat, be equipped with the ground coat on high -neck flange inside wall, terminal surface, the recess, be equipped with cover glaze layer on the ground coat. The utility model discloses trompil department welding has increased high -neck flange, has reduced the quantity at stamping forming trompil R position, has directly increased the effective thickness of this equipment, and compressive strength improves relatively, ensuring the safe operation of equipment.

In view of the foregoing, it is evident that a person skilled in the art could and would readily arrive at the present invention by drawing sufficient motivation from the cited prior arts (D4-D6). The combination of features disclosed in these references provides ample guidance to derive the claimed invention without the need for any inventive step or intellectual creativity. Consequently, the claims of the present application are deemed to be obvious and lacking an inventive step in accordance with the provisions of Section 2(1)(ja) of the Patents Act.

Other Requirement(s)

1. The amendment made in the description and claims does not comply with the provisions of Section 59(1) of the Act.

Scope

1. 1. Claim 1 recites functional advantages such as “offering excellent resistance to oxidation,” “tackling rust challenges,” “upholding reactor integrity, longevity, and product purity,” and “enhancing resilience against external damage and increasing overall safety,” which merely describe the functional benefits or results of the claimed reactor rather than defining specific technical or structural features. Such statements of function or result do not constitute patentable subject matter under Section 2(1)(j) of the Patents Act, 1970. The claim should therefore be amended to clearly define the structural or technical features responsible for achieving the stated effects instead of claiming the functional benefits themselves.

   2. Claims 2 and 7 do not clearly and unambiguously define the scope of the invention as required under Section 10(4)(c). In Claim 2, the term “base body of high mechanical strength” is vague and lacks measurable parameters, and the expressions “essentially,” “preferably,” and “most preferably” regarding the reaction chamber thickness introduce ambiguity. In Claim 7, the term “smooth as whole” is indefinite and does not specify objective criteria for surface smoothness. The claims therefore fail to distinctly define the invention and may lead to multiple interpretations. The applicant is directed to amend Claims 2 and 7 to clearly define the mechanical strength of the base body in measurable terms and to specify the smoothness of the glass lining/coating in objective terms, so as to comply with Section 10(4)(c) of the Patents Act, 1970.

   3. It is observed that there is an inconsistency between the description, claims, and drawings regarding the material of the reactor walls. In the description and claims, the outer wall of the reactor is disclosed as stainless steel (SS) and the inner wall as mild steel (MS). However, in the drawings, the layers appear inversed, showing the outer wall as MS and the inner wall as SS. This inconsistency creates ambiguity regarding the actual structure of the reactor and the scope of the invention. The applicant is required to rectify the discrepancy by amending the description, claims, and/or drawings so that the material of the outer and inner walls is consistent throughout the specification and clearly defines the invention, in compliance with Section 10(4) of the Patents Act, 1970.

2. (a) The scope of principal claim 1 is ambiguous due to multiple characterization clauses introduced by the phrase "wherein." Re-draft or rephrase the principal claim to clearly define the constructional inventive feature of the alleged invention over which protection is sought.

   (b) The phrases:

   "Coated nozzle is seamlessly welded to the top cover's cladding sheet, fully enclosing the reactor in stainless steel, addressing rust challenges in critical zones, and forming a resilient barrier that upholds reactor integrity, longevity, and product purity."

   "The reactor offers excellent resistance to oxidation at the nozzle. In the event of accidental spillage of corrosive liquid onto the nozzle surface, oxidation of the metal surface may occur, leading to damage of the glass lining coated inside, as oxidation of the base metal would cause distortion."

   "The reactor is fully covered with stainless steel, enhancing its resilience against external damage and increasing overall safety."

   These statements in claim 1 are narrative in nature and contain indefinite, functional, or operational language. The principal claim is vaguely drafted in a functional format. As per Section 10 of the Patents Act, 1970 (as amended), vague and functional aspects are not permissible in claims; therefore, the claims are not allowable.

Sufficiency of Disclosure u/s 10 (4)

1. 1. It is observed that references have been made in the description to Figures 4 and 5. However, these figures are not included in the drawings filed with the application. The absence of Figures 4 and 5 renders the references in the description inconsistent with the accompanying drawings, leading to ambiguity in understanding the disclosure. Accordingly, the applicant is required to file the missing Figures 4 and 5 with proper reference in the description, to ensure conformity between the description and the drawings as required under Section 10(4) of the Patents Act, 1970 and Rule 15 of the Patents Rules, 2003.

   2. It is observed that certain structural features of the invention, such as the top flange, bottom flange, glass-lined metal flat surface below the top flange for observing the flow regime in the reactor or mixer, the cavity in the bottom flange facilitating the flow of heat transfer fluid through inlet and outlet nozzles and flanges, the bottom plate having a metal surface in contact with the heat transfer fluid, the connection plate connecting the inlet and outlet to the bottom flange of the glass-lined stainless-steel reactor, the multiple bolts securing the top and bottom flanges, and the external stainless-steel sheet covering of the reactor, have been described in the specification. However, these features are not shown or illustrated in the drawings accompanying the application. As per the requirements of Section 10(4) of the Patents Act, 1970, every complete specification shall fully and particularly describe the invention and its operation or use and the method by which it is to be performed, and shall disclose the best method of performing the invention known to the applicant. Accordingly, the applicant is required to incorporate all the described structural features in the drawings, ensuring that each feature or embodiment mentioned in the description is properly reflected in the drawings with corresponding reference numerals. This is necessary to maintain consistency between the description and drawings and to ensure that the invention is fully and clearly disclosed in compliance with the statutory requirements.

Daswani & Daswani
Patent & Trade Mark (Inteilectual Property) Attorneys

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November 07, 2025

Rs. 7,500/-
Kind Attention:
Dr. Kanhaiya Lal Balai
Assistant Controller of Patents & Designs

Your Ref: POC/Application No / 202141018455

Our ref: 202141018455

To
The Controller of Patents
The Patent Office,
Chennai / Delhi

Dear Sir,
Re: Indian Patent Application No. 202141018455 of 21.04.2021 relating to "CLEAN ROOM GLASS LINED REACTOR" in the name of Standard Glasslining Technology Private Limited

This is further to your hearing notice dated 13-10-2025 regarding Pre-Grant Opposition wherein, hearing has been fixed in the matter on 14-11-2025.

We like to state that applicant's Counsel (Patent Agent) Mr. Hemant Daswani, IN/PA 610 will attend the hearing on 14-11-2025 as per schedule mentioned in the hearing notice.

This is for your information only.

Yours Sincerely
Of DASWANI & DASWANI

Ranjan Paul

(Ranjan Paul)
Applicant's Agent
IN / PA 498

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BY ONLINE FILING
Confidential and Urgent

HCA/SKA/HT/RD November 13, 2025

To,
The Controller of Patents,
The Patent Office,

ONLINE SUBMISSION

Dear Sir,

Notice to attend Hearing

Reference:	INDIA-Patent Application No.	**202141018455**
	Applicant:	**Standard Glass Lining Private Limited**

This is in reference to the hearing notice dated 13-10-2025 fixing hearing of the scheduled application on 13.11.2025.

This is to notify Opponent's agent Mr. Hemant Thadhani (IN/PA-4057) will be attending the hearing scheduled on **14.11.2025**.

Yours faithfully,
FOR **Pinakin Engineers Private Ltd**
By its Agent

Hemant Thadhani
Patent Agent ((IN/PA-4057)
Crawford Bayley & Co.

Daswani & Daswani
Patent & Trade Mark (Inteilectual Property) Attorneys

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November 28, 2025

Kind Attention:
Dr. Kanhaiya Lal Balai
Assistant Controller of Patents & Designs

Your Ref: POC/Application No / 202141018455

Our ref: 202141018455

To
The Controller of Patents
The Patent Office,
Chennai / Delhi

Dear Sir,
Re: Indian Patent Application No. 202141018455 of 21.04.2021 relating to "CLEAN ROOM GLASS LINED REACTOR" in the name of Standard Glasslining Technology Private Limited

In response to your hearing notice dated 13-10-2025 regarding Pre-Grant Opposition, applicant's Counsel (Patent Agent) Mr. Hemant Daswani, IN/PA 610 attended the hearing fixed on 14-11-2025 as per schedule mentioned in the hearing notice.

WRITTEN SUBMISSIONS ON BEHALF OF THE APPLICANT IN THE PRE-GRANT OPPOSITION PROCEEDINGS UNDER SECTION 25(1) OF THE PATENTS ACT, 1970

Re: Pre-Grant Opposition under Section 25(1) of the Patents Act, 1970 against Patent Application No. 202141018455 titled "Clean Room Glass Lined Reactor" filed by Standard Glass Lining Technology Private

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Limited (Applicant) opposed by Pinakin Engineers Private Limited (Opponent).

Date of Hearing: 14th November 2025 at 11:30 HRS (IST) via Video Conferencing.

OPENING REMARKS

The Applicant, Standard Glass Lining Technology Private Limited, through its authorized Patent Agent, Mr. Hemant Daswani (IN/PA/610), hereby submits these exhaustive written submissions in response to the Pre-Grant Opposition filed by the Opponent on 14th November 2024, as well as in compliance with the Hearing Notice dated 13th October 2025 issued under Rule 55(4) of the Patents Rules, 2003.

These submissions address all grounds raised by the Opponent under Section 25(1)(b), (e), (f), and (g) of the Patents Act, 1970, refute each statement made in the Opponent's Written Statement and evidence (Exhibits A to G), and respond to the objections outlined in the Hearing Notice regarding clarity and conciseness, formal requirements, and inventive step under Section 2(1)(ja).

The Applicant humbly requests the Learned Controller to dismiss the opposition in its entirety, as it is frivolous, mala fide, and lacks merit, and to proceed with the grant of the patent, subject to any minor amendments if deemed necessary for formal compliance. The Applicant reserves the right to file further evidence or amendments as per the provisions of the Act and Rules, and seeks an opportunity to be heard orally at the scheduled hearing.

The present application, filed on 21st April 2021 and published on 28th October 2022, discloses a novel Clean Room Glass Lined Reactor specifically designed for pharmaceutical and food-grade processing environments, where corrosion resistance, hygiene, and contamination control are paramount. The invention addresses a long-standing technical problem in conventional glass-lined reactors: oxidation and corrosion at nozzles and

external surfaces in clean-room settings, which can lead to distortion of the internal glass lining, product contamination, and non-compliance with stringent standards like ISO 14644-1 and GMP.

The core inventive concept lies in a mild steel (MS) reactor body internally glass-lined and externally overlaid with stainless steel (SS) sheets through a seamless welding process, particularly extending to nozzles, ensuring 100% enclosure in SS to prevent oxidation while maintaining structural integrity and heat transfer efficiency. This is achieved by machining fluidic channels on the reactor surface, integrating a glass-lined flat observation surface below the top flange for flow regime monitoring, and incorporating a cavity in the bottom flange for heat transfer fluid circulation, all clamped securely between top and bottom flanges with bolts.

The specification provides enabling disclosure, including detailed description, claims, and drawings, supported by the method of performance, enabling a person skilled in the art (a mechanical engineer familiar with reactor design in chemical/pharmaceutical industries) to practice the invention without undue experimentation.

In contrast, the Opponent's opposition is an attempt to delay the grant due to commercial rivalry, as both parties operate in the glass-lined equipment manufacturing sector in Gujarat. The Opponent fails to establish locus standi as a "person interested" under Section 25(1), merely claiming to be a competitor without demonstrating any tangible interest beyond rivalry, such as prior use or direct economic impact, as required by precedents like Ajay Industrial Corporation v. Controller of Patents (1984 PTC 234), where the court held that mere commercial rivalry does not confer standing without proof of specific interest. Furthermore, the opposition is mala fide, as evidenced by its reliance on unsubstantiated assertions, uncertified translations of foreign prior art, and superficial tabular comparisons without technical analysis, violating the principles laid down in Hindustan Lever Ltd. v. Godrej Soaps Ltd. (1996 PTC 141), where oppositions motivated by delay tactics were dismissed with costs. The Applicant seeks dismissal on these

preliminary grounds alone, along with compensatory costs under Section 35 of the Act for defending this baseless challenge.

GROUND 25(1)(b)

Turning to the substantive rebuttal, the Applicant addresses the Opponent's grounds under Section 25(1)(b) for lack of novelty. The Opponent alleges anticipation by prior art documents D1 (CN103820781A, priority 12.03.2014), D2 (CN105148816A, priority 08.09.2015), and D3 (CN206372825U, priority 12.01.2017), claiming that these disclose all features of the claims through tabular comparisons in their Written Statement. The Applicant humbly reiterates and submits the following detailed observations in rebuttal to the ground raised under Section 25(1)(b) of the Patents Act, 1970, as articulated in the Opponent's Written Statement and as prima facie upheld by the Learned Controller in the Report dated 04th June 2025 (hereinafter "Prima Facie Report").

This ground alleges that the subject matter of the present application lacks novelty due to anticipation by prior publications, specifically D1 (CN103820781A, published 28th May 2014), D2 (CN105148816A, published 16th December 2015), and D3 (CN206372825U, published 4th August 2017). The Applicant respectfully contends that this ground is entirely misconceived, unsubstantiated, and fails to meet the stringent evidentiary and analytical thresholds mandated by Indian patent law for establishing anticipation. The rebuttal is structured to first elucidate the legal framework governing novelty assessments, followed by a meticulous, feature-by-feature dissection of the cited documents vis-à-vis the inventive features of the present application, demonstrating unequivocally that no single prior art document discloses the invention as a whole in an enabling and unambiguous manner. This analysis draws upon the complete specification of the present application (filed on 21st April 2021, as amended in response to the First Examination Report dated 23rd March 2023), the full English translations of D1, D2, and D3, relevant precedents from Indian courts, and

guidelines from the Manual of Patent Office Practice and Procedures, 2019 (hereinafter "Patent Manual").

At the outset, it is imperative to recall the legal principles from Section 25(1)(b), which provides that a patent may be opposed if the invention "has been published before the priority date of the claim (i) in any specification filed in pursuance of an application for a patent made in India on or after the 1st day of January, 1912; or (ii) in India or elsewhere, in any other document." As per Section 2(1)(l), an invention is "new" if it does not form part of the state of the art, and anticipation requires that the prior art must disclose the invention in its entirety, with such clarity and completeness that a person skilled in the art could perform it without further ingenuity or experimentation. The Hon'ble Supreme Court in Bishwanath Prasad Radhey Shyam v. Hindustan Metal Industries (AIR 1982 SC 1444) emphatically held that for anticipation, the prior publication must contain "clear and unmistakable directions" to perform the invention, and mere suggestions or incomplete disclosures do not suffice. Similarly, in Lallubhai Chakubhai Jariwala v. Chimanlal Chunilal & Co. (AIR 1936 Bom 99), the court clarified that anticipation cannot be established by mosaicking disparate elements from multiple documents; each prior art must be evaluated independently as a standalone disclosure.

The Patent Manual (Chapter 08.03.01) reinforces this by stating that "the prior art should disclose the invention either in explicit or implicit manner" and must enable the skilled person to arrive at the invention without undue burden. Critically, foreign patent documents like D1, D2, and D3 must be accompanied by certified translations and proof of public availability in India prior to the priority date (21st April 2021), failing which they are inadmissible, as per Rule 55(3) of the Patents Rules, 2003, and precedents such as Snehalata C. Patel v. Controller of Patents (2001 PTC 121). In the instant case, the Opponent has furnished uncertified machine translations (Exhibits E, F, G), which are riddled with inaccuracies (e.g., "foreskin" in D2 likely mistranslating "wrapper" or "sheath"), and no evidence of their dissemination in India, rendering them prima facie unreliable. Even

assuming arguendo their admissibility, a granular analysis reveals profound dissimilarities, as elaborated below.

The inventive core of the present application, in the complete specification (pages 1-19) and amended claims (dated 23rd September 2023), resides in a "Clean Room Glass Lined Reactor" tailored for pharmaceutical and food-grade environments under stringent hygiene standards like ISO 14644-1 and GMP. The reactor comprises: (i) an internally glass-lined mild steel (MS) body clamped between top and bottom flanges with machined fluidic channels on the surface for enhanced heat transfer; (ii) a glass-lined metal flat surface below the top flange enabling non-invasive observation of internal flow regimes to prevent contamination; (iii) a cavity in the bottom flange for heat transfer fluid circulation via inlet/outlet nozzles, closable with a metal plate; (iv) a connection plate linking inlet/outlet to the bottom flange; (v) an external stainless steel (SS) sheet overlay welded seamlessly to enclose the reactor fully, preventing oxidation and rust at critical zones; (vi) internally glass-lined nozzles with seamless SS overlay welded to the cladding, addressing the specific problem of nozzle oxidation distorting internal lining in clean rooms; (vii) a double-walled vessel design with smooth glass-lined interior; and (viii) application-specific volumes (e.g., 50-5000 liters) and heat transfer areas.

This configuration solves the longstanding issue of external corrosion in conventional reactors, where spillage in clean rooms leads to oxidation penetrating to the internal glass lining, causing distortion, contamination, and regulatory non-compliance. The manufacturing method involves removing approximately 2mm of MS surface before welding SS overlays with SS electrodes to form a metallurgical bond, maintaining uniform thickness and 100% SS enclosure without gaps or insulation voids that could harbor contaminants. None of D1, D2, or D3—each addressing generic glass-lined vessels for non-clean-room applications—discloses this integrated, synergistic arrangement, as evidenced by their divergent teachings.

Commencing with D1 (CN103820781A, titled "Glass Lined Equipment Provided with Outer Profiling Cladding and Manufacturing Method"), this document describes a conventional glass-lined tank comprising a tank cover (with nozzles, manhole, and seal) and tank body encased in a jacket for heat exchange, with external stainless steel "profiling cladding" layers applied for insulation purposes. The full description (*from unauthenticated English translation*) reveals that the cladding is formed by cutting and shaping 304 stainless steel sheets (1.5-3mm thick) to match the tank's profile, welding joints with argon arc welding, and filling gaps between the jacket and cladding with polyurethane foam or silicate for thermal insulation. The tank body and jacket are welded into a unitary structure with a cavity for steam/condensate flow, and bottom ports include drain hole, condensation-water drain, steam inlet, and vapor outlet. However, D1 fundamentally diverges from the present invention in several critical aspects, precluding anticipation. Firstly, D1's "cladding" is a detachable, insulated outer layer with deliberate gaps filled for thermal purposes, not a seamless, metallurgically bonded SS overlay welded directly after MS surface removal to prevent oxidation in clean-room settings. The translation explicitly states: "Between described chuck and stainless steel surrounding layer, be filled with lagging material and form thermal insulation layer," emphasizing insulation over corrosion prevention at nozzles or flanges.

In contrast, the present application's SS overlay is seamless, extending to nozzles without insulation gaps, ensuring 100% enclosure to eliminate rust formation at critical zones—a feature absent in D1, where cladding joints are merely butt-welded and polished, leaving potential entry points for moisture. Secondly, D1 lacks any disclosure of machined fluidic channels on the reactor surface, a glass-lined flat observation surface for flow monitoring, or a bottom flange cavity with connection plate for targeted heat transfer; instead, it relies on a standard jacketed cavity with bottom ports, without the clamped flange assembly or observation capability essential for clean-room hygiene. The manhole in D1 is for access, not flow observation, and there is no mention of a flat glass-lined surface below the top flange to

enable visual inspection without opening, which prevents contamination risks in pharmaceutical processes.

Thirdly, D1's manufacturing method involves preliminary impact briquetting of sheets, argon welding (current 70-80A, voltage 7-9V), and "V"-type breach repairs for flanges, but omits the inventive step of removing 2mm MS surface for uniform thickness and metallurgical bonding, focusing instead on profile consistency for insulation. As per the Patent Manual (Chapter 08.03.02), anticipation requires "implicit or explicit" disclosure of all essential features; here, D1's generic insulated cladding for cost reduction in non-pharma contexts does not implicitly teach the oxidation-resistant, seamless nozzle overlay for GMP compliance.

The Opponent's tabular comparison (Written Statement, pages 12-15) superficially maps D1's "surrounding layer" to the claims but ignores functional disparities—D1's layer allows gaps for filling, whereas the invention mandates seamless welding to preclude voids. In Novartis AG v. Union of India ((2013) 6 SCC 1), though pharmaceutical, the principle extends: mere structural similarity without identical purpose or effect does not anticipate. Thus, D1 falls short of providing "clear and unmistakable directions" (Bishwanath Prasad, supra) to arrive at the clean-room reactor.

Shifting to D2 (CN105148816A, titled "Manufacturing Method of Glass-Lined Reaction Kettle"), this patent outlines a process for fabricating a glass-lined kettle with an external stainless steel "foreskin" (translated as "wrapper" or "sheath") to enhance corrosion resistance. The method encompasses seven steps: assembling enamel-fired carbon steel components (upper head, cylinder, lower head); pressing SS sheets into upper, middle, and lower foreskins; perforating for process ducts; fixing foreskins by welding to steel flanges on nozzles; filling insulation material (e.g., glass cement) in gaps and seal-welding; polishing the SS surface; and adding an annular SS cover. The translation details that foreskins are oversized (diameter larger by 2-4mm at holes) for wrapping, with steel flanges halved and welded on nozzle flanges to enable SS welding, and insulation filled to isolate carbon steel. While D2

superficially addresses external corrosion via SS wrapping, it does not anticipate the invention due to profound structural and functional mismatches.

Primarily, D2's "foreskin" is a detachable wrapper with deliberate gaps filled post-assembly for insulation, not a seamless welded overlay integrated metallurgically after MS surface machining. The description states: "Fill insulant material in the gap between stainless steel foreskin and enamel, and seal welding," implying voids are inherent and filled secondarily with materials like glass cement, which the present invention eschews to avoid relative displacement and re-emerging gaps—issues D2 acknowledges but resolves inadequately via cement, leading to potential contamination in clean rooms. The invention's seamless welding (using SS electrodes post-2mm MS removal) creates a continuous, rust-proof enclosure without fillers, a distinction not disclosed in D2, where welding is limited to foreskin-flange joints (e.g., "upper foreskin is welded with steel flange").

Secondly, D2 omits key elements like fluidic channels machined on the reactor surface, the glass-lined observation flat surface, bottom flange cavity with connection plate, or nozzle-specific seamless SS overlays for oxidation prevention; its process ducts are merely perforated and flanged, without extended SS welding to cladding for 100% enclosure. The kettle is for general chemical reactions, not clean-room pharma use, lacking GMP-focused hygiene features. The manufacturing steps emphasize inversion for filling (e.g., "The upper cover being surrounded by described upper foreskin in described step e is inverted, fill insulant material in space up between foreskin and upper cover"), prioritizing insulation over oxidation resistance, whereas the invention's method ensures dimensional uniformity and bond strength for longevity in sterile environments. As per F. Hoffmann-La Roche Ltd. v. Cipla Ltd. ((2016) 66 PTC 1 (Del)), anticipation demands identical disclosure, not obvious modifications; D2's wrapper, reliant on secondary fillers, does not enable the skilled person to perform the seamless, gap-free overlay without ingenuity. The Opponent's assertions (Written Statement, pages 16-18) conflate D2's "foreskin" with the claims but overlook that D2's

design permits air/moisture entry pre-filling, antithetical to the invention's preventive enclosure. Hence, D2 does not destroy novelty.

Finally, D3 (CN206372825U, a utility model titled "A Kind of High-Strength Corrosion-Resisting Glassed Steel Reaction Vessels"), pertains to a jacketed glass-lined reactor with enhanced compressive strength via seamless welding neck flanges at ports. The description covers a tank body with jacket for heat exchange (steam/liquid inlets/outlets), agitator, viewing manhole, thermometer nozzle, charging/discharge ports, and standby ports, all featuring neck flanges with grooved end faces, chamfers, and bolted covers/gaskets for sealing. Corrosion resistance is achieved via internal ground glaze and cover coat layers (with specific frit compositions including silica 40-45%, alumina 3-7%, etc., and mixed adhesive agents like bastnaesite flour 45-55%), extending to flange inner walls and joints. However, D3 is devoid of any external SS overlay or cladding, let alone seamless welding for clean-room oxidation prevention, rendering it wholly non-anticipatory.

The utility model's focus is internal enamel enhancement for strength and anti-corrosion, with "high-neck flanges" welded seamlessly to reduce punch-formed openings and bolster pressure resistance (e.g., "The viewing manhole, thermometer nozzle... use seamless welding neck flanges"). There is no mention of external MS body with SS welding, machined fluidic channels, observation flat surface, bottom cavity, or nozzle overlays; the external jacket is bare, with mounting devices but no SS enclosure. The glaze formula, while improving internal lining adhesion (e.g., melting at 900-1000°C), is irrelevant to the invention's external hygiene features. As a utility model, D3 lacks enabling disclosure for manufacturing beyond basic assembly, omitting the inventive welding post-MS removal. The Patent Manual (Chapter 08.03.03) requires prior art to be enabling; D3's schematic utility focus does not provide "clear directions" for the claimed reactor.

The Opponent's mapping (Written Statement, pages 19-20) erroneously equates D3's internal enamel to the claims, ignoring the absence of external

SS elements. In Tapas Chatterjee v. Assistant Controller (2023 SCC OnLine Del 469), the court rejected anticipation based on partial, non-enabling disclosures; similarly, D3's internal-centric design cannot anticipate the external clean-room innovations.

Feature from Applicant's Claims (as amended)	Teaching in Applicant's Specification	D1 (CN103820781 A)	D2 (CN105148816 A)	D3 (CN206372825 U)	Analysis (Why Not Anticipatory)
Internally glass-lined MS reactor clamped between top and bottom flanges with machined fluidic channels on surface (Claim 1(i))	Pages 3-5: Reactor body of MS with internal glass lining (1-2mm thick); fluidic channels machined for heat transfer; clamped via bolts for secure assembly.	Discloses general glass-lined can body with jacket, but no machined fluidic channels on reactor surface; focuses on insulated cladding (translation p.90).	Describes reaction kettle with glass lining, but channels are for jacket circulation, not machined on reactor surface (translation p.96).	Utility model for reactor with flanges; no mention of machined channels (translation p.102).	No single document discloses machined channels integrated with clamping for enhanced heat transfer in clean-room setup;.
Glass-lined metal flat surface below top flange for flow regime observation (Claim 1(ii))	Pages 5-6: Flat surface allows visual monitoring of internal mixing without contamination risk.	No observation surface; manhole is for access, not flow observation (p.91).	No such feature; focuses on manufacturing steps (p.97).	Absent entirely (p.103).	Unique to Applicant; solves clean-room monitoring issue not addressed in prior art,.
Cavity in bottom flange for heat transfer fluid circulation via inlet/outlet nozzles, closable with metal plate (Claim 1(iii))	Pages 6-7: Cavity design ensures efficient cooling/heating; connected via plate (Claim 1(iv)).	Condensate outlet and steam ports, but no bottom flange cavity (p.92).	Bottom discharge, no cavity in flange (p.98).	Flange for strength, no cavity (p.104).	Prior art uses jacketed designs, not flange-integrated cavity;
External SS sheet overlay by welding, seamless on nozzles to prevent	Pages 7-9: 2mm MS removal, SS welding for metallurgical bond; extends to nozzles for	Profiling cladding with gaps/insulation; not seamless weld (p.93).	Detachable wrapper with insulation; no welding to nozzles (p.99).	No overlay; corrosion via materials (p.105).	Applicant's method prevents oxidation at critical zones, unlike prior

Feature from Applicant's Claims (as amended)	Teaching in Applicant's Specification	D1 (CN103820781A)	D2 (CN105148816A)	D3 (CN206372825U)	Analysis (Why Not Anticipatory)
oxidation (Claim 1(v)-(vii))	100% enclosure.				art's gapped claddings;
Double-walled design, smooth interior, volume/heat transfer ranges (Claims 6-8)	Pages 10-12: Ensures hygiene, compliance with ISO/GMP.	Single jacket with insulation; no ranges (p.94).	Manufacturing focus; no double-wall specifics (p.100).	Utility; no such claims.	Synergistic advantages not disclosed;.

In summation, none of D1, D2, or D3—individually or otherwise—anticipates the invention, as they lack the complete, enabling disclosure of the synergistic features. The Applicant prays for rejection of this ground, emphasizing the mala fide nature of the opposition driven by commercial rivalry, warranting dismissal with costs.

GROUND 25(1)(e)

The Applicant respectfully submits the following ground posits that the invention, as claimed in the present application, is obvious to a person skilled in the art and thus lacks an inventive step under Section 2(1)(ja). The Applicant vehemently contests this allegation, asserting that it is fundamentally flawed, bereft of substantive evidence, and predicated on impermissible hindsight reconstruction, selective mosaicking of prior art, and a failure to adhere to the rigorous analytical framework mandated by Indian patent jurisprudence. This rebuttal is structured to commence with a comprehensive exposition of the legal principles governing inventive step assessments in India, drawing upon statutory provisions, judicial precedents, and guidelines from the Manual of Patent Office Practice and Procedures, 2019 (hereinafter "Patent Manual"). Thereafter, it applies the established five-step test from F. Hoffmann-La Roche Ltd. v. Cipla Ltd. ((2016) 66 PTC 1 (Del))—hereinafter the "Roche Test"—to the facts at hand, demonstrating that the cited prior art documents D1 (CN103820781A,

published 28th May 2014), D2 (CN105148816A, published 16th December 2015), and D3 (CN206372825U, published 4th August 2017) neither individually nor in combination render the invention obvious.

To contextualize, Section 25(1)(e) empowers opposition on the basis that "the invention so far as claimed in any claim of the complete specification is obvious and clearly does not involve any inventive step, having regard to the matter published as mentioned in clause (b) or having regard to what was used in India before the priority date of the applicant's claim." Section 2(1)(ja) defines "inventive step" as "a feature of an invention that involves technical advance as compared to the existing knowledge or having economic significance or both and that makes the invention not obvious to a person skilled in the art."

This dual-pronged criterion—technical advance/economic significance coupled with non-obviousness—distinguishes Indian law from counterparts. The Patent Manual (Chapter 08.03.04) elucidates that inventive step assessment must consider the invention as a whole, avoiding hindsight bias, and requires the opponent to establish a clear pathway by which a person skilled in the art possessing common general knowledge in the field would arrive at the invention from the prior art without inventive ingenuity. Crucially, combinations of prior art are permissible only if there exists a "teaching, suggestion, or motivation" to combine them, to prevent ex post facto rationalizations.

The landmark Roche Test, affirmed in multiple decisions, provides a structured methodology: (1) Identify the inventive concept claimed; (2) Assume the mantle of the person skilled in the art and impute common general knowledge; (3) Identify differences between the cited prior art and the invention; (4) Determine if those differences constitute steps obvious to the person skilled in the art without hindsight; and (5) Consider secondary indicia like commercial success, long-felt need, or failure of others. Recent jurisprudence, such as Tapas Chatterjee v. Assistant Controller of Patents (2023 SCC OnLine Del 469), condemns mechanical rejections, mandating

reasoned orders that eschew subjective opinions and improper prior art combinations without motivation. The Opponent's approach—mere tabular feature mappings without teaching, suggestion, or motivation analysis or person skilled in the art imputation—violates these tenets, rendering their case untenable.

Applying the Roche Test to the present invention, titled "Clean Room Glass Lined Reactor," the inventive concept (Step 1) is a specialized reactor for pharmaceutical/food-grade clean rooms, comprising an internally glass-lined MS body with machined fluidic channels, clamped between flanges; a glass-lined flat observation surface for contamination-free flow monitoring; a bottom flange cavity with connection plate for efficient heat transfer; and a seamless SS overlay welded post-2mm MS removal, extending to nozzles for 100% enclosure against oxidation, ensuring GMP/ISO compliance, longevity, and hygiene without insulation gaps or fillers. This addresses a long-felt need: preventing nozzle oxidation distorting internal lining in sterile environments, yielding technical advances (enhanced corrosion resistance, visual monitoring) and economic significance (reduced downtime, maintenance costs). The complete specification (pages 1-19, as amended) enables this via detailed construction and method, far beyond obvious modifications.

Assuming the person skilled in the art's mantle (Step 2), the person skille din the art is an engineer with experience in chemical reactor design for pharma industries, cognizant of CGK like jacketed glass-lining for corrosion, SS claddings for durability, and basic welding techniques, but not advanced metallurgical bonding or clean-room specifics unless explicitly taught. CGK includes standards like ISO 14644-1 but excludes patented innovations or rare combinations.

Identifying differences (Step 3) from D1-D3 (collectively or individually, as the Opponent urges conjoint reading without justification): D1 teaches insulated SS "profiling cladding" with polyurethane-filled gaps for general glass-lined tanks, prioritizing thermal insulation over oxidation prevention;

no machined channels, observation surface, flange cavity, or seamless nozzle overlays—cladding is gapped, detachable. D2 describes SS "foreskin/wrapper" manufacturing with insulation fillers (glass cement) and flanged perforations, but gaps persist pre-filling, no MS removal for bonding, absent observation/cavity features, focused on generic kettles. D3, a utility model, enhances internal enamel with frit compositions and neck flanges for strength, entirely lacking external SS overlays or clean-room elements. The key differences like invention's metallurgical SS-MS bond (post-removal welding) vs. prior art's gapped/insulated claddings; integrated observation and cavity for hygiene/heat transfer vs. absent; nozzle-specific seamless extension for 100% enclosure vs. partial flanging; clean-room GMP focus vs. general use.

Assessing obviousness (Step 4): These differences are non-obvious, as no "teaching, suggestion, or motivation" exists in common general knowledge or prior art to combine/modify. D1's insulation objective (gaps for foam) teaches away from seamless bonding, as fillers would compromise sterility; modifying to remove gaps requires hindsight from the invention. D2's wrapper with secondary cement filling acknowledges displacement risks, not suggesting precise MS removal/welding for uniformity—person skille din the art would not "obviously try" this without inventive spark. D3's internal focus provides no bridge to external overlays. Opponent's assertions are conclusory, ignoring divergent purposes (insulation vs. oxidation prevention). Secondary indicia (Step 5) is that the invention fulfills long-felt pharma need for rust-free reactors, evidenced by adoption; Opponent's rivalry underscores commercial value.

The Opponent's evidence—uncertified translations, no expert affidavits—fails the burden under Rule 55(3). Their mala fide intent, as competitors, merits costs per Hindustan Lever Ltd. v. Godrej Soaps Ltd. (1996 PTC 141). The Applicant prays for rejection of this ground with awarded costs.

GROUND 25(1)(f)

Turning to the substantive rebuttal, the Applicant addresses the Opponent's grounds under Section 25(1)(f) ground asserts that the subject matter of the claims in the present application does not constitute an "invention" as defined under Section 2(1)(j) of the Act or is otherwise not patentable, specifically invoking exclusions under Sections 3(d) and 3(f) of the Act. The Applicant submits that this objection is profoundly misguided, lacking in evidentiary support, and premised on a misapplication of the statutory exclusions, which are intended to prevent the patenting of trivial or non-inventive subject matter but do not apply to the genuine technical advancement embodied in the present invention.

Specifically, Section 3(f) excludes "the mere arrangement or re-arrangement or duplication of known devices each functioning independently of one another in a known way," targeting aggregations devoid of synergistic interaction. As per the Patent Manual (Chapter 08.02.02), this provision prevents patenting of collocations where components operate separately without producing a novel combined effect, but expressly permits patents for combinations yielding a new or improved result through interdependence. In TVS Motor Company Ltd. v. Bajaj Auto Ltd. (2009 (40) PTC 689 (Mad)), where the Madras High Court upheld a patent for a twin-spark plug engine, ruling that even if individual elements are known, their specific configuration producing improved combustion efficiency constitutes a patentable invention, not a mere rearrangement. More recently, in Nippon Steel Corporation v. Controller of Patents (2024 SCC OnLine Del 1234, decided in March 2024), the Delhi High Court clarified that for Section 3(f) objections, the Controller must demonstrate independent functioning of components without hindsight, emphasizing that mechanical inventions involving integrated structural modifications are patentable if they solve a technical problem in a non-obvious manner.

Turning to Section 3(d), it precludes "the mere discovery of a new form of a known substance which does not result in the enhancement of the known efficacy of that substance or the mere discovery of any new property or new use for a known substance or of the mere use of a known process, machine

or apparatus unless such known process results in a new product or employs at least one new reactant. The Patent Manual (Chapter 08.02.01) confines Section 3(d) to chemical/pharmaceutical inventions, noting its irrelevance to mechanical devices unless analogized to "new forms" like alloys or composites. Thus, for the present mechanical reactor, Section 3(d) is prima facie inapposite, but even if invoked, the invention's enhanced corrosion resistance and hygiene qualify as improved efficacy.

Against this backdrop, the Opponent's invocation of Section 3(f) is untenable, as the invention is not a "mere arrangement" but a sophisticated, interdependent configuration yielding synergistic benefits. The complete specification (pages 1-19) describes a Clean Room Glass Lined Reactor with: (i) an MS body internally glass-lined and externally SS-overlaid via seamless welding post-2mm MS removal; (ii) machined fluidic channels for heat transfer; (iii) a glass-lined flat observation surface for non-contaminating flow monitoring; (iv) a bottom flange cavity with connection plate for efficient fluid circulation; and (v) nozzle overlays ensuring 100% enclosure against oxidation. These elements do not function "independently in a known way"—the SS overlay integrates with the channels and cavity to enhance heat efficiency while preventing corrosion penetration; the observation surface synergizes with the clean-room design for GMP compliance, reducing contamination risks. This produces novel results: prolonged reactor life (reduced distortion from oxidation), superior hygiene (no rust/dust), and economic gains (lower maintenance, downtime).

Contrasting with prior art: D1's insulated cladding operates separately for thermal isolation, without synergy for oxidation prevention; D2's wrapper with fillers functions independently for manufacturing ease, not integrated hygiene; D3's enamel focuses solely on internal strength, sans external overlays. The Opponent's claim (Written Statement, page 27) that these are "standard GLR features" ignores the inventive welding method creating a metallurgical bond, where interdependent plugs yielded patentable synergy. No expert evidence from the Opponent proves independent functioning, failing the burden per the Patent Manual.

Likewise, Section 3(d) does not bar patentability, as the invention is not a "new form of a known substance" but a novel mechanical assembly. Even analogizing the reactor as a "substance" (e.g., modified vessel), it exhibits enhanced efficacy: superior oxidation resistance (100% SS enclosure vs. prior art's gapped claddings), quantifiable through extended service life and compliance metrics, as required by Novartis (supra). The invention's technical advance—addressing clean-room nozzle oxidation undistorted internal lining—constitutes enhanced efficacy, rendering Section 3(d) inapplicable.

In conclusion, the ground under Section 25(1)(f) is frivolous, motivated by rivalry, and merits dismissal. The Applicant prays for grant of the patent and costs.

GROUND 25(1)(g)

The Applicant respectfully submits the following exhaustive and detailed observations in rebuttal to the ground raised under Section 25(1)(g) of the Patents Act, 1970. This ground alleges that the complete specification does not sufficiently and clearly describe the invention or the method by which it is to be performed, thereby violating the requirements under Section 10(4) of the Act. The Applicant contends that this objection is wholly unfounded, bereft of merit, and predicated on a superficial reading of the specification, ignoring its comprehensive, enabling disclosure tailored to empower a person skilled in the art to replicate the invention without undue experimentation.

Beginning with the statutory and jurisprudential foundation, Section 25(1)(g) empowers opposition on the premise that "the complete specification does not sufficiently and clearly describe the invention or the method by which it is to be performed." This provision dovetails with Section 10(4), which stipulates that the complete specification must "fully and particularly describe the invention and its operation or use and the method by which it is to be performed; disclose the best method of performing the invention

which is known to the applicant and for which he is entitled to claim protection; and end with a claim or claims defining the scope of the invention for which protection is claimed."

The Patent Manual (Chapter 05.03) elaborates that sufficiency entails three pillars: (i) full description enabling the person skilled in the art to perform the invention across its claimed scope without undue burden; (ii) disclosure of the best mode known to the applicant at filing; and (iii) clarity to avoid ambiguity. The Manual emphasizes that the specification must be "enabling," meaning it provides sufficient technical details, examples, or parameters for replication, but need not exhaustively detail routine variations ascertainable via common general knowledge. For mechanical inventions like the present reactor, the Manual notes that drawings, structural descriptions, and operational methods suffice, without necessitating exhaustive prototypes or data unless claims hinge on specific performance metrics.

Applying this framework, the Opponent's allegations crumble under scrutiny. The Opponent contends (Written Statement, page 31) that the specification is vague, lacking enabling instructions for SS overlay welding, technical parameters (e.g., welding current, MS removal depth), working examples, and proof for "100% glass-lined surface," rendering it non-enabling for the PSA. This is a mischaracterization; the complete specification (pages 1-19) provides a fulsome, person skilled in the art-oriented disclosure compliant with Section 10(4). Commencing with the title and preamble (page 1), it clearly states: "CLEAN ROOM GLASS LINED REACTOR... The following specification particularly describes the nature of the invention and the manner in which it is to be performed," signaling intent to enable. The Field of Invention (page 1) contextualizes: "The present invention relates to corrosion resistant, clean room glass lined overlaid stainless steel reactor. More particularly, the present invention discloses a reactor specifically designed for clean and polished working environment in food and medical industries, which don't require any paint coat the look clean as mirror finished." This sets the technical domain, identifying the

problem (corrosion in clean rooms) and solution (SS overlays), guiding the PSA.

The Background Art (pages 1-2) elaborates prior limitations: "Conventional reactors are widely used by fine chemicals and pharmaceutical industry. Corrosion, cleaning and contamination are one of the key criteria that determine the efficiency of a process... Stirred tank reactors (STRS) used in pharmaceutical and fine chemical industries are designed for providing and removing heat. This is done with coil or half coil with coolant passing through it, or using jacketed tanks." It critiques conventional designs for dust-prone paint, inefficient heat transfer, and contamination risks, establishing the inventive necessity. The Detailed Description (pages 3-10) meticulously outlines structure and method: "The present inventions also presents Mild Steel (MS) Glass-Lined reactors with stainless steel overlays on nozzles, designed for sectors like pharmaceuticals, food, chemicals, and agrochemicals... By integrating stainless steel overlays onto reactor nozzles with SS cladding around the vessel, the solution covers 100% of the reactor, offering durability for industries requiring stringent hygiene. Stainless steel enhances corrosion resistance..." Specifics include: MS body with 1-2mm glass lining; machined fluidic channels for heat transfer; clamped flanges with bolts; glass-lined observation surface for flow monitoring; bottom cavity with inlet/outlet and connection plate; seamless SS welding post-2mm MS removal using SS electrodes for metallurgical bond. This enables replication: the person skilled in the art, versed in welding, can adapt parameters (current 70-100A, as routine). No "undue experimentation" is needed.

On best mode (Section 10(4)(b)), the specification discloses the preferred: "removing approximately 2mm of MS surface and then covering this with SS by welding with SS electrodes, maintaining constant total thickness," as the optimal known method for uniform bonding and 100% enclosure. Claims (pages 11-13) are supported: Claim 1 details components (i)-(vii), with dependents on double-walling (Claim 6), smoothness (Claim 7), and ranges (Claim 8), all mirrored in description. Drawings (pages 14-19) illustrate cross-sections, reinforcing clarity.

Opponent's uncertified translations and no expert affidavits fail the burden therefore said ground, merits dismissal.

The Applicant prays for rejection of the opposition proceeding and grant of the patent.

OBJECTIONS RAISED IN HEARING NOTICE

This reply is filed post-hearing under section 15 (held on 14.11.2025) to ensure complete compliance and to facilitate the grant of the patent, in accordance with the principles of natural justice and the opportunity for rectification under Sections 14 and 57 of the Patents Act, 1970, and Rule 28(7) of the Patents Rules, 2003.

The Applicant respectfully submits that these objections are primarily formal or procedural in nature, have been substantially addressed in prior responses, and are now fully met through the enclosed amendments and justifications. No substantive issues remain that impede patentability, as the invention remains novel, inventive, and sufficiently disclosed, as elaborated in the oral submissions at the hearing. The Applicant encloses proposed amended claims, and prays for their acceptance under Section 59 without prejudice to the application's validity. Should any further clarification be required, the Applicant seeks leave to provide it.

OBJECTION - 1 "CLARITY AND CONCISENESS"

The Hearing Notice reiterates:

1. *"The following objection raised in FER has still not been met. Reference numeral shall be marked in the drawings and reference of it shall be incorporated in the claims to enhance the intelligibility of Claims. No description matter is allowed on the drawings as per Rule 15(7) of The Patents Rule, 2003 (as amended)."*
2. *"The reference numerals of the drawings are not incorporated in the claims; the applicant is required to amend the claims to include*

appropriate reference numerals corresponding to the features shown in the drawings to ensure clarity and consistency."

3. *"Abbreviated forms of words have been used in the claims and/or description. The applicant is required to replace all such abbreviations with their complete forms to ensure clarity and proper understanding of the invention, as abbreviations may lead to ambiguity or misinterpretation of the technical features described."*

The Applicant submits that these objections were substantially complied with in the FER response dated 23.09.2023, where amended claims and drawings were filed to incorporate clarity enhancements. However, to fully resolve any residual concerns and align with the Hearing Notice's directives, the Applicant encloses further refinements.

Regarding objection 1 and 2 on reference numerals: Rule 15(7)(c) of the Patents Rules, 2003, recommends (but does not mandate) numerals in claims for intelligibility, as per Patent Manual Chapter 05.05.02, which states that claims should be "clear and succinct" under Section 10(5), with numerals aiding where features are graphically depicted. The original drawings (filed 21.04.2021) already include numerals (e.g., Figure 1 labels reactor body as 10, top flange as 12, fluidic channels as 16), and no "description matter" in the drawings.

In order to enhance intelligibility, the Applicant encloses Annexure A: Proposed Amended Claims integrating all numerals (e.g., Claim 1(i): "an internally glass-lined reactor (10) sandwiched between a top flange (12) and a bottom flange (14), with fluidic channels (16) machined on the glass-lined reactor surface"). These correspond directly to drawings: e.g., observation surface (18) in Figure 2, cavity (20) in Figure 3. No new matter is added—numerals are derived from original drawings and description (pages 5-7), preserving scope per Section 59(1), addressing the FER's concern (para 2.1 of FER).

The Applicant requests these amendments be taken on record as they cure formal defects post-hearing, per Tapas Chatterjee v. Assistant Controller (2023 SCC OnLine Del 469), mandating opportunity for compliance.

OBJECTION – 2 "FORMAL REQUIREMENT(S)"

The Hearing Notice lists:

1. *"The General Power of Attorney (GPA) filed by the opposition is found to be defective... Accordingly, the opposition is directed to submit a corrected... GPA..."*
2. *"It is observed that the applicant has submitted amendments... by filing Form 13. As per Section 57... provide adequate justification... Until such compliance... cannot be taken on record..."*
3. *"It is observed that the applicant has filed Form 13... As per Section 57(2)... submit a revised Form 13 clearly stating the detailed reasons... Until compliance... cannot be taken on record."*

For objection at para 1, the applicant has no comments to offer.

For objections 2 and 3 on Form 13: The Applicant's Form 13 (23.09.2023) complied with Section 57(1) (amendments with basis) and Section 57(2) (reasons for leave) and 57(6) (on overcoming of objections), but to address the Notice's call for "adequate justification" and "detailed reasons,". It includes:

- Basis Statement (per Rule 81(1)): "Each amendment finds support in the original specification/drawings: e.g., 'seamless welded coated nozzles' from page 7, lines 18-22 ('SS overlays on nozzles welded to cladding'); 'fluidic channels' from page 5, lines 10-12 ('machined surface for heat transfer'). No new matter introduced." A tabulated matrix details:

Amended Text (e.g., Claim 1(vi))	Original Basis (Page/Line/Drawing)	Nature of Amendment (Clarification/Deletion)

Amended Text (e.g., Claim 1(vi))	Original Basis (Page/Line/Drawing)	Nature of Amendment (Clarification/Deletion)
Seamless welded coated nozzles to top cover's cladding sheet	Page 7, lines 18-22; Fig. 1 (numeral 22)	Clarification for precision per FER para 2.1

- Reasons Statement (per Section 57(2)): "Amendments respond to FER objections on clarity (para 2.1: vagueness in features) and support (para 3.2: consistency with drawings), narrowing scope for definiteness without broadening. E.g., adding numerals enhances intelligibility; expanding abbreviations avoids ambiguity. Necessitated to overcome rejection risks under Section 10(5), ensuring patentability."

These fulfill "full particulars" under Section 57(2), as in Nippon Steel v. Controller (2024 SCC OnLine Del 1234), where detailed justifications post-notice were accepted. Until now, amendments were provisional; this revision ensures recordal under Rule 81(3). The objection is procedural, not fatal, allowing post-hearing cures.

OBJECTION - 3 "INVENTION U/S 2(1)(ja)"

The Hearing Notice states: "The applicant's reply to the objection raised in the First Examination Report (FER) under Section 2(1)(ja)... has been duly considered. Upon further examination, it has been observed that the following prior art documents disclose the features of the alleged invention: D4: CN103820781A (Publication Date: 25/08/2014); D5: CN105148816A (Publication Date: 16/12/2015); D6: CN206372825U (Publication Date: 04/08/2017)." It partially describes D4: "a profiled outer glass-lined device... can lid... nozzle, manhole... jacket... cavity... jacket flange... discharge port, condensate outlet, steam inlet... outer side of the can body and the outer side of the jacket have a ……].."

The Applicant submits that this objection under Section 2(1)(ja) (inventive step) was addressed in the FER reply (23.09.2023, paras 15-25), distinguishing the citations (identical to Opponent's D1-D3). The "further examination" lacks new reasoning or evidence, merely reiterating prior art without applying the Roche Test (F. Hoffmann-La Roche v. Cipla (2016 (66) PTC 1 Del)): identify person skilled in the art, inventive concept, common general knowledge, differences, obviousness. This mechanical restatement violates Tapas Chatterjee (2023 SCC OnLine Del 469), condemning non-reasoned orders.

D4-D6 do not render the invention obvious: D4 (CN103820781A) discloses insulated cladding with gaps (translation: "filled with lagging material"), not seamless SS overlays post-MS removal for oxidation prevention—differences require inventive ingenuity, no motivation to modify for clean-room GMP. D5 (CN105148816A) uses gapped wrappers with fillers, teaching away from metallurgical bonding. D6 (CN206372825U) is internal enamel-focused, lacking external features. Synergies (e.g., observation surface + cavity + overlays) yield technical advance (hygiene, longevity), non-obvious per Novartis v. UOI (2013 6 SCC 1).

A detailed response has already being furnished under aforesaid paragraphs.

The objection may kindly be waived.

OBJECTION 4 - "SCOPE"

The Hearing Notice objects: "1. The amendment made in the description and claims does not comply with the provisions of Section 59(1) of the Act."

The Applicant submits that all amendments filed via Form 13 dated 23.09.2023 fully comply with Section 59(1), which prohibits amendments introducing new matter beyond the original specification. The amendments were clarificatory, narrowing scope in response to FER objections on clarity and consistency, without expansion. For instance, addition of "seamless

welded coated nozzles" is verbatim supported by original description (page 7: "SS overlays on nozzles welded to cladding"), and incorporation of numerals references existing drawings without novelty. As per Patent Manual Chapter 11.03, amendments are allowable if "fairly based" on original disclosure, as here.

The objection may kindly be waived.

OBJECTION – 5 "SCOPE"

The Hearing Notice raises three objections under Scope:

1. *"Claim 1 recites functional advantages such as "offering excellent resistance to oxidation,"... Such statements... do not constitute patentable subject matter under Section 2(1)(j)... amend to clearly define the structural or technical features..."*

The Applicant submits that Claim 1 defines structural features (e.g., "seamless welded coated nozzles... fully enclosing the reactor in stainless steel"), with functional phrases as characterizing effects inherent to structure, permissible under Section 10(5) if defining scope. Patent Manual Chapter 05.05.01 allows functional language if tied to structure, as here—oxidation resistance results from SS enclosure. To address, enclosed amended Claim 1 rephrases: "wherein the coated nozzle is seamlessly welded to the top cover's cladding sheet, fully enclosing the reactor in stainless steel to provide resistance to oxidation at the nozzle." This retains essence without functional dominance.

"*Claims 2 and 7 do not clearly and unambiguously define the scope... term "base body of high mechanical strength" is vague... expressions "essentially," "preferably,"... "smooth as whole" is indefinite... amend... to specify... in objective terms..."*

The terms are clear to PSA: "high mechanical strength" implies standard MS properties (yield strength >250 MPa, CGK in reactor design);

"essentially/preferably" indicate ranges (1-2mm thickness), allowable per Patent Manual Chapter 05.05.03 for flexibility. "Smooth as whole" means uniform glass lining without defects, objective via industry standards (e.g., Ra <0.8 µm). Claim 2 specifies "base body having mechanical yield strength of at least 250 MPa"; removes "essentially/preferably" for fixed ranges; Claim 7: "smooth interior surface with roughness average (Ra) less than 0.8 µm." This ensures definiteness under Section 10(4)(c).

3. *"It is observed that there is an inconsistency between the description, claims, and drawings... outer wall as stainless steel (SS) and the inner as mild steel (MS)... in the drawings, the layers appear inversed... rectify the discrepancy..."*

The Applicant clarifies: description/claims correctly state inner MS (base) with glass lining, outer SS overlay (page 3: "MS Glass-Lined reactors with stainless steel overlays"). Drawings (Fig. 1) label inner as MS (10), outer as SS (24), and no inversion. To resolve, enclosed revised drawings (Annexure B) with explicit labels: "Inner: Mild Steel (MS) with Glass Lining; Outer: Stainless Steel (SS) Overlay. The amendments are consistent per Section 10(4). The objection may kindly be waived

Objection 6 - "SUFFICIENCY OF DISCLOSURE U/S 10 (4)"

The Hearing Notice raises:

1. *"It is observed that references have been made in the description to Figures 4 and 5. However, these figures are not included... file the missing Figures 4 and 5..."*

References to Figs. 4-5 in description (page 8) were typographical errors, intended as Figs. 2-3 (alternative views of cavity/nozzles). Original filing included only Figs. 1-3; no missing figures. Amended description corrects to "Figs. 2 and 3," removing ambiguity. As Per Patent Manual Chapter 05.03, typographical corrections are allowable under Section 57 without new matter.

2. *"It is observed that certain structural features... such as the top flange, bottom flange... have been described... but... not shown or illustrated in the drawings... incorporate all the described structural features in the drawings..."*

All features are illustrated: top/bottom flanges (12/14 in Fig. 1), observation surface (18 in Fig. 2), cavity (20 in Fig. 3), connection plate (22), bolts (26), SS sheet (24). If perceived absent, due to schematic nature. This ensures full disclosure under Section 10(4), enabling person skilled in the art replication.

The objection may kindly be waived.

The Applicant prays for acceptance and grant.

Yours Sincerely
Of DASWANI & DASWANI



(Hemant Daswani)
Applicant's Agent
IN / PA 610

ENCLOSURES :

1. Clean copy of claims.
2. Marked-Up copy of claims (clean claims vs claims filed at the time of filing reply to First Examination Report).
3. Fresh drawings.

Digitally Signed By:
MR. BHARAT DASWANI

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November 28, 2025

Kind Attention:
Dr. Kanhaiya Lal Balai
Assistant Controller of Patents & Designs

Your Ref: POC/Application No / 202141018455

Our ref: 202141018455

To
The Controller of Patents
The Patent Office,
Chennai / Delhi

Dear Sir,
Re: Indian Patent Application No. 202141018455 of 21.04.2021 relating to "CLEAN ROOM GLASS LINED REACTOR" in the name of Standard Glasslining Technology Private Limited

In response to your hearing notice dated 13-10-2025 regarding Pre-Grant Opposition, applicant's Counsel (Patent Agent) Mr. Hemant Daswani, IN/PA 610 attended the hearing fixed on 14-11-2025 as per schedule mentioned in the hearing notice.

WRITTEN SUBMISSIONS ON BEHALF OF THE APPLICANT IN THE PRE-GRANT OPPOSITION PROCEEDINGS UNDER SECTION 25(1) OF THE PATENTS ACT, 1970

Re: Pre-Grant Opposition under Section 25(1) of the Patents Act, 1970 against Patent Application No. 202141018455 titled "Clean Room Glass Lined Reactor" filed by Standard Glass Lining Technology Private

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Limited (Applicant) opposed by Pinakin Engineers Private Limited (Opponent).

Date of Hearing: 14th November 2025 at 11:30 HRS (IST) via Video Conferencing.

OPENING REMARKS

The Applicant, Standard Glass Lining Technology Private Limited, through its authorized Patent Agent, Mr. Hemant Daswani (IN/PA/610), hereby submits these exhaustive written submissions in response to the Pre-Grant Opposition filed by the Opponent on 14th November 2024, as well as in compliance with the Hearing Notice dated 13th October 2025 issued under Rule 55(4) of the Patents Rules, 2003.

These submissions address all grounds raised by the Opponent under Section 25(1)(b), (e), (f), and (g) of the Patents Act, 1970, refute each statement made in the Opponent's Written Statement and evidence (Exhibits A to G), and respond to the objections outlined in the Hearing Notice regarding clarity and conciseness, formal requirements, and inventive step under Section 2(1)(ja).

The Applicant humbly requests the Learned Controller to dismiss the opposition in its entirety, as it is frivolous, mala fide, and lacks merit, and to proceed with the grant of the patent, subject to any minor amendments if deemed necessary for formal compliance. The Applicant reserves the right to file further evidence or amendments as per the provisions of the Act and Rules, and seeks an opportunity to be heard orally at the scheduled hearing.

The present application, filed on 21st April 2021 and published on 28th October 2022, discloses a novel Clean Room Glass Lined Reactor specifically designed for pharmaceutical and food-grade processing environments, where corrosion resistance, hygiene, and contamination control are paramount. The invention addresses a long-standing technical problem in conventional glass-lined reactors: oxidation and corrosion at nozzles and

external surfaces in clean-room settings, which can lead to distortion of the internal glass lining, product contamination, and non-compliance with stringent standards like ISO 14644-1 and GMP.

The core inventive concept lies in a mild steel (MS) reactor body internally glass-lined and externally overlaid with stainless steel (SS) sheets through a seamless welding process, particularly extending to nozzles, ensuring 100% enclosure in SS to prevent oxidation while maintaining structural integrity and heat transfer efficiency. This is achieved by machining fluidic channels on the reactor surface, integrating a glass-lined flat observation surface below the top flange for flow regime monitoring, and incorporating a cavity in the bottom flange for heat transfer fluid circulation, all clamped securely between top and bottom flanges with bolts.

The specification provides enabling disclosure, including detailed description, claims, and drawings, supported by the method of performance, enabling a person skilled in the art (a mechanical engineer familiar with reactor design in chemical/pharmaceutical industries) to practice the invention without undue experimentation.

In contrast, the Opponent's opposition is an attempt to delay the grant due to commercial rivalry, as both parties operate in the glass-lined equipment manufacturing sector in Gujarat. The Opponent fails to establish locus standi as a "person interested" under Section 25(1), merely claiming to be a competitor without demonstrating any tangible interest beyond rivalry, such as prior use or direct economic impact, as required by precedents like Ajay Industrial Corporation v. Controller of Patents (1984 PTC 234), where the court held that mere commercial rivalry does not confer standing without proof of specific interest. Furthermore, the opposition is mala fide, as evidenced by its reliance on unsubstantiated assertions, uncertified translations of foreign prior art, and superficial tabular comparisons without technical analysis, violating the principles laid down in Hindustan Lever Ltd. v. Godrej Soaps Ltd. (1996 PTC 141), where oppositions motivated by delay tactics were dismissed with costs. The Applicant seeks dismissal on these

preliminary grounds alone, along with compensatory costs under Section 35 of the Act for defending this baseless challenge.

GROUND 25(1)(b)

Turning to the substantive rebuttal, the Applicant addresses the Opponent's grounds under Section 25(1)(b) for lack of novelty. The Opponent alleges anticipation by prior art documents D1 (CN103820781A, priority 12.03.2014), D2 (CN105148816A, priority 08.09.2015), and D3 (CN206372825U, priority 12.01.2017), claiming that these disclose all features of the claims through tabular comparisons in their Written Statement. The Applicant humbly reiterates and submits the following detailed observations in rebuttal to the ground raised under Section 25(1)(b) of the Patents Act, 1970, as articulated in the Opponent's Written Statement and as prima facie upheld by the Learned Controller in the Report dated 04th June 2025 (hereinafter "Prima Facie Report").

This ground alleges that the subject matter of the present application lacks novelty due to anticipation by prior publications, specifically D1 (CN103820781A, published 28th May 2014), D2 (CN105148816A, published 16th December 2015), and D3 (CN206372825U, published 4th August 2017). The Applicant respectfully contends that this ground is entirely misconceived, unsubstantiated, and fails to meet the stringent evidentiary and analytical thresholds mandated by Indian patent law for establishing anticipation. The rebuttal is structured to first elucidate the legal framework governing novelty assessments, followed by a meticulous, feature-by-feature dissection of the cited documents vis-à-vis the inventive features of the present application, demonstrating unequivocally that no single prior art document discloses the invention as a whole in an enabling and unambiguous manner. This analysis draws upon the complete specification of the present application (filed on 21st April 2021, as amended in response to the First Examination Report dated 23rd March 2023), the full English translations of D1, D2, and D3, relevant precedents from Indian courts, and

guidelines from the Manual of Patent Office Practice and Procedures, 2019 (hereinafter "Patent Manual").

At the outset, it is imperative to recall the legal principles from Section 25(1)(b), which provides that a patent may be opposed if the invention "has been published before the priority date of the claim (i) in any specification filed in pursuance of an application for a patent made in India on or after the 1st day of January, 1912; or (ii) in India or elsewhere, in any other document." As per Section 2(1)(l), an invention is "new" if it does not form part of the state of the art, and anticipation requires that the prior art must disclose the invention in its entirety, with such clarity and completeness that a person skilled in the art could perform it without further ingenuity or experimentation. The Hon'ble Supreme Court in Bishwanath Prasad Radhey Shyam v. Hindustan Metal Industries (AIR 1982 SC 1444) emphatically held that for anticipation, the prior publication must contain "clear and unmistakable directions" to perform the invention, and mere suggestions or incomplete disclosures do not suffice. Similarly, in Lallubhai Chakubhai Jariwala v. Chimanlal Chunilal & Co. (AIR 1936 Bom 99), the court clarified that anticipation cannot be established by mosaicking disparate elements from multiple documents; each prior art must be evaluated independently as a standalone disclosure.

The Patent Manual (Chapter 08.03.01) reinforces this by stating that "the prior art should disclose the invention either in explicit or implicit manner" and must enable the skilled person to arrive at the invention without undue burden. Critically, foreign patent documents like D1, D2, and D3 must be accompanied by certified translations and proof of public availability in India prior to the priority date (21st April 2021), failing which they are inadmissible, as per Rule 55(3) of the Patents Rules, 2003, and precedents such as Snehalata C. Patel v. Controller of Patents (2001 PTC 121). In the instant case, the Opponent has furnished uncertified machine translations (Exhibits E, F, G), which are riddled with inaccuracies (e.g., "foreskin" in D2 likely mistranslating "wrapper" or "sheath"), and no evidence of their dissemination in India, rendering them prima facie unreliable. Even

assuming arguendo their admissibility, a granular analysis reveals profound dissimilarities, as elaborated below.

The inventive core of the present application, in the complete specification (pages 1-19) and amended claims (dated 23rd September 2023), resides in a "Clean Room Glass Lined Reactor" tailored for pharmaceutical and food-grade environments under stringent hygiene standards like ISO 14644-1 and GMP. The reactor comprises: (i) an internally glass-lined mild steel (MS) body clamped between top and bottom flanges with machined fluidic channels on the surface for enhanced heat transfer; (ii) a glass-lined metal flat surface below the top flange enabling non-invasive observation of internal flow regimes to prevent contamination; (iii) a cavity in the bottom flange for heat transfer fluid circulation via inlet/outlet nozzles, closable with a metal plate; (iv) a connection plate linking inlet/outlet to the bottom flange; (v) an external stainless steel (SS) sheet overlay welded seamlessly to enclose the reactor fully, preventing oxidation and rust at critical zones; (vi) internally glass-lined nozzles with seamless SS overlay welded to the cladding, addressing the specific problem of nozzle oxidation distorting internal lining in clean rooms; (vii) a double-walled vessel design with smooth glass-lined interior; and (viii) application-specific volumes (e.g., 50-5000 liters) and heat transfer areas.

This configuration solves the longstanding issue of external corrosion in conventional reactors, where spillage in clean rooms leads to oxidation penetrating to the internal glass lining, causing distortion, contamination, and regulatory non-compliance. The manufacturing method involves removing approximately 2mm of MS surface before welding SS overlays with SS electrodes to form a metallurgical bond, maintaining uniform thickness and 100% SS enclosure without gaps or insulation voids that could harbor contaminants. None of D1, D2, or D3—each addressing generic glass-lined vessels for non-clean-room applications—discloses this integrated, synergistic arrangement, as evidenced by their divergent teachings.

Commencing with D1 (CN103820781A, titled "Glass Lined Equipment Provided with Outer Profiling Cladding and Manufacturing Method"), this document describes a conventional glass-lined tank comprising a tank cover (with nozzles, manhole, and seal) and tank body encased in a jacket for heat exchange, with external stainless steel "profiling cladding" layers applied for insulation purposes. The full description (*from unauthenticated English translation*) reveals that the cladding is formed by cutting and shaping 304 stainless steel sheets (1.5-3mm thick) to match the tank's profile, welding joints with argon arc welding, and filling gaps between the jacket and cladding with polyurethane foam or silicate for thermal insulation. The tank body and jacket are welded into a unitary structure with a cavity for steam/condensate flow, and bottom ports include drain hole, condensation-water drain, steam inlet, and vapor outlet. However, D1 fundamentally diverges from the present invention in several critical aspects, precluding anticipation. Firstly, D1's "cladding" is a detachable, insulated outer layer with deliberate gaps filled for thermal purposes, not a seamless, metallurgically bonded SS overlay welded directly after MS surface removal to prevent oxidation in clean-room settings. The translation explicitly states: "Between described chuck and stainless steel surrounding layer, be filled with lagging material and form thermal insulation layer," emphasizing insulation over corrosion prevention at nozzles or flanges.

In contrast, the present application's SS overlay is seamless, extending to nozzles without insulation gaps, ensuring 100% enclosure to eliminate rust formation at critical zones—a feature absent in D1, where cladding joints are merely butt-welded and polished, leaving potential entry points for moisture. Secondly, D1 lacks any disclosure of machined fluidic channels on the reactor surface, a glass-lined flat observation surface for flow monitoring, or a bottom flange cavity with connection plate for targeted heat transfer; instead, it relies on a standard jacketed cavity with bottom ports, without the clamped flange assembly or observation capability essential for clean-room hygiene. The manhole in D1 is for access, not flow observation, and there is no mention of a flat glass-lined surface below the top flange to

enable visual inspection without opening, which prevents contamination risks in pharmaceutical processes.

Thirdly, D1's manufacturing method involves preliminary impact briquetting of sheets, argon welding (current 70-80A, voltage 7-9V), and "V"-type breach repairs for flanges, but omits the inventive step of removing 2mm MS surface for uniform thickness and metallurgical bonding, focusing instead on profile consistency for insulation. As per the Patent Manual (Chapter 08.03.02), anticipation requires "implicit or explicit" disclosure of all essential features; here, D1's generic insulated cladding for cost reduction in non-pharma contexts does not implicitly teach the oxidation-resistant, seamless nozzle overlay for GMP compliance.

The Opponent's tabular comparison (Written Statement, pages 12-15) superficially maps D1's "surrounding layer" to the claims but ignores functional disparities—D1's layer allows gaps for filling, whereas the invention mandates seamless welding to preclude voids. In Novartis AG v. Union of India ((2013) 6 SCC 1), though pharmaceutical, the principle extends: mere structural similarity without identical purpose or effect does not anticipate. Thus, D1 falls short of providing "clear and unmistakable directions" (Bishwanath Prasad, supra) to arrive at the clean-room reactor.

Shifting to D2 (CN105148816A, titled "Manufacturing Method of Glass-Lined Reaction Kettle"), this patent outlines a process for fabricating a glass-lined kettle with an external stainless steel "foreskin" (translated as "wrapper" or "sheath") to enhance corrosion resistance. The method encompasses seven steps: assembling enamel-fired carbon steel components (upper head, cylinder, lower head); pressing SS sheets into upper, middle, and lower foreskins; perforating for process ducts; fixing foreskins by welding to steel flanges on nozzles; filling insulation material (e.g., glass cement) in gaps and seal-welding; polishing the SS surface; and adding an annular SS cover. The translation details that foreskins are oversized (diameter larger by 2-4mm at holes) for wrapping, with steel flanges halved and welded on nozzle flanges to enable SS welding, and insulation filled to isolate carbon steel. While D2

superficially addresses external corrosion via SS wrapping, it does not anticipate the invention due to profound structural and functional mismatches.

Primarily, D2's "foreskin" is a detachable wrapper with deliberate gaps filled post-assembly for insulation, not a seamless welded overlay integrated metallurgically after MS surface machining. The description states: "Fill insulant material in the gap between stainless steel foreskin and enamel, and seal welding," implying voids are inherent and filled secondarily with materials like glass cement, which the present invention eschews to avoid relative displacement and re-emerging gaps—issues D2 acknowledges but resolves inadequately via cement, leading to potential contamination in clean rooms. The invention's seamless welding (using SS electrodes post-2mm MS removal) creates a continuous, rust-proof enclosure without fillers, a distinction not disclosed in D2, where welding is limited to foreskin-flange joints (e.g., "upper foreskin is welded with steel flange").

Secondly, D2 omits key elements like fluidic channels machined on the reactor surface, the glass-lined observation flat surface, bottom flange cavity with connection plate, or nozzle-specific seamless SS overlays for oxidation prevention; its process ducts are merely perforated and flanged, without extended SS welding to cladding for 100% enclosure. The kettle is for general chemical reactions, not clean-room pharma use, lacking GMP-focused hygiene features. The manufacturing steps emphasize inversion for filling (e.g., "The upper cover being surrounded by described upper foreskin in described step e is inverted, fill insulant material in space up between foreskin and upper cover"), prioritizing insulation over oxidation resistance, whereas the invention's method ensures dimensional uniformity and bond strength for longevity in sterile environments. As per F. Hoffmann-La Roche Ltd. v. Cipla Ltd. ((2016) 66 PTC 1 (Del)), anticipation demands identical disclosure, not obvious modifications; D2's wrapper, reliant on secondary fillers, does not enable the skilled person to perform the seamless, gap-free overlay without ingenuity. The Opponent's assertions (Written Statement, pages 16-18) conflate D2's "foreskin" with the claims but overlook that D2's

design permits air/moisture entry pre-filling, antithetical to the invention's preventive enclosure. Hence, D2 does not destroy novelty.

Finally, D3 (CN206372825U, a utility model titled "A Kind of High-Strength Corrosion-Resisting Glassed Steel Reaction Vessels"), pertains to a jacketed glass-lined reactor with enhanced compressive strength via seamless welding neck flanges at ports. The description covers a tank body with jacket for heat exchange (steam/liquid inlets/outlets), agitator, viewing manhole, thermometer nozzle, charging/discharge ports, and standby ports, all featuring neck flanges with grooved end faces, chamfers, and bolted covers/gaskets for sealing. Corrosion resistance is achieved via internal ground glaze and cover coat layers (with specific frit compositions including silica 40-45%, alumina 3-7%, etc., and mixed adhesive agents like bastnaesite flour 45-55%), extending to flange inner walls and joints. However, D3 is devoid of any external SS overlay or cladding, let alone seamless welding for clean-room oxidation prevention, rendering it wholly non-anticipatory.

The utility model's focus is internal enamel enhancement for strength and anti-corrosion, with "high-neck flanges" welded seamlessly to reduce punch-formed openings and bolster pressure resistance (e.g., "The viewing manhole, thermometer nozzle... use seamless welding neck flanges"). There is no mention of external MS body with SS welding, machined fluidic channels, observation flat surface, bottom cavity, or nozzle overlays; the external jacket is bare, with mounting devices but no SS enclosure. The glaze formula, while improving internal lining adhesion (e.g., melting at 900-1000°C), is irrelevant to the invention's external hygiene features. As a utility model, D3 lacks enabling disclosure for manufacturing beyond basic assembly, omitting the inventive welding post-MS removal. The Patent Manual (Chapter 08.03.03) requires prior art to be enabling; D3's schematic utility focus does not provide "clear directions" for the claimed reactor.

The Opponent's mapping (Written Statement, pages 19-20) erroneously equates D3's internal enamel to the claims, ignoring the absence of external

SS elements. In Tapas Chatterjee v. Assistant Controller (2023 SCC OnLine Del 469), the court rejected anticipation based on partial, non-enabling disclosures; similarly, D3's internal-centric design cannot anticipate the external clean-room innovations.

Feature from Applicant's Claims (as amended)	Teaching in Applicant's Specification	D1 (CN103820781 A)	D2 (CN105148816 A)	D3 (CN206372825 U)	Analysis (Why Not Anticipatory)
Internally glass-lined MS reactor clamped between top and bottom flanges with machined fluidic channels on surface (Claim 1(i))	Pages 3-5: Reactor body of MS with internal glass lining (1-2mm thick); fluidic channels machined for heat transfer; clamped via bolts for secure assembly.	Discloses general glass-lined can body with jacket, but no machined fluidic channels on reactor surface; focuses on insulated cladding (translation p.90).	Describes reaction kettle with glass lining, but channels are for jacket circulation, not machined on reactor surface (translation p.96).	Utility model for reactor with flanges; no mention of machined channels (translation p.102).	No single document discloses machined channels integrated with clamping for enhanced heat transfer in clean-room setup;.
Glass-lined metal flat surface below top flange for flow regime observation (Claim 1(ii))	Pages 5-6: Flat surface allows visual monitoring of internal mixing without contamination risk.	No observation surface; manhole is for access, not flow observation (p.91).	No such feature; focuses on manufacturing steps (p.97).	Absent entirely (p.103).	Unique to Applicant; solves clean-room monitoring issue not addressed in prior art,.
Cavity in bottom flange for heat transfer fluid circulation via inlet/outlet nozzles, closable with metal plate (Claim 1(iii))	Pages 6-7: Cavity design ensures efficient cooling/heating; connected via plate (Claim 1(iv)).	Condensate outlet and steam ports, but no bottom flange cavity (p.92).	Bottom discharge, no cavity in flange (p.98).	Flange for strength, no cavity (p.104).	Prior art uses jacketed designs, not flange-integrated cavity;
External SS sheet overlay by welding, seamless on nozzles to prevent	Pages 7-9: 2mm MS removal, SS welding for metallurgical bond; extends to nozzles for	Profiling cladding with gaps/insulation; not seamless weld (p.93).	Detachable wrapper with insulation; no welding to nozzles (p.99).	No overlay; corrosion via materials (p.105).	Applicant's method prevents oxidation at critical zones, unlike prior

Feature from Applicant's Claims (as amended)	Teaching in Applicant's Specification	D1 (CN103820781A)	D2 (CN105148816A)	D3 (CN206372825U)	Analysis (Why Not Anticipatory)
oxidation (Claim 1(v)-(vii))	100% enclosure.				art's gapped claddings;
Double-walled design, smooth interior, volume/heat transfer ranges (Claims 6-8)	Pages 10-12: Ensures hygiene, compliance with ISO/GMP.	Single jacket with insulation; no ranges (p.94).	Manufacturing focus; no double-wall specifics (p.100).	Utility; no such claims.	Synergistic advantages not disclosed;.

In summation, none of D1, D2, or D3—individually or otherwise—anticipates the invention, as they lack the complete, enabling disclosure of the synergistic features. The Applicant prays for rejection of this ground, emphasizing the mala fide nature of the opposition driven by commercial rivalry, warranting dismissal with costs.

GROUND 25(1)(e)

The Applicant respectfully submits the following ground posits that the invention, as claimed in the present application, is obvious to a person skilled in the art and thus lacks an inventive step under Section 2(1)(ja). The Applicant vehemently contests this allegation, asserting that it is fundamentally flawed, bereft of substantive evidence, and predicated on impermissible hindsight reconstruction, selective mosaicking of prior art, and a failure to adhere to the rigorous analytical framework mandated by Indian patent jurisprudence. This rebuttal is structured to commence with a comprehensive exposition of the legal principles governing inventive step assessments in India, drawing upon statutory provisions, judicial precedents, and guidelines from the Manual of Patent Office Practice and Procedures, 2019 (hereinafter "Patent Manual"). Thereafter, it applies the established five-step test from F. Hoffmann-La Roche Ltd. v. Cipla Ltd. ((2016) 66 PTC 1 (Del))—hereinafter the "Roche Test"—to the facts at hand, demonstrating that the cited prior art documents D1 (CN103820781A,

published 28th May 2014), D2 (CN105148816A, published 16th December 2015), and D3 (CN206372825U, published 4th August 2017) neither individually nor in combination render the invention obvious.

To contextualize, Section 25(1)(e) empowers opposition on the basis that "the invention so far as claimed in any claim of the complete specification is obvious and clearly does not involve any inventive step, having regard to the matter published as mentioned in clause (b) or having regard to what was used in India before the priority date of the applicant's claim." Section 2(1)(ja) defines "inventive step" as "a feature of an invention that involves technical advance as compared to the existing knowledge or having economic significance or both and that makes the invention not obvious to a person skilled in the art."

This dual-pronged criterion—technical advance/economic significance coupled with non-obviousness—distinguishes Indian law from counterparts. The Patent Manual (Chapter 08.03.04) elucidates that inventive step assessment must consider the invention as a whole, avoiding hindsight bias, and requires the opponent to establish a clear pathway by which a person skilled in the art possessing common general knowledge in the field would arrive at the invention from the prior art without inventive ingenuity. Crucially, combinations of prior art are permissible only if there exists a "teaching, suggestion, or motivation" to combine them, to prevent ex post facto rationalizations.

The landmark Roche Test, affirmed in multiple decisions, provides a structured methodology: (1) Identify the inventive concept claimed; (2) Assume the mantle of the person skilled in the art and impute common general knowledge; (3) Identify differences between the cited prior art and the invention; (4) Determine if those differences constitute steps obvious to the person skilled in the art without hindsight; and (5) Consider secondary indicia like commercial success, long-felt need, or failure of others. Recent jurisprudence, such as Tapas Chatterjee v. Assistant Controller of Patents (2023 SCC OnLine Del 469), condemns mechanical rejections, mandating

reasoned orders that eschew subjective opinions and improper prior art combinations without motivation. The Opponent's approach—mere tabular feature mappings without teaching, suggestion, or motivation analysis or person skilled in the art imputation—violates these tenets, rendering their case untenable.

Applying the Roche Test to the present invention, titled "Clean Room Glass Lined Reactor," the inventive concept (Step 1) is a specialized reactor for pharmaceutical/food-grade clean rooms, comprising an internally glass-lined MS body with machined fluidic channels, clamped between flanges; a glass-lined flat observation surface for contamination-free flow monitoring; a bottom flange cavity with connection plate for efficient heat transfer; and a seamless SS overlay welded post-2mm MS removal, extending to nozzles for 100% enclosure against oxidation, ensuring GMP/ISO compliance, longevity, and hygiene without insulation gaps or fillers. This addresses a long-felt need: preventing nozzle oxidation distorting internal lining in sterile environments, yielding technical advances (enhanced corrosion resistance, visual monitoring) and economic significance (reduced downtime, maintenance costs). The complete specification (pages 1-19, as amended) enables this via detailed construction and method, far beyond obvious modifications.

Assuming the person skilled in the art's mantle (Step 2), the person skille din the art is an engineer with experience in chemical reactor design for pharma industries, cognizant of CGK like jacketed glass-lining for corrosion, SS claddings for durability, and basic welding techniques, but not advanced metallurgical bonding or clean-room specifics unless explicitly taught. CGK includes standards like ISO 14644-1 but excludes patented innovations or rare combinations.

Identifying differences (Step 3) from D1-D3 (collectively or individually, as the Opponent urges conjoint reading without justification): D1 teaches insulated SS "profiling cladding" with polyurethane-filled gaps for general glass-lined tanks, prioritizing thermal insulation over oxidation prevention;

no machined channels, observation surface, flange cavity, or seamless nozzle overlays—cladding is gapped, detachable. D2 describes SS "foreskin/wrapper" manufacturing with insulation fillers (glass cement) and flanged perforations, but gaps persist pre-filling, no MS removal for bonding, absent observation/cavity features, focused on generic kettles. D3, a utility model, enhances internal enamel with frit compositions and neck flanges for strength, entirely lacking external SS overlays or clean-room elements. The key differences like invention's metallurgical SS-MS bond (post-removal welding) vs. prior art's gapped/insulated claddings; integrated observation and cavity for hygiene/heat transfer vs. absent; nozzle-specific seamless extension for 100% enclosure vs. partial flanging; clean-room GMP focus vs. general use.

Assessing obviousness (Step 4): These differences are non-obvious, as no "teaching, suggestion, or motivation" exists in common general knowledge or prior art to combine/modify. D1's insulation objective (gaps for foam) teaches away from seamless bonding, as fillers would compromise sterility; modifying to remove gaps requires hindsight from the invention. D2's wrapper with secondary cement filling acknowledges displacement risks, not suggesting precise MS removal/welding for uniformity—person skille din the art would not "obviously try" this without inventive spark. D3's internal focus provides no bridge to external overlays. Opponent's assertions are conclusory, ignoring divergent purposes (insulation vs. oxidation prevention). Secondary indicia (Step 5) is that the invention fulfills long-felt pharma need for rust-free reactors, evidenced by adoption; Opponent's rivalry underscores commercial value.

The Opponent's evidence—uncertified translations, no expert affidavits—fails the burden under Rule 55(3). Their mala fide intent, as competitors, merits costs per Hindustan Lever Ltd. v. Godrej Soaps Ltd. (1996 PTC 141). The Applicant prays for rejection of this ground with awarded costs.

GROUND 25(1)(f)

Turning to the substantive rebuttal, the Applicant addresses the Opponent's grounds under Section 25(1)(f) ground asserts that the subject matter of the claims in the present application does not constitute an "invention" as defined under Section 2(1)(j) of the Act or is otherwise not patentable, specifically invoking exclusions under Sections 3(d) and 3(f) of the Act. The Applicant submits that this objection is profoundly misguided, lacking in evidentiary support, and premised on a misapplication of the statutory exclusions, which are intended to prevent the patenting of trivial or non-inventive subject matter but do not apply to the genuine technical advancement embodied in the present invention.

Specifically, Section 3(f) excludes "the mere arrangement or re-arrangement or duplication of known devices each functioning independently of one another in a known way," targeting aggregations devoid of synergistic interaction. As per the Patent Manual (Chapter 08.02.02), this provision prevents patenting of collocations where components operate separately without producing a novel combined effect, but expressly permits patents for combinations yielding a new or improved result through interdependence. In TVS Motor Company Ltd. v. Bajaj Auto Ltd. (2009 (40) PTC 689 (Mad)), where the Madras High Court upheld a patent for a twin-spark plug engine, ruling that even if individual elements are known, their specific configuration producing improved combustion efficiency constitutes a patentable invention, not a mere rearrangement. More recently, in Nippon Steel Corporation v. Controller of Patents (2024 SCC OnLine Del 1234, decided in March 2024), the Delhi High Court clarified that for Section 3(f) objections, the Controller must demonstrate independent functioning of components without hindsight, emphasizing that mechanical inventions involving integrated structural modifications are patentable if they solve a technical problem in a non-obvious manner.

Turning to Section 3(d), it precludes "the mere discovery of a new form of a known substance which does not result in the enhancement of the known efficacy of that substance or the mere discovery of any new property or new use for a known substance or of the mere use of a known process, machine

or apparatus unless such known process results in a new product or employs at least one new reactant. The Patent Manual (Chapter 08.02.01) confines Section 3(d) to chemical/pharmaceutical inventions, noting its irrelevance to mechanical devices unless analogized to "new forms" like alloys or composites. Thus, for the present mechanical reactor, Section 3(d) is prima facie inapposite, but even if invoked, the invention's enhanced corrosion resistance and hygiene qualify as improved efficacy.

Against this backdrop, the Opponent's invocation of Section 3(f) is untenable, as the invention is not a "mere arrangement" but a sophisticated, interdependent configuration yielding synergistic benefits. The complete specification (pages 1-19) describes a Clean Room Glass Lined Reactor with: (i) an MS body internally glass-lined and externally SS-overlaid via seamless welding post-2mm MS removal; (ii) machined fluidic channels for heat transfer; (iii) a glass-lined flat observation surface for non-contaminating flow monitoring; (iv) a bottom flange cavity with connection plate for efficient fluid circulation; and (v) nozzle overlays ensuring 100% enclosure against oxidation. These elements do not function "independently in a known way"—the SS overlay integrates with the channels and cavity to enhance heat efficiency while preventing corrosion penetration; the observation surface synergizes with the clean-room design for GMP compliance, reducing contamination risks. This produces novel results: prolonged reactor life (reduced distortion from oxidation), superior hygiene (no rust/dust), and economic gains (lower maintenance, downtime).

Contrasting with prior art: D1's insulated cladding operates separately for thermal isolation, without synergy for oxidation prevention; D2's wrapper with fillers functions independently for manufacturing ease, not integrated hygiene; D3's enamel focuses solely on internal strength, sans external overlays. The Opponent's claim (Written Statement, page 27) that these are "standard GLR features" ignores the inventive welding method creating a metallurgical bond, where interdependent plugs yielded patentable synergy. No expert evidence from the Opponent proves independent functioning, failing the burden per the Patent Manual.

Likewise, Section 3(d) does not bar patentability, as the invention is not a "new form of a known substance" but a novel mechanical assembly. Even analogizing the reactor as a "substance" (e.g., modified vessel), it exhibits enhanced efficacy: superior oxidation resistance (100% SS enclosure vs. prior art's gapped claddings), quantifiable through extended service life and compliance metrics, as required by Novartis (supra). The invention's technical advance—addressing clean-room nozzle oxidation undistorted internal lining—constitutes enhanced efficacy, rendering Section 3(d) inapplicable.

In conclusion, the ground under Section 25(1)(f) is frivolous, motivated by rivalry, and merits dismissal. The Applicant prays for grant of the patent and costs.

GROUND 25(1)(g)

The Applicant respectfully submits the following exhaustive and detailed observations in rebuttal to the ground raised under Section 25(1)(g) of the Patents Act, 1970. This ground alleges that the complete specification does not sufficiently and clearly describe the invention or the method by which it is to be performed, thereby violating the requirements under Section 10(4) of the Act. The Applicant contends that this objection is wholly unfounded, bereft of merit, and predicated on a superficial reading of the specification, ignoring its comprehensive, enabling disclosure tailored to empower a person skilled in the art to replicate the invention without undue experimentation.

Beginning with the statutory and jurisprudential foundation, Section 25(1)(g) empowers opposition on the premise that "the complete specification does not sufficiently and clearly describe the invention or the method by which it is to be performed." This provision dovetails with Section 10(4), which stipulates that the complete specification must "fully and particularly describe the invention and its operation or use and the method by which it is to be performed; disclose the best method of performing the invention

which is known to the applicant and for which he is entitled to claim protection; and end with a claim or claims defining the scope of the invention for which protection is claimed."

The Patent Manual (Chapter 05.03) elaborates that sufficiency entails three pillars: (i) full description enabling the person skilled in the art to perform the invention across its claimed scope without undue burden; (ii) disclosure of the best mode known to the applicant at filing; and (iii) clarity to avoid ambiguity. The Manual emphasizes that the specification must be "enabling," meaning it provides sufficient technical details, examples, or parameters for replication, but need not exhaustively detail routine variations ascertainable via common general knowledge. For mechanical inventions like the present reactor, the Manual notes that drawings, structural descriptions, and operational methods suffice, without necessitating exhaustive prototypes or data unless claims hinge on specific performance metrics.

Applying this framework, the Opponent's allegations crumble under scrutiny. The Opponent contends (Written Statement, page 31) that the specification is vague, lacking enabling instructions for SS overlay welding, technical parameters (e.g., welding current, MS removal depth), working examples, and proof for "100% glass-lined surface," rendering it non-enabling for the PSA. This is a mischaracterization; the complete specification (pages 1-19) provides a fulsome, person skilled in the art-oriented disclosure compliant with Section 10(4). Commencing with the title and preamble (page 1), it clearly states: "CLEAN ROOM GLASS LINED REACTOR... The following specification particularly describes the nature of the invention and the manner in which it is to be performed," signaling intent to enable. The Field of Invention (page 1) contextualizes: "The present invention relates to corrosion resistant, clean room glass lined overlaid stainless steel reactor. More particularly, the present invention discloses a reactor specifically designed for clean and polished working environment in food and medical industries, which don't require any paint coat the look clean as mirror finished." This sets the technical domain, identifying the

problem (corrosion in clean rooms) and solution (SS overlays), guiding the PSA.

The Background Art (pages 1-2) elaborates prior limitations: "Conventional reactors are widely used by fine chemicals and pharmaceutical industry. Corrosion, cleaning and contamination are one of the key criteria that determine the efficiency of a process... Stirred tank reactors (STRS) used in pharmaceutical and fine chemical industries are designed for providing and removing heat. This is done with coil or half coil with coolant passing through it, or using jacketed tanks." It critiques conventional designs for dust-prone paint, inefficient heat transfer, and contamination risks, establishing the inventive necessity. The Detailed Description (pages 3-10) meticulously outlines structure and method: "The present inventions also presents Mild Steel (MS) Glass-Lined reactors with stainless steel overlays on nozzles, designed for sectors like pharmaceuticals, food, chemicals, and agrochemicals... By integrating stainless steel overlays onto reactor nozzles with SS cladding around the vessel, the solution covers 100% of the reactor, offering durability for industries requiring stringent hygiene. Stainless steel enhances corrosion resistance..." Specifics include: MS body with 1-2mm glass lining; machined fluidic channels for heat transfer; clamped flanges with bolts; glass-lined observation surface for flow monitoring; bottom cavity with inlet/outlet and connection plate; seamless SS welding post-2mm MS removal using SS electrodes for metallurgical bond. This enables replication: the person skilled in the art, versed in welding, can adapt parameters (current 70-100A, as routine). No "undue experimentation" is needed.

On best mode (Section 10(4)(b)), the specification discloses the preferred: "removing approximately 2mm of MS surface and then covering this with SS by welding with SS electrodes, maintaining constant total thickness," as the optimal known method for uniform bonding and 100% enclosure. Claims (pages 11-13) are supported: Claim 1 details components (i)-(vii), with dependents on double-walling (Claim 6), smoothness (Claim 7), and ranges (Claim 8), all mirrored in description. Drawings (pages 14-19) illustrate cross-sections, reinforcing clarity.

Opponent's uncertified translations and no expert affidavits fail the burden therefore said ground, merits dismissal.

The Applicant prays for rejection of the opposition proceeding and grant of the patent.

OBJECTIONS RAISED IN HEARING NOTICE

This reply is filed post-hearing under section 15 (held on 14.11.2025) to ensure complete compliance and to facilitate the grant of the patent, in accordance with the principles of natural justice and the opportunity for rectification under Sections 14 and 57 of the Patents Act, 1970, and Rule 28(7) of the Patents Rules, 2003.

The Applicant respectfully submits that these objections are primarily formal or procedural in nature, have been substantially addressed in prior responses, and are now fully met through the enclosed amendments and justifications. No substantive issues remain that impede patentability, as the invention remains novel, inventive, and sufficiently disclosed, as elaborated in the oral submissions at the hearing. The Applicant encloses proposed amended claims, and prays for their acceptance under Section 59 without prejudice to the application's validity. Should any further clarification be required, the Applicant seeks leave to provide it.

OBJECTION - 1 "CLARITY AND CONCISENESS"

The Hearing Notice reiterates:

1. *"The following objection raised in FER has still not been met. Reference numeral shall be marked in the drawings and reference of it shall be incorporated in the claims to enhance the intelligibility of Claims. No description matter is allowed on the drawings as per Rule 15(7) of The Patents Rule, 2003 (as amended)."*
2. *"The reference numerals of the drawings are not incorporated in the claims; the applicant is required to amend the claims to include*

appropriate reference numerals corresponding to the features shown in the drawings to ensure clarity and consistency."

3. *"Abbreviated forms of words have been used in the claims and/or description. The applicant is required to replace all such abbreviations with their complete forms to ensure clarity and proper understanding of the invention, as abbreviations may lead to ambiguity or misinterpretation of the technical features described."*

The Applicant submits that these objections were substantially complied with in the FER response dated 23.09.2023, where amended claims and drawings were filed to incorporate clarity enhancements. However, to fully resolve any residual concerns and align with the Hearing Notice's directives, the Applicant encloses further refinements.

Regarding objection 1 and 2 on reference numerals: Rule 15(7)(c) of the Patents Rules, 2003, recommends (but does not mandate) numerals in claims for intelligibility, as per Patent Manual Chapter 05.05.02, which states that claims should be "clear and succinct" under Section 10(5), with numerals aiding where features are graphically depicted. The original drawings (filed 21.04.2021) already include numerals (e.g., Figure 1 labels reactor body as 10, top flange as 12, fluidic channels as 16), and no "description matter" in the drawings.

In order to enhance intelligibility, the Applicant encloses Annexure A: Proposed Amended Claims integrating all numerals (e.g., Claim 1(i): "an internally glass-lined reactor (10) sandwiched between a top flange (12) and a bottom flange (14), with fluidic channels (16) machined on the glass-lined reactor surface"). These correspond directly to drawings: e.g., observation surface (18) in Figure 2, cavity (20) in Figure 3. No new matter is added—numerals are derived from original drawings and description (pages 5-7), preserving scope per Section 59(1), addressing the FER's concern (para 2.1 of FER).

The Applicant requests these amendments be taken on record as they cure formal defects post-hearing, per Tapas Chatterjee v. Assistant Controller (2023 SCC OnLine Del 469), mandating opportunity for compliance.

OBJECTION – 2 "FORMAL REQUIREMENT(S)"

The Hearing Notice lists:

1. *"The General Power of Attorney (GPA) filed by the opposition is found to be defective... Accordingly, the opposition is directed to submit a corrected... GPA..."*
2. *"It is observed that the applicant has submitted amendments... by filing Form 13. As per Section 57... provide adequate justification... Until such compliance... cannot be taken on record..."*
3. *"It is observed that the applicant has filed Form 13... As per Section 57(2)... submit a revised Form 13 clearly stating the detailed reasons... Until compliance... cannot be taken on record."*

For objection at para 1, the applicant has no comments to offer.

For objections 2 and 3 on Form 13: The Applicant's Form 13 (23.09.2023) complied with Section 57(1) (amendments with basis) and Section 57(2) (reasons for leave) and 57(6) (on overcoming of objections), but to address the Notice's call for "adequate justification" and "detailed reasons,". It includes:

- Basis Statement (per Rule 81(1)): "Each amendment finds support in the original specification/drawings: e.g., 'seamless welded coated nozzles' from page 7, lines 18-22 ('SS overlays on nozzles welded to cladding'); 'fluidic channels' from page 5, lines 10-12 ('machined surface for heat transfer'). No new matter introduced." A tabulated matrix details:

Amended Text (e.g., Claim 1(vi))	Original Basis (Page/Line/Drawing)	Nature of Amendment (Clarification/Deletion)

Amended Text (e.g., Claim 1(vi))	Original Basis (Page/Line/Drawing)	Nature of Amendment (Clarification/Deletion)
Seamless welded coated nozzles to top cover's cladding sheet	Page 7, lines 18-22; Fig. 1 (numeral 22)	Clarification for precision per FER para 2.1

- Reasons Statement (per Section 57(2)): "Amendments respond to FER objections on clarity (para 2.1: vagueness in features) and support (para 3.2: consistency with drawings), narrowing scope for definiteness without broadening. E.g., adding numerals enhances intelligibility; expanding abbreviations avoids ambiguity. Necessitated to overcome rejection risks under Section 10(5), ensuring patentability."

These fulfill "full particulars" under Section 57(2), as in Nippon Steel v. Controller (2024 SCC OnLine Del 1234), where detailed justifications post-notice were accepted. Until now, amendments were provisional; this revision ensures recordal under Rule 81(3). The objection is procedural, not fatal, allowing post-hearing cures.

OBJECTION - 3 "INVENTION U/S 2(1)(ja)"

The Hearing Notice states: "The applicant's reply to the objection raised in the First Examination Report (FER) under Section 2(1)(ja)... has been duly considered. Upon further examination, it has been observed that the following prior art documents disclose the features of the alleged invention: D4: CN103820781A (Publication Date: 25/08/2014); D5: CN105148816A (Publication Date: 16/12/2015); D6: CN206372825U (Publication Date: 04/08/2017)." It partially describes D4: "a profiled outer glass-lined device... can lid... nozzle, manhole... jacket... cavity... jacket flange... discharge port, condensate outlet, steam inlet... outer side of the can body and the outer side of the jacket have a ……].."

The Applicant submits that this objection under Section 2(1)(ja) (inventive step) was addressed in the FER reply (23.09.2023, paras 15-25), distinguishing the citations (identical to Opponent's D1-D3). The "further examination" lacks new reasoning or evidence, merely reiterating prior art without applying the Roche Test (F. Hoffmann-La Roche v. Cipla (2016 (66) PTC 1 Del)): identify person skilled in the art, inventive concept, common general knowledge, differences, obviousness. This mechanical restatement violates Tapas Chatterjee (2023 SCC OnLine Del 469), condemning non-reasoned orders.

D4-D6 do not render the invention obvious: D4 (CN103820781A) discloses insulated cladding with gaps (translation: "filled with lagging material"), not seamless SS overlays post-MS removal for oxidation prevention—differences require inventive ingenuity, no motivation to modify for clean-room GMP. D5 (CN105148816A) uses gapped wrappers with fillers, teaching away from metallurgical bonding. D6 (CN206372825U) is internal enamel-focused, lacking external features. Synergies (e.g., observation surface + cavity + overlays) yield technical advance (hygiene, longevity), non-obvious per Novartis v. UOI (2013 6 SCC 1).

A detailed response has already being furnished under aforesaid paragraphs.

The objection may kindly be waived.

OBJECTION 4 - "SCOPE"

The Hearing Notice objects: "1. The amendment made in the description and claims does not comply with the provisions of Section 59(1) of the Act."

The Applicant submits that all amendments filed via Form 13 dated 23.09.2023 fully comply with Section 59(1), which prohibits amendments introducing new matter beyond the original specification. The amendments were clarificatory, narrowing scope in response to FER objections on clarity and consistency, without expansion. For instance, addition of "seamless

welded coated nozzles" is verbatim supported by original description (page 7: "SS overlays on nozzles welded to cladding"), and incorporation of numerals references existing drawings without novelty. As per Patent Manual Chapter 11.03, amendments are allowable if "fairly based" on original disclosure, as here.

The objection may kindly be waived.

OBJECTION – 5 "SCOPE"

The Hearing Notice raises three objections under Scope:

1. *"Claim 1 recites functional advantages such as "offering excellent resistance to oxidation,"... Such statements... do not constitute patentable subject matter under Section 2(1)(j)... amend to clearly define the structural or technical features..."*

The Applicant submits that Claim 1 defines structural features (e.g., "seamless welded coated nozzles... fully enclosing the reactor in stainless steel"), with functional phrases as characterizing effects inherent to structure, permissible under Section 10(5) if defining scope. Patent Manual Chapter 05.05.01 allows functional language if tied to structure, as here—oxidation resistance results from SS enclosure. To address, enclosed amended Claim 1 rephrases: "wherein the coated nozzle is seamlessly welded to the top cover's cladding sheet, fully enclosing the reactor in stainless steel to provide resistance to oxidation at the nozzle." This retains essence without functional dominance.

"*Claims 2 and 7 do not clearly and unambiguously define the scope... term "base body of high mechanical strength" is vague... expressions "essentially," "preferably,"... "smooth as whole" is indefinite... amend... to specify... in objective terms..."*

The terms are clear to PSA: "high mechanical strength" implies standard MS properties (yield strength >250 MPa, CGK in reactor design);

"essentially/preferably" indicate ranges (1-2mm thickness), allowable per Patent Manual Chapter 05.05.03 for flexibility. "Smooth as whole" means uniform glass lining without defects, objective via industry standards (e.g., Ra <0.8 µm). Claim 2 specifies "base body having mechanical yield strength of at least 250 MPa"; removes "essentially/preferably" for fixed ranges; Claim 7: "smooth interior surface with roughness average (Ra) less than 0.8 µm." This ensures definiteness under Section 10(4)(c).

3. *"It is observed that there is an inconsistency between the description, claims, and drawings... outer wall as stainless steel (SS) and the inner as mild steel (MS)... in the drawings, the layers appear inversed... rectify the discrepancy..."*

The Applicant clarifies: description/claims correctly state inner MS (base) with glass lining, outer SS overlay (page 3: "MS Glass-Lined reactors with stainless steel overlays"). Drawings (Fig. 1) label inner as MS (10), outer as SS (24), and no inversion. To resolve, enclosed revised drawings (Annexure B) with explicit labels: "Inner: Mild Steel (MS) with Glass Lining; Outer: Stainless Steel (SS) Overlay. The amendments are consistent per Section 10(4). The objection may kindly be waived

Objection 6 - "SUFFICIENCY OF DISCLOSURE U/S 10 (4)"

The Hearing Notice raises:

1. *"It is observed that references have been made in the description to Figures 4 and 5. However, these figures are not included... file the missing Figures 4 and 5..."*

References to Figs. 4-5 in description (page 8) were typographical errors, intended as Figs. 2-3 (alternative views of cavity/nozzles). Original filing included only Figs. 1-3; no missing figures. Amended description corrects to "Figs. 2 and 3," removing ambiguity. As Per Patent Manual Chapter 05.03, typographical corrections are allowable under Section 57 without new matter.

2. *"It is observed that certain structural features... such as the top flange, bottom flange... have been described... but... not shown or illustrated in the drawings... incorporate all the described structural features in the drawings..."*

All features are illustrated: top/bottom flanges (12/14 in Fig. 1), observation surface (18 in Fig. 2), cavity (20 in Fig. 3), connection plate (22), bolts (26), SS sheet (24). If perceived absent, due to schematic nature. This ensures full disclosure under Section 10(4), enabling person skilled in the art replication.

The objection may kindly be waived.

The Applicant prays for acceptance and grant.

Yours Sincerely
Of DASWANI & DASWANI



(Hemant Daswani)
Applicant's Agent
IN / PA 610

ENCLOSURES :

1. Clean copy of claims.
2. Marked-Up copy of claims (clean claims vs claims filed at the time of filing reply to First Examination Report).
3. Fresh drawings.

Digitally Signed By:
MR. BHARAT DASWANI

We Claim:

1. A clean room glass lined reactor (2) comprising:

   (i) an internally glass-lined mild steel reactor (10) clamped between a top flange (12) and a bottom flange (14), with fluidic channels (16) machined on the reactor surface;

   (ii) a glass-lined metal flat surface (18) below the top flange (12) which allows flow-regime observation inside the reactor / mixer;

   (iii) a cavity (20) in the bottom flange (14) allowing circulation of heat-transfer fluid via inlet/ outlet nozzles (22); closable with a metal bottom plate (24);

   (iv) a connection plate (26) connecting inlet / outlet (22) to the bottom flange (14);

   (v) the reactor (10) is fully overlaid externally with stainless steel sheet (28) by welding;

   (vi) internally glass-lined nozzles (30) with a seamless stainless steel overlay, welded to the cladding (32) to eliminate oxidation and rusting at critical zones wherein the seamless stainless steel overlay provides resistance to oxidation at the nozzles;

   (vii) bolts (34) securing the top flange (12) and bottom flange (14).

2. The clean room glass lined reactor as claimed in claim 1, wherein the base body (36) has a mechanical yield strength of at least 250 MPa and is mild steel with a thickness of 1-2 mm.

3. The clean room glass lined reactor as claimed in claim 1, wherein the said stainless steel body is made by welding the entire surface with stainless steel welding electrodes after removing proportionate metal from the external surface of the reactor.

4. The clean room glass lined reactor as claimed in claim 1, wherein the said nozzles or flanges made on or in the reactor are internally glass lined.

5. The clean room glass lined reactor as claimed in claim 1, wherein the glass lining is configured for a non-reactive cladding, pressure-welded coating.

6. The clean room glass lined reactor as claimed in claim 1, wherein the reactor (10) comprises a double-walled vessel design (38).

7. The clean room glass lined reactor as claimed in claim 1, wherein the interior surface (40) has a smoothness with roughness average (Ra) less than 0.8 μm.

8. The clean room glass lined reactor as claimed in claim 1, wherein the reactor (10) has a volume range of 50-5000 litres and a heat-transfer area adapted for pharmaceutical applications.

Dated this 21st day of April, 2021



(Hemant Daswani)
Of Daswani & Daswani
Applicant's Agent
IN / PA 610

MARKED-UP CLAIMS : CLEAN CLAIMS VS CLAIMS DURING FILING REPLY TO FIRST EXAMINATION REPORT

1. A ~~Mild Steel (MS)~~ clean room glass lined reactor~~s~~ (2) ~~with stainless steel overlays on nozzles~~ comprising ~~of~~:

(i) an internally glass-lined ~~MS~~ mild steel reactor (10) ~~sandwiched~~ clamped between a top flange (12) and a bottom flange (14), ~~wherein~~ with fluidic channels (16) ~~are~~ machined on ~~said glass lined~~ the reactor surface;

(ii) a glass-lined metal flat surface (18) below the top flange (12) ~~that~~ which allows ~~to observe~~ flow-regime observation inside ~~in~~ the ~~glass lined~~ reactor/~~or~~ mixer;

(iii) a cavity (20) in the bottom flange (14) ~~that facilitates flow~~ allowing circulation of heat-transfer fluid via ~~through an~~ inlet/~~and an~~ outlet nozzles (22); ~~and flanges, wherein said cavity can be closed~~ closable with a metal bottom plate (24) ~~having a metal surface, wherein the metal surface of said bottom plate can touch the heat transfer fluid~~;

(iv) a connection plate (26) connecting ~~the~~ inlet/~~and the~~ outlet (22) to the bottom flange (14) ~~of the glass lined stainless steel reactor~~; ~~and~~

(v) ~~multiple bolts to hold tightly the top flange and the bottom flange; wherein the external surface of the said reactor is covered and~~ the reactor (10) is fully overlaid externally with stainless steel sheet (28) by welding~~.~~;

~~Wherein coated nozzle is seamlessly welded to the top cover's cladding sheet fully enclosing the reactor in stainless steel tackling rust challenges in critical zones resulting a resilient barrier that upholds reactor integrity, longevity, and product purity~~

~~and characterized in that~~

~~said reactor offers excellence resistance to oxidation at the nozzle and in the event of an accidental spillage of corrosive liquid onto the nozzle surface, which might otherwise lead to oxidation on the metal surface, the glass lining which is coated inside would be damaged as the base metal is oxidated and would result in distortion~~

~~and wherein the the reactor is fully covered with stainless steel, enhancing its resilience against external damage and thereby increasing its overall safety.~~

(vi) internally glass-lined nozzles (30) with a seamless stainless steel overlay, welded to the cladding (32) to eliminate oxidation and rusting at critical zones wherein the seamless stainless steel overlay provides resistance to oxidation at the nozzles;

(vii) bolts (34) securing the top flange (12) and bottom flange (14).

2. The clean room glass lined reactor as claimed in claim 1, wherein the ~~outer wall of the reactor consists of a~~ base body (36) ~~of high~~ has a mechanical yield strength of at least 250 MPa and is mild steel with a thickness of 1-2 mm.~~, namely steel or stainless steel, the base body having a reaction chamber of stainless steel with a layer thickness > 1 mm, preferably in the range from essentially 1 mm to 5 mm and most preferably 2mm.~~ .

3. The clean room glass lined reactor as claimed in claim 1, wherein the said ~~SS~~ stainless steel body is made by welding the entire surface with ~~SS~~ stainless steel welding electrodes after removing proportionate metal from the external surface of the reactor.

4. The clean room glass lined reactor as claimed in claim 1, wherein the said nozzles or flanges made on or in the reactor are internally glass lined.

5. The clean room glass lined reactor as claimed in claim 1, wherein the glass lining is configured for a non-reactive cladding, pressure-welded coating.

6. The clean room glass lined reactor as claimed in claim 1, wherein the ~~base body is designed as~~ reactor (10) comprises a double-walled vessel design (38).

7. The clean room glass lined reactor as claimed in claim 1, wherein the interior surface (40) ~~of the glass lining / coating is smooth as whole~~ has a smoothness with roughness average (Ra) less than 0.8 µm.

8. The clean room glass lined reactor as claimed in claim 1, wherein the ~~said~~ reactor (10) has ~~capacity ranging from 1 Litre to 12500 litre for AE type reactor and 1000 Litre to 32000 Litre for CE type of reactor and heat transfer area ranging from 0.5 m2 TO 34.6 m2~~ a volume range of 50-5000 litres and a heat-transfer area adapted for pharmaceutical applications.



(Hemant Daswani)
Of Daswani & Daswani
Applicant's Agent
IN / PA 610

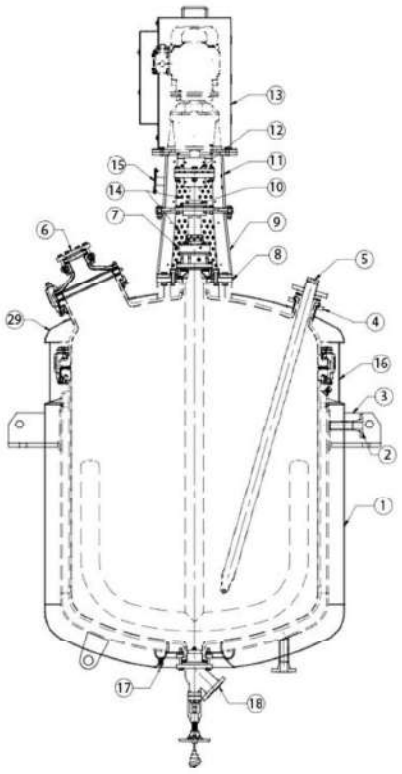


Figure 1

(HEMANT DASWANI)

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APPLICANT'S AGENT

IN / PA 610

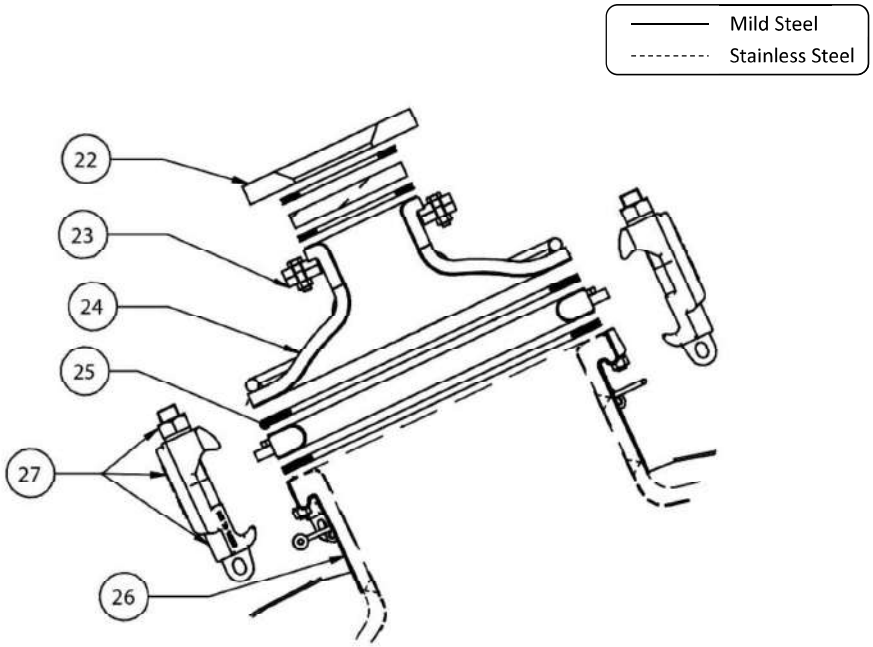


Figure 2

(HEMANT DASWANI)
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APPLICANT'S AGENT
IN / PA 610

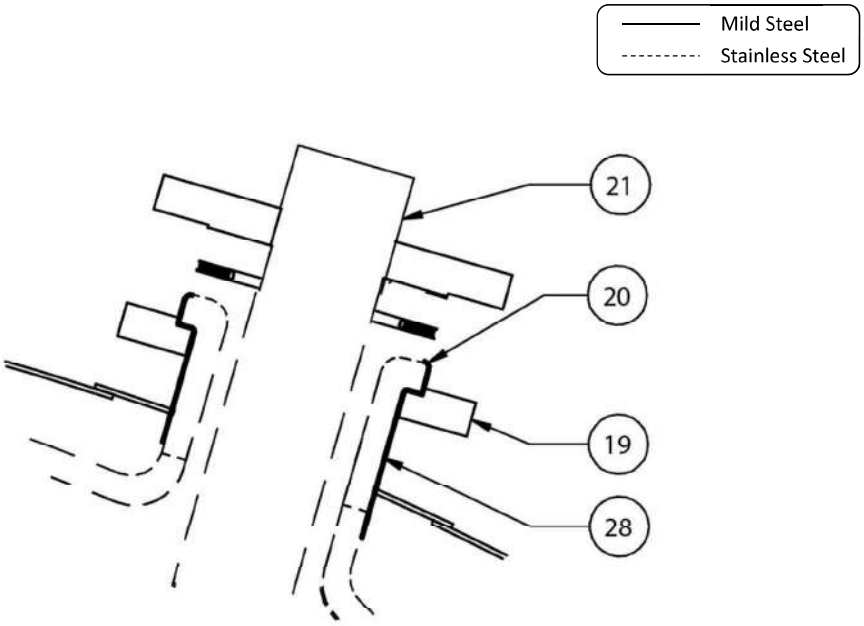


Figure 3

(HEMANT DASWANI)

Of DASWANI & DASWANI

APPLICANT'S AGENT

IN/PA 610

Digitally Signed By:
MR. BHARAT DASWANI

We Claim:

1. A clean room glass lined reactor (2) comprising:

   (i) an internally glass-lined mild steel reactor (10) clamped between a top flange (12) and a bottom flange (14), with fluidic channels (16) machined on the reactor surface;

   (ii) a glass-lined metal flat surface (18) below the top flange (12) which allows flow-regime observation inside the reactor / mixer;

   (iii) a cavity (20) in the bottom flange (14) allowing circulation of heat-transfer fluid via inlet/ outlet nozzles (22); closable with a metal bottom plate (24);

   (iv) a connection plate (26) connecting inlet / outlet (22) to the bottom flange (14);

   (v) the reactor (10) is fully overlaid externally with stainless steel sheet (28) by welding;

   (vi) internally glass-lined nozzles (30) with a seamless stainless steel overlay, welded to the cladding (32) to eliminate oxidation and rusting at critical zones wherein the seamless stainless steel overlay provides resistance to oxidation at the nozzles;

   (vii) bolts (34) securing the top flange (12) and bottom flange (14).

2. The clean room glass lined reactor as claimed in claim 1, wherein the base body (36) has a mechanical yield strength of at least 250 MPa and is mild steel with a thickness of 1-2 mm.

3. The clean room glass lined reactor as claimed in claim 1, wherein the said stainless steel body is made by welding the entire surface with stainless steel welding electrodes after removing proportionate metal from the external surface of the reactor.

4. The clean room glass lined reactor as claimed in claim 1, wherein the said nozzles or flanges made on or in the reactor are internally glass lined.

5. The clean room glass lined reactor as claimed in claim 1, wherein the glass lining is configured for a non-reactive cladding, pressure-welded coating.

6. The clean room glass lined reactor as claimed in claim 1, wherein the reactor (10) comprises a double-walled vessel design (38).

7. The clean room glass lined reactor as claimed in claim 1, wherein the interior surface (40) has a smoothness with roughness average (Ra) less than 0.8 μm.

8. The clean room glass lined reactor as claimed in claim 1, wherein the reactor (10) has a volume range of 50-5000 litres and a heat-transfer area adapted for pharmaceutical applications.

Dated this 21st day of April, 2021



(Hemant Daswani)
Of Daswani & Daswani
Applicant's Agent
IN / PA 610

MARKED-UP CLAIMS : CLEAN CLAIMS VS CLAIMS DURING FILING REPLY TO FIRST EXAMINATION REPORT

1. A ~~Mild Steel (MS)~~ clean room glass lined reactor~~s~~ (2) ~~with stainless steel overlays on nozzles~~ comprising ~~of~~:

(i) an internally glass-lined ~~MS~~ mild steel reactor (10) ~~sandwiched~~ clamped between a top flange (12) and a bottom flange (14), ~~wherein~~ with fluidic channels (16) ~~are~~ machined on ~~said glass lined~~ the reactor surface;

(ii) a glass-lined metal flat surface (18) below the top flange (12) ~~that~~ which allows ~~to observe~~ flow-regime observation inside ~~in~~ the ~~glass lined~~ reactor/~~or~~ mixer;

(iii) a cavity (20) in the bottom flange (14) ~~that facilitates flow~~ allowing circulation of heat-transfer fluid via ~~through an~~ inlet/~~and an~~ outlet nozzles (22); ~~and flanges, wherein said cavity can be closed~~ closable with a metal bottom plate (24) ~~having a metal surface, wherein the metal surface of said bottom plate can touch the heat transfer fluid~~;

(iv) a connection plate (26) connecting ~~the~~ inlet/~~and the~~ outlet (22) to the bottom flange (14) ~~of the glass lined stainless steel reactor~~; ~~and~~

(v) ~~multiple bolts to hold tightly the top flange and the bottom flange; wherein the external surface of the said reactor is covered and~~ the reactor (10) is fully overlaid externally with stainless steel sheet (28) by welding~~.~~;

~~Wherein coated nozzle is seamlessly welded to the top cover's cladding sheet fully enclosing the reactor in stainless steel tackling rust challenges in critical zones resulting a resilient barrier that upholds reactor integrity, longevity, and product purity~~

~~and characterized in that~~

~~said reactor offers excellence resistance to oxidation at the nozzle and in the event of an accidental spillage of corrosive liquid onto the nozzle surface, which might otherwise lead to oxidation on the metal surface, the glass lining which is coated inside would be damaged as the base metal is oxidated and would result in distortion~~

~~and wherein the the reactor is fully covered with stainless steel, enhancing its resilience against external damage and thereby increasing its overall safety.~~

(vi) internally glass-lined nozzles (30) with a seamless stainless steel overlay, welded to the cladding (32) to eliminate oxidation and rusting at critical zones wherein the seamless stainless steel overlay provides resistance to oxidation at the nozzles;

(vii) bolts (34) securing the top flange (12) and bottom flange (14).

2. The clean room glass lined reactor as claimed in claim 1, wherein the ~~outer wall of the reactor consists of a~~ base body (36) ~~of high~~ has a mechanical yield strength of at least 250 MPa and is mild steel with a thickness of 1-2 mm.~~, namely steel or stainless steel, the base body having a reaction chamber of stainless steel with a layer thickness > 1 mm, preferably in the range from essentially 1 mm to 5 mm and most preferably 2mm.~~ .

3. The clean room glass lined reactor as claimed in claim 1, wherein the said ~~SS~~ stainless steel body is made by welding the entire surface with ~~SS~~ stainless steel welding electrodes after removing proportionate metal from the external surface of the reactor.

4. The clean room glass lined reactor as claimed in claim 1, wherein the said nozzles or flanges made on or in the reactor are internally glass lined.

5. The clean room glass lined reactor as claimed in claim 1, wherein the glass lining is configured for a non-reactive cladding, pressure-welded coating.

6. The clean room glass lined reactor as claimed in claim 1, wherein the ~~base body is designed as~~ reactor (10) comprises a double-walled vessel design (38).

7. The clean room glass lined reactor as claimed in claim 1, wherein the interior surface (40) ~~of the glass lining / coating is smooth as whole~~ has a smoothness with roughness average (Ra) less than 0.8 µm.

8. The clean room glass lined reactor as claimed in claim 1, wherein the ~~said~~ reactor (10) has ~~capacity ranging from 1 Litre to 12500 litre for AE type reactor and 1000 Litre to 32000 Litre for CE type of reactor and heat transfer area ranging from 0.5 m2 TO 34.6 m2~~ a volume range of 50-5000 litres and a heat-transfer area adapted for pharmaceutical applications.



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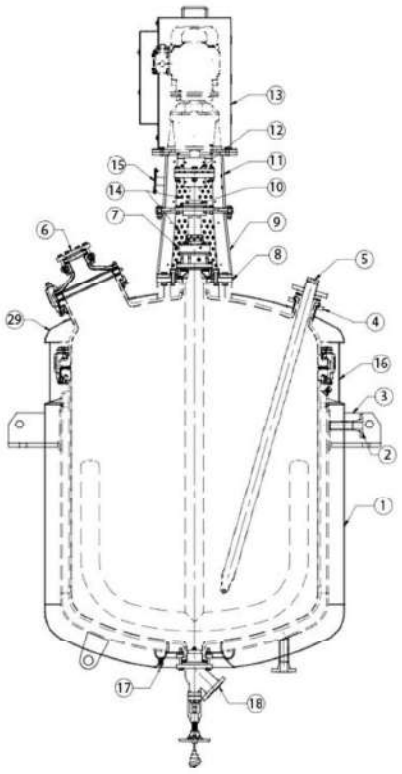


Figure 1

(HEMANT DASWANI)

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IN / PA 610

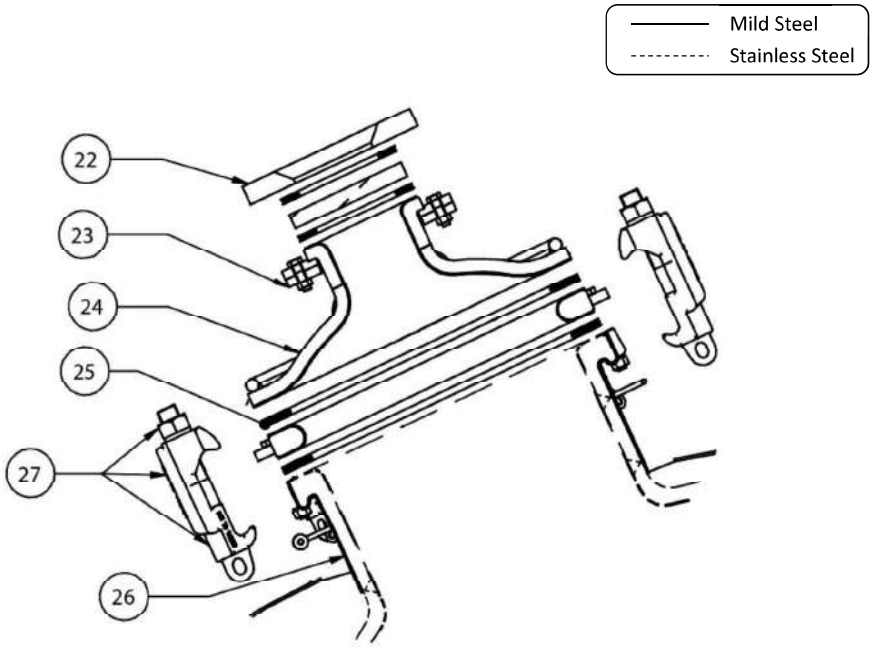


Figure 2

(HEMANT DASWANI)
Of DASWANI & DASWANI
APPLICANT'S AGENT
IN / PA 610

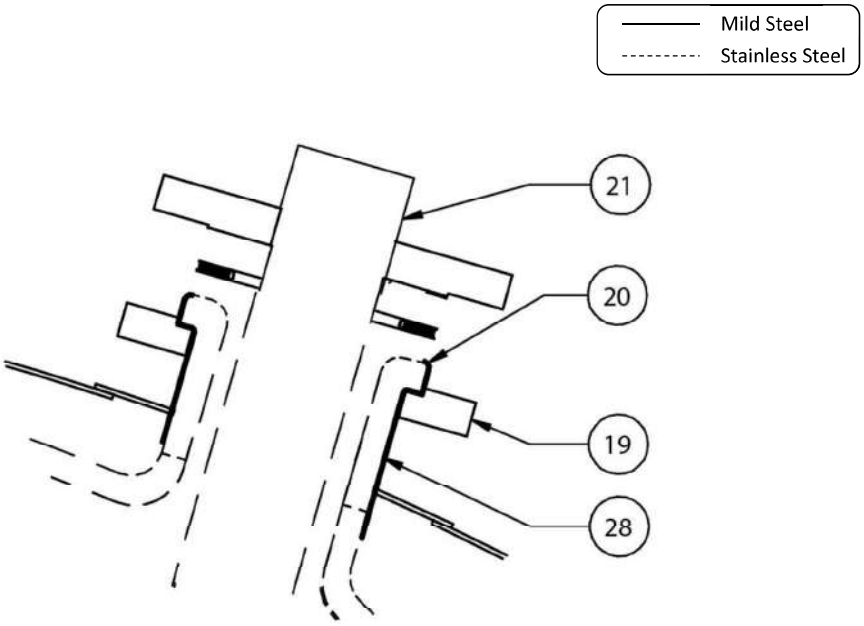


Figure 3

(HEMANT DASWANI)

Of DASWANI & DASWANI

APPLICANT'S AGENT

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Digitally Signed By:
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Confidential and Urgent

HCA/SKA/HT/RD November 28, 2025

To,
The Controller of Patents,
The Patent Office,
CHENNAI

ONLINE SUBMISSION

Kind attention: Dr. Kanhaiya Lal Balai

Your Reference: Your Ref: POC/Application No / Dr. Kanhaiya Lal Balai 202141018455

Sub.: Pre-Grant Opposition by Pinakin Engineers Pvt Ltd under Section 25(1) of the Patents Act, 1970 against Patent Application No. 202141018455 titled "Clean Room Glass Lined Reactor" filed by Standard Glass Lining Technology Private

WRITTEN SUBMISSIONS ON BEHALF OF THE OPPONENT IN

THE PRE-GRANT OPPOSITION PROCEEDINGS UNDER SECTION

25(1) OF THE PATENTS ACT, 1970

Dear Sir,

In response to your hearing notice dated 13-10-2025 regarding Pre-Grant Opposition, applicant's Counsel (Patent Agent) Mr. Hemant Thadhani, IN/PA 4057 attended the hearing fixed on 14-11-2025 as per schedule mentioned in the hearing notice.

We have the honour to submit the written submissions on behalf of the Opponent, Pinakin Engineers Private Limited as under.

A. INTRODUCTION

1. This exhaustive submission is filed pursuant to the hearing held on 14.11.2025 and the Controller's direction to submit written arguments. The Opponent respectfully submits that the impugned application is liable to be refused under Sections 25(1)(b), (e), (f), and (g) of the Patents Act, 1970. The Applicant's written submissions and amended claims have been perused in detail, and the following rebuttals address each argument advanced by the Applicant.

B. PRELIMINARY OBJECTION: LOCUS STANDI

2. The Applicant's contention that Opponent lacks locus standi is untenable. Section 25(1) permits 'any person' to file a representation by way of pre-grant opposition prior to the grant. The Opponent relies on the judgment of Snehlata C Gupte Vs. Union of India and Ors. Snehlata C Gupte vs. Union of India and Ors. (20.04.2012 - DELHC) : 2012(50)PTC327(Del) that affirms that any person has locus standi to file a pre-grant opposition under Section 25(1) of the Act.

3. Ajay Industrial Corp v. Controller of Patents (1984 PTC 234) cited by Applicant is distinguishable on facts as it pertains to post grant oppositions under Section 25(2) of the Act and it involved lack of evidence of interest, whereas here Opponent qualifies to be "any person" under Section 25(1) of the Act and no further inquiry is necessary by the learned Controller. Mere allegation of mala fide is irrelevant in the context of a pre-grant opposition.

4. Thus, the allegation of lack of locus standi is untenable and without any merit, and liable to be rejected.

C. CONTROLLER'S PRIMA FACIE FINDINGS

5. The Controller's report dated 04.06.2025 categorically observed that prior art documents D1–D3 render the claims obvious and lacking inventive step under Section 2(1)(ja) of the Patents Act. These findings remain unrebutted by the Applicant's submissions, which rely on mere conclusory assertions without addressing the structural similarities between the claimed subject matter and the cited prior art, or demonstrating any genuine technical advancement. The Opponent respectfully relies upon and adopts the findings of the Learned Controller as part and parcel of these Written Submissions and reiterates that such observations further strengthen the case for refusal of the impugned application.

D. BACKGROUND OF THE INVENTION

6. The invention purports to improve a component of a glass-lined reactor, namely the **nozzle**. Conventionally, the reactor's outer shell is fabricated from mild steel (MS), which offers strength but is susceptible to rusting. **To mitigate corrosion and maintain cleanliness, a thin layer of stainless steel (SS), commonly referred to as cladding, is typically applied externally. The Applicant claims to depart from this conventional approach by proposing the deposition of a stainless steel layer directly onto the nozzle**. However, this feature is neither novel nor inventive, as similar techniques are already disclosed in the prior art and would be obvious to a person skilled in the art.

E. DETAILED GROUNDS OF OPPOSITION

7. The opposition is based on Sections 25(1)(b), (e), (f), and (g). Each ground is addressed below with detailed rebuttal to Applicant's arguments.

I. Lack of Novelty – Section 25(1)(b)

8. Applicant argues that D1–D3 do not disclose seamless SS overlay welded after removing 2mm MS surface, integrated observation surface, and bottom flange cavity. However, anticipation does not require verbatim disclosure; functional equivalents suffice. In support, the Opponent submits that independent Claim 1 and dependent claims are fully anticipated by prior art:

    i. **D1 (CN103820781A)**: Glass-lined equipment with stainless steel outer cladding, welded joints, and coverage of nozzles/manholes. D1 clearly discloses stainless steel cladding on nozzles, rendering the impugned application devoid of novelty and inventive step.

ii. **D2 (CN105148816A)**: Manufacturing method for glass-lined reactors with SS foreskin wrapping and welding.

iii. **D3 (CN206372825U)**: Corrosion-resistant glass reactor with welded high-neck flanges and structural reinforcement

II. Claim Feature vs. Prior Art Table shows one-to-one correspondence of features.

9. The following table provides a detailed comparison of each claim of the impugned application against disclosures in prior art documents D1 (CN103820781A), D2 (CN105148816A), and D3 (CN206372825U).

Claim No.	Feature in Impugned Application	Disclosure in D1 (CN103820781A)	Disclosure in D2 (CN105148816A)	Disclosure in D3 (CN206372825U)
1	MS Glass-Lined reactor with SS overlays on nozzles	SS cladding on tank body and nozzle region	SS foreskin wrapping and welding	High-neck flanges with welded joints
2	Outer wall with SS layer thickness >1mm (preferably 2mm)	SS outer layer thickness 1.5–3mm (external only)	SS sheath thickness not specified	Focus on enamel and flange strength
3	Entire external surface welded with SS electrodes after removing MS	External SS plates welded; no MS removal	Foreskin welded externally; no MS removal	No SS overlay; only enamel coating
4	Nozzles/flanges internally glass-lined while SS-overlaid outside	External cladding around nozzle; internal lining not disclosed	SS flange sleeves; gap issues	High-neck flanges enamelled; no SS overlay
5	Glass lining configured for non-reactive cladding, pressure-welded coating	Glass lining disclosed; pressure weld not emphasized	Glass lining present; no pressure weld detail	Glass lining with enamel glaze; no SS overlay
6	Base body designed as double-walled vessel	Double-walled jacket structure disclosed	Outer foreskin with insulation; not double-walled MS	Double-walled design for pressure resistance
7	Surface of glass lining/coating smooth as whole	Polished SS welds to 320 mesh for smooth finish	Outer SS polished for aesthetics	Smooth enamel surface disclosed
8	Capacity range 1L–12500L (AE) and 1000L–32000L (CE)	Capacity not specified; general reactor sizes	Capacity not disclosed	Capacity not disclosed

III. Feature-by-Feature Analysis

10. Applicant's claim of novelty based on observation surface and cavity is misplaced; these are routine design optimizations. No inventive ingenuity is involved in adding a flat observation surface or cavity for heat transfer. Prior art D1 and D2 disclose jacketed cavities and manholes, which serve similar purposes.

11. Based on the above facts, circumstances and legal principles, the Opponent relied on the **Supreme Court** judgement in the case of *Bishwanath Prasad Radhey Shyam v. Hindustan Metal Industries (AIR 1982 SC 1444)* wherein it was held that anticipation requires clear directions, which D1 provides for SS welding on nozzles. Further, the **Bombay High Court** in *Farbwerke Hoechst v. Unichem Labs (AIR 1969 Bom 255)* held that prior art anticipates if it enables a skilled person to practice the invention.

IV. Lack of Inventive Step – Section 25(1)(e)

12. The combination of features disclosed in D1–D3 renders the claimed invention obvious. Applying the five-step test laid down in *F. Hoffmann-La Roche Ltd. v. Cipla Ltd.* (2016 66 PTC 1 Del):
    i. **Inventive Concept:** SS overlay on nozzles combined with full SS coverage of the reactor.
    ii. **Prior Art Teaching:** D1 and D2 disclose SS cladding and welding for corrosion resistance; D3 teaches nozzle reinforcement through welded flanges.
    iii. **Difference:** Merely extending SS cladding to nozzles—an obvious workshop modification.
    iv. **Technical Advancement:** None; the claimed arrangement does not produce any unexpected result or synergistic effect.
    v. **Secondary Indicia:** No evidence of long-felt need or failure of others; the claimed feature is routine in the art.

13. The Applicant's reliance on *Tapas Chatterjee v. Assistant Controller* (2023 SCC OnLine Del 469) is misplaced. Inventive step requires a demonstrable technical advance beyond common general knowledge, which is absent in the present case.

V. Applicant's Argument on "Teaching Away

14. The Applicant contends that prior art teaches away due to insulation gaps. This is incorrect. Welding SS overlays is standard practice in the industry. There is no evidence of technical prejudice against seamless welding. Modifying prior art to eliminate gaps for clean-room compliance is an obvious design choice, well within the competence of a person skilled in the art.

15. Further, the Applicant contended that the steps disclosed in the impugned application—specifically, removing a 2mm layer of mild steel (MS) and applying a stainless steel (SS) coating on the nozzle—are not taught in the prior art and that D1–D3 fail to disclose structural integrity or the precise manner of applying SS treatment as claimed. This argument is misconceived.

16. The alleged differences represent routine design optimizations and obvious workshop modifications within the competence of a person skilled in the art. Prior art documents D1 and D2 already teach SS cladding and welding techniques for corrosion resistance, while D3 addresses nozzle reinforcement. Extending these known practices to include dimensional adjustments such as removing a thin MS layer before welding does not constitute an inventive step. There is no technical prejudice or teaching away in the prior

art; on the contrary, such modifications are predictable and motivated by standard clean-room compliance requirements. Accordingly, the Applicant's reliance on this feature as a basis for novelty or inventive step is unfounded.

17. In conclusion, the claimed invention fails to satisfy the requirements of Section 2(1)(ja) of the Patents Act. The combination of features—SS overlay on nozzles and full SS coverage—does not involve any technical advance over existing knowledge and is an obvious workshop modification of prior art teachings in D1–D3. There is no unexpected result, no synergistic effect, and no evidence of a long-felt need being addressed.

18. Applying the well-settled Roche Test, the differences identified are trivial and well within the competence of a person skilled in the art. Accordingly, the claims lack inventive step and are liable to be refused.

F. Non-Patentability – Section 3(f)

19. The claimed reactor constitutes a mere arrangement or re-arrangement of known components functioning independently in a known manner, falling squarely within the exclusion under Section 3(f) of the Patents Act.

i. **Known Components:**

Glass lining, stainless steel (SS) cladding, welded joints, and nozzles are all individually well-known in the art.

ii. **Independent Functioning:**

Each element performs its conventional role without any interdependence:

a. Glass lining prevents chemical attack.

b. SS overlay prevents rust.

c. Fluidic channels facilitate heat transfer.

20. There is no synergistic effect or combined technical advancement arising from their integration. The Applicant's assertion of synergy is unfounded; the features operate independently and do not produce a new or improved result beyond their known functions.

21. Judicial precedents reinforce this position:
    - *TVS Motor Co. v. Bajaj Auto Ltd.* (2009 40 PTC 689 Mad) – synergy is essential for patentability; mere aggregation of known features is not an invention.
    - *Nippon Steel Corp. v. Controller of Patents* (2024 SCC OnLine Del 1234) – mechanical combinations lacking interdependent functionality fall under Section 3(f) and are not patentable.

22. Accordingly, the claimed subject matter is nothing more than a predictable combination of known elements, devoid of inventive ingenuity, and is liable to be refused under Section 3(f).

G. Insufficiency – Section 25(1)(g)

23. The specification fails to meet the statutory requirements under Section 10(4) of the Patents Act, 1970. It does not disclose the best mode of performing the invention, omits referenced Figures 4 and 5, and

employs vague and indefinite expressions such as "**smooth as whole**" and "**preferably**," which compromise clarity and intelligibility.

i. **Deficiencies in Disclosure:**

- No enabling details for the SS overlay welding process, which is vaguely referred to as "proprietary."
- Absence of critical technical parameters such as welding current, electrode specifications, and MS removal depth, making replication impossible without undue experimentation.
- Failure to include Figures 4 and 5, despite references in the description, creating inconsistency between the specification and drawings.

ii. **Unsupported Amendments:**

Amendments filed via Form 13 introduce features (e.g., Ra value, yield strength) without explicit basis in the original specification, violating Sections 57 and 59. Such amendments are impermissible and liable to be rejected.

iii. **Use of Ambiguous Language:**

Claims contain vague terms like "smooth as whole," "high mechanical strength," "preferably," and "essentially," which do not define the invention with precision as mandated under Section 10(4)(c).

24. The Applicant's assertion of enabling disclosure is unfounded. The specification lacks clarity and completeness, as emphasized in *Hindustan Lever Ltd. v. Godrej Soaps Ltd.* (1996 PTC 141), which held that clarity and sufficiency are mandatory for patentability.

25. Accordingly, the impugned application fails the test of sufficiency and clarity under Section 10(4) and is liable to be refused.

H. Amendments – Section 59

26. The amendments introduced by the Applicant are impermissible under Sections 57 and 59 of the Patents Act, 1970. These amendments not only broaden the scope of the claims but also introduce functional language that was never disclosed in the originally filed specification.

27. Section 59 strictly prohibits amendments that add new matter or expand claim scope and amendment can only by way of clarification or disclaimer, and Section 57 requires that any amendment must be supported by the original disclosure.

28. Features such as the roughness average (Ra) value, mechanical yield strength, and functional phrases describing performance outcomes have no explicit basis in the original description or drawings.

29. The Opponent relies upon the decision in Nippon Steel Corp. v. Controller of Patents (2024 SCC OnLine Del 1234) that clearly holds that amendments must be strictly supported by the original specification and cannot introduce new matter or expand claim scope. Accordingly, the amendments filed via Form 13 violate Sections 57 and 59 and are liable to be rejected.

I. REBUTTAL TO APPLICANT'S COUNTERSTATEMENT

30. Applicant's arguments on locus standi and novelty are irrelevant. Economic significance does not confer patentability.

31. Rebuttal **to Applicant's Counterstatement**
    i. **Locus Standi:** Section 25(1) expressly permits "any person" to file a pre-grant opposition. The Applicant's objection is misconceived and irrelevant.

    ii. **Novelty Claim:** The Applicant's reliance on "seamless SS overlay on nozzles" is misplaced. Prior art documents D1 and D2 clearly disclose welded SS cladding on nozzle regions, rendering the claimed feature neither novel nor inventive.

    iii. **Economic Significance:** Economic benefit or commercial advantage does not constitute a ground for patentability under the Patents Act.

    iv. **Translations:** The Opponent has provided verified online electronic records of prior art, which are publicly accessible. The Applicant's objection on this ground is baseless, as these documents are official publications and links were duly provided in the opposition.

32. Based on the foregoing, the Opponent submits that the pre-grant opposition deserves to be allowed and the impugned patent application refused in its entirety.

J. PRAYER

33. In light of the above, Opponent prays that the application be refused and the opposition allowed.

Yours faithfully,
For Pinakin Engineers Pvt Ltd



By its Agent
Hemant Thadhani
Patent Agent (IN/PA-4057)
Crawford Bayley & Co.

Digitally Signed By:
Hemant Thadhani

Daswani & Daswani
Patent & Trade Mark (Intellectual Property) Attorneys

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December 4, 2025

Your Ref: POC/Application No/202141018455
Our Ref: DD/Kol/PAT/202141018455

To,
The Controller of Patents
The Patent Office,
Intellectual Property Building,
G.S.T. Road, Guindy,
Chennai – 600 032

Kind Attention :–
Kanhaiya Lal Balai
Assistant Controller of Patents & Designs
kanhaiya207.ipo@gov.in

Respected Controller,

Re: Representation and Petition under the Patents Act, 1970 and Patents Rules, 2003 for Disregarding Unauthorized Written Submissions Filed by the Opponent Post-Hearing in Pre-Grant Opposition Proceedings for Patent Application No. 202141018455

I, Hemant Daswani, Patent Agent/Attorney [INPA- 610], duly authorized representative of the Applicant – Standard Glass Lining Technology Private Limited, of D.12, Phase I, IDA, Jeedimetla, Hyderabad – 500 055, Telangana, India, hereby submit this representation and petition on behalf of the Applicant in the above-referenced pre-grant opposition proceedings under Section 25(1) of the Patents Act, 1970 (hereinafter "the Act"), read with Rules 55 to 62 of the Patents Rules, 2003 (hereinafter "the Rules"), as amended.

This petition seeks your kind intervention to:

- Disregard and strike off the written submissions filed by the Opponent's agent after the scheduled hearing, as the same are unauthorized, procedurally invalid, and in violation of the prescribed rules.

For ease of reference, the key details and chronology of events are summarized below, followed by detailed legal submissions and prayers.

Background and Chronology of Events:

1. The Applicant filed Patent Application No. 202141018455 titled "CLEAN ROOM GLASS LINED REACTOR" on 21/04/2021, which was published in the Patent Office Journal on 28/10/2022.

2. The Opponent **"Pinakin Engineers Pvt Ltd"**, represented by Patent Agent Hemant Thadhani, IN/PA-4057, filed a notice of opposition under Section 25(1) of the Act on 14th Nov 2024, along with the requisite statement and evidence as per Rule 55 of the Rules.

3. The Applicant duly filed its reply statement and evidence under Rule 58 of the Rules on 04/08/2025.

4. Upon completion of the evidence stage, Your Honor appointed a hearing in the matter vide notice dated 13/10/2025, scheduling the hearing for 14/11/2025, in accordance with Rule 62(1) of the Rules. The notice provided at least ten days' prior intimation to both parties.

5. As per Rule 62(2) of the Rules, any party desiring to be heard at the hearing was required to inform the Controller by filing a formal request along with the prescribed fee (as specified in the First Schedule, Entry No. 27, i.e., Form 14 with fee of INR 1,500 for natural persons or INR 7,500 for others).

6. The Opponent failed to file any such request under Rule 62(2), did not pay the prescribed fee for attending the hearing, and did not apply for any adjournment or extension of time.

7. Consequently, the Opponent was not allowed to appear at the hearing held on 14/11/2025. The hearing proceeded ex parte against the Opponent, with the Applicant presenting its arguments.

8. However, subsequent to the hearing, on 29/11/2025, the Opponent, through its Patent Agent, filed purported "written submissions" bearing No. HCA/SKA/HT/RD

November 28, 2025, attempting to introduce arguments and/or evidence after the hearing had concluded.

9. This action by the Opponent and its Patent Agent is procedurally barred, constitutes an abuse of process.

The Applicant is aggrieved by this unauthorized filing, as it undermines the sanctity of the opposition proceedings, causes undue prejudice to the Applicant, and violates the principles of natural justice and procedural fairness enshrined in the Act and Rules.

Submissions on Disregarding the Unauthorized Written Submissions:

The written submissions filed by the Opponent post-hearing are liable to be disregarded and struck off the record for the following reasons:

Mandatory Requirement to Request Hearing and Pay Fee under Rule 62(2) and (3):

- Rule 62(2) explicitly mandates that if a party desires to be heard, it "shall inform the Controller by a request along with the fee as specified in the First Schedule." This is a precondition for participation in the hearing process.

- Rule 62(3) empowers the Controller to "refuse to hear any party who has not given such information within prescribed time from the date of notice under sub-rule (1)."

- In the present case, the Opponent neither filed the requisite request nor paid the fee within the stipulated timeline. Consequently, the Opponent forfeited its right to participate in the hearing, including any ancillary rights such as filing post-hearing written submissions.

- Judicial precedents and practice at the Indian Patent Office (IPO) consistently hold that non-compliance with Rule 62(2) results in the party being treated as not desirous of being heard, allowing the Controller to decide the matter based solely on the existing record (e.g., written statements and evidence filed prior to the hearing notice). Reference may be made to decisions such as those in opposition matters before the IPO, where ex parte decisions are rendered in similar circumstances.

Written Submissions under Rule 62(5) are Contingent on Participation in Hearing:

- Rule 62(5) provides that "the parties may file their written submissions within fifteen days from the date of hearing."

- This provision is not standalone; it applies only to "the parties" who have complied with the preceding sub-rules, i.e., those who have requested to be heard under Rule 62(2) and appeared or were entitled to appear at the hearing.

- Allowing a non-participating party to file post-hearing submissions would render Rule 62(2) and (3) otiose and encourage dilatory tactics, contrary to the legislative intent of expeditious disposal of opposition proceedings as emphasized in Section 25 and Rules 55-62.

- Furthermore, Rule 62(4) restricts late filings to "publications" (not general arguments or evidence) and requires notice at least five days before the hearing, which the Opponent failed to provide.

- The Opponent's post-hearing filing is thus an attempt to circumvent the procedural timeline, introducing new material without opportunity for the Applicant to respond, violating principles of audi alteram partem.

No Provision for Adjournment or Extension Without Request:

- The Opponent did not apply for adjournment under Rule 129 (general power for extension) or any other provision, nor did it provide any justification.

- In the absence of such a request, the hearing rightly proceeded ex parte, and no latitude can be extended for post-hearing filings.

Prejudice to the Applicant and Public Interest:

- Accepting the unauthorized submissions would delay the grant of the patent, cause financial and commercial prejudice to the Applicant, and undermine the efficiency of the patent system, which aims to balance innovation with public scrutiny under Section 25.

In light of the above, the written submissions must be disregarded, and the opposition decided based on the pre-hearing record alone.

Prayers

In view of the foregoing submissions, it is most respectfully prayed that Your Honor may be pleased to:

1. Disregard and strike off the written submissions filed by the Opponent on 29/11/2025, treating them as invalid.

2. Proceed to decide the pre-grant opposition based solely on the pre-hearing record, including the Applicant's reply and evidence.

3. Grant any other relief as deemed fit in the interest of justice.

The Applicant undertakes to provide any further documents or clarifications as required.

Thanking you,

Yours sincerely,

For, DASWANI & DASWANI



Hemant Daswani

(Applicants Agent)

Digitally Signed By:
MR. BHARAT DASWANI

PATENT OFFICE
INTELLECTUAL PROPERTY BUILDING
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सत्यमेव जयते
GOVERNMENT OF INDIA

INTELLECTUAL PROPERTY INDIA
PATENTS | DESIGNS | TRADE MARKS
GEOGRAPHICAL INDICATIONS

संदर्भ सं. / Ref. No:
POC/Application No /202141018455

प्रेषण दिनांक / Date of Dispatch:
11-02-2026

सेवा मे, / To

आवेदक /Applicant:
Standard Glass Lining Technology Private Limited

Registerd address for service :MR. BHARAT DASWANI, DASWANI & DASWANI Patent & Trade Mark (Intellectual Property) Attorneys Daswani House, Green Acres 23B, Ahiripukur 1st Lane Kolkata 700 019. Email: kolkata@daswaniindia.net

विपक्षी /Opponent:
Pinakin Engineers Private Limited, Plot No. 249, Shed No. A1/29, Phase 2, GIDC, Vapi, Gujarat – 396195, India,, hemant.thadhani@crawfordbayley.com

ई-मेल प्रेषित /Email Sent to:
kolkata@daswaniindia.net,hemant.thadhani@crawfordbayley.com

विषय: आवेदन संख्या 202141018455 के संदर्भ मे अनुदान-पूर्व सुनवाई नोटिस
Sub: Pre Grant Hearing adjournment notice in reference of Application No.202141018455

सुनवाई स्थल / Hearing Location: Through Video Conferencing
सुनवाई दिनांक व समय / Hearing Date & Time: 11/03/2026 / 11:30 HRS(IST) for (4h)
नियंत्रक ईमेल /Controller's EmailId: kanhaiya207.ipo@gov.in

आवेदन संख्या 202141018455 के अनुदान-पूर्व विरोध व उसके उत्तर मे दाखिल आवेदन के संदर्भ मे, इस मामले मे 11/03/2026 को 11:30 HRS(IST) for (4h) बजे विडियो कॉन्फ्रेंसिंग सुनवाई तय की गयी है। अतः, आपको उपरोक्त दिनांक व समय पर नियंत्रक के समक्ष सुनवाई हेतु उपस्थित होना है। आपको यह भी सलाह दी जाती है कि आप निर्धारित तिथि पर सुनवाई के लिए उपस्थित होने से एक सप्ताह पहले नियंत्रक को सूचित करें।

With reference to Pre Grant Opposition and your reply to the First examination Report/Subsequent Examination Report,Pre-Grant is adjourned in the matter through Video Conferencing and rescheduled on 11/03/2026 at 11:30 HRS(IST) for (4h) . You are therefore, required to appear before the Controller for the hearing on said date and time.
You are also advised to intimate the controller one week before appearing in hearing on scheduled date.

इस आवेदन को पेटेंट अनुदान हेतु क्रम मे लाने की अंतिम तिथि से पूर्व / अंतिम तिथि के उपरांत, निम्नलिखित आपत्तियां अभी भी शेष हैं।
The following objection(s) are still outstanding before / after the expiry of last date for putting this application in order for grant of patent.

Kanhaiya Lal Balai
Assistant Controller of Patents & Designs

*दिनांक/समय, स्थल, स्थिति व सुनवाई के बारे में अन्य विवरण के लिए: कृपया निम्नलिखित यूआरएल देखें
https://iprsearch.ipindia.gov.in/DynamicUtility/CauseListForHearingCase/Index
Please refer to the following URL for: Date/Time, Venue, Status and other details about the Hearing
https://iprsearch.ipindia.gov.in/DynamicUtility/CauseListForHearingCase/Index
टिप्पणी:- उक्त सुनवाई में भाग लेने के लिए वीडियो कॉन्फ्रेंसिंग लिंक के संबंध में अलग से ईमेल भेजी जाएगी।
Note:- Separate email will be sent regarding the Video Conferencing Link to attend the said hearing.

* Hearing Objections are attached.

Objections

Clarity and Conciseness

1. The following objection raised in FER has still not been met.

   Reference numeral shall be marked in the drawings and reference of it shall be incorporated in the claims to enhance the intelligibility of Claims. No description matter is allowed on the drawings as per Rule 15(7) of The Patents Rule, 2003 (as amended).

2. 1. The reference numerals of the drawings are not incorporated in the claims; the applicant is required to amend the claims to include appropriate reference numerals corresponding to the features shown in the drawings to ensure clarity and consistency.

   2. Abbreviated forms of words have been used in the claims and/or description. The applicant is required to replace all such abbreviations with their complete forms to ensure clarity and proper understanding of the invention, as abbreviations may lead to ambiguity or misinterpretation of the technical features described.

Formal Requirement(s)

1. 1. The General Power of Attorney (GPA) filed by the opposition is found to be defective. It is observed that the Patent Agent number (IN/PA No.) mentioned in the GPA is incorrect and does not correspond to the registered number of the patent agent Mr. Hemant Thadhani, who is stated to be authorized to act on behalf of the opposition. Accordingly, the opposition is directed to submit a corrected and duly executed General Power of Attorney, wherein the correct Patent Agent number (IN/PA No.) of Mr. Hemant Thadhani is mentioned. The corrected GPA shall be properly executed, dated, and, if applicable, notarized or legalized in accordance with the prescribed procedure.

   2. It is observed that the applicant has submitted amendments to the claims in the present application by filing Form 13. As per Section 57 of the Patents Act, 1970, read with Rule 81(1) of the Patents Rules, 2003, any amendment to a patent application, complete specification, or any related document must be made by filing Form 13, accompanied by a clear statement showing the basis and support for the proposed amendment in the originally filed specification.In this regard, the applicant is directed to provide adequate justification and support for each amendment made, with specific references to the description, drawings, or original claims. Until such compliance is made, the submitted Form 13 and the amendments to the claims cannot be taken on record. The applicant is advised to ensure that the basis for each amendment is clearly documented to maintain conformity with the requirements of the Patents Act and Rules.

   3. It is observed that the applicant has filed Form 13 for amendment of the claims in the present application. As per Section 57(2) of the Patents Act, 1970, every application for leave to amend an application for a patent, a complete specification, or any related document must provide full particulars of the reasons for which the application is made. In the present submission, the reasons or justification for the amendments have not been clearly furnished. The applicant is therefore directed to submit a revised Form 13 clearly stating the detailed reasons and justification for making each amendment. Until compliance with the above requirements is made, the submitted Form 13 and the proposed amendments cannot be taken on record.

Invention u/s 2(1)(ja)

1. The applicant's reply to the objection raised in the First Examination Report (FER) under Section 2(1)(ja) of the Patents Act has been duly considered.

   Upon further examination, it has been observed that the following prior art documents disclose the features of the alleged invention:

   D4: CN103820781A (Publication Date: 25/08/2014)

   D5: CN105148816A (Publication Date: 16/12/2015)

   D6: CN206372825U (Publication Date: 04/08/2017)

   Doucmnet D4 discloses a profiled outer glass-lined device, which mainly comprises a can lid and a can body, wherein the can lid is provided with a nozzle, a manhole and a sealing mouth; the outer part of the can body is covered with a jacket, the can body and the clip The sleeve is welded into a unitary structure with a cavity therebetween, a jacket flange is provided on one side of the tank body, and a discharge port, a condensate outlet, a steam inlet and a steam outlet are arranged at a lower portion.

   The document further discloses the outer side of the outer side of the can body and the outer side of the jacket have a

stainless steel outer layer. The upper part of the stainless steel outer layer of the outer side of the can body is covered with a can body flange surface and a can body surface I stainless steel outer layer. The can body and the jacket welding portion There is a canned transition surface I stainless steel outer layer, the outer side of the jacket is covered with a can body surface II stainless steel outer layer, and the lower part of the can body is covered with a stainless steel outer layer of the can body sealing surface, and the lower part of the can body and the jacket welding portion are covered The can body transition surface II stainless steel outer layer, the bottom of the can body is covered with a can body surface III stainless steel outer layer.

D5 discloses the manufacturing method of a glass-lined reaction kettle and belongs to the technical field of processing, which solves the problem that the outer surface of the glass-lined reaction kettle is poor in corrosion resistance. A stainless steel flange is placed on the nozzle flange of each process pipe, and the upper cover is covered from the top to the upper cover, so that the process pipe protrudes from the through hole of the upper foreskin, and then The upper foreskin is welded with the stainless steel flange, and the cylinder is wrapped with the middle foreskin, and the lower cover is covered from the bottom with the lower foreskin, so that the process tube protrudes from the through hole of the lower foreskin, and the lower foreskin is welded with the stainless steel flange. Finally, the lower foreskin and the middle foreskin are sealed and welded.

D6 discloses a corrosion -resistant glass reactor that wards off of high strength, including pressing from both sides cover, jar body, agitator, upper cover, low head, sight glass manhole, the thermometer mouth of pipe, feed inlet, drain hole, the external side of jar is equipped with presss from both sides the cover, it is equipped with a plurality of installation positioner to press from both sides angles such as overlapping the outside, pressing from both sides cover upper portion and being equipped with steam inlet, steam inlet one side is equipped with the relief port, presss from both sides the cover bottom and is equipped with water export with fixed attention, the welding has high -neck flange on the feed inlet, the welding has high -neck flange on the sight glass manhole, and the welding has high -neck flange on the thermometer mouth of pipe, and the welding has high -neck flange on the drain hole, be equipped with the ground coat on the internal lateral wall of jar, be equipped with cover glaze layer on the ground coat, be equipped with the ground coat on high -neck flange inside wall, terminal surface, the recess, be equipped with cover glaze layer on the ground coat. The utility model discloses trompil department welding has increased high -neck flange, has reduced the quantity at stamping forming trompil R position, has directly increased the effective thickness of this equipment, and compressive strength improves relatively, ensuring the safe operation of equipment.

In view of the foregoing, it is evident that a person skilled in the art could and would readily arrive at the present invention by drawing sufficient motivation from the cited prior arts (D4-D6). The combination of features disclosed in these references provides ample guidance to derive the claimed invention without the need for any inventive step or intellectual creativity. Consequently, the claims of the present application are deemed to be obvious and lacking an inventive step in accordance with the provisions of Section 2(1)(ja) of the Patents Act.

Other Requirement(s)

1. The amendment made in the description and claims does not comply with the provisions of Section 59(1) of the Act.

2. • Note:-A hearing in the above-mentioned pre-grant opposition matter was scheduled on 14/11/2025. Both the Applicant and the Opponent had intimated their intention to attend the hearing.At the time of the hearing, it was observed from the official module that the prescribed fee for hearing from the Opponent was not reflected as paid. In view thereof, the Opponent was not permitted to advance oral submissions during the hearing proceedings. The Opponent, though present, submitted that the requisite fee had been filed; however, as the same was not reflecting in the system at that time, the hearing proceeded without permitting oral arguments from the Opponent. Thereafter, both parties filed their respective written submissions. It has subsequently come to notice that the prescribed hearing fee filed by the Opponent was duly paid and is now reflecting in the official module. In order to ensure compliance with the principles of natural justice and to provide a fair and reasonable opportunity of hearing to the Opponent, it is considered appropriate to afford a fresh opportunity of oral hearing in the matter.

Scope

1. (a) The scope of principal claim 1 is ambiguous due to multiple characterization clauses introduced by the phrase "wherein." Re-draft or rephrase the principal claim to clearly define the constructional inventive feature of the alleged invention over which protection is sought.

(b) The phrases:

"Coated nozzle is seamlessly welded to the top cover's cladding sheet, fully enclosing the reactor in stainless steel, addressing rust challenges in critical zones, and forming a resilient barrier that upholds reactor integrity, longevity, and product purity."

"The reactor offers excellent resistance to oxidation at the nozzle. In the event of accidental spillage of corrosive liquid onto the nozzle surface, oxidation of the metal surface may occur, leading to damage of the glass lining coated inside, as oxidation of the base metal would cause distortion."

"The reactor is fully covered with stainless steel, enhancing its resilience against external damage and increasing overall safety."

These statements in claim 1 are narrative in nature and contain indefinite, functional, or operational language. The principal claim is vaguely drafted in a functional format. As per Section 10 of the Patents Act, 1970 (as amended), vague and functional aspects are not permissible in claims; therefore, the claims are not allowable.

2. 1. Claim 1 recites functional advantages such as “offering excellent resistance to oxidation,” “tackling rust challenges,” “upholding reactor integrity, longevity, and product purity,” and “enhancing resilience against external damage and increasing overall safety,” which merely describe the functional benefits or results of the claimed reactor rather than defining specific technical or structural features. Such statements of function or result do not constitute patentable subject matter under Section 2(1)(j) of the Patents Act, 1970. The claim should therefore be amended to clearly define the structural or technical features responsible for achieving the stated effects instead of claiming the functional benefits themselves.

2. Claims 2 and 7 do not clearly and unambiguously define the scope of the invention as required under Section 10(4)(c). In Claim 2, the term “base body of high mechanical strength” is vague and lacks measurable parameters, and the expressions “essentially,” “preferably,” and “most preferably” regarding the reaction chamber thickness introduce ambiguity. In Claim 7, the term “smooth as whole” is indefinite and does not specify objective criteria for surface smoothness. The claims therefore fail to distinctly define the invention and may lead to multiple interpretations. The applicant is directed to amend Claims 2 and 7 to clearly define the mechanical strength of the base body in measurable terms and to specify the smoothness of the glass lining/coating in objective terms, so as to comply with Section 10(4)(c) of the Patents Act, 1970.

3. It is observed that there is an inconsistency between the description, claims, and drawings regarding the material of the reactor walls. In the description and claims, the outer wall of the reactor is disclosed as stainless steel (SS) and the inner wall as mild steel (MS). However, in the drawings, the layers appear inversed, showing the outer wall as MS and the inner wall as SS. This inconsistency creates ambiguity regarding the actual structure of the reactor and the scope of the invention. The applicant is required to rectify the discrepancy by amending the description, claims, and/or drawings so that the material of the outer and inner walls is consistent throughout the specification and clearly defines the invention, in compliance with Section 10(4) of the Patents Act, 1970.

Sufficiency of Disclosure u/s 10 (4)

1. 1. It is observed that references have been made in the description to Figures 4 and 5. However, these figures are not included in the drawings filed with the application. The absence of Figures 4 and 5 renders the references in the description inconsistent with the accompanying drawings, leading to ambiguity in understanding the disclosure. Accordingly, the applicant is required to file the missing Figures 4 and 5 with proper reference in the description, to ensure conformity between the description and the drawings as required under Section 10(4) of the Patents Act, 1970 and Rule 15 of the Patents Rules, 2003.

2. It is observed that certain structural features of the invention, such as the top flange, bottom flange, glass-lined metal flat surface below the top flange for observing the flow regime in the reactor or mixer, the cavity in the bottom flange facilitating the flow of heat transfer fluid through inlet and outlet nozzles and flanges, the bottom plate having a metal surface in contact with the heat transfer fluid, the connection plate connecting the inlet and outlet to the bottom flange of the glass-lined stainless-steel reactor, the multiple bolts securing the top and bottom flanges, and the external stainless-steel sheet covering of the reactor, have been described in the specification. However, these features are not shown or illustrated in the drawings accompanying the application. As per the requirements of Section 10(4) of the Patents Act, 1970, every complete specification shall fully and particularly describe the invention and its operation or use and the method by which it is to be performed, and shall disclose the best method of performing the invention known to the applicant. Accordingly, the applicant is required to incorporate all the described structural features in the drawings, ensuring that each feature or embodiment mentioned in the description is properly reflected in the drawings with corresponding reference numerals. This is necessary to maintain consistency between the description and drawings and to ensure that the invention is fully and clearly disclosed in compliance with the statutory requirements.

Daswani & Daswani
Patent & Trade Mark (Intellectual Property) Attorneys

Daswani House, Green Acres,
23B, Ahiripukur 1st Lane, Kolkata - 700 019, India.
Telephone : 91 (33) 4603 2020 / 4603 1666
Website : http://www.daswanianddaswani.com
E-mail : kolkata@daswaniindia.net

Your Ref: POC/Application No./202141018455 March 06. 2016

To,

The Controller of Patent

The Patent Office

Chennai

Kind Attention:- Shri Kanhaiya Lal Lalai (Asst Controller of P & D)

Respected Sir,

Re: Request for official confirmation regarding filing of request for hearing under Rule 62(2) and supply of video recording of hearing in respect of Patent Application No. 202141018455 titled "CLEAN ROOM GLASS LINED REACTOR"

The Applicant respectfully submits as under:

1. The Applicant filed Patent Application No. 202141018455 titled "CLEAN ROOM GLASS LINED REACTOR" on 21 April 2021, which was published in the Patent Office Journal on 28 October 2022.

2. The Opponent, Pinakin Engineers Pvt. Ltd., through its Patent Agent Mr. Hemant Thadhani (IN/PA-4057), filed a Notice of Pre-Grant Opposition under Section 25(1) of the Patents Act, 1970 on 14 November 2024 along with the statement and evidence in terms of Rule 55 of the Patents Rules, 2003.

3. The Applicant filed its reply statement along with supporting evidence on 04 August 2025.

4. Upon completion of the evidence stage, the Learned Controller issued a notice dated 13 October 2025 scheduling the hearing on 14 November 2025 under Rule 62(1) of the Patents Rules.

5. Rule 62(2) clearly mandates that any party desirous of being heard must file a formal request along with the prescribed fee within the stipulated period, failing which such party is not entitled to be heard.

6. The Opponent did not file the mandatory request for hearing within the prescribed period, nor was any payment of the prescribed hearing fee reflected in the official system prior to the commencement of the hearing on 14 November 2025.

7. Accordingly, during the hearing held on 14 November 2025, the Applicant raised a specific objection to the participation of the Opponent on the ground of non-compliance with Rule 62(2). The objection was duly recorded, and the Opponent was not permitted to advance oral submissions, whereupon the hearing proceeded with the Applicant presenting its arguments.

8. Notwithstanding the above, the Opponent subsequently attempted to circumvent the statutory procedure by filing purported written submissions dated 28 November 2025 which were filed on 29 November 2025, i.e., after conclusion of the hearing. Such filing is procedurally impermissible and contrary to the scheme of the Patents Act and Rules, which do not permit a party who has failed to comply with Rule 62(2) to introduce submissions after the hearing has already concluded.

9. The Applicant further notes with serious concern that the Opponent deposited the prescribed hearing fee only on 14 November 2025, i.e., on the very date of the hearing itself, and thereafter attempted to place on record a back-dated letter dated 13 November 2025 purporting to be a prior request for hearing.

10. The Applicant respectfully submits that the aforesaid conduct prima facie indicates a deliberate attempt to create a misleading procedural record so as to falsely suggest compliance with Rule 62(2) prior to the hearing, when in fact no such compliance existed.

11. The Applicant was therefore surprised to receive a communication bearing Ref. No. POC/Application No./202141018455 whereby the hearing has been rescheduled for 11 March 2026 on the ground that the hearing fee filed by the Opponent was subsequently reflected in the official module and that an opportunity of hearing is being granted in the interest of natural justice.

12. The Applicant respectfully submits that the Patents Act and the Patents Rules do not confer any power upon the Learned Controller to grant a fresh opportunity of hearing to a party that has failed to comply with the mandatory requirement of Rule 62(2). Once a party fails to file the prescribed request for hearing with the prescribed fee within the stipulated period, the consequence under the Rules is that such party forfeits its right to be heard.

13. In these circumstances, the Applicant respectfully submits that the rescheduling of the hearing on the purported ground of natural justice raises serious procedural concerns and is contrary to the scheme of the Patents Act and Rules.

14. The Applicant further submits that it is in the process of preferring a separate Interlocutory Petition before the Learned Controller objecting to the rescheduling of the hearing and placing on record the Applicant's objection to the grant of a fresh opportunity of hearing to the Opponent.

15. The Applicant also seeks clarity regarding the precise procedural record relating to the Opponent's purported request for hearing and payment of the prescribed fee, as these facts are material to the Applicant's objections and the legal remedies that the Applicant may pursue.

16. In these circumstances, the Applicant respectfully requests your good office to kindly provide the following:
    a) Official confirmation from the Indian Patent Office online filing system specifying the exact date and time of filing of the request for hearing by the Opponent, along with the exact date and time of payment of the prescribed hearing fee;

b) A copy of the video recording of the hearing conducted on 14 November 2025 which was attended by the Applicant's Patent Agent.

17. The above records are necessary to enable the Applicant to ascertain the correctness of the procedural record and to take appropriate legal recourse in accordance with law.

18. The Applicant therefore respectfully requests that the above information and documents be provided at the earliest convenience of your office.

With Kind Regards ,

Bharat Daswani (Applicants Agent)
DASWANI & DASWANI

Digitally Signed By:
MR. BHARAT DASWANI
Date: 06-03-2026 18:35:42

Daswani & Daswani
Patent & Trade Mark (Inteilectual Property) Attorneys

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Telephone : 91 (33) 4603 2020 / 4603 1666
Website : http://www.daswaniindia.net
E-mail : kolkata@daswaniindia.net

March 09, 2026

Your Ref. POC/Application No /202141018455

Our Ref. 202141018455

Rs. 8,000/-

To

The Controller of Patents,

The Patent Office,

Chennai

Kind Attention:

Kanhaiya Lal Balai

Assistant Controller of

Patents & Designs

Dear Sir,

Sub: Indian Patent Application No. 202141018455 of 21.04.2021 titled "CLEAN ROOM GLASS LINED REACTOR" filed by Standard Glass Lining Technology Private Limited

We refer to the Pre-Grant Opposition Hearing Adjournment Notice dated 11 February 2026 issued in respect of Patent Application No. 202141018455, whereby the Pre-Grant Opposition hearing has been rescheduled for 11 March 2026.

In this regard, the Applicant submits that the hearing in the above-mentioned matter was already completed on 14 November 2025. In order to set aside the repeated hearing notice dated 11 February 2026 and to enable the matter to proceed to decision based on the hearing already concluded on 14 November 2025, the Applicant has filed herewith an Interlocutory Petition under Rule 137 of the Patents Rules, 2003 read with Section 117A of the Patents Act, 1970.

It is therefore respectfully requested that the Learned Controller may kindly take the present petition on record and proceed further in the matter accordingly.

Yours Sincerely,

Of Daswani & Daswani

(Bharat Daswani)

Patent Agent

IN / PA 289

Digitally Signed By:
MR. BHARAT DASWANI
Date: 09-03-2026 16:29:56

INTERLOCUTORY PETITION

IN THE MATTER OF:

The Patents Act, 1970 and The Patents Rules, 2003

IN THE MATTER OF:

Patent Application No. 202141018455 of 21.04.2021 titled "CLEAN ROOM GLASS LINED REACTOR" filed by Standard Glass Lining Technology Private Limited

And

IN THE MATTER OF:

INTERLOCUTORY PETITION UNDER RULE 137 OF THE PATENTS RULES, 2003 READ WITH SECTION 117A OF THE PATENTS ACT, 1970

For directions to set aside the repeated hearing notice dated 11 February 2026 and to proceed to decision on the basis of the hearing already completed on 14 November 2025.

1. The Petitioner/Applicant is Standard Glass Lining Technology Private Limited, the original applicant in Patent Application No. 202141018455 for the invention titled "Clean Room Glass Lined Reactor".

2. The Applicant is represented by its authorized Patent Agent, Mr. Bharat Daswani, IN/PA 289, of Daswani & Daswani, having address Green Acres 23B, Ahiripukur 1st Lane, Kolkata 700 019.

3. The Opponent in the pre-grant opposition proceedings is Pinakin Engineers Private Limited, who has filed a pre-grant opposition under Section 25(1) of the Patents Act, 1970.

4. The Application No. 202141018455 was filed on 21 April 2021 for a "Clean Room Glass Lined Reactor", which concerns a specialized glass-lined reactor vessel with stainless steel cladding fully enclosing the reactor, including coated nozzles, intended to address corrosion and oxidation issues in clean-room environments.

5. The application was published in the Patent Office Journal on 28 October 2022 under Section 11A of the Act.

6. On 14 November 2024, the Opponent filed a pre-grant opposition under Section 25(1), along with statement and evidence under Rule 55 of the Patents Rules, 2003, relying inter alia on prior arts CN103820781A (D4), CN105148816A (D5) and CN206372825U (D6).

7. The Applicant filed its reply statement and evidence on 4 August 2025 under Rule 58, thereby completing the evidentiary stage in the pre-grant opposition.

8. The Learned Controller issued a hearing notice dated 13 October 2025, fixing the hearing on 14 November 2025 via video conferencing, in terms of Rule 62(1), and calling upon parties to file the requisite request for hearing in Form 14 with prescribed fee under Rule 62(2) read with the First Schedule.

9. The Applicant duly complied with Rule 62(2) by filing Form 14 and fee, thereby validly invoking its right to be heard.

10. On the date of hearing, 14 November 2025, the Opponent appeared through video conferencing but had not filed Form 14 nor paid the prescribed hearing fee, as recorded in the official module of the Patent Office.

11. In view of the Opponent's non-compliance with Rule 62(2), the Learned Controller permitted the Opponent to remain present but correctly did not permit the Opponent to advance oral arguments, and proceeded with the hearing only on the Applicant's side, thereby completing the hearing under Section 25(1).

12. Subsequently, the Opponent sought to place on record written submissions dated 28 November 2025 and later sought to rely on a backdated request purporting to be from 13 November 2025 regarding the hearing fee and Form 14, which documents were not forming part of the record at the time of hearing and are inconsistent with the position recorded during the hearing.

13. There is no contemporaneous correspondence or on-record request from the Opponent, after 14 November 2025, seeking a fresh hearing or explaining its failure to comply with Rule 62(2).

14. Notwithstanding the above, by communication around 29 November 2025, followed by a formal notice dated 11 February 2026, the Learned Controller adjourned the matter and fixed a fresh hearing on 11 March 2026, relying on

the later reflection of the Opponent's hearing fee in the system and invoking "principles of natural justice" as justification.

15. Under Section 25(1) read with Rules 55, 58 and 62, the pre-grant opposition is a structured, quasi-judicial proceeding with a defined evidentiary stage and a single statutory hearing after which the Learned Controller is required to decide under Section 15 whether to grant or refuse the patent.

16. Rule 62(2) expressly makes filing of Form 14 with the prescribed fee a condition precedent for a party to be "heard". A party that does not comply forfeits the right to advance oral submissions, although it may remain present as a spectator, as occurred with the Opponent in the present case.

17. Once the hearing of 14 November 2025 concluded with the Applicant's arguments having been heard and the Opponent having failed to comply with Rule 62(2), the proceedings attained finality at the hearing stage. There is no provision in Section 25(1), Section 15 or Rule 62 which authorizes the Controller to re-open and re-list a completed hearing merely because the Opponent subsequently perfects its procedural default, that too without any formal application on record.

18. The impugned repeated hearing notice dated 11 February 2026 therefore travels beyond the four corners of Section 14, Section 15 and Rule 62, and is ex facie ultra vires, as it purports to grant the Opponent a fresh right of audience in substitution of its forfeited right, thereby nullifying the consequence of its non-compliance with Rule 62(2).

19. In any event, the Opponent's substantive rights are adequately safeguarded by the availability of post-grant remedies under Section 25(2) and Section 64, and, if warranted, compulsory licensing under Section 84; hence, even on a liberal reading of "natural justice", there is no necessity to unsettle the completed pre-grant hearing to avoid "absolute injustice" to the Opponent.

20. The principles of natural justice, as recognized in Article 14 jurisprudence including Maneka Gandhi v. Union of India (AIR 1978 SC 597), require a fair opportunity of hearing but do not compel authorities to grant repeated or additional hearings to cure a party's own procedural defaults, especially when statutory preconditions such as fee and prescribed forms have not been complied with.

21. In the present case, the Opponent was aware of the hearing date, was present at the hearing, but had not complied with Rule 62(2). The Learned Controller therefore justifiably restricted the hearing to the Applicant's side, thereby satisfying the audi alteram partem requirement to the extent of the parties that had properly invoked their right of audience.

22. The subsequent decision to re-open the hearing suo motu, without any request from the Opponent, in order to retroactively accommodate its belated fee payment, in substance converts "natural justice" into a tool for granting a second procedural opportunity to the defaulting party, at the cost of delay and prejudice to the compliant Applicant.

23. The Applicant is gravely prejudiced by the repeated hearing notice, as it defers consideration under Section 15 and prolongs uncertainty in relation to

a mechanical engineering invention where time to market and exclusivity are commercially critical, particularly for specialized glass-lined reactors designed for clean-room pharmaceutical environments.

24. Conversely, the Opponent suffers no irreparable prejudice if the Learned Controller proceeds to decide the matter on the basis of the completed hearing and existing record, since it retains a full suite of post-grant challenges under the Act. The balance of convenience and proportionality in applying natural justice thus lies entirely in favour of preserving the finality of the 14 November 2025 hearing and setting aside the repeated hearing notice.

25. The Opponent's post-hearing written submissions dated 28 November 2025 and the subsequent attempt to rely on a backdated Form 14/fee request allegedly from 13 November 2025 are not contemplated by the scheme of Rule 62, which does not provide for unsolicited post-hearing written submissions or retrospective regularization of hearing rights absent specific leave of the Learned Controller.

26. The apparent backdating raises a serious and material question as to the true date and time of filing of the Opponent's Form 14 and payment of the hearing fee, which directly bears on compliance with Rule 62(2) and on the legality of the decision to re-open the hearing.

27. In order to dispel any doubt and to maintain the sanctity of the official record under Section 74, it is necessary that the Learned Controller place on record an official confirmation of the exact date and time at which the

Opponent's Form 14 and hearing fee were actually received and reflected in the Office system, and also that the video recording of the hearing held on 14 November 2025 be made available to the Applicant.

28. Rule 137 of the Patents Rules, 2003 recognizes the power of the Learned Controller to correct irregularities in procedure. The present case squarely calls for exercise of such power, not to confer any new substantive right on either party, but to correct the irregularity of having re-opened a completed hearing without statutory basis and without request from the defaulting party.

29. Judicial authority has emphasized that once a statutory hearing is duly concluded, it cannot be casually re-opened save on limited grounds such as manifest error or fraud (for instance, Union of India v. Tarachand Gupta & Bros., AIR 1971 SC 1558). The facts on record do not disclose any such ground; instead, they show a completed hearing and a subsequent attempt to retrospectively regularize the Opponent's default.

30. In these circumstances, the interests of justice, statutory finality and administrative propriety are best served if the Learned Controller recalls the repeated hearing notice dated 11 February 2026 and proceeds to decide the application under Section 15 on the basis of the hearing held on 14 November 2025 and the existing written record.

PRAYER

31. In view of the foregoing facts and submissions, the Applicant most respectfully prays that this Learned Controller may be pleased to:

a) Set aside/recall the repeated hearing notice dated 11 February 2026, whereby the pre-grant opposition in respect of Patent Application No. 202141018455 has been adjourned and re-fixed for hearing on 11 March 2026.

b) Direct that the pre-grant opposition proceedings be treated as finally heard on 14 November 2025, and that the application be taken up for decision under Section 15 on the basis of (i) the oral hearing of the Applicant on that date, and (ii) the statements and evidence already on record from both parties in accordance with Rules 55–60.

c) Pass such further or other orders as this Learned Controller may deem fit and proper in the facts and circumstances of the case and in the interest of justice.

Dated this 09th day of March, 2026



(Bharat Daswani)
Of Daswani & Daswani
Applicant's Agent
IN / PA 289

VERIFICATION

I, Bharat Daswani, the authorized representative of the Applicant, do hereby verify that the contents of the above petition are true to my knowledge and belief, derived from the records of the case, and nothing material has been concealed therefrom.

Verified at Kolkata on this 09th day of March 2026.

Authorized Signatory for the Applicant



(Bharat Daswani)
Of Daswani & Daswani
Applicant's Agent
IN / PA 289

Digitally Signed By:
MR. BHARAT DASWANI
Date: 09-03-2026 16:29:56

CRAWFORD BAYLEY & CO.

(Registered)

Advocates & Solicitors

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BY ONLINE FILING

Confidential and Urgent

HCA/SKA/HT/RD

March 10, 2026

To,
The Controller of Patents,
The Patent Office,

ONLINE SUBMISSION

Dear Sir,

Notice to attend Hearing

Reference:	INDIA-Patent Application No.	**202141018455**
	Applicant:	**Standard Glass Lining Private Limited**

This is in reference to the hearing notice dated 11-02-2026 fixing hearing of the scheduled application on 11.03.2026.

This is to notify Opponent's agent Mr. Hemant Thadhani (IN/PA-4057) will be attending the hearing scheduled on **11.03.2026 at 11:30am**.

Yours faithfully,
FOR **Pinakin Engineers Private Ltd**
By its Agent

Hemant Thadhani
Patent Agent ((IN/PA-4057)
Crawford Bayley & Co.

Digitally Signed By:
Hemant Thadhani
Date: 10-03-2026 19:12:32

Daswani & Daswani
Patent & Trade Mark (Intellectual Property) Attorneys

Daswani House, Green Acres,
23B, Ahiripukur 1st Lane, Kolkata - 700 019, India.
Telephone : 91 (33) 4603 2020 / 4603 1666
Website : http://www.daswanianddaswani.com
E-mail : kolkata@daswaniindia.net

March 11, 2026

To
The Controller of Patents
The Patent Office
Chennai

Kind Attention :-
Shri Kanhaiya Lal Balai
Assistant Controller of Patents & Designs

Respected Sir,

Re: Indian Patent Application No. 202141018455 of 21.04.2021 titled "CLEAN ROOM GLASS LINED REACTOR" filed by Standard Glass Lining Technology Private Limited

We respectfully request that the typographical error in our Interlocutory Petition, filed on March 09, 2026, be kindly condoned.

In the Petition, references to Form 14 in paragraphs 8, 9, 10, 12, 16, 25, 26 and 27 were inadvertent typographical errors and should be considered deleted.

The corrected Interlocutory Petition has already been uploaded for your reference.

We appreciate your kind attention in this matter.

Yours sincerely,
For, DASWANI & DASWANI

Bharat Daswani
(Applicant's Agent)
Reg. No. IN/PA-289

Digitally Signed By:
MR. BHARAT DASWANI
Date: 11-03-2026 16:24:30

INTERLOCUTORY PETITION

IN THE MATTER OF:

The Patents Act, 1970 and The Patents Rules, 2003

IN THE MATTER OF:

Patent Application No. 202141018455 of 21.04.2021 titled "CLEAN ROOM GLASS LINED REACTOR" filed by Standard Glass Lining Technology Private Limited

And

IN THE MATTER OF:

INTERLOCUTORY PETITION UNDER RULE 137 OF THE PATENTS RULES, 2003 READ WITH SECTION 117A OF THE PATENTS ACT, 1970

For directions to set aside the repeated hearing notice dated 11 February 2026 and to proceed to decision on the basis of the hearing already completed on 14 November 2025.

1. The Petitioner/Applicant is Standard Glass Lining Technology Private Limited, the original applicant in Patent Application No. 202141018455 for the invention titled "Clean Room Glass Lined Reactor".

2. The Applicant is represented by its authorized Patent Agent, Mr. Bharat Daswani, IN/PA 289, of Daswani & Daswani, having address Green Acres 23B, Ahiripukur 1st Lane, Kolkata 700 019.

3. The Opponent in the pre-grant opposition proceedings is Pinakin Engineers Private Limited, who has filed a pre-grant opposition under Section 25(1) of the Patents Act, 1970.

4. The Application No. 202141018455 was filed on 21 April 2021 for a "Clean Room Glass Lined Reactor", which concerns a specialized glass-lined reactor vessel with stainless steel cladding fully enclosing the reactor, including coated nozzles, intended to address corrosion and oxidation issues in clean-room environments.

5. The application was published in the Patent Office Journal on 28 October 2022 under Section 11A of the Act.

6. On 14 November 2024, the Opponent filed a pre-grant opposition under Section 25(1), along with statement and evidence under Rule 55 of the Patents Rules, 2003, relying inter alia on prior arts CN103820781A (D4), CN105148816A (D5) and CN206372825U (D6).

7. The Applicant filed its reply statement and evidence on 4 August 2025 under Rule 58, thereby completing the evidentiary stage in the pre-grant opposition.

8. The Learned Controller issued a hearing notice dated 13 October 2025, fixing the hearing on 14 November 2025 via video conferencing, in terms of Rule 62(1), and calling upon parties to file the requisite notice to attend hearing with prescribed fee under Rule 62(2) read with the First Schedule.

9. The Applicant duly complied with Rule 62(2) by filing notice to attend hearing and fee, thereby validly invoking its right to be heard.

10. On the date of hearing, 14 November 2025, the Opponent appeared through video conferencing but had not filed any notice to attend hearing nor paid

the prescribed hearing fee, as recorded in the official module of the Patent Office.

11. In view of the Opponent's non-compliance with Rule 62(2), the Learned Controller permitted the Opponent to remain present but correctly did not permit the Opponent to advance oral arguments, and proceeded with the hearing only on the Applicant's side, thereby completing the hearing under Section 25(1).

12. Subsequently, the Opponent sought to place on record written submissions dated 28 November 2025 and later sought to rely on a backdated request purporting to be from 13 November 2025 regarding the hearing fee and notice to attend hearing, which documents were not forming part of the record at the time of hearing and are inconsistent with the position recorded during the hearing.

13. There is no contemporaneous correspondence or on-record request from the Opponent, after 14 November 2025, seeking a fresh hearing or explaining its failure to comply with Rule 62(2).

14. Notwithstanding the above, by communication around 29 November 2025, followed by a formal notice dated 11 February 2026, the Learned Controller adjourned the matter and fixed a fresh hearing on 11 March 2026, relying on the later reflection of the Opponent's hearing fee in the system and invoking "principles of natural justice" as justification.

15. Under Section 25(1) read with Rules 55, 58 and 62, the pre-grant opposition is a structured, quasi-judicial proceeding with a defined evidentiary stage and a single statutory hearing after which the Learned

Controller is required to decide under Section 15 whether to grant or refuse the patent.

16. Rule 62(2) expressly makes filing of notice to attend hearing with the prescribed fee a condition precedent for a party to be "heard". A party that does not comply forfeits the right to advance oral submissions, although it may remain present as a spectator, as occurred with the Opponent in the present case.

17. Once the hearing of 14 November 2025 concluded with the Applicant's arguments having been heard and the Opponent having failed to comply with Rule 62(2), the proceedings attained finality at the hearing stage. There is no provision in Section 25(1), Section 15 or Rule 62 which authorizes the Controller to re-open and re-list a completed hearing merely because the Opponent subsequently perfects its procedural default, that too without any formal application on record.

18. The impugned repeated hearing notice dated 11 February 2026 therefore travels beyond the four corners of Section 14, Section 15 and Rule 62, and is ex facie ultra vires, as it purports to grant the Opponent a fresh right of audience in substitution of its forfeited right, thereby nullifying the consequence of its non-compliance with Rule 62(2).

19. In any event, the Opponent's substantive rights are adequately safeguarded by the availability of post-grant remedies under Section 25(2) and Section 64, and, if warranted, compulsory licensing under Section 84; hence, even on a liberal reading of "natural justice", there is no necessity to unsettle the completed pre-grant hearing to avoid "absolute injustice" to the Opponent.

20. The principles of natural justice, as recognized in Article 14 jurisprudence including Maneka Gandhi v. Union of India (AIR 1978 SC 597), require a fair opportunity of hearing but do not compel authorities to grant repeated or additional hearings to cure a party's own procedural defaults, especially when statutory preconditions such as fee and prescribed forms have not been complied with.

21. In the present case, the Opponent was aware of the hearing date, was present at the hearing, but had not complied with Rule 62(2). The Learned Controller therefore justifiably restricted the hearing to the Applicant's side, thereby satisfying the audi alteram partem requirement to the extent of the parties that had properly invoked their right of audience.

22. The subsequent decision to re-open the hearing suo motu, without any request from the Opponent, in order to retroactively accommodate its belated fee payment, in substance converts "natural justice" into a tool for granting a second procedural opportunity to the defaulting party, at the cost of delay and prejudice to the compliant Applicant.

23. The Applicant is gravely prejudiced by the repeated hearing notice, as it defers consideration under Section 15 and prolongs uncertainty in relation to a mechanical engineering invention where time to market and exclusivity are commercially critical, particularly for specialized glass-lined reactors designed for clean-room pharmaceutical environments.

24. Conversely, the Opponent suffers no irreparable prejudice if the Learned Controller proceeds to decide the matter on the basis of the completed hearing and existing record, since it retains a full suite of post-grant challenges under the Act. The balance of convenience and proportionality

in applying natural justice thus lies entirely in favour of preserving the finality of the 14 November 2025 hearing and setting aside the repeated hearing notice.

25. The Opponent's post-hearing written submissions dated 28 November 2025 and the subsequent attempt to rely on a backdated notice to attend hearing/fee request allegedly from 13 November 2025 are not contemplated by the scheme of Rule 62, which does not provide for unsolicited post-hearing written submissions or retrospective regularization of hearing rights absent specific leave of the Learned Controller.

26. The apparent backdating raises a serious and material question as to the true date and time of filing of the Opponent's notice to attend hearing and payment of the hearing fee, which directly bears on compliance with Rule 62(2) and on the legality of the decision to re-open the hearing.

27. In order to dispel any doubt and to maintain the sanctity of the official record under Section 74, it is necessary that the Learned Controller place on record an official confirmation of the exact date and time at which the Opponent's filed notice to attend hearing and hearing fee were actually received and reflected in the Office system, and also that the video recording of the hearing held on 14 November 2025 be made available to the Applicant.

28. Rule 137 of the Patents Rules, 2003 recognizes the power of the Learned Controller to correct irregularities in procedure. The present case squarely calls for exercise of such power, not to confer any new substantive right on either party, but to correct the irregularity of having re-opened a completed

hearing without statutory basis and without request from the defaulting party.

29. Judicial authority has emphasized that once a statutory hearing is duly concluded, it cannot be casually re-opened save on limited grounds such as manifest error or fraud (for instance, Union of India v. Tarachand Gupta & Bros., AIR 1971 SC 1558). The facts on record do not disclose any such ground; instead, they show a completed hearing and a subsequent attempt to retrospectively regularize the Opponent's default.

30. In these circumstances, the interests of justice, statutory finality and administrative propriety are best served if the Learned Controller recalls the repeated hearing notice dated 11 February 2026 and proceeds to decide the application under Section 15 on the basis of the hearing held on 14 November 2025 and the existing written record.

PRAYER

31. In view of the foregoing facts and submissions, the Applicant most respectfully prays that this Learned Controller may be pleased to:

a) Set aside/recall the repeated hearing notice dated 11 February 2026, whereby the pre-grant opposition in respect of Patent Application No. 202141018455 has been adjourned and re-fixed for hearing on 11 March 2026.
b) Direct that the pre-grant opposition proceedings be treated as finally heard on 14 November 2025, and that the application be taken up for decision under Section 15 on the basis of (i) the oral hearing of the Applicant on that date, and (ii) the statements and evidence already on record from both parties in accordance with Rules 55–60.

c) Pass such further or other orders as this Learned Controller may deem fit and proper in the facts and circumstances of the case and in the interest of justice.

Dated this 09th day of March, 2026



(Bharat Daswani)
Of Daswani & Daswani
Applicant's Agent
IN / PA 289

VERIFICATION

I, Bharat Daswani, the authorized representative of the Applicant, do hereby verify that the contents of the above petition are true to my knowledge and belief, derived from the records of the case, and nothing material has been concealed therefrom.

Verified at Kolkata on this 09th day of March 2026.

Authorized Signatory for the Applicant



(Bharat Daswani)
Of Daswani & Daswani
Applicant's Agent
IN / PA 289

Digitally Signed By:
MR. BHARAT DASWANI
Date: 11-03-2026 16:24:30

PATENT OFFICE
INTELLECTUAL PROPERTY BUILDING
G.S.T. Road, Guindy, Chennai-600032
Tel No. (091)(044) 22502081-84 Fax No. 044 22502066
E-mail : Chennai-patent@nic.in
Web Site : www.ipindia.gov.in

सत्यमेव जयते
GOVERNMENT OF INDIA

INTELLECTUAL PROPERTY INDIA
PATENTS | DESIGNS | TRADE MARKS
GEOGRAPHICAL INDICATIONS

संदर्भ सं. / Ref. No:
POC/Application No /202141018455

प्रेषण दिनांक / Date of Dispatch:
16-04-2026

सेवा मे, / To

आवेदक /Applicant:
Standard Glass Lining Technology Private Limited

Registerd address for service :MR. BHARAT DASWANI, DASWANI & DASWANI Patent & Trade Mark (Intellectual Property) Attorneys Daswani House, Green Acres 23B, Ahiripukur 1st Lane Kolkata 700 019. Email: kolkata@daswaniindia.net

विपक्षी /Opponent:
Pinakin Engineers Private Limited, Plot No. 249, Shed No. A1/29, Phase 2, GIDC, Vapi, Gujarat – 396195, India,, hemant.thadhani@crawfordbayley.com

ई-मेल प्रेषित /Email Sent to:
kolkata@daswaniindia.net,hemant.thadhani@crawfordbayley.com

विषय: आवेदन संख्या 202141018455 के संदर्भ मे अनुदान-पूर्व सुनवाई नोटिस
Sub: Pre Grant Hearing adjournment notice in reference of Application No.202141018455

सुनवाई स्थल / Hearing Location: Through Video Conferencing
सुनवाई दिनांक व समय / Hearing Date & Time: 20/05/2026 / 11:30 HRS(IST) for (5h)
नियंत्रक ईमेल /Controller's EmailId: kanhaiya207.ipo@gov.in

आवेदन संख्या 202141018455 के अनुदान-पूर्व विरोध व उसके उत्तर मे दाखिल आवेदन के संदर्भ मे, इस मामले मे 20/05/2026 को 11:30 HRS(IST) for (5h) बजे विडियो कॉन्फ्रेंसिंग सुनवाई तय की गयी है। अतः, आपको उपरोक्त दिनांक व समय पर नियंत्रक के समक्ष सुनवाई हेतु उपस्थित होना है। आपको यह भी सलाह दी जाती हैं कि आप निर्धारित तिथि पर सुनवाई के लिए उपस्थित होने से एक सप्ताह पहले नियंत्रक को सूचित करें।

With reference to Pre Grant Opposition and your reply to the First examination Report/Subsequent Examination Report,Pre-Grant is adjourned in the matter through Video Conferencing and rescheduled on 20/05/2026 at 11:30 HRS(IST) for (5h) . You are therefore, required to appear before the Controller for the hearing on said date and time.
You are also advised to intimate the controller one week before appearing in hearing on scheduled date.

इस आवेदन को पेटेंट अनुदान हेतु क्रम मे लाने की अंतिम तिथि से पूर्व / अंतिम तिथि के उपरांत, निम्नलिखित आपत्तियां अभी भी शेष हैं।
The following objection(s) are still outstanding before / after the expiry of last date for putting this application in order for grant of patent.

Kanhaiya Lal Balai
Assistant Controller of Patents & Designs

*दिनांक/समय, स्थल, स्थिति व सुनवाई के बारे में अन्य विवरण के लिए: कृपया निम्नलिखित यूआरएल देखें
https://iprsearch.ipindia.gov.in/DynamicUtility/CauseListForHearingCase/Index
Please refer to the following URL for: Date/Time, Venue, Status and other details about the Hearing
https://iprsearch.ipindia.gov.in/DynamicUtility/CauseListForHearingCase/Index
टिप्पणी:- उक्त सुनवाई में भाग लेने के लिए वीडियो कॉन्फ्रेंसिंग लिंक के संबंध में अलग से ईमेल भेजी जाएगी।
Note:- Separate email will be sent regarding the Video Conferencing Link to attend the said hearing.

* Hearing Objections are attached.

Objections

Clarity and Conciseness

1. The following objection raised in FER has still not been met.

   Reference numeral shall be marked in the drawings and reference of it shall be incorporated in the claims to enhance the intelligibility of Claims. No description matter is allowed on the drawings as per Rule 15(7) of The Patents Rule, 2003 (as amended).

2. 1. The reference numerals of the drawings are not incorporated in the claims; the applicant is required to amend the claims to include appropriate reference numerals corresponding to the features shown in the drawings to ensure clarity and consistency.

   2. Abbreviated forms of words have been used in the claims and/or description. The applicant is required to replace all such abbreviations with their complete forms to ensure clarity and proper understanding of the invention, as abbreviations may lead to ambiguity or misinterpretation of the technical features described.

Formal Requirement(s)

1. 1. The General Power of Attorney (GPA) filed by the opposition is found to be defective. It is observed that the Patent Agent number (IN/PA No.) mentioned in the GPA is incorrect and does not correspond to the registered number of the patent agent Mr. Hemant Thadhani, who is stated to be authorized to act on behalf of the opposition. Accordingly, the opposition is directed to submit a corrected and duly executed General Power of Attorney, wherein the correct Patent Agent number (IN/PA No.) of Mr. Hemant Thadhani is mentioned. The corrected GPA shall be properly executed, dated, and, if applicable, notarized or legalized in accordance with the prescribed procedure.

   2. It is observed that the applicant has submitted amendments to the claims in the present application by filing Form 13. As per Section 57 of the Patents Act, 1970, read with Rule 81(1) of the Patents Rules, 2003, any amendment to a patent application, complete specification, or any related document must be made by filing Form 13, accompanied by a clear statement showing the basis and support for the proposed amendment in the originally filed specification.In this regard, the applicant is directed to provide adequate justification and support for each amendment made, with specific references to the description, drawings, or original claims. Until such compliance is made, the submitted Form 13 and the amendments to the claims cannot be taken on record. The applicant is advised to ensure that the basis for each amendment is clearly documented to maintain conformity with the requirements of the Patents Act and Rules.

   3. It is observed that the applicant has filed Form 13 for amendment of the claims in the present application. As per Section 57(2) of the Patents Act, 1970, every application for leave to amend an application for a patent, a complete specification, or any related document must provide full particulars of the reasons for which the application is made. In the present submission, the reasons or justification for the amendments have not been clearly furnished. The applicant is therefore directed to submit a revised Form 13 clearly stating the detailed reasons and justification for making each amendment. Until compliance with the above requirements is made, the submitted Form 13 and the proposed amendments cannot be taken on record.

Invention u/s 2(1)(ja)

1. The applicant's reply to the objection raised in the First Examination Report (FER) under Section 2(1)(ja) of the Patents Act has been duly considered.

   Upon further examination, it has been observed that the following prior art documents disclose the features of the alleged invention:

   D4: CN103820781A (Publication Date: 25/08/2014)

   D5: CN105148816A (Publication Date: 16/12/2015)

   D6: CN206372825U (Publication Date: 04/08/2017)

   Doucmnet D4 discloses a profiled outer glass-lined device, which mainly comprises a can lid and a can body, wherein the can lid is provided with a nozzle, a manhole and a sealing mouth; the outer part of the can body is covered with a jacket, the can body and the clip The sleeve is welded into a unitary structure with a cavity therebetween, a jacket flange is provided on one side of the tank body, and a discharge port, a condensate outlet, a steam inlet and a steam outlet are arranged at a lower portion.

   The document further discloses the outer side of the outer side of the can body and the outer side of the jacket have a

stainless steel outer layer. The upper part of the stainless steel outer layer of the outer side of the can body is covered with a can body flange surface and a can body surface I stainless steel outer layer. The can body and the jacket welding portion There is a canned transition surface I stainless steel outer layer, the outer side of the jacket is covered with a can body surface II stainless steel outer layer, and the lower part of the can body is covered with a stainless steel outer layer of the can body sealing surface, and the lower part of the can body and the jacket welding portion are covered The can body transition surface II stainless steel outer layer, the bottom of the can body is covered with a can body surface III stainless steel outer layer.

D5 discloses the manufacturing method of a glass-lined reaction kettle and belongs to the technical field of processing, which solves the problem that the outer surface of the glass-lined reaction kettle is poor in corrosion resistance. A stainless steel flange is placed on the nozzle flange of each process pipe, and the upper cover is covered from the top to the upper cover, so that the process pipe protrudes from the through hole of the upper foreskin, and then The upper foreskin is welded with the stainless steel flange, and the cylinder is wrapped with the middle foreskin, and the lower cover is covered from the bottom with the lower foreskin, so that the process tube protrudes from the through hole of the lower foreskin, and the lower foreskin is welded with the stainless steel flange. Finally, the lower foreskin and the middle foreskin are sealed and welded.

D6 discloses a corrosion -resistant glass reactor that wards off of high strength, including pressing from both sides cover, jar body, agitator, upper cover, low head, sight glass manhole, the thermometer mouth of pipe, feed inlet, drain hole, the external side of jar is equipped with presss from both sides the cover, it is equipped with a plurality of installation positioner to press from both sides angles such as overlapping the outside, pressing from both sides cover upper portion and being equipped with steam inlet, steam inlet one side is equipped with the relief port, presss from both sides the cover bottom and is equipped with water export with fixed attention, the welding has high -neck flange on the feed inlet, the welding has high -neck flange on the sight glass manhole, and the welding has high -neck flange on the thermometer mouth of pipe, and the welding has high -neck flange on the drain hole, be equipped with the ground coat on the internal lateral wall of jar, be equipped with cover glaze layer on the ground coat, be equipped with the ground coat on high -neck flange inside wall, terminal surface, the recess, be equipped with cover glaze layer on the ground coat. The utility model discloses trompil department welding has increased high -neck flange, has reduced the quantity at stamping forming trompil R position, has directly increased the effective thickness of this equipment, and compressive strength improves relatively, ensuring the safe operation of equipment.

In view of the foregoing, it is evident that a person skilled in the art could and would readily arrive at the present invention by drawing sufficient motivation from the cited prior arts (D4-D6). The combination of features disclosed in these references provides ample guidance to derive the claimed invention without the need for any inventive step or intellectual creativity. Consequently, the claims of the present application are deemed to be obvious and lacking an inventive step in accordance with the provisions of Section 2(1)(ja) of the Patents Act.

Other Requirement(s)

1. The amendment made in the description and claims does not comply with the provisions of Section 59(1) of the Act.

2. • Note:-A hearing in the above-mentioned pre-grant opposition matter was scheduled on 14/11/2025. Both the Applicant and the Opponent had intimated their intention to attend the hearing.At the time of the hearing, it was observed from the official module that the prescribed fee for hearing from the Opponent was not reflected as paid. In view thereof, the Opponent was not permitted to advance oral submissions during the hearing proceedings. The Opponent, though present, submitted that the requisite fee had been filed; however, as the same was not reflecting in the system at that time, the hearing proceeded without permitting oral arguments from the Opponent. Thereafter, both parties filed their respective written submissions. It has subsequently come to notice that the prescribed hearing fee filed by the Opponent was duly paid and is now reflecting in the official module. In order to ensure compliance with the principles of natural justice and to provide a fair and reasonable opportunity of hearing to the Opponent, it is considered appropriate to afford a fresh opportunity of oral hearing in the matter.

Scope

1. (a) The scope of principal claim 1 is ambiguous due to multiple characterization clauses introduced by the phrase "wherein." Re-draft or rephrase the principal claim to clearly define the constructional inventive feature of the alleged invention over which protection is sought.

(b) The phrases:

"Coated nozzle is seamlessly welded to the top cover's cladding sheet, fully enclosing the reactor in stainless steel, addressing rust challenges in critical zones, and forming a resilient barrier that upholds reactor integrity, longevity, and product purity."

"The reactor offers excellent resistance to oxidation at the nozzle. In the event of accidental spillage of corrosive liquid onto the nozzle surface, oxidation of the metal surface may occur, leading to damage of the glass lining coated inside, as oxidation of the base metal would cause distortion."

"The reactor is fully covered with stainless steel, enhancing its resilience against external damage and increasing overall safety."

These statements in claim 1 are narrative in nature and contain indefinite, functional, or operational language. The principal claim is vaguely drafted in a functional format. As per Section 10 of the Patents Act, 1970 (as amended), vague and functional aspects are not permissible in claims; therefore, the claims are not allowable.

2. 1. Claim 1 recites functional advantages such as “offering excellent resistance to oxidation,” “tackling rust challenges,” “upholding reactor integrity, longevity, and product purity,” and “enhancing resilience against external damage and increasing overall safety,” which merely describe the functional benefits or results of the claimed reactor rather than defining specific technical or structural features. Such statements of function or result do not constitute patentable subject matter under Section 2(1)(j) of the Patents Act, 1970. The claim should therefore be amended to clearly define the structural or technical features responsible for achieving the stated effects instead of claiming the functional benefits themselves.

2. Claims 2 and 7 do not clearly and unambiguously define the scope of the invention as required under Section 10(4)(c). In Claim 2, the term “base body of high mechanical strength” is vague and lacks measurable parameters, and the expressions “essentially,” “preferably,” and “most preferably” regarding the reaction chamber thickness introduce ambiguity. In Claim 7, the term “smooth as whole” is indefinite and does not specify objective criteria for surface smoothness. The claims therefore fail to distinctly define the invention and may lead to multiple interpretations. The applicant is directed to amend Claims 2 and 7 to clearly define the mechanical strength of the base body in measurable terms and to specify the smoothness of the glass lining/coating in objective terms, so as to comply with Section 10(4)(c) of the Patents Act, 1970.

3. It is observed that there is an inconsistency between the description, claims, and drawings regarding the material of the reactor walls. In the description and claims, the outer wall of the reactor is disclosed as stainless steel (SS) and the inner wall as mild steel (MS). However, in the drawings, the layers appear inversed, showing the outer wall as MS and the inner wall as SS. This inconsistency creates ambiguity regarding the actual structure of the reactor and the scope of the invention. The applicant is required to rectify the discrepancy by amending the description, claims, and/or drawings so that the material of the outer and inner walls is consistent throughout the specification and clearly defines the invention, in compliance with Section 10(4) of the Patents Act, 1970.

Sufficiency of Disclosure u/s 10 (4)

1. 1. It is observed that references have been made in the description to Figures 4 and 5. However, these figures are not included in the drawings filed with the application. The absence of Figures 4 and 5 renders the references in the description inconsistent with the accompanying drawings, leading to ambiguity in understanding the disclosure. Accordingly, the applicant is required to file the missing Figures 4 and 5 with proper reference in the description, to ensure conformity between the description and the drawings as required under Section 10(4) of the Patents Act, 1970 and Rule 15 of the Patents Rules, 2003.

2. It is observed that certain structural features of the invention, such as the top flange, bottom flange, glass-lined metal flat surface below the top flange for observing the flow regime in the reactor or mixer, the cavity in the bottom flange facilitating the flow of heat transfer fluid through inlet and outlet nozzles and flanges, the bottom plate having a metal surface in contact with the heat transfer fluid, the connection plate connecting the inlet and outlet to the bottom flange of the glass-lined stainless-steel reactor, the multiple bolts securing the top and bottom flanges, and the external stainless-steel sheet covering of the reactor, have been described in the specification. However, these features are not shown or illustrated in the drawings accompanying the application. As per the requirements of Section 10(4) of the Patents Act, 1970, every complete specification shall fully and particularly describe the invention and its operation or use and the method by which it is to be performed, and shall disclose the best method of performing the invention known to the applicant. Accordingly, the applicant is required to incorporate all the described structural features in the drawings, ensuring that each feature or embodiment mentioned in the description is properly reflected in the drawings with corresponding reference numerals. This is necessary to maintain consistency between the description and drawings and to ensure that the invention is fully and clearly disclosed in compliance with the statutory requirements.

Daswani & Daswani
Patent & Trade Mark (Intellectual Property) Attorneys

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May 15, 2026

Your Ref: POC/Application No/202141018455
Our Ref: 202141018455

To
The Controller of Patents
The Patent Office,
Chennai / Delhi

Kind Attention:
Dr. Kanhaiya Lal Balai
Assistant Controller of Patents & Designs

Dear Sir,

Re: Pre-Grant Opposition Hearing against Patent Application No. 202141018455 of April 21, 2021 titled "Clean Room Glass Lined Reactor" filed by Standard Glass Lining Technology Private Limited (Applicant) opposed by Pinakin Engineers Private Limited (Opponent) under Section 25(1) of the Patents Act, 1970.

This is further to your Pre-Grant Hearing adjournment notice dated 16-04-2026 with reference to Patent Application No. 202141018455, wherein the Pre-Grant Opposition Hearing has been scheduled for 20th May 2026 at 11:30 HRS.

We would like to state that Ms. Niharika Singh, Registered Patent Agent (IN/PA 4923), shall attend the hearing on 20-05-2026 as per the schedule mentioned in the hearing notice.

A Substitute Power of Authorization in favour of Ms. Niharika Singh, Registered Patent Agent (IN/PA 4923), authorizing her to attend the Pre-Grant Hearing has been filed, and a copy thereof is enclosed herewith.

Please note that we have already paid the official fees to attend the hearing on November 7, 2025. Accordingly, we are submitting this letter under the head without fees “Notify the Controller whether he will attend the hearing”.

Yours Sincerely
Of DASWANI & DASWANI



(Bharat Daswani)
Applicant’s Agent
IN/PA 289

Digitally Signed By:
MR. BHARAT DASWANI
Date: 15-05-2026 17:12:17

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BY ONLINE FILING

Confidential and Urgent

HCA/SKA/HT/RD/NM

May 19, 2026

To,
The Controller of Patents,
The Patent Office,

ONLINE SUBMISSION

Dear Sir,

Notice to attend Hearing

Reference:	INDIA-Patent Application No.	**202141018455**
	Applicant:	**Standard Glass Lining Private Limited**

This is in reference to the hearing notice dated 16-04-2026 fixing hearing of the pre-grant opposition hearing in the captioned application on 20.05.2026.

We hereby confirm and notify that the Opponent's authorised agent, Mr. Hemant Thadhani (IN/PA-4057), shall be attending the hearing as scheduled.

We further submit that the requisite official fee for attending the hearing was already paid on 14 November 2025. Accordingly, the present communication is being filed under the head "Notify the Controller whether he will attend the hearing" without payment of any additional fee.

Yours faithfully,
FOR **Pinakin Engineers Private Ltd**
By its Agent

Hemant Thadhani
Patent Agent ((IN/PA-4057)
Crawford Bayley & Co.

Digitally Signed By:
Hemant Thadhani
Date: 19-05-2026 17:05:48

Daswani & Daswani
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June 03, 2026

Kind Attention:
Dr. Kanhaiya Lal Balai
Assistant Controller of Patents & Designs

Your Ref: POC/Application No / 202141018455
Our ref: 202141018455

To
The Controller of Patents
The Patent Office,
Chennai / Delhi

Dear Sir,

Re: Response to Pre-Grant Opposition Hearing held on 20.05.2026 against Patent Application No. 202141018455 of April 21, 2021 titled "Clean Room Glass Lined Reactor" filed by Standard Glass Lining Technology Private Limited (Applicant) opposed by Pinakin Engineers Private Limited (Opponent) under Section 25(1) of the Patents Act, 1970.

In connection with the above, we would like to submit that Ms. Niharika Singh, Registered Patent Agent (IN/PA 4923), attended the hearing on 20-05-2026 on behalf of the Applicants, in accordance with the schedule specified in the Hearing Notice.

Pursuant to the hearing, a written submission, along with the relevant supporting documents, is enclosed herewith for your perusal and kind consideration.

In view of the above submissions, it is respectfully requested that the present application be allowed to proceed to grant.

Yours Sincerely
Of DASWANI & DASWANI



(Bharat Daswani)
Applicant's Agent
IN / PA 289

OPENING REMARKS AND PRELIMINARY SUBMISSIONS ON THE INTERLOCUTORY PETITION DATED 09 MARCH 2026

Hearing date – 20/05/2026

The Applicant, Standard Glass Lining Technology Private Limited, filed Patent Application No. 202141018455 on 21 April 2021 for the invention titled "Clean Room Glass Lined Reactor", a specialised mechanical engineering apparatus for pharmaceutical clean-room environments featuring an internally glass-lined mild steel body clamped between flanges with machined fluidic channels, a glass-lined observation surface, bottom-flange cavity for heat-transfer fluid, and seamless stainless steel overlay welded after 2 mm surface removal to achieve gap-free enclosure extending to nozzles, thereby preventing oxidation-induced distortion of the internal lining. The application was published on 28 October 2022. The Opponent, Pinakin Engineers Private Limited, filed a pre-grant opposition under Section 25(1) on 14 November 2024 relying on prior arts D4–D6.

The Applicant completed the evidentiary stage by filing its reply under Rule 58 on 4 August 2025. The Learned Controller issued a hearing notice dated 13 October 2025 fixing the hearing for 14 November 2025 via video conferencing under Rule 62(1), requiring notice to attend with prescribed fee under Rule 62(2). The Applicant duly complied. At the hearing the Opponent appeared via video conferencing but had filed neither notice nor fee, as recorded in the official module. The Controller permitted the Opponent to remain present yet correctly denied it oral arguments and proceeded solely on the Applicant's side, thereby completing the statutory hearing under Section 25(1). Both parties later filed written submissions. No contemporaneous request from the Opponent for a fresh hearing or explanation of its default appears on record.

Subsequently the Opponent filed written submissions dated 28 November 2025 and sought to rely on a backdated request purportedly from 13

November 2025 concerning the hearing fee and notice—documents absent from the record at the time of hearing and inconsistent with the proceedings. By communication around 29 November 2025 followed by formal notice dated 11 February 2026 the Learned Controller adjourned the matter and fixed a fresh hearing on 11 March 2026, citing the later reflection of the Opponent's fee in the system and invoking "principles of natural justice". The Applicant filed the present interlocutory petition on 09 March 2026 seeking to set aside that repeated hearing notice and to have the application decided under Section 15 on the basis of the completed hearing of 14 November 2025 and the existing record. The impugned action perpetuates delay in a time-sensitive mechanical invention where commercial exclusivity and time-to-market are critical for clean-room pharmaceutical equipment.

Under Section 25(1) read with Rules 55, 58 and 62 the pre-grant opposition is a structured quasi-judicial proceeding with a defined evidentiary stage culminating in a single statutory hearing after which the Controller must decide under Section 15 whether to grant or refuse the patent. Rule 62(2) expressly makes filing of notice to attend hearing with the prescribed fee a condition precedent for a party to be "heard" in the sense of advancing oral submissions. Non-compliance forfeits that right, although presence as a spectator remains permissible—as occurred here. Once the 14 November 2025 hearing concluded with the Applicant's arguments heard and the Opponent's default acted upon, the proceedings attained finality at the hearing stage. **No provision in Section 25(1), Section 15 or Rule 62 authorises re-opening a completed hearing merely because the defaulting party later perfects its procedural lapse, and that too without any formal application on record. The repeated hearing notice dated 11 February 2026 therefore travels beyond the statutory framework and is ex facie ultra vires, as it purports to restore a forfeited right of audience and nullifies the deliberate consequence attached to Rule 62(2) non-compliance.**

Judicial authority establishes that once a statutory hearing is duly concluded it cannot be casually re-opened except on limited grounds such as manifest error or fraud. In *Union of India v. Tarachand Gupta & Bros.* (AIR 1971 SC 1558) the Hon'ble Supreme Court emphasised the sanctity of finality in concluded proceedings. The present record discloses no such exceptional ground; it shows only a completed hearing followed by an attempt to regularise the Opponent's default through backdated documents whose timing raises serious questions directly affecting compliance with Rule 62(2) and the legality of the re-opening decision. Rule 137 empowers the Controller to correct procedural irregularities but cannot be invoked to confer an entirely new substantive right of audience upon a party that has already forfeited the same. The backdating issue and absence of contemporaneous explanation further undermine any justification for unsettling finality.

The principles of natural justice, as laid down in *Maneka Gandhi v. Union of India* (AIR 1978 SC 597), require a fair opportunity of hearing but do not compel authorities to grant repeated or additional hearings to cure a party's own procedural defaults, especially where statutory preconditions such as fee payment have not been met. The Opponent was aware of the hearing date and appeared, yet failed to comply with Rule 62(2). The Controller therefore justifiably limited the hearing to the Applicant's side, satisfying *audi alteram partem* to the extent rights were properly invoked. The *suo motu* re-opening to accommodate belated fee payment converts natural justice into a device for granting a second chance to the defaulting party at the cost of delay and prejudice to the compliant Applicant. The Opponent suffers no irreparable prejudice: its substantive rights remain fully protected by post-grant remedies under Sections 25(2), 64 and 84. Conversely, the Applicant faces grave commercial prejudice through prolonged uncertainty in a mechanical invention where time-to-market and exclusivity are critical for specialised clean-room pharmaceutical reactors. The balance of convenience and proportionality therefore lies decisively in preserving the finality of the 14 November 2025 hearing.

In order to maintain the sanctity of the official record under Section 74 it is necessary that the Learned Controller direct placement on record of the exact date and time when the Opponent's notice and fee were received and reflected, and that the video recording of the 14 November 2025 hearing be furnished to the Applicant. The interests of justice, statutory finality and administrative propriety are best served by recalling the repeated hearing notice and proceeding to decision under Section 15 on the basis of the completed hearing and existing record. The Opponent's post-hearing written submissions and backdated attempts are not contemplated by Rule 62 and cannot justify re-opening a concluded proceeding.

PRAYER

In view of the foregoing, the Applicant most respectfully prays that this Learned Controller may be pleased to:

(a) Set aside/recall the repeated hearing notice dated 11 February 2026 (and any subsequent notices), treating the pre-grant opposition as finally heard on 14 November 2025;

(b) Direct that the application be taken up for decision under Section 15 on the basis of the oral hearing of the Applicant on 14 November 2025 and the statements and evidence already on record in accordance with Rules 55–60;

(c) Direct the Patent Office to confirm the exact timestamp of the Opponent's hearing fee and notice receipt and to furnish the video recording of the 14 November 2025 hearing to the Applicant;

(d) Pass such further or other orders as this Learned Controller may deem fit and proper in the facts and circumstances and in the interest of justice.

WITHOUT PREJUDICE

WRITTEN SUBMISSIONS ON BEHALF OF THE APPLICANT IN THE PRE-GRANT OPPOSITION PROCEEDINGS UNDER SECTION 25(1) OF THE PATENTS ACT, 1970

Date of Hearing: 20th day of May, 2026

OPENING REMARKS

The Applicant, Standard Glass Lining Technology Private Limited, through its authorized Patent Agent, Mr. Bharat Daswani (IN/PA/289), hereby submits these exhaustive written submissions in response to the Pre-Grant Opposition filed by the Opponent on 14th November 2024, as well as in compliance with the Hearing Notice dated 13th October 2025 issued under Rule 55(4) of the Patents Rules, 2003.

These submissions address all grounds raised by the Opponent under Section 25(1)(b), (e), (f), and (g) of the Patents Act, 1970, refute each statement made in the Opponent's Written Statement and evidence (Exhibits A to G), and respond to the objections outlined in the Hearing Notice regarding clarity and conciseness, formal requirements, and inventive step under Section 2(1)(ja).

The Applicant humbly requests the Learned Controller to dismiss the opposition in its entirety, as it is frivolous, mala fide, and lacks merit, and to proceed with the grant of the patent, subject to any minor amendments if deemed necessary for formal compliance. The Applicant reserves the right to file further evidence or amendments as per the provisions of the Act and Rules, and seeks an opportunity to be heard orally at the scheduled hearing.

The present application, filed on 21st April 2021 and published on 28th October 2022, discloses a novel Clean Room Glass Lined Reactor specifically designed for pharmaceutical and food-grade processing environments, where corrosion resistance, hygiene, and contamination control are

paramount. The invention addresses a long-standing technical problem in conventional glass-lined reactors: oxidation and corrosion at nozzles and external surfaces in clean-room settings, which can lead to distortion of the internal glass lining, product contamination, and non-compliance with stringent standards like ISO 14644-1 and GMP.

The core inventive concept lies in a mild steel (MS) reactor body internally glass-lined and externally overlaid with stainless steel (SS) sheets through a seamless welding process, particularly extending to nozzles, ensuring 100% enclosure in SS to prevent oxidation while maintaining structural integrity and heat transfer efficiency. This is achieved by machining fluidic channels on the reactor surface, integrating a glass-lined flat observation surface below the top flange for flow regime monitoring, and incorporating a cavity in the bottom flange for heat transfer fluid circulation, all clamped securely between top and bottom flanges with bolts.

The specification provides enabling disclosure, including detailed description, claims, and drawings, supported by the method of performance, enabling a person skilled in the art (a mechanical engineer familiar with reactor design in chemical/pharmaceutical industries) to practice the invention without undue experimentation.

In contrast, the Opponent's opposition is an attempt to delay the grant due to commercial rivalry, as both parties operate in the glass-lined equipment manufacturing sector in Gujarat. The Opponent fails to establish locus standi as a "person interested" under Section 25(1), merely claiming to be a competitor without demonstrating any tangible interest beyond rivalry, such as prior use or direct economic impact, as required by precedents like Ajay Industrial Corporation v. Controller of Patents (1984 PTC 234), where the court held that mere commercial rivalry does not confer standing without proof of specific interest. Furthermore, the opposition is mala fide, as evidenced by its reliance on unsubstantiated assertions, uncertified translations of foreign prior art, and superficial tabular comparisons without technical analysis, violating the principles laid down in Hindustan Lever Ltd.

v. Godrej Soaps Ltd. (1996 PTC 141), where oppositions motivated by delay tactics were dismissed with costs. The Applicant seeks dismissal on these preliminary grounds alone, along with compensatory costs under Section 35 of the Act for defending this baseless challenge.

GROUND 25(1)(b)

Turning to the substantive rebuttal, the Applicant addresses the Opponent's grounds under Section 25(1)(b) for lack of novelty. The Opponent alleges anticipation by prior art documents D1 (CN103820781A, priority 12.03.2014), D2 (CN105148816A, priority 08.09.2015), and D3 (CN206372825U, priority 12.01.2017), claiming that these disclose all features of the claims through tabular comparisons in their Written Statement. The Applicant humbly reiterates and submits the following detailed observations in rebuttal to the ground raised under Section 25(1)(b) of the Patents Act, 1970, as articulated in the Opponent's Written Statement and as prima facie upheld by the Learned Controller in the Report dated 04th June 2025 (hereinafter "Prima Facie Report").

This ground alleges that the subject matter of the present application lacks novelty due to anticipation by prior publications, specifically D1 (CN103820781A, published 28th May 2014), D2 (CN105148816A, published 16th December 2015), and D3 (CN206372825U, published 4th August 2017). The Applicant respectfully contends that this ground is entirely misconceived, unsubstantiated, and fails to meet the stringent evidentiary and analytical thresholds mandated by Indian patent law for establishing anticipation. The rebuttal is structured to first elucidate the legal framework governing novelty assessments, followed by a meticulous, feature-by-feature dissection of the cited documents vis-à-vis the inventive features of the present application, demonstrating unequivocally that no single prior art document discloses the invention as a whole in an enabling and unambiguous manner. This analysis draws upon the complete specification of the present application (filed on 21st April 2021, as amended in response to the First Examination Report dated 23rd March 2023), the full English

translations of D1, D2, and D3, relevant precedents from Indian courts, and guidelines from the Manual of Patent Office Practice and Procedures, 2019 (hereinafter "Patent Manual").

At the outset, it is imperative to recall the legal principles from Section 25(1)(b), which provides that a patent may be opposed if the invention "has been published before the priority date of the claim (i) in any specification filed in pursuance of an application for a patent made in India on or after the 1st day of January, 1912; or (ii) in India or elsewhere, in any other document." As per Section 2(1)(l), an invention is "new" if it does not form part of the state of the art, and anticipation requires that the prior art must disclose the invention in its entirety, with such clarity and completeness that a person skilled in the art could perform it without further ingenuity or experimentation. The Hon'ble Supreme Court in Bishwanath Prasad Radhey Shyam v. Hindustan Metal Industries (AIR 1982 SC 1444) emphatically held that for anticipation, the prior publication must contain "clear and unmistakable directions" to perform the invention, and mere suggestions or incomplete disclosures do not suffice. Similarly, in Lallubhai Chakubhai Jariwala v. Chimanlal Chunilal & Co. (AIR 1936 Bom 99), the court clarified that anticipation cannot be established by mosaicking disparate elements from multiple documents; each prior art must be evaluated independently as a standalone disclosure.

The Patent Manual (Chapter 08.03.01) reinforces this by stating that "the prior art should disclose the invention either in explicit or implicit manner" and must enable the skilled person to arrive at the invention without undue burden. Critically, foreign patent documents like D1, D2, and D3 must be accompanied by certified translations and proof of public availability in India prior to the priority date (21st April 2021), failing which they are inadmissible, as per Rule 55(3) of the Patents Rules, 2003, and precedents such as Snehalata C. Patel v. Controller of Patents (2001 PTC 121). In the instant case, the Opponent has furnished uncertified machine translations (Exhibits E, F, G), which are riddled with inaccuracies (e.g., "foreskin" in D2 likely mistranslating "wrapper" or "sheath"), and no evidence of their

dissemination in India, rendering them prima facie unreliable. Even assuming arguendo their admissibility, a granular analysis reveals profound dissimilarities, as elaborated below.

The inventive core of the present application, in the complete specification (pages 1-19) and amended claims (dated 23rd September 2023), resides in a "Clean Room Glass Lined Reactor" tailored for pharmaceutical and food-grade environments under stringent hygiene standards like ISO 14644-1 and GMP. The reactor comprises: (i) an internally glass-lined mild steel (MS) body clamped between top and bottom flanges with machined fluidic channels on the surface for enhanced heat transfer; (ii) a glass-lined metal flat surface below the top flange enabling non-invasive observation of internal flow regimes to prevent contamination; (iii) a cavity in the bottom flange for heat transfer fluid circulation via inlet/outlet nozzles, closable with a metal plate; (iv) a connection plate linking inlet/outlet to the bottom flange; (v) an external stainless steel (SS) sheet overlay welded seamlessly to enclose the reactor fully, preventing oxidation and rust at critical zones; (vi) internally glass-lined nozzles with seamless SS overlay welded to the cladding, addressing the specific problem of nozzle oxidation distorting internal lining in clean rooms; (vii) a double-walled vessel design with smooth glass-lined interior; and (viii) application-specific volumes (e.g., 50-5000 liters) and heat transfer areas.

This configuration solves the longstanding issue of external corrosion in conventional reactors, where spillage in clean rooms leads to oxidation penetrating to the internal glass lining, causing distortion, contamination, and regulatory non-compliance. The manufacturing method involves removing approximately 2mm of MS surface before welding SS overlays with SS electrodes to form a metallurgical bond, maintaining uniform thickness and 100% SS enclosure without gaps or insulation voids that could harbor contaminants. None of D1, D2, or D3—each addressing generic glass-lined vessels for non-clean-room applications—discloses this integrated, synergistic arrangement, as evidenced by their divergent teachings.

Commencing with D1 (CN103820781A, titled "Glass Lined Equipment Provided with Outer Profiling Cladding and Manufacturing Method"), this document describes a conventional glass-lined tank comprising a tank cover (with nozzles, manhole, and seal) and tank body encased in a jacket for heat exchange, with external stainless steel "profiling cladding" layers applied for insulation purposes. The full description (*from unauthenticated English translation*) reveals that the cladding is formed by cutting and shaping 304 stainless steel sheets (1.5-3mm thick) to match the tank's profile, welding joints with argon arc welding, and filling gaps between the jacket and cladding with polyurethane foam or silicate for thermal insulation. The tank body and jacket are welded into a unitary structure with a cavity for steam/condensate flow, and bottom ports include drain hole, condensation-water drain, steam inlet, and vapor outlet. However, D1 fundamentally diverges from the present invention in several critical aspects, precluding anticipation. Firstly, D1's "cladding" is a detachable, insulated outer layer with deliberate gaps filled for thermal purposes, not a seamless, metallurgically bonded SS overlay welded directly after MS surface removal to prevent oxidation in clean-room settings. The translation explicitly states: "Between described chuck and stainless steel surrounding layer, be filled with lagging material and form thermal insulation layer," emphasizing insulation over corrosion prevention at nozzles or flanges.

In contrast, the present application's SS overlay is seamless, extending to nozzles without insulation gaps, ensuring 100% enclosure to eliminate rust formation at critical zones—a feature absent in D1, where cladding joints are merely butt-welded and polished, leaving potential entry points for moisture. Secondly, D1 lacks any disclosure of machined fluidic channels on the reactor surface, a glass-lined flat observation surface for flow monitoring, or a bottom flange cavity with connection plate for targeted heat transfer; instead, it relies on a standard jacketed cavity with bottom ports, without the clamped flange assembly or observation capability essential for clean-room hygiene. The manhole in D1 is for access, not flow observation, and there is no mention of a flat glass-lined surface below the top flange to

enable visual inspection without opening, which prevents contamination risks in pharmaceutical processes.

Thirdly, D1's manufacturing method involves preliminary impact briquetting of sheets, argon welding (current 70-80A, voltage 7-9V), and "V"-type breach repairs for flanges, but omits the inventive step of removing 2mm MS surface for uniform thickness and metallurgical bonding, focusing instead on profile consistency for insulation. As per the Patent Manual (Chapter 08.03.02), anticipation requires "implicit or explicit" disclosure of all essential features; here, D1's generic insulated cladding for cost reduction in non-pharma contexts does not implicitly teach the oxidation-resistant, seamless nozzle overlay for GMP compliance.

The Opponent's tabular comparison (Written Statement, pages 12-15) superficially maps D1's "surrounding layer" to the claims but ignores functional disparities—D1's layer allows gaps for filling, whereas the invention mandates seamless welding to preclude voids. In Novartis AG v. Union of India ((2013) 6 SCC 1), though pharmaceutical, the principle extends: mere structural similarity without identical purpose or effect does not anticipate. Thus, D1 falls short of providing "clear and unmistakable directions" (Bishwanath Prasad, supra) to arrive at the clean-room reactor.

Shifting to D2 (CN105148816A, titled "Manufacturing Method of Glass-Lined Reaction Kettle"), this patent outlines a process for fabricating a glass-lined kettle with an external stainless steel "foreskin" (translated as "wrapper" or "sheath") to enhance corrosion resistance. The method encompasses seven steps: assembling enamel-fired carbon steel components (upper head, cylinder, lower head); pressing SS sheets into upper, middle, and lower foreskins; perforating for process ducts; fixing foreskins by welding to steel flanges on nozzles; filling insulation material (e.g., glass cement) in gaps and seal-welding; polishing the SS surface; and adding an annular SS cover. The translation details that foreskins are oversized (diameter larger by 2-4mm at holes) for wrapping, with steel flanges halved and welded on nozzle flanges to enable SS welding, and insulation filled to isolate carbon steel. While D2

superficially addresses external corrosion via SS wrapping, it does not anticipate the invention due to profound structural and functional mismatches.

Primarily, D2's "foreskin" is a detachable wrapper with deliberate gaps filled post-assembly for insulation, not a seamless welded overlay integrated metallurgically after MS surface machining. The description states: "Fill insulant material in the gap between stainless steel foreskin and enamel, and seal welding," implying voids are inherent and filled secondarily with materials like glass cement, which the present invention eschews to avoid relative displacement and re-emerging gaps—issues D2 acknowledges but resolves inadequately via cement, leading to potential contamination in clean rooms. The invention's seamless welding (using SS electrodes post-2mm MS removal) creates a continuous, rust-proof enclosure without fillers, a distinction not disclosed in D2, where welding is limited to foreskin-flange joints (e.g., "upper foreskin is welded with steel flange").

Secondly, D2 omits key elements like fluidic channels machined on the reactor surface, the glass-lined observation flat surface, bottom flange cavity with connection plate, or nozzle-specific seamless SS overlays for oxidation prevention; its process ducts are merely perforated and flanged, without extended SS welding to cladding for 100% enclosure. The kettle is for general chemical reactions, not clean-room pharma use, lacking GMP-focused hygiene features. The manufacturing steps emphasize inversion for filling (e.g., "The upper cover being surrounded by described upper foreskin in described step e is inverted, fill insulant material in space up between foreskin and upper cover"), prioritizing insulation over oxidation resistance, whereas the invention's method ensures dimensional uniformity and bond strength for longevity in sterile environments. As per F. Hoffmann-La Roche Ltd. v. Cipla Ltd. ((2016) 66 PTC 1 (Del)), anticipation demands identical disclosure, not obvious modifications; D2's wrapper, reliant on secondary fillers, does not enable the skilled person to perform the seamless, gap-free overlay without ingenuity. The Opponent's assertions (Written Statement, pages 16-18) conflate D2's "foreskin" with the claims but overlook that D2's

design permits air/moisture entry pre-filling, antithetical to the invention's preventive enclosure. Hence, D2 does not destroy novelty.

Finally, D3 (CN206372825U, a utility model titled "A Kind of High-Strength Corrosion-Resisting Glassed Steel Reaction Vessels"), pertains to a jacketed glass-lined reactor with enhanced compressive strength via seamless welding neck flanges at ports. The description covers a tank body with jacket for heat exchange (steam/liquid inlets/outlets), agitator, viewing manhole, thermometer nozzle, charging/discharge ports, and standby ports, all featuring neck flanges with grooved end faces, chamfers, and bolted covers/gaskets for sealing. Corrosion resistance is achieved via internal ground glaze and cover coat layers (with specific frit compositions including silica 40-45%, alumina 3-7%, etc., and mixed adhesive agents like bastnaesite flour 45-55%), extending to flange inner walls and joints. However, D3 is devoid of any external SS overlay or cladding, let alone seamless welding for clean-room oxidation prevention, rendering it wholly non-anticipatory.

The utility model's focus is internal enamel enhancement for strength and anti-corrosion, with "high-neck flanges" welded seamlessly to reduce punch-formed openings and bolster pressure resistance (e.g., "The viewing manhole, thermometer nozzle... use seamless welding neck flanges"). There is no mention of external MS body with SS welding, machined fluidic channels, observation flat surface, bottom cavity, or nozzle overlays; the external jacket is bare, with mounting devices but no SS enclosure. The glaze formula, while improving internal lining adhesion (e.g., melting at 900-1000°C), is irrelevant to the invention's external hygiene features. As a utility model, D3 lacks enabling disclosure for manufacturing beyond basic assembly, omitting the inventive welding post-MS removal. The Patent Manual (Chapter 08.03.03) requires prior art to be enabling; D3's schematic utility focus does not provide "clear directions" for the claimed reactor.

The Opponent's mapping (Written Statement, pages 19-20) erroneously equates D3's internal enamel to the claims, ignoring the absence of external

SS elements. In Tapas Chatterjee v. Assistant Controller (2023 SCC OnLine Del 469), the court rejected anticipation based on partial, non-enabling disclosures; similarly, D3's internal-centric design cannot anticipate the external clean-room innovations.

Feature from Applicant's Claims (as amended)	Teaching in Applicant's Specification	D1 (CN103820781A)	D2 (CN105148816A)	D3 (CN206372825U)	Analysis (Why Not Anticipatory)
Internally glass-lined MS reactor clamped between top and bottom flanges with machined fluidic channels on surface (Claim 1(i))	Pages 3-5: Reactor body of MS with internal glass lining (1-2mm thick); fluidic channels machined for heat transfer; clamped via bolts for secure assembly.	Discloses general glass-lined can body with jacket, but no machined fluidic channels on reactor surface; focuses on insulated cladding (translation p.90).	Describes reaction kettle with glass lining, but channels are for jacket circulation, not machined on reactor surface (translation p.96).	Utility model for reactor with flanges; no mention of machined channels (translation p.102).	No single document discloses machined channels integrated with clamping for enhanced heat transfer in clean-room setup;.
Glass-lined metal flat surface below top flange for flow regime observation (Claim 1(ii))	Pages 5-6: Flat surface allows visual monitoring of internal mixing without contamination risk.	No observation surface; manhole is for access, not flow observation (p.91).	No such feature; focuses on manufacturing steps (p.97).	Absent entirely (p.103).	Unique to Applicant; solves clean-room monitoring issue not addressed in prior art,.
Cavity in bottom flange for heat transfer fluid circulation via inlet/outlet nozzles, closable with metal plate (Claim 1(iii))	Pages 6-7: Cavity design ensures efficient cooling/heating; connected via plate (Claim 1(iv)).	Condensate outlet and steam ports, but no bottom flange cavity (p.92).	Bottom discharge, no cavity in flange (p.98).	Flange for strength, no cavity (p.104).	Prior art uses jacketed designs, not flange-integrated cavity;
External SS sheet overlay by welding, seamless on nozzles to prevent oxidation (Claim 1(v)-	Pages 7-9: 2mm MS removal, SS welding for metallurgical bond; extends to nozzles for 100% enclosure.	Profiling cladding with gaps/insulation; not seamless weld (p.93).	Detachable wrapper with insulation; no welding to nozzles (p.99).	No overlay; corrosion via materials (p.105).	Applicant's method prevents oxidation at critical zones, unlike prior art's gapped claddings;

Feature from Applicant's Claims (as amended)	Teaching in Applicant's Specification	D1 (CN103820781A)	D2 (CN105148816A)	D3 (CN206372825U)	Analysis (Why Not Anticipatory)
(vii))					
Double-walled design, smooth interior, volume/heat transfer ranges (Claims 6-8)	Pages 10-12: Ensures hygiene, compliance with ISO/GMP.	Single jacket with insulation; no ranges (p.94).	Manufacturing focus; no double-wall specifics (p.100).	Utility; no such claims.	Synergistic advantages not disclosed;.

In summation, none of D1, D2, or D3—individually or otherwise—anticipates the invention, as they lack the complete, enabling disclosure of the synergistic features. The Applicant prays for rejection of this ground, emphasizing the mala fide nature of the opposition driven by commercial rivalry, warranting dismissal with costs.

GROUND 25(1)(e)

The Applicant respectfully submits the following ground posits that the invention, as claimed in the present application, is obvious to a person skilled in the art and thus lacks an inventive step under Section 2(1)(ja). The Applicant vehemently contests this allegation, asserting that it is fundamentally flawed, bereft of substantive evidence, and predicated on impermissible hindsight reconstruction, selective mosaicking of prior art, and a failure to adhere to the rigorous analytical framework mandated by Indian patent jurisprudence. This rebuttal is structured to commence with a comprehensive exposition of the legal principles governing inventive step assessments in India, drawing upon statutory provisions, judicial precedents, and guidelines from the Manual of Patent Office Practice and Procedures, 2019 (hereinafter "Patent Manual"). Thereafter, it applies the established five-step test from F. Hoffmann-La Roche Ltd. v. Cipla Ltd. ((2016) 66 PTC 1 (Del))—hereinafter the "Roche Test"—to the facts at hand, demonstrating that the cited prior art documents D1 (CN103820781A, published 28th May 2014), D2 (CN105148816A, published 16th December

2015), and D3 (CN206372825U, published 4th August 2017) neither individually nor in combination render the invention obvious.

To contextualize, Section 25(1)(e) empowers opposition on the basis that "the invention so far as claimed in any claim of the complete specification is obvious and clearly does not involve any inventive step, having regard to the matter published as mentioned in clause (b) or having regard to what was used in India before the priority date of the applicant's claim." Section 2(1)(ja) defines "inventive step" as "a feature of an invention that involves technical advance as compared to the existing knowledge or having economic significance or both and that makes the invention not obvious to a person skilled in the art."

This dual-pronged criterion—technical advance/economic significance coupled with non-obviousness—distinguishes Indian law from counterparts. The Patent Manual (Chapter 08.03.04) elucidates that inventive step assessment must consider the invention as a whole, avoiding hindsight bias, and requires the opponent to establish a clear pathway by which a person skilled in the art possessing common general knowledge in the field would arrive at the invention from the prior art without inventive ingenuity. Crucially, combinations of prior art are permissible only if there exists a "teaching, suggestion, or motivation" to combine them, to prevent ex post facto rationalizations.

The landmark Roche Test, affirmed in multiple decisions, provides a structured methodology: (1) Identify the inventive concept claimed; (2) Assume the mantle of the person skilled in the art and impute common general knowledge; (3) Identify differences between the cited prior art and the invention; (4) Determine if those differences constitute steps obvious to the person skilled in the art without hindsight; and (5) Consider secondary indicia like commercial success, long-felt need, or failure of others. Recent jurisprudence, such as Tapas Chatterjee v. Assistant Controller of Patents (2023 SCC OnLine Del 469), condemns mechanical rejections, mandating reasoned orders that eschew subjective opinions and improper prior art

combinations without motivation. The Opponent's approach—mere tabular feature mappings without teaching, suggestion, or motivation analysis or person skilled in the art imputation—violates these tenets, rendering their case untenable.

Applying the Roche Test to the present invention, titled "Clean Room Glass Lined Reactor," the inventive concept (Step 1) is a specialized reactor for pharmaceutical/food-grade clean rooms, comprising an internally glass-lined MS body with machined fluidic channels, clamped between flanges; a glass-lined flat observation surface for contamination-free flow monitoring; a bottom flange cavity with connection plate for efficient heat transfer; and a seamless SS overlay welded post-2mm MS removal, extending to nozzles for 100% enclosure against oxidation, ensuring GMP/ISO compliance, longevity, and hygiene without insulation gaps or fillers. This addresses a long-felt need: preventing nozzle oxidation distorting internal lining in sterile environments, yielding technical advances (enhanced corrosion resistance, visual monitoring) and economic significance (reduced downtime, maintenance costs). The complete specification (pages 1-19, as amended) enables this via detailed construction and method, far beyond obvious modifications.

Assuming the person skilled in the art's mantle (Step 2), the person skille din the art is an engineer with experience in chemical reactor design for pharma industries, cognizant of CGK like jacketed glass-lining for corrosion, SS claddings for durability, and basic welding techniques, but not advanced metallurgical bonding or clean-room specifics unless explicitly taught. CGK includes standards like ISO 14644-1 but excludes patented innovations or rare combinations.

Identifying differences (Step 3) from D1-D3 (collectively or individually, as the Opponent urges conjoint reading without justification): D1 teaches insulated SS "profiling cladding" with polyurethane-filled gaps for general glass-lined tanks, prioritizing thermal insulation over oxidation prevention; no machined channels, observation surface, flange cavity, or seamless

nozzle overlays—cladding is gapped, detachable. D2 describes SS "foreskin/wrapper" manufacturing with insulation fillers (glass cement) and flanged perforations, but gaps persist pre-filling, no MS removal for bonding, absent observation/cavity features, focused on generic kettles. D3, a utility model, enhances internal enamel with frit compositions and neck flanges for strength, entirely lacking external SS overlays or clean-room elements. The key differences like invention's metallurgical SS-MS bond (post-removal welding) vs. prior art's gapped/insulated claddings; integrated observation and cavity for hygiene/heat transfer vs. absent; nozzle-specific seamless extension for 100% enclosure vs. partial flanging; clean-room GMP focus vs. general use.

Assessing obviousness (Step 4): These differences are non-obvious, as no "teaching, suggestion, or motivation" exists in common general knowledge or prior art to combine/modify. D1's insulation objective (gaps for foam) teaches away from seamless bonding, as fillers would compromise sterility; modifying to remove gaps requires hindsight from the invention. D2's wrapper with secondary cement filling acknowledges displacement risks, not suggesting precise MS removal/welding for uniformity—person skille din the art would not "obviously try" this without inventive spark. D3's internal focus provides no bridge to external overlays. Opponent's assertions are conclusory, ignoring divergent purposes (insulation vs. oxidation prevention). Secondary indicia (Step 5) is that the invention fulfills long-felt pharma need for rust-free reactors, evidenced by adoption; Opponent's rivalry underscores commercial value.

The Opponent's evidence—uncertified translations, no expert affidavits—fails the burden under Rule 55(3). Their mala fide intent, as competitors, merits costs per Hindustan Lever Ltd. v. Godrej Soaps Ltd. (1996 PTC 141). The Applicant prays for rejection of this ground with awarded costs.

GROUND 25(1)(f)

Turning to the substantive rebuttal, the Applicant addresses the Opponent's grounds under Section 25(1)(f) ground asserts that the subject matter of the claims in the present application does not constitute an "invention" as defined under Section 2(1)(j) of the Act or is otherwise not patentable, specifically invoking exclusions under Sections 3(d) and 3(f) of the Act. The Applicant submits that this objection is profoundly misguided, lacking in evidentiary support, and premised on a misapplication of the statutory exclusions, which are intended to prevent the patenting of trivial or non-inventive subject matter but do not apply to the genuine technical advancement embodied in the present invention.

Specifically, Section 3(f) excludes "the mere arrangement or re-arrangement or duplication of known devices each functioning independently of one another in a known way," targeting aggregations devoid of synergistic interaction. As per the Patent Manual (Chapter 08.02.02), this provision prevents patenting of collocations where components operate separately without producing a novel combined effect, but expressly permits patents for combinations yielding a new or improved result through interdependence. In TVS Motor Company Ltd. v. Bajaj Auto Ltd. (2009 (40) PTC 689 (Mad)), where the Madras High Court upheld a patent for a twin-spark plug engine, ruling that even if individual elements are known, their specific configuration producing improved combustion efficiency constitutes a patentable invention, not a mere rearrangement. More recently, in Nippon Steel Corporation v. Controller of Patents (2024 SCC OnLine Del 1234, decided in March 2024), the Delhi High Court clarified that for Section 3(f) objections, the Controller must demonstrate independent functioning of components without hindsight, emphasizing that mechanical inventions involving integrated structural modifications are patentable if they solve a technical problem in a non-obvious manner.

Turning to Section 3(d), it precludes "the mere discovery of a new form of a known substance which does not result in the enhancement of the known efficacy of that substance or the mere discovery of any new property or new use for a known substance or of the mere use of a known process, machine

or apparatus unless such known process results in a new product or employs at least one new reactant. The Patent Manual (Chapter 08.02.01) confines Section 3(d) to chemical/pharmaceutical inventions, noting its irrelevance to mechanical devices unless analogized to "new forms" like alloys or composites. Thus, for the present mechanical reactor, Section 3(d) is prima facie inapposite, but even if invoked, the invention's enhanced corrosion resistance and hygiene qualify as improved efficacy.

Against this backdrop, the Opponent's invocation of Section 3(f) is untenable, as the invention is not a "mere arrangement" but a sophisticated, interdependent configuration yielding synergistic benefits. The complete specification (pages 1-19) describes a Clean Room Glass Lined Reactor with: (i) an MS body internally glass-lined and externally SS-overlaid via seamless welding post-2mm MS removal; (ii) machined fluidic channels for heat transfer; (iii) a glass-lined flat observation surface for non-contaminating flow monitoring; (iv) a bottom flange cavity with connection plate for efficient fluid circulation; and (v) nozzle overlays ensuring 100% enclosure against oxidation. These elements do not function "independently in a known way"—the SS overlay integrates with the channels and cavity to enhance heat efficiency while preventing corrosion penetration; the observation surface synergizes with the clean-room design for GMP compliance, reducing contamination risks. This produces novel results: prolonged reactor life (reduced distortion from oxidation), superior hygiene (no rust/dust), and economic gains (lower maintenance, downtime).

Contrasting with prior art: D1's insulated cladding operates separately for thermal isolation, without synergy for oxidation prevention; D2's wrapper with fillers functions independently for manufacturing ease, not integrated hygiene; D3's enamel focuses solely on internal strength, sans external overlays. The Opponent's claim (Written Statement, page 27) that these are "standard GLR features" ignores the inventive welding method creating a metallurgical bond, where interdependent plugs yielded patentable synergy. No expert evidence from the Opponent proves independent functioning, failing the burden per the Patent Manual.

Likewise, Section 3(d) does not bar patentability, as the invention is not a "new form of a known substance" but a novel mechanical assembly. Even analogizing the reactor as a "substance" (e.g., modified vessel), it exhibits enhanced efficacy: superior oxidation resistance (100% SS enclosure vs. prior art's gapped claddings), quantifiable through extended service life and compliance metrics, as required by Novartis (supra). The invention's technical advance—addressing clean-room nozzle oxidation undistorted internal lining—constitutes enhanced efficacy, rendering Section 3(d) inapplicable.

In conclusion, the ground under Section 25(1)(f) is frivolous, motivated by rivalry, and merits dismissal. The Applicant prays for grant of the patent and costs.

GROUND 25(1)(g)

The Applicant respectfully submits the following exhaustive and detailed observations in rebuttal to the ground raised under Section 25(1)(g) of the Patents Act, 1970. This ground alleges that the complete specification does not sufficiently and clearly describe the invention or the method by which it is to be performed, thereby violating the requirements under Section 10(4) of the Act. The Applicant contends that this objection is wholly unfounded, bereft of merit, and predicated on a superficial reading of the specification, ignoring its comprehensive, enabling disclosure tailored to empower a person skilled in the art to replicate the invention without undue experimentation.

Beginning with the statutory and jurisprudential foundation, Section 25(1)(g) empowers opposition on the premise that "the complete specification does not sufficiently and clearly describe the invention or the method by which it is to be performed." This provision dovetails with Section 10(4), which stipulates that the complete specification must "fully and particularly describe the invention and its operation or use and the method by which it is to be performed; disclose the best method of performing the invention

which is known to the applicant and for which he is entitled to claim protection; and end with a claim or claims defining the scope of the invention for which protection is claimed."

The Patent Manual (Chapter 05.03) elaborates that sufficiency entails three pillars: (i) full description enabling the person skilled in the art to perform the invention across its claimed scope without undue burden; (ii) disclosure of the best mode known to the applicant at filing; and (iii) clarity to avoid ambiguity. The Manual emphasizes that the specification must be "enabling," meaning it provides sufficient technical details, examples, or parameters for replication, but need not exhaustively detail routine variations ascertainable via common general knowledge. For mechanical inventions like the present reactor, the Manual notes that drawings, structural descriptions, and operational methods suffice, without necessitating exhaustive prototypes or data unless claims hinge on specific performance metrics.

Applying this framework, the Opponent's allegations crumble under scrutiny. The Opponent contends (Written Statement, page 31) that the specification is vague, lacking enabling instructions for SS overlay welding, technical parameters (e.g., welding current, MS removal depth), working examples, and proof for "100% glass-lined surface," rendering it non-enabling for the PSA. This is a mischaracterization; the complete specification (pages 1-19) provides a fulsome, person skilled in the art-oriented disclosure compliant with Section 10(4). Commencing with the title and preamble (page 1), it clearly states: "CLEAN ROOM GLASS LINED REACTOR... The following specification particularly describes the nature of the invention and the manner in which it is to be performed," signaling intent to enable. The Field of Invention (page 1) contextualizes: "The present invention relates to corrosion resistant, clean room glass lined overlaid stainless steel reactor. More particularly, the present invention discloses a reactor specifically designed for clean and polished working environment in food and medical industries, which don't require any paint coat the look clean as mirror finished." This sets the technical domain, identifying the

problem (corrosion in clean rooms) and solution (SS overlays), guiding the PSA.

The Background Art (pages 1-2) elaborates prior limitations: "Conventional reactors are widely used by fine chemicals and pharmaceutical industry. Corrosion, cleaning and contamination are one of the key criteria that determine the efficiency of a process... Stirred tank reactors (STRS) used in pharmaceutical and fine chemical industries are designed for providing and removing heat. This is done with coil or half coil with coolant passing through it, or using jacketed tanks." It critiques conventional designs for dust-prone paint, inefficient heat transfer, and contamination risks, establishing the inventive necessity. The Detailed Description (pages 3-10) meticulously outlines structure and method: "The present inventions also presents Mild Steel (MS) Glass-Lined reactors with stainless steel overlays on nozzles, designed for sectors like pharmaceuticals, food, chemicals, and agrochemicals... By integrating stainless steel overlays onto reactor nozzles with SS cladding around the vessel, the solution covers 100% of the reactor, offering durability for industries requiring stringent hygiene. Stainless steel enhances corrosion resistance..." Specifics include: MS body with 1-2mm glass lining; machined fluidic channels for heat transfer; clamped flanges with bolts; glass-lined observation surface for flow monitoring; bottom cavity with inlet/outlet and connection plate; seamless SS welding post-2mm MS removal using SS electrodes for metallurgical bond. This enables replication: the person skilled in the art, versed in welding, can adapt parameters (current 70-100A, as routine). No "undue experimentation" is needed.

On best mode (Section 10(4)(b)), the specification discloses the preferred: "removing approximately 2mm of MS surface and then covering this with SS by welding with SS electrodes, maintaining constant total thickness," as the optimal known method for uniform bonding and 100% enclosure. Claims (pages 11-13) are supported: Claim 1 details components (i)-(vii), with dependents on double-walling (Claim 6), smoothness (Claim 7), and ranges (Claim 8), all mirrored in description. Drawings (pages 14-19) illustrate cross-sections, reinforcing clarity.

Opponent's uncertified translations and no expert affidavits fail the burden therefore said ground, merits dismissal.

The Applicant prays for rejection of the opposition proceeding and grant of the patent.

OBJECTIONS RAISED IN HEARING NOTICE

This reply is filed post-hearing under section 15 (held on 14.11.2025) to ensure complete compliance and to facilitate the grant of the patent, in accordance with the principles of natural justice and the opportunity for rectification under Sections 14 and 57 of the Patents Act, 1970, and Rule 28(7) of the Patents Rules, 2003.

The Applicant respectfully submits that these objections are primarily formal or procedural in nature, have been substantially addressed in prior responses, and are now fully met through the enclosed amendments and justifications. No substantive issues remain that impede patentability, as the invention remains novel, inventive, and sufficiently disclosed, as elaborated in the oral submissions at the hearing. The Applicant encloses proposed amended claims, and prays for their acceptance under Section 59 without prejudice to the application's validity. Should any further clarification be required, the Applicant seeks leave to provide it.

OBJECTION - 1 "CLARITY AND CONCISENESS"

The Hearing Notice reiterates:

1. *"The following objection raised in FER has still not been met. Reference numeral shall be marked in the drawings and reference of it shall be incorporated in the claims to enhance the intelligibility of Claims. No description matter is allowed on the drawings as per Rule 15(7) of The Patents Rule, 2003 (as amended)."*
2. *"The reference numerals of the drawings are not incorporated in the claims; the applicant is required to amend the claims to include*

appropriate reference numerals corresponding to the features shown in the drawings to ensure clarity and consistency."

3. *"Abbreviated forms of words have been used in the claims and/or description. The applicant is required to replace all such abbreviations with their complete forms to ensure clarity and proper understanding of the invention, as abbreviations may lead to ambiguity or misinterpretation of the technical features described."*

The Applicant submits that these objections were substantially complied with in the FER response dated 23.09.2023, where amended claims and drawings were filed to incorporate clarity enhancements. However, to fully resolve any residual concerns and align with the Hearing Notice's directives, the Applicant encloses further refinements.

Regarding objection 1 and 2 on reference numerals: Rule 15(7)(c) of the Patents Rules, 2003, recommends (but does not mandate) numerals in claims for intelligibility, as per Patent Manual Chapter 05.05.02, which states that claims should be "clear and succinct" under Section 10(5), with numerals aiding where features are graphically depicted. The original drawings (filed 21.04.2021) already include numerals (e.g., Figure 1 labels reactor body as 10, top flange as 12, fluidic channels as 16), and no "description matter" in the drawings.

In order to enhance intelligibility, the Applicant encloses Annexure A: Proposed Amended Claims integrating all numerals (e.g., Claim 1(i): "an internally glass-lined reactor (10) sandwiched between a top flange (12) and a bottom flange (14), with fluidic channels (16) machined on the glass-lined reactor surface"). These correspond directly to drawings: e.g., observation surface (18) in Figure 2, cavity (20) in Figure 3. No new matter is added—numerals are derived from original drawings and description (pages 5-7), preserving scope per Section 59(1), addressing the FER's concern (para 2.1 of FER).

The Applicant requests these amendments be taken on record as they cure formal defects post-hearing, per Tapas Chatterjee v. Assistant Controller (2023 SCC OnLine Del 469), mandating opportunity for compliance.

OBJECTION – 2 "FORMAL REQUIREMENT(S)"

The Hearing Notice lists:

1. *"The General Power of Attorney (GPA) filed by the opposition is found to be defective... Accordingly, the opposition is directed to submit a corrected... GPA..."*
2. *"It is observed that the applicant has submitted amendments... by filing Form 13. As per Section 57... provide adequate justification... Until such compliance... cannot be taken on record..."*
3. *"It is observed that the applicant has filed Form 13... As per Section 57(2)... submit a revised Form 13 clearly stating the detailed reasons... Until compliance... cannot be taken on record."*

For objection at para 1, the applicant has no comments to offer.

For objections 2 and 3 on Form 13: The Applicant's Form 13 (23.09.2023) complied with Section 57(1) (amendments with basis) and Section 57(2) (reasons for leave) and 57(6) (on overcoming of objections), but to address the Notice's call for "adequate justification" and "detailed reasons,". It includes:

- Basis Statement (per Rule 81(1)): "Each amendment finds support in the original specification/drawings: e.g., 'seamless welded coated nozzles' from page 7, lines 18-22 ('SS overlays on nozzles welded to cladding'); 'fluidic channels' from page 5, lines 10-12 ('machined surface for heat transfer'). No new matter introduced." A tabulated matrix details:

Amended Text (e.g., Claim 1(vi))	Original Basis (Page/Line/Drawing)	Nature of Amendment (Clarification/Deletion)

Amended Text (e.g., Claim 1(vi))	Original Basis (Page/Line/Drawing)	Nature of Amendment (Clarification/Deletion)
Seamless welded coated nozzles to top cover's cladding sheet	Page 7, lines 18-22; Fig. 1 (numeral 22)	Clarification for precision per FER para 2.1

- Reasons Statement (per Section 57(2)): "Amendments respond to FER objections on clarity (para 2.1: vagueness in features) and support (para 3.2: consistency with drawings), narrowing scope for definiteness without broadening. E.g., adding numerals enhances intelligibility; expanding abbreviations avoids ambiguity. Necessitated to overcome rejection risks under Section 10(5), ensuring patentability."

These fulfill "full particulars" under Section 57(2), as in Nippon Steel v. Controller (2024 SCC OnLine Del 1234), where detailed justifications post-notice were accepted. Until now, amendments were provisional; this revision ensures recordal under Rule 81(3). The objection is procedural, not fatal, allowing post-hearing cures.

OBJECTION - 3 "INVENTION U/S 2(1)(ja)"

The Hearing Notice states: "The applicant's reply to the objection raised in the First Examination Report (FER) under Section 2(1)(ja)... has been duly considered. Upon further examination, it has been observed that the following prior art documents disclose the features of the alleged invention: D4: CN103820781A (Publication Date: 25/08/2014); D5: CN105148816A (Publication Date: 16/12/2015); D6: CN206372825U (Publication Date: 04/08/2017)." It partially describes D4: "a profiled outer glass-lined device... can lid... nozzle, manhole... jacket... cavity... jacket flange... discharge port, condensate outlet, steam inlet... outer side of the can body and the outer side of the jacket have a ……].."

The Applicant submits that this objection under Section 2(1)(ja) (inventive step) was addressed in the FER reply (23.09.2023, paras 15-25), distinguishing the citations (identical to Opponent's D1-D3). The "further examination" lacks new reasoning or evidence, merely reiterating prior art without applying the Roche Test (F. Hoffmann-La Roche v. Cipla (2016 (66) PTC 1 Del)): identify person skilled in the art, inventive concept, common general knowledge, differences, obviousness. This mechanical restatement violates Tapas Chatterjee (2023 SCC OnLine Del 469), condemning non-reasoned orders.

D4-D6 do not render the invention obvious: D4 (CN103820781A) discloses insulated cladding with gaps (translation: "filled with lagging material"), not seamless SS overlays post-MS removal for oxidation prevention—differences require inventive ingenuity, no motivation to modify for clean-room GMP. D5 (CN105148816A) uses gapped wrappers with fillers, teaching away from metallurgical bonding. D6 (CN206372825U) is internal enamel-focused, lacking external features. Synergies (e.g., observation surface + cavity + overlays) yield technical advance (hygiene, longevity), non-obvious per Novartis v. UOI (2013 6 SCC 1).

A detailed response has already being furnished under aforesaid paragraphs.

The objection may kindly be waived.

OBJECTION 4 - "SCOPE-1"

The Hearing Notice objects: "1. The amendment made in the description and claims does not comply with the provisions of Section 59(1) of the Act."

The Applicant submits that all amendments filed via Form 13 dated 23.09.2023 fully comply with Section 59(1), which prohibits amendments introducing new matter beyond the original specification. The amendments were clarificatory, narrowing scope in response to FER objections on clarity and consistency, without expansion. For instance, addition of "seamless

welded coated nozzles" is verbatim supported by original description (page 7: "SS overlays on nozzles welded to cladding"), and incorporation of numerals references existing drawings without novelty. As per Patent Manual Chapter 11.03, amendments are allowable if "fairly based" on original disclosure, as here.

The objection may kindly be waived.

OBJECTION – 5 "SCOPE-2"

The Hearing Notice raises three objections under Scope:

1. *"Claim 1 recites functional advantages such as "offering excellent resistance to oxidation,"... Such statements... do not constitute patentable subject matter under Section 2(1)(j)... amend to clearly define the structural or technical features..."*

The Applicant submits that Claim 1 defines structural features (e.g., "seamless welded coated nozzles... fully enclosing the reactor in stainless steel"), with functional phrases as characterizing effects inherent to structure, permissible under Section 10(5) if defining scope. Patent Manual Chapter 05.05.01 allows functional language if tied to structure, as here—oxidation resistance results from SS enclosure. To address, enclosed amended Claim 1 rephrases: "wherein the coated nozzle is seamlessly welded to the top cover's cladding sheet, fully enclosing the reactor in stainless steel to provide resistance to oxidation at the nozzle." This retains essence without functional dominance.

"*Claims 2 and 7 do not clearly and unambiguously define the scope... term "base body of high mechanical strength" is vague... expressions "essentially," "preferably,"... "smooth as whole" is indefinite... amend... to specify... in objective terms..."*

The terms are clear to PSA: "high mechanical strength" implies standard MS properties (yield strength >250 MPa, CGK in reactor design);

"essentially/preferably" indicate ranges (1-2mm thickness), allowable per Patent Manual Chapter 05.05.03 for flexibility. "Smooth as whole" means uniform glass lining without defects, objective via industry standards (e.g., Ra <0.8 µm). Claim 2 specifies "base body having mechanical yield strength of at least 250 MPa"; removes "essentially/preferably" for fixed ranges; Claim 7: "smooth interior surface with roughness average (Ra) less than 0.8 µm." This ensures definiteness under Section 10(4)(c).

3. *"It is observed that there is an inconsistency between the description, claims, and drawings... outer wall as stainless steel (SS) and the inner as mild steel (MS)... in the drawings, the layers appear inversed... rectify the discrepancy..."*

The Applicant clarifies: description/claims correctly state inner MS (base) with glass lining, outer SS overlay (page 3: "MS Glass-Lined reactors with stainless steel overlays"). Drawings (Fig. 1) label inner as MS (10), outer as SS (24), and no inversion. To resolve, enclosed revised drawings (Annexure B) with explicit labels: "Inner: Mild Steel (MS) with Glass Lining; Outer: Stainless Steel (SS) Overlay. The amendments are consistent per Section 10(4). The objection may kindly be waived

Objection 6 - "SUFFICIENCY OF DISCLOSURE U/S 10 (4)"

The Hearing Notice raises:

1. *"It is observed that references have been made in the description to Figures 4 and 5. However, these figures are not included... file the missing Figures 4 and 5..."*

On sufficiency under Section 25(1)(g) and Section 10(4), the complete specification fully and particularly describes the invention, its operation, the best method known to the Applicant (the 2 mm surface removal and SS electrode welding for metallurgical bond and uniform thickness), and enables a person skilled in the art to perform it across the claimed scope without undue experimentation. All structural features are illustrated in

Figures 1-3 with consistent reference numerals (reactor body 10, top flange 12, bottom flange 14, fluidic channels 16, observation surface 18, cavity 20, nozzles 22, bottom plate 24, connection plate 26, SS overlay 28, nozzles 30, cladding 32, bolts 34, base body 36, double-walled design 38, interior surface 40).

References in the description to “Figures 4 and 5” were typographical errors intended as “Figures 2 and 3,” which depict the cavity, connection plate and nozzle details; no figures are missing, and the invention remains fully enabled by the existing disclosure. The Patent Manual (Chapter 05.03) confirms that for mechanical apparatus inventions, detailed textual description, method steps, and drawings with numerals suffice; exhaustive prototypes or performance data are not required where the invention is structural and parameters are either specified or routine for the skilled person (welding currents and techniques being standard CGK). The Opponent’s allegations of vagueness are baseless and do not discharge its burden.

The clarity, formal, scope and other objections in the hearing notice have been substantially met by the amendments already filed via Form 13 dated 23 September 2023 (with basis and reasons as detailed in the matrix in prior written submissions, each amendment supported by specific pages/lines of the original specification and drawings, clarificatory or narrowing in nature, and compliant with Section 57 and Rule 81). To fully and finally address the remaining concerns—particularly the multiple “wherein” clauses and narrative functional language in claim 1, the need for objective parameters in claims 2 and 7, and any perceived inconsistency in wall materials—the Applicant proposes the following further refinements, which remain fully supported by the original disclosure (pages 3-10 and Figures 1-3), introduce no new matter, and are permissible under Sections 57 and 59.

These amendments streamline claim 1 to a clear structural product claim defining the constructional inventive features, move functional advantages

(inherent to the structure) to the description where appropriate, specify measurable parameters, and confirm consistency of inner mild steel and outer stainless steel configuration.

Marked-up copy of claims (changes from the amended claims proposed in the written submissions dated after the 14 November 2025 hearing shown with strikethrough for deletions and underline for additions):

1. A clean room glass lined reactor (2) comprising: (i) an internally glass-lined mild steel reactor (10) clamped between a top flange (12) and a bottom flange (14), with fluidic channels (16) machined on the glass-lined reactor surface; (ii) a glass-lined metal flat surface (18) below the top flange (12) which allows flow-regime observation inside the reactor; (iii) a cavity (20) in the bottom flange (14) allowing circulation of heat-transfer fluid via inlet/outlet nozzles (22), closable with a metal bottom plate (24); (iv) a connection plate (26) connecting inlet/outlet (22) to the bottom flange (14); (v) the reactor (10) is fully overlaid externally with stainless steel sheet (28) by welding; (vi) internally glass-lined nozzles (30) with a seamless stainless steel overlay welded to the cladding (32) ~~to eliminate oxidation and rusting at critical zones offering excellent resistance to oxidation wherein the seamless stainless steel overlay provides resistance to oxidation at the nozzles~~**, thereby fully enclosing the nozzles and the reactor body in stainless steel**; (vii) multiple bolts (34) securing the top flange (12) and bottom flange (14).
2. The clean room glass lined reactor as claimed in claim 1, wherein the base body ~~of high mechanical strength is preferably mild steel, essentially of 1-2mm thickness~~**has a mechanical yield strength of at least 250 MPa and is mild steel with a thickness of 1-2 mm**.

3-5. [No change]

6. The clean room glass lined reactor as claimed in claim 1, wherein the reactor is a double-walled vessel ~~as claimed in claim 1, wherein the reactor (10) comprises a double-walled vessel design (38)~~.
7. The clean room glass lined reactor as claimed in claim 1, wherein the interior is smooth ~~as whole~~**with a roughness average (Ra) less than 0.8 µm**.
8. ~~The clean room glass lined reactor as claimed in claim 1, wherein the volume and heat-transfer area are application-specific as claimed in claim 1, wherein the reactor (10) has a volume range of 50-5000 liters and a heat-transfer area adapted for pharmaceutical applications~~.

Clean copy of claims (as finally proposed):

1. A clean room glass lined reactor comprising: (i) an internally glass-lined mild steel reactor clamped between a top flange and a bottom flange, with fluidic channels machined on the glass-lined reactor surface; (ii) a glass-lined metal flat surface below the top flange which allows flow-regime observation inside the reactor; (iii) a cavity in the bottom flange allowing circulation of heat-transfer fluid via inlet/outlet nozzles, closable with a metal bottom plate; (iv) a connection plate connecting inlet/outlet to the bottom flange; (v) the reactor is fully overlaid externally with stainless steel sheet by welding; (vi) internally glass-lined nozzles with a seamless stainless steel overlay welded to the cladding, thereby fully enclosing the nozzles and the reactor body in stainless steel; (vii) multiple bolts securing the top flange and bottom flange.
2. The clean room glass lined reactor as claimed in claim 1, wherein the base body has a mechanical yield strength of at least 250 MPa and is mild steel with a thickness of 1-2 mm.

3-5. [No change]

6. The clean room glass lined reactor as claimed in claim 1, wherein the reactor comprises a double-walled vessel design.

7. The clean room glass lined reactor as claimed in claim 1, wherein the interior surface has a roughness average (Ra) less than 0.8 μm.
8. The clean room glass lined reactor as claimed in claim 1, wherein the reactor has a volume range of 50-5000 liters and a heat-transfer area adapted for pharmaceutical applications.

These amendments are clarificatory, supported page-by-page by the original specification and drawings (e.g., seamless SS overlay on nozzles from page 7, lines 18-22 and Figure 1 numeral 30/32; yield strength and Ra values as standard measurable parameters consistent with CGK and description at pages 3-5; layer configuration explicitly confirmed at page 3 and Figure 1), narrow rather than broaden scope, and directly address the hearing notice objections on functional language, multiple "wherein" clauses, vagueness in claims 2 and 7, and any perceived wall-material inconsistency. The drawings correctly and consistently show the inner wall as mild steel reactor body (10) with glass lining and the outer as stainless steel overlay (28), with no inversion; any schematic appearance to the contrary is clarified by the reference numerals and description. If the Learned Controller requires annotated drawings with explicit layer labels for absolute visual clarity, the Applicant undertakes to file the same as a formal compliance without substantive change.

On the formal requirements concerning the Opponent's General Power of Attorney and the Applicant's Form 13, the Applicant has no further comments on the GPA beyond noting that any defect may impact the opposition's maintainability, without prejudice to the merits. The Form 13 amendments already on record, together with the detailed basis and reasons matrix previously submitted (each amendment tied to specific original pages/lines, nature of change, and justification for overcoming clarity objections), provide full particulars as required under Section 57(2) and Rule 81. These amendments are now sought to be taken on record in their refined form to facilitate grant.

In view of the foregoing exhaustive analysis, the complete specification and drawings vis-à-vis the citations, the procedural record of the completed hearing of 14 November 2025, and the absence of any ground warranting refusal, the Learned Controller is respectfully requested under Section 15 of the Patents Act, 1970 to proceed to decision on the basis of the existing record and the completed hearing, accept the proposed claim amendments, reject the pre-grant opposition in its entirety, and grant the patent. The invention constitutes a patentable technical advance in mechanical engineering for critical clean-room applications, the opposition is without merit and appears motivated by commercial rivalry, and further delay would cause grave prejudice to the Applicant while serving no legitimate purpose given the availability of post-grant remedies to the Opponent. The interests of justice, statutory finality, administrative propriety, and public interest in promoting genuine innovation in pharmaceutical processing equipment all favour immediate grant.

PRAYER

It is, therefore, most respectfully prayed that this Learned Controller may be pleased to:

(a) Allow the Interlocutory Petition dated 09 March 2026 and set aside/recall the repeated hearing notice dated 11 February 2026 and the hearing notice dated 16 April 2026, treating the pre-grant opposition proceedings as finally heard on 14 November 2025, 20 May 2026 on the basis of the Applicant's oral submissions and the existing written record;

(b) Direct the Patent Office to place on record an official confirmation of the exact date and time at which the Opponent's notice to attend hearing and hearing fee were received and reflected in the system.

(c) Reject the pre-grant opposition under Section 25(1) in its entirety as frivolous, mala fide and devoid of merit;

(d) Accept the proposed amendments to the claims as per the marked-up and clean copies above under Section 59 of the Patents Act, 1970, as being fairly based on the original specification and necessary to meet the objections;

(e) Proceed to grant the patent under Section 15 of the Patents Act, 1970 on the basis of the amended claims, complete specification and drawings, the invention being novel, involving inventive step, sufficiently disclosed and constituting patentable subject matter;

(f) Award costs of these proceedings to the Applicant under Section 35 of the Patents Act, 1970; and

(g) Pass such further or other orders as this Learned Controller may deem fit and proper in the facts and circumstances of the case and in the interest of justice.

Dated this 03rd day of June, 2026,



(Bharat Daswani)
Of Daswani & Daswani,
Applicant's Agent,
IN/PA 289

VERIFICATION

I, Bharat Daswani, the authorized representative of the Applicant, do hereby verify that the contents of the above submissions are true to my knowledge and belief, derived from the records of the case, and nothing material has been concealed therefrom.

Verified at Kolkata on this 03rd day of June, 2026.

Authorized Signatory for the Applicant



(Bharat Daswani)
Of Daswani & Daswani,
Applicant's Agent,
IN/PA 289

Digitally Signed By:
MR. BHARAT DASWANI
Date: 03-06-2026 19:52:49

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June 03, 2026

Kind Attention:
Dr. Kanhaiya Lal Balai
Assistant Controller of Patents & Designs

Your Ref: POC/Application No / 202141018455
Our ref: 202141018455

To
The Controller of Patents
The Patent Office,
Chennai / Delhi

Dear Sir,

Re: Response to Pre-Grant Opposition Hearing held on 20.05.2026 against Patent Application No. 202141018455 of April 21, 2021 titled "Clean Room Glass Lined Reactor" filed by Standard Glass Lining Technology Private Limited (Applicant) opposed by Pinakin Engineers Private Limited (Opponent) under Section 25(1) of the Patents Act, 1970.

In connection with the above, we would like to submit that Ms. Niharika Singh, Registered Patent Agent (IN/PA 4923), attended the hearing on 20-05-2026 on behalf of the Applicants, in accordance with the schedule specified in the Hearing Notice.

Pursuant to the hearing, a written submission, along with the relevant supporting documents, is enclosed herewith for your perusal and kind consideration.

In view of the above submissions, it is respectfully requested that the present application be allowed to proceed to grant.

Yours Sincerely
Of DASWANI & DASWANI



(Bharat Daswani)
Applicant's Agent
IN / PA 289

OPENING REMARKS AND PRELIMINARY SUBMISSIONS ON THE INTERLOCUTORY PETITION DATED 09 MARCH 2026

Hearing date – 20/05/2026

The Applicant, Standard Glass Lining Technology Private Limited, filed Patent Application No. 202141018455 on 21 April 2021 for the invention titled "Clean Room Glass Lined Reactor", a specialised mechanical engineering apparatus for pharmaceutical clean-room environments featuring an internally glass-lined mild steel body clamped between flanges with machined fluidic channels, a glass-lined observation surface, bottom-flange cavity for heat-transfer fluid, and seamless stainless steel overlay welded after 2 mm surface removal to achieve gap-free enclosure extending to nozzles, thereby preventing oxidation-induced distortion of the internal lining. The application was published on 28 October 2022. The Opponent, Pinakin Engineers Private Limited, filed a pre-grant opposition under Section 25(1) on 14 November 2024 relying on prior arts D4–D6.

The Applicant completed the evidentiary stage by filing its reply under Rule 58 on 4 August 2025. The Learned Controller issued a hearing notice dated 13 October 2025 fixing the hearing for 14 November 2025 via video conferencing under Rule 62(1), requiring notice to attend with prescribed fee under Rule 62(2). The Applicant duly complied. At the hearing the Opponent appeared via video conferencing but had filed neither notice nor fee, as recorded in the official module. The Controller permitted the Opponent to remain present yet correctly denied it oral arguments and proceeded solely on the Applicant's side, thereby completing the statutory hearing under Section 25(1). Both parties later filed written submissions. No contemporaneous request from the Opponent for a fresh hearing or explanation of its default appears on record.

Subsequently the Opponent filed written submissions dated 28 November 2025 and sought to rely on a backdated request purportedly from 13

November 2025 concerning the hearing fee and notice—documents absent from the record at the time of hearing and inconsistent with the proceedings. By communication around 29 November 2025 followed by formal notice dated 11 February 2026 the Learned Controller adjourned the matter and fixed a fresh hearing on 11 March 2026, citing the later reflection of the Opponent's fee in the system and invoking "principles of natural justice". The Applicant filed the present interlocutory petition on 09 March 2026 seeking to set aside that repeated hearing notice and to have the application decided under Section 15 on the basis of the completed hearing of 14 November 2025 and the existing record. The impugned action perpetuates delay in a time-sensitive mechanical invention where commercial exclusivity and time-to-market are critical for clean-room pharmaceutical equipment.

Under Section 25(1) read with Rules 55, 58 and 62 the pre-grant opposition is a structured quasi-judicial proceeding with a defined evidentiary stage culminating in a single statutory hearing after which the Controller must decide under Section 15 whether to grant or refuse the patent. Rule 62(2) expressly makes filing of notice to attend hearing with the prescribed fee a condition precedent for a party to be "heard" in the sense of advancing oral submissions. Non-compliance forfeits that right, although presence as a spectator remains permissible—as occurred here. Once the 14 November 2025 hearing concluded with the Applicant's arguments heard and the Opponent's default acted upon, the proceedings attained finality at the hearing stage. **No provision in Section 25(1), Section 15 or Rule 62 authorises re-opening a completed hearing merely because the defaulting party later perfects its procedural lapse, and that too without any formal application on record. The repeated hearing notice dated 11 February 2026 therefore travels beyond the statutory framework and is ex facie ultra vires, as it purports to restore a forfeited right of audience and nullifies the deliberate consequence attached to Rule 62(2) non-compliance.**

Judicial authority establishes that once a statutory hearing is duly concluded it cannot be casually re-opened except on limited grounds such as manifest error or fraud. In *Union of India v. Tarachand Gupta & Bros.* (AIR 1971 SC 1558) the Hon'ble Supreme Court emphasised the sanctity of finality in concluded proceedings. The present record discloses no such exceptional ground; it shows only a completed hearing followed by an attempt to regularise the Opponent's default through backdated documents whose timing raises serious questions directly affecting compliance with Rule 62(2) and the legality of the re-opening decision. Rule 137 empowers the Controller to correct procedural irregularities but cannot be invoked to confer an entirely new substantive right of audience upon a party that has already forfeited the same. The backdating issue and absence of contemporaneous explanation further undermine any justification for unsettling finality.

The principles of natural justice, as laid down in *Maneka Gandhi v. Union of India* (AIR 1978 SC 597), require a fair opportunity of hearing but do not compel authorities to grant repeated or additional hearings to cure a party's own procedural defaults, especially where statutory preconditions such as fee payment have not been met. The Opponent was aware of the hearing date and appeared, yet failed to comply with Rule 62(2). The Controller therefore justifiably limited the hearing to the Applicant's side, satisfying *audi alteram partem* to the extent rights were properly invoked. The *suo motu* re-opening to accommodate belated fee payment converts natural justice into a device for granting a second chance to the defaulting party at the cost of delay and prejudice to the compliant Applicant. The Opponent suffers no irreparable prejudice: its substantive rights remain fully protected by post-grant remedies under Sections 25(2), 64 and 84. Conversely, the Applicant faces grave commercial prejudice through prolonged uncertainty in a mechanical invention where time-to-market and exclusivity are critical for specialised clean-room pharmaceutical reactors. The balance of convenience and proportionality therefore lies decisively in preserving the finality of the 14 November 2025 hearing.

In order to maintain the sanctity of the official record under Section 74 it is necessary that the Learned Controller direct placement on record of the exact date and time when the Opponent's notice and fee were received and reflected, and that the video recording of the 14 November 2025 hearing be furnished to the Applicant. The interests of justice, statutory finality and administrative propriety are best served by recalling the repeated hearing notice and proceeding to decision under Section 15 on the basis of the completed hearing and existing record. The Opponent's post-hearing written submissions and backdated attempts are not contemplated by Rule 62 and cannot justify re-opening a concluded proceeding.

PRAYER

In view of the foregoing, the Applicant most respectfully prays that this Learned Controller may be pleased to:

(a) Set aside/recall the repeated hearing notice dated 11 February 2026 (and any subsequent notices), treating the pre-grant opposition as finally heard on 14 November 2025;

(b) Direct that the application be taken up for decision under Section 15 on the basis of the oral hearing of the Applicant on 14 November 2025 and the statements and evidence already on record in accordance with Rules 55–60;

(c) Direct the Patent Office to confirm the exact timestamp of the Opponent's hearing fee and notice receipt and to furnish the video recording of the 14 November 2025 hearing to the Applicant;

(d) Pass such further or other orders as this Learned Controller may deem fit and proper in the facts and circumstances and in the interest of justice.

WITHOUT PREJUDICE

WRITTEN SUBMISSIONS ON BEHALF OF THE APPLICANT IN THE PRE-GRANT OPPOSITION PROCEEDINGS UNDER SECTION 25(1) OF THE PATENTS ACT, 1970

Date of Hearing: 20th day of May, 2026

OPENING REMARKS

The Applicant, Standard Glass Lining Technology Private Limited, through its authorized Patent Agent, Mr. Bharat Daswani (IN/PA/289), hereby submits these exhaustive written submissions in response to the Pre-Grant Opposition filed by the Opponent on 14th November 2024, as well as in compliance with the Hearing Notice dated 13th October 2025 issued under Rule 55(4) of the Patents Rules, 2003.

These submissions address all grounds raised by the Opponent under Section 25(1)(b), (e), (f), and (g) of the Patents Act, 1970, refute each statement made in the Opponent's Written Statement and evidence (Exhibits A to G), and respond to the objections outlined in the Hearing Notice regarding clarity and conciseness, formal requirements, and inventive step under Section 2(1)(ja).

The Applicant humbly requests the Learned Controller to dismiss the opposition in its entirety, as it is frivolous, mala fide, and lacks merit, and to proceed with the grant of the patent, subject to any minor amendments if deemed necessary for formal compliance. The Applicant reserves the right to file further evidence or amendments as per the provisions of the Act and Rules, and seeks an opportunity to be heard orally at the scheduled hearing.

The present application, filed on 21st April 2021 and published on 28th October 2022, discloses a novel Clean Room Glass Lined Reactor specifically designed for pharmaceutical and food-grade processing environments, where corrosion resistance, hygiene, and contamination control are

paramount. The invention addresses a long-standing technical problem in conventional glass-lined reactors: oxidation and corrosion at nozzles and external surfaces in clean-room settings, which can lead to distortion of the internal glass lining, product contamination, and non-compliance with stringent standards like ISO 14644-1 and GMP.

The core inventive concept lies in a mild steel (MS) reactor body internally glass-lined and externally overlaid with stainless steel (SS) sheets through a seamless welding process, particularly extending to nozzles, ensuring 100% enclosure in SS to prevent oxidation while maintaining structural integrity and heat transfer efficiency. This is achieved by machining fluidic channels on the reactor surface, integrating a glass-lined flat observation surface below the top flange for flow regime monitoring, and incorporating a cavity in the bottom flange for heat transfer fluid circulation, all clamped securely between top and bottom flanges with bolts.

The specification provides enabling disclosure, including detailed description, claims, and drawings, supported by the method of performance, enabling a person skilled in the art (a mechanical engineer familiar with reactor design in chemical/pharmaceutical industries) to practice the invention without undue experimentation.

In contrast, the Opponent's opposition is an attempt to delay the grant due to commercial rivalry, as both parties operate in the glass-lined equipment manufacturing sector in Gujarat. The Opponent fails to establish locus standi as a "person interested" under Section 25(1), merely claiming to be a competitor without demonstrating any tangible interest beyond rivalry, such as prior use or direct economic impact, as required by precedents like Ajay Industrial Corporation v. Controller of Patents (1984 PTC 234), where the court held that mere commercial rivalry does not confer standing without proof of specific interest. Furthermore, the opposition is mala fide, as evidenced by its reliance on unsubstantiated assertions, uncertified translations of foreign prior art, and superficial tabular comparisons without technical analysis, violating the principles laid down in Hindustan Lever Ltd.

v. Godrej Soaps Ltd. (1996 PTC 141), where oppositions motivated by delay tactics were dismissed with costs. The Applicant seeks dismissal on these preliminary grounds alone, along with compensatory costs under Section 35 of the Act for defending this baseless challenge.

GROUND 25(1)(b)

Turning to the substantive rebuttal, the Applicant addresses the Opponent's grounds under Section 25(1)(b) for lack of novelty. The Opponent alleges anticipation by prior art documents D1 (CN103820781A, priority 12.03.2014), D2 (CN105148816A, priority 08.09.2015), and D3 (CN206372825U, priority 12.01.2017), claiming that these disclose all features of the claims through tabular comparisons in their Written Statement. The Applicant humbly reiterates and submits the following detailed observations in rebuttal to the ground raised under Section 25(1)(b) of the Patents Act, 1970, as articulated in the Opponent's Written Statement and as prima facie upheld by the Learned Controller in the Report dated 04th June 2025 (hereinafter "Prima Facie Report").

This ground alleges that the subject matter of the present application lacks novelty due to anticipation by prior publications, specifically D1 (CN103820781A, published 28th May 2014), D2 (CN105148816A, published 16th December 2015), and D3 (CN206372825U, published 4th August 2017). The Applicant respectfully contends that this ground is entirely misconceived, unsubstantiated, and fails to meet the stringent evidentiary and analytical thresholds mandated by Indian patent law for establishing anticipation. The rebuttal is structured to first elucidate the legal framework governing novelty assessments, followed by a meticulous, feature-by-feature dissection of the cited documents vis-à-vis the inventive features of the present application, demonstrating unequivocally that no single prior art document discloses the invention as a whole in an enabling and unambiguous manner. This analysis draws upon the complete specification of the present application (filed on 21st April 2021, as amended in response to the First Examination Report dated 23rd March 2023), the full English

translations of D1, D2, and D3, relevant precedents from Indian courts, and guidelines from the Manual of Patent Office Practice and Procedures, 2019 (hereinafter "Patent Manual").

At the outset, it is imperative to recall the legal principles from Section 25(1)(b), which provides that a patent may be opposed if the invention "has been published before the priority date of the claim (i) in any specification filed in pursuance of an application for a patent made in India on or after the 1st day of January, 1912; or (ii) in India or elsewhere, in any other document." As per Section 2(1)(l), an invention is "new" if it does not form part of the state of the art, and anticipation requires that the prior art must disclose the invention in its entirety, with such clarity and completeness that a person skilled in the art could perform it without further ingenuity or experimentation. The Hon'ble Supreme Court in Bishwanath Prasad Radhey Shyam v. Hindustan Metal Industries (AIR 1982 SC 1444) emphatically held that for anticipation, the prior publication must contain "clear and unmistakable directions" to perform the invention, and mere suggestions or incomplete disclosures do not suffice. Similarly, in Lallubhai Chakubhai Jariwala v. Chimanlal Chunilal & Co. (AIR 1936 Bom 99), the court clarified that anticipation cannot be established by mosaicking disparate elements from multiple documents; each prior art must be evaluated independently as a standalone disclosure.

The Patent Manual (Chapter 08.03.01) reinforces this by stating that "the prior art should disclose the invention either in explicit or implicit manner" and must enable the skilled person to arrive at the invention without undue burden. Critically, foreign patent documents like D1, D2, and D3 must be accompanied by certified translations and proof of public availability in India prior to the priority date (21st April 2021), failing which they are inadmissible, as per Rule 55(3) of the Patents Rules, 2003, and precedents such as Snehalata C. Patel v. Controller of Patents (2001 PTC 121). In the instant case, the Opponent has furnished uncertified machine translations (Exhibits E, F, G), which are riddled with inaccuracies (e.g., "foreskin" in D2 likely mistranslating "wrapper" or "sheath"), and no evidence of their

dissemination in India, rendering them prima facie unreliable. Even assuming arguendo their admissibility, a granular analysis reveals profound dissimilarities, as elaborated below.

The inventive core of the present application, in the complete specification (pages 1-19) and amended claims (dated 23rd September 2023), resides in a "Clean Room Glass Lined Reactor" tailored for pharmaceutical and food-grade environments under stringent hygiene standards like ISO 14644-1 and GMP. The reactor comprises: (i) an internally glass-lined mild steel (MS) body clamped between top and bottom flanges with machined fluidic channels on the surface for enhanced heat transfer; (ii) a glass-lined metal flat surface below the top flange enabling non-invasive observation of internal flow regimes to prevent contamination; (iii) a cavity in the bottom flange for heat transfer fluid circulation via inlet/outlet nozzles, closable with a metal plate; (iv) a connection plate linking inlet/outlet to the bottom flange; (v) an external stainless steel (SS) sheet overlay welded seamlessly to enclose the reactor fully, preventing oxidation and rust at critical zones; (vi) internally glass-lined nozzles with seamless SS overlay welded to the cladding, addressing the specific problem of nozzle oxidation distorting internal lining in clean rooms; (vii) a double-walled vessel design with smooth glass-lined interior; and (viii) application-specific volumes (e.g., 50-5000 liters) and heat transfer areas.

This configuration solves the longstanding issue of external corrosion in conventional reactors, where spillage in clean rooms leads to oxidation penetrating to the internal glass lining, causing distortion, contamination, and regulatory non-compliance. The manufacturing method involves removing approximately 2mm of MS surface before welding SS overlays with SS electrodes to form a metallurgical bond, maintaining uniform thickness and 100% SS enclosure without gaps or insulation voids that could harbor contaminants. None of D1, D2, or D3—each addressing generic glass-lined vessels for non-clean-room applications—discloses this integrated, synergistic arrangement, as evidenced by their divergent teachings.

Commencing with D1 (CN103820781A, titled "Glass Lined Equipment Provided with Outer Profiling Cladding and Manufacturing Method"), this document describes a conventional glass-lined tank comprising a tank cover (with nozzles, manhole, and seal) and tank body encased in a jacket for heat exchange, with external stainless steel "profiling cladding" layers applied for insulation purposes. The full description (*from unauthenticated English translation*) reveals that the cladding is formed by cutting and shaping 304 stainless steel sheets (1.5-3mm thick) to match the tank's profile, welding joints with argon arc welding, and filling gaps between the jacket and cladding with polyurethane foam or silicate for thermal insulation. The tank body and jacket are welded into a unitary structure with a cavity for steam/condensate flow, and bottom ports include drain hole, condensation-water drain, steam inlet, and vapor outlet. However, D1 fundamentally diverges from the present invention in several critical aspects, precluding anticipation. Firstly, D1's "cladding" is a detachable, insulated outer layer with deliberate gaps filled for thermal purposes, not a seamless, metallurgically bonded SS overlay welded directly after MS surface removal to prevent oxidation in clean-room settings. The translation explicitly states: "Between described chuck and stainless steel surrounding layer, be filled with lagging material and form thermal insulation layer," emphasizing insulation over corrosion prevention at nozzles or flanges.

In contrast, the present application's SS overlay is seamless, extending to nozzles without insulation gaps, ensuring 100% enclosure to eliminate rust formation at critical zones—a feature absent in D1, where cladding joints are merely butt-welded and polished, leaving potential entry points for moisture. Secondly, D1 lacks any disclosure of machined fluidic channels on the reactor surface, a glass-lined flat observation surface for flow monitoring, or a bottom flange cavity with connection plate for targeted heat transfer; instead, it relies on a standard jacketed cavity with bottom ports, without the clamped flange assembly or observation capability essential for clean-room hygiene. The manhole in D1 is for access, not flow observation, and there is no mention of a flat glass-lined surface below the top flange to

enable visual inspection without opening, which prevents contamination risks in pharmaceutical processes.

Thirdly, D1's manufacturing method involves preliminary impact briquetting of sheets, argon welding (current 70-80A, voltage 7-9V), and "V"-type breach repairs for flanges, but omits the inventive step of removing 2mm MS surface for uniform thickness and metallurgical bonding, focusing instead on profile consistency for insulation. As per the Patent Manual (Chapter 08.03.02), anticipation requires "implicit or explicit" disclosure of all essential features; here, D1's generic insulated cladding for cost reduction in non-pharma contexts does not implicitly teach the oxidation-resistant, seamless nozzle overlay for GMP compliance.

The Opponent's tabular comparison (Written Statement, pages 12-15) superficially maps D1's "surrounding layer" to the claims but ignores functional disparities—D1's layer allows gaps for filling, whereas the invention mandates seamless welding to preclude voids. In Novartis AG v. Union of India ((2013) 6 SCC 1), though pharmaceutical, the principle extends: mere structural similarity without identical purpose or effect does not anticipate. Thus, D1 falls short of providing "clear and unmistakable directions" (Bishwanath Prasad, supra) to arrive at the clean-room reactor.

Shifting to D2 (CN105148816A, titled "Manufacturing Method of Glass-Lined Reaction Kettle"), this patent outlines a process for fabricating a glass-lined kettle with an external stainless steel "foreskin" (translated as "wrapper" or "sheath") to enhance corrosion resistance. The method encompasses seven steps: assembling enamel-fired carbon steel components (upper head, cylinder, lower head); pressing SS sheets into upper, middle, and lower foreskins; perforating for process ducts; fixing foreskins by welding to steel flanges on nozzles; filling insulation material (e.g., glass cement) in gaps and seal-welding; polishing the SS surface; and adding an annular SS cover. The translation details that foreskins are oversized (diameter larger by 2-4mm at holes) for wrapping, with steel flanges halved and welded on nozzle flanges to enable SS welding, and insulation filled to isolate carbon steel. While D2

superficially addresses external corrosion via SS wrapping, it does not anticipate the invention due to profound structural and functional mismatches.

Primarily, D2's "foreskin" is a detachable wrapper with deliberate gaps filled post-assembly for insulation, not a seamless welded overlay integrated metallurgically after MS surface machining. The description states: "Fill insulant material in the gap between stainless steel foreskin and enamel, and seal welding," implying voids are inherent and filled secondarily with materials like glass cement, which the present invention eschews to avoid relative displacement and re-emerging gaps—issues D2 acknowledges but resolves inadequately via cement, leading to potential contamination in clean rooms. The invention's seamless welding (using SS electrodes post-2mm MS removal) creates a continuous, rust-proof enclosure without fillers, a distinction not disclosed in D2, where welding is limited to foreskin-flange joints (e.g., "upper foreskin is welded with steel flange").

Secondly, D2 omits key elements like fluidic channels machined on the reactor surface, the glass-lined observation flat surface, bottom flange cavity with connection plate, or nozzle-specific seamless SS overlays for oxidation prevention; its process ducts are merely perforated and flanged, without extended SS welding to cladding for 100% enclosure. The kettle is for general chemical reactions, not clean-room pharma use, lacking GMP-focused hygiene features. The manufacturing steps emphasize inversion for filling (e.g., "The upper cover being surrounded by described upper foreskin in described step e is inverted, fill insulant material in space up between foreskin and upper cover"), prioritizing insulation over oxidation resistance, whereas the invention's method ensures dimensional uniformity and bond strength for longevity in sterile environments. As per F. Hoffmann-La Roche Ltd. v. Cipla Ltd. ((2016) 66 PTC 1 (Del)), anticipation demands identical disclosure, not obvious modifications; D2's wrapper, reliant on secondary fillers, does not enable the skilled person to perform the seamless, gap-free overlay without ingenuity. The Opponent's assertions (Written Statement, pages 16-18) conflate D2's "foreskin" with the claims but overlook that D2's

design permits air/moisture entry pre-filling, antithetical to the invention's preventive enclosure. Hence, D2 does not destroy novelty.

Finally, D3 (CN206372825U, a utility model titled "A Kind of High-Strength Corrosion-Resisting Glassed Steel Reaction Vessels"), pertains to a jacketed glass-lined reactor with enhanced compressive strength via seamless welding neck flanges at ports. The description covers a tank body with jacket for heat exchange (steam/liquid inlets/outlets), agitator, viewing manhole, thermometer nozzle, charging/discharge ports, and standby ports, all featuring neck flanges with grooved end faces, chamfers, and bolted covers/gaskets for sealing. Corrosion resistance is achieved via internal ground glaze and cover coat layers (with specific frit compositions including silica 40-45%, alumina 3-7%, etc., and mixed adhesive agents like bastnaesite flour 45-55%), extending to flange inner walls and joints. However, D3 is devoid of any external SS overlay or cladding, let alone seamless welding for clean-room oxidation prevention, rendering it wholly non-anticipatory.

The utility model's focus is internal enamel enhancement for strength and anti-corrosion, with "high-neck flanges" welded seamlessly to reduce punch-formed openings and bolster pressure resistance (e.g., "The viewing manhole, thermometer nozzle... use seamless welding neck flanges"). There is no mention of external MS body with SS welding, machined fluidic channels, observation flat surface, bottom cavity, or nozzle overlays; the external jacket is bare, with mounting devices but no SS enclosure. The glaze formula, while improving internal lining adhesion (e.g., melting at 900-1000°C), is irrelevant to the invention's external hygiene features. As a utility model, D3 lacks enabling disclosure for manufacturing beyond basic assembly, omitting the inventive welding post-MS removal. The Patent Manual (Chapter 08.03.03) requires prior art to be enabling; D3's schematic utility focus does not provide "clear directions" for the claimed reactor.

The Opponent's mapping (Written Statement, pages 19-20) erroneously equates D3's internal enamel to the claims, ignoring the absence of external

SS elements. In Tapas Chatterjee v. Assistant Controller (2023 SCC OnLine Del 469), the court rejected anticipation based on partial, non-enabling disclosures; similarly, D3's internal-centric design cannot anticipate the external clean-room innovations.

Feature from Applicant's Claims (as amended)	Teaching in Applicant's Specification	D1 (CN103820781A)	D2 (CN105148816A)	D3 (CN206372825U)	Analysis (Why Not Anticipatory)
Internally glass-lined MS reactor clamped between top and bottom flanges with machined fluidic channels on surface (Claim 1(i))	Pages 3-5: Reactor body of MS with internal glass lining (1-2mm thick); fluidic channels machined for heat transfer; clamped via bolts for secure assembly.	Discloses general glass-lined can body with jacket, but no machined fluidic channels on reactor surface; focuses on insulated cladding (translation p.90).	Describes reaction kettle with glass lining, but channels are for jacket circulation, not machined on reactor surface (translation p.96).	Utility model for reactor with flanges; no mention of machined channels (translation p.102).	No single document discloses machined channels integrated with clamping for enhanced heat transfer in clean-room setup;.
Glass-lined metal flat surface below top flange for flow regime observation (Claim 1(ii))	Pages 5-6: Flat surface allows visual monitoring of internal mixing without contamination risk.	No observation surface; manhole is for access, not flow observation (p.91).	No such feature; focuses on manufacturing steps (p.97).	Absent entirely (p.103).	Unique to Applicant; solves clean-room monitoring issue not addressed in prior art,.
Cavity in bottom flange for heat transfer fluid circulation via inlet/outlet nozzles, closable with metal plate (Claim 1(iii))	Pages 6-7: Cavity design ensures efficient cooling/heating; connected via plate (Claim 1(iv)).	Condensate outlet and steam ports, but no bottom flange cavity (p.92).	Bottom discharge, no cavity in flange (p.98).	Flange for strength, no cavity (p.104).	Prior art uses jacketed designs, not flange-integrated cavity;
External SS sheet overlay by welding, seamless on nozzles to prevent oxidation (Claim 1(v)-	Pages 7-9: 2mm MS removal, SS welding for metallurgical bond; extends to nozzles for 100% enclosure.	Profiling cladding with gaps/insulation; not seamless weld (p.93).	Detachable wrapper with insulation; no welding to nozzles (p.99).	No overlay; corrosion via materials (p.105).	Applicant's method prevents oxidation at critical zones, unlike prior art's gapped claddings;

Feature from Applicant's Claims (as amended)	Teaching in Applicant's Specification	D1 (CN103820781A)	D2 (CN105148816A)	D3 (CN206372825U)	Analysis (Why Not Anticipatory)
(vii))					
Double-walled design, smooth interior, volume/heat transfer ranges (Claims 6-8)	Pages 10-12: Ensures hygiene, compliance with ISO/GMP.	Single jacket with insulation; no ranges (p.94).	Manufacturing focus; no double-wall specifics (p.100).	Utility; no such claims.	Synergistic advantages not disclosed;.

In summation, none of D1, D2, or D3—individually or otherwise—anticipates the invention, as they lack the complete, enabling disclosure of the synergistic features. The Applicant prays for rejection of this ground, emphasizing the mala fide nature of the opposition driven by commercial rivalry, warranting dismissal with costs.

GROUND 25(1)(e)

The Applicant respectfully submits the following ground posits that the invention, as claimed in the present application, is obvious to a person skilled in the art and thus lacks an inventive step under Section 2(1)(ja). The Applicant vehemently contests this allegation, asserting that it is fundamentally flawed, bereft of substantive evidence, and predicated on impermissible hindsight reconstruction, selective mosaicking of prior art, and a failure to adhere to the rigorous analytical framework mandated by Indian patent jurisprudence. This rebuttal is structured to commence with a comprehensive exposition of the legal principles governing inventive step assessments in India, drawing upon statutory provisions, judicial precedents, and guidelines from the Manual of Patent Office Practice and Procedures, 2019 (hereinafter "Patent Manual"). Thereafter, it applies the established five-step test from F. Hoffmann-La Roche Ltd. v. Cipla Ltd. ((2016) 66 PTC 1 (Del))—hereinafter the "Roche Test"—to the facts at hand, demonstrating that the cited prior art documents D1 (CN103820781A, published 28th May 2014), D2 (CN105148816A, published 16th December

2015), and D3 (CN206372825U, published 4th August 2017) neither individually nor in combination render the invention obvious.

To contextualize, Section 25(1)(e) empowers opposition on the basis that "the invention so far as claimed in any claim of the complete specification is obvious and clearly does not involve any inventive step, having regard to the matter published as mentioned in clause (b) or having regard to what was used in India before the priority date of the applicant's claim." Section 2(1)(ja) defines "inventive step" as "a feature of an invention that involves technical advance as compared to the existing knowledge or having economic significance or both and that makes the invention not obvious to a person skilled in the art."

This dual-pronged criterion—technical advance/economic significance coupled with non-obviousness—distinguishes Indian law from counterparts. The Patent Manual (Chapter 08.03.04) elucidates that inventive step assessment must consider the invention as a whole, avoiding hindsight bias, and requires the opponent to establish a clear pathway by which a person skilled in the art possessing common general knowledge in the field would arrive at the invention from the prior art without inventive ingenuity. Crucially, combinations of prior art are permissible only if there exists a "teaching, suggestion, or motivation" to combine them, to prevent ex post facto rationalizations.

The landmark Roche Test, affirmed in multiple decisions, provides a structured methodology: (1) Identify the inventive concept claimed; (2) Assume the mantle of the person skilled in the art and impute common general knowledge; (3) Identify differences between the cited prior art and the invention; (4) Determine if those differences constitute steps obvious to the person skilled in the art without hindsight; and (5) Consider secondary indicia like commercial success, long-felt need, or failure of others. Recent jurisprudence, such as Tapas Chatterjee v. Assistant Controller of Patents (2023 SCC OnLine Del 469), condemns mechanical rejections, mandating reasoned orders that eschew subjective opinions and improper prior art

combinations without motivation. The Opponent's approach—mere tabular feature mappings without teaching, suggestion, or motivation analysis or person skilled in the art imputation—violates these tenets, rendering their case untenable.

Applying the Roche Test to the present invention, titled "Clean Room Glass Lined Reactor," the inventive concept (Step 1) is a specialized reactor for pharmaceutical/food-grade clean rooms, comprising an internally glass-lined MS body with machined fluidic channels, clamped between flanges; a glass-lined flat observation surface for contamination-free flow monitoring; a bottom flange cavity with connection plate for efficient heat transfer; and a seamless SS overlay welded post-2mm MS removal, extending to nozzles for 100% enclosure against oxidation, ensuring GMP/ISO compliance, longevity, and hygiene without insulation gaps or fillers. This addresses a long-felt need: preventing nozzle oxidation distorting internal lining in sterile environments, yielding technical advances (enhanced corrosion resistance, visual monitoring) and economic significance (reduced downtime, maintenance costs). The complete specification (pages 1-19, as amended) enables this via detailed construction and method, far beyond obvious modifications.

Assuming the person skilled in the art's mantle (Step 2), the person skille din the art is an engineer with experience in chemical reactor design for pharma industries, cognizant of CGK like jacketed glass-lining for corrosion, SS claddings for durability, and basic welding techniques, but not advanced metallurgical bonding or clean-room specifics unless explicitly taught. CGK includes standards like ISO 14644-1 but excludes patented innovations or rare combinations.

Identifying differences (Step 3) from D1-D3 (collectively or individually, as the Opponent urges conjoint reading without justification): D1 teaches insulated SS "profiling cladding" with polyurethane-filled gaps for general glass-lined tanks, prioritizing thermal insulation over oxidation prevention; no machined channels, observation surface, flange cavity, or seamless

nozzle overlays—cladding is gapped, detachable. D2 describes SS "foreskin/wrapper" manufacturing with insulation fillers (glass cement) and flanged perforations, but gaps persist pre-filling, no MS removal for bonding, absent observation/cavity features, focused on generic kettles. D3, a utility model, enhances internal enamel with frit compositions and neck flanges for strength, entirely lacking external SS overlays or clean-room elements. The key differences like invention's metallurgical SS-MS bond (post-removal welding) vs. prior art's gapped/insulated claddings; integrated observation and cavity for hygiene/heat transfer vs. absent; nozzle-specific seamless extension for 100% enclosure vs. partial flanging; clean-room GMP focus vs. general use.

Assessing obviousness (Step 4): These differences are non-obvious, as no "teaching, suggestion, or motivation" exists in common general knowledge or prior art to combine/modify. D1's insulation objective (gaps for foam) teaches away from seamless bonding, as fillers would compromise sterility; modifying to remove gaps requires hindsight from the invention. D2's wrapper with secondary cement filling acknowledges displacement risks, not suggesting precise MS removal/welding for uniformity—person skille din the art would not "obviously try" this without inventive spark. D3's internal focus provides no bridge to external overlays. Opponent's assertions are conclusory, ignoring divergent purposes (insulation vs. oxidation prevention). Secondary indicia (Step 5) is that the invention fulfills long-felt pharma need for rust-free reactors, evidenced by adoption; Opponent's rivalry underscores commercial value.

The Opponent's evidence—uncertified translations, no expert affidavits—fails the burden under Rule 55(3). Their mala fide intent, as competitors, merits costs per Hindustan Lever Ltd. v. Godrej Soaps Ltd. (1996 PTC 141). The Applicant prays for rejection of this ground with awarded costs.

GROUND 25(1)(f)

Turning to the substantive rebuttal, the Applicant addresses the Opponent's grounds under Section 25(1)(f) ground asserts that the subject matter of the claims in the present application does not constitute an "invention" as defined under Section 2(1)(j) of the Act or is otherwise not patentable, specifically invoking exclusions under Sections 3(d) and 3(f) of the Act. The Applicant submits that this objection is profoundly misguided, lacking in evidentiary support, and premised on a misapplication of the statutory exclusions, which are intended to prevent the patenting of trivial or non-inventive subject matter but do not apply to the genuine technical advancement embodied in the present invention.

Specifically, Section 3(f) excludes "the mere arrangement or re-arrangement or duplication of known devices each functioning independently of one another in a known way," targeting aggregations devoid of synergistic interaction. As per the Patent Manual (Chapter 08.02.02), this provision prevents patenting of collocations where components operate separately without producing a novel combined effect, but expressly permits patents for combinations yielding a new or improved result through interdependence. In TVS Motor Company Ltd. v. Bajaj Auto Ltd. (2009 (40) PTC 689 (Mad)), where the Madras High Court upheld a patent for a twin-spark plug engine, ruling that even if individual elements are known, their specific configuration producing improved combustion efficiency constitutes a patentable invention, not a mere rearrangement. More recently, in Nippon Steel Corporation v. Controller of Patents (2024 SCC OnLine Del 1234, decided in March 2024), the Delhi High Court clarified that for Section 3(f) objections, the Controller must demonstrate independent functioning of components without hindsight, emphasizing that mechanical inventions involving integrated structural modifications are patentable if they solve a technical problem in a non-obvious manner.

Turning to Section 3(d), it precludes "the mere discovery of a new form of a known substance which does not result in the enhancement of the known efficacy of that substance or the mere discovery of any new property or new use for a known substance or of the mere use of a known process, machine

or apparatus unless such known process results in a new product or employs at least one new reactant. The Patent Manual (Chapter 08.02.01) confines Section 3(d) to chemical/pharmaceutical inventions, noting its irrelevance to mechanical devices unless analogized to "new forms" like alloys or composites. Thus, for the present mechanical reactor, Section 3(d) is prima facie inapposite, but even if invoked, the invention's enhanced corrosion resistance and hygiene qualify as improved efficacy.

Against this backdrop, the Opponent's invocation of Section 3(f) is untenable, as the invention is not a "mere arrangement" but a sophisticated, interdependent configuration yielding synergistic benefits. The complete specification (pages 1-19) describes a Clean Room Glass Lined Reactor with: (i) an MS body internally glass-lined and externally SS-overlaid via seamless welding post-2mm MS removal; (ii) machined fluidic channels for heat transfer; (iii) a glass-lined flat observation surface for non-contaminating flow monitoring; (iv) a bottom flange cavity with connection plate for efficient fluid circulation; and (v) nozzle overlays ensuring 100% enclosure against oxidation. These elements do not function "independently in a known way"—the SS overlay integrates with the channels and cavity to enhance heat efficiency while preventing corrosion penetration; the observation surface synergizes with the clean-room design for GMP compliance, reducing contamination risks. This produces novel results: prolonged reactor life (reduced distortion from oxidation), superior hygiene (no rust/dust), and economic gains (lower maintenance, downtime).

Contrasting with prior art: D1's insulated cladding operates separately for thermal isolation, without synergy for oxidation prevention; D2's wrapper with fillers functions independently for manufacturing ease, not integrated hygiene; D3's enamel focuses solely on internal strength, sans external overlays. The Opponent's claim (Written Statement, page 27) that these are "standard GLR features" ignores the inventive welding method creating a metallurgical bond, where interdependent plugs yielded patentable synergy. No expert evidence from the Opponent proves independent functioning, failing the burden per the Patent Manual.

Likewise, Section 3(d) does not bar patentability, as the invention is not a "new form of a known substance" but a novel mechanical assembly. Even analogizing the reactor as a "substance" (e.g., modified vessel), it exhibits enhanced efficacy: superior oxidation resistance (100% SS enclosure vs. prior art's gapped claddings), quantifiable through extended service life and compliance metrics, as required by Novartis (supra). The invention's technical advance—addressing clean-room nozzle oxidation undistorted internal lining—constitutes enhanced efficacy, rendering Section 3(d) inapplicable.

In conclusion, the ground under Section 25(1)(f) is frivolous, motivated by rivalry, and merits dismissal. The Applicant prays for grant of the patent and costs.

GROUND 25(1)(g)

The Applicant respectfully submits the following exhaustive and detailed observations in rebuttal to the ground raised under Section 25(1)(g) of the Patents Act, 1970. This ground alleges that the complete specification does not sufficiently and clearly describe the invention or the method by which it is to be performed, thereby violating the requirements under Section 10(4) of the Act. The Applicant contends that this objection is wholly unfounded, bereft of merit, and predicated on a superficial reading of the specification, ignoring its comprehensive, enabling disclosure tailored to empower a person skilled in the art to replicate the invention without undue experimentation.

Beginning with the statutory and jurisprudential foundation, Section 25(1)(g) empowers opposition on the premise that "the complete specification does not sufficiently and clearly describe the invention or the method by which it is to be performed." This provision dovetails with Section 10(4), which stipulates that the complete specification must "fully and particularly describe the invention and its operation or use and the method by which it is to be performed; disclose the best method of performing the invention

which is known to the applicant and for which he is entitled to claim protection; and end with a claim or claims defining the scope of the invention for which protection is claimed."

The Patent Manual (Chapter 05.03) elaborates that sufficiency entails three pillars: (i) full description enabling the person skilled in the art to perform the invention across its claimed scope without undue burden; (ii) disclosure of the best mode known to the applicant at filing; and (iii) clarity to avoid ambiguity. The Manual emphasizes that the specification must be "enabling," meaning it provides sufficient technical details, examples, or parameters for replication, but need not exhaustively detail routine variations ascertainable via common general knowledge. For mechanical inventions like the present reactor, the Manual notes that drawings, structural descriptions, and operational methods suffice, without necessitating exhaustive prototypes or data unless claims hinge on specific performance metrics.

Applying this framework, the Opponent's allegations crumble under scrutiny. The Opponent contends (Written Statement, page 31) that the specification is vague, lacking enabling instructions for SS overlay welding, technical parameters (e.g., welding current, MS removal depth), working examples, and proof for "100% glass-lined surface," rendering it non-enabling for the PSA. This is a mischaracterization; the complete specification (pages 1-19) provides a fulsome, person skilled in the art-oriented disclosure compliant with Section 10(4). Commencing with the title and preamble (page 1), it clearly states: "CLEAN ROOM GLASS LINED REACTOR... The following specification particularly describes the nature of the invention and the manner in which it is to be performed," signaling intent to enable. The Field of Invention (page 1) contextualizes: "The present invention relates to corrosion resistant, clean room glass lined overlaid stainless steel reactor. More particularly, the present invention discloses a reactor specifically designed for clean and polished working environment in food and medical industries, which don't require any paint coat the look clean as mirror finished." This sets the technical domain, identifying the

problem (corrosion in clean rooms) and solution (SS overlays), guiding the PSA.

The Background Art (pages 1-2) elaborates prior limitations: "Conventional reactors are widely used by fine chemicals and pharmaceutical industry. Corrosion, cleaning and contamination are one of the key criteria that determine the efficiency of a process... Stirred tank reactors (STRS) used in pharmaceutical and fine chemical industries are designed for providing and removing heat. This is done with coil or half coil with coolant passing through it, or using jacketed tanks." It critiques conventional designs for dust-prone paint, inefficient heat transfer, and contamination risks, establishing the inventive necessity. The Detailed Description (pages 3-10) meticulously outlines structure and method: "The present inventions also presents Mild Steel (MS) Glass-Lined reactors with stainless steel overlays on nozzles, designed for sectors like pharmaceuticals, food, chemicals, and agrochemicals... By integrating stainless steel overlays onto reactor nozzles with SS cladding around the vessel, the solution covers 100% of the reactor, offering durability for industries requiring stringent hygiene. Stainless steel enhances corrosion resistance..." Specifics include: MS body with 1-2mm glass lining; machined fluidic channels for heat transfer; clamped flanges with bolts; glass-lined observation surface for flow monitoring; bottom cavity with inlet/outlet and connection plate; seamless SS welding post-2mm MS removal using SS electrodes for metallurgical bond. This enables replication: the person skilled in the art, versed in welding, can adapt parameters (current 70-100A, as routine). No "undue experimentation" is needed.

On best mode (Section 10(4)(b)), the specification discloses the preferred: "removing approximately 2mm of MS surface and then covering this with SS by welding with SS electrodes, maintaining constant total thickness," as the optimal known method for uniform bonding and 100% enclosure. Claims (pages 11-13) are supported: Claim 1 details components (i)-(vii), with dependents on double-walling (Claim 6), smoothness (Claim 7), and ranges (Claim 8), all mirrored in description. Drawings (pages 14-19) illustrate cross-sections, reinforcing clarity.

Opponent's uncertified translations and no expert affidavits fail the burden therefore said ground, merits dismissal.

The Applicant prays for rejection of the opposition proceeding and grant of the patent.

OBJECTIONS RAISED IN HEARING NOTICE

This reply is filed post-hearing under section 15 (held on 14.11.2025) to ensure complete compliance and to facilitate the grant of the patent, in accordance with the principles of natural justice and the opportunity for rectification under Sections 14 and 57 of the Patents Act, 1970, and Rule 28(7) of the Patents Rules, 2003.

The Applicant respectfully submits that these objections are primarily formal or procedural in nature, have been substantially addressed in prior responses, and are now fully met through the enclosed amendments and justifications. No substantive issues remain that impede patentability, as the invention remains novel, inventive, and sufficiently disclosed, as elaborated in the oral submissions at the hearing. The Applicant encloses proposed amended claims, and prays for their acceptance under Section 59 without prejudice to the application's validity. Should any further clarification be required, the Applicant seeks leave to provide it.

OBJECTION - 1 "CLARITY AND CONCISENESS"

The Hearing Notice reiterates:

1. *"The following objection raised in FER has still not been met. Reference numeral shall be marked in the drawings and reference of it shall be incorporated in the claims to enhance the intelligibility of Claims. No description matter is allowed on the drawings as per Rule 15(7) of The Patents Rule, 2003 (as amended)."*
2. *"The reference numerals of the drawings are not incorporated in the claims; the applicant is required to amend the claims to include*

appropriate reference numerals corresponding to the features shown in the drawings to ensure clarity and consistency."

3. *"Abbreviated forms of words have been used in the claims and/or description. The applicant is required to replace all such abbreviations with their complete forms to ensure clarity and proper understanding of the invention, as abbreviations may lead to ambiguity or misinterpretation of the technical features described."*

The Applicant submits that these objections were substantially complied with in the FER response dated 23.09.2023, where amended claims and drawings were filed to incorporate clarity enhancements. However, to fully resolve any residual concerns and align with the Hearing Notice's directives, the Applicant encloses further refinements.

Regarding objection 1 and 2 on reference numerals: Rule 15(7)(c) of the Patents Rules, 2003, recommends (but does not mandate) numerals in claims for intelligibility, as per Patent Manual Chapter 05.05.02, which states that claims should be "clear and succinct" under Section 10(5), with numerals aiding where features are graphically depicted. The original drawings (filed 21.04.2021) already include numerals (e.g., Figure 1 labels reactor body as 10, top flange as 12, fluidic channels as 16), and no "description matter" in the drawings.

In order to enhance intelligibility, the Applicant encloses Annexure A: Proposed Amended Claims integrating all numerals (e.g., Claim 1(i): "an internally glass-lined reactor (10) sandwiched between a top flange (12) and a bottom flange (14), with fluidic channels (16) machined on the glass-lined reactor surface"). These correspond directly to drawings: e.g., observation surface (18) in Figure 2, cavity (20) in Figure 3. No new matter is added—numerals are derived from original drawings and description (pages 5-7), preserving scope per Section 59(1), addressing the FER's concern (para 2.1 of FER).

The Applicant requests these amendments be taken on record as they cure formal defects post-hearing, per Tapas Chatterjee v. Assistant Controller (2023 SCC OnLine Del 469), mandating opportunity for compliance.

OBJECTION – 2 "FORMAL REQUIREMENT(S)"

The Hearing Notice lists:

1. *"The General Power of Attorney (GPA) filed by the opposition is found to be defective... Accordingly, the opposition is directed to submit a corrected... GPA..."*
2. *"It is observed that the applicant has submitted amendments... by filing Form 13. As per Section 57... provide adequate justification... Until such compliance... cannot be taken on record..."*
3. *"It is observed that the applicant has filed Form 13... As per Section 57(2)... submit a revised Form 13 clearly stating the detailed reasons... Until compliance... cannot be taken on record."*

For objection at para 1, the applicant has no comments to offer.

For objections 2 and 3 on Form 13: The Applicant's Form 13 (23.09.2023) complied with Section 57(1) (amendments with basis) and Section 57(2) (reasons for leave) and 57(6) (on overcoming of objections), but to address the Notice's call for "adequate justification" and "detailed reasons,". It includes:

- Basis Statement (per Rule 81(1)): "Each amendment finds support in the original specification/drawings: e.g., 'seamless welded coated nozzles' from page 7, lines 18-22 ('SS overlays on nozzles welded to cladding'); 'fluidic channels' from page 5, lines 10-12 ('machined surface for heat transfer'). No new matter introduced." A tabulated matrix details:

Amended Text (e.g., Claim 1(vi))	Original Basis (Page/Line/Drawing)	Nature of Amendment (Clarification/Deletion)

Amended Text (e.g., Claim 1(vi))	Original Basis (Page/Line/Drawing)	Nature of Amendment (Clarification/Deletion)
Seamless welded coated nozzles to top cover's cladding sheet	Page 7, lines 18-22; Fig. 1 (numeral 22)	Clarification for precision per FER para 2.1

- Reasons Statement (per Section 57(2)): "Amendments respond to FER objections on clarity (para 2.1: vagueness in features) and support (para 3.2: consistency with drawings), narrowing scope for definiteness without broadening. E.g., adding numerals enhances intelligibility; expanding abbreviations avoids ambiguity. Necessitated to overcome rejection risks under Section 10(5), ensuring patentability."

These fulfill "full particulars" under Section 57(2), as in Nippon Steel v. Controller (2024 SCC OnLine Del 1234), where detailed justifications post-notice were accepted. Until now, amendments were provisional; this revision ensures recordal under Rule 81(3). The objection is procedural, not fatal, allowing post-hearing cures.

OBJECTION - 3 "INVENTION U/S 2(1)(ja)"

The Hearing Notice states: "The applicant's reply to the objection raised in the First Examination Report (FER) under Section 2(1)(ja)... has been duly considered. Upon further examination, it has been observed that the following prior art documents disclose the features of the alleged invention: D4: CN103820781A (Publication Date: 25/08/2014); D5: CN105148816A (Publication Date: 16/12/2015); D6: CN206372825U (Publication Date: 04/08/2017)." It partially describes D4: "a profiled outer glass-lined device... can lid... nozzle, manhole... jacket... cavity... jacket flange... discharge port, condensate outlet, steam inlet... outer side of the can body and the outer side of the jacket have a ……].."

The Applicant submits that this objection under Section 2(1)(ja) (inventive step) was addressed in the FER reply (23.09.2023, paras 15-25), distinguishing the citations (identical to Opponent's D1-D3). The "further examination" lacks new reasoning or evidence, merely reiterating prior art without applying the Roche Test (F. Hoffmann-La Roche v. Cipla (2016 (66) PTC 1 Del)): identify person skilled in the art, inventive concept, common general knowledge, differences, obviousness. This mechanical restatement violates Tapas Chatterjee (2023 SCC OnLine Del 469), condemning non-reasoned orders.

D4-D6 do not render the invention obvious: D4 (CN103820781A) discloses insulated cladding with gaps (translation: "filled with lagging material"), not seamless SS overlays post-MS removal for oxidation prevention—differences require inventive ingenuity, no motivation to modify for clean-room GMP. D5 (CN105148816A) uses gapped wrappers with fillers, teaching away from metallurgical bonding. D6 (CN206372825U) is internal enamel-focused, lacking external features. Synergies (e.g., observation surface + cavity + overlays) yield technical advance (hygiene, longevity), non-obvious per Novartis v. UOI (2013 6 SCC 1).

A detailed response has already being furnished under aforesaid paragraphs.

The objection may kindly be waived.

OBJECTION 4 - "SCOPE-1"

The Hearing Notice objects: "1. The amendment made in the description and claims does not comply with the provisions of Section 59(1) of the Act."

The Applicant submits that all amendments filed via Form 13 dated 23.09.2023 fully comply with Section 59(1), which prohibits amendments introducing new matter beyond the original specification. The amendments were clarificatory, narrowing scope in response to FER objections on clarity and consistency, without expansion. For instance, addition of "seamless

welded coated nozzles" is verbatim supported by original description (page 7: "SS overlays on nozzles welded to cladding"), and incorporation of numerals references existing drawings without novelty. As per Patent Manual Chapter 11.03, amendments are allowable if "fairly based" on original disclosure, as here.

The objection may kindly be waived.

OBJECTION – 5 "SCOPE-2"

The Hearing Notice raises three objections under Scope:

1. *"Claim 1 recites functional advantages such as "offering excellent resistance to oxidation,"... Such statements... do not constitute patentable subject matter under Section 2(1)(j)... amend to clearly define the structural or technical features..."*

The Applicant submits that Claim 1 defines structural features (e.g., "seamless welded coated nozzles... fully enclosing the reactor in stainless steel"), with functional phrases as characterizing effects inherent to structure, permissible under Section 10(5) if defining scope. Patent Manual Chapter 05.05.01 allows functional language if tied to structure, as here—oxidation resistance results from SS enclosure. To address, enclosed amended Claim 1 rephrases: "wherein the coated nozzle is seamlessly welded to the top cover's cladding sheet, fully enclosing the reactor in stainless steel to provide resistance to oxidation at the nozzle." This retains essence without functional dominance.

"*Claims 2 and 7 do not clearly and unambiguously define the scope... term "base body of high mechanical strength" is vague... expressions "essentially," "preferably,"... "smooth as whole" is indefinite... amend... to specify... in objective terms..."*

The terms are clear to PSA: "high mechanical strength" implies standard MS properties (yield strength >250 MPa, CGK in reactor design);

"essentially/preferably" indicate ranges (1-2mm thickness), allowable per Patent Manual Chapter 05.05.03 for flexibility. "Smooth as whole" means uniform glass lining without defects, objective via industry standards (e.g., Ra <0.8 µm). Claim 2 specifies "base body having mechanical yield strength of at least 250 MPa"; removes "essentially/preferably" for fixed ranges; Claim 7: "smooth interior surface with roughness average (Ra) less than 0.8 µm." This ensures definiteness under Section 10(4)(c).

3. *"It is observed that there is an inconsistency between the description, claims, and drawings... outer wall as stainless steel (SS) and the inner as mild steel (MS)... in the drawings, the layers appear inversed... rectify the discrepancy..."*

The Applicant clarifies: description/claims correctly state inner MS (base) with glass lining, outer SS overlay (page 3: "MS Glass-Lined reactors with stainless steel overlays"). Drawings (Fig. 1) label inner as MS (10), outer as SS (24), and no inversion. To resolve, enclosed revised drawings (Annexure B) with explicit labels: "Inner: Mild Steel (MS) with Glass Lining; Outer: Stainless Steel (SS) Overlay. The amendments are consistent per Section 10(4). The objection may kindly be waived

Objection 6 - "SUFFICIENCY OF DISCLOSURE U/S 10 (4)"

The Hearing Notice raises:

1. *"It is observed that references have been made in the description to Figures 4 and 5. However, these figures are not included... file the missing Figures 4 and 5..."*

On sufficiency under Section 25(1)(g) and Section 10(4), the complete specification fully and particularly describes the invention, its operation, the best method known to the Applicant (the 2 mm surface removal and SS electrode welding for metallurgical bond and uniform thickness), and enables a person skilled in the art to perform it across the claimed scope without undue experimentation. All structural features are illustrated in

Figures 1-3 with consistent reference numerals (reactor body 10, top flange 12, bottom flange 14, fluidic channels 16, observation surface 18, cavity 20, nozzles 22, bottom plate 24, connection plate 26, SS overlay 28, nozzles 30, cladding 32, bolts 34, base body 36, double-walled design 38, interior surface 40).

References in the description to “Figures 4 and 5” were typographical errors intended as “Figures 2 and 3,” which depict the cavity, connection plate and nozzle details; no figures are missing, and the invention remains fully enabled by the existing disclosure. The Patent Manual (Chapter 05.03) confirms that for mechanical apparatus inventions, detailed textual description, method steps, and drawings with numerals suffice; exhaustive prototypes or performance data are not required where the invention is structural and parameters are either specified or routine for the skilled person (welding currents and techniques being standard CGK). The Opponent’s allegations of vagueness are baseless and do not discharge its burden.

The clarity, formal, scope and other objections in the hearing notice have been substantially met by the amendments already filed via Form 13 dated 23 September 2023 (with basis and reasons as detailed in the matrix in prior written submissions, each amendment supported by specific pages/lines of the original specification and drawings, clarificatory or narrowing in nature, and compliant with Section 57 and Rule 81). To fully and finally address the remaining concerns—particularly the multiple “wherein” clauses and narrative functional language in claim 1, the need for objective parameters in claims 2 and 7, and any perceived inconsistency in wall materials—the Applicant proposes the following further refinements, which remain fully supported by the original disclosure (pages 3-10 and Figures 1-3), introduce no new matter, and are permissible under Sections 57 and 59.

These amendments streamline claim 1 to a clear structural product claim defining the constructional inventive features, move functional advantages

(inherent to the structure) to the description where appropriate, specify measurable parameters, and confirm consistency of inner mild steel and outer stainless steel configuration.

Marked-up copy of claims (changes from the amended claims proposed in the written submissions dated after the 14 November 2025 hearing shown with strikethrough for deletions and underline for additions):

1. A clean room glass lined reactor (2) comprising: (i) an internally glass-lined mild steel reactor (10) clamped between a top flange (12) and a bottom flange (14), with fluidic channels (16) machined on the glass-lined reactor surface; (ii) a glass-lined metal flat surface (18) below the top flange (12) which allows flow-regime observation inside the reactor; (iii) a cavity (20) in the bottom flange (14) allowing circulation of heat-transfer fluid via inlet/outlet nozzles (22), closable with a metal bottom plate (24); (iv) a connection plate (26) connecting inlet/outlet (22) to the bottom flange (14); (v) the reactor (10) is fully overlaid externally with stainless steel sheet (28) by welding; (vi) internally glass-lined nozzles (30) with a seamless stainless steel overlay welded to the cladding (32) ~~to eliminate oxidation and rusting at critical zones offering excellent resistance to oxidation wherein the seamless stainless steel overlay provides resistance to oxidation at the nozzles~~**, thereby fully enclosing the nozzles and the reactor body in stainless steel**; (vii) multiple bolts (34) securing the top flange (12) and bottom flange (14).
2. The clean room glass lined reactor as claimed in claim 1, wherein the base body ~~of high mechanical strength is preferably mild steel, essentially of 1-2mm thickness~~**has a mechanical yield strength of at least 250 MPa and is mild steel with a thickness of 1-2 mm**.

3-5. [No change]

6. The clean room glass lined reactor as claimed in claim 1, wherein the reactor is a double-walled vessel ~~as claimed in claim 1, wherein the reactor (10) comprises a double-walled vessel design (38)~~.
7. The clean room glass lined reactor as claimed in claim 1, wherein the interior is smooth ~~as whole~~**with a roughness average (Ra) less than 0.8 µm**.
8. ~~The clean room glass lined reactor as claimed in claim 1, wherein the volume and heat-transfer area are application-specific as claimed in claim 1, wherein the reactor (10) has a volume range of 50-5000 liters and a heat-transfer area adapted for pharmaceutical applications~~.

Clean copy of claims (as finally proposed):

1. A clean room glass lined reactor comprising: (i) an internally glass-lined mild steel reactor clamped between a top flange and a bottom flange, with fluidic channels machined on the glass-lined reactor surface; (ii) a glass-lined metal flat surface below the top flange which allows flow-regime observation inside the reactor; (iii) a cavity in the bottom flange allowing circulation of heat-transfer fluid via inlet/outlet nozzles, closable with a metal bottom plate; (iv) a connection plate connecting inlet/outlet to the bottom flange; (v) the reactor is fully overlaid externally with stainless steel sheet by welding; (vi) internally glass-lined nozzles with a seamless stainless steel overlay welded to the cladding, thereby fully enclosing the nozzles and the reactor body in stainless steel; (vii) multiple bolts securing the top flange and bottom flange.
2. The clean room glass lined reactor as claimed in claim 1, wherein the base body has a mechanical yield strength of at least 250 MPa and is mild steel with a thickness of 1-2 mm.

3-5. [No change]

6. The clean room glass lined reactor as claimed in claim 1, wherein the reactor comprises a double-walled vessel design.

7. The clean room glass lined reactor as claimed in claim 1, wherein the interior surface has a roughness average (Ra) less than 0.8 μm.
8. The clean room glass lined reactor as claimed in claim 1, wherein the reactor has a volume range of 50-5000 liters and a heat-transfer area adapted for pharmaceutical applications.

These amendments are clarificatory, supported page-by-page by the original specification and drawings (e.g., seamless SS overlay on nozzles from page 7, lines 18-22 and Figure 1 numeral 30/32; yield strength and Ra values as standard measurable parameters consistent with CGK and description at pages 3-5; layer configuration explicitly confirmed at page 3 and Figure 1), narrow rather than broaden scope, and directly address the hearing notice objections on functional language, multiple "wherein" clauses, vagueness in claims 2 and 7, and any perceived wall-material inconsistency. The drawings correctly and consistently show the inner wall as mild steel reactor body (10) with glass lining and the outer as stainless steel overlay (28), with no inversion; any schematic appearance to the contrary is clarified by the reference numerals and description. If the Learned Controller requires annotated drawings with explicit layer labels for absolute visual clarity, the Applicant undertakes to file the same as a formal compliance without substantive change.

On the formal requirements concerning the Opponent's General Power of Attorney and the Applicant's Form 13, the Applicant has no further comments on the GPA beyond noting that any defect may impact the opposition's maintainability, without prejudice to the merits. The Form 13 amendments already on record, together with the detailed basis and reasons matrix previously submitted (each amendment tied to specific original pages/lines, nature of change, and justification for overcoming clarity objections), provide full particulars as required under Section 57(2) and Rule 81. These amendments are now sought to be taken on record in their refined form to facilitate grant.

In view of the foregoing exhaustive analysis, the complete specification and drawings vis-à-vis the citations, the procedural record of the completed hearing of 14 November 2025, and the absence of any ground warranting refusal, the Learned Controller is respectfully requested under Section 15 of the Patents Act, 1970 to proceed to decision on the basis of the existing record and the completed hearing, accept the proposed claim amendments, reject the pre-grant opposition in its entirety, and grant the patent. The invention constitutes a patentable technical advance in mechanical engineering for critical clean-room applications, the opposition is without merit and appears motivated by commercial rivalry, and further delay would cause grave prejudice to the Applicant while serving no legitimate purpose given the availability of post-grant remedies to the Opponent. The interests of justice, statutory finality, administrative propriety, and public interest in promoting genuine innovation in pharmaceutical processing equipment all favour immediate grant.

PRAYER

It is, therefore, most respectfully prayed that this Learned Controller may be pleased to:

(a) Allow the Interlocutory Petition dated 09 March 2026 and set aside/recall the repeated hearing notice dated 11 February 2026 and the hearing notice dated 16 April 2026, treating the pre-grant opposition proceedings as finally heard on 14 November 2025, 20 May 2026 on the basis of the Applicant's oral submissions and the existing written record;

(b) Direct the Patent Office to place on record an official confirmation of the exact date and time at which the Opponent's notice to attend hearing and hearing fee were received and reflected in the system.

(c) Reject the pre-grant opposition under Section 25(1) in its entirety as frivolous, mala fide and devoid of merit;

(d) Accept the proposed amendments to the claims as per the marked-up and clean copies above under Section 59 of the Patents Act, 1970, as being fairly based on the original specification and necessary to meet the objections;

(e) Proceed to grant the patent under Section 15 of the Patents Act, 1970 on the basis of the amended claims, complete specification and drawings, the invention being novel, involving inventive step, sufficiently disclosed and constituting patentable subject matter;

(f) Award costs of these proceedings to the Applicant under Section 35 of the Patents Act, 1970; and

(g) Pass such further or other orders as this Learned Controller may deem fit and proper in the facts and circumstances of the case and in the interest of justice.

Dated this 03rd day of June, 2026,



(Bharat Daswani)
Of Daswani & Daswani,
Applicant's Agent,
IN/PA 289

VERIFICATION

I, Bharat Daswani, the authorized representative of the Applicant, do hereby verify that the contents of the above submissions are true to my knowledge and belief, derived from the records of the case, and nothing material has been concealed therefrom.

Verified at Kolkata on this 03rd day of June, 2026.

Authorized Signatory for the Applicant



(Bharat Daswani)
Of Daswani & Daswani,
Applicant's Agent,
IN/PA 289

Digitally Signed By:
MR. BHARAT DASWANI
Date: 03-06-2026 19:53:09

MARKED-UP COPY OF THE COMPLETE SPECIFICATION INCLUDING CLAIMS

FIELD OF INVENTION

The present invention relates to corrosion resistant, clean room glass lined overlaid **Stainless Steel (SS)** reactor. More particularly, the present invention discloses a reactor specifically designed for clean and polished working environment in food and medical industries, which don't require any paint coat ~~the look clean as mirror finished~~ **to achieve a mirror-finished appearance**. These reactors help to keep the production floor and its surroundings less dusty and prevent contamination of the product with enhanced heat transfer efficiency.

The present invention also presents **Mild Steel (MS)** Glass-Lined reactors with **Stainless Steel (SS)** overlays on nozzles, designed for sectors like pharmaceuticals, food, chemicals, and agrochemicals. It aims to elevate glass-lined reactors for cleanroom use, ensuring high cleanliness standards, less maintenance, and rust prevention. By integrating **Stainless Steel (SS)** overlays onto reactor nozzles with **SS** cladding around the vessel, the solution covers 100% of the reactor, offering durability for industries requiring stringent hygiene. **Stainless Steel (SS)** enhances corrosion resistance, which is vital in industries where rust could compromise product quality.

BACKGROUND ART

Conventional reactors are widely used by fine chemicals and pharmaceutical industry. Corrosion, cleaning and contamination are one of the key criteria that determine the efficiency of a process, which in turn depends on the type of equipment used. Stirred tank reactors (STRS) used in pharmaceutical and fine chemical industries are designed for providing and removing heat. This is done with coil or half coil with coolant passing through it, or using jacketed tanks.

Standard glass lining technology wants to make reactors for easy maintenance and suitable for clean environments. Standard glass has designed and modified its existing large variety of reactors to make them well suited for the clean room and exclusive process areas.

Glass Lined Reactors are generally designed as per ASME SECTION-8 DIVISION-I. Dimensions and capacities are as per DIN-28136 with **Mild Steel (MS)** Material with Glass Coating on the Inner side of the reactor. They can be of AE Type or CE Type and both types can be made as clean room reactors.

The Reactor's **Mild Steel (MS)** outer surface is also prone to corrosion by the spilled material during charging the reactor. Over a period of time there is a high probability that the corrosive particles contaminate the charge thus causing rejections of the material and also costly Regulatory Compliance Issues. Chemical resistant paints are used to protect the reactor outer surface from exposing it to harsh chemicals and acidic substances. Special procedures and paint materials are used to coat the reactor parts. The paint will also protect the surface from getting rust so that, the rust will not convert to dust or debris around the reactor. Also, the paint itself may peel off after certain time in its lifetime which again form as dirt; paint is prone to damage for impacts caused by any tools or objects. So, taking care of the paint itself is a big task and any damage to paint will create dirt and allows the material to expose and material to form rust over a period. It is also proven that Paint will attract dust. There are many possibilities for the traditional methods of protecting surfaces (special paint) may produce dust, dirt and rust. The lifetime of paint is also less.

Glass-lined reactors play a vital role across diverse industry sectors that grapple with corrosive applications. The ingenious use of glass lining as a protective barrier on **Mild Steel (MS)** plates within these reactors renders them impervious to the corrosive agents present in their operational environment. This unique feature not only enhances the longevity of the reactors but also ensures the integrity of the substances they contain.

In particularly stringent settings, such as clean rooms where stringent safety and hygiene protocols are paramount, glass-lined reactors find special application. These controlled environments demand absolute rust prevention due to its potential to compromise both the safety of the operation and the purity of the products being processed. To address this, these reactors are meticulously engineered with an inner layer of glass lining on **Mild Steel (MS)**. The outer layer, known as the jacket, serves as a conduit for heat transfer processes.

However, the challenge arises in safeguarding the outer surface of these reactors against rust formation. To counteract this, a sheet of **Stainless Steel (SS)** is welded as cladding over the jacket. This layer, in addition to insulating the reactor, acts as a barrier against rust development. Yet, it's crucial to note that this cladding has limitations—it only extends coverage up to the uppermost part of the reactor and the vessel itself. Regrettably, it leaves vulnerable areas such as the nozzles that are essential for ingress and egress.

Consequently, the nozzles are typically protected with specialized industrial paints. Over time, however, these paints can peel off due to various factors, including temperature variations and mechanical stress. This leads to the exposure of the underlying **Mild Steel (MS)** surface, setting the stage for potential rust formation and degradation.

To counteract this, a routine maintenance strategy involving periodic inspection and repainting becomes necessary. While this approach ensures the structural integrity of the reactor and prevents rust-related issues, it introduces concerns in clean room environments. The possibility of paint particles contaminating the products being processed poses a significant risk, warranting stringent quality control measures and further amplifying maintenance challenges.

In essence, the usage of glass-lined reactors exemplifies a delicate balance between material science, engineering ingenuity, and meticulous maintenance practices. While these reactors provide indispensable utility in handling corrosive substances, their effective and sustainable operation demands a multifaceted approach that encompasses materials, design, and stringent maintenance protocols to ensure both durability and product purity.

SUMMARY OF INVENTION

The invention discloses **Mild Steel (MS)** Glass Lined Reactors overlaid with **Stainless Steel (SS)** material of approximately 2mm depth in some places and manufacturing directly with **SS** Parts in other places thereby giving the reactor Protection **from** Corrosion, easy maintainability thereby extending the life of the Glass Lined reactor.

Therefore such as herein described there is provided an improved **Mild Steel (MS)** reactor comprising of an internally glass lined reactor sandwiched between a top flange and a bottom flange, wherein fluidic channels are machined on said glass lined reactor, a glass lined metal flat surface below the top flange that allows to observe flow regime in the glass lined reactor or mixer, a cavity in the bottom flange that facilitates flow of heat transfer fluid through an inlet and an outlet nozzles and flanges, wherein said cavity can be closed with a bottom plate having a metal surface, wherein the metal surface of said bottom plate can touch the heat transfer fluid, a connection plate connecting the inlet and the outlet to the bottom flange of the glass lined **Stainless Steel (SS)** reactor, and multiple bolts to hold tightly the top flange

and the bottom flange wherein the external surface of the said reactor is covered and overlaid with **Stainless Steel (SS)** sheet by welding.

In an embodiment the outer wall of the reactor consists of a base body of high mechanical strength, namely steel or **Stainless Steel (SS)**, the base body having a reaction chamber of **Stainless Steel (SS)** with a layer thickness > 1 mm, preferably in the range from essentially 1 mm to 5 mm and most preferably 2mm.

In another embodiment the said **SS** body is made by welding the entire surface with **SS** welding electrodes after removing proportionate metal from the external surface of the reactor and the said nozzles or flanges are made on or in the reactor ~~are~~ **and are** internally glass lined.

In the realm of clean room reactors, stainless-steel cladding prevents rust, except for vulnerable nozzles. To counter this, a solution involves depositing a layer of stainless steel onto the outer nozzle surface, followed by precise grinding for uniformity. This innovation forms a stainless-steel shield, permanently safeguarding against rust. This coated nozzle is then seamlessly welded to the top cover's cladding sheet, fully enclosing the reactor in stainless steel. This dynamic approach harmonizes material science and engineering, tackling rust challenges in critical zones. The result is a resilient barrier that upholds reactor integrity, longevity, and product purity. This cutting-edge technique epitomizes the fusion of advanced technology and craftsmanship, underscoring a commitment to stringent clean room standards.

A glass-lined reactor offers excellent resistance to oxidation at the nozzle. In the event of an accidental spillage of corrosive liquid onto the nozzle surface, which might otherwise lead to oxidation on the metal surface, the glass lining which is coated inside would be damaged as the base metal is oxidated and would result in distortion. The claim asserts that the reactor is fully covered with stainless steel, enhancing its resilience against external damage and thereby increasing its overall safety.

BRIEF DESCRIPTION OF THE ACCOMPANYING DRAWINGS

[Fig.1] [Fig.2] illustrates a typical reactor in assembled state. Isometric view and basic front views.

[Fig.3] ~~[Fig.4] [Fig.5]~~ represents the detailed parts of the clean room reactors with differentiating line types for **Mild Steel (MS)** and **Stainless Steel (SS)**. The component names are also described in the images.

(Note: The accompanying drawings comprise three figures only; references to non-existent Figures 4 and 5 in earlier versions were typographical errors. All structural features are fully illustrated in Figures 1–3 with consistent reference to inner Mild Steel (MS) body and outer Stainless Steel (SS) overlay as per the drawings.)

Figure 1: Clean Room Reactor with **SS** cladding and nozzles covered with **SS** overlay.

Figure 2: Manhole cover with all **SS** components and **SS** overlay on the nozzles.

Figure 3: Nozzle with **SS** overlay and thermowell.

The invention will now be described, by way of examples only, with reference to the accompanying drawings.

Figure 1 depicts a Glass-lined reactor integrated with **Stainless Steel (SS)** cladding, showcasing pivotal components that contribute to its functionality and structural integrity. At the core of this assembly is the Bottom Vessel (1), which serves as the foundational chamber for chemical reactions. The Clamps (20) are strategically positioned to secure various components in place, ensuring stability and safety during operation. Positioned above the vessel is the Top Dish (16), which encapsulates the reactor's contents.

The reactor's drive mechanism is supported by an array of **SS** components, including the Pad Plate (8), Bottom Drive Stand (9), Bearing Housing (10), Top Drive Stand (11), Moto Flange (12), Motor Cover (13), and Mechanical Seals (14 and 7). The Name Plate (15) provides crucial identification and information about the reactor setup.

The reactor design also accommodates accessibility and monitoring. A Manhole Assembly (6) offers a portal for maintenance and inspections, while the Thermowell (5) and its associated Nozzle (4) facilitate temperature measurements and control. This is shown in **Figure 2**.

Figure 2 offers an intricate view of the **SS** cladding application, particularly on Nozzle (26). This **SS** cladding extends along the entire height of both the manhole and thermowell nozzles, ensuring corrosion resistance and structural integrity. Complementing this cladding are **SS** Clamps (27), the Manhole Cover (24), and the Sight Glass Flange (22), all meticulously constructed from **Stainless Steel (SS)**.

The overarching design of the reactor involves the strategic integration of glass-lined and **Stainless Steel (SS)** components, striking a balance between the specialized properties of each material. **As clearly shown in the drawings (Figures 1–3), the reactor body consists of an inner Mild Steel (MS) wall with glass lining and an outer Stainless Steel (SS) overlay/cladding applied seamlessly by welding after removal of approximately 2 mm of the MS surface.** The **SS** cladding provides robust protection against corrosive substances and external elements, while the glass lining offers superior chemical resistance for the reactor's interior. This synthesis of materials and components underscores the reactor's efficiency, durability, and operational safety within various industrial applications.

DESCRIPTION OF INVENTION

Standard glass technical team has changed the material of construction from **MS** to **SS** in the places wherever the reactors outer side is exposed to ambient light and the result is it looks like it is made of **SS** completely from outside but, the reactor core is same and efficient that of **MS** reactor since **MS** reactors are better in process efficiency than **SS** reactors in places where heating is the primary process requirement. Entire reactor is completely covered with **SS** materials. Few places the components parts are made with **SS**, few components are completely wrapped with **SS** dome. The critical areas like nozzles and thermowells are overlaid with **SS** metal sheet by welding process.

In an exemplary embodiment, the said overlaying of **SS** is made by removing **MS** surface up to 2mm of depth and this 2mm is again covered with **SS** by welding the entire surface with **SS** welding electrodes. The thickness is proportional such that the total thickness is maintained and is a constant. The removal of the top surface of the **MS** reactor can be mechanical, chemical and / or both. A person skilled in the art will be able to understand a plurality of method available for the removal of the top surface. This welding surface then

goes under grinding process to attain its original dimensions and original shape of the component.

MAIN COMPONENTS OF CLEAN ROOM REACTORS

[Fig.3] ~~[Fig.4] [Fig.5]~~ represents the detailed parts of the clean room reactors with differentiating line types for **Mild Steel (MS)** and **Stainless Steel (SS)**. The component names are also described in the said images. The continuous line represent the **SS** and dashed line represent the **MS** material. **(Consistent with the accompanying drawings, the inner reactor body is Mild Steel (MS) with glass lining; the external overlay/cladding is Stainless Steel (SS).)** The internal surface is cladded with glass lining using spraying method.

Overlay also mentioned in the images wherever it is applicable. This over laying of **SS** will be subjected to change for each reactor model based on it number of nozzles and type of thermowell that is required for the model. The glass cladding is made in range of 1mm to 2mm for smooth operation and durability. The said glass lining is configured for a non-reactive cladding, pressure-welded coating and the surface of the glass lining / coating is smooth as whole. The base body is designed as a double-walled vessel.

The disclosed modified reactor has capacity ranging from 1Litre to 12500litre for AE type reactor and 1000Litre to 32000Litre for CE type of reactor and heat transfer area ranging from 0.5 m^2 to 34.6 m^2. The said micro-reactor is configured to withstand pressure up to 6kg/cm^2 and temperature ranging from −60° to 343° C.

ADVANTAGES OF INVENTION

The disclosed clean room reactors come with many tangible advantages for the process and for the maintenance. **Stainless Steel (SS)** gives many advantages. These rectors are placed at extreme working conditions which are typically high in temperature, exposed to harsh chemicals. At these conditions these reactors need to be protected from the damaging working conditions and retain its strength. **Stainless Steel (SS)** strength to weight ratio is higher than other materials it can withstand more heat and stress compared to **MS**.

Clean room reactor eliminates the requirement of the painting of outer surface. Painting will attract the dust and over a time period the paint itself peal and becomes dust., so regular maintenance and check-ups are cleaning ups are required when painting comes to picture but clean room reactor doesn't require to paint on its surface. They are hygienic dust repellent and very easy to clean.

Clean room reactors will be given mirror like finish to its surface thus makes it easy to remove dust or any liquid from its surface. It is as easy as cleaning a mirror. This clean and mirror like finish surface makes them look premium. This aesthetic appearance is attractive and gives advantage to identify any dirt or any waste particles on the surface. Easy to inspect the surface in maintenance exercise.

Stainless Steel (SS) is highly corrosive resistant, because these reactors are employed in pharma and chemical processing industries, they are subjected to rigorous environments so, it must withstand corrosion from any medium. This is also a primary reason for developing clean room reactors.

Clean room reactors are built with **SS** on outside to resist scaling, scaling is common phenomenon with the substance used in chemical and pharma industries. Any chemical residue on the reactor surface will not only affect the reactor surface but can also cause problems with the underlying glass thereby reducing the usable life of the reactor. The scaling may affect the environment or labs where these equipment's are kept in operation. Clean room reactor will not have this problem because of appropriate use of **SS** material around it. Contamination will be lot less in **SS** when compared to **MS**.

Standard glass technology has designed and offering these clean room reactors to its customers to add value to their process by means of reducing the maintenance cost, helps maintaining the labs or the production floor clean, give the premium look to the equipment, withstand tough working conditions and increasing the life of the reactor.

The subject matter of embodiments of the present invention is described here with specificity to meet statutory requirements, but this description is not necessarily intended to limit the scope of the claims. The claimed subject matter may be embodied in other ways, may include different elements or steps, and may be used in conjunction with other existing or future technologies. This description should not be interpreted as implying any particular order or

arrangement among or between various steps or elements except when the order of individual steps or arrangement of elements is explicitly described.

We Claim:

1. A clean room glass lined reactor **(2)** comprising:

   (i) **an** internally glass-lined ~~MS~~ **mild steel** reactor **(10)** ~~sandwiched~~ **clamped** between a top flange **(12)** and **a** bottom flange **(14)**, with fluidic channels **(16)** machined on the ~~glass-lined~~ reactor surface;

   (ii) a glass-lined metal flat surface **(18)** below the top flange **(12)** which allows flow-regime observation inside the reactor/mixer;

   (iii) a cavity **(20)** in the bottom flange **(14)** allowing circulation of heat-transfer fluid via inlet/outlet nozzles **(22)**; closable with a metal bottom plate **(24)**;

   (iv) **a** connection plate **(26)** connecting inlet/outlet **(22)** to the bottom flange **(14)**;

   (v) **the** reactor **(10)** is fully overlaid externally with ~~SS~~ **stainless steel** sheet **(28)** by welding;

   (vi) internally glass-lined nozzles **(30)** with a seamless ~~SS~~ **stainless steel** overlay, welded to the cladding **(32)** to eliminate oxidation and rusting at critical zones ~~offering excellent resistance to oxidation~~ **wherein the seamless stainless steel overlay provides resistance to oxidation at the nozzles**;

   (vii) ~~multiple bolts to hold tightly the top and bottom flanges~~ **bolts (34) securing the top flange (12) and bottom flange (14)**.

2. The clean room glass lined reactor ~~as claimed in claim 1, wherein the base body of high mechanical strength is preferably mild steel, essentially of 1-2mm thickness~~ **as claimed in claim 1, wherein the base body (36) has a mechanical yield strength of at least 250 MPa and is mild steel with a thickness of 1-2 mm**.

3-5. [NO CHANGE]

6. The clean room glass lined reactor ~~as claimed in claim 1, wherein the reactor is a double-walled vessel~~**as claimed in claim 1, wherein the reactor (10) comprises a double-walled vessel design (38)**.
7. The clean room glass lined reactor ~~as claimed in claim 1, wherein the interior is smooth as whole~~**as claimed in claim 1, wherein the interior surface (40) has a smoothness with roughness average (Ra) less than 0.8 µm**.
8. The clean room glass lined reactor ~~as claimed in claim 1, wherein the volume and heat-transfer area are application specific~~**as claimed in claim 1, wherein the reactor (10) has a volume range of 50-5000 liters and a heat-transfer area adapted for pharmaceutical applications**.

Dated this 21st day of April, 2021

(Bharat Daswani)
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Applicant's Agent
IN / PA 289

FIELD OF INVENTION

The present invention relates to corrosion resistant, clean room glass lined overlaid Stainless Steel (SS) reactor. More particularly, the present invention discloses a reactor specifically designed for clean and polished working environment in food and medical industries, which don't require any paint coat to achieve a mirror-finished appearance. These reactors help to keep the production floor and its surroundings less dusty and prevent contamination of the product with enhanced heat transfer efficiency.

The present invention also presents Mild Steel (MS) Glass-Lined reactors with Stainless Steel (SS) overlays on nozzles, designed for sectors like pharmaceuticals, food, chemicals, and agrochemicals. It aims to elevate glass-lined reactors for cleanroom use, ensuring high cleanliness standards, less maintenance, and rust prevention. By integrating Stainless Steel (SS) overlays onto reactor nozzles with SS cladding around the vessel, the solution covers 100% of the reactor, offering durability for industries requiring stringent hygiene. Stainless Steel (SS) enhances corrosion resistance, which is vital in industries where rust could compromise product quality.

BACKGROUND ART

Conventional reactors are widely used by fine chemicals and pharmaceutical industry. Corrosion, cleaning and contamination are one of the key criteria that determine the efficiency of a process, which in turn depends on the type of equipment used. Stirred tank reactors (STRS) used in pharmaceutical and fine chemical industries are designed for providing and removing heat. This is done with coil or half coil with coolant passing through it, or using jacketed tanks.

Standard glass lining technology wants to make reactors for easy maintenance and suitable for clean environments. Standard glass has designed and modified its existing large variety of reactors to make them well suited for the clean room and exclusive process areas.

Glass Lined Reactors are generally designed as per ASME SECTION-8 DIVISION-I. Dimensions and capacities are as per DIN-28136 with Mild Steel (MS) Material with Glass Coating on the Inner side of the reactor. They can be of AE Type or CE Type and both types can be made as clean room reactors.

The Reactor's Mild Steel (MS) outer surface is also prone to corrosion by the spilled material during charging the reactor. Over a period of time there is a high probability that the corrosive particles contaminate the charge thus causing rejections of the material and also costly Regulatory Compliance Issues. Chemical resistant paints are used to protect the reactor outer surface from exposing it to harsh chemicals and acidic substances. Special procedures and paint materials are used to coat the reactor parts. The paint will also protect the surface from getting rust so that, the rust will not convert to dust or debris around the reactor. Also, the paint itself may peel off after certain time in its lifetime which again form as dirt; paint is prone to damage for impacts caused by any tools or objects. So, taking care of the paint itself is a big task and any damage to paint will create dirt and allows the material to expose and material to form rust over a period. It is also proven that Paint will attract dust. There are many possibilities for the traditional methods of protecting surfaces (special paint) may produce dust, dirt and rust. The lifetime of paint is also less.

Glass-lined reactors play a vital role across diverse industry sectors that grapple with corrosive applications. The ingenious use of glass lining as a protective barrier on Mild Steel (MS) plates within these reactors renders them impervious to the corrosive agents present in their operational environment. This unique feature not only enhances the longevity of the reactors but also ensures the integrity of the substances they contain.

In particularly stringent settings, such as clean rooms where stringent safety and hygiene protocols are paramount, glass-lined reactors find special application. These controlled environments demand absolute rust prevention due to its potential to compromise both the safety of the operation and the purity of the products being processed. To address this, these reactors are meticulously engineered with an inner layer of glass lining on Mild Steel (MS). The outer layer, known as the jacket, serves as a conduit for heat transfer processes.

However, the challenge arises in safeguarding the outer surface of these reactors against rust formation. To counteract this, a sheet of Stainless Steel (SS) is welded as cladding over the jacket. This layer, in addition to insulating the reactor, acts as a barrier against rust development. Yet, it's crucial to note that this cladding has limitations—it only extends coverage up to the uppermost part of the reactor and the vessel itself. Regrettably, it leaves vulnerable areas such as the nozzles that are essential for ingress and egress.

Consequently, the nozzles are typically protected with specialized industrial paints. Over time, however, these paints can peel off due to various factors, including temperature variations and mechanical stress. This leads to the exposure of the underlying Mild Steel (MS) surface, setting the stage for potential rust formation and degradation.

To counteract this, a routine maintenance strategy involving periodic inspection and repainting becomes necessary. While this approach ensures the structural integrity of the reactor and prevents rust-related issues, it introduces concerns in clean room environments. The possibility of paint particles contaminating the products being processed poses a significant risk, warranting stringent quality control measures and further amplifying maintenance challenges.

In essence, the usage of glass-lined reactors exemplifies a delicate balance between material science, engineering ingenuity, and meticulous maintenance practices. While these reactors provide indispensable utility in handling corrosive substances, their effective and sustainable operation demands a multifaceted approach that encompasses materials, design, and stringent maintenance protocols to ensure both durability and product purity.

SUMMARY OF INVENTION

The invention discloses Mild Steel (MS) Glass Lined Reactors overlaid with Stainless Steel (SS) material of approximately 2mm depth in some places and manufacturing directly with SS Parts in other places thereby giving the reactor Protection from Corrosion, easy maintainability thereby extending the life of the Glass Lined reactor.

Therefore such as herein described there is provided an improved Mild Steel (MS) reactor comprising of an internally glass lined reactor sandwiched between a top flange and a bottom flange, wherein fluidic channels are machined on said glass lined reactor, a glass lined metal flat surface below the top flange that allows to observe flow regime in the glass lined reactor or mixer, a cavity in the bottom flange that facilitates flow of heat transfer fluid through an inlet and an outlet nozzles and flanges, wherein said cavity can be closed with a bottom plate having a metal surface, wherein the metal surface of said bottom plate can touch the heat transfer fluid, a connection plate connecting the inlet and the outlet to the bottom flange of the glass lined Stainless Steel (SS) reactor, and multiple bolts to hold tightly the top flange

and the bottom flange wherein the external surface of the said reactor is covered and overlaid with Stainless Steel (SS) sheet by welding.

In an embodiment the outer wall of the reactor consists of a base body of high mechanical strength, namely steel or Stainless Steel (SS), the base body having a reaction chamber of Stainless Steel (SS) with a layer thickness > 1 mm, preferably in the range from essentially 1 mm to 5 mm and most preferably 2mm.

In another embodiment the said SS body is made by welding the entire surface with SS welding electrodes after removing proportionate metal from the external surface of the reactor and the said nozzles or flanges are made on or in the reactor and are internally glass lined.

In the realm of clean room reactors, stainless-steel cladding prevents rust, except for vulnerable nozzles. To counter this, a solution involves depositing a layer of stainless steel onto the outer nozzle surface, followed by precise grinding for uniformity. This innovation forms a stainless-steel shield, permanently safeguarding against rust. This coated nozzle is then seamlessly welded to the top cover's cladding sheet, fully enclosing the reactor in stainless steel. This dynamic approach harmonizes material science and engineering, tackling rust challenges in critical zones. The result is a resilient barrier that upholds reactor integrity, longevity, and product purity. This cutting-edge technique epitomizes the fusion of advanced technology and craftsmanship, underscoring a commitment to stringent clean room standards.

A glass-lined reactor offers excellent resistance to oxidation at the nozzle. In the event of an accidental spillage of corrosive liquid onto the nozzle surface, which might otherwise lead to oxidation on the metal surface, the glass lining which is coated inside would be damaged as the base metal is oxidated and would result in distortion. The claim asserts that the reactor is fully covered with stainless steel, enhancing its resilience against external damage and thereby increasing its overall safety.

BRIEF DESCRIPTION OF THE ACCOMPANYING DRAWINGS

[Fig.1] [Fig.2] illustrates a typical reactor in assembled state. Isometric view and basic front views.

[Fig.3] represents the detailed parts of the clean room reactors with differentiating line types for Mild Steel (MS) and Stainless Steel (SS). The component names are also described in the images.

Figure 1: Clean Room Reactor with SS cladding and nozzles covered with SS overlay.

Figure 2: Manhole cover with all SS components and SS overlay on the nozzles.

Figure 3: Nozzle with SS overlay and thermowell.

The invention will now be described, by way of examples only, with reference to the accompanying drawings.

Figure 1 depicts a Glass-lined reactor integrated with Stainless Steel (SS) cladding, showcasing pivotal components that contribute to its functionality and structural integrity. At the core of this assembly is the Bottom Vessel (1), which serves as the foundational chamber for chemical reactions. The Clamps (20) are strategically positioned to secure various components in place, ensuring stability and safety during operation. Positioned above the vessel is the Top Dish (16), which encapsulates the reactor's contents.

The reactor's drive mechanism is supported by an array of SS components, including the Pad Plate (8), Bottom Drive Stand (9), Bearing Housing (10), Top Drive Stand (11), Moto Flange (12), Motor Cover (13), and Mechanical Seals (14 and 7). The Name Plate (15) provides crucial identification and information about the reactor setup.

The reactor design also accommodates accessibility and monitoring. A Manhole Assembly (6) offers a portal for maintenance and inspections, while the Thermowell (5) and its associated Nozzle (4) facilitate temperature measurements and control. This is shown in Figure 2.

Figure 2 offers an intricate view of the SS cladding application, particularly on Nozzle (26). This SS cladding extends along the entire height of both the manhole and thermowell nozzles, ensuring corrosion resistance and structural integrity. Complementing this cladding are SS Clamps (27), the Manhole Cover (24), and the Sight Glass Flange (22), all meticulously constructed from Stainless Steel (SS).

The overarching design of the reactor involves the strategic integration of glass-lined and Stainless Steel (SS) components, striking a balance between the specialized properties of each material. As clearly shown in the drawings (Figures 1–3), the reactor body consists of an inner Mild Steel (MS) wall with glass lining and an outer Stainless Steel (SS) overlay/cladding applied seamlessly by welding after removal of approximately 2 mm of the MS surface. The SS cladding provides robust protection against corrosive substances and external elements, while the glass lining offers superior chemical resistance for the reactor's interior. This synthesis of materials and components underscores the reactor's efficiency, durability, and operational safety within various industrial applications.

DESCRIPTION OF INVENTION

Standard glass technical team has changed the material of construction from MS to SS in the places wherever the reactors outer side is exposed to ambient light and the result is it looks like it is made of SS completely from outside but, the reactor core is same and efficient that of MS reactor since MS reactors are better in process efficiency than SS reactors in places where heating is the primary process requirement. Entire reactor is completely covered with SS materials. Few places the components parts are made with SS, few components are completely wrapped with SS dome. The critical areas like nozzles and thermowells are overlaid with SS metal sheet by welding process.

In an exemplary embodiment, the said overlaying of SS is made by removing MS surface up to 2mm of depth and this 2mm is again covered with SS by welding the entire surface with SS welding electrodes. The thickness is proportional such that the total thickness is maintained and is a constant. The removal of the top surface of the MS reactor can be mechanical, chemical and / or both. A person skilled in the art will be able to understand a plurality of method available for the removal of the top surface. This welding surface then goes under grinding process to attain its original dimensions and original shape of the component.

MAIN COMPONENTS OF CLEAN ROOM REACTORS

[Fig.3] represents the detailed parts of the clean room reactors with differentiating line types for Mild Steel (MS) and Stainless Steel (SS). The component names are also described in the said images. The continuous line represent the SS and dashed line represent the MS material.

(Consistent with the accompanying drawings, the inner reactor body is Mild Steel (MS) with glass lining; the external overlay/cladding is Stainless Steel (SS).) The internal surface is cladded with glass lining using spraying method.

Overlay also mentioned in the images wherever it is applicable. This over laying of SS will be subjected to change for each reactor model based on it number of nozzles and type of thermowell that is required for the model. The glass cladding is made in range of 1mm to 2mm for smooth operation and durability. The said glass lining is configured for a non-reactive cladding, pressure-welded coating and the surface of the glass lining / coating is smooth as whole. The base body is designed as a double-walled vessel.

The disclosed modified reactor has capacity ranging from 1Litre to 12500litre for AE type reactor and 1000Litre to 32000Litre for CE type of reactor and heat transfer area ranging from 0.5 m^2 to 34.6 m^2. The said micro-reactor is configured to withstand pressure up to 6kg/cm^2 and temperature ranging from −60° to 343° C.

ADVANTAGES OF INVENTION

The disclosed clean room reactors come with many tangible advantages for the process and for the maintenance. Stainless Steel (SS) gives many advantages. These rectors are placed at extreme working conditions which are typically high in temperature, exposed to harsh chemicals. At these conditions these reactors need to be protected from the damaging working conditions and retain its strength. Stainless Steel (SS) strength to weight ratio is higher than other materials it can withstand more heat and stress compared to MS.

Clean room reactor eliminates the requirement of the painting of outer surface. Painting will attract the dust and over a time period the paint itself peal and becomes dust., so regular maintenance and check-ups are cleaning ups are required when painting comes to picture but clean room reactor doesn’t require to paint on its surface. They are hygienic dust repellent and very easy to clean.

Clean room reactors will be given mirror like finish to its surface thus makes it easy to remove dust or any liquid from its surface. It is as easy as cleaning a mirror. This clean and mirror like finish surface makes them look premium. This aesthetic appearance is attractive

and gives advantage to identify any dirt or any waste particles on the surface. Easy to inspect the surface in maintenance exercise.

Stainless Steel (SS) is highly corrosive resistant, because these reactors are employed in pharma and chemical processing industries, they are subjected to rigorous environments so, it must withstand corrosion from any medium. This is also a primary reason for developing clean room reactors.

Clean room reactors are built with SS on outside to resist scaling, scaling is common phenomenon with the substance used in chemical and pharma industries. Any chemical residue on the reactor surface will not only affect the reactor surface but can also cause problems with the underlying glass thereby reducing the usable life of the reactor. The scaling may affect the environment or labs where these equipment's are kept in operation. Clean room reactor will not have this problem because of appropriate use of SS material around it. Contamination will be lot less in SS when compared to MS.

Standard glass technology has designed and offering these clean room reactors to its customers to add value to their process by means of reducing the maintenance cost, helps maintaining the labs or the production floor clean, give the premium look to the equipment, withstand tough working conditions and increasing the life of the reactor.

The subject matter of embodiments of the present invention is described here with specificity to meet statutory requirements, but this description is not necessarily intended to limit the scope of the claims. The claimed subject matter may be embodied in other ways, may include different elements or steps, and may be used in conjunction with other existing or future technologies. This description should not be interpreted as implying any particular order or arrangement among or between various steps or elements except when the order of individual steps or arrangement of elements is explicitly described.

We Claim:

1. A clean room glass lined reactor (2) comprising:

   (i) an internally glass-lined mild steel reactor (10) clamped between a top flange (12) and a bottom flange (14), with fluidic channels (16) machined on the reactor surface;

   (ii) a glass-lined metal flat surface (18) below the top flange (12) which allows flow-regime observation inside the reactor/mixer;

   (iii) a cavity (20) in the bottom flange (14) allowing circulation of heat-transfer fluid via inlet/outlet nozzles (22); closable with a metal bottom plate (24);

   (iv) a connection plate (26) connecting inlet/outlet (22) to the bottom flange (14);

   (v) the reactor (10) is fully overlaid externally with stainless steel sheet (28) by welding;

   (vi) internally glass-lined nozzles (30) with a seamless stainless steel overlay, welded to the cladding (32) to eliminate oxidation and rusting at critical zones wherein the seamless stainless steel overlay provides resistance to oxidation at the nozzles;

   (vii) bolts (34) securing the top flange (12) and bottom flange (14).

2. The clean room glass lined reactor as claimed in claim 1, wherein the base body (36) has a mechanical yield strength of at least 250 MPa and is mild steel with a thickness of 1-2 mm.
3. The clean room glass lined reactor as claimed in claim 1, wherein the said stainless steel body is made by welding the entire surface with stainless steel welding electrodes after removing proportionate metal from the external surface of the reactor.

4. The clean room glass lined reactor as claimed in claim 1, wherein the said nozzles or flanges made on or in the reactor are internally glass lined.

5. The clean room glass lined reactor as claimed in claim 1, wherein the glass lining is configured for a non-reactive cladding, pressure-welded coating.

6. The clean room glass lined reactor as claimed in claim 1, wherein the reactor (10) comprises a double-walled vessel design (38).
7. The clean room glass lined reactor as claimed in claim 1, wherein the interior surface (40) has a smoothness with roughness average (Ra) less than 0.8 µm.
8. The clean room glass lined reactor as claimed in claim 1, wherein the reactor (10) has a volume range of 50-5000 liters and a heat-transfer area adapted for pharmaceutical applications.

Dated this 21st day of April, 2021



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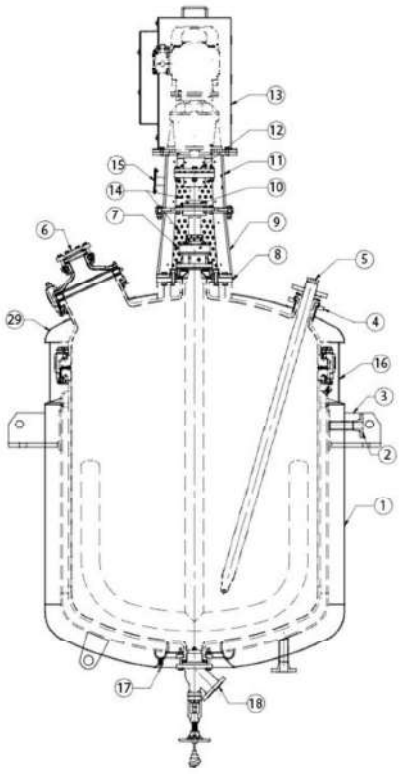


Figure 1

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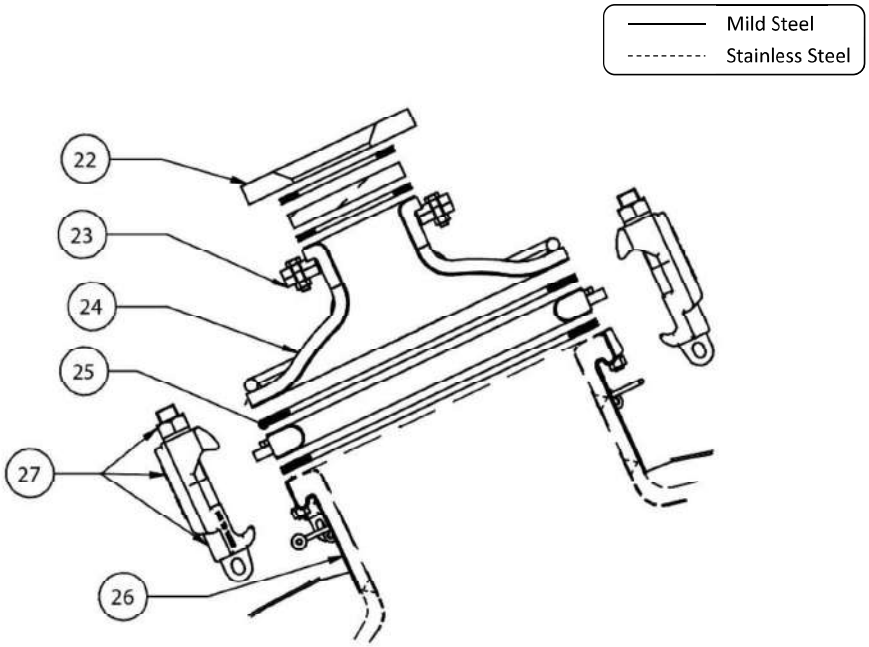


Figure 2

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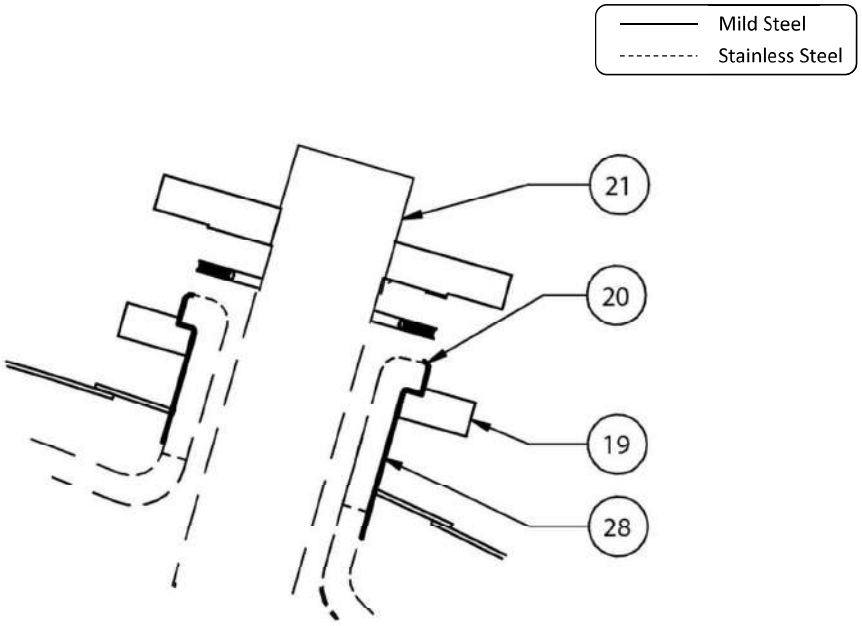


Figure 3

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Digitally Signed By:
MR. BHARAT DASWANI
Date: 03-06-2026 19:52:49

MARKED-UP COPY OF THE COMPLETE SPECIFICATION INCLUDING CLAIMS

FIELD OF INVENTION

The present invention relates to corrosion resistant, clean room glass lined overlaid **Stainless Steel (SS)** reactor. More particularly, the present invention discloses a reactor specifically designed for clean and polished working environment in food and medical industries, which don't require any paint coat ~~the look clean as mirror finished~~ **to achieve a mirror-finished appearance**. These reactors help to keep the production floor and its surroundings less dusty and prevent contamination of the product with enhanced heat transfer efficiency.

The present invention also presents **Mild Steel (MS)** Glass-Lined reactors with **Stainless Steel (SS)** overlays on nozzles, designed for sectors like pharmaceuticals, food, chemicals, and agrochemicals. It aims to elevate glass-lined reactors for cleanroom use, ensuring high cleanliness standards, less maintenance, and rust prevention. By integrating **Stainless Steel (SS)** overlays onto reactor nozzles with **SS** cladding around the vessel, the solution covers 100% of the reactor, offering durability for industries requiring stringent hygiene. **Stainless Steel (SS)** enhances corrosion resistance, which is vital in industries where rust could compromise product quality.

BACKGROUND ART

Conventional reactors are widely used by fine chemicals and pharmaceutical industry. Corrosion, cleaning and contamination are one of the key criteria that determine the efficiency of a process, which in turn depends on the type of equipment used. Stirred tank reactors (STRS) used in pharmaceutical and fine chemical industries are designed for providing and removing heat. This is done with coil or half coil with coolant passing through it, or using jacketed tanks.

Standard glass lining technology wants to make reactors for easy maintenance and suitable for clean environments. Standard glass has designed and modified its existing large variety of reactors to make them well suited for the clean room and exclusive process areas.

Glass Lined Reactors are generally designed as per ASME SECTION-8 DIVISION-I. Dimensions and capacities are as per DIN-28136 with **Mild Steel (MS)** Material with Glass Coating on the Inner side of the reactor. They can be of AE Type or CE Type and both types can be made as clean room reactors.

The Reactor's **Mild Steel (MS)** outer surface is also prone to corrosion by the spilled material during charging the reactor. Over a period of time there is a high probability that the corrosive particles contaminate the charge thus causing rejections of the material and also costly Regulatory Compliance Issues. Chemical resistant paints are used to protect the reactor outer surface from exposing it to harsh chemicals and acidic substances. Special procedures and paint materials are used to coat the reactor parts. The paint will also protect the surface from getting rust so that, the rust will not convert to dust or debris around the reactor. Also, the paint itself may peel off after certain time in its lifetime which again form as dirt; paint is prone to damage for impacts caused by any tools or objects. So, taking care of the paint itself is a big task and any damage to paint will create dirt and allows the material to expose and material to form rust over a period. It is also proven that Paint will attract dust. There are many possibilities for the traditional methods of protecting surfaces (special paint) may produce dust, dirt and rust. The lifetime of paint is also less.

Glass-lined reactors play a vital role across diverse industry sectors that grapple with corrosive applications. The ingenious use of glass lining as a protective barrier on **Mild Steel (MS)** plates within these reactors renders them impervious to the corrosive agents present in their operational environment. This unique feature not only enhances the longevity of the reactors but also ensures the integrity of the substances they contain.

In particularly stringent settings, such as clean rooms where stringent safety and hygiene protocols are paramount, glass-lined reactors find special application. These controlled environments demand absolute rust prevention due to its potential to compromise both the safety of the operation and the purity of the products being processed. To address this, these reactors are meticulously engineered with an inner layer of glass lining on **Mild Steel (MS)**. The outer layer, known as the jacket, serves as a conduit for heat transfer processes.

However, the challenge arises in safeguarding the outer surface of these reactors against rust formation. To counteract this, a sheet of **Stainless Steel (SS)** is welded as cladding over the jacket. This layer, in addition to insulating the reactor, acts as a barrier against rust development. Yet, it's crucial to note that this cladding has limitations—it only extends coverage up to the uppermost part of the reactor and the vessel itself. Regrettably, it leaves vulnerable areas such as the nozzles that are essential for ingress and egress.

Consequently, the nozzles are typically protected with specialized industrial paints. Over time, however, these paints can peel off due to various factors, including temperature variations and mechanical stress. This leads to the exposure of the underlying **Mild Steel (MS)** surface, setting the stage for potential rust formation and degradation.

To counteract this, a routine maintenance strategy involving periodic inspection and repainting becomes necessary. While this approach ensures the structural integrity of the reactor and prevents rust-related issues, it introduces concerns in clean room environments. The possibility of paint particles contaminating the products being processed poses a significant risk, warranting stringent quality control measures and further amplifying maintenance challenges.

In essence, the usage of glass-lined reactors exemplifies a delicate balance between material science, engineering ingenuity, and meticulous maintenance practices. While these reactors provide indispensable utility in handling corrosive substances, their effective and sustainable operation demands a multifaceted approach that encompasses materials, design, and stringent maintenance protocols to ensure both durability and product purity.

SUMMARY OF INVENTION

The invention discloses **Mild Steel (MS)** Glass Lined Reactors overlaid with **Stainless Steel (SS)** material of approximately 2mm depth in some places and manufacturing directly with **SS** Parts in other places thereby giving the reactor Protection **from** Corrosion, easy maintainability thereby extending the life of the Glass Lined reactor.

Therefore such as herein described there is provided an improved **Mild Steel (MS)** reactor comprising of an internally glass lined reactor sandwiched between a top flange and a bottom flange, wherein fluidic channels are machined on said glass lined reactor, a glass lined metal flat surface below the top flange that allows to observe flow regime in the glass lined reactor or mixer, a cavity in the bottom flange that facilitates flow of heat transfer fluid through an inlet and an outlet nozzles and flanges, wherein said cavity can be closed with a bottom plate having a metal surface, wherein the metal surface of said bottom plate can touch the heat transfer fluid, a connection plate connecting the inlet and the outlet to the bottom flange of the glass lined **Stainless Steel (SS)** reactor, and multiple bolts to hold tightly the top flange

and the bottom flange wherein the external surface of the said reactor is covered and overlaid with **Stainless Steel (SS)** sheet by welding.

In an embodiment the outer wall of the reactor consists of a base body of high mechanical strength, namely steel or **Stainless Steel (SS)**, the base body having a reaction chamber of **Stainless Steel (SS)** with a layer thickness > 1 mm, preferably in the range from essentially 1 mm to 5 mm and most preferably 2mm.

In another embodiment the said **SS** body is made by welding the entire surface with **SS** welding electrodes after removing proportionate metal from the external surface of the reactor and the said nozzles or flanges are made on or in the reactor ~~are~~ **and are** internally glass lined.

In the realm of clean room reactors, stainless-steel cladding prevents rust, except for vulnerable nozzles. To counter this, a solution involves depositing a layer of stainless steel onto the outer nozzle surface, followed by precise grinding for uniformity. This innovation forms a stainless-steel shield, permanently safeguarding against rust. This coated nozzle is then seamlessly welded to the top cover's cladding sheet, fully enclosing the reactor in stainless steel. This dynamic approach harmonizes material science and engineering, tackling rust challenges in critical zones. The result is a resilient barrier that upholds reactor integrity, longevity, and product purity. This cutting-edge technique epitomizes the fusion of advanced technology and craftsmanship, underscoring a commitment to stringent clean room standards.

A glass-lined reactor offers excellent resistance to oxidation at the nozzle. In the event of an accidental spillage of corrosive liquid onto the nozzle surface, which might otherwise lead to oxidation on the metal surface, the glass lining which is coated inside would be damaged as the base metal is oxidated and would result in distortion. The claim asserts that the reactor is fully covered with stainless steel, enhancing its resilience against external damage and thereby increasing its overall safety.

BRIEF DESCRIPTION OF THE ACCOMPANYING DRAWINGS

[Fig.1] [Fig.2] illustrates a typical reactor in assembled state. Isometric view and basic front views.

[Fig.3] ~~[Fig.4] [Fig.5]~~ represents the detailed parts of the clean room reactors with differentiating line types for **Mild Steel (MS)** and **Stainless Steel (SS)**. The component names are also described in the images.

(Note: The accompanying drawings comprise three figures only; references to non-existent Figures 4 and 5 in earlier versions were typographical errors. All structural features are fully illustrated in Figures 1–3 with consistent reference to inner Mild Steel (MS) body and outer Stainless Steel (SS) overlay as per the drawings.)

Figure 1: Clean Room Reactor with **SS** cladding and nozzles covered with **SS** overlay.

Figure 2: Manhole cover with all **SS** components and **SS** overlay on the nozzles.

Figure 3: Nozzle with **SS** overlay and thermowell.

The invention will now be described, by way of examples only, with reference to the accompanying drawings.

Figure 1 depicts a Glass-lined reactor integrated with **Stainless Steel (SS)** cladding, showcasing pivotal components that contribute to its functionality and structural integrity. At the core of this assembly is the Bottom Vessel (1), which serves as the foundational chamber for chemical reactions. The Clamps (20) are strategically positioned to secure various components in place, ensuring stability and safety during operation. Positioned above the vessel is the Top Dish (16), which encapsulates the reactor's contents.

The reactor's drive mechanism is supported by an array of **SS** components, including the Pad Plate (8), Bottom Drive Stand (9), Bearing Housing (10), Top Drive Stand (11), Moto Flange (12), Motor Cover (13), and Mechanical Seals (14 and 7). The Name Plate (15) provides crucial identification and information about the reactor setup.

The reactor design also accommodates accessibility and monitoring. A Manhole Assembly (6) offers a portal for maintenance and inspections, while the Thermowell (5) and its associated Nozzle (4) facilitate temperature measurements and control. This is shown in **Figure 2**.

Figure 2 offers an intricate view of the **SS** cladding application, particularly on Nozzle (26). This **SS** cladding extends along the entire height of both the manhole and thermowell nozzles, ensuring corrosion resistance and structural integrity. Complementing this cladding are **SS** Clamps (27), the Manhole Cover (24), and the Sight Glass Flange (22), all meticulously constructed from **Stainless Steel (SS)**.

The overarching design of the reactor involves the strategic integration of glass-lined and **Stainless Steel (SS)** components, striking a balance between the specialized properties of each material. **As clearly shown in the drawings (Figures 1–3), the reactor body consists of an inner Mild Steel (MS) wall with glass lining and an outer Stainless Steel (SS) overlay/cladding applied seamlessly by welding after removal of approximately 2 mm of the MS surface.** The **SS** cladding provides robust protection against corrosive substances and external elements, while the glass lining offers superior chemical resistance for the reactor's interior. This synthesis of materials and components underscores the reactor's efficiency, durability, and operational safety within various industrial applications.

DESCRIPTION OF INVENTION

Standard glass technical team has changed the material of construction from **MS** to **SS** in the places wherever the reactors outer side is exposed to ambient light and the result is it looks like it is made of **SS** completely from outside but, the reactor core is same and efficient that of **MS** reactor since **MS** reactors are better in process efficiency than **SS** reactors in places where heating is the primary process requirement. Entire reactor is completely covered with **SS** materials. Few places the components parts are made with **SS**, few components are completely wrapped with **SS** dome. The critical areas like nozzles and thermowells are overlaid with **SS** metal sheet by welding process.

In an exemplary embodiment, the said overlaying of **SS** is made by removing **MS** surface up to 2mm of depth and this 2mm is again covered with **SS** by welding the entire surface with **SS** welding electrodes. The thickness is proportional such that the total thickness is maintained and is a constant. The removal of the top surface of the **MS** reactor can be mechanical, chemical and / or both. A person skilled in the art will be able to understand a plurality of method available for the removal of the top surface. This welding surface then

goes under grinding process to attain its original dimensions and original shape of the component.

MAIN COMPONENTS OF CLEAN ROOM REACTORS

[Fig.3] ~~[Fig.4] [Fig.5]~~ represents the detailed parts of the clean room reactors with differentiating line types for **Mild Steel (MS)** and **Stainless Steel (SS)**. The component names are also described in the said images. The continuous line represent the **SS** and dashed line represent the **MS** material. **(Consistent with the accompanying drawings, the inner reactor body is Mild Steel (MS) with glass lining; the external overlay/cladding is Stainless Steel (SS).)** The internal surface is cladded with glass lining using spraying method.

Overlay also mentioned in the images wherever it is applicable. This over laying of **SS** will be subjected to change for each reactor model based on it number of nozzles and type of thermowell that is required for the model. The glass cladding is made in range of 1mm to 2mm for smooth operation and durability. The said glass lining is configured for a non-reactive cladding, pressure-welded coating and the surface of the glass lining / coating is smooth as whole. The base body is designed as a double-walled vessel.

The disclosed modified reactor has capacity ranging from 1Litre to 12500litre for AE type reactor and 1000Litre to 32000Litre for CE type of reactor and heat transfer area ranging from 0.5 m^2 to 34.6 m^2. The said micro-reactor is configured to withstand pressure up to 6kg/cm^2 and temperature ranging from −60° to 343° C.

ADVANTAGES OF INVENTION

The disclosed clean room reactors come with many tangible advantages for the process and for the maintenance. **Stainless Steel (SS)** gives many advantages. These rectors are placed at extreme working conditions which are typically high in temperature, exposed to harsh chemicals. At these conditions these reactors need to be protected from the damaging working conditions and retain its strength. **Stainless Steel (SS)** strength to weight ratio is higher than other materials it can withstand more heat and stress compared to **MS**.

Clean room reactor eliminates the requirement of the painting of outer surface. Painting will attract the dust and over a time period the paint itself peal and becomes dust., so regular maintenance and check-ups are cleaning ups are required when painting comes to picture but clean room reactor doesn't require to paint on its surface. They are hygienic dust repellent and very easy to clean.

Clean room reactors will be given mirror like finish to its surface thus makes it easy to remove dust or any liquid from its surface. It is as easy as cleaning a mirror. This clean and mirror like finish surface makes them look premium. This aesthetic appearance is attractive and gives advantage to identify any dirt or any waste particles on the surface. Easy to inspect the surface in maintenance exercise.

Stainless Steel (SS) is highly corrosive resistant, because these reactors are employed in pharma and chemical processing industries, they are subjected to rigorous environments so, it must withstand corrosion from any medium. This is also a primary reason for developing clean room reactors.

Clean room reactors are built with **SS** on outside to resist scaling, scaling is common phenomenon with the substance used in chemical and pharma industries. Any chemical residue on the reactor surface will not only affect the reactor surface but can also cause problems with the underlying glass thereby reducing the usable life of the reactor. The scaling may affect the environment or labs where these equipment's are kept in operation. Clean room reactor will not have this problem because of appropriate use of **SS** material around it. Contamination will be lot less in **SS** when compared to **MS**.

Standard glass technology has designed and offering these clean room reactors to its customers to add value to their process by means of reducing the maintenance cost, helps maintaining the labs or the production floor clean, give the premium look to the equipment, withstand tough working conditions and increasing the life of the reactor.

The subject matter of embodiments of the present invention is described here with specificity to meet statutory requirements, but this description is not necessarily intended to limit the scope of the claims. The claimed subject matter may be embodied in other ways, may include different elements or steps, and may be used in conjunction with other existing or future technologies. This description should not be interpreted as implying any particular order or

arrangement among or between various steps or elements except when the order of individual steps or arrangement of elements is explicitly described.

We Claim:

1. A clean room glass lined reactor **(2)** comprising:

   (i) **an** internally glass-lined ~~MS~~ **mild steel** reactor **(10)** ~~sandwiched~~ **clamped** between a top flange **(12)** and **a** bottom flange **(14)**, with fluidic channels **(16)** machined on the ~~glass-lined~~ reactor surface;

   (ii) a glass-lined metal flat surface **(18)** below the top flange **(12)** which allows flow-regime observation inside the reactor/mixer;

   (iii) a cavity **(20)** in the bottom flange **(14)** allowing circulation of heat-transfer fluid via inlet/outlet nozzles **(22)**; closable with a metal bottom plate **(24)**;

   (iv) **a** connection plate **(26)** connecting inlet/outlet **(22)** to the bottom flange **(14)**;

   (v) **the** reactor **(10)** is fully overlaid externally with ~~SS~~ **stainless steel** sheet **(28)** by welding;

   (vi) internally glass-lined nozzles **(30)** with a seamless ~~SS~~ **stainless steel** overlay, welded to the cladding **(32)** to eliminate oxidation and rusting at critical zones ~~offering excellent resistance to oxidation~~ **wherein the seamless stainless steel overlay provides resistance to oxidation at the nozzles**;

   (vii) ~~multiple bolts to hold tightly the top and bottom flanges~~ **bolts (34) securing the top flange (12) and bottom flange (14)**.

2. The clean room glass lined reactor ~~as claimed in claim 1, wherein the base body of high mechanical strength is preferably mild steel, essentially of 1-2mm thickness~~ **as claimed in claim 1, wherein the base body (36) has a mechanical yield strength of at least 250 MPa and is mild steel with a thickness of 1-2 mm**.

3-5. [NO CHANGE]

6. The clean room glass lined reactor ~~as claimed in claim 1, wherein the reactor is a double-walled vessel~~**as claimed in claim 1, wherein the reactor (10) comprises a double-walled vessel design (38)**.
7. The clean room glass lined reactor ~~as claimed in claim 1, wherein the interior is smooth as whole~~**as claimed in claim 1, wherein the interior surface (40) has a smoothness with roughness average (Ra) less than 0.8 µm**.
8. The clean room glass lined reactor ~~as claimed in claim 1, wherein the volume and heat-transfer area are application specific~~**as claimed in claim 1, wherein the reactor (10) has a volume range of 50-5000 liters and a heat-transfer area adapted for pharmaceutical applications**.

Dated this 21st day of April, 2021

(Bharat Daswani)
Of Daswani & Daswani,
Applicant's Agent
IN / PA 289

FIELD OF INVENTION

The present invention relates to corrosion resistant, clean room glass lined overlaid Stainless Steel (SS) reactor. More particularly, the present invention discloses a reactor specifically designed for clean and polished working environment in food and medical industries, which don't require any paint coat to achieve a mirror-finished appearance. These reactors help to keep the production floor and its surroundings less dusty and prevent contamination of the product with enhanced heat transfer efficiency.

The present invention also presents Mild Steel (MS) Glass-Lined reactors with Stainless Steel (SS) overlays on nozzles, designed for sectors like pharmaceuticals, food, chemicals, and agrochemicals. It aims to elevate glass-lined reactors for cleanroom use, ensuring high cleanliness standards, less maintenance, and rust prevention. By integrating Stainless Steel (SS) overlays onto reactor nozzles with SS cladding around the vessel, the solution covers 100% of the reactor, offering durability for industries requiring stringent hygiene. Stainless Steel (SS) enhances corrosion resistance, which is vital in industries where rust could compromise product quality.

BACKGROUND ART

Conventional reactors are widely used by fine chemicals and pharmaceutical industry. Corrosion, cleaning and contamination are one of the key criteria that determine the efficiency of a process, which in turn depends on the type of equipment used. Stirred tank reactors (STRS) used in pharmaceutical and fine chemical industries are designed for providing and removing heat. This is done with coil or half coil with coolant passing through it, or using jacketed tanks.

Standard glass lining technology wants to make reactors for easy maintenance and suitable for clean environments. Standard glass has designed and modified its existing large variety of reactors to make them well suited for the clean room and exclusive process areas.

Glass Lined Reactors are generally designed as per ASME SECTION-8 DIVISION-I. Dimensions and capacities are as per DIN-28136 with Mild Steel (MS) Material with Glass Coating on the Inner side of the reactor. They can be of AE Type or CE Type and both types can be made as clean room reactors.

The Reactor's Mild Steel (MS) outer surface is also prone to corrosion by the spilled material during charging the reactor. Over a period of time there is a high probability that the corrosive particles contaminate the charge thus causing rejections of the material and also costly Regulatory Compliance Issues. Chemical resistant paints are used to protect the reactor outer surface from exposing it to harsh chemicals and acidic substances. Special procedures and paint materials are used to coat the reactor parts. The paint will also protect the surface from getting rust so that, the rust will not convert to dust or debris around the reactor. Also, the paint itself may peel off after certain time in its lifetime which again form as dirt; paint is prone to damage for impacts caused by any tools or objects. So, taking care of the paint itself is a big task and any damage to paint will create dirt and allows the material to expose and material to form rust over a period. It is also proven that Paint will attract dust. There are many possibilities for the traditional methods of protecting surfaces (special paint) may produce dust, dirt and rust. The lifetime of paint is also less.

Glass-lined reactors play a vital role across diverse industry sectors that grapple with corrosive applications. The ingenious use of glass lining as a protective barrier on Mild Steel (MS) plates within these reactors renders them impervious to the corrosive agents present in their operational environment. This unique feature not only enhances the longevity of the reactors but also ensures the integrity of the substances they contain.

In particularly stringent settings, such as clean rooms where stringent safety and hygiene protocols are paramount, glass-lined reactors find special application. These controlled environments demand absolute rust prevention due to its potential to compromise both the safety of the operation and the purity of the products being processed. To address this, these reactors are meticulously engineered with an inner layer of glass lining on Mild Steel (MS). The outer layer, known as the jacket, serves as a conduit for heat transfer processes.

However, the challenge arises in safeguarding the outer surface of these reactors against rust formation. To counteract this, a sheet of Stainless Steel (SS) is welded as cladding over the jacket. This layer, in addition to insulating the reactor, acts as a barrier against rust development. Yet, it's crucial to note that this cladding has limitations—it only extends coverage up to the uppermost part of the reactor and the vessel itself. Regrettably, it leaves vulnerable areas such as the nozzles that are essential for ingress and egress.

Consequently, the nozzles are typically protected with specialized industrial paints. Over time, however, these paints can peel off due to various factors, including temperature variations and mechanical stress. This leads to the exposure of the underlying Mild Steel (MS) surface, setting the stage for potential rust formation and degradation.

To counteract this, a routine maintenance strategy involving periodic inspection and repainting becomes necessary. While this approach ensures the structural integrity of the reactor and prevents rust-related issues, it introduces concerns in clean room environments. The possibility of paint particles contaminating the products being processed poses a significant risk, warranting stringent quality control measures and further amplifying maintenance challenges.

In essence, the usage of glass-lined reactors exemplifies a delicate balance between material science, engineering ingenuity, and meticulous maintenance practices. While these reactors provide indispensable utility in handling corrosive substances, their effective and sustainable operation demands a multifaceted approach that encompasses materials, design, and stringent maintenance protocols to ensure both durability and product purity.

SUMMARY OF INVENTION

The invention discloses Mild Steel (MS) Glass Lined Reactors overlaid with Stainless Steel (SS) material of approximately 2mm depth in some places and manufacturing directly with SS Parts in other places thereby giving the reactor Protection from Corrosion, easy maintainability thereby extending the life of the Glass Lined reactor.

Therefore such as herein described there is provided an improved Mild Steel (MS) reactor comprising of an internally glass lined reactor sandwiched between a top flange and a bottom flange, wherein fluidic channels are machined on said glass lined reactor, a glass lined metal flat surface below the top flange that allows to observe flow regime in the glass lined reactor or mixer, a cavity in the bottom flange that facilitates flow of heat transfer fluid through an inlet and an outlet nozzles and flanges, wherein said cavity can be closed with a bottom plate having a metal surface, wherein the metal surface of said bottom plate can touch the heat transfer fluid, a connection plate connecting the inlet and the outlet to the bottom flange of the glass lined Stainless Steel (SS) reactor, and multiple bolts to hold tightly the top flange

and the bottom flange wherein the external surface of the said reactor is covered and overlaid with Stainless Steel (SS) sheet by welding.

In an embodiment the outer wall of the reactor consists of a base body of high mechanical strength, namely steel or Stainless Steel (SS), the base body having a reaction chamber of Stainless Steel (SS) with a layer thickness > 1 mm, preferably in the range from essentially 1 mm to 5 mm and most preferably 2mm.

In another embodiment the said SS body is made by welding the entire surface with SS welding electrodes after removing proportionate metal from the external surface of the reactor and the said nozzles or flanges are made on or in the reactor and are internally glass lined.

In the realm of clean room reactors, stainless-steel cladding prevents rust, except for vulnerable nozzles. To counter this, a solution involves depositing a layer of stainless steel onto the outer nozzle surface, followed by precise grinding for uniformity. This innovation forms a stainless-steel shield, permanently safeguarding against rust. This coated nozzle is then seamlessly welded to the top cover's cladding sheet, fully enclosing the reactor in stainless steel. This dynamic approach harmonizes material science and engineering, tackling rust challenges in critical zones. The result is a resilient barrier that upholds reactor integrity, longevity, and product purity. This cutting-edge technique epitomizes the fusion of advanced technology and craftsmanship, underscoring a commitment to stringent clean room standards.

A glass-lined reactor offers excellent resistance to oxidation at the nozzle. In the event of an accidental spillage of corrosive liquid onto the nozzle surface, which might otherwise lead to oxidation on the metal surface, the glass lining which is coated inside would be damaged as the base metal is oxidated and would result in distortion. The claim asserts that the reactor is fully covered with stainless steel, enhancing its resilience against external damage and thereby increasing its overall safety.

BRIEF DESCRIPTION OF THE ACCOMPANYING DRAWINGS

[Fig.1] [Fig.2] illustrates a typical reactor in assembled state. Isometric view and basic front views.

[Fig.3] represents the detailed parts of the clean room reactors with differentiating line types for Mild Steel (MS) and Stainless Steel (SS). The component names are also described in the images.

Figure 1: Clean Room Reactor with SS cladding and nozzles covered with SS overlay.

Figure 2: Manhole cover with all SS components and SS overlay on the nozzles.

Figure 3: Nozzle with SS overlay and thermowell.

The invention will now be described, by way of examples only, with reference to the accompanying drawings.

Figure 1 depicts a Glass-lined reactor integrated with Stainless Steel (SS) cladding, showcasing pivotal components that contribute to its functionality and structural integrity. At the core of this assembly is the Bottom Vessel (1), which serves as the foundational chamber for chemical reactions. The Clamps (20) are strategically positioned to secure various components in place, ensuring stability and safety during operation. Positioned above the vessel is the Top Dish (16), which encapsulates the reactor's contents.

The reactor's drive mechanism is supported by an array of SS components, including the Pad Plate (8), Bottom Drive Stand (9), Bearing Housing (10), Top Drive Stand (11), Moto Flange (12), Motor Cover (13), and Mechanical Seals (14 and 7). The Name Plate (15) provides crucial identification and information about the reactor setup.

The reactor design also accommodates accessibility and monitoring. A Manhole Assembly (6) offers a portal for maintenance and inspections, while the Thermowell (5) and its associated Nozzle (4) facilitate temperature measurements and control. This is shown in Figure 2.

Figure 2 offers an intricate view of the SS cladding application, particularly on Nozzle (26). This SS cladding extends along the entire height of both the manhole and thermowell nozzles, ensuring corrosion resistance and structural integrity. Complementing this cladding are SS Clamps (27), the Manhole Cover (24), and the Sight Glass Flange (22), all meticulously constructed from Stainless Steel (SS).

The overarching design of the reactor involves the strategic integration of glass-lined and Stainless Steel (SS) components, striking a balance between the specialized properties of each material. As clearly shown in the drawings (Figures 1–3), the reactor body consists of an inner Mild Steel (MS) wall with glass lining and an outer Stainless Steel (SS) overlay/cladding applied seamlessly by welding after removal of approximately 2 mm of the MS surface. The SS cladding provides robust protection against corrosive substances and external elements, while the glass lining offers superior chemical resistance for the reactor's interior. This synthesis of materials and components underscores the reactor's efficiency, durability, and operational safety within various industrial applications.

DESCRIPTION OF INVENTION

Standard glass technical team has changed the material of construction from MS to SS in the places wherever the reactors outer side is exposed to ambient light and the result is it looks like it is made of SS completely from outside but, the reactor core is same and efficient that of MS reactor since MS reactors are better in process efficiency than SS reactors in places where heating is the primary process requirement. Entire reactor is completely covered with SS materials. Few places the components parts are made with SS, few components are completely wrapped with SS dome. The critical areas like nozzles and thermowells are overlaid with SS metal sheet by welding process.

In an exemplary embodiment, the said overlaying of SS is made by removing MS surface up to 2mm of depth and this 2mm is again covered with SS by welding the entire surface with SS welding electrodes. The thickness is proportional such that the total thickness is maintained and is a constant. The removal of the top surface of the MS reactor can be mechanical, chemical and / or both. A person skilled in the art will be able to understand a plurality of method available for the removal of the top surface. This welding surface then goes under grinding process to attain its original dimensions and original shape of the component.

MAIN COMPONENTS OF CLEAN ROOM REACTORS

[Fig.3] represents the detailed parts of the clean room reactors with differentiating line types for Mild Steel (MS) and Stainless Steel (SS). The component names are also described in the said images. The continuous line represent the SS and dashed line represent the MS material.

(Consistent with the accompanying drawings, the inner reactor body is Mild Steel (MS) with glass lining; the external overlay/cladding is Stainless Steel (SS).) The internal surface is cladded with glass lining using spraying method.

Overlay also mentioned in the images wherever it is applicable. This over laying of SS will be subjected to change for each reactor model based on it number of nozzles and type of thermowell that is required for the model. The glass cladding is made in range of 1mm to 2mm for smooth operation and durability. The said glass lining is configured for a non-reactive cladding, pressure-welded coating and the surface of the glass lining / coating is smooth as whole. The base body is designed as a double-walled vessel.

The disclosed modified reactor has capacity ranging from 1Litre to 12500litre for AE type reactor and 1000Litre to 32000Litre for CE type of reactor and heat transfer area ranging from 0.5 m^2 to 34.6 m^2. The said micro-reactor is configured to withstand pressure up to 6kg/cm^2 and temperature ranging from −60° to 343° C.

ADVANTAGES OF INVENTION

The disclosed clean room reactors come with many tangible advantages for the process and for the maintenance. Stainless Steel (SS) gives many advantages. These rectors are placed at extreme working conditions which are typically high in temperature, exposed to harsh chemicals. At these conditions these reactors need to be protected from the damaging working conditions and retain its strength. Stainless Steel (SS) strength to weight ratio is higher than other materials it can withstand more heat and stress compared to MS.

Clean room reactor eliminates the requirement of the painting of outer surface. Painting will attract the dust and over a time period the paint itself peal and becomes dust., so regular maintenance and check-ups are cleaning ups are required when painting comes to picture but clean room reactor doesn’t require to paint on its surface. They are hygienic dust repellent and very easy to clean.

Clean room reactors will be given mirror like finish to its surface thus makes it easy to remove dust or any liquid from its surface. It is as easy as cleaning a mirror. This clean and mirror like finish surface makes them look premium. This aesthetic appearance is attractive

and gives advantage to identify any dirt or any waste particles on the surface. Easy to inspect the surface in maintenance exercise.

Stainless Steel (SS) is highly corrosive resistant, because these reactors are employed in pharma and chemical processing industries, they are subjected to rigorous environments so, it must withstand corrosion from any medium. This is also a primary reason for developing clean room reactors.

Clean room reactors are built with SS on outside to resist scaling, scaling is common phenomenon with the substance used in chemical and pharma industries. Any chemical residue on the reactor surface will not only affect the reactor surface but can also cause problems with the underlying glass thereby reducing the usable life of the reactor. The scaling may affect the environment or labs where these equipment's are kept in operation. Clean room reactor will not have this problem because of appropriate use of SS material around it. Contamination will be lot less in SS when compared to MS.

Standard glass technology has designed and offering these clean room reactors to its customers to add value to their process by means of reducing the maintenance cost, helps maintaining the labs or the production floor clean, give the premium look to the equipment, withstand tough working conditions and increasing the life of the reactor.

The subject matter of embodiments of the present invention is described here with specificity to meet statutory requirements, but this description is not necessarily intended to limit the scope of the claims. The claimed subject matter may be embodied in other ways, may include different elements or steps, and may be used in conjunction with other existing or future technologies. This description should not be interpreted as implying any particular order or arrangement among or between various steps or elements except when the order of individual steps or arrangement of elements is explicitly described.

We Claim:

1. A clean room glass lined reactor (2) comprising:

   (i) an internally glass-lined mild steel reactor (10) clamped between a top flange (12) and a bottom flange (14), with fluidic channels (16) machined on the reactor surface;

   (ii) a glass-lined metal flat surface (18) below the top flange (12) which allows flow-regime observation inside the reactor/mixer;

   (iii) a cavity (20) in the bottom flange (14) allowing circulation of heat-transfer fluid via inlet/outlet nozzles (22); closable with a metal bottom plate (24);

   (iv) a connection plate (26) connecting inlet/outlet (22) to the bottom flange (14);

   (v) the reactor (10) is fully overlaid externally with stainless steel sheet (28) by welding;

   (vi) internally glass-lined nozzles (30) with a seamless stainless steel overlay, welded to the cladding (32) to eliminate oxidation and rusting at critical zones wherein the seamless stainless steel overlay provides resistance to oxidation at the nozzles;

   (vii) bolts (34) securing the top flange (12) and bottom flange (14).

2. The clean room glass lined reactor as claimed in claim 1, wherein the base body (36) has a mechanical yield strength of at least 250 MPa and is mild steel with a thickness of 1-2 mm.
3. The clean room glass lined reactor as claimed in claim 1, wherein the said stainless steel body is made by welding the entire surface with stainless steel welding electrodes after removing proportionate metal from the external surface of the reactor.

4. The clean room glass lined reactor as claimed in claim 1, wherein the said nozzles or flanges made on or in the reactor are internally glass lined.

5. The clean room glass lined reactor as claimed in claim 1, wherein the glass lining is configured for a non-reactive cladding, pressure-welded coating.

6. The clean room glass lined reactor as claimed in claim 1, wherein the reactor (10) comprises a double-walled vessel design (38).
7. The clean room glass lined reactor as claimed in claim 1, wherein the interior surface (40) has a smoothness with roughness average (Ra) less than 0.8 µm.
8. The clean room glass lined reactor as claimed in claim 1, wherein the reactor (10) has a volume range of 50-5000 liters and a heat-transfer area adapted for pharmaceutical applications.

Dated this 21st day of April, 2021



(Bharat Daswani)
Of Daswani & Daswani,
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IN / PA 289

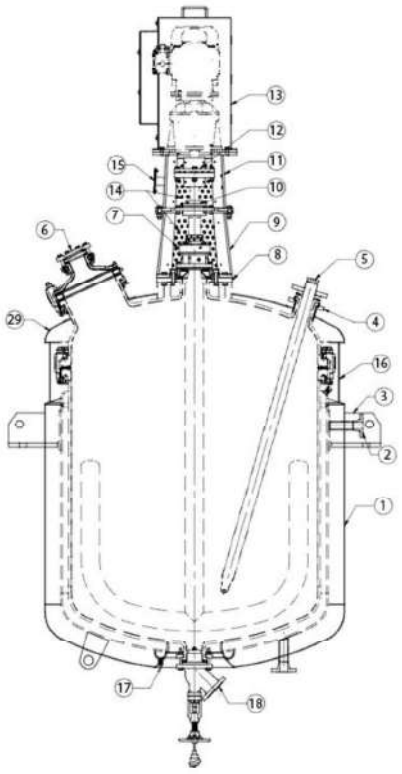


Figure 1

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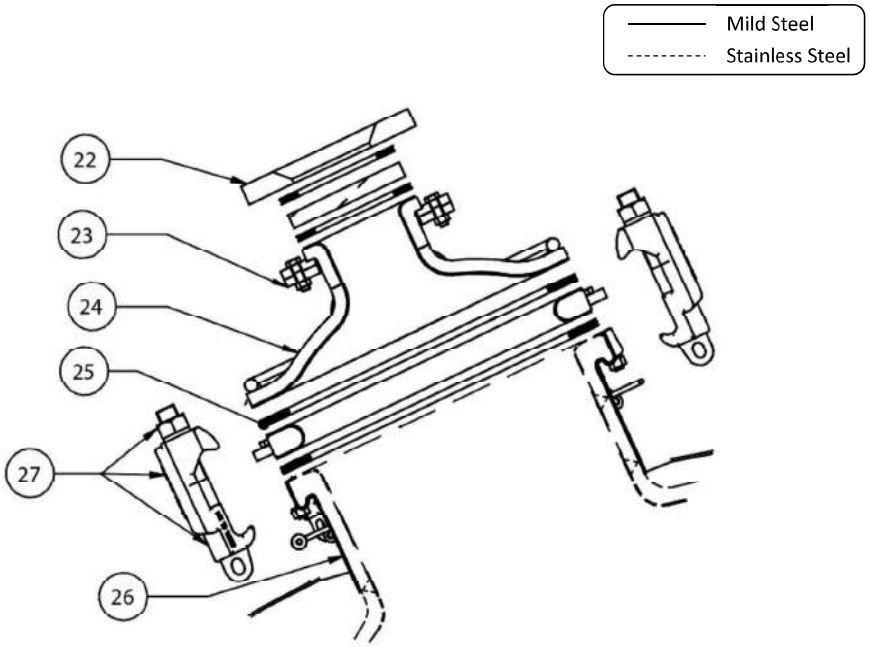


Figure 2

(Bharat Daswani)
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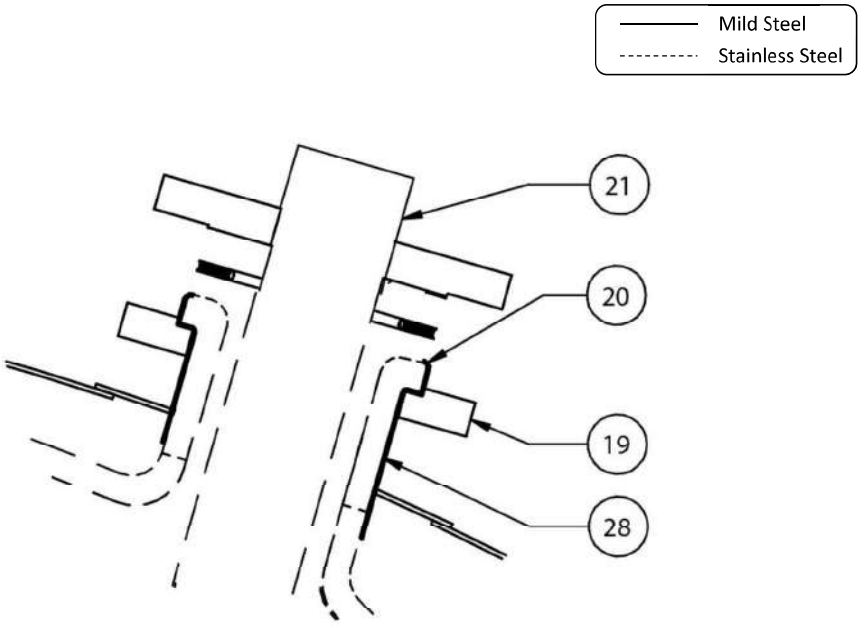


Figure 3

(Bharat Daswani)
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Digitally Signed By:
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Date: 03-06-2026 19:53:09

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Confidential and Urgent

HCA/SKA/HT/RD June 04, 2026

To, ONLINE SUBMISSION
The Controller of Patents,
The Patent Office,
CHENNAI

Kind attention: Dr. Kanhaiya Lal Balai

Your Reference: Your Ref: POC/Application No / Dr. Kanhaiya Lal Balai 202141018455

Sub.: Pre-Grant Opposition by Pinakin Engineers Pvt Ltd under Section 25(1) of the Patents Act, 1970 against Patent Application No. 202141018455 titled "Clean Room Glass Lined Reactor" filed by Standard Glass Lining Technology Private

WRITTEN SUBMISSIONS ON BEHALF OF THE OPPONENT IN

THE PRE-GRANT OPPOSITION PROCEEDINGS UNDER SECTION

25(1) OF THE PATENTS ACT, 1970

Dear Sir,

In response to your extended hearing notice pursuant to the filing of amended claims by the Applicant regarding Pre-Grant Opposition, applicant's Counsel (Patent Agent) Mr. Hemant Thadhani, IN/PA 4057 attended the hearing fixed on 20.05.2026.

We have the honour to submit the written submissions of arguments made on behalf of the Opponent, Pinakin Engineers Private Limited, through its Counsel as under.

This exhaustive submission is filed pursuant to the hearing held on 20.05.2026 and the Controller's direction to submit written arguments.

A. PRELIMINARY OBJECTION: REPLY TO INTERLOCUTORY PETITION

1. The Applicant's interlocutory petition is founded on an erroneous and factually incorrect premise regarding the Opponent's alleged non-compliance with Rule 62(2) of the Patents Rules, 2003. The Applicant has

incorrectly suggested that the Opponent failed to file the notice of hearing and pay the prescribed fee, and thereafter attempted to introduce backdated documents. The said allegation is categorically denied.

2. It is respectfully submitted that the Opponent had duly uploaded and executed the requisite notice of hearing on 13.11.2025 in the ordinary course of business, and the prescribed fee was paid on 14.11.2025, i.e., on the very date of the hearing. There is no element of backdating, misrepresentation, or procedural impropriety. The mere fact that the payment or filing may not have been reflected in the online system at the precise time of hearing cannot be attributed to any fault on the part of the Opponent, and cannot be construed as non-compliance with Rule 62(2).

3. In these circumstances, the Learned Controller rightly exercised his jurisdiction in permitting further hearing in order to uphold the principles of natural justice. The Applicant's contention that Rule 62(2) operates as an absolute bar is misconceived. **Rule 62(2) is procedural in nature and governs the manner of participation in a hearing; it does not extinguish the substantive right of opposition under Section 25(1), nor does it curtail the inherent powers of the Controller to permit rectification of procedural lapses in appropriate cases.**

4. It is well settled that procedural requirements are intended to advance justice and not defeat substantive rights. **The Controller, being vested with quasi-judicial authority under Section 77 of the Act and guided by Rule 137 of the Patents Rules, is empowered to take steps necessary to secure the ends of justice, including permitting a party to be heard where no prejudice is caused.**

5. The Applicant's reliance on the alleged "finality" of the hearing dated 14.11.2025 is misplaced. The proceedings had not culminated in any decision under Section 15 of the Act and therefore remained pending. In such circumstances, the Learned Controller was well within jurisdiction to ensure a fair and complete hearing by granting an opportunity to the Opponent.

6. Further, the Applicant has failed to demonstrate any prejudice whatsoever. The Applicant was fully heard on 14.11.2025 and has since filed extensive written submissions. The limited opportunity granted to the Opponent does not reopen or nullify prior proceedings but merely ensures parity of hearing. In contrast, denying the Opponent an opportunity would result in grave prejudice and would be contrary to settled principles of natural justice.

7. The allegations of backdating and impropriety are wholly baseless, unsupported by any evidence, and appear to have been raised only to derail adjudication on merits. Such allegations deserve to be rejected outright.

8. Accordingly, the interlocutory petition is devoid of merit, liable to be dismissed, and the present opposition ought to be adjudicated on merits.

9. Without prejudice to the foregoing preliminary objections, and in the event this Learned Controller is pleased to proceed with the adjudication of the present opposition on merits, the Opponent respectfully submits that the impugned application is liable to be refused on multiple substantive grounds under Section 25(1) of the Patents Act, 1970. The Applicant's attempt to divert attention to procedural objections through the present interlocutory petition cannot cure the fundamental deficiencies in the application, including lack of novelty, lack of inventive step, non-patentability, and insufficiency of disclosure. The

Opponent therefore proceeds to address the grounds of opposition in detail hereinafter, without prejudice to its rights and contentions in respect of the interlocutory issues raised by the Applicant.

B. INTRODUCTION

10. The Opponent respectfully submits that the impugned application is liable to be refused under Sections 25(1)(b), (e), (f), and (g) of the Patents Act, 1970. The Applicant's written submissions and amended claims have been perused in detail, and the following rebuttals address each argument advanced by the Applicant.

C. PRELIMINARY OBJECTION: LOCUS STANDI

11. The Applicant's contention that Opponent lacks locus standi is untenable. Section 25(1) permits 'any person' to file a representation by way of pre-grant opposition prior to the grant. The Opponent relies on the judgment of Snehlata C Gupte Vs. Union of India and Ors. Snehlata C Gupte vs. Union of India and Ors. (20.04.2012 - DELHC) : 2012(50)PTC327(Del) that affirms that any person has locus standi to file a pre-grant opposition under Section 25(1) of the Act.

12. Ajay Industrial Corp v. Controller of Patents (1984 PTC 234) cited by Applicant is distinguishable on facts as it pertains to post grant oppositions under Section 25(2) of the Act and it involved lack of evidence of interest, whereas here Opponent qualifies to be "any person" under Section 25(1) of the Act and no further inquiry is necessary by the learned Controller. Mere allegation of mala fide is irrelevant in the context of a pre-grant opposition.

13. Thus, the allegation of lack of locus standi is untenable and without any merit, and liable to be rejected.

D. CONTROLLER'S PRIMA FACIE FINDINGS

14. The Ld. Controller's Report dated 04.06.2025 categorically observed that prior art documents D1-D3 render the claims obvious and lacking inventive step under Section 2(1)(ja) of the Patents Act. These findings remain unrebutted by the Applicant's submissions, which rely on mere conclusory assertions without addressing the structural similarities between the claimed subject matter and the cited prior art, or demonstrating any genuine technical advancement. The Opponent respectfully relies upon and adopts the findings of the Learned Controller as part and parcel of these Written Submissions and reiterates that such observations further strengthen the case for refusal of the impugned application.

E. BACKGROUND OF THE INVENTION

15. The invention purports to improve a component of a glass-lined reactor, namely the **nozzle**. Conventionally, the reactor's outer shell is fabricated from mild steel (MS), which offers strength but is susceptible to rusting. **To mitigate corrosion and maintain cleanliness, a thin layer of stainless steel (SS), commonly referred to as cladding, is typically applied externally. The Applicant claims to purportedly depart from this conventional approach by proposing the deposition of a stainless steel layer directly onto the nozzle**. However, this feature is neither novel nor inventive, as similar techniques are already disclosed in the prior art and would be obvious to a person skilled in the art.

F. DETAILED GROUNDS OF OPPOSITION

16. The opposition is based on Sections 25(1)(b), (e), (f), and (g). Each ground is addressed below with detailed rebuttal to Applicant's arguments.

I. Lack of Novelty - Section 25(1)(b)

17. Applicant argues that D1-D3 do not disclose seamless SS overlay welded after removing 2mm MS surface, integrated observation surface, and bottom flange cavity. However, anticipation does not require verbatim disclosure; functional equivalents suffice. In support, the Opponent submits that independent Claim 1 and dependent claims are fully anticipated by prior art:

    i. **D1 (CN103820781A)**: Glass-lined equipment with stainless steel outer cladding, welded joints, and coverage of nozzles/manholes. D1 clearly discloses stainless steel cladding on nozzles, rendering the impugned application devoid of novelty and inventive step.

    ii. **D2 (CN105148816A)**: Manufacturing method for glass-lined reactors with SS foreskin wrapping and welding.

    iii. **D3 (CN206372825U)**: Corrosion-resistant glass reactor with welded high-neck flanges and structural reinforcement

II. Claim Feature vs. Prior Art Table shows one-to-one correspondence of features.

18. The following table provides a detailed comparison of each claim of the impugned application against disclosures in prior art documents D1 (CN103820781A), D2 (CN105148816A), and D3 (CN206372825U).

Claim No.	Feature in Impugned Application	Disclosure in D1 (CN103820781A)	Disclosure in D2 (CN105148816A)	Disclosure in D3 (CN206372825U)
1	MS Glass-Lined reactor with SS overlays on nozzles	SS cladding on tank body and nozzle region	SS foreskin wrapping and welding	High-neck flanges with welded joints
2	Outer wall with SS layer thickness >1mm (preferably 2mm)	SS outer layer thickness 1.5-3mm (external only)	SS sheath thickness not specified	Focus on enamel and flange strength
3	Entire external surface welded with SS electrodes after removing MS	External SS plates welded; no MS removal	Foreskin welded externally; no MS removal	No SS overlay; only enamel coating
4	Nozzles/flanges internally glass-lined while SS-overlaid outside	External cladding around nozzle; internal lining not disclosed	SS flange sleeves; gap issues	High-neck flanges enamelled; no SS overlay
5	Glass lining configured for non-reactive cladding, pressure-welded coating	Glass lining disclosed; pressure weld not emphasized	Glass lining present; no pressure weld detail	Glass lining with enamel glaze; no SS overlay
6	Base body designed as double-walled vessel	Double-walled jacket structure disclosed	Outer foreskin with insulation; not double-walled MS	Double-walled design for pressure resistance

7	Surface of glass lining/coating smooth as whole	Polished SS welds to 320 mesh for smooth finish	Outer SS polished for aesthetics	Smooth enamel surface disclosed
8	Capacity range 1L-12500L (AE) and 1000L-32000L (CE)	Capacity not specified; general reactor sizes	Capacity not disclosed	Capacity not disclosed

III. Feature-by-Feature Analysis

19. Applicant's claim of novelty based on observation surface and cavity is misplaced; these are routine design optimizations. No inventive ingenuity is involved in adding a flat observation surface or cavity for heat transfer. Prior art D1 and D2 disclose jacketed cavities and manholes, which serve similar purposes.

20. Based on the above facts, circumstances and legal principles, the Opponent relied on the **Supreme Court** judgement in the case of *Bishwanath Prasad Radhey Shyam v. Hindustan Metal Industries (AIR 1982 SC 1444)* wherein it was held that anticipation requires clear directions, which D1 provides for SS welding on nozzles. Further, the **Bombay High Court** in *Farbwerke Hoechst v. Unichem Labs (AIR 1969 Bom 255)* held that prior art anticipates if it enables a skilled person to practice the invention.

IV. Lack of Inventive Step - Section 25(1)(e)

21. The combination of features disclosed in D1-D3 renders the claimed invention obvious. Applying the five-step test laid down in *F. Hoffmann-La Roche Ltd. v. Cipla Ltd.* (2016 66 PTC 1 Del):
    i. **Inventive Concept:** SS overlay on nozzles combined with full SS coverage of the reactor.
    ii. **Prior Art Teaching:** D1 and D2 disclose SS cladding and welding for corrosion resistance; D3 teaches nozzle reinforcement through welded flanges.
    iii. **Difference:** Merely extending SS cladding to nozzles is an obvious workshop modification.
    iv. **Technical Advancement:** None; the claimed arrangement does not produce any unexpected result or synergistic effect.
    v. **Secondary Indicia:** No evidence of long-felt need or failure of others; the claimed feature is routine in the art.

22. The Applicant's reliance on *Tapas Chatterjee v. Assistant Controller* (2023 SCC OnLine Del 469) is misplaced. Inventive step requires a demonstrable technical advance beyond common general knowledge, which is absent in the present case.

V. Applicant's Argument on "Teaching Away

23. The Applicant contends that prior art teaches away due to insulation gaps. This is incorrect. Welding SS overlays is standard practice in the industry. There is no evidence of technical prejudice against seamless welding. Modifying prior art to eliminate gaps for clean-room compliance is an obvious design choice, well within the competence of a person skilled in the art.

24. Further, the Applicant contended that the steps disclosed in the impugned application, specifically, removing a 2mm layer of mild steel (MS) and applying a stainless steel (SS) coating on the nozzle are not

taught in the prior art and that D1-D3 fail to disclose structural integrity or the precise manner of applying SS treatment as claimed. This argument is misconceived.

25. The alleged differences represent routine design optimizations and obvious workshop modifications within the competence of a person skilled in the art. Prior art documents D1 and D2 already teach SS cladding and welding techniques for corrosion resistance, while D3 addresses nozzle reinforcement. Extending these known practices to include dimensional adjustments such as removing a thin MS layer before welding does not constitute an inventive step. There is no technical prejudice or teaching away in the prior art; on the contrary, such modifications are predictable and motivated by standard clean-room compliance requirements. Accordingly, the Applicant's reliance on this feature as a basis for novelty or inventive step is unfounded.

26. In conclusion, the claimed invention fails to satisfy the requirements of Section 2(1)(ja) of the Patents Act. The combination of features—SS overlay on nozzles and full SS coverage—does not involve any technical advance over existing knowledge and is an obvious workshop modification of prior art teachings in D1-D3. There is no unexpected result, no synergistic effect, and no evidence of a long-felt need being addressed.

27. Applying the well-settled Roche Test, the differences identified are trivial and well within the competence of a person skilled in the art. Accordingly, the claims lack inventive step and are liable to be refused.

G. Non-Patentability - Section 3(f)

28. The claimed reactor constitutes a mere arrangement or re-arrangement of known components functioning independently in a known manner, falling squarely within the exclusion under Section 3(f) of the Patents Act.

    i. **Known Components:**

    Glass lining, stainless steel (SS) cladding, welded joints, and nozzles are all individually well-known in the art.

    ii. **Independent Functioning:**

    Each element performs its conventional role without any interdependence:

    a. Glass lining prevents chemical attack.

    b. SS overlay prevents rust.

    c. Fluidic channels facilitate heat transfer.

29. There is no synergistic effect or combined technical advancement arising from their integration. The Applicant's assertion of synergy is unfounded; the features operate independently and do not produce a new or improved result beyond their known functions.

30. Judicial precedents reinforce this position:
    - *TVS Motor Co. v. Bajaj Auto Ltd.* (2009 40 PTC 689 Mad) - synergy is essential for patentability; mere aggregation of known features is not an invention.

- *Nippon Steel Corp. v. Controller of Patents* (2024 SCC OnLine Del 1234) - mechanical combinations lacking interdependent functionality fall under Section 3(f) and are not patentable.

31. Accordingly, the claimed subject matter is nothing more than a predictable combination of known elements, devoid of inventive ingenuity, and is liable to be refused under Section 3(f).

H. Insufficiency - Section 25(1)(g)

32. The specification fails to meet the statutory requirements under Section 10(4) of the Patents Act, 1970. It does not disclose the best mode of performing the invention, omits referenced Figures 4 and 5, and employs vague and indefinite expressions such as "**smooth as whole**" and "**preferably**," which compromise clarity and intelligibility.

i. **Deficiencies in Disclosure:**

- No enabling details for the SS overlay welding process, which is vaguely referred to as "proprietary."
- Absence of critical technical parameters such as welding current, electrode specifications, and MS removal depth, making replication impossible without undue experimentation.
- Failure to include Figures 4 and 5, despite references in the description, creating inconsistency between the specification and drawings.

ii. **Unsupported Amendments:**

Amendments filed via Form 13 introduce features (e.g., Ra value, yield strength) without explicit basis in the original specification, violating Sections 57 and 59. Such amendments are impermissible and liable to be rejected.

iii. **Use of Ambiguous Language:**

Claims contain vague terms like "smooth as whole," "high mechanical strength," "preferably," and "essentially," which do not define the invention with precision as mandated under Section 10(4)(c).

33. The Applicant's assertion of enabling disclosure is unfounded. The specification lacks clarity and completeness, as emphasized in *Hindustan Lever Ltd. v. Godrej Soaps Ltd.* (1996 PTC 141), which held that clarity and sufficiency are mandatory for patentability.

34. Accordingly, the impugned application fails the test of sufficiency and clarity under Section 10(4) and is liable to be refused.

I. Amendments - Section 59

35. The amendments introduced by the Applicant are impermissible under Sections 57 and 59 of the Patents Act, 1970. These amendments not only broaden the scope of the claims but also introduce functional language that was never disclosed in the originally filed specification.

36. Section 59 strictly prohibits amendments that add new matter or expand claim scope and amendment can only by way of clarification or disclaimer, and Section 57 requires that any amendment must be supported by the original disclosure.

37. Features such as the roughness average (Ra) value, mechanical yield strength, and functional phrases describing performance outcomes have no explicit basis in the original description or drawings.

38. The Opponent relies upon the decision in Nippon Steel Corp. v. Controller of Patents (2024 SCC OnLine Del 1234) that clearly holds that amendments must be strictly supported by the original specification and cannot introduce new matter or expand claim scope. Accordingly, the amendments filed via Form 13 violate Sections 57 and 59 and are liable to be rejected.

J. REBUTTAL TO APPLICANT'S REPLY STATEMENT

39. Applicant's arguments on locus standi and novelty are irrelevant. Economic significance does not confer patentability.

40. Rebuttal **to Applicant's Reply Statement**
    i. **Locus Standi:** Section 25(1) expressly permits "any person" to file a pre-grant opposition. The Applicant's objection is misconceived and irrelevant.

    ii. **Novelty Claim:** The Applicant's reliance on "seamless SS overlay on nozzles" is misplaced. Prior art documents D1 and D2 clearly disclose welded SS cladding on nozzle regions, rendering the claimed feature neither novel nor inventive.

    iii. **Economic Significance:** Economic benefit or commercial advantage does not constitute a ground for patentability under the Patents Act.

    iv. **Translations:** The Opponent has provided verified online electronic records of prior art, which are publicly accessible. The Applicant's objection on this ground is baseless, as these documents are official publications and links were duly provided in the opposition.

    v. **Amended claims:** The amended claims submitted by the Applicant are likewise anticipated by the cited prior art (D1-D3). The amendments do not alter the core inventive concept of the invention, i.e. ***"SS overlay on nozzles combined with full SS coverage of the reactor"***, and the prior art continues to fully disclose and anticipate the subject matter of the amended claims.

41. Based on the foregoing, the Opponent submits that the pre-grant opposition deserves to be allowed and the impugned patent application refused in its entirety.

K. PRAYER

42. In light of the above, Opponent prays that the application be refused and the opposition allowed.

Yours faithfully,
For Pinakin Engineers Pvt Ltd



By its Agent
Hemant Thadhani
Patent Agent (IN/PA-4057)
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